



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2025

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
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- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

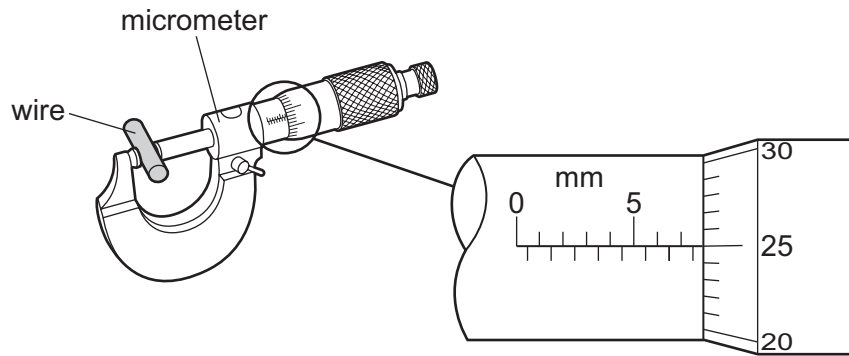
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

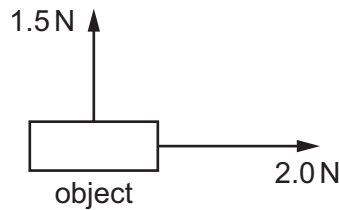


- 1 A micrometer is used to measure the diameter of a piece of wire.



What is the reading shown on the micrometer?

- A** 7.25 mm **B** 7.75 mm **C** 10.25 mm **D** 25.77 mm
- 2 Two forces act on an object. 1.5 N acts vertically upwards and 2.0 N acts horizontally.



What is the magnitude of the resultant force?

- A** 0.50 N **B** 2.5 N **C** 3.0 N **D** 3.5 N
- 3 Which equation defines acceleration?
- A** $\frac{\text{change in distance}}{\text{time taken}}$
- B** $\frac{\text{change in distance in a fixed direction}}{\text{time taken}}$
- C** $\frac{\text{change in speed}}{\text{time taken}}$
- D** $\frac{\text{change in velocity}}{\text{time taken}}$
- 4 The acceleration of free fall on the surface of the Moon is $\frac{1}{6}$ the acceleration of free fall on the surface of the Earth.

What is the approximate weight of a 1 kg mass on the Moon?

- A** 0.98 N **B** 1.6 N **C** 9.8 N **D** 16 N

- 5 A ship uses ultrasound to measure the depth of the sea.

There is a time interval of 0.52 s between transmitting and detecting a pulse of ultrasound.

The speed of ultrasound in sea water is 1500 m/s.

What is the depth of the sea?

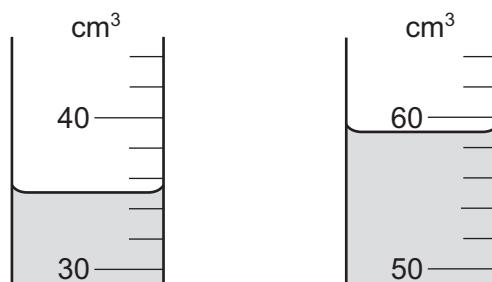
- A** 390 m **B** 780 m **C** 1600 m **D** 5800 m

- 6 Which row shows a definition of the mass of an object and an instrument used to measure mass?

	definition of mass	measuring instrument
A	force experienced by an object	electronic balance
B	force experienced by an object	force meter
C	quantity of matter in an object	electronic balance
D	quantity of matter in an object	force meter

- 7 A quantity of water is poured into a measuring cylinder. A small piece of rock is then added carefully.

The diagrams show the water levels and the measuring cylinder scales.

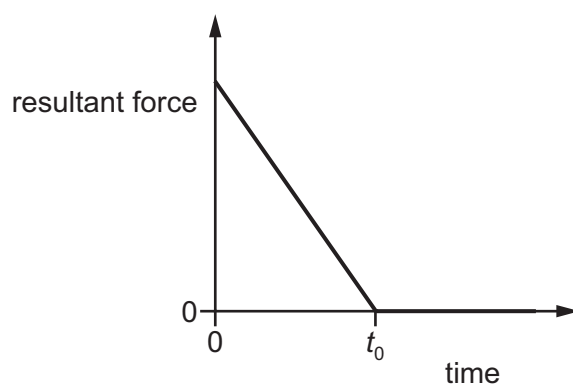


What are the correct values for the volumes of water and rock?

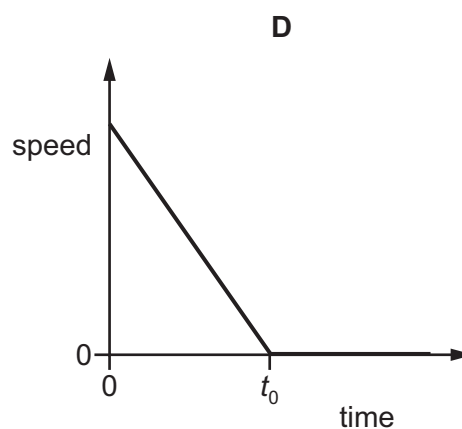
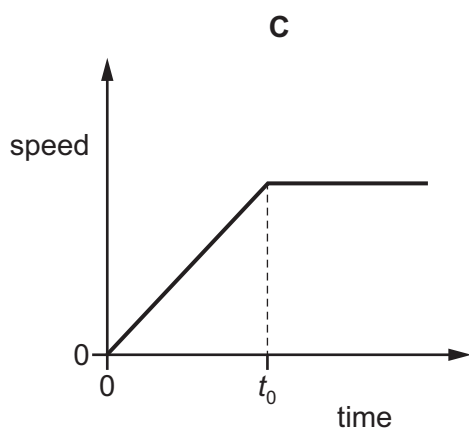
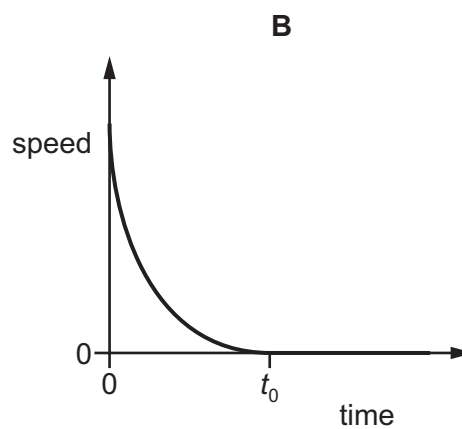
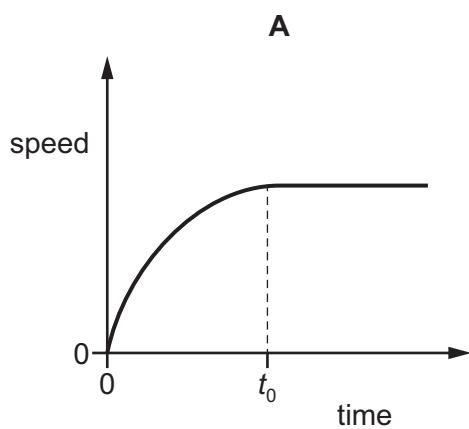
	volume of water / cm ³	volume of rock / cm ³
A	32.5	22.0
B	32.5	54.5
C	35.0	24.0
D	35.0	59.0

- 8 Which list contains only quantities that can be changed by a force?
- A** mass, shape, velocity
B mass, shape, volume
C mass, velocity, volume
D shape, velocity, volume
- 9 A resultant force acts on an object and causes it to move in a straight line.

The graph shows how the resultant force varies with time.



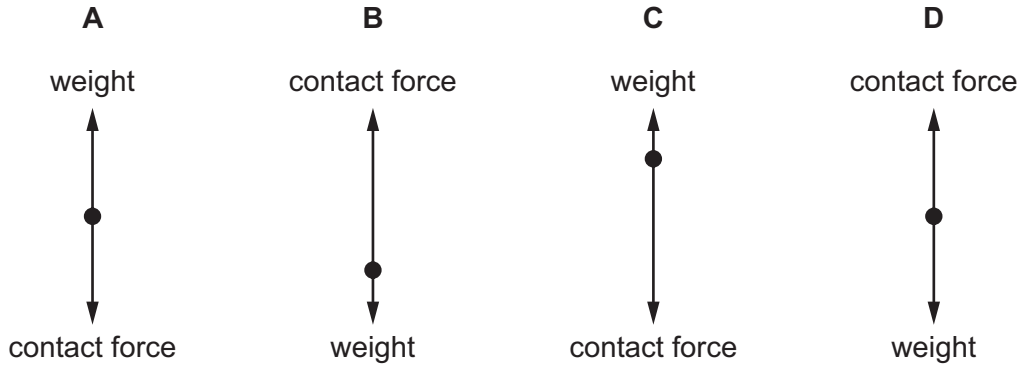
Which graph is the speed–time graph for the object?



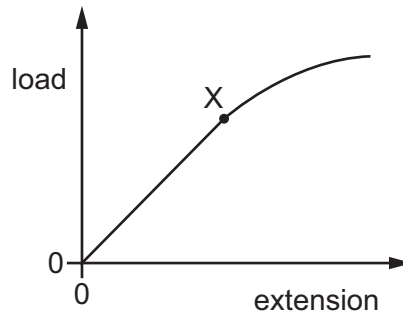
- 10** An object is at rest on a table.

The length of the arrows in the free-body diagrams represent the magnitude and the direction of the forces acting on the object.

Which diagram shows the free-body diagram for the object?



- 11** The load–extension graph for a spring is shown.

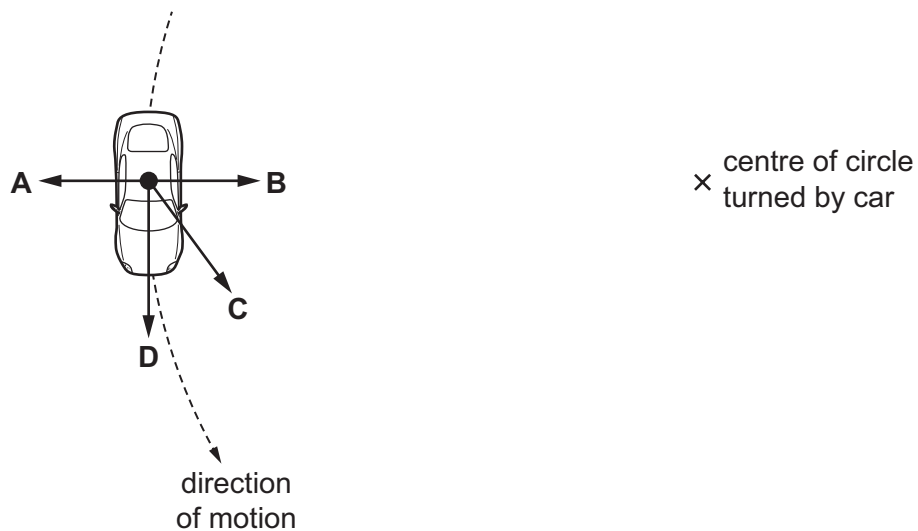


What does point X show?

- A** the elastic deformation
- B** the limit of proportionality
- C** the maximum extension
- D** the spring constant

- 12** A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?



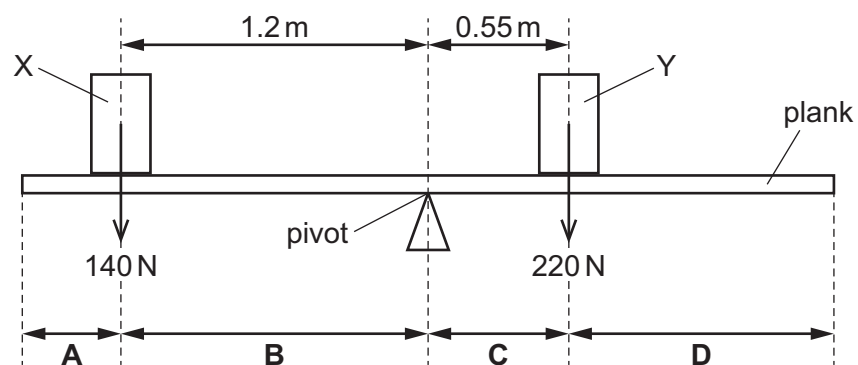
- 13** A long plank is pivoted at its centre of gravity.

Two objects, X and Y, are then placed on the plank.

X has a weight of 140 N and is 1.2 m to the left of the pivot. Y has a weight of 220 N and is 0.55 m to the right of the pivot.

A third object of weight 110 N is placed on the plank so that the plank is in equilibrium.

On which part of the plank is the object placed?



- 14** Which statement about centre of gravity is correct?

- A** Objects with a centre of gravity at the same height are less stable when the base is larger.
- B** Objects with a centre of gravity at the same height are more stable when the base is larger.
- C** Objects with higher centres of gravity and smaller bases are more stable.
- D** Objects with identical bases are more stable when the centre of gravity is higher.

- 15 Energy is stored in batteries, fuel and food.

What is the name of this energy store?

- A chemical
- B electrostatic
- C nuclear
- D thermal

- 16 A rocket has mass M when empty. The rocket carries an additional mass M of fuel.

The rocket and fuel travel at a speed v . When the engine of the rocket is fired, all of the fuel is expelled and the speed of the rocket increases to $2v$.

What happens to the kinetic energy of the **rocket**?

- A It doubles.
- B It halves.
- C It increases by a factor of four.
- D It stays the same.

- 17 What is the definition of efficiency?

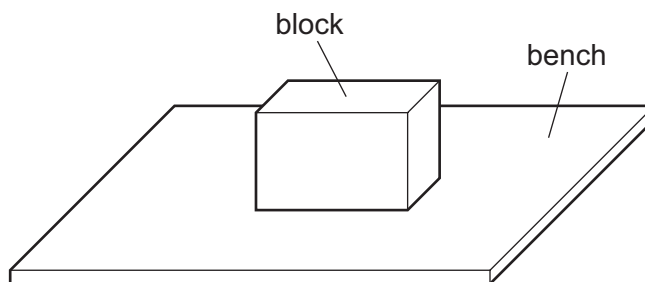
- A $\text{efficiency} = \frac{\text{useful energy input}}{\text{total energy output}}$
- B $\text{efficiency} = \frac{\text{total energy input}}{\text{total energy output}}$
- C $\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$
- D $\text{efficiency} = \frac{\text{total energy output}}{\text{total energy input}}$

- 18 An object of mass m is lifted through a vertical distance d in time t .

Which equation is used to calculate the power P required to lift the object?

- A $P = \frac{md}{gt}$
- B $P = \frac{mgd}{t}$
- C $P = \frac{dt}{mg}$
- D $P = \frac{m}{gdt}$

- 19 A block of wood is placed on a bench.



The pressure exerted on the bench due to the block is 1.0 Pa .

The area of the block in contact with the base is A .

Which statement is possible?

- A** The mass of the block is 1.0 kg and A is 1.0 cm^2 .
 - B** The mass of the block is 1.0 kg and A is 1.0 m^2 .
 - C** The weight of the block is 1.0 N and A is 1.0 cm^2 .
 - D** The weight of the block is 1.0 N and A is 1.0 m^2 .
- 20 Different liquids are poured into four different containers.

In which container is the force exerted by the liquid on the bottom of the container the greatest?

	area of base / cm^2	density of liquid / g/cm^3	depth of liquid / cm
A	10	1.3	50
B	20	0.80	80
C	40	1.0	60
D	50	0.92	75

21 A student compares the properties of the three states of matter.

The partially completed table in the student's notebook for one of the states is shown.

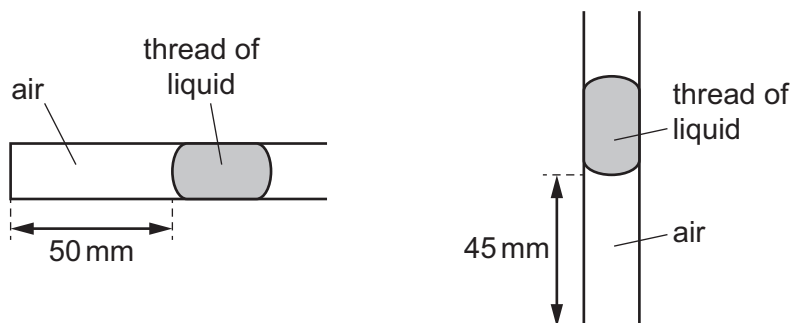
state	1
arrangement of particles	random
motion of particles	moving freely
forces between particles	2
expansion when heated	3

Which words complete the table?

	1	2	3
A	gas	weakest	smallest
B	gas	weakest	greatest
C	gas	strongest	greatest
D	liquid	strongest	smallest

22 Air is trapped in a glass tube by a liquid.

When the tube is horizontal, the length of the trapped air is 50 mm and its pressure is $1.0 \times 10^5 \text{ Pa}$.



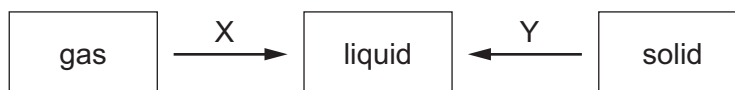
When the tube is vertical, with the closed end downwards, the length of the trapped air is 45 mm.

The temperature of the air remains constant.

What is the pressure of the trapped air when the tube is vertical?

- A** $0.50 \times 10^5 \text{ Pa}$
- B** $0.90 \times 10^5 \text{ Pa}$
- C** $1.1 \times 10^5 \text{ Pa}$
- D** $1.9 \times 10^5 \text{ Pa}$

- 23 The diagram shows two changes of state, X and Y.



Which row gives the names of the two changes of state?

	X	Y
A	boiling	freezing
B	boiling	melting
C	condensation	freezing
D	condensation	melting

- 24 Energy can be transferred by thermal radiation.

Four surfaces have different colours and textures. The surfaces are at equal temperatures and have equal surface areas.

Which surface transfers energy by thermal radiation at the fastest rate?

- A** black and dull surface
- B** black and shiny surface
- C** white and dull surface
- D** white and shiny surface

- 25 Four copper spheres have identical surface colours and identical surface textures. They have different surface temperatures and surface areas.

Which sphere emits infrared radiation at the greatest rate?

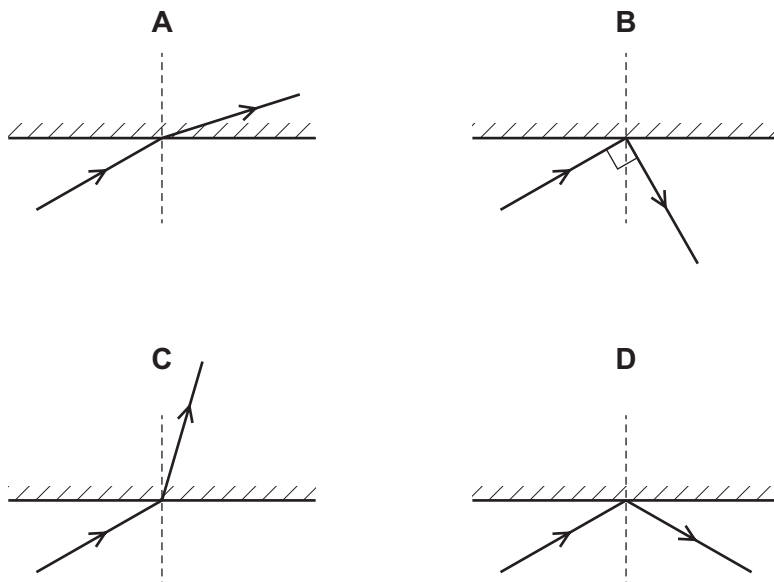
	surface temperature / °C	surface area / cm ²
A	500	120
B	500	480
C	1000	120
D	1000	480

- 26 The table shows the speed and the wavelength for different sound waves in four different materials.

wave	speed of sound m/s	wavelength /m
1	340	0.19
2	1500	0.050
3	4000	0.23
4	6700	0.31

Which waves are ultrasound waves?

- A 1 and 3 B 1 only C 2 and 4 D 4 only
- 27 Light passes into a transparent material that has a refractive index n .
The angle of incidence at the surface of the transparent material is i .
Which equation is used to calculate the angle of refraction r ?
- A $r = n \sin i$ B $i = n \sin r$ C $\sin r = \frac{n}{\sin i}$ D $\sin r = \frac{\sin i}{n}$
- 28 Which diagram shows a ray of light that is reflected by a plane mirror?



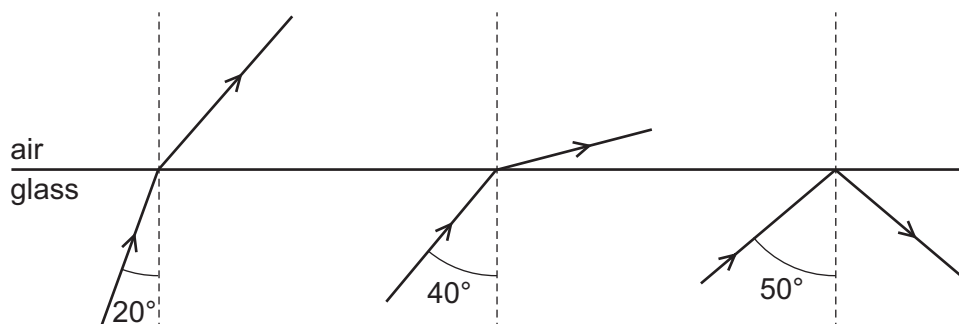
- 29 An object is placed in front of a plane mirror.

Which statement describes the image produced?

- A real and smaller than the object
- B real and the same size as the object
- C virtual and smaller than the object
- D virtual and the same size as the object

- 30 Three rays of light are incident on the boundary between a glass block and air.

The angles of incidence are different.



What is a possible critical angle for light in the glass?

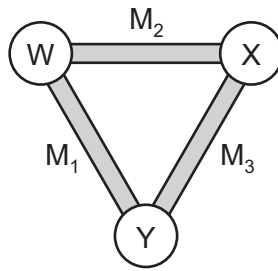
- A 15°
 - B 30°
 - C 45°
 - D 60°
- 31 A student does an investigation using four thin lenses. The lenses are used to produce focused images of four different objects.

The student measures the length of the image and the length of the object. The results are shown in the table.

In which case is the magnification produced the greatest?

	image length / cm	object length / cm
A	3.5	0.80
B	4.7	0.60
C	4.8	0.70
D	5.2	0.90

- 32 Magnets M_1 , M_2 and M_3 are joined by iron balls W, X and Y as shown.



Magnetic poles are induced in the iron balls W and X as shown.

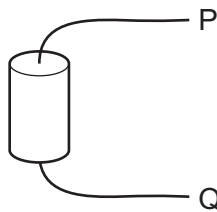


Which diagram shows the poles induced in ball Y?



- 33 A cardboard tube has two wires, P and Q, connected to its ends. Inside the tube, wire P and wire Q are connected to two resistors.

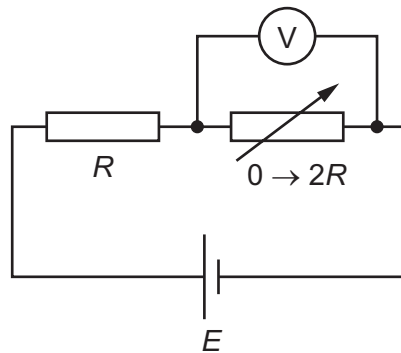
The potential difference (p.d.) between wire P and wire Q is 9.0 V and the current in the wires is 0.75 A.



Which combination of resistors is inside the tube?

- A a 0.75Ω resistor and a 6.0Ω resistor in parallel
- B a 0.75Ω resistor and a 6.0Ω resistor in series
- C a 20Ω resistor and a 30Ω resistor in parallel
- D a 20Ω resistor and a 30Ω resistor in series

- 34 A resistor of resistance R is connected to a variable resistor, a voltmeter and a power supply of electromotive force (e.m.f.) E .



The resistance of the variable resistor increases from 0 to $2R$.

What happens to the reading on the voltmeter?

- A** It increases from 0 to $\frac{1}{3}E$.
- B** It increases from 0 to $\frac{1}{2}E$.
- C** It increases from 0 to $\frac{2}{3}E$.
- D** It increases from 0 to E .
- 35 Which device works due to electromagnetic induction?
- A** a battery
- B** an electromagnet
- C** a motor
- D** a transformer
- 36 A beta-particle (β -particle) travels to the right at a constant speed.



Which field deflects the particle towards the top of the page?

- A** a horizontal magnetic field directed out of the page
- B** a horizontal magnetic field directed into the page
- C** a horizontal electric field directed to the left
- D** a horizontal electric field directed to the right
- 37 Which nuclide is produced when thorium-223, ${}^{223}_{90}\text{Th}$, emits an alpha-particle?

- A** ${}^{219}_{88}\text{Ra}$ **B** ${}^{219}_{92}\text{U}$ **C** ${}^{227}_{88}\text{Ra}$ **D** ${}^{227}_{92}\text{U}$

- 38** A radioactive source is placed 2.0 cm away from a Geiger–Müller (G.M.) tube that is connected to a counter.

A sheet of paper is placed between the source and the G.M. tube. The count rate measured by the counter decreases to a value greater than the background radiation count rate.

A sheet of aluminium replaces the paper and the count rate decreases to the background radiation count rate.

What is the radiation emitted by the source?

- A** alpha-particles (α -particles), beta-particles (β -particles) and gamma radiation (γ -radiation)
 - B** alpha-particles (α -particles) and beta-particles (β -particles) only
 - C** alpha-particles (α -particles) and gamma radiation (γ -radiation) only
 - D** beta-particles (β -particles) and gamma radiation (γ -radiation) only
- 39** Which statement about the Earth is correct?
- A** It has an elliptical orbit around the Sun.
 - B** It is a star that takes approximately 365 days to orbit the Sun.
 - C** It rotates on its axis once every 28 hours.
 - D** It orbits the Moon exactly once a year.

- 40** A teacher asks two students to describe redshift.

Student 1 says: It is a decrease in the observed wavelength of electromagnetic radiation from stars.

Student 2 says: It occurs because stars and galaxies move away from the Earth.

Which students are correct?

- A** both student 1 and student 2
- B** student 1 only
- C** student 2 only
- D** neither student 1 nor student 2

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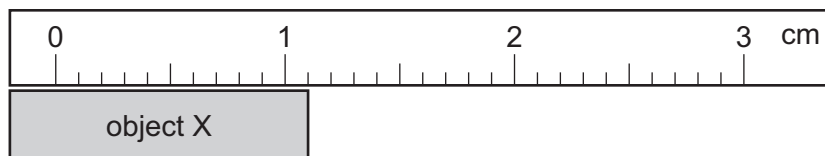
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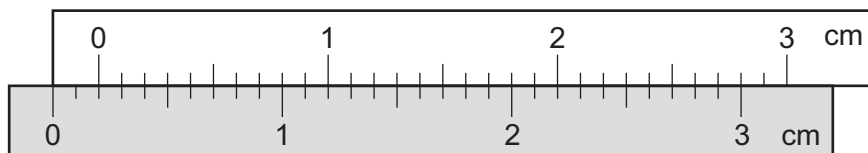
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- 1 Object X is placed against a ruler.



This ruler is placed against another ruler to measure the end correction.



What is the length of the object X?

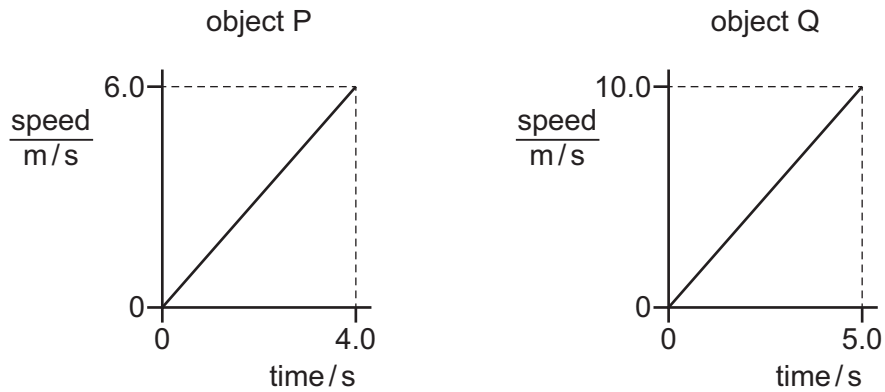
- A** 0.9 cm **B** 1.1 cm **C** 1.3 cm **D** 1.5 cm

- 2 Quantities can be described as scalar or vector.

Which row shows scalar quantities and vector quantities?

	scalar quantities	vector quantities
A	displacement, force, temperature	distance, weight, mass
B	displacement, mass, force	distance, weight, temperature
C	distance, force, weight	displacement, mass, temperature
D	distance, mass, temperature	displacement, force, weight

- 3 The diagram shows the speed–time graphs for two objects moving with uniform acceleration.

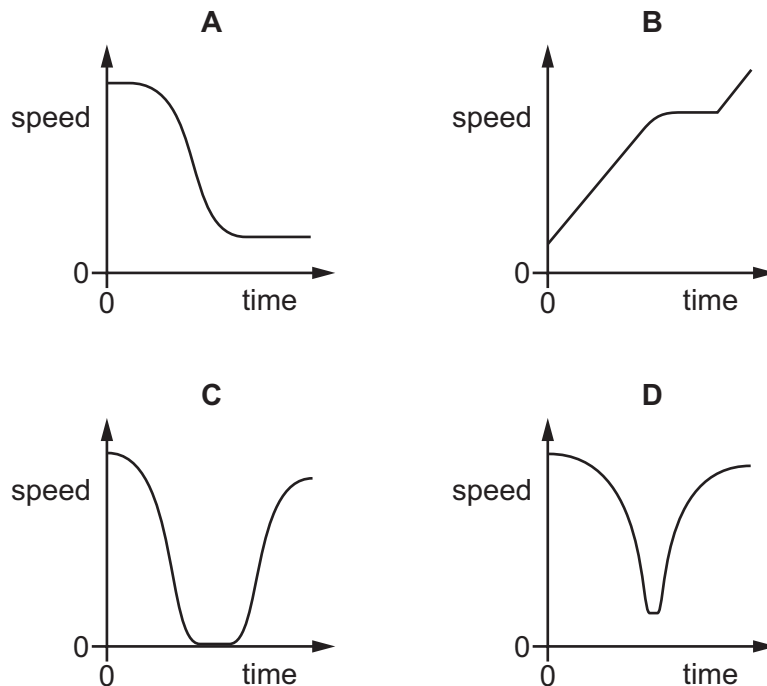


A student uses the graphs to make two conclusions.

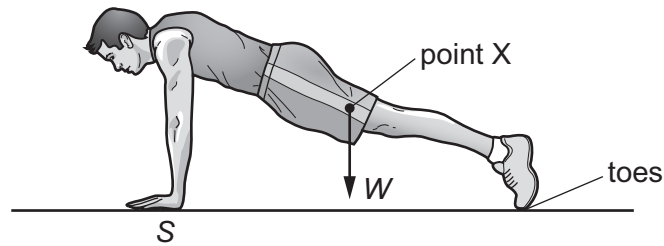
- 1 The distance travelled by object P is 26 m less than the distance travelled by object Q.
- 2 The acceleration of object P is 0.50 m/s^2 less than the acceleration of object Q.

Which conclusions are correct?

- A** neither 1 nor 2
B 1 only
C 2 only
D both 1 and 2
- 4 Which speed–time graph represents the motion of a railway train making a short stop at a station?



- 5 The diagram shows an athlete doing press-ups on a floor. The athlete is using both hands. The area of contact of each hand with the floor is S . The athlete's toes act as a pivot during each press-up.



The athlete's weight W acts through point X. The distance between his toes and point X is three quarters of the distance between his toes and his shoulders.

What is the pressure exerted on the palm of each of the athlete's hands?

- A $\frac{W}{8S}$ B $\frac{3W}{8S}$ C $\frac{W}{4S}$ D $\frac{3W}{4S}$

- 6 A parachutist falls towards the ground at constant speed with a fully open parachute.

Which statement is correct?

- A There are **no** forces acting on the parachutist.
 B The upward force on the parachute is equal to the weight of the parachutist.
 C The upward force on the parachute is greater than the weight of the parachutist.
 D The upward force on the parachute is less than the weight of the parachutist.

- 7 A student measures the density of an irregularly shaped stone.

Which items of equipment are needed?

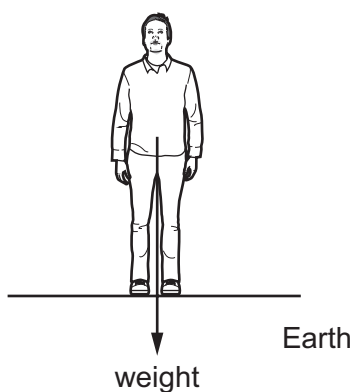
- A a balance and a measuring cylinder containing water
 B a balance and a ruler
 C a ruler and a measuring cylinder containing water
 D a measuring cylinder containing water only

- 8 The table shows how the extension of a spring varies with load.

load / N	0	2	4	6	8	10	12	14	16
extension / cm	0	3	6	9	12	15	20	27	38

Between which two loads is the limit of proportionality?

- A 0 N and 2 N
 B 8 N and 10 N
 C 10 N and 12 N
 D 14 N and 16 N
- 9 A man stands at rest on the surface of the Earth.



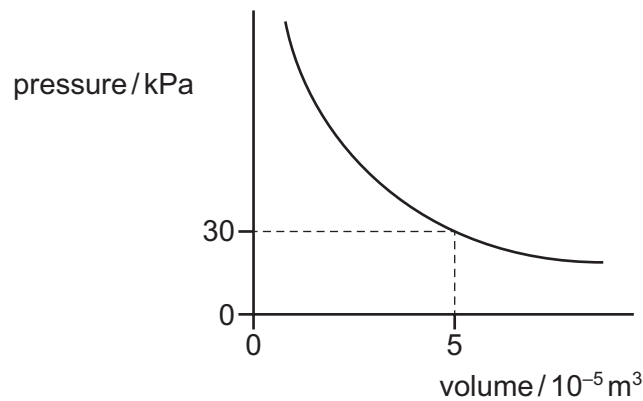
The weight of the man is the gravitational force that acts on him.

According to Newton's third law, the weight of the man is one force in a pair of forces.

Which force is the other force in the pair?

- A a force downwards on the Earth
 B a force downwards on the man
 C a force upwards on the Earth
 D a force upwards on the man

- 10 The pressure and volume of a sample of gas at constant temperature are measured and plotted on a graph.

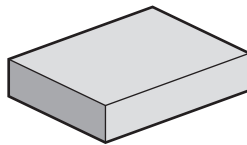


The pressure is decreased to 12.5 kPa.

What is the new volume of this sample of gas?

- A** $1.2 \times 10^{-4} \text{ m}^3$ **B** $4.8 \times 10^{-4} \text{ m}^3$ **C** $1.2 \times 10^{-2} \text{ m}^3$ **D** $4.8 \times 10^4 \text{ m}^3$

- 11 The diagram shows a solid metal block.

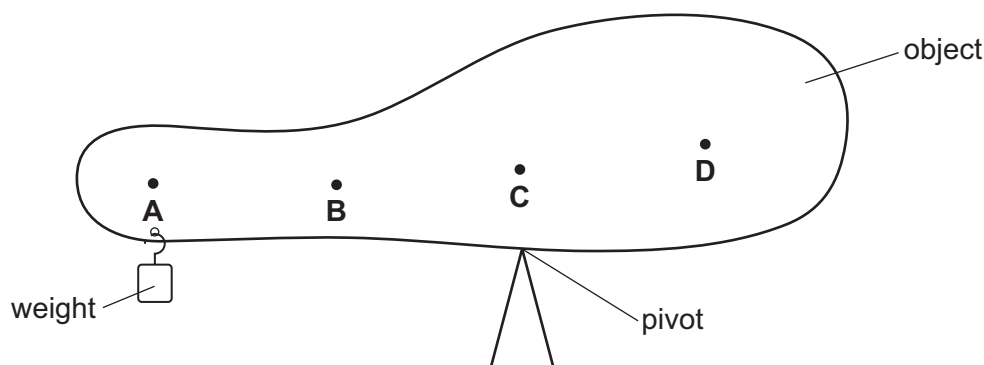


Which equation is used to calculate the density of the metal from which the block is made?

- A** density = mass $\times$ volume
- B** density = $\frac{\text{mass}}{\text{volume}}$
- C** density = $\frac{\text{weight}}{\text{area}}$
- D** density = $\frac{\text{weight}}{\text{volume}}$

- 12** A student balances a non-uniform object on a pivot. To do this, a weight is suspended near the left-hand end of the object, as shown.

Where is the centre of gravity of the object?



- 13** Seven energy sources are listed.

fossil fuel hydroelectric solar nuclear geothermal wind tides

How many of these energy sources are renewable energy sources?

- A 3 B 4 C 5 D 6**

- 14** A student heats a substance in which the particles are in fixed positions and vibrate.

After a short period of heating, the particles start to flow past each other, but the forces between the particles remain strong.

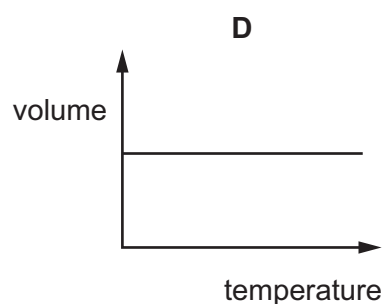
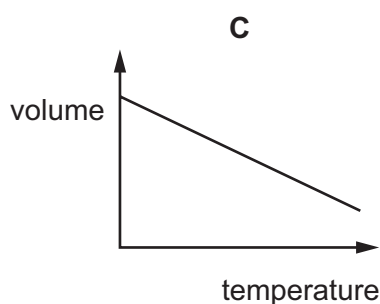
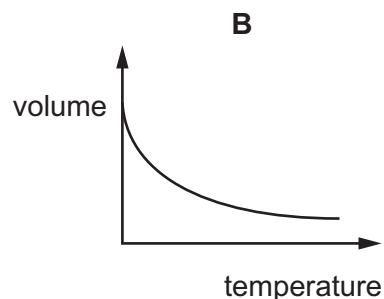
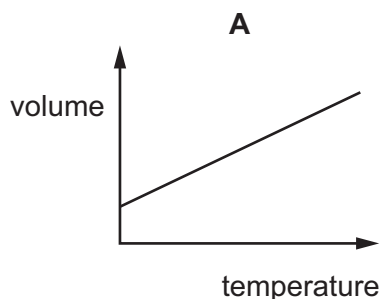
What has happened to the substance?

- A** It has condensed.
- B** It has evaporated.
- C** It has frozen.
- D** It has melted.

- 15** A student investigates how changing the temperature of a gas affects the volume of the gas.

The gas is kept at a constant pressure.

Which graph shows the relationship between the temperature and the volume of a sample of gas at constant pressure?



- 16** A sample of a solid melts and becomes a liquid.

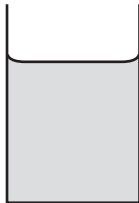
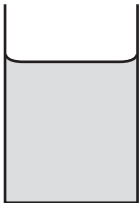
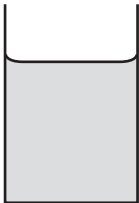
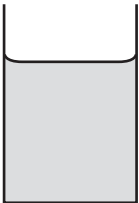
Which statement about the sample during the process of melting is correct?

- A** Its temperature changes from a negative to a positive value.
- B** Its temperature is constant.
- C** The forces between its particles become stronger.
- D** The kinetic energy of its particles increases.

- 17** Four metal cans are identical except for the colour and the texture of their outer surfaces.

100 cm³ of water at 70 °C is poured into each can.

Which can cools the most rapidly?

	A	B	C	D
				
description of outer surface:	black, rough	black, shiny	white, rough	white, shiny

- 18** The sentences describe sound waves. Three words have been omitted.

Sound waves are longitudinal waves that consist of regions of higher pressure called1..... and regions of lower pressure called2..... .

An echo occurs when a sound wave is3..... .

Which words complete gaps 1, 2 and 3?

	1	2	3
A	compressions	rarefactions	reflected
B	compressions	rarefactions	refracted
C	rarefactions	compressions	refracted
D	rarefactions	compressions	reflected

- 19** A ray of light strikes a plane mirror at an angle of incidence of 20°.

The angle of incidence is then increased by 5°.

What is the new angle between the incident ray and the reflected ray?

- A** 10° **B** 25° **C** 45° **D** 50°

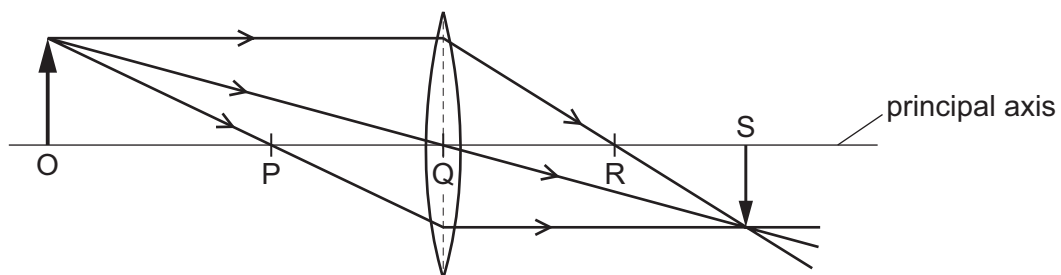
20 The following lists show colours of the spectrum.

Which list shows these colours in order of increasing frequency?

- A blue → violet → red → orange → yellow → green
- B green → blue → violet → red → orange → yellow
- C red → orange → violet → yellow → green → blue
- D red → orange → yellow → green → blue → violet

21 An object O is placed near to a thin converging lens.

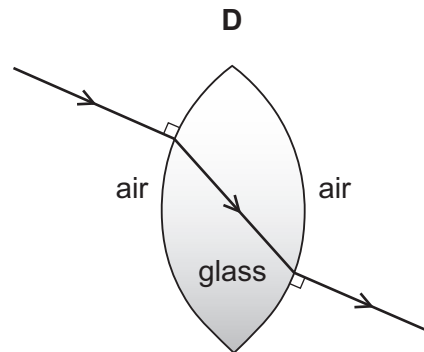
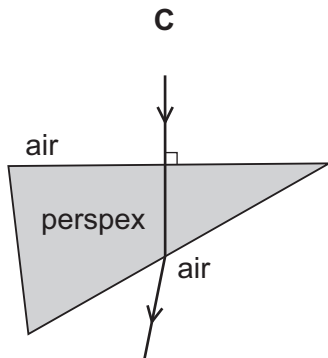
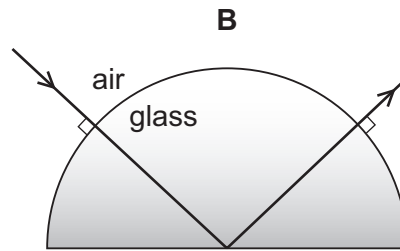
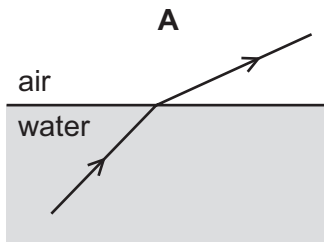
The diagram shows three rays of light from the object passing through the lens.



Which row describes the image formed and gives the position of a principal focus (focal point) of the lens?

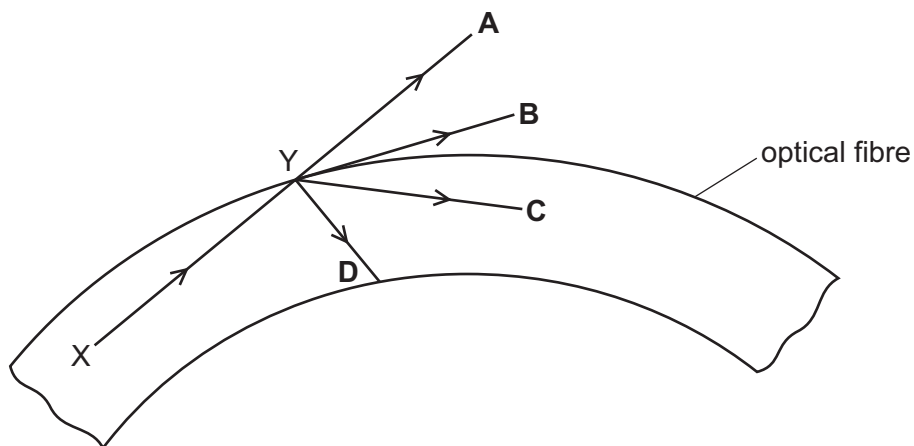
	image formed	position of principal focus
A	real	R
B	real	Q
C	virtual	P
D	virtual	S

22 In which diagram is the path of the light ray **not** correct?



23 A ray of light travels from X to Y along an optical fibre. The angle of incidence at Y is greater than the critical angle.

In which direction does the ray of light travel after reaching point Y?



24 A firework is launched vertically from the ground.

When the firework reaches a height of 360 m, the firework explodes, producing coloured light.

A student on the ground below the firework incorrectly measures the time between seeing the coloured light and hearing the explosion. The student obtains a value of 1.8 s.

What is the speed of sound that the student's value suggests?

- A** 200 m/s **B** 360 m/s **C** 400 m/s **D** 650 m/s

- 25** A loudspeaker produces a sound wave of frequency 50 Hz.

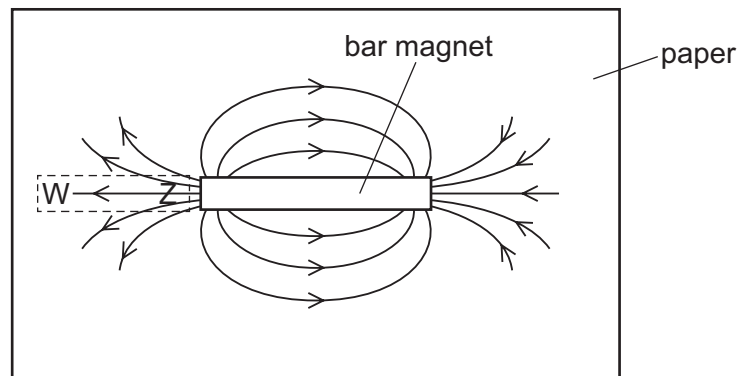
The amplitude of the sound wave is increased.

What is heard?

- A** a louder sound of a higher pitch
- B** a louder sound of the same pitch
- C** a sound of higher pitch but the same loudness
- D** a sound of the same pitch and same loudness

- 26** A bar magnet is placed on a sheet of paper.

The diagram shows the field lines due to the magnet.



A soft-iron bar is placed on the paper on the rectangle marked WZ. The soft-iron bar becomes an induced magnet.

What are the magnetic poles at the ends of the soft-iron bar?

	pole at W	pole at Z
A	N	N
B	N	S
C	S	N
D	S	S

- 27** Which material is an electrical conductor at room temperature?

- A** aluminium
- B** glass
- C** plastic
- D** rubber

- 28** A student investigates electrostatic charge using a balloon and a cloth. The balloon and cloth are initially uncharged.

The student rubs the balloon with the cloth. The balloon gains a negative charge of magnitude Q .

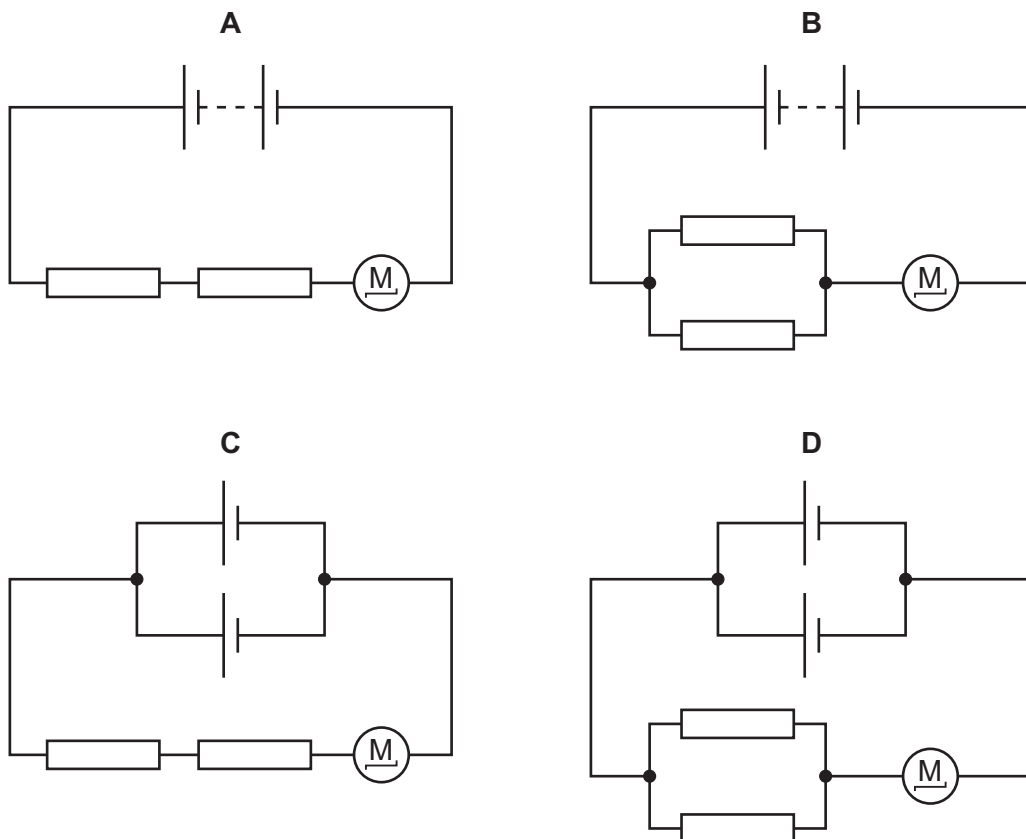
The cloth becomes charged and a force acts between the balloon and the cloth.

Which row correctly describes the magnitude and sign of the charge on the cloth and the type of force between the balloon and the cloth?

	magnitude of charge on cloth	sign of charge on cloth	type of force
A	equal to Q	positive	attraction
B	equal to Q	negative	repulsion
C	greater than Q	positive	repulsion
D	greater than Q	negative	attraction

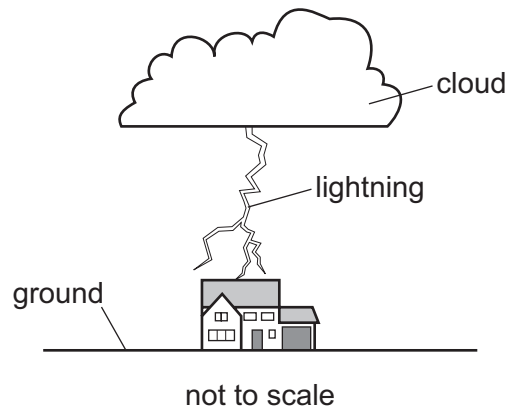
- 29** A student connects four circuits using two identical cells, two identical resistors and a motor.

In which circuit does the coil of the motor rotate the fastest?



- 30 The diagram shows a lightning strike.

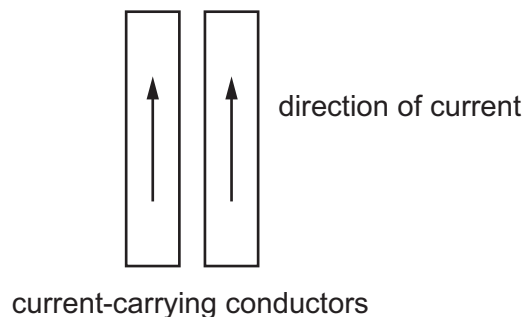
Charge flows between the cloud and objects on the ground.



During the lightning strike, there is a current of $2.8 \times 10^4 \text{ A}$ between the cloud and the house for 52 ms.

How much charge flows during the lightning strike?

- A** 1500 C **B** 15 000 C **C** 24 000 C **D** 54 000 C
- 31 Two current-carrying conductors are placed next to each other. The current in the conductors is in the same direction.



Which statement about the forces experienced by the conductors is correct?

- A** The forces are in the same direction, and the conductors are forced apart.
B The forces are in opposite directions, and the conductors are forced apart.
C The forces are in the same direction, and the conductors are forced towards each other.
D The forces are in opposite directions, and the conductors are forced towards each other.
- 32 Which component is **not** needed to make a simple d.c. motor work?
- A** battery
B brushes
C diode
D split-ring commutator

33 Which component is used in an electric circuit to detect changes in temperature?



34 A kettle of power 2.0 kW is powered by the 230 V mains supply.

What is the appropriate rating for the fuse in the circuit?

- A** 1 A **B** 5 A **C** 13 A **D** 20 A

35 A student uses a simple iron-cored transformer. The primary coil has 250 turns. The secondary coil has 200 turns.

The input voltage to the primary coil is 12 V.

What is the output voltage?

- A** 0.069 V **B** 0.11 V **C** 9.6 V **D** 15 V

36 Four statements about an atom are listed.

- 1 The atom has a nucleus surrounded by positively charged electrons.
- 2 Most of the atom is empty space.
- 3 The nucleus is positively charged.
- 4 The nucleus is very large compared with the rest of the atom.

Which statements are correct?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

37 A smoke alarm contains a radioactive source, S. The useful emissions from S are blocked by a few centimetres of air.

What is the useful radioactive radiation emitted by S?

- A** alpha and gamma
B alpha only
C beta and gamma
D beta only

38 The planets in the Solar System orbit the Sun.

Which statement is correct?

- A** There is a force on each planet away from the Sun.
- B** There is a force on each planet in the direction in which it travels.
- C** There is a force on each planet opposite to the direction in which it travels.
- D** There is a force on each planet towards the Sun.

39 A star is a distance of 4.2 light-years from the Earth.

What is 4.2 light-years in metres?

- A** $1.1 \times 10^{13} \text{ m}$ **B** $6.6 \times 10^{14} \text{ m}$ **C** $9.5 \times 10^{15} \text{ m}$ **D** $4.0 \times 10^{16} \text{ m}$

40 What is a protostar?

- A** a cloud of gas that is collapsing and decreasing in temperature
- B** a cloud of gas that is collapsing and increasing in temperature
- C** a star that is expanding and decreasing in temperature
- D** a star that is expanding and increasing in temperature

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Cambridge O Level

PHYSICS**5054/12**

Paper 1 Multiple Choice

October/November 2025

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	B	1
2	B	1
3	D	1
4	B	1
5	A	1
6	C	1
7	C	1
8	D	1
9	A	1
10	D	1
11	B	1
12	B	1
13	C	1
14	B	1
15	A	1
16	C	1
17	C	1
18	B	1
19	D	1
20	D	1
21	B	1
22	C	1
23	D	1
24	A	1
25	D	1
26	C	1
27	D	1
28	D	1

Question	Answer	Marks
29	D	1
30	C	1
31	B	1
32	B	1
33	C	1
34	C	1
35	D	1
36	A	1
37	A	1
38	B	1
39	A	1
40	C	1

Cambridge O Level

PHYSICS**5054/11**

Paper 1 Multiple Choice

October/November 2025**MARK SCHEME**Maximum Mark: 40

Published

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	D	1
3	C	1
4	C	1
5	B	1
6	B	1
7	A	1
8	C	1
9	C	1
10	A	1
11	B	1
12	D	1
13	C	1
14	D	1
15	A	1
16	B	1
17	A	1
18	A	1
19	D	1
20	D	1
21	A	1
22	D	1
23	C	1
24	A	1
25	B	1
26	B	1
27	A	1
28	A	1

Question	Answer	Marks
29	B	1
30	A	1
31	D	1
32	C	1
33	B	1
34	C	1
35	C	1
36	C	1
37	B	1
38	D	1
39	D	1
40	B	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2024

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

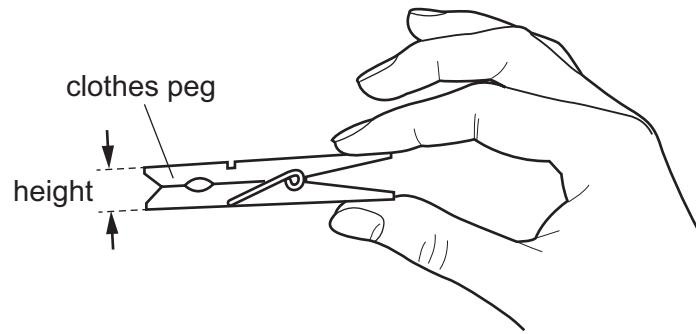
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

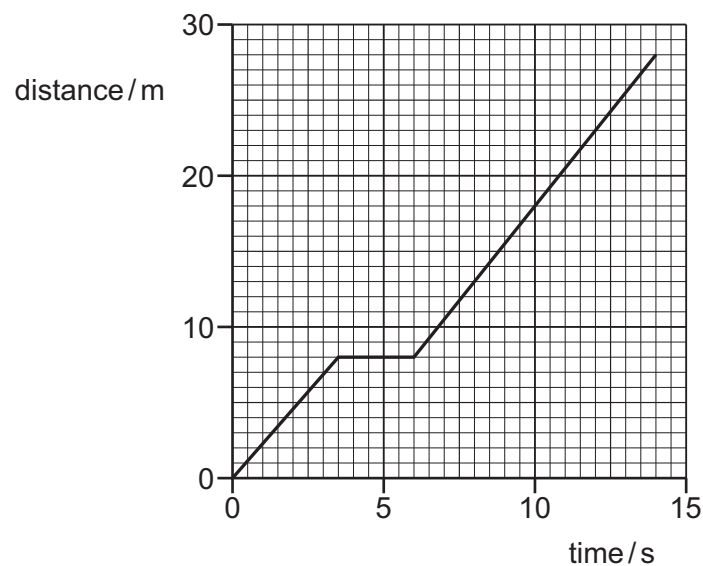


- 1 The diagram shows a clothes peg.



Which device is suitable for obtaining an accurate value for the height of the clothes peg?

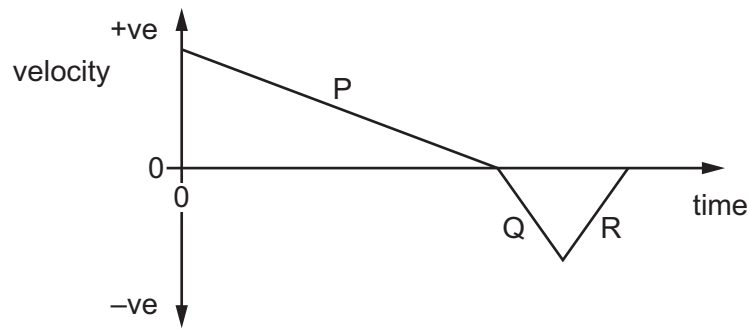
- A a tape
 - B a metre rule
 - C a micrometer
 - D a measuring cylinder
- 2 Which set of quantities are all vectors?
- A acceleration, displacement, velocity
 - B chemical energy, mass, power
 - C extension, force, gravitational potential energy
 - D weight, kinetic energy, work
- 3 The graph shows the motion of a cyclist during a ride that lasts for 14 s.



What is the average speed of the cyclist for the ride?

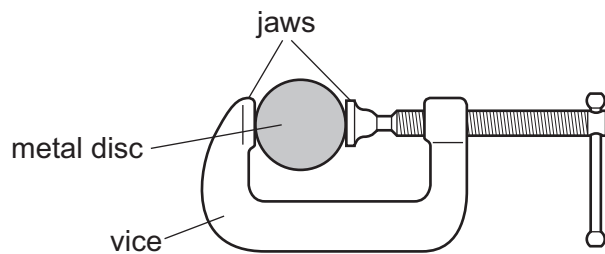
- A 2.0 m/s
- B 2.3 m/s
- C 2.4 m/s
- D 2.5 m/s

- 4 The diagram shows how the velocity of a train varies with time.



Which sections of the graph show where the magnitude of the velocity of the train is decreasing?

- A** P and Q **B** P and R **C** P only **D** R only
- 5 A metal disc is squeezed in the jaws of a vice.



Which property of the metal disc is unchanged when it is squeezed?

- A** mass
B radius
C distance between its atoms
D forces between its atoms
- 6 The gravitational field strength in space is smaller than on the Earth's surface.

How are the mass and the weight of a satellite affected as the satellite moves away from the surface of the Earth and into space?

- A** Both the mass and the weight are unaffected.
B The mass decreases and the weight decreases.
C The mass increases and the weight is unaffected.
D The mass is unaffected and the weight decreases.

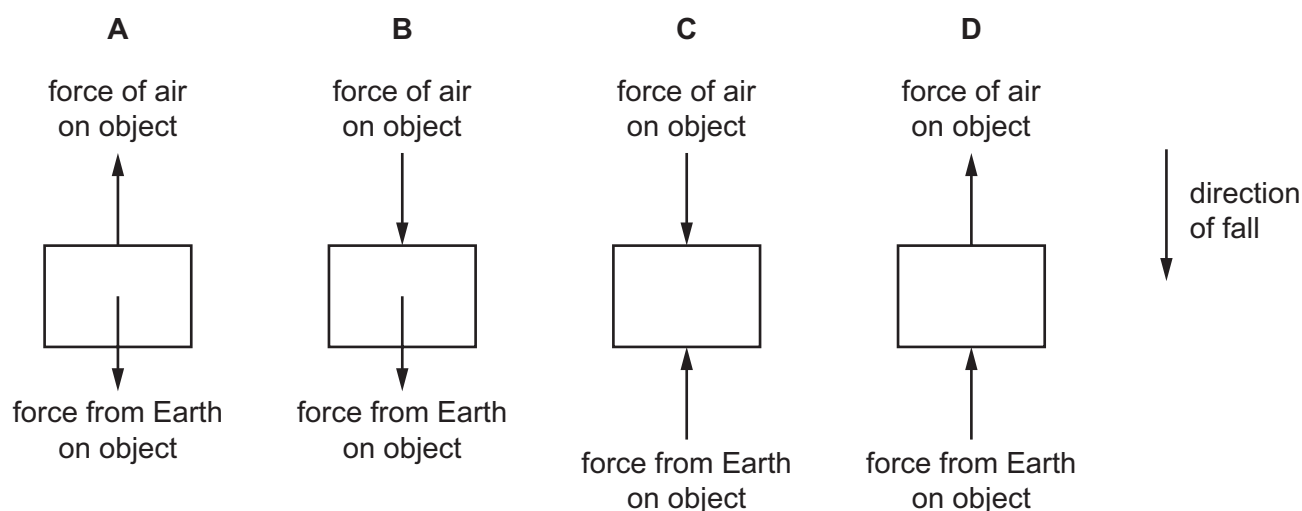
- 7 A material has a density of 2.4 g/cm^3 .

What is the density in kg/m^3 ?

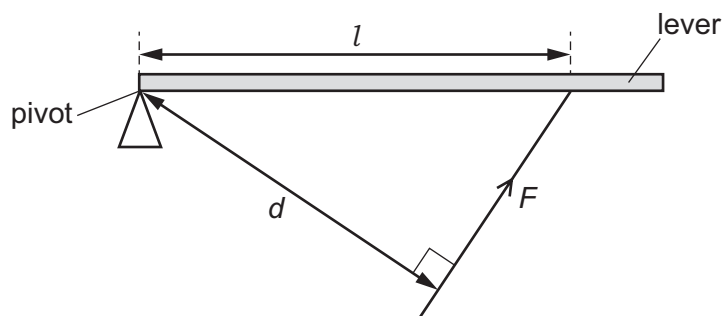
- A 0.0024 kg/m^3
 B 0.24 kg/m^3
 C 24 kg/m^3
 D 2400 kg/m^3

- 8 An object falls vertically through the air.

Which diagram represents the forces acting on the object?



- 9 A force F is applied to a lever at distance l from the pivot as shown.



Length d is the perpendicular distance between the force and the pivot.

Which expression is the moment of force F about the pivot?

- A $\frac{F}{d}$ B $\frac{F}{l}$ C $F \times d$ D $F \times l$

- 10 A rocket is launched vertically upwards by a constant force from its engines.

The data for the launch is shown.

weight of rocket	1.9 MN
average air resistance	0.10 MN
force from engine	3.5 MN

How much of the work done on the rocket is used to increase its speed in the first 100 m?

- A** 150 MJ **B** 170 MJ **C** 350 MJ **D** 550 MJ

- 11 Which equation is used to calculate the efficiency of an energy transfer?

A $\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$

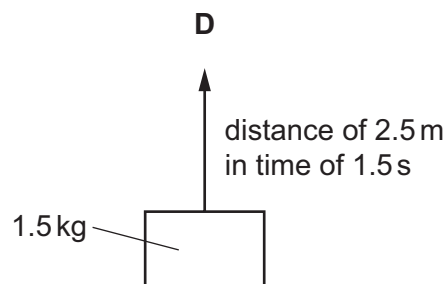
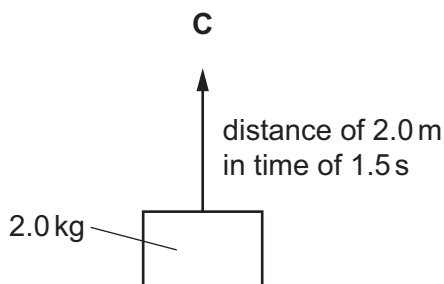
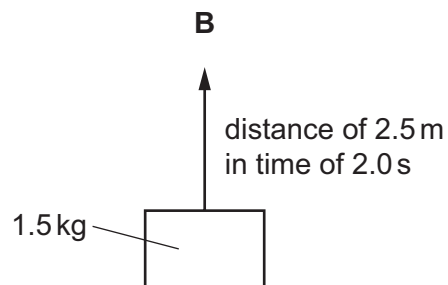
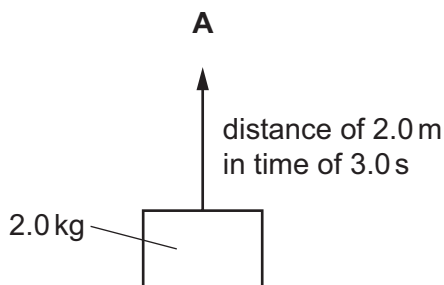
B $\text{efficiency} = \frac{\text{total energy input}}{\text{useful energy output}}$

C $\text{efficiency} = \frac{\text{total energy output}}{\text{total power input}}$

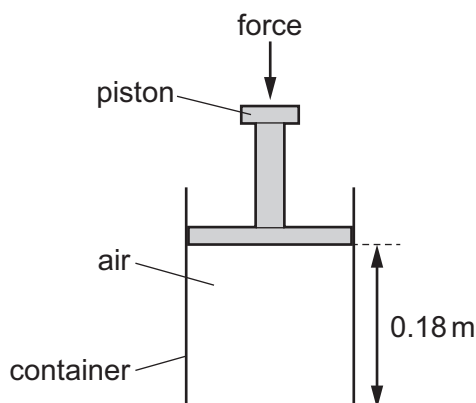
D $\text{efficiency} = (\text{useful energy output}) \times (\text{total energy input})$

- 12 Four objects of different masses are lifted through different distances in different amounts of time.

In which combination of mass, distance and time is the greatest power transferred?



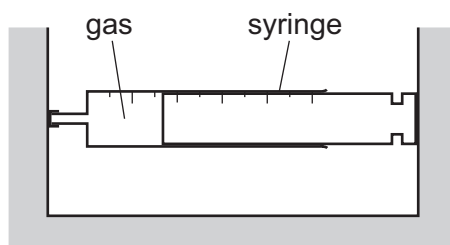
- 13 Air at a pressure of $1.0 \times 10^5 \text{ N/m}^2$ is trapped in a container by a piston.



A force is applied to move the piston down into the container. The temperature of the air remains constant. The distance between the base of the container and the piston decreases from 0.18 m to 0.15 m.

What is the new air pressure inside the container?

- A $0.69 \times 10^5 \text{ N/m}^2$
 B $0.83 \times 10^5 \text{ N/m}^2$
 C $1.2 \times 10^5 \text{ N/m}^2$
 D $1.4 \times 10^5 \text{ N/m}^2$
- 14 A sealed gas syringe contains a gas that is kept at constant volume.



Which row describes how a change in the temperature of the gas affects its pressure?

	temperature	pressure of gas
A	decreases	increases
B	decreases	unchanged
C	increases	increases
D	increases	unchanged

- 15** The melting temperature of gold is 1064°C .

What is the melting temperature of gold on the Kelvin scale?

- A** 791 K **B** 964 K **C** 1164 K **D** 1337 K

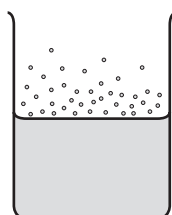
- 16** The table shows the melting and boiling temperatures of three elements.

element	melting temperature / $^{\circ}\text{C}$	boiling temperature / $^{\circ}\text{C}$
argon	-189	-185
nitrogen	-210	-196
oxygen	-218	-183

At which temperature are the three elements in different states?

- A** -220°C **B** -205°C **C** -190°C **D** -175°C

- 17** The diagram shows a liquid at room temperature in a beaker.



What reduces the rate of loss of liquid by evaporation?

- A** blowing air across the top of the beaker
B heating the liquid
C putting a lid on the beaker
D transferring the liquid to a beaker with a larger diameter

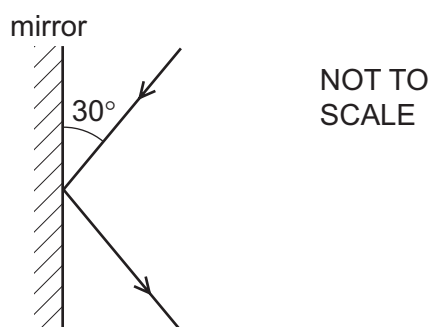
- 18** What is the colour and what is the texture of the best absorber of infrared radiation?

- A** black and shiny
B black and dull
C white and shiny
D white and dull

- 19 Which row gives the velocity and wavelength of a wave that is an ultrasound wave?

	velocity m/s	wavelength /cm
A	330	6.6
B	890	89
C	1500	6.0
D	4000	25

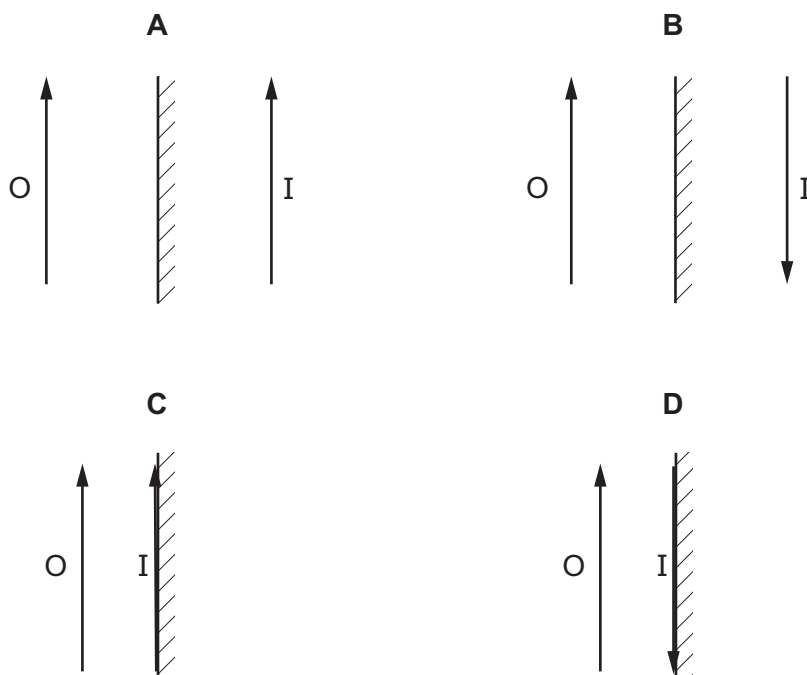
- 20 The diagram shows a ray of light incident on a plane mirror.



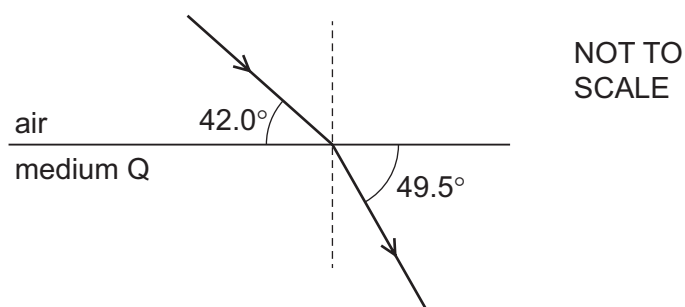
What is the angle of reflection?

- A** 30° **B** 60° **C** 90° **D** 120°
- 21 An object O is placed in front of a plane mirror.

Which diagram shows the image I formed by the mirror?



22 The diagram shows a ray of light travelling from air to medium Q.



What is the critical angle for the boundary between medium Q and air?

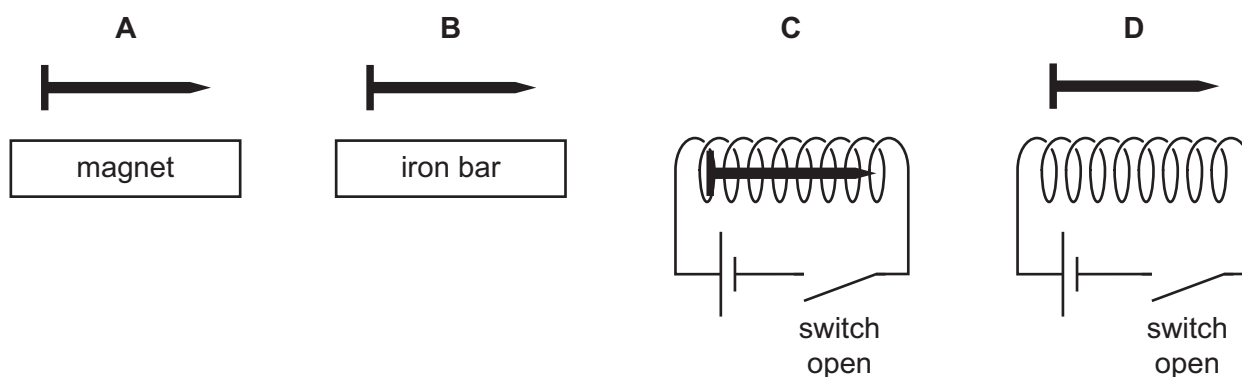
- A 57.5° B 58.0° C 60.9° D 61.6°

23 Which two waves are components of the electromagnetic spectrum?

- A light and sound
B ultrasound and ultraviolet
C water waves and infrared
D X-rays and microwaves

24 The diagrams show an iron nail in four different situations.

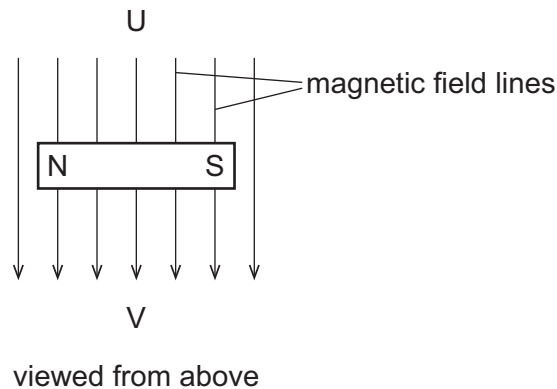
In which diagram is the nail an induced magnet?



25 Which row correctly shows a magnetic metal and a non-magnetic metal?

	magnetic metal	non-magnetic metal
A	aluminium	copper
B	copper	steel
C	iron	copper
D	steel	iron

- 26** A freely suspended magnet is held perpendicular to a uniform magnetic field as shown. The magnet is viewed from above.



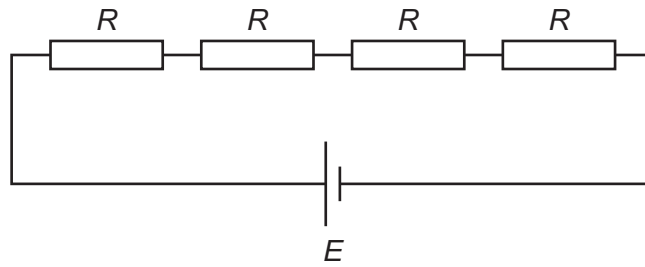
What happens to the magnet when it is released?

- A** It turns through 90° clockwise and then stops moving.
 - B** It turns through 90° anticlockwise and then stops moving.
 - C** It turns through 90° clockwise and moves towards U.
 - D** It turns through 90° anticlockwise and moves towards V.
- 27** A plastic rod becomes charged when it is rubbed with a cloth.

Which statement gives the reason for the charge on the rod?

- A** The plastic rod loses protons and becomes negatively charged.
- B** The plastic rod loses electrons and becomes negatively charged.
- C** The plastic rod gains protons and becomes positively charged.
- D** The plastic rod loses electrons and becomes positively charged.

- 28 Four resistors, each of resistance R , are connected in series with a cell of e.m.f. E .



From left to right, what is the current in each of the resistors?

- A** $\frac{E}{4R}$ $\frac{E}{4R}$ $\frac{E}{4R}$ $\frac{E}{4R}$
- B** $\frac{E}{4R}$ $\frac{E}{3R}$ $\frac{E}{2R}$ $\frac{E}{R}$
- C** $\frac{E}{R}$ $\frac{E}{2R}$ $\frac{E}{3R}$ $\frac{E}{4R}$
- D** $\frac{E}{R}$ $\frac{E}{R}$ $\frac{E}{R}$ $\frac{E}{R}$

- 29 How can one volt also be expressed?

- A** one coulomb per ampere
- B** one coulomb per joule
- C** one joule per ampere
- D** one joule per coulomb

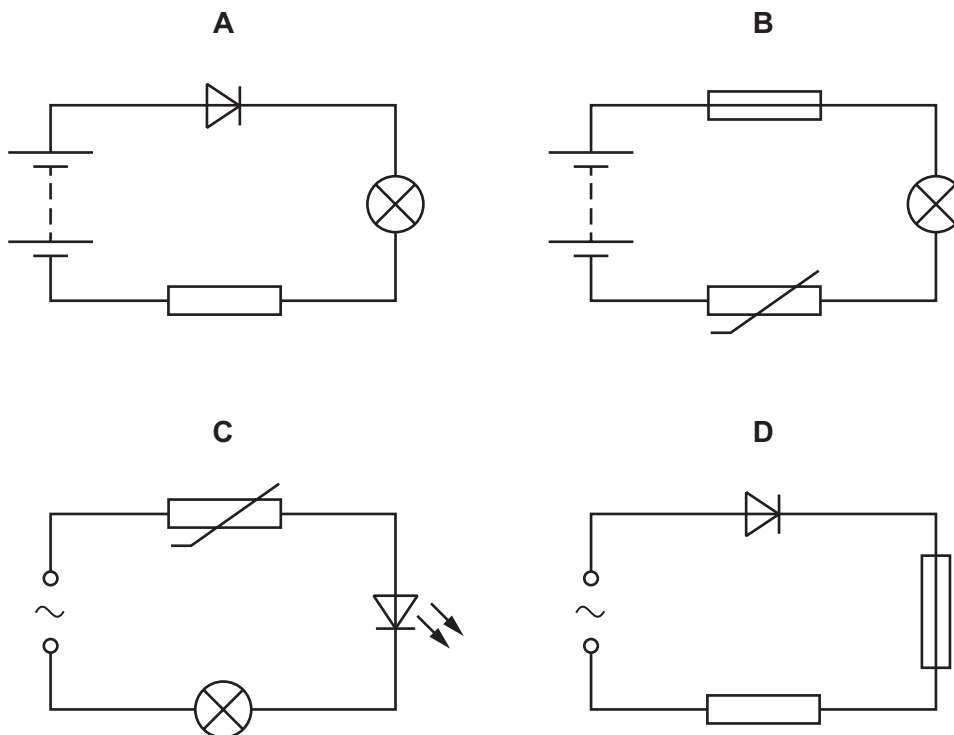
- 30 X is a piece of wire with length l , area a and resistance R .

Wire Y is made from the same material and is connected to wire X to produce a component with a total resistance of $\frac{R}{2}$.

What are the dimensions of Y and how is it connected to X ?

	length	area	connection
A	$2l$	$0.25a$	in parallel
B	$2l$	$0.25a$	in series
C	$4l$	$4a$	in parallel
D	$4l$	$4a$	in series

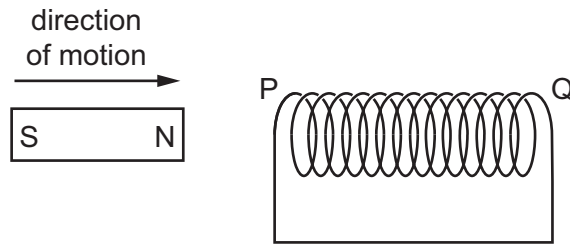
31 Which circuit contains a fuse and a diode?



32 How is energy transferred within a kettle that is powered by the mains supply?

- A by electrical work done to chemical energy
- B by electrical work done to thermal energy
- C by mechanical work done to chemical energy
- D by mechanical work done to thermal energy

33 A magnet moves through a coil of wire as shown.

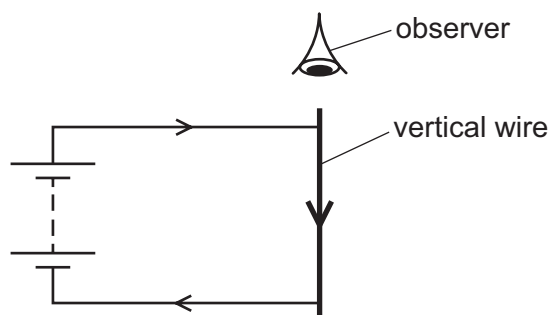


Magnetic poles are produced at the two ends of the coil as the magnet enters and as the magnet leaves the coil.

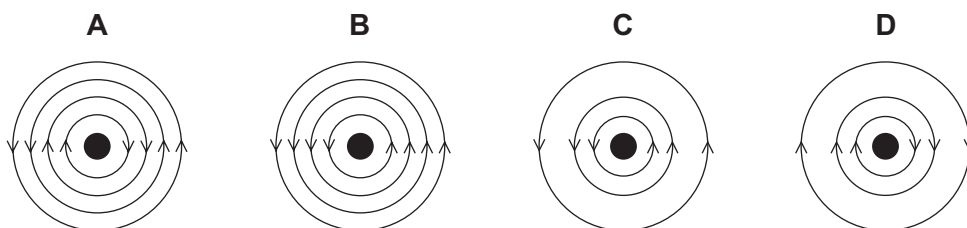
Which pole is induced at P when the magnet enters the coil and which pole is induced at Q when the magnet leaves the coil?

	pole at P when magnet enters the coil	pole at Q when magnet leaves the coil
A	N	N
B	N	S
C	S	N
D	S	S

34 A straight vertical wire carries a current in the direction shown.



What is the pattern of the magnetic field around the wire seen by an observer viewing the wire from the position shown?



35 Which part of a simple d.c. motor reverses the direction of current in the coil once every half rotation of the coil?

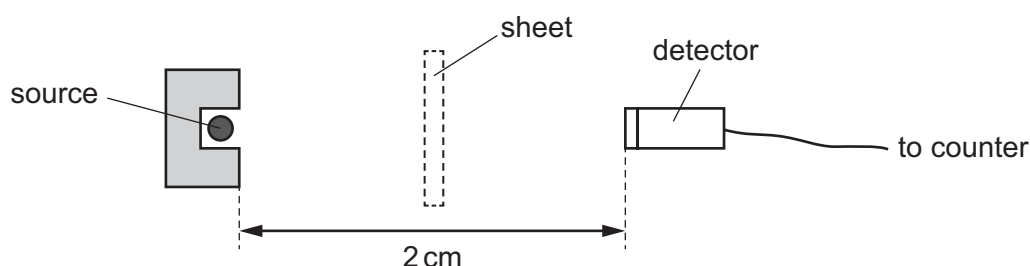
- A the armature
- B the brushes
- C the slip rings
- D the split-ring commutator

36 Which statement about the nuclide notation R_ZX is correct?

- A R is the nucleon number which is different for different isotopes of X .
- B R is the proton number which is the same for different isotopes of X .
- C Z is the nucleon number which is the same for different isotopes of X .
- D Z is the proton number which is different for different isotopes of X .

37 During a radioactivity experiment, the background reading is 28 counts per minute.

A radioactive source is placed 2 cm in front of a detector which is connected to a counter.



Counter readings are taken with and without sheets of different materials placed between the source and the detector.

type of sheet	reading / counts per minute
no sheet	750
thin card	750
5 mm thick aluminium	478

Which types of radiation are being emitted by the source?

- A alpha particles, beta particles and gamma rays
- B alpha particles and beta particles only
- C alpha particles and gamma rays only
- D beta particles and gamma rays only

- 38** A radioactive isotope is injected into a patient as part of a medical test. The radiation emitted is detected outside the body.

What are suitable properties for this isotope?

	radiation type	half-life
A	alpha particles	a few hours
B	beta particles	a few years
C	gamma rays	a few hours
D	gamma rays	a few years

- 39** A student writes four statements about the Solar System.

- 1 Light takes approximately 3.0×10^8 seconds to travel from the Sun to the Earth.
- 2 The Earth takes approximately 24 hours to rotate on its axis once.
- 3 The Moon takes approximately 1 month to orbit the Earth.
- 4 The Sun takes approximately 365 days to orbit the Earth.

Which statements are correct?

- A** 1 and 2 **B** 2, 3 and 4 **C** 2 and 3 only **D** 3 and 4 only

- 40** What is **never** formed by the explosion of a red supergiant?

- A** a black hole
B a neutron star
C a white dwarf
D new heavier elements

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

October/November 2024

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 A crane lifts a heavy load at a steady rate.

Which unit is suitable for the output power of the crane's motor?

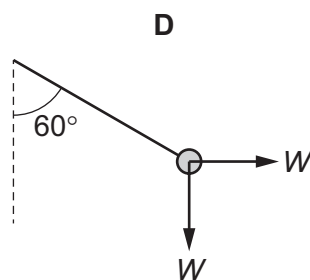
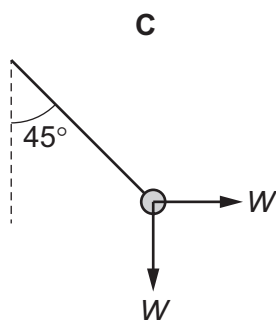
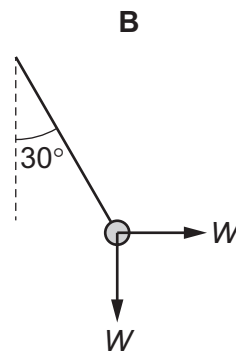
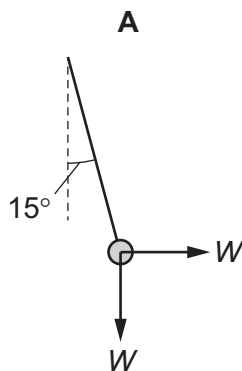
- A kN B kW C kN/s D kWh

- 2 Which statement about mass is correct?

- A Mass is a scalar and its unit is the kilogram.
 B Mass is a scalar and its unit is the newton.
 C Mass is a vector and its unit is the kilogram.
 D Mass is a vector and its unit is the newton.

- 3 The bob of a simple pendulum is pulled horizontally by a force that is equal to weight W of the bob.

Which diagram shows the angle between the pendulum thread and the vertical?



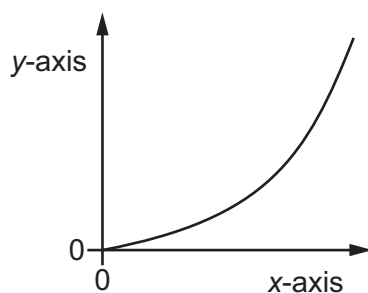
- 4 A student takes 20 minutes to walk from X to Y at a speed of 3.0 km/h.

She turns 90° at Y and walks at a speed of 4.0 km/h for 15 minutes to reach Z.

What is her average speed from X to Z?

- A 2.4 km/h B 3.4 km/h C 3.5 km/h D 5.0 km/h

- 5 A car is at rest by a red traffic light. When the light changes to green, the car starts moving. The graph represents the motion of the car.



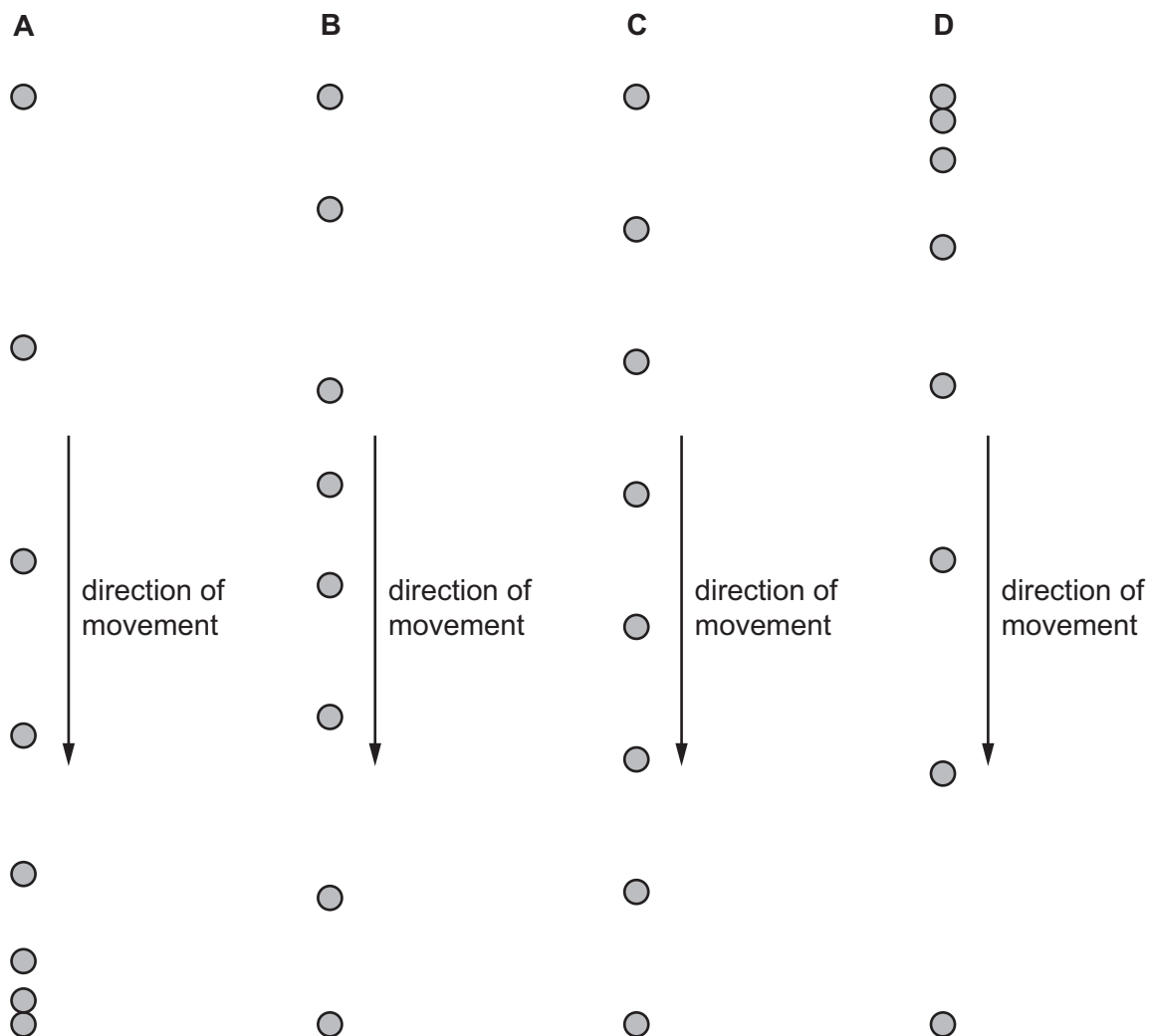
The car has a constant acceleration.

Which quantity is plotted on the x-axis and which quantity is plotted on the y-axis?

	x-axis	y-axis
A	distance	time
B	speed	time
C	time	distance
D	time	speed

- 6 An object falls without air resistance. Its position at equal time intervals is shown by a sequence of dots.

Which sequence of dots is correct?



- 7 Some gas trapped in a container is compressed at constant temperature by a piston.

Which property of the gas does **not** change?

- A pressure
 - B mass
 - C frequency of particle collision
 - D density
- 8 A car has a weight of 8000 N.

What is the mass of the car?

- A 810 kg
- B 820 kg
- C 8000 kg
- D 78 000 kg

- 9 A lorry is travelling along a straight, horizontal road. The constant driving force on the lorry is 80 kN and the total resistive force is 20 kN.

Which row describes the motion of the lorry and gives the resultant force acting on it?

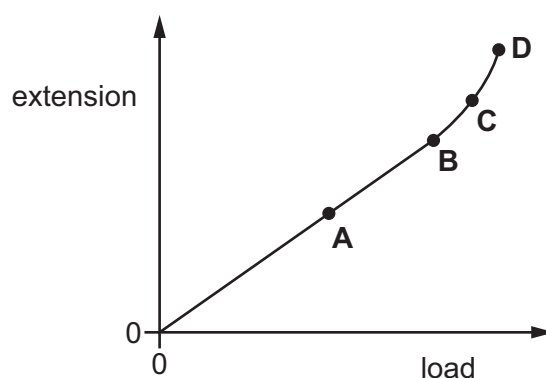
	motion of lorry	resultant force / kN
A	acceleration	60
B	acceleration	100
C	deceleration	60
D	deceleration	100

- 10 As a cyclist uses the brakes on a bicycle, energy is transferred from the kinetic store to the thermal store.

What causes the transfer of energy?

- A** friction
B mass
C thrust
D weight
- 11 The extension of a spring is measured as the load stretching it is increased. The graph shows the results.

Which point is the spring's limit of proportionality?

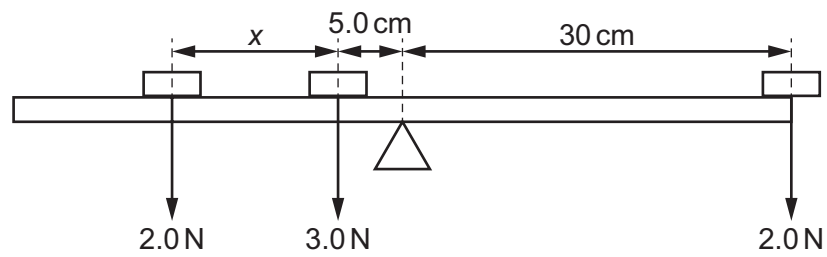


- 12 A satellite orbits the Earth.

What is the direction of the force on the satellite causing this circular motion?

- A** away from the centre of the Earth
B in the direction of motion of the satellite
C in the opposite direction to the motion of the satellite
D towards the centre of the Earth

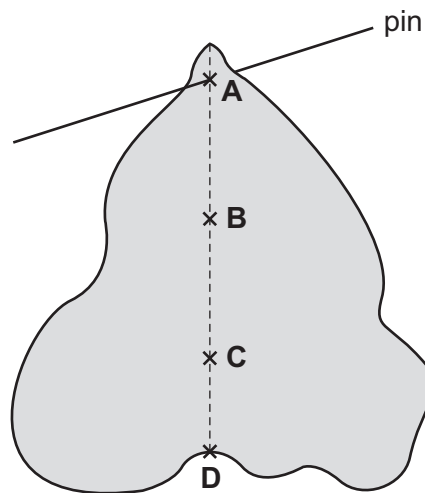
- 13 A uniform beam is pivoted at its centre and balanced by three weights as shown.



What is distance x ?

- A** 17.5 cm **B** 22.5 cm **C** 35.0 cm **D** 37.5 cm
- 14 A piece of card of uniform thickness and density is suspended freely from a horizontal pin. The diagram shows the card at rest.

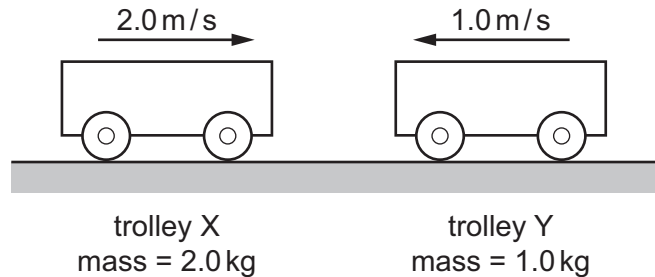
Which cross is the centre of gravity of the card?



- 15 Trolley X of mass 2.0 kg is moving to the right at 2.0 m/s.

Trolley Y of mass 1.0 kg is moving to the left at 1.0 m/s.

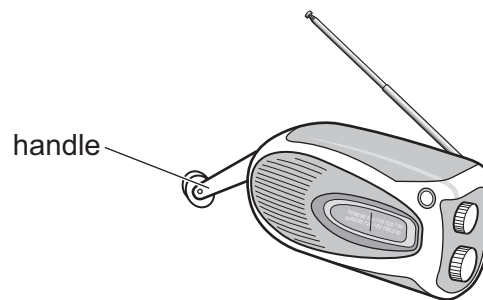
The trolleys collide and stick together.



The collision takes a total time of 0.20 s.

What is the average force acting on trolley X during the collision?

- A 3.3 N B 10 N C 15 N D 25 N
- 16 An object of mass 5.0 kg is moving at a constant speed. It has 10 J of kinetic energy.
- What is the speed of the object?
- A 0.25 m/s B 0.5 m/s C 2.0 m/s D 4.0 m/s
- 17 A wind-up radio is used in areas where mains electricity is not available.



A person winds a spring in the radio using the handle. When the spring unwinds, it makes a small coil rotate in a magnetic field.

Which energy transfer is **not** useful for this radio?

- A elastic potential energy → kinetic energy
- B kinetic energy → thermal energy
- C chemical energy → elastic potential energy
- D kinetic energy → sound energy

- 18 Energy is transferred in a filament lamp at a rate of 60 W. The lamp wastes energy at a rate of 46 W.

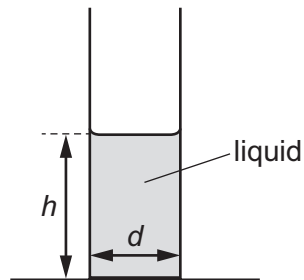
Which ratio is used to calculate the efficiency of the lamp?

- A $\frac{14}{74}$ B $\frac{14}{60}$ C $\frac{14}{46}$ D $\frac{46}{60}$

- 19 A liquid is held in a cylindrical container.

The diameter of the base of the cylinder is d .

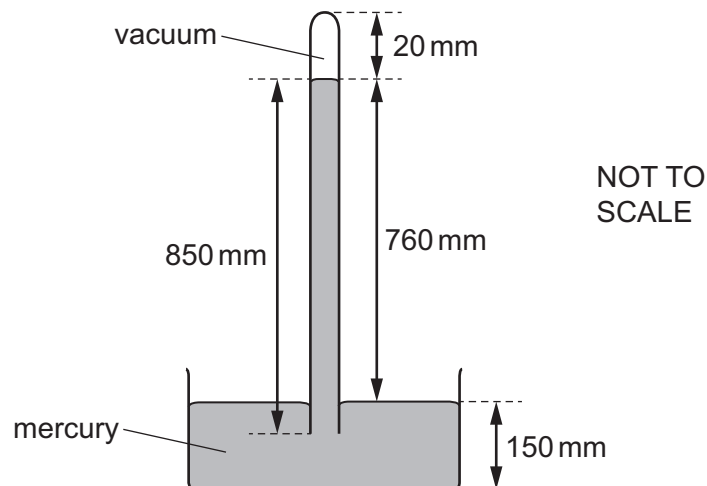
The mass of the liquid is m and its depth is h .



Which expression gives the pressure due to the liquid on the bottom of the cylinder?

- A $\frac{mg}{\pi d^2}$ B $\frac{mgh}{\pi d^2}$ C $\frac{4mg}{\pi d^2}$ D $\frac{4mgh}{\pi d^2}$

- 20 The diagram shows an instrument that is used to measure the atmospheric pressure.

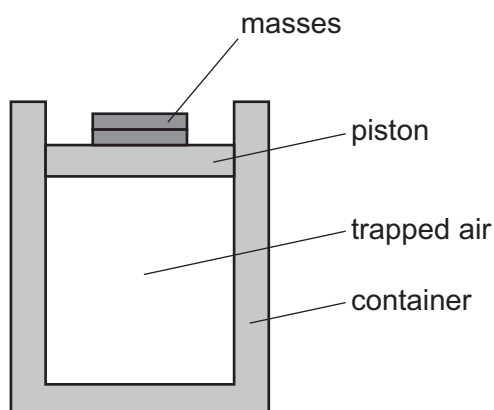


The density of mercury is $1.4 \times 10^4 \text{ kg/m}^3$.

What is the atmospheric pressure?

- A 2.7 kPa B 21 kPa C 104 kPa D 117 kPa

- 21** Air at atmospheric pressure and room temperature is trapped in an insulated container by a frictionless piston of negligible weight. The air can be compressed by placing masses on top of the piston.



A mass of 1.0 kg is placed on the piston and the volume of the trapped air decreases from 100 cm³ to 90 cm³.

Which statement is correct?

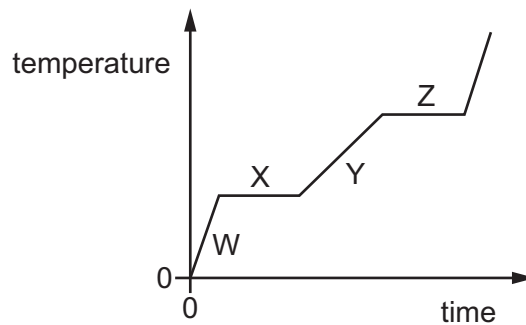
- A** Adding another 1.0 kg mass decreases the volume of the air by less than 10 cm³.
 - B** Adding an additional 9.0 kg mass reduces the volume of the air to zero.
 - C** The pressure of the trapped air is less than atmospheric pressure when its volume is 90 cm³.
 - D** The temperature of the air decreases as the 1.0 kg mass is added.
- 22** A substance has a melting temperature of -17°C and a boiling temperature of 117°C .

In which state does the substance exist at -10°C and at 110°C ?

	at -10°C	at 110°C
A	solid	liquid
B	solid	gas
C	liquid	liquid
D	liquid	gas

- 23** Thermal energy is supplied at a constant rate to a fixed mass of a substance that is initially solid.

The graph shows how the temperature of the fixed mass of the substance varies with time.



In which sections of the graph does the substance absorb latent heat?

- A** W and X **B** W and Y **C** X and Z **D** Y and Z
- 24** A piece of paper torn out of an exercise book is shown.

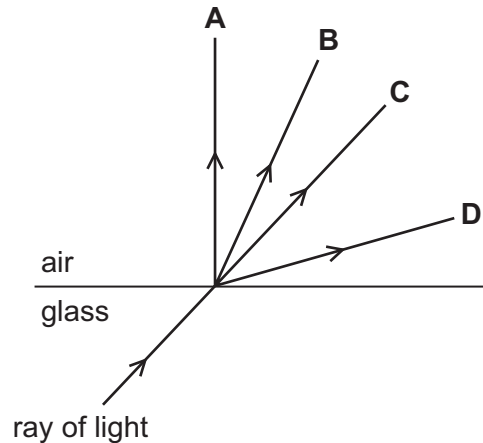
	As the average kinetic energy of the
	molecules decreases, the space between them
	decreases and they move downwards.

Which process is being described?

- A** conduction
B convection
C evaporation
D radiation

25 A ray of light passes from glass to air.

Which arrow shows the direction of the ray in air?



26 A short-sighted student looks at a distant object.

Which row shows the position of an image in a short-sighted eye and the type of lens used to correct short-sightedness?

	position of image in short-sighted eye	lens used to correct short-sightedness
A	in front of retina	diverging
B	behind retina	converging
C	on retina	converging
D	behind retina	diverging

27 A student makes four statements about infrared waves.

- 1 They can be seen by the human eye.
- 2 They travel at 330 m/s in air.
- 3 They are transverse waves.
- 4 They are used by intruder alarms.

Which statements are correct for infrared waves?

- A** 1 and 3 **B** 2, 3 and 4 **C** 2 only **D** 3 and 4 only

28 Which waves are longitudinal?

- A** sound waves in water
- B** ultraviolet waves in air
- C** waves on the surface of water
- D** X-rays in a vacuum

29 A person hears an echo after a sound hits a solid cliff face.

What causes the echo?

- A** absorption
- B** dispersion
- C** reflection
- D** refraction

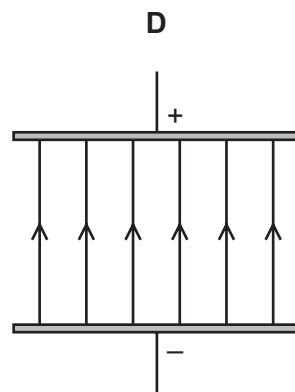
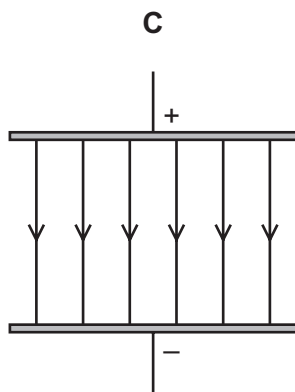
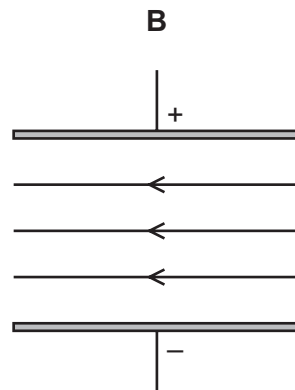
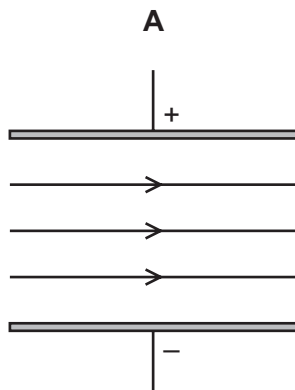
30 A plastic strip is rubbed with a cloth and the plastic becomes negatively charged.

Which statement explains this?

- A** Electrons are transferred from earth to the plastic.
- B** Electrons are transferred from the cloth to the plastic.
- C** Electrons move from one side of the plastic to the other side of the plastic.
- D** Protons are transferred from the plastic to the cloth.

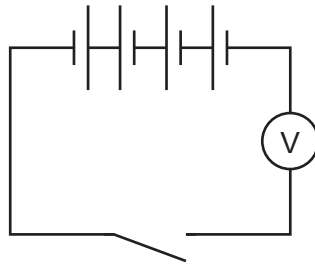
- 31** Two parallel, conducting plates are charged. The charge on the upper plate is positive. The charge on the lower plate is equal in size but negative.

Which diagram shows the pattern and the direction of the electric field between the two plates?



- 32** Which quantity is defined as the electrical work done by a source in moving a unit charge around a complete circuit?
- A** power
 - B** potential difference
 - C** electromotive force
 - D** energy

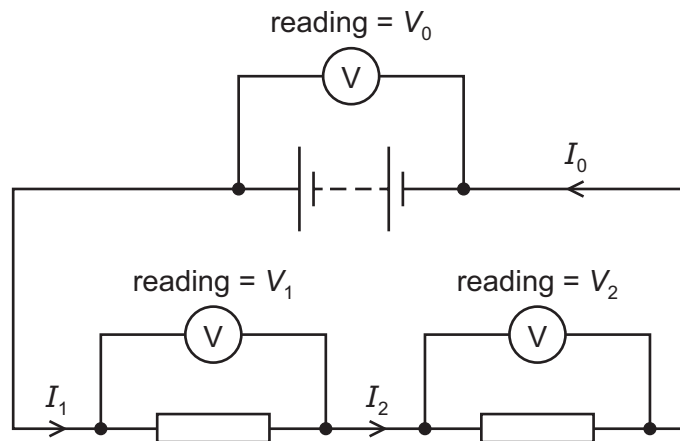
- 33 Four 1.5 V cells are connected in a circuit as shown.



The switch is closed.

What is the reading on the voltmeter?

- A 0 V B 1.5 V C 3.0 V D 4.5 V
- 34 Two resistors are connected in series with a battery, as shown. The readings on three voltmeters V_0 , V_1 and V_2 and the currents I_0 , I_1 and I_2 are also shown.



Which row gives values for the current I_0 and the potential difference V_0 ?

	I_0 is equal to	V_0 is equal to
A	I_1	V_2
B	I_2	the sum of V_1 and V_2
C	the sum of I_1 and I_2	V_1
D	the sum of I_1 and I_2	the sum of V_1 and V_2

- 35 The current in a home computer varies between 0.40 A and 1.2 A when the computer is connected to the mains supply.

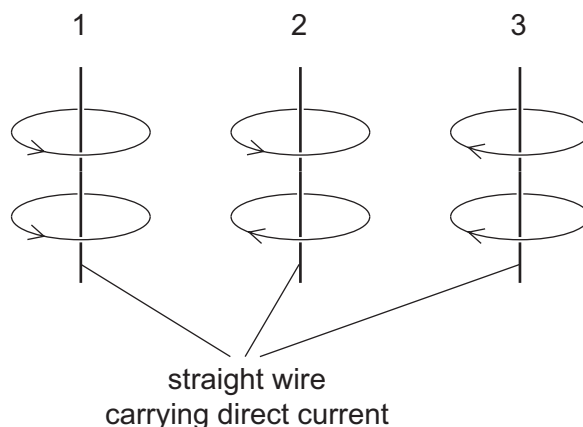
What is the rating of the **most** suitable fuse to use to protect the computer?

- A 0.5 A B 1.2 A C 1.5 A D 13 A

- 36 Students are asked to draw the pattern and direction of the magnetic field due to a direct current (d.c.) in a straight wire.

Diagrams 1, 2 and 3 are three of their answers.

The students do not know the direction of the current in the wire.



Which diagrams show possible patterns and directions for the magnetic field due to the current in the wire?

- A** 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 only
- 37 Which row identifies the positive and negative particles in the atom?

	positive particle	negative particle
A	proton	electron
B	electron	proton
C	proton	neutron
D	neutron	proton

- 38 The nuclide notation of a certain isotope of element Q is ${}^R_Z\text{Q}$.

What is the total number of protons, neutrons and electrons in a neutral atom of this isotope?

- A** Z **B** R **C** $Z + R$ **D** $2Z + R$
- 39 Which list shows the planets in order of increasing distance from the Sun?
- A** Venus, Mercury, Earth, Mars, Saturn, Jupiter, Uranus, Neptune
B Mercury, Mars, Earth, Venus, Jupiter, Saturn, Uranus, Neptune
C Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune
D Mercury, Mars, Earth, Venus, Uranus, Saturn, Jupiter, Neptune

- 40** A student spills ink on the page in a textbook that defines redshift. The ink covers the three important words 1, 2 and 3.

Redshift is1..... in the observed2..... of electromagnetic radiation emitted from3..... stars and galaxies.

What are words 1, 2 and 3?

	1	2	3
A	an increase	frequency	receding
B	a decrease	frequency	approaching
C	an increase	wavelength	receding
D	an increase	wavelength	approaching

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Cambridge O Level

PHYSICS**5054/12**

Paper 1 Multiple Choice

October/November 2024

MARK SCHEME

Maximum Mark: 40

Published

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Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	A	1
3	A	1
4	B	1
5	A	1
6	D	1
7	D	1
8	A	1
9	C	1
10	A	1
11	A	1
12	C	1
13	C	1
14	C	1
15	D	1
16	C	1
17	C	1
18	B	1
19	C	1
20	B	1
21	A	1
22	C	1
23	D	1
24	A	1
25	C	1
26	B	1
27	D	1
28	A	1

Question	Answer	Marks
29	D	1
30	C	1
31	D	1
32	B	1
33	A	1
34	D	1
35	D	1
36	A	1
37	D	1
38	C	1
39	C	1
40	C	1

Cambridge O Level

PHYSICS**5054/11**

Paper 1 Multiple Choice

October/November 2024

MARK SCHEME

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Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	A	1
3	C	1
4	B	1
5	C	1
6	D	1
7	B	1
8	B	1
9	A	1
10	A	1
11	B	1
12	D	1
13	A	1
14	C	1
15	B	1
16	C	1
17	B	1
18	B	1
19	C	1
20	C	1
21	A	1
22	C	1
23	C	1
24	B	1
25	D	1
26	A	1
27	D	1
28	A	1

Question	Answer	Marks
29	C	1
30	B	1
31	C	1
32	C	1
33	C	1
34	B	1
35	C	1
36	B	1
37	A	1
38	C	1
39	C	1
40	C	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2023

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

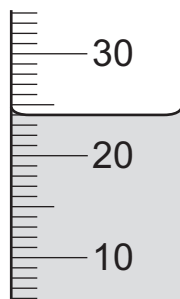
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

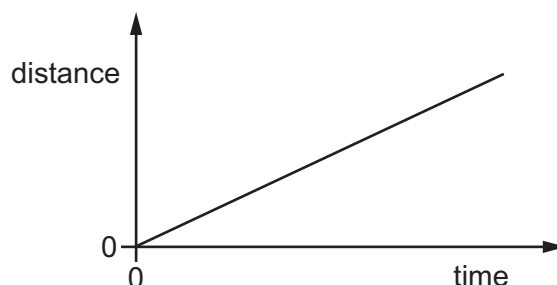


- 1 The diagram shows the level of liquid in a measuring cylinder. The scale is marked in cm^3 .



What is the volume of the liquid?

- A** 20.4 cm^3 **B** 24.0 cm^3 **C** 24.5 cm^3 **D** 25.0 cm^3
- 2 The diagram shows the distance–time graph of a car.



The car is travelling along a straight road up a hill.

Which quantity for the car is constant and greater than zero?

- A** acceleration
B displacement
C gravitational potential energy
D kinetic energy
- 3 A distance–time graph and a speed–time graph show the motion of an object.
- How are the speed and the acceleration of the object determined from these graphs?

	speed	acceleration
A	area under the distance–time graph	area under the speed–time graph
B	area under the distance–time graph	gradient of the speed–time graph
C	gradient of the distance–time graph	area under the speed–time graph
D	gradient of the distance–time graph	gradient of the speed–time graph

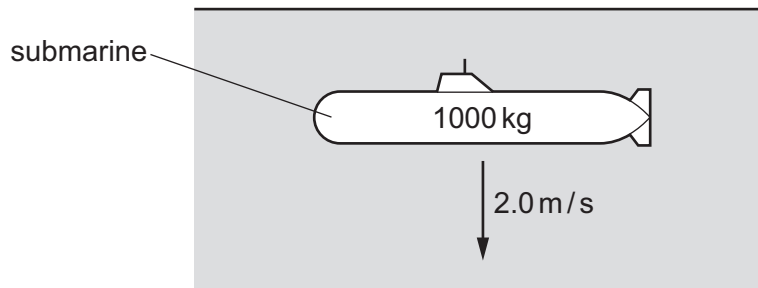
- 4 A student finds that the mass of a rock is 65 g.

She pours 30 cm^3 of water into a measuring cylinder.

She places the rock in the water so that the water covers the rock completely. The water level rises to 55 cm^3 .

What is the density of the rock?

- A** 0.38 g/cm^3 **B** 1.2 g/cm^3 **C** 2.2 g/cm^3 **D** 2.6 g/cm^3
- 5 A small submarine of mass 1000 kg sinks in water with a constant speed of 2.0 m/s .



What is the resultant force exerted on the submarine as it sinks?

- A** 0
B 500 N downwards
C 2000 N downwards
D 9800 N downwards
- 6 A student drops, from rest, a light ball in air. The ball experiences some air resistance as it falls.

What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

7 The stopping distance of a car travelling at a certain speed is measured in four different situations.

- on a dry road with new tyres
- on a dry road with old, worn tyres
- on a wet road with new tyres
- on a wet road with old, worn tyres

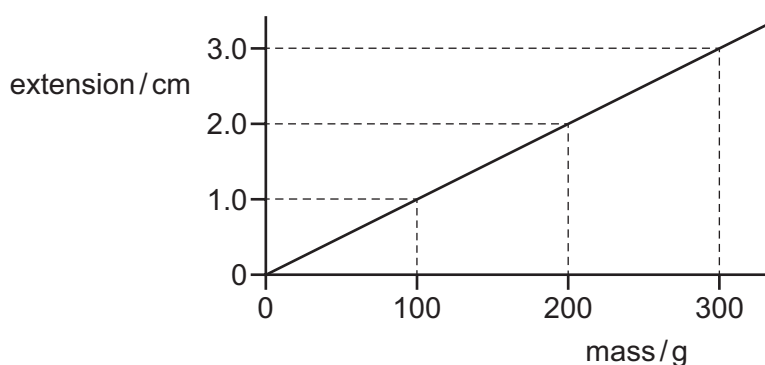
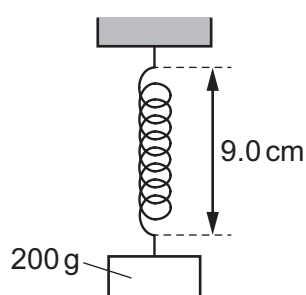
The results are shown.

	W	X	Y	Z
stopping distance / m	12	16	17	40

Which two results are possible measurements on the dry road with old, worn tyres?

- A** W and X **B** X and Y **C** X and Z **D** Y and Z

8 A spring has a length of 9.0 cm when loaded with a 200 g mass. The extension–mass graph for the spring is shown.



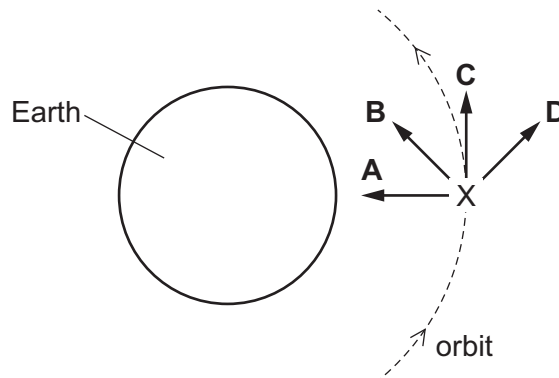
The 200 g mass is replaced with a 100 g mass.

What is the new length of the spring?

- A** 7.0 cm **B** 8.0 cm **C** 9.0 cm **D** 10 cm

- 9 The International Space Station orbits the Earth.

Which arrow shows the direction of the force on the International Space Station when it is at point X?



- 10 Which expression is equal to impulse?

- A change in momentum $\div$ time
- B change in momentum
- C force $\times$ distance moved in the direction of the force
- D force $\div$ area

- 11 A box with mass 4.0 kg is pulled from rest along a horizontal floor by a constant force F .

A constant frictional force of 1.0 N acts on the box.

When the box has travelled 1.0 m its velocity is 1.5 m/s.

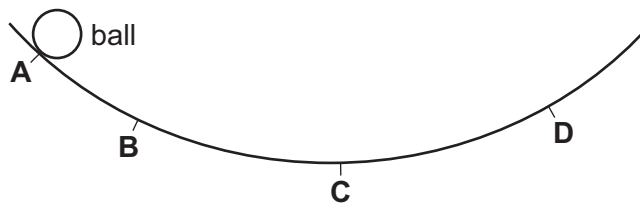
The kinetic energy of the box is equal to the work done on the box.

What is force F ?

- A 2.0 N B 3.5 N C 4.0 N D 5.5 N

- 12 The diagram shows a curved track. A ball is released from the position shown.

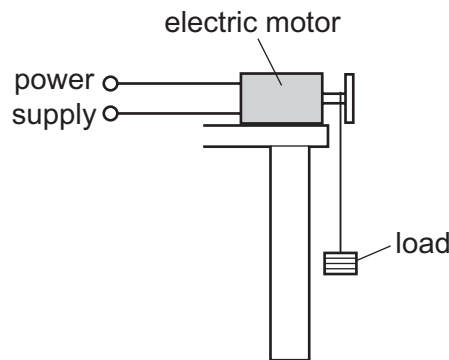
At which point is the gravitational potential energy stored in the ball the maximum?



13 How does an oil-fired power station differ from a nuclear power station?

- A** Gases emitted by hot fuel are emitted into the atmosphere.
- B** Steam is produced in a boiler using hot fuel.
- C** The hot steam is used to turn a turbine.
- D** Turbines are used to drive an electric generator.

14 An electric motor is used to lift a load at a constant speed.



The gravitational potential energy (GPE) stored in the load increases.

The moving load also has stored kinetic energy (KE).

Which expression is used to calculate the efficiency of the motor when the load is being lifted at a constant rate?

- A** $\text{efficiency} = \frac{\text{GPE gained per second}}{\text{KE gained per second}}$
- B** $\text{efficiency} = \frac{\text{GPE gained per second}}{\text{power supplied}}$
- C** $\text{efficiency} = \frac{\text{power supplied}}{\text{GPE gained per second}}$
- D** $\text{efficiency} = \frac{\text{power supplied}}{\text{KE gained per second}}$

15 A battery delivers 3000 J of energy to a small 2.0 W electric motor before the battery stops working.

For how long is the motor switched on?

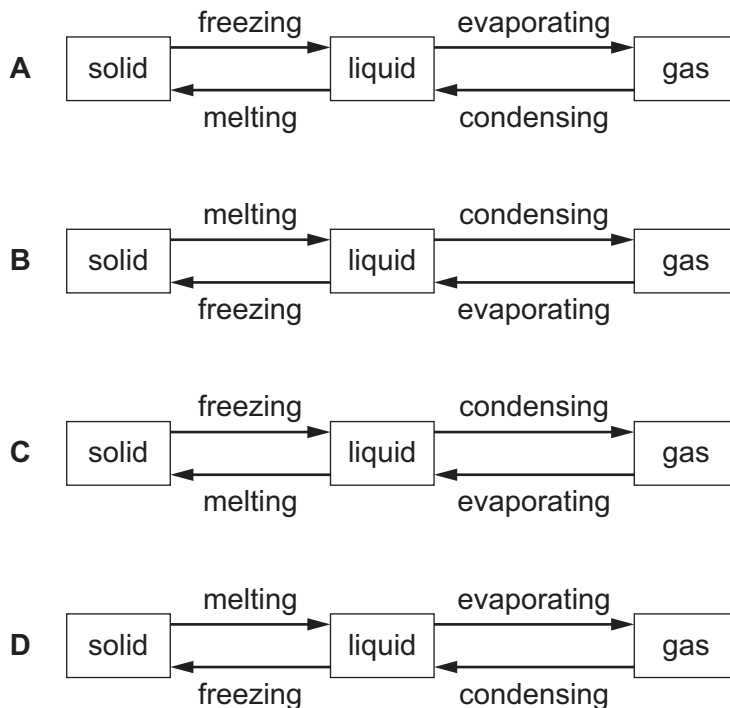
- A** 1500 minutes
- B** 100 minutes
- C** 50 minutes
- D** 25 minutes

- 16** A man finds it difficult to hammer a wooden post into the ground.

How could he make the post go in more easily?

- A** Make the bottom of the post pointed.
- B** Make the top of the post pointed.
- C** Use a longer post.
- D** Use a wider post.

- 17** Which diagram shows the names of the changes of state that correspond to the directions of the arrows?



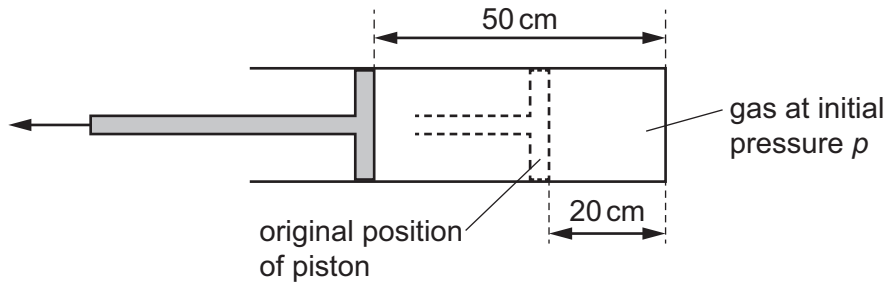
- 18** A sealed cylinder containing gas is left in direct sunlight.

Eventually the cylinder explodes.

How is this explained in terms of the behaviour of the gas particles?

- A** Each particle gets bigger and they collide with each other more often.
- B** Each particle gets bigger and they collide harder with the cylinder.
- C** The particles gain kinetic energy and hit the cylinder harder and more often.
- D** The particles lose kinetic energy and they hit the cylinder harder and more often.

- 19 A gas is trapped inside a cylinder by a movable piston. The length of the gas column is 20 cm and the pressure inside the cylinder is p .



The piston is pulled out a distance of 30 cm, so that the length of the gas column is now 50 cm. The temperature of the gas does not change.

What is the new pressure of the gas?

- A $0.40p$ B $0.60p$ C $1.5p$ D $2.5p$
- 20 Pot X and pot Y are both in a cool room.

Pot X contains hot water below its boiling point.

Pot Y is identical to pot X but contains water that is kept boiling by a hotplate underneath it.



What happens to the average kinetic energy of the particles in the liquid water in pot X and to the average kinetic energy of the particles in the liquid water in pot Y?

	kinetic energy of liquid particles in X	kinetic energy of liquid particles in Y
A	decreases	increases
B	decreases	does not change
C	stays the same	increases
D	stays the same	does not change

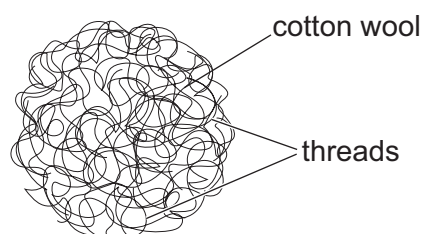
21 Solid metals are good thermal conductors.

How closely packed are their particles and do the metals contain free electrons?

	packing of particles	contains free electrons
A	close together	no
B	close together	yes
C	far apart	no
D	far apart	yes

22 Cotton wool is a good thermal insulator.

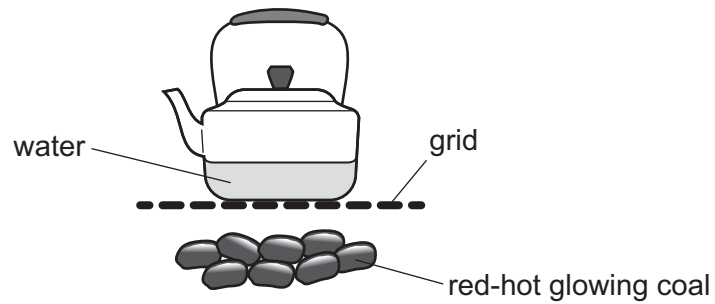
It consists of many threads of cotton which are tangled together to make a material which can easily be pulled apart or compressed. The diagram shows the structure of cotton wool.



Why is cotton wool a good thermal insulator?

- A** Each thread is very thin.
- B** It has a large surface area because it is made of many threads.
- C** It has a low density because of the trapped air.
- D** It traps air and prevents convection.

23 A metal kettle is heated on a grid above red-hot glowing coal.

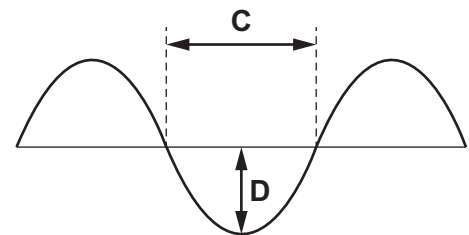
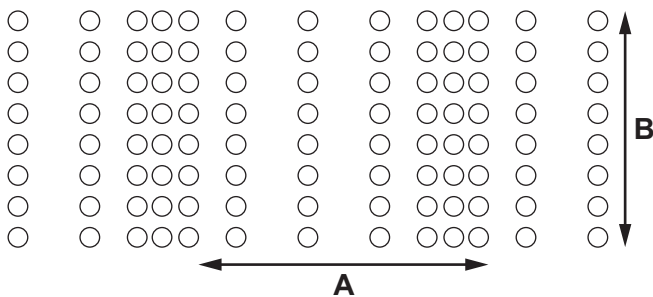


What are the main thermal energy transfers through the bottom of the kettle and within the water?

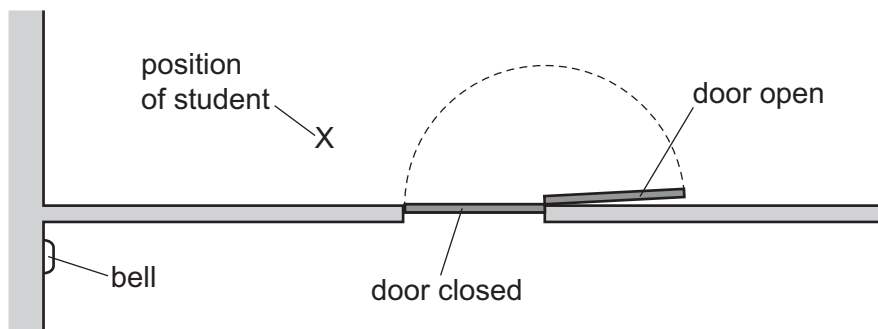
	through kettle	within water
A	conduction	conduction
B	radiation	convection
C	conduction	convection
D	radiation	conduction

24 The diagram shows two waveforms.

Which arrow shows the wavelength of a longitudinal wave?



- 25 A bell, on a wall outside a classroom, rings. A student is standing at point X shown on the view from above.

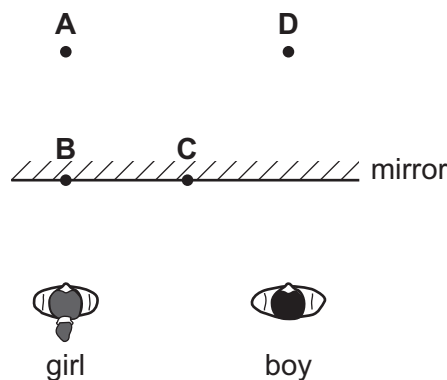


The door of the classroom is closed and the student in the room at X hears the bell ring.

The door is opened, and the sound heard by the student from the bell is louder.

What causes the sound reaching the student to be louder?

- A diffraction
 - B dispersion
 - C reflection
 - D refraction
- 26 A boy stands beside a girl in front of a large vertical plane mirror. They are equal distances from the mirror, as shown. The boy sees an image of the girl.



- 27 Light travelling in air is incident on a water surface.

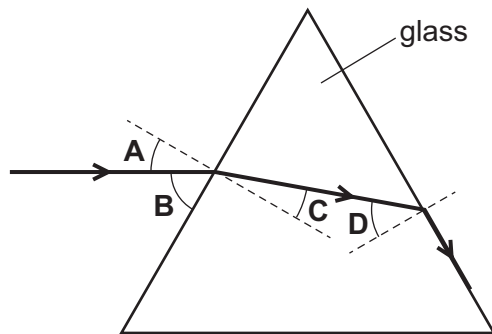
The refractive index of water is 1.3 and the angle of refraction in the water is 40° .

What is the angle of incidence in the air?

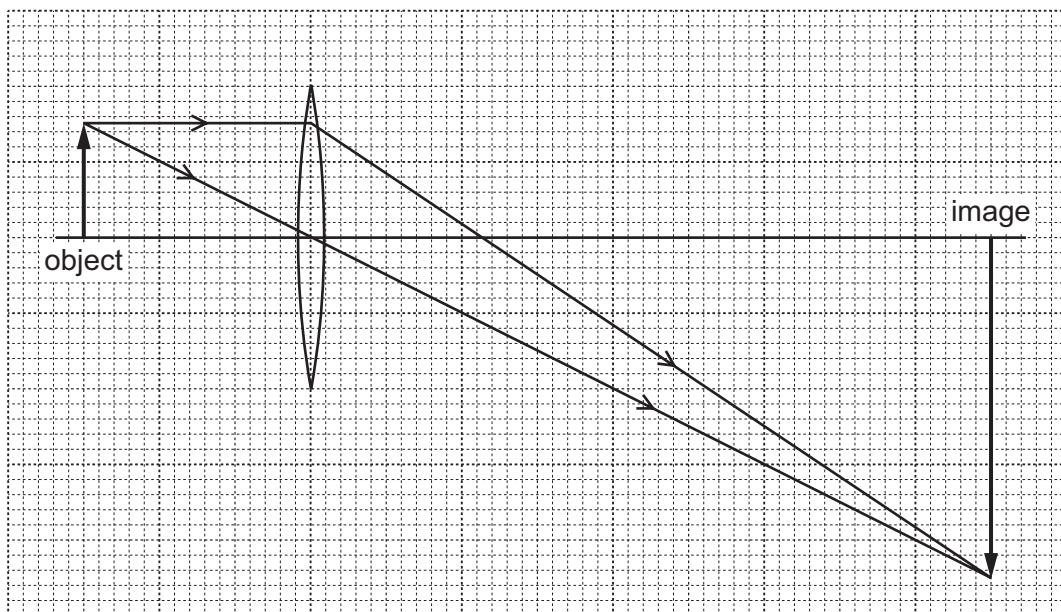
- A 30°
- B 50°
- C 52°
- D 57°

28 The diagram shows a ray of light passing through a triangular glass block.

Which angle is the critical angle of light in glass?



29 A lens is used to produce a magnified image, as shown in the scale diagram.



What is the linear magnification produced by the lens?

- A** 0.33 **B** 3.0 **C** 4.0 **D** 6.0

30 Ultraviolet radiation is a component of the electromagnetic spectrum.

Which application uses ultraviolet radiation?

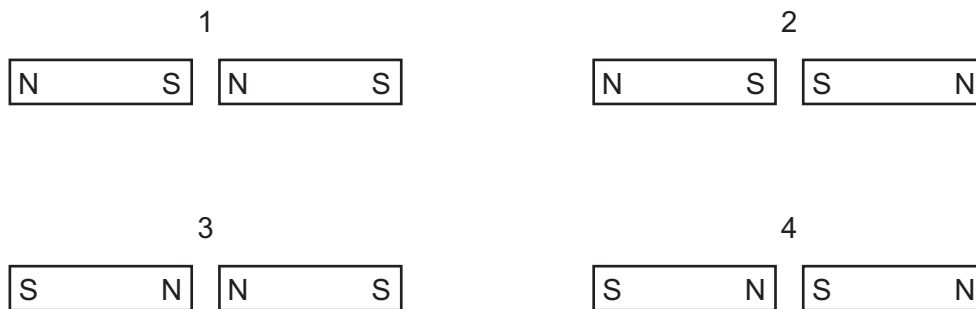
- A** Bluetooth technology
B prenatal scanning
C sterilising water
D thermal imaging

- 31 A student rubs different rods held in his hand.

Which rod gains a large electrostatic charge?

- A a polythene rod rubbed with a steel magnet
- B a polythene rod rubbed with a woollen duster
- C an iron rod rubbed with a steel magnet
- D an iron rod rubbed with a woollen duster

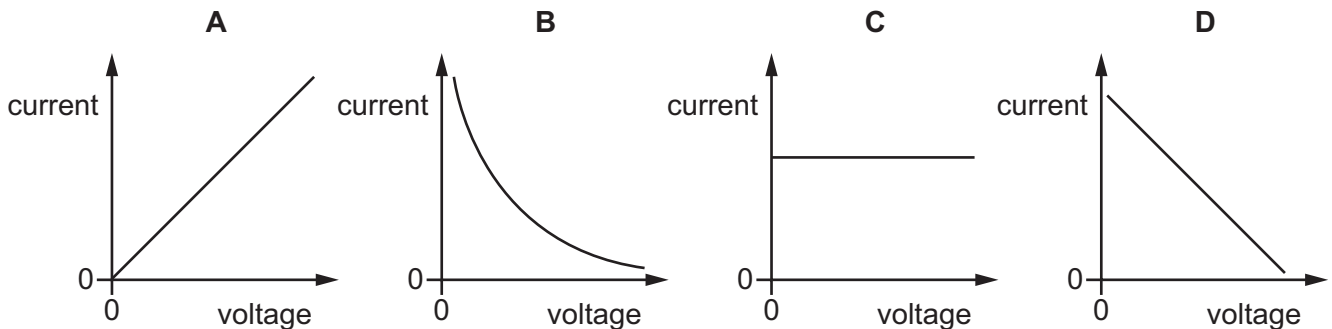
- 32 Two permanent magnets are placed side by side in four different arrangements.



In which arrangements do the magnets repel each other?

- A 1 and 2
- B 1 and 4
- C 2 and 3
- D 3 and 4

- 33 Which graph shows how the current changes when the voltage across a resistor of constant resistance is varied?



- 34 Two identical resistors are connected first in series and then in parallel.

The two resistors in series are connected to a power supply and the current is I_s . The parallel combination is connected to the same power supply and the current supplied is I_p .

What is the ratio $\frac{I_s}{I_p}$?

- A 0.25
- B 0.50
- C 2.0
- D 4.0

- 35** The output of an a.c. generator is connected to a lamp.

The coil of the generator is rotated at a constant rate which makes the lamp flash twice every second.

The coil is then rotated at half the rate of rotation.

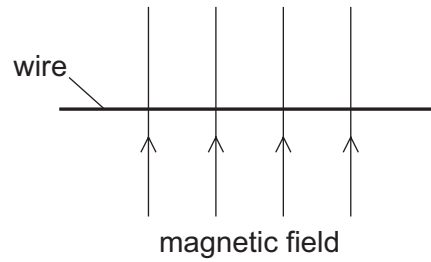
What happens to the maximum brightness of the lamp and how many times does the lamp flash every second?

	brightness	number of flashes every second
A	decreases	1
B	decreases	4
C	does not change	1
D	does not change	4

- 36** Which device uses the magnetic effect of an electric current?

- A** compass
- B** fuse
- C** loudspeaker
- D** magnifying glass

37 A wire is placed at right angles to a uniform magnetic field.



There is a current in the wire and it experiences a force.

Beams of alpha-particles, beta-particles and neutrons are fired into the magnetic field in the same direction as the conventional current in the wire.

Which beams experience a force that is initially in the same direction as the force experienced by the current-carrying wire?

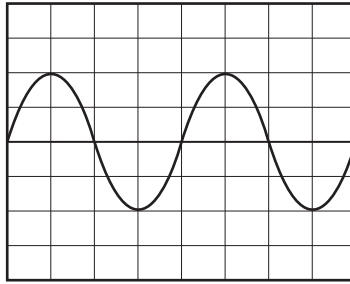
	alpha-particles	beta-particles	neutrons
A	✓	✗	✓
B	✓	✗	✗
C	✗	✓	✓
D	✗	✓	✗

key

✓ = beam experiences a force in the same direction

✗ = beam does not experience a force in the same direction

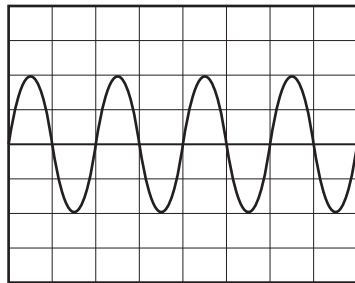
- 38 The diagram shows the trace produced on the screen of an oscilloscope due to an alternating voltage input.



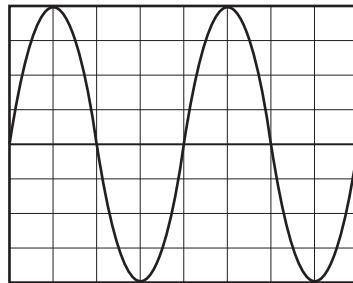
The time-base on the oscilloscope is set to 4 ms/cm.

What does the trace look like when the time base is set to 2 ms/cm?

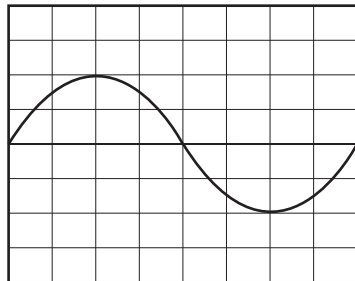
A



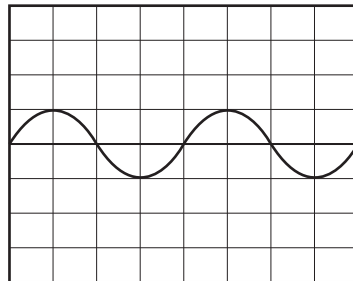
B



C



D



- 39 The half-life of a radioactive isotope is 2.5 minutes.

This isotope is the only radioactive isotope in a radioactive sample.

Between time $t = 0$ and $t = 7.5$ minutes, the mass of this isotope in the sample decreases by $70\ \mu\text{g}$.

What is the mass of this isotope at $t = 0$?

- A** $75\ \mu\text{g}$ **B** $80\ \mu\text{g}$ **C** $210\ \mu\text{g}$ **D** $560\ \mu\text{g}$

- 40** When observed on the Earth, the redshift of the light from galaxy X is smaller than the redshift of the light from galaxy Y.

Which galaxy is closer to the Earth and which galaxy is receding from the Earth faster?

	closer to the Earth	faster recession
A	X	X
B	X	Y
C	Y	X
D	Y	Y

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

October/November 2023

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
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Soft pencil (type B or HB is recommended)

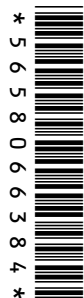
INSTRUCTIONS

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- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

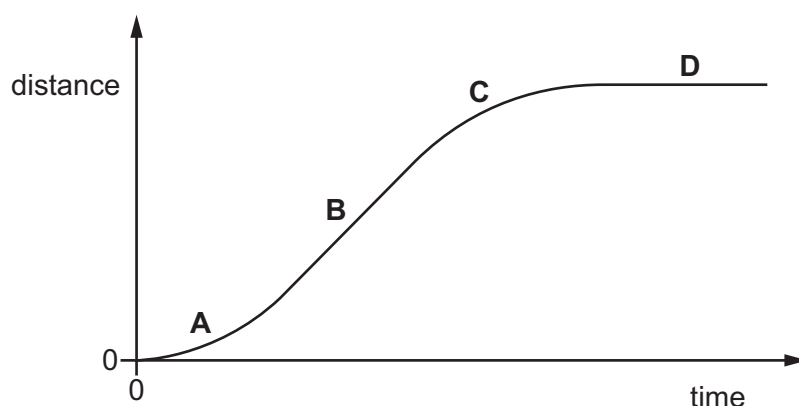


1 Which description is of a scalar quantity?

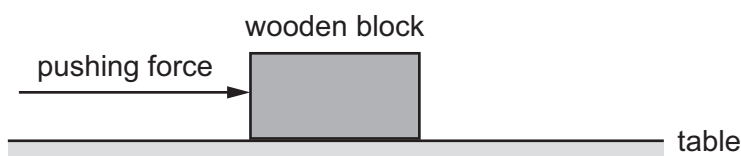
- A It has a direction only.
- B It has direction and unit only.
- C It has direction, magnitude and unit.
- D It has magnitude and unit only.

2 The diagram shows a distance–time graph for a car travelling in a straight line.

In which region is the car decelerating?



3 A wooden block is pushed across a table at constant speed.



Which statement is correct?

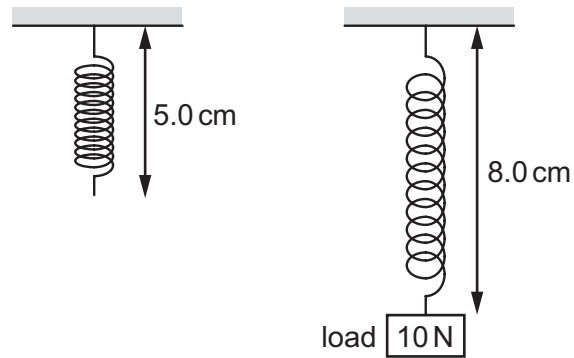
- A The frictional force increases as the block moves at constant speed.
- B The frictional force is equal and opposite to the pushing force.
- C The frictional force is greater than the pushing force.
- D The frictional force is less than the pushing force.

4 A car begins to move. It speeds up until it reaches a constant speed.

What happens to the acceleration and what happens to the velocity of the car?

- A Both the acceleration and the velocity change.
- B Only the acceleration changes.
- C Only the velocity changes.
- D Neither the acceleration nor the velocity changes.

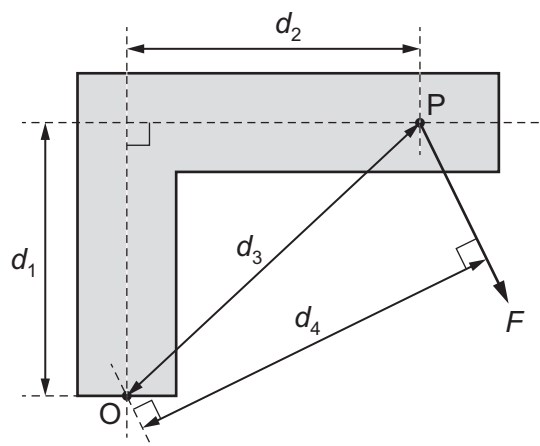
- 5 The diagram shows how the length of a spring changes when a load of 10 N is suspended from it.



The 10 N load is replaced by a 20 N load.

What is the new length of the spring?

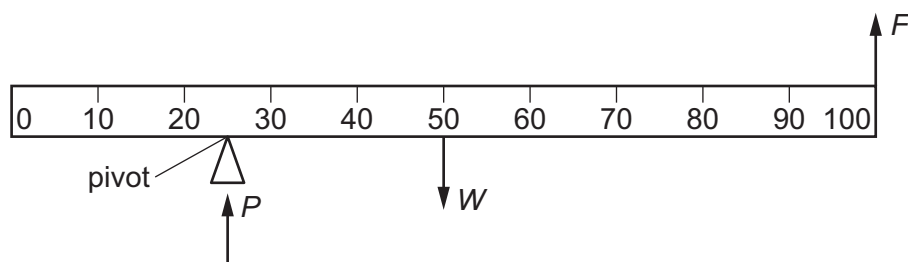
- A** 6.0 cm **B** 11 cm **C** 14 cm **D** 16 cm
- 6 A force F causes a moment about point O on an L-shaped bar. The force F acts at point P.



What is the moment of F about O?

- A** Fd_1 **B** Fd_2 **C** Fd_3 **D** Fd_4

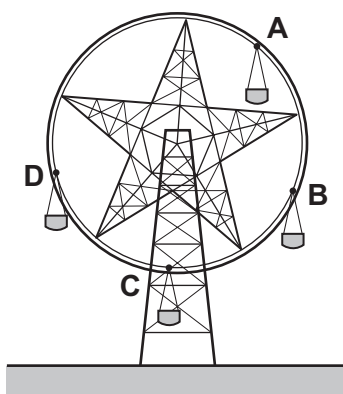
- 7 A uniform metre rule of weight W is pivoted at the 25 cm mark and held horizontal by a force F applied upwards at the 100 cm mark. The rule is supported by a vertical force P acting at the pivot.



What is the magnitude of P ?

- A $\frac{1}{3} W$ B $\frac{1}{2} W$ C $\frac{2}{3} W$ D $\frac{4}{3} W$
- 8 The diagram shows a stationary fairground ride with four chairs of equal mass.

Which chair has the most gravitational potential energy?



- 9 Hydroelectric and tidal power stations generate electrical energy.

Do these use renewable sources of energy?

	hydroelectric	tidal
A	no	no
B	no	yes
C	yes	no
D	yes	yes

10 Which energy resource is correctly described?

	energy resource	renewable	availability
A	biofuels	yes	at all times
B	nuclear	yes	intermittent
C	solar radiation	no	at all times
D	wind	no	intermittent

11 A small motor has an input power rating of 10 W and is switched on for 5.0 minutes.

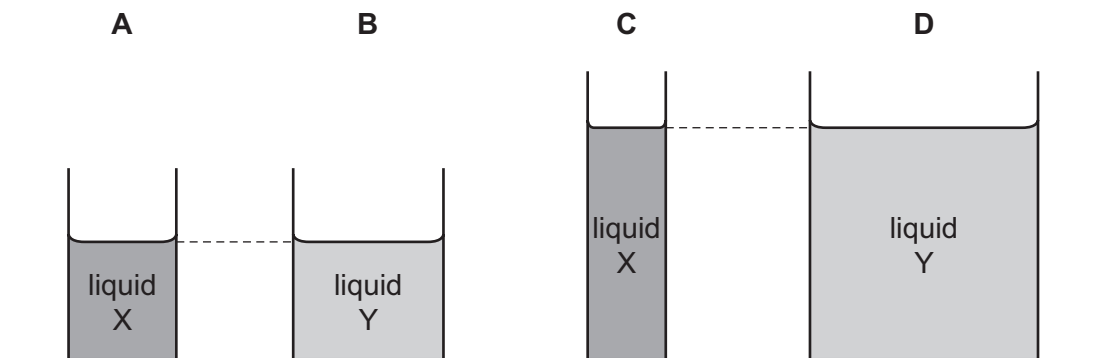
What is the electrical energy input to the motor in this time?

- A** 2.0 J **B** 50 J **C** 300 J **D** 3000 J

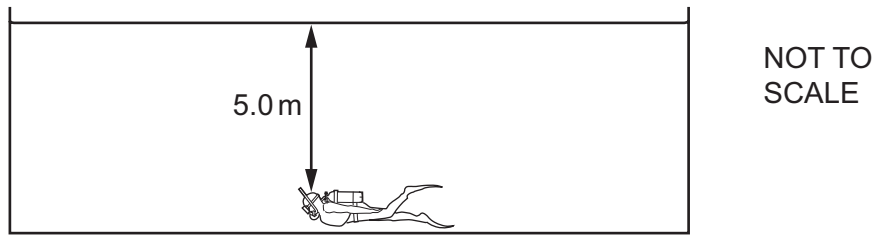
12 Two liquids, X and Y, are poured into beakers of different sizes.

Liquid X has a density of 1000 kg/m^3 and liquid Y has a density of 900 kg/m^3 .

On the base of which beaker is the pressure greatest?



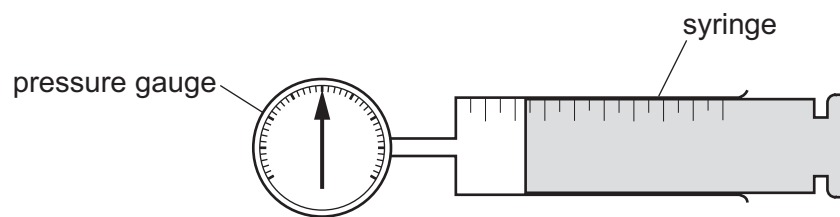
- 13 A swimmer is at the bottom of a pool. The density of the water in the pool is 1000 kg/m^3 and the gravitational field strength is 9.8 N/kg .



Which expression gives the pressure exerted on the swimmer due to the water?

- A $\frac{5.0 \times 1000}{9.8} \text{ Pa}$
- B $5.0 \times 1000 \times 9.8 \text{ Pa}$
- C $\frac{1000}{5.0} \text{ Pa}$
- D $\frac{1000 \times 9.8}{5.0} \text{ Pa}$

- 14 Some gas is trapped in a large syringe.



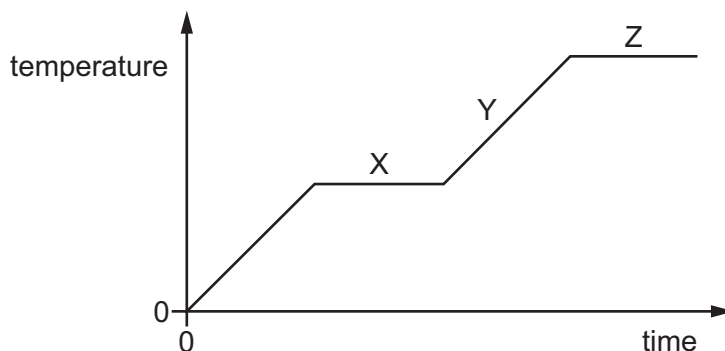
The atmospheric pressure is 100 kPa . The gas pressure is 200 kPa above atmospheric pressure.

The piston moves outwards and the volume of the trapped gas doubles. The temperature remains constant.

What is the new gas pressure?

- A 100 kPa B 150 kPa C 200 kPa D 400 kPa

- 15 A graph of temperature against time is shown for a material being heated from its solid state.



Which row describes what is happening at X, Y and Z?

	X	Y	Z
A	boiling	solid becomes hotter	melting
B	liquid becomes hotter	boiling	gas becomes hotter
C	solid becomes hotter	melting	liquid becomes hotter
D	melting	liquid becomes hotter	boiling

- 16 A student on a camping expedition cools a sealed bottle of water which is at the same temperature as the surrounding air.

Which method cools the water at the greatest rate?

- A** Wrap the bottle in aluminium foil and place it in a shady place.
- B** Wrap the bottle in dry, white paper and put it in a sunny place.
- C** Wrap the bottle in foam and put it in a breeze.
- D** Wrap the bottle in wet paper and put it in a breeze.
- 17 A student suggests three factors that affect the rate of emission of thermal energy by radiation from a hot object. These are:
- 1 the surface temperature of the object
 - 2 the surface area of the object
 - 3 the surface colour of the object.

Which suggestions are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 18** A plane mirror on a vertical wall forms an image of an object placed in front of it.

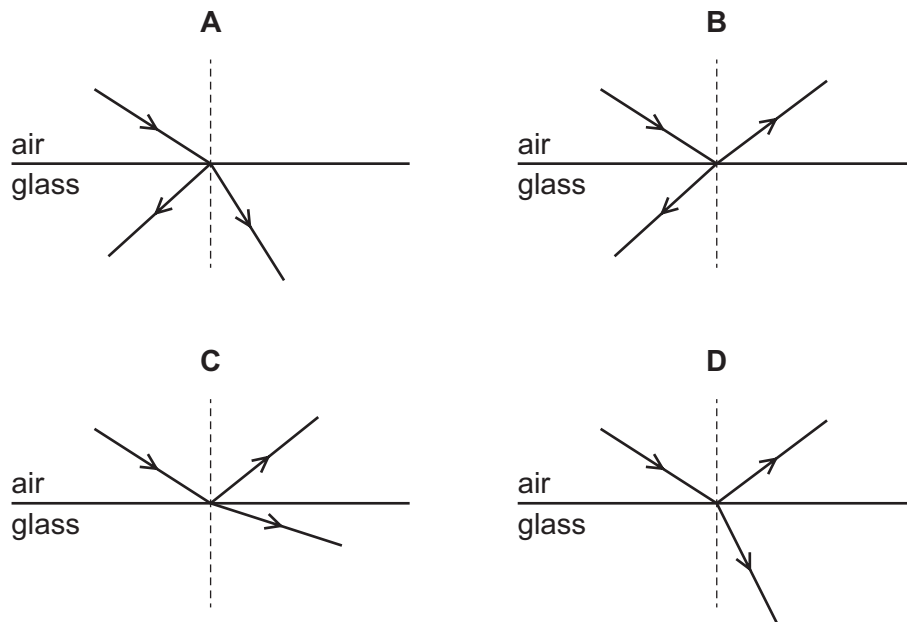
Which characteristics describe the image?

- A** real and inverted
- B** real and upright
- C** virtual and inverted
- D** virtual and upright

- 19** The diagrams show light travelling in air incident on the surface of a glass block.

Some light is reflected and some light is refracted.

Which diagram shows the reflection and refraction of the light?

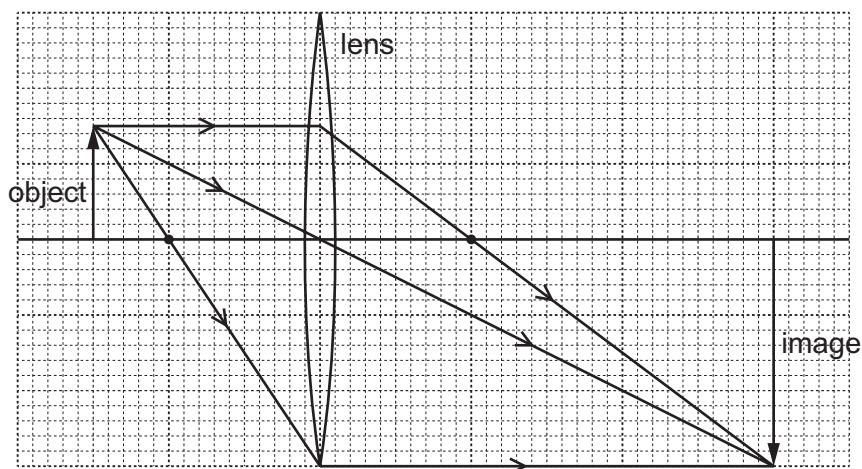


- 20** Which statement is correct?

- A** Total internal reflection only occurs when light travels from air into glass.
- B** The larger the refractive index of glass, the larger is the critical angle.
- C** When total internal reflection occurs, the angle of incidence is equal to the angle of reflection.
- D** When total internal reflection occurs, the angle of incidence is less than the critical angle.

- 21 An object is placed in front of a converging lens of focal length 4.0 cm. The height of the image is 6.0 cm.

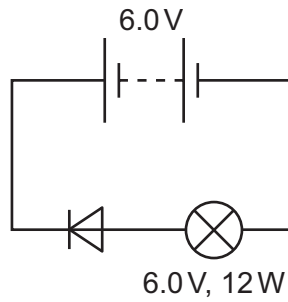
The arrangement is shown on the scale diagram.



What is the linear magnification produced by the lens?

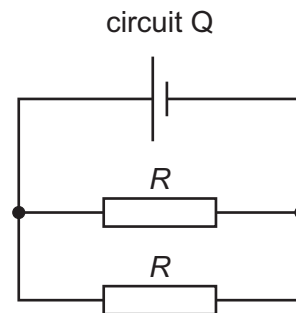
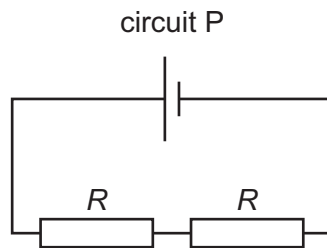
- A** 0.50 **B** 1.5 **C** 2.0 **D** 6.0
- 22 Which range of sound frequencies includes only frequencies of sound that can be heard by a healthy human ear?
- A** 0.5–50 Hz **B** 5–500 Hz **C** 50–5000 Hz **D** 500–50 000 Hz
- 23 An aircraft flying through the air becomes electrically charged.
- What causes this to happen?
- A** background radiation in the aircraft
B friction of the aircraft with the air
C heating of the aircraft surface
D radio waves emitted by the aircraft
- 24 Two metal spheres are brought close to each other. The two spheres repel each other.
- Which statement explains this?
- A** Both spheres are charged and both are positive.
B Both spheres are charged but one is positive and the other is negative.
C The spheres have the same number of electrons.
D The spheres have the same number of protons.

- 25 A student sets up the circuit shown. The lamp does not light.



What makes the lamp light?

- A** changing the supply to a 12 V battery
 - B** inserting a fuse
 - C** reversing the battery connections
 - D** reversing the connections to the lamp
- 26 Circuits P and Q each contain two identical resistors, each of resistance R .



What is the resistance of circuit P and what is the resistance of circuit Q?

	resistance of circuit P	resistance of circuit Q
A	R	$\frac{R}{2}$
B	R	$2R$
C	$2R$	$\frac{R}{2}$
D	$2R$	$2R$

- 27 The current in an electric motor is 2.0 A.

It transfers 20 J of energy in 5.0 s.

What is the potential difference (p.d.) across the motor?

- A** 0.50 V
- B** 2.0 V
- C** 8.0 V
- D** 50 V

28 Three students make comments about appliances with double insulation.

student P Appliances with double insulation do not need an earth wire.

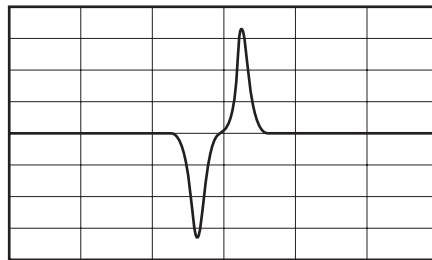
student Q Double insulation protects the user of the appliance from an electric shock.

student R The circuits of appliances with double insulation do not contain fuses.

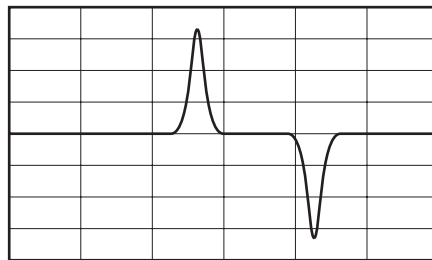
Which students are correct?

A P, Q and R **B** P and Q only **C** P and R only **D** Q and R only

29 A magnet moves at constant speed through a coil of wire. The coil is connected to an oscilloscope. The diagram shows the trace produced on the screen.



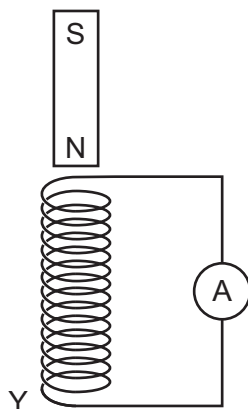
The experiment is changed and a new trace is produced.



Which statement explains why the new trace is different from the original trace?

- A** The coil is longer and the other pole of the magnet enters the coil first.
- B** The coil is longer and the same pole of the magnet enters the coil first.
- C** The coil is shorter and the other pole of the magnet enters the coil first.
- D** The coil is shorter and the same pole of the magnet enters the coil first.

- 30** A bar magnet is dropped through a vertical solenoid that is connected to a sensitive ammeter.

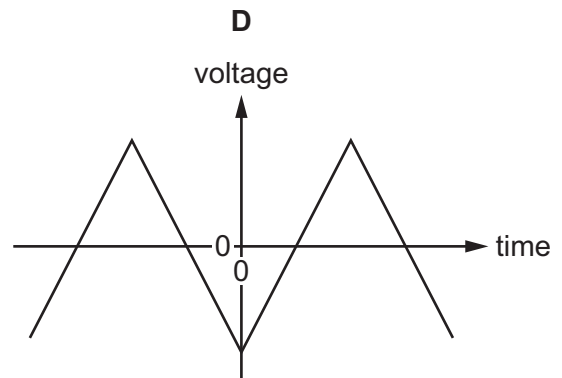
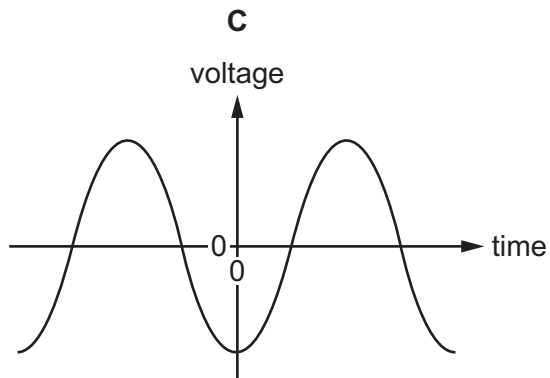
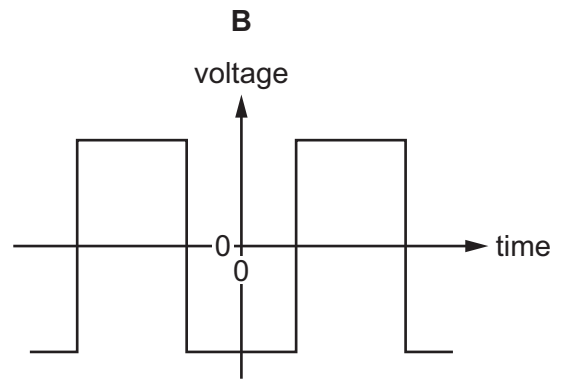
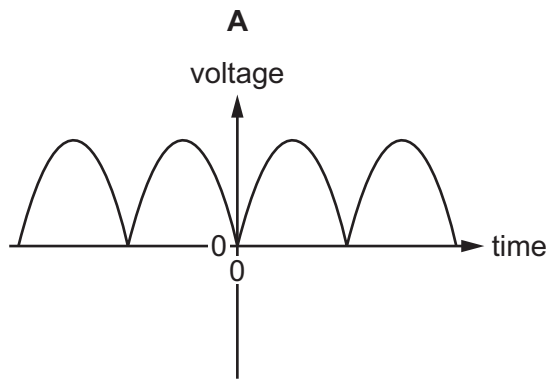


A current is induced in the solenoid and the solenoid becomes an electromagnet.

Which row describes the polarity of end Y of the solenoid and the force between the solenoid and the bar magnet when the bar magnet leaves the solenoid?

	polarity of end Y of the solenoid	force between solenoid and bar magnet
A	north	attraction
B	north	repulsion
C	south	attraction
D	south	repulsion

31 Which graph shows the voltage output against time for a simple a.c. generator?



32 A charger for a mobile phone (cellphone) uses a transformer which steps the voltage down from 230 V to 5.0 V.

The primary coil in the transformer has 2500 turns.

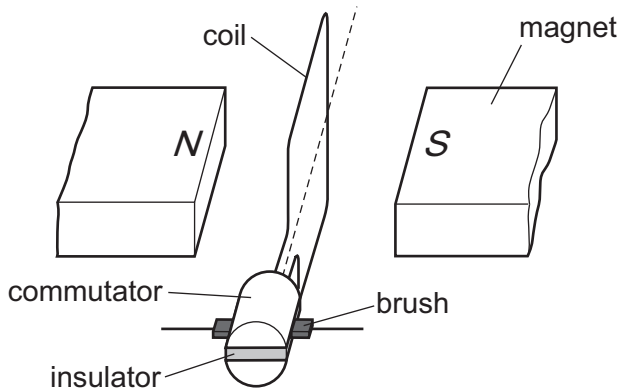
How many turns are on the secondary coil?

- A** 11 **B** 54 **C** 500 **D** 11 500

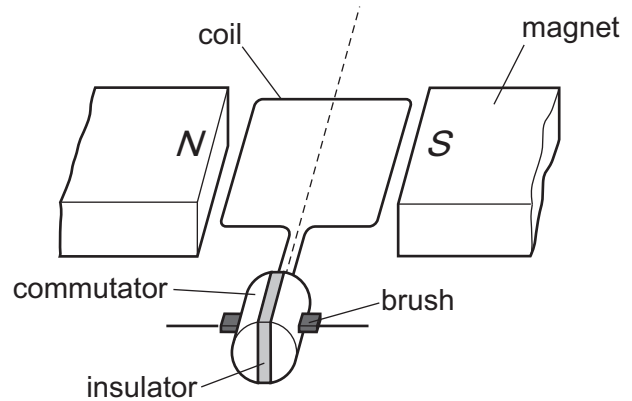
33 A commutator is connected to the coil of a d.c. motor.

The commutator rotates and connects the coil to the power supply.

Once every half-turn, the current in the coil reverses.



coil perpendicular to magnetic field

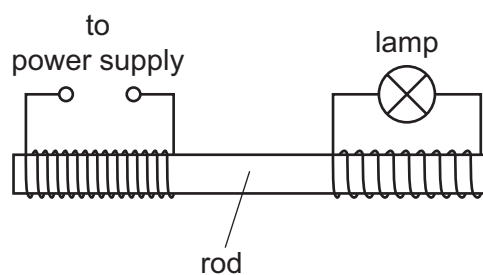


coil parallel to magnetic field

What is the name of the commutator and what is the orientation of the plane of the coil as the current reverses?

	name	orientation
A	slip ring	parallel to the magnetic field
B	slip ring	perpendicular to the magnetic field
C	split ring	parallel to the magnetic field
D	split ring	perpendicular to the magnetic field

34 A girl makes a simple transformer to illuminate a low voltage lamp. The transformer is shown.



What is the material of the rod and what is the type of power supply?

	material of rod	power supply
A	plastic	d.c. supply
B	plastic	a.c. supply
C	iron	d.c. supply
D	iron	a.c. supply

35 Which particles are in the nucleus of a large atom?

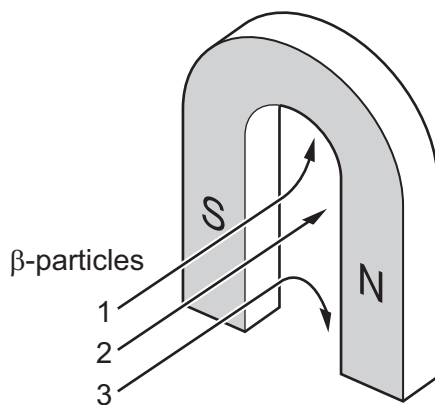
- A neutrons and electrons only
- B protons and electrons only
- C protons and neutrons only
- D protons, neutrons and electrons

36 Alpha (α) particles and beta (β) particles are emitted during radioactive decay.

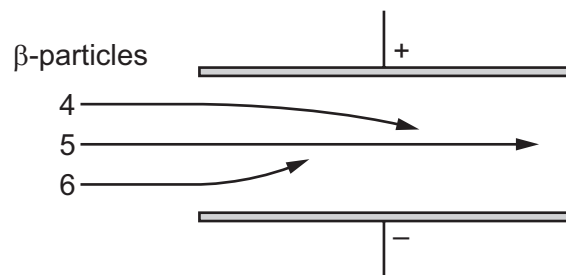
Which particle has the weaker ionising effect and which has the weaker penetrating power?

	weaker ionising effect	weaker penetrating power
A	α	α
B	α	β
C	β	α
D	β	β

37 A beam of beta (β) particles passes into a magnetic field and into an electric field.



into magnetic field



into electric field

Which numbers show the paths of the beam through the fields?

- A 1 and 4
- B 1 and 6
- C 2 and 5
- D 3 and 6

38 The isotope radon-222 has a half-life of 3.8 days.

Which statement about a sample containing 1.0 mg of radon-222 is correct?

- A** After 1.9 days there is 0.50 mg of radon-222 left.
- B** After 3.8 days there is 0.50 mg of radon-222 left.
- C** Its half-life is half that of a 2.0 mg sample.
- D** There is no radon-222 left after 7.6 days.

39 What is the first stage in the life cycle of stars?

- A** a black hole
- B** a cloud of gas and dust
- C** a red supergiant
- D** a supernova

40 What is the definition of redshift?

- A** the increase in observed frequency of light from galaxies that are moving away from the Earth
- B** the increase in observed wavelength of light from galaxies that are moving away from the Earth
- C** the light emitted by distant galaxies that is blue when it reaches the Earth
- D** the light reaching the Earth that is red when it is emitted by distant galaxies

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Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2023

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	B	1
2	D	1
3	D	1
4	D	1
5	A	1
6	C	1
7	B	1
8	B	1
9	A	1
10	B	1
11	D	1
12	A	1
13	A	1
14	B	1
15	D	1
16	A	1
17	D	1
18	C	1
19	A	1
20	B	1
21	B	1
22	D	1
23	C	1
24	A	1
25	A	1
26	A	1
27	D	1
28	D	1

Question	Answer	Marks
29	B	1
30	C	1
31	B	1
32	C	1
33	A	1
34	A	1
35	A	1
36	C	1
37	B	1
38	C	1
39	B	1
40	B	1



Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

October/November 2023

MARK SCHEME

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This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	C	1
3	B	1
4	A	1
5	B	1
6	D	1
7	C	1
8	A	1
9	D	1
10	A	1
11	D	1
12	C	1
13	B	1
14	B	1
15	D	1
16	D	1
17	A	1
18	D	1
19	D	1
20	C	1
21	C	1
22	C	1
23	B	1
24	A	1
25	C	1
26	C	1
27	B	1
28	B	1

Question	Answer	Marks
29	A	1
30	A	1
31	C	1
32	B	1
33	D	1
34	D	1
35	C	1
36	C	1
37	D	1
38	B	1
39	B	1
40	B	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2022

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

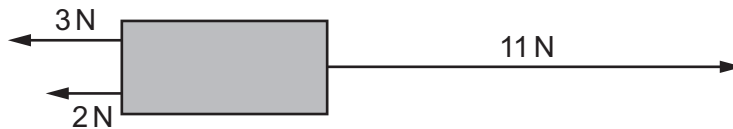
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 The diagram shows three forces acting on a block. The resultant force is 6 N to the right.

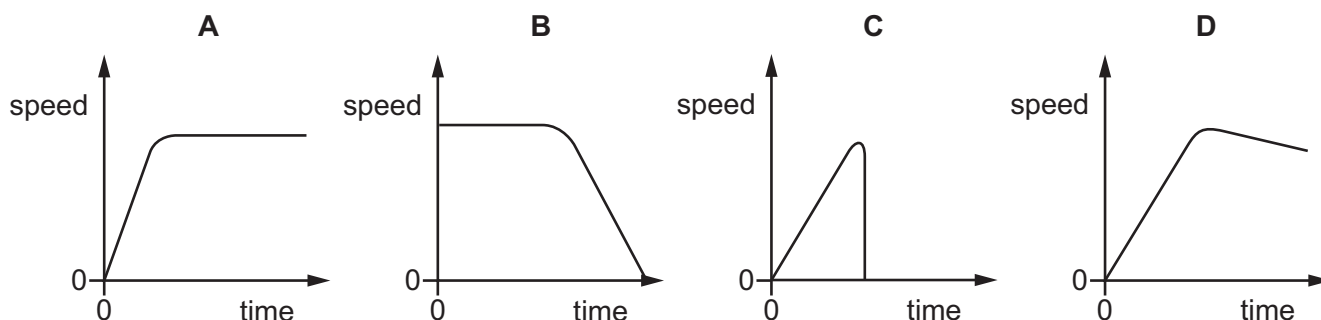


Which additional force produces a resultant force of 3 N to the left?

- A** 3 N to the left
B 9 N to the left
C 6 N to the right
D 13 N to the right
- 2 A teacher measures the length of her classroom.
What is the most appropriate instrument to use?
- A** a 30 cm ruler
B a caliper
C a micrometer
D a tape
- 3 Which value is one-thousandth of a metre?
- A** 0.0001 cm **B** 0.001 cm **C** 0.01 cm **D** 0.1 cm
- 4 A car of mass 1000 kg is travelling down a steep hill. The brakes fail and the driver uses a horizontal sand-filled safety road to stop the car.
The car enters the sand at a speed of 10 m/s and experiences a constant stopping force of 2500 N.
How far does the car travel in the sand before coming to rest?
- A** 2.0 m **B** 4.0 m **C** 20 m **D** 40 m

- 5 A ball starts from rest and rolls down a steep slope. The ball then rolls along rough horizontal ground.

Which graph shows the speed of the ball at different times?



- 6 A block of wood is placed on a table. External forces act on the block but the block remains stationary.

A student suggests three conditions for the block to remain stationary.

condition P The resultant force on the block **must** equal zero.

condition Q The resultant moment on the block **must** equal zero.

condition R The external forces **must** act through the centre of mass of the block.

Which conditions are necessary for the block to remain stationary?

A P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

- 7 A car travels at 60 km/h on a straight road. The road is dry. The driver applies the brakes suddenly. The table shows the thinking distance, the braking distance and the stopping distance.

thinking distance /m	braking distance /m	stopping distance /m
27	22	49

The same car is driven by the same driver at 60 km/h on the same road when it is wet. The driver applies the brakes suddenly again.

What is the effect of the wet road on the thinking distance, the braking distance and the stopping distance?

	thinking distance	braking distance	stopping distance
A	no change	increases	increases
B	no change	no change	no change
C	increases	increases	increases
D	increases	no change	increases

- 8 A car travels on a horizontal road around a bend at constant speed.

What is the direction of the resultant force on the car?

- A backwards
- B forwards
- C towards the centre of the bend
- D towards the road

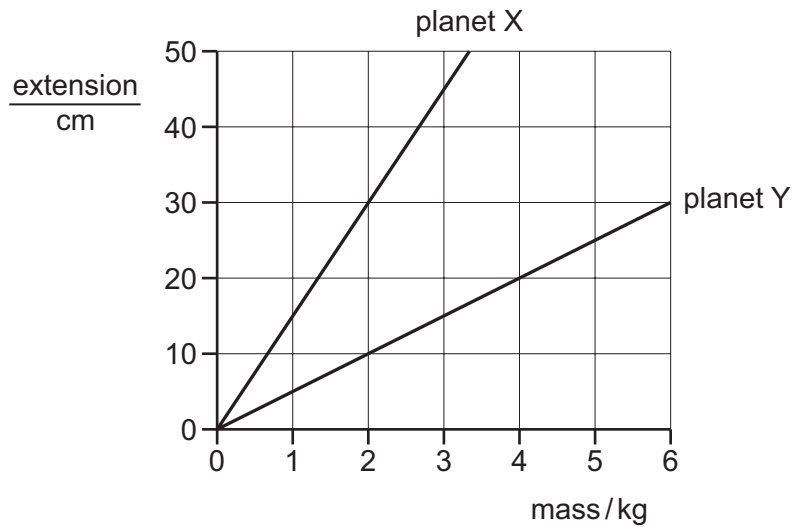
- 9 A student writes two statements about mass and weight.

- 1 Mass is the property of a body which resists change in its state of rest or motion.
- 2 Weight is the amount of substance in a body.

Which statements are correct?

- A statement 1 only
- B statement 2 only
- C statement 1 and statement 2
- D neither statement 1 nor statement 2

- 10 The graph shows how the extension of a spring changes with the masses suspended from it when the spring is on planet X and when the spring is on planet Y.



Which conclusion can be drawn from these graphs?

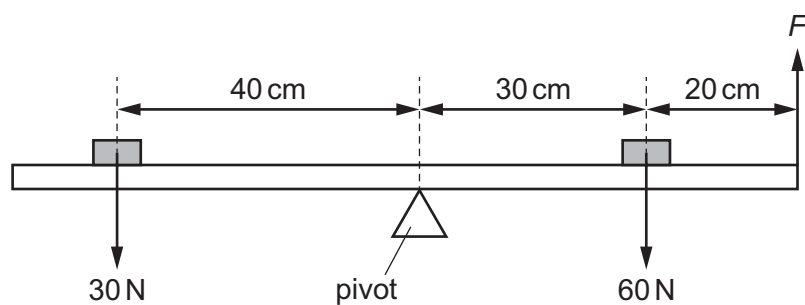
- A** It is not possible to compare the gravitational field strengths on planets X and Y.
- B** The gravitational field strength on planet X is equal to the gravitational field strength on planet Y.
- C** The gravitational field strength on planet X is one third of the gravitational field strength on planet Y.
- D** The gravitational field strength on planet X is three times the gravitational field strength on planet Y.
- 11 A student finds the density of an irregularly shaped object. He chooses his equipment from this list.

- 1 stop-watch
- 2 measuring cylinder
- 3 balance

Which equipment does the student need to use?

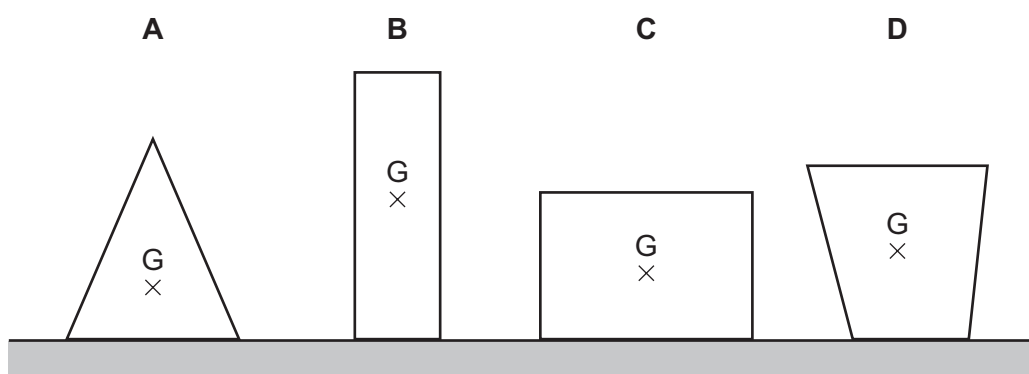
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 12 A uniform beam is pivoted at its centre. Two weights are placed on the beam in the positions shown and the beam is balanced by an upward force F .



What is the size of F ?

- A 6.0 N B 12 N C 30 N D 60 N
- 13 Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.
- Which object is the least stable?



- 14 Each tyre of a car has an area of 100 cm^2 in contact with the ground.

The car has a mass of 1600 kg. The weight of the car is equally distributed amongst the four tyres.

The gravitational field strength g is 10 N/kg .

What is the pressure exerted on the ground?

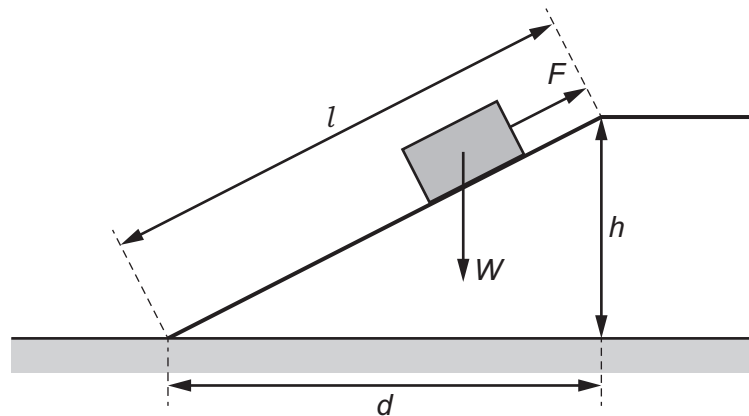
- A 4.0 N/cm^2 B 16 N/cm^2 C 40 N/cm^2 D 160 N/cm^2
- 15 A gas syringe contains a volume V_1 of gas at pressure p_1 .

The volume is reduced to V_2 without any change of temperature.

What is the expression for the **change** in pressure?

- A $\frac{p_1 V_1}{V_2} + p_1$ B $p_1 + \frac{p_1 V_2}{V_1}$ C $\frac{p_1 V_1}{V_2} - p_1$ D $\frac{p_1 V_2}{V_1} - p_1$

- 16 A constant force F pulls a block of weight W up the slope shown.



How much work is done by F in pulling the block up the slope?

- A** $F \times h$ **B** $F \times l$ **C** $W \times d$ **D** $W \times l$

- 17 The input power to a motor is 12 W.

The motor wastes 590 J of energy in 1.0 min.

What is the efficiency of the motor?

- A** 18% **B** 22% **C** 55% **D** 82%

- 18 Electric motors have an efficiency of about 90% when used in an electric train.

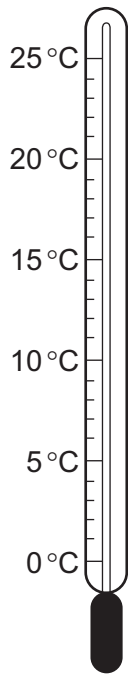
Which forms of wasted energy are produced?

- A** sound only
B thermal only
C thermal and chemical
D thermal and sound

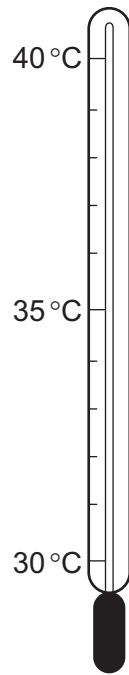
- 19 What is a unit of power?

- A** Js **B** J/s **C** Ns **D** N/s

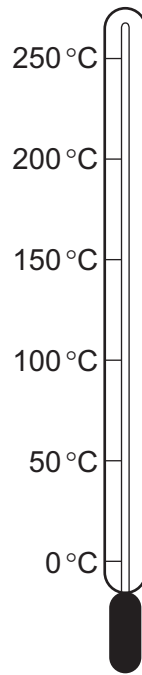
20 The diagrams represent four thermometers.



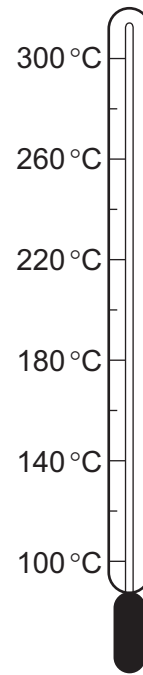
P



Q



R



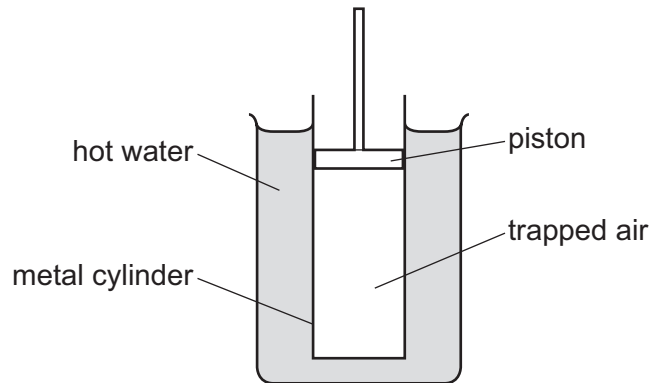
S

Which thermometer has the greatest sensitivity and which thermometer has the greatest range?

	greatest sensitivity	greatest range
A	P	R
B	P	S
C	Q	R
D	Q	S

- 21** Air is trapped in a metal cylinder by a piston. The piston is free to move and the trapped air is at atmospheric pressure.

The cylinder is in hot water.

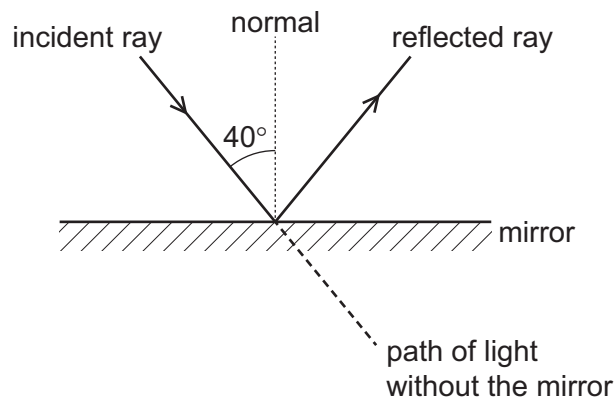


The cylinder is taken out of the water and left to cool.

What happens to the mass of the air in the cylinder and its pressure as it cools?

	mass of air	pressure of air
A	does not change	does not change
B	does not change	changes
C	changes	does not change
D	changes	changes

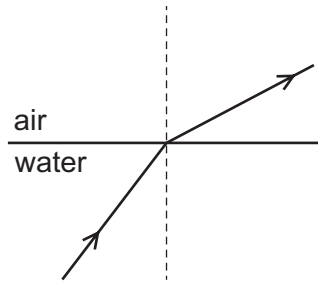
- 22** A mirror is placed in the path of a ray of light.



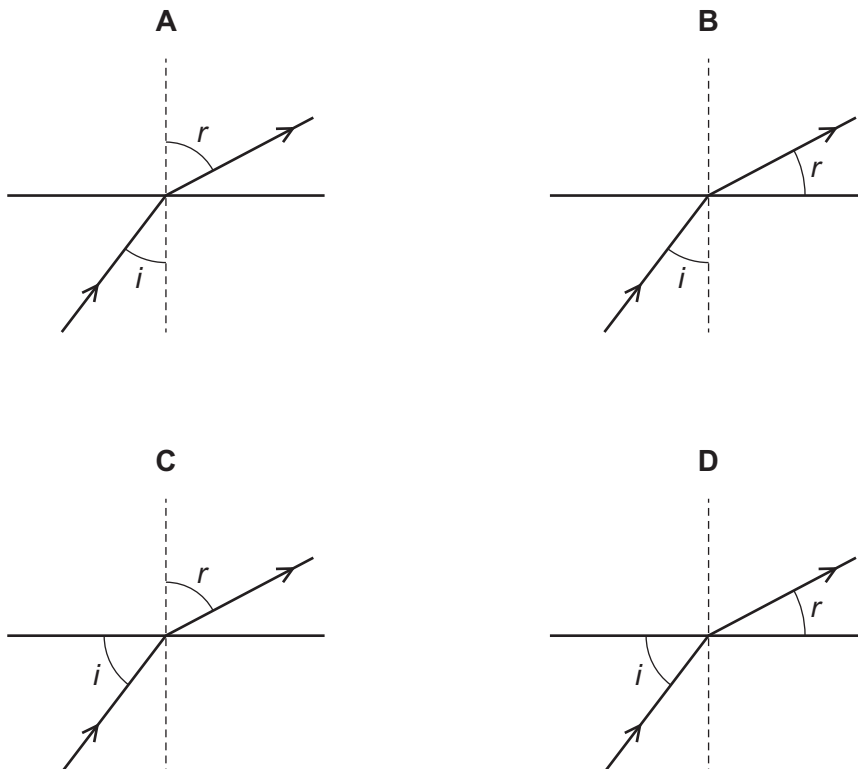
Through which angle does the direction of the ray of light change?

- A** 40° **B** 90° **C** 100° **D** 140°

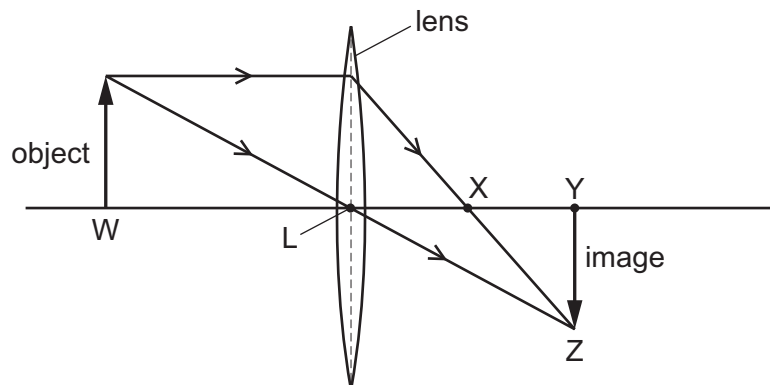
- 23 A ray of light in water is refracted at the surface into air.



Which diagram shows the angle of incidence i and the angle of refraction r ?



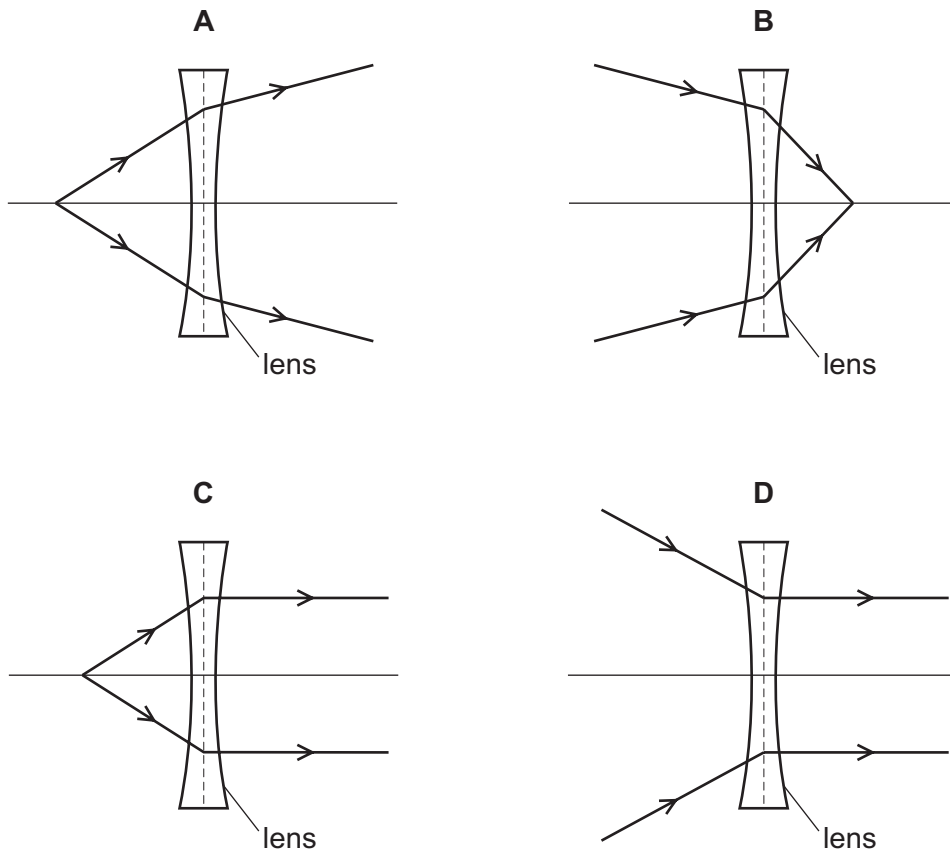
- 24 A thin converging lens forms a real, focused image of an object, as shown.



Which distance is equal to the focal length of the lens?

- A LW B LX C LY D LZ

25 Which ray diagram shows the action of a diverging lens?



26 Many devices produce electromagnetic waves when operating.

Which device produces electromagnetic waves of the highest frequency?

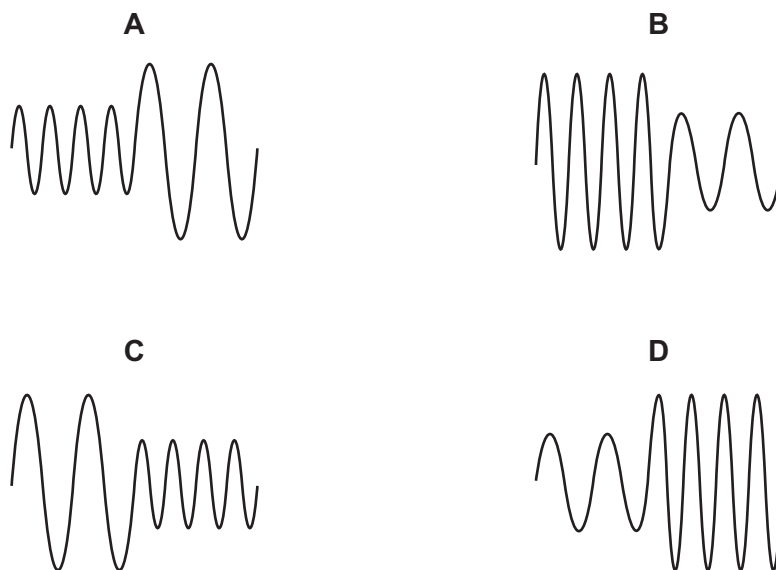
- A mobile phone
- B sunbed
- C television controller
- D toaster

27 Which two frequencies are both outside the range of audible frequencies for a healthy human ear?

- A 2.0 Hz and 500 Hz
- B 2.0 Hz and 25 kHz
- C 25 Hz and 25 kHz
- D 500 Hz and 25 kHz

28 Sound waves are displayed as a trace on the screen of an oscilloscope.

Which trace shows a sound that becomes quieter with a higher pitch?



29 The N pole of a permanent magnet is placed in turn close to a piece of iron, the S pole of another magnet, and a piece of copper.

Which objects are attracted to the N pole of the permanent magnet?

	piece of iron	S pole	piece of copper
A	✓	✓	x
B	x	✓	x
C	x	✓	✓
D	x	x	✓

key

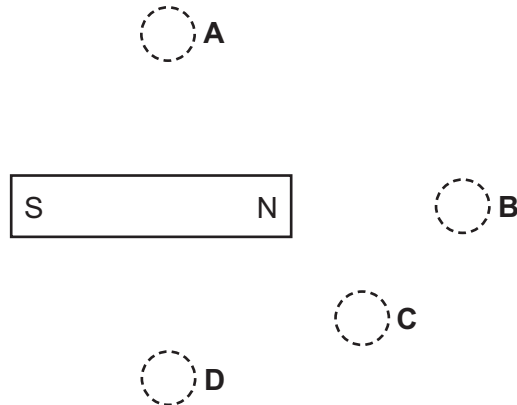
✓ = attracted

x = not attracted

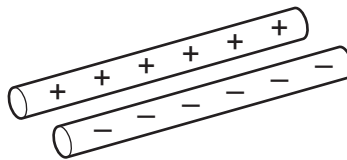
- 30** Four plotting compasses are placed near a bar magnet. Ignore any effects of the Earth's magnetic field.

One compass appears like this .

What is a possible position for this compass?



- 31** Two oppositely charged plastic rods are placed next to each other.



Which row describes and explains how the rods move?

	movement	explanation
A	away from each other	opposite charges attract
B	away from each other	opposite charges repel
C	towards each other	opposite charges attract
D	towards each other	opposite charges repel

- 32** A meter that measures the rate of flow of charge in a resistor is placed in a circuit.

What is the name of this meter and how is it connected to the resistor?

	name	connection
A	ammeter	in parallel
B	ammeter	in series
C	voltmeter	in parallel
D	voltmeter	in series

33 Which device is used to convert electrical energy to kinetic energy?

- A** iron
- B** kettle
- C** lamp
- D** motor

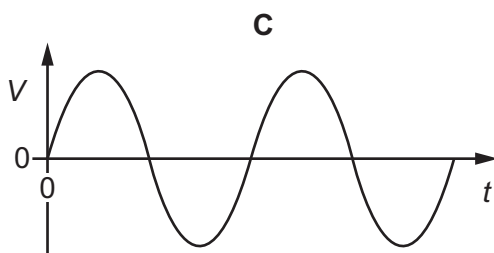
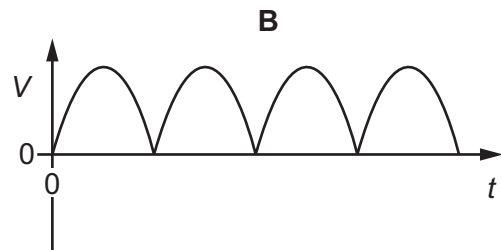
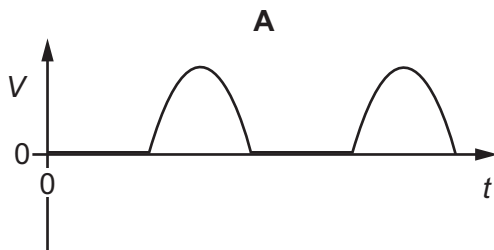
34 Why is a fuse used in an electrical appliance?

- A** to earth the appliance
- B** to protect the appliance
- C** to change the power of the appliance
- D** to change the voltage supplied to the appliance

35 A coil of wire is rotated at a constant rate between the poles of a U-shaped magnet.

The two ends of the coil are connected to different slip rings.

Which graph shows how the voltage V between the slip rings varies with time t ?



36 Which statement about a transformer is correct?

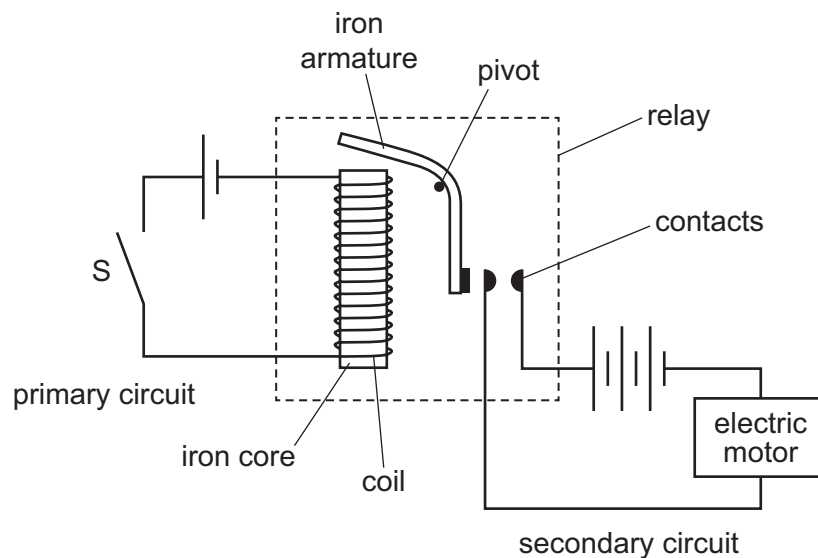
- A** The changing magnetic field in the transformer induces an e.m.f. in the secondary coil.
- B** The core of the transformer is made of iron because iron is a good electrical conductor.
- C** The transformer converts alternating current to direct current.
- D** The transformer converts direct current to alternating current.

- 37 Three types of radiation emitted by unstable nuclei are helium nuclei, short wavelength electromagnetic waves and electrons.

What are these three types of radiation?

	helium nuclei	electromagnetic waves	electrons
A	alpha	gamma	beta
B	beta	gamma	alpha
C	gamma	alpha	beta
D	gamma	beta	alpha

- 38 The diagram shows a relay used to switch on an electric motor.



A student makes five statements to explain how the relay switches on the electric motor. The statements are **not** in the correct order.

- 1 The current in the coil magnetises the electromagnet.
- 2 The armature closes the contacts.
- 3 The current in the secondary circuit makes the motor turn.
- 4 The electromagnet attracts the iron armature.
- 5 The switch S in the primary circuit is closed.

What is the correct order of the statements?

- A** 1 → 5 → 2 → 4 → 3
- B** 3 → 2 → 4 → 1 → 5
- C** 3 → 4 → 1 → 5 → 2
- D** 5 → 1 → 4 → 2 → 3

39 Different radioactive emissions have different characteristics.

A student lists the following characteristics for an alpha-particle.

characteristic P It has a positive charge.

characteristic Q It has a high penetrating ability.

characteristic R It is a component of the electromagnetic spectrum.

Which characteristics are correct?

- A** P only **B** R only **C** P and Q **D** Q and R

40 A wooden object is believed to be 6000 years old.

Which isotope, found in the object, is used to determine the age of the object?

- A** ^{12}C **B** ^{13}C **C** ^{14}C **D** ^{15}C

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

October/November 2022

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

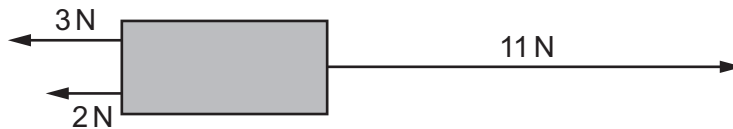
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

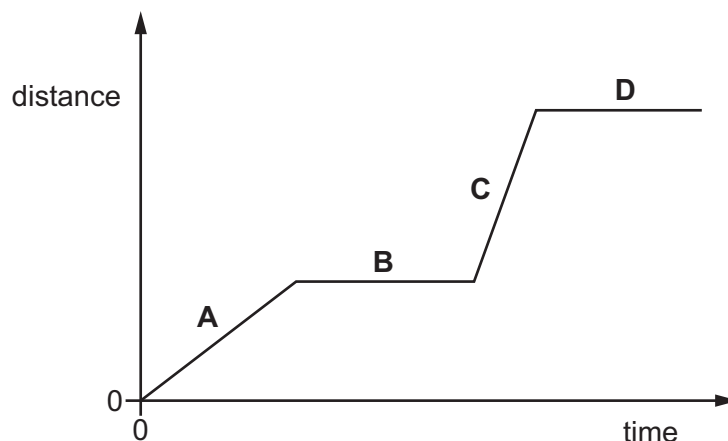


- 1 The diagram shows three forces acting on a block. The resultant force is 6 N to the right.

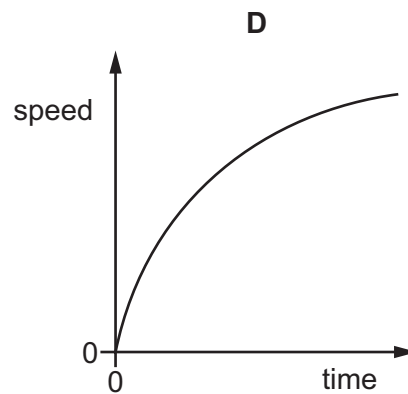
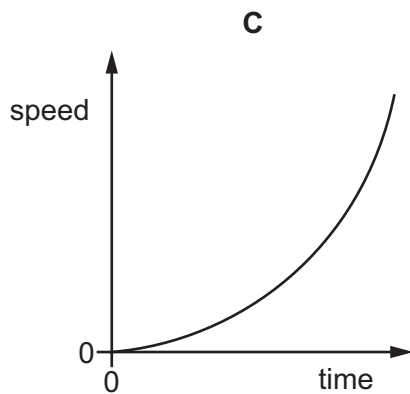
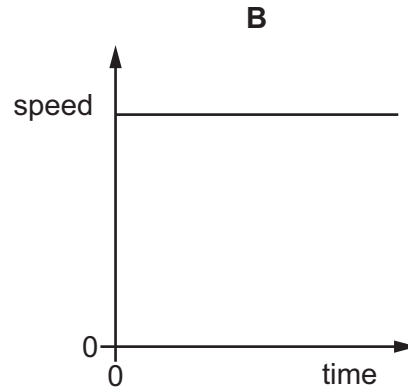
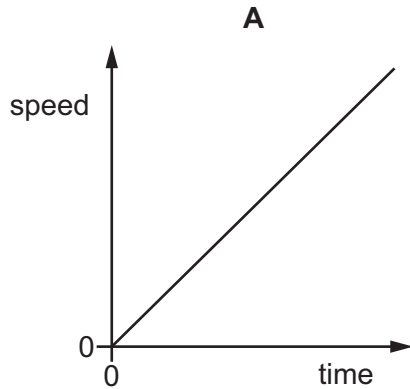


Which additional force produces a resultant force of 3 N to the left?

- A** 3 N to the left
B 9 N to the left
C 6 N to the right
D 13 N to the right
- 2 Velocity is given by the change in displacement divided by the change in time.
 How many vector quantities appear in this statement?
- A** 0 **B** 1 **C** 2 **D** 3
- 3 A car of mass 1000 kg is travelling down a steep hill. The brakes fail and the driver uses a horizontal sand-filled safety road to stop the car.
 The car enters the sand at a speed of 10 m/s and experiences a constant stopping force of 2500 N.
 How far does the car travel in the sand before coming to rest?
- A** 2.0 m **B** 4.0 m **C** 20 m **D** 40 m
- 4 Which part of the distance–time graph shows the fastest constant speed?

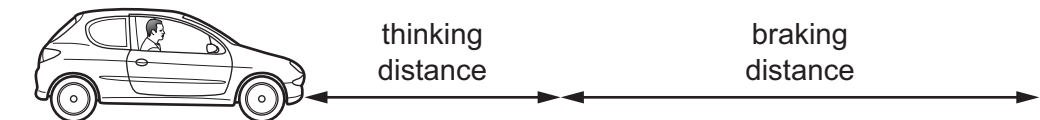


- 5 An object is accelerating along a straight, horizontal track. The acceleration is uniform. Which speed–time graph represents the motion of the object?



- 6 What causes the frictional effects experienced by a cyclist to increase?
- A** using a lighter bicycle
 - B** travelling at greater speed
 - C** a smoother tyre surface
 - D** a road surface covered in oil

- 7 The diagram shows a car at the moment the driver sees a hazard ahead.

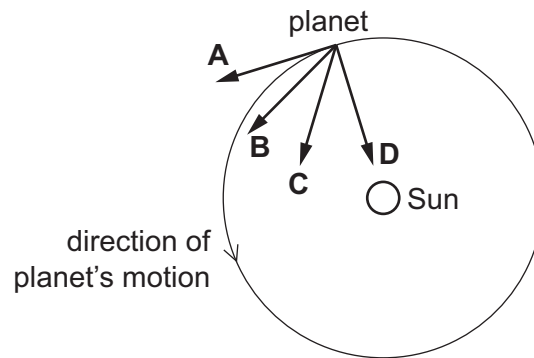


Which expression is equal to the stopping distance?

- A** braking distance + thinking distance
- B** braking distance – thinking distance
- C** braking distance $\times$ thinking distance
- D** braking distance $\div$ thinking distance

- 8 A planet orbits the Sun. The planet moves in a circular path at constant speed, but is accelerating because it is changing direction.

In which direction does the resultant force act to cause this acceleration?

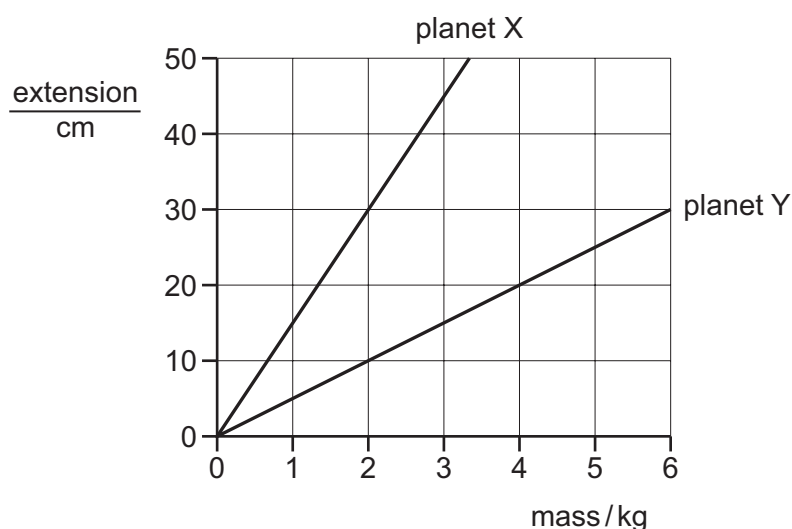


- 9 X is the result of Y divided by Z.

What are X, Y and Z?

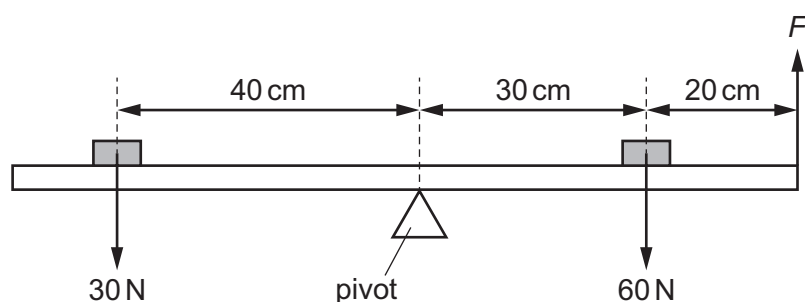
	X	Y	Z
A	mass	weight	gravitational field strength
B	weight	mass	gravitational field strength
C	mass	gravitational field strength	weight
D	weight	gravitational field strength	mass

- 10 The graph shows how the extension of a spring changes with the masses suspended from it when the spring is on planet X and when the spring is on planet Y.



Which conclusion can be drawn from these graphs?

- A** It is not possible to compare the gravitational field strengths on planets X and Y.
- B** The gravitational field strength on planet X is equal to the gravitational field strength on planet Y.
- C** The gravitational field strength on planet X is one third of the gravitational field strength on planet Y.
- D** The gravitational field strength on planet X is three times the gravitational field strength on planet Y.
- 11 A uniform beam is pivoted at its centre. Two weights are placed on the beam in the positions shown and the beam is balanced by an upward force F .

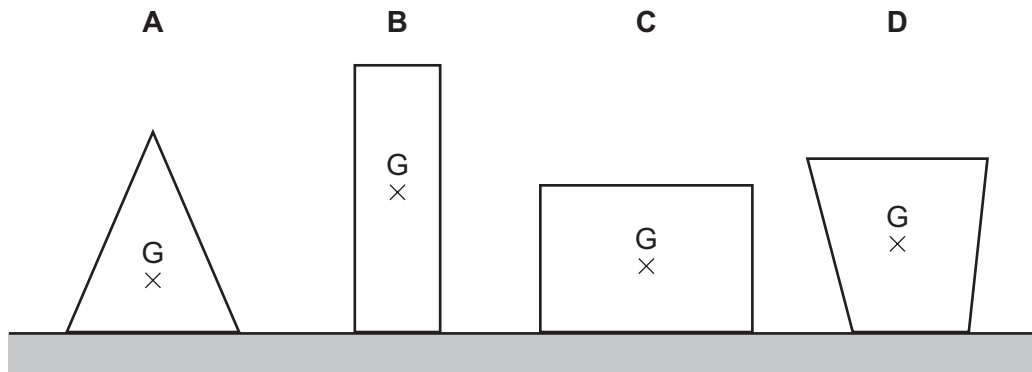


What is the size of F ?

- A** 6.0 N **B** 12 N **C** 30 N **D** 60 N

- 12 Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

Which object is the least stable?



- 13 Each tyre of a car has an area of 100 cm^2 in contact with the ground.

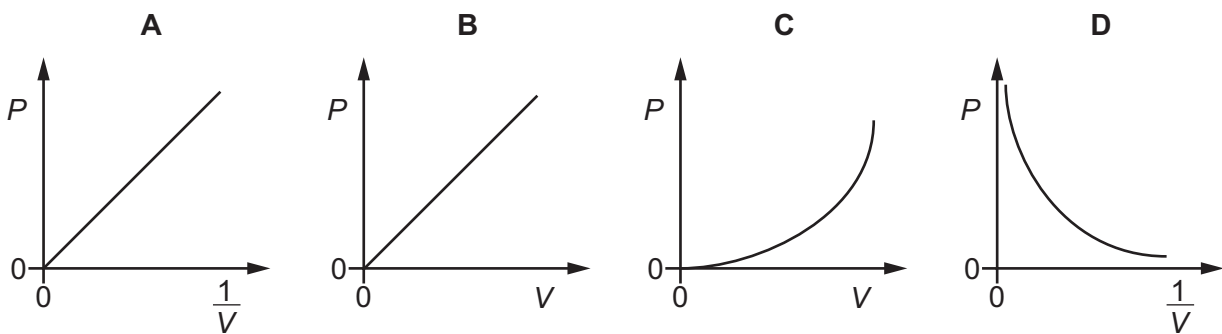
The car has a mass of 1600 kg . The weight of the car is equally distributed amongst the four tyres.

The gravitational field strength g is 10 N/kg .

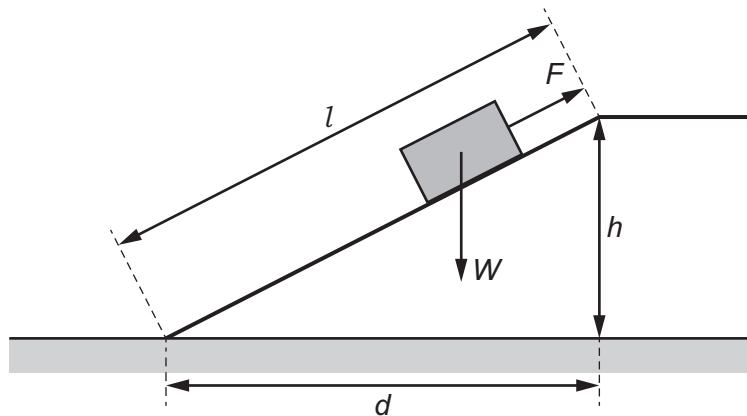
What is the pressure exerted on the ground?

- A 4.0 N/cm^2 B 16 N/cm^2 C 40 N/cm^2 D 160 N/cm^2
- 14 The pressure P of some trapped air is varied. The mass and the temperature of the trapped air remain constant.

Which graph shows how the volume V of the air varies with the pressure P ?

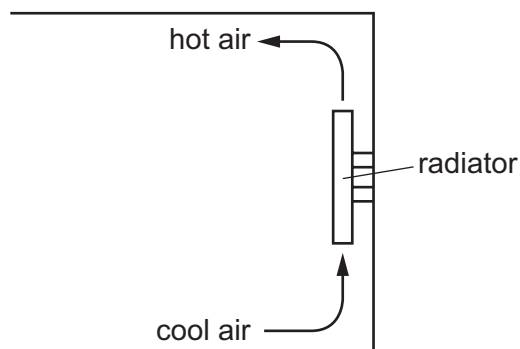


- 15 A constant force F pulls a block of weight W up the slope shown.



How much work is done by F in pulling the block up the slope?

- A** $F \times h$ **B** $F \times l$ **C** $W \times d$ **D** $W \times l$
- 16 X and Y are both 40 W, 240 V lamps. Lamp X is more efficient than lamp Y.
- Which statement correctly compares X with Y?
- A** X is more expensive to run.
B X has a lower electrical resistance.
C X produces less light energy.
D X produces less thermal energy.
- 17 The diagram shows a radiator heating the air in a room.



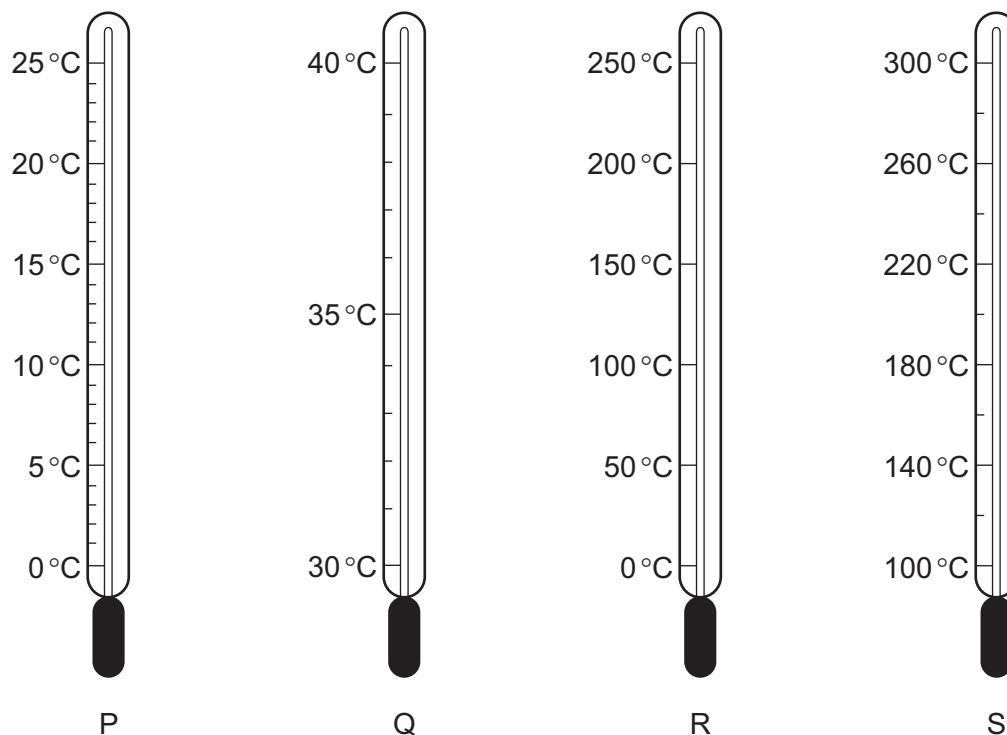
What is the name of this process?

- A** conduction
B convection
C evaporation
D expansion

18 What is a process of heat transfer that can take place in a vacuum?

- A conduction
- B convection
- C evaporation
- D radiation

19 The diagrams represent four thermometers.



Which thermometer has the greatest sensitivity and which thermometer has the greatest range?

	greatest sensitivity	greatest range
A	P	R
B	P	S
C	Q	R
D	Q	S

- 20** A piece of iron of mass m is placed in a mixture of ice and water.

Its temperature decreases from θ_1 to θ_2 .

How much thermal energy is lost by the piece of iron?

- A** $m \times \text{specific heat capacity of iron} \times (\theta_1 - \theta_2)$
- B** $m \times \text{specific heat capacity of iron} \times \theta_2$
- C** $m \times \text{heat capacity of piece of iron} \times (\theta_1 - \theta_2)$
- D** $m \times \text{heat capacity of piece of iron} \times \theta_2$

- 21** A kilogram of a substance has a fixed shape.

Thermal energy supplied to the substance weakens the forces of attraction between its molecules.

What is happening?

- A** A gas is being heated.
- B** A liquid is at its boiling point.
- C** A liquid is evaporating.
- D** A solid is at its melting point.

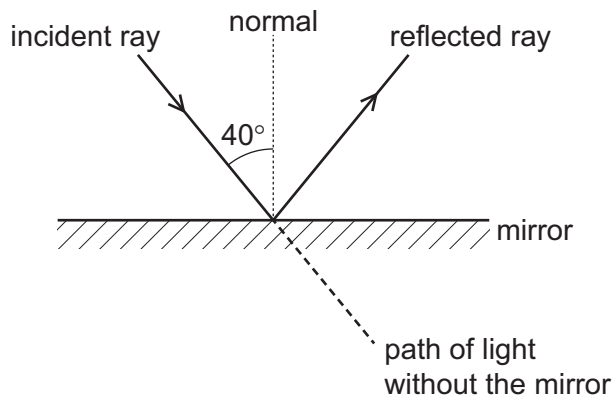
- 22** Which conditions produce the greatest rate of evaporation from a liquid?

- A** high temperature, draught and large surface area of liquid
- B** high temperature, no draught and large surface area of liquid
- C** low temperature, draught and small surface area of liquid
- D** low temperature, no draught and small surface area of liquid

- 23** Which statement is correct?

- A** Both longitudinal waves and transverse waves can travel at the speed of light.
- B** Both longitudinal waves and transverse waves of similar wavelengths must have similar frequencies.
- C** Both longitudinal waves and transverse waves can travel in a vacuum.
- D** Both longitudinal waves and transverse waves can travel in water.

- 24 A mirror is placed in the path of a ray of light.

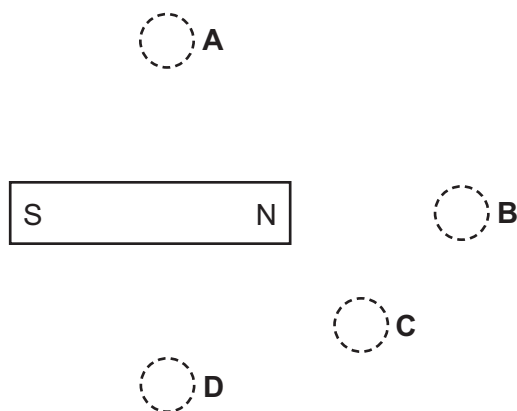


Through which angle does the direction of the ray of light change?

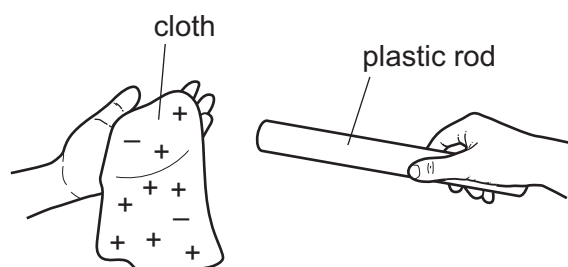
- A** 40° **B** 90° **C** 100° **D** 140°
- 25 An oscilloscope is used to demonstrate a sound.
- Which feature of the oscilloscope trace is affected by the quality (timbre) of the sound?
- A** amplitude
B frequency
C shape
D wavelength
- 26 Four plotting compasses are placed near a bar magnet. Ignore any effects of the Earth's magnetic field.

One compass appears like this .

What is a possible position for this compass?



- 27 A plastic rod and a cloth are initially uncharged. The rod is then rubbed with the cloth and the cloth becomes positively charged.



What is the sign of the charge on the plastic rod and why does it become charged?

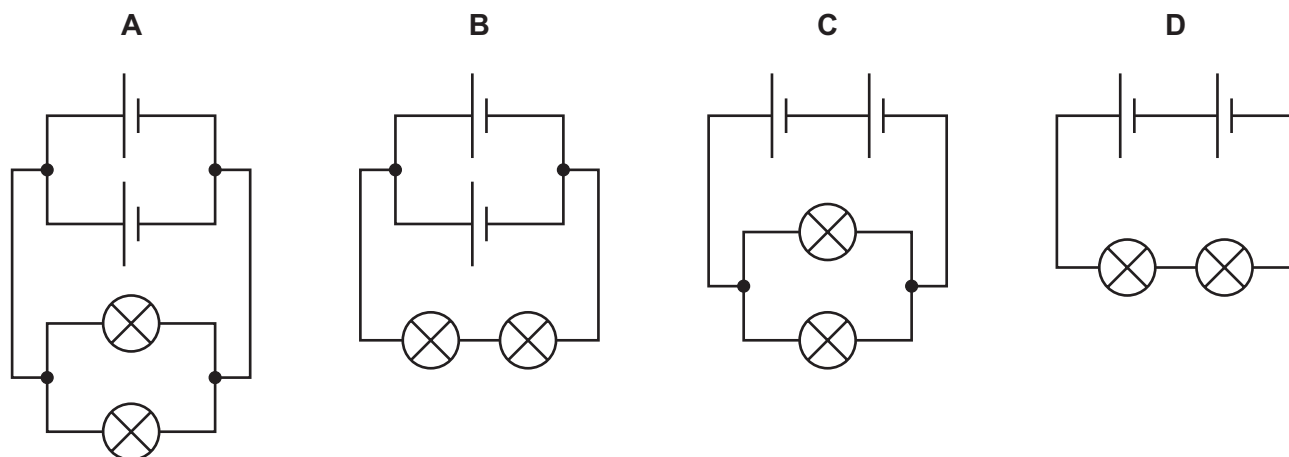
	charge on rod	reason why rod becomes charged
A	negative	electrons move from the cloth to the rod
B	negative	protons move from the rod to the cloth
C	positive	electrons move from the rod to the cloth
D	positive	protons move from the cloth to the rod

- 28 Which expression gives the electromotive force (e.m.f.) of a battery?

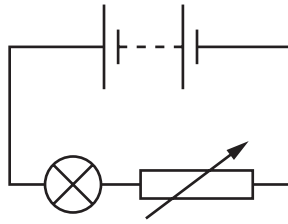
- A** work done $\times$ time
B work done $\times$ charge
C $\frac{\text{work done}}{\text{time}}$
D $\frac{\text{work done}}{\text{charge}}$

- 29 A circuit contains two cells and two lamps. The cells are new and are fully charged.

In which arrangement of the cells and lamps do the cells discharge the most quickly?



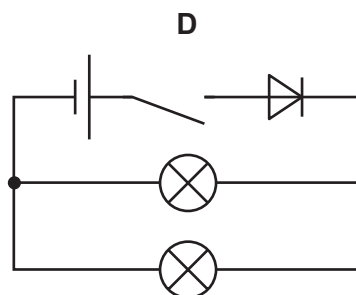
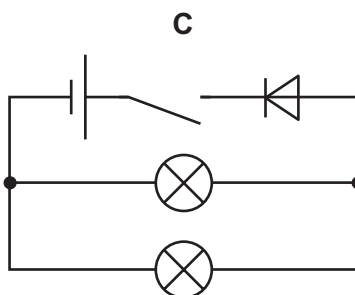
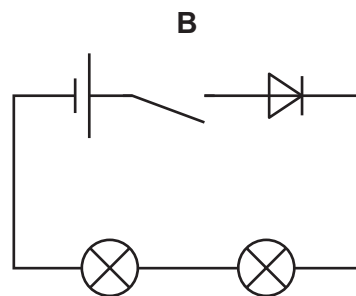
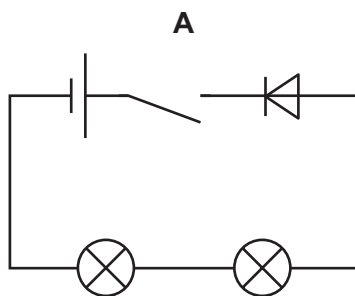
- 30** A lamp is connected in series with a battery and a variable resistor. The resistance of the variable resistor is decreased gradually.



Which statement is correct?

- A** The current in the lamp increases.
 - B** The current in the variable resistor decreases.
 - C** The e.m.f. of the battery decreases.
 - D** The total resistance in the circuit increases.
- 31** In the circuits shown all the cells are identical and all the lamps are identical. The switches are closed.

In which circuit are both lamps the brightest?



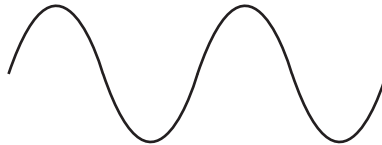
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- A** to earth the appliance
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- A The changing magnetic field in the transformer induces an e.m.f. in the secondary coil.
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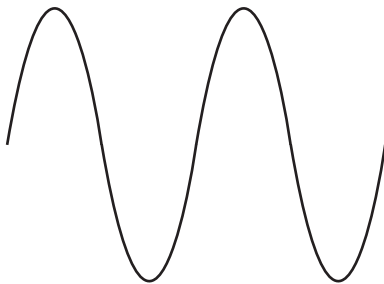
34 The diagram shows the output from an a.c. generator displayed on an oscilloscope screen.



The generator is turned at a slower rate. The settings on the oscilloscope controls are not changed.

Which diagram shows the new output?

A



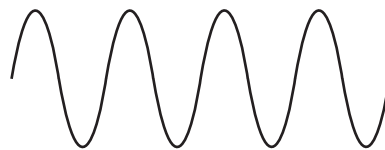
B



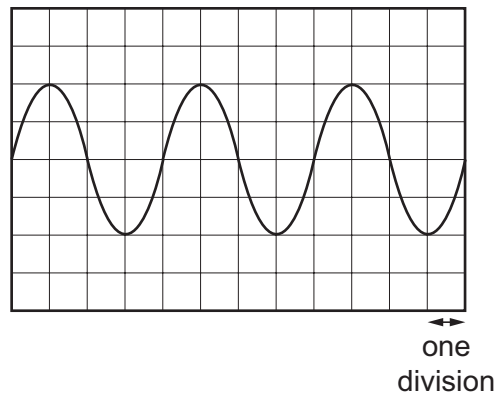
C



D



- 35** The sound wave produced by a note is displayed on the screen of a cathode-ray oscilloscope.
- The time-base is set at 5 ms for one division.



What is the frequency of the note?

- A** 20 Hz **B** 50 Hz **C** 60 Hz **D** 150 Hz
- 36** Three types of radiation emitted by unstable nuclei are helium nuclei, short wavelength electromagnetic waves and electrons.

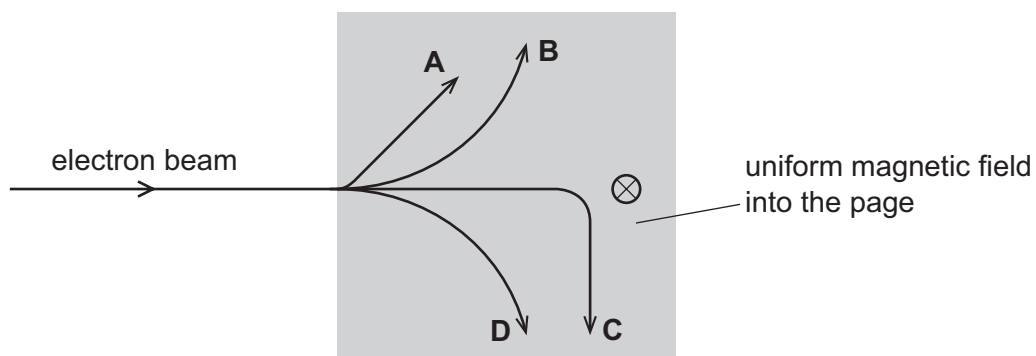
What are these three types of radiation?

	helium nuclei	electromagnetic waves	electrons
A	alpha	gamma	beta
B	beta	gamma	alpha
C	gamma	alpha	beta
D	gamma	beta	alpha

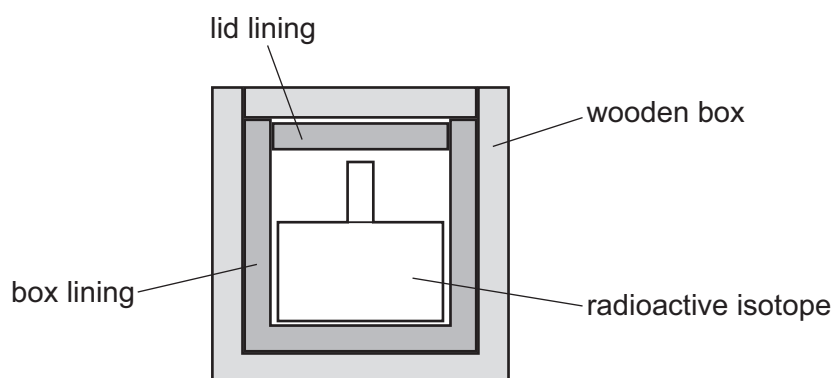
- 37** A beam of electrons is fired into a uniform magnetic field as shown.

The direction of the magnetic field is into the page.

Which path do the electrons follow?



- 38 Which statement about all three types of radioactive emissions is correct?
- A They are all completely absorbed by a thin aluminium sheet.
 - B They are all deflected by electric fields.
 - C They are all deflected by magnetic fields.
 - D They all cause ionisation of gases.
- 39 What is meant by the half-life of a radioactive substance?
- A half the time for all the unstable nuclei to decay
 - B half the time it takes an unstable nucleus to emit radiation
 - C the time for half the unstable nuclei to decay
 - D the time for the nucleon number to halve
- 40 Radioactive isotopes that emit alpha-particles, beta-particles and gamma rays are stored in wooden boxes that are lined with a material that prevents most radiation escaping from the box.



Which material is used for the lining?

- A copper
- B glass
- C lead
- D plastic

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Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	D	1
3	D	1
4	C	1
5	D	1
6	A	1
7	A	1
8	C	1
9	A	1
10	D	1
11	D	1
12	B	1
13	B	1
14	C	1
15	C	1
16	B	1
17	A	1
18	D	1
19	B	1
20	C	1
21	A	1
22	C	1
23	A	1
24	B	1
25	D	1
26	B	1
27	B	1
28	C	1

Question	Answer	Marks
29	A	1
30	C	1
31	C	1
32	B	1
33	D	1
34	B	1
35	C	1
36	A	1
37	A	1
38	D	1
39	A	1
40	C	1



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9	A	1
10	D	1
11	B	1
12	B	1
13	C	1
14	A	1
15	B	1
16	D	1
17	B	1
18	D	1
19	C	1
20	A	1
21	D	1
22	A	1
23	D	1
24	C	1
25	C	1
26	C	1
27	A	1
28	D	1

Question	Answer	Marks
29	C	1
30	A	1
31	D	1
32	B	1
33	A	1
34	B	1
35	B	1
36	A	1
37	D	1
38	D	1
39	C	1
40	C	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2021

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

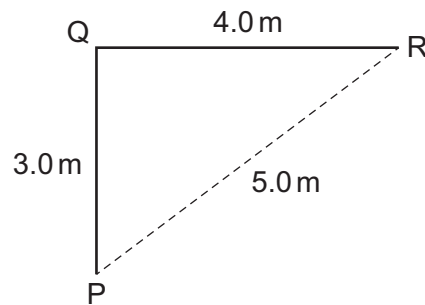
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



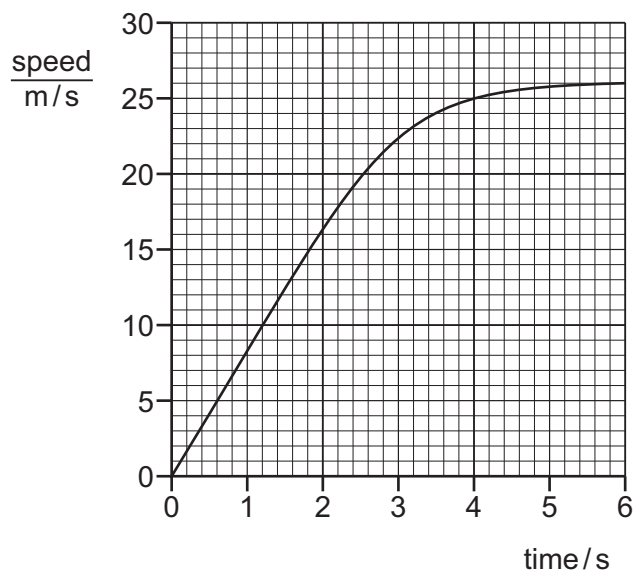
- 1 A boy starts at P and walks 3.0 m due north from P to Q and then 4.0 m due east from Q to R.



What is the shortest distance that he must now walk to have an overall displacement of zero?

- A** 3.0 m **B** 4.0 m **C** 5.0 m **D** 7.0 m
- 2 A student investigates the motion of a ball falling through the air.
- Which quantity is a vector?
- A** the diameter of the ball
- B** the gravitational force on the ball
- C** the distance from which the ball is dropped
- D** the speed at which the ball hits the ground
- 3 A skier slides down a slope.

The diagram shows how his speed varies with time.

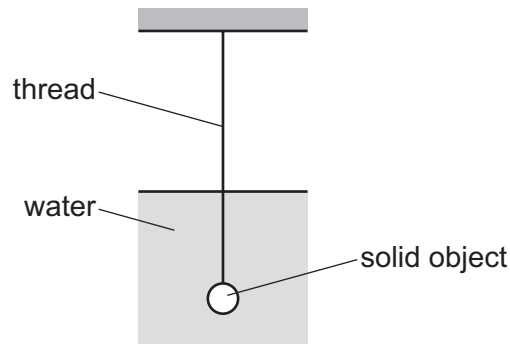


What is his average acceleration during the 6.0 s?

- A** 2.2 m/s^2 **B** 4.3 m/s^2 **C** 8.0 m/s^2 **D** 13.0 m/s^2

- 4 A solid object, immersed in water, hangs from an elastic thread. Three forces act on the object: its weight W , the tension in the thread T , and a force F from the water.

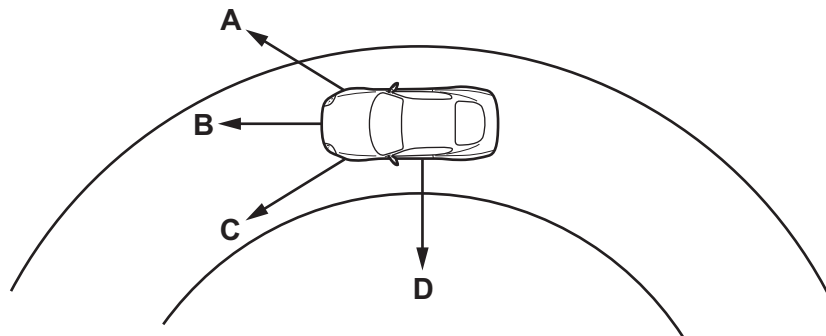
The force F acts upwards.



Which equation is correct when the object is stationary?

- A $F + W = 0$
 - B $F - T = 0$
 - C $F - W - T = 0$
 - D $F - W + T = 0$
- 5 The diagram shows a car going around a circular track at constant speed.

Which arrow shows the direction of the resultant force on the car?

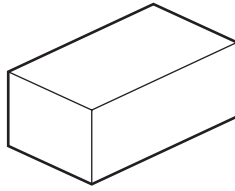


- 6 An object in a laboratory is used to determine the gravitational field strength g .

Which quantity is equal to g ?

- A the weight of the object
- B the mass of the object
- C the $\frac{\text{weight}}{\text{mass}}$ of the object
- D the $\frac{\text{mass}}{\text{weight}}$ of the object

- 7 A student takes measurements to find the density of a regularly shaped block.



Which formula is used to calculate the density of the block?

A $\text{density} = \frac{\text{mass}}{\text{volume}}$

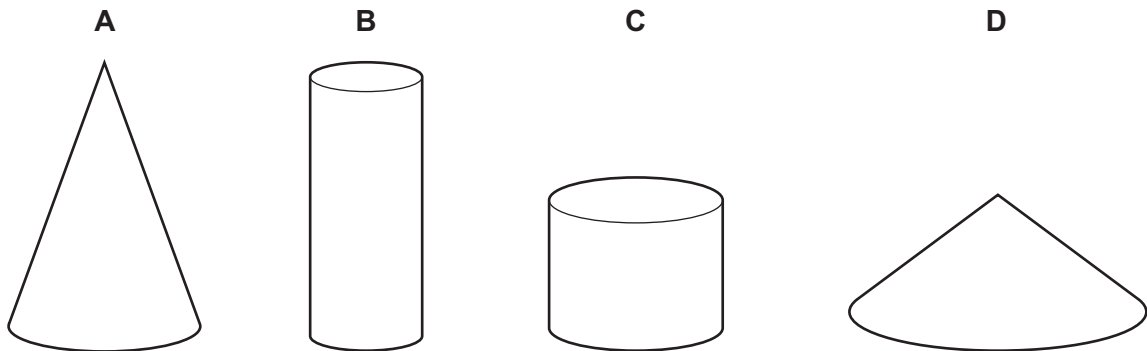
B $\text{density} = \frac{\text{volume}}{\text{mass}}$

C $\text{density} = \frac{\text{weight}}{\text{area}}$

D $\text{density} = \frac{\text{weight}}{\text{volume}}$

- 8 Four solid objects are made from the same material and have equal volumes.

Which object is the most stable?



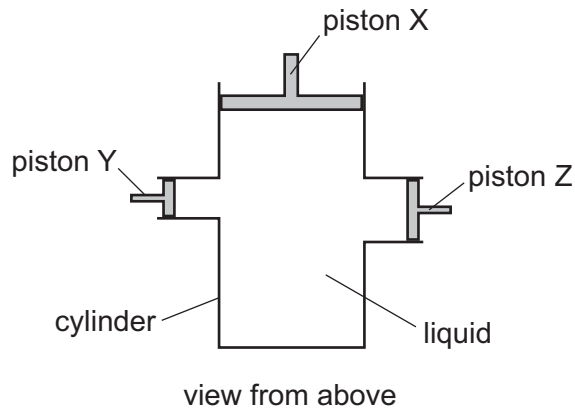
- 9 A metal wire is stretched by a force F up to the limit of proportionality.

Which statement describes what happens when F is increased slightly?

- A** The wire breaks.
- B** The wire continues to extend but not in direct proportion to F .
- C** The wire continues to extend in direct proportion to F .
- D** The wire does not extend any further.

- 10 Piston X is pushed into a hydraulic cylinder. Piston X produces a pressure P_X in the liquid in the cylinder.

The diagram shows the cylinder viewed from above.



There are two other pistons, Y and Z, in the cylinder.

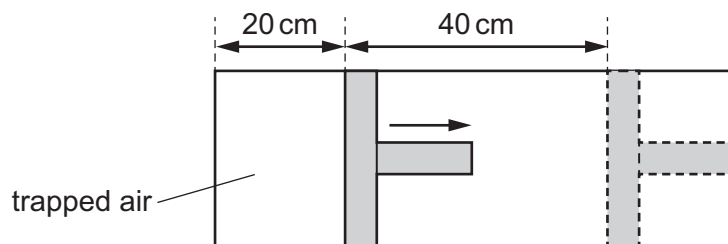
The pressures on piston Y and Z are P_Y and P_Z .

What is the relationship between P_X , P_Y and P_Z ?

- A** $P_X = P_Y + P_Z$
- B** $P_X > P_Z > P_Y$
- C** $P_X < P_Z < P_Y$
- D** $P_X = P_Y = P_Z$
- 11 Air is trapped in a cylinder by a piston. The pressure of the air is p and the length of the air column is 20 cm.

The piston is moved outwards. The length of the air column increases by 40 cm.

The temperature of the air remains constant.



What is the new air pressure?

- A** $\frac{p}{2}$ **B** $\frac{p}{3}$ **C** $2p$ **D** $3p$

- 12** A 60 kg ice-skater, travelling initially at 3.0 m/s, glides across a horizontal icy surface. The skater brakes and comes to rest after 5.0 m.

The work done against the braking force is the change in the kinetic energy of the skater.

What is the braking force?

- A** 18 N **B** 27 N **C** 54 N **D** 108 N

- 13** Which equation is used to find the efficiency of an energy transfer?

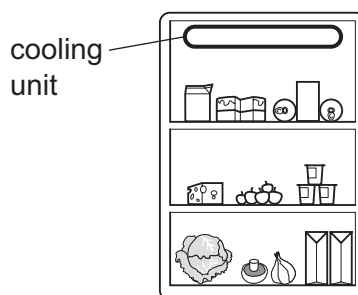
A $\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$

B $\text{efficiency} = \frac{\text{total energy input}}{\text{useful energy output}}$

C $\text{efficiency} = \frac{\text{energy converted to the required form}}{\text{total power input}}$

D $\text{efficiency} = (\text{energy converted to the required form}) \times (\text{total energy input})$

- 14** The diagram shows the inside of a refrigerator.



When the refrigerator is first switched on, what happens to the air near the cooling unit?

	the molecules of this air	the density of this air
A	become smaller	decreases
B	become smaller	increases
C	move closer together	decreases
D	move closer together	increases

- 15** What is the colour and what is the texture of a good absorber of infrared radiation?

- A** black and shiny
B black and dull
C white and shiny
D white and dull

- 16 Which temperatures are used as the lower and upper fixed points for a mercury-in-glass thermometer?

	lower fixed point	upper fixed point
A	freezing point of mercury	boiling point of mercury
B	freezing point of water	boiling point of water
C	room temperature	boiling point of mercury
D	room temperature	boiling point of water

- 17 Two liquid-in-glass thermometers have identical dimensions.

One contains ethanol and the other an equal volume of mercury.

Ethanol expands more than mercury for the same temperature rise.

How do the sensitivity and range of the ethanol thermometer compare to those of the mercury thermometer?

	sensitivity of ethanol thermometer	range of ethanol thermometer
A	greater	greater
B	greater	smaller
C	smaller	greater
D	smaller	smaller

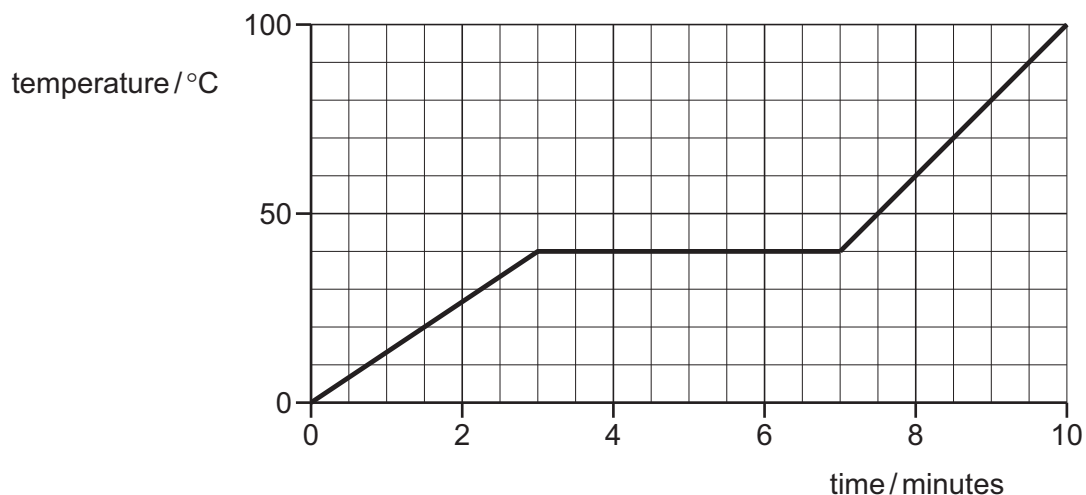
- 18 Solar panels are used to heat water with a mass of 500 kg.

The total area of the solar panels is 10 m^2 and the average power output from each square metre of the panels is 300 W. The specific heat capacity of water is $4200 \text{ J/(kg } ^\circ\text{C)}$.

What is the increase in the water temperature after 8.0 hours? (Assume there is no heat loss.)

- A** 0.69°C **B** 4.1°C **C** 5.1°C **D** 41°C

- 19 The graph is the temperature–time graph for a sample of wax that is heated so that it melts.



The mass of the wax is 200 g.

Thermal energy is supplied to the wax at a constant rate of 12 000 J/minute.

What is the specific latent heat of fusion of the wax?

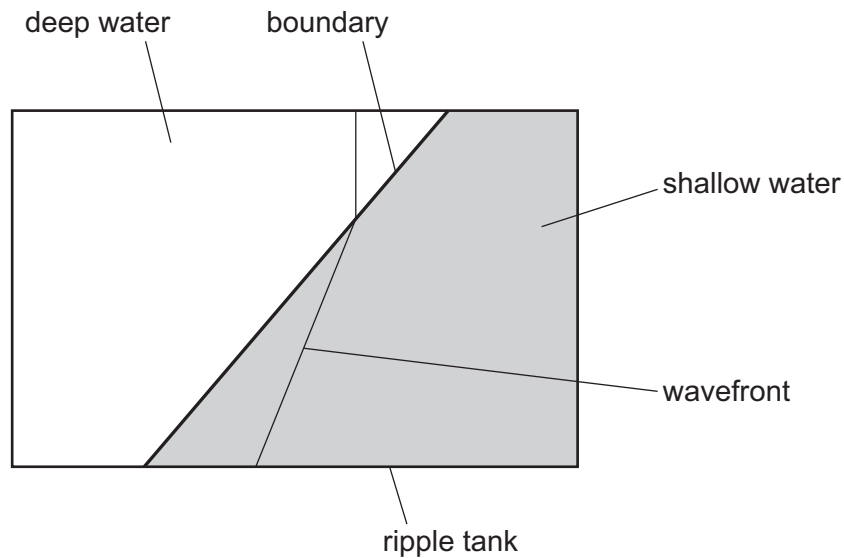
- A** 180 J/g **B** 240 J/g **C** 480 J/g **D** 600 J/g

- 20 The temperatures of the water in four beakers are different and areas of the surfaces of the water are also different.

In which beaker is the rate of evaporation of the water greatest?

	temperature of water / °C	surface area of water / cm ²
A	20	50
B	20	100
C	25	50
D	25	100

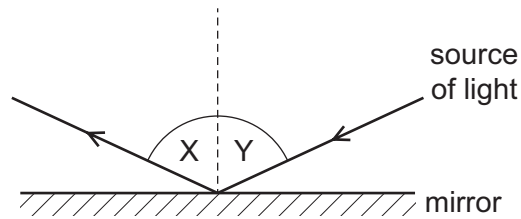
- 21** The diagram shows one wavefront of a wave as it travels from deep water to shallow water in a ripple tank.



What happens as the wavefront moves into the shallow water?

- A** The speed of the wavefront increases.
 - B** The speed of the wavefront decreases.
 - C** The wavelength of the wave remains constant.
 - D** The wavelength of the wave increases.
- 22** Which example describes a transverse wave being refracted?
- A** infrared waves travelling in an optical fibre
 - B** light waves from an object entering a magnifying glass
 - C** ultrasound waves being used to clean jewellery
 - D** water waves hitting a barrier

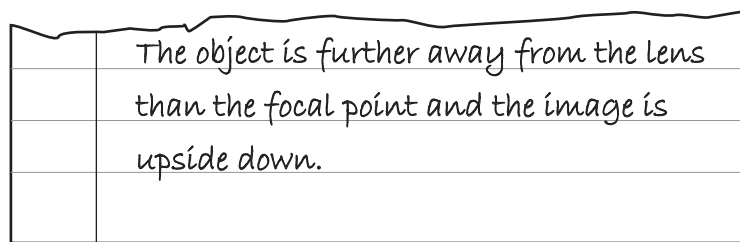
23 Light reflects from a plane mirror as shown.



Which row is always correct?

	symbol of angle X	relationship between X and Y
A	i	$X = Y$
B	i	$X + Y = 90^\circ$
C	r	$X = Y$
D	r	$X + Y = 90^\circ$

24 A piece of paper torn from an exercise book is shown.



Which process is being described?

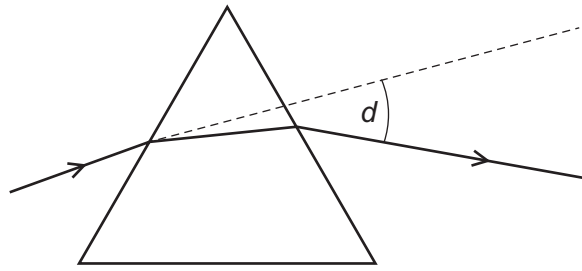
- A** the formation of a virtual image by a diverging lens
- B** the formation of a virtual image by a converging lens
- C** the formation of a real image by a diverging lens
- D** the formation of a real image by a converging lens

25 A girl is long-sighted.

Which statement is correct?

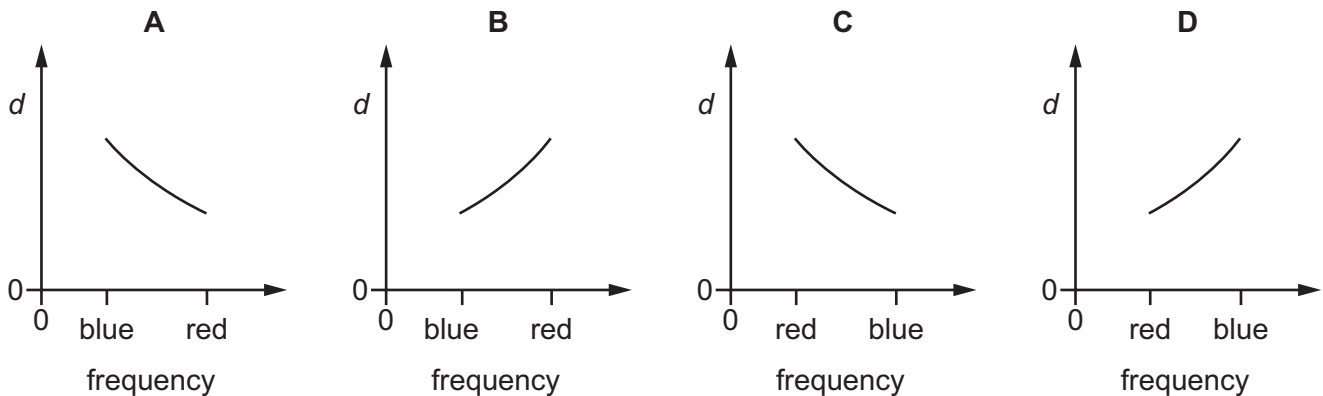
- A** She sees close objects less clearly than a person with normal vision.
- B** She sees distant objects more clearly than a person with normal vision.
- C** The fault is corrected with a diverging lens.
- D** The image of a close object is formed in front of her retina.

26 Light rays are deviated by a prism.



The deviation angle d is measured for light rays of different frequency, including blue light and red light.

Which graph of d against frequency is correct?



27 A satellite orbits the Earth at a height of 300 km above the surface. The speed of electromagnetic waves is 3.0×10^8 m/s.

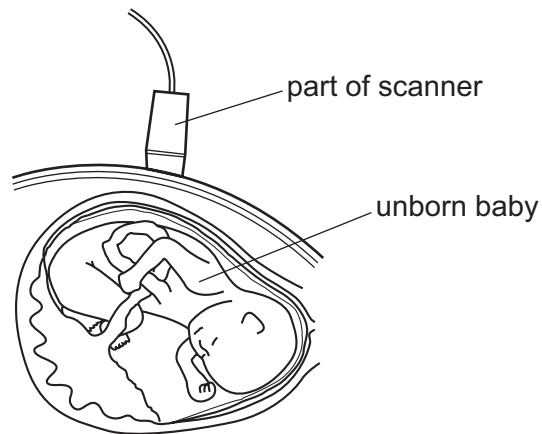
What is the minimum time taken for a radio signal transmitted from the Earth to reach the satellite?

- A** 1.0×10^{-6} s **B** 2.0×10^{-6} s **C** 1.0×10^{-3} s **D** 2.0×10^{-3} s

28 Which of the following is most closely associated with echoes?

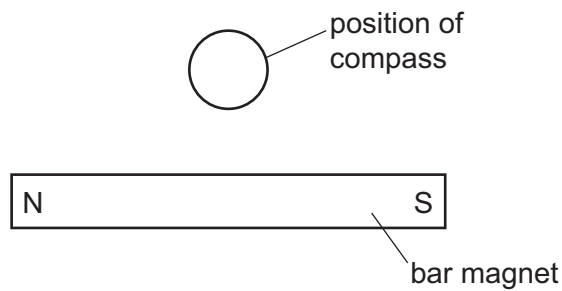
- A** focussing
B reflection
C refraction
D total internal reflection

- 29 An ultrasound scanner produces an image of an unborn baby.

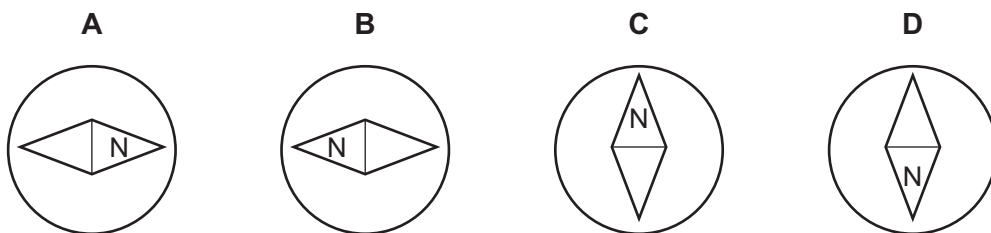


What does the scanner use to form an image of the baby?

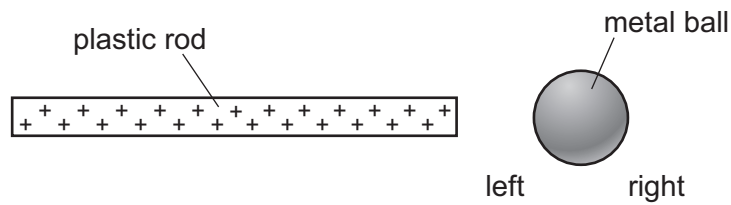
- A ultrasound absorbed by the baby
 - B ultrasound emitted by the baby
 - C ultrasound reflected by the baby
 - D ultrasound refracted by the baby
- 30 A small plotting compass is placed near to a bar magnet as shown.



Which diagram shows the direction in which the compass needle points?



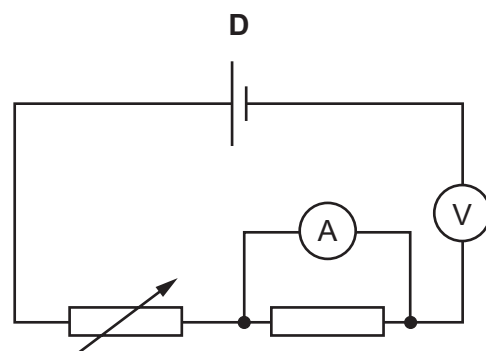
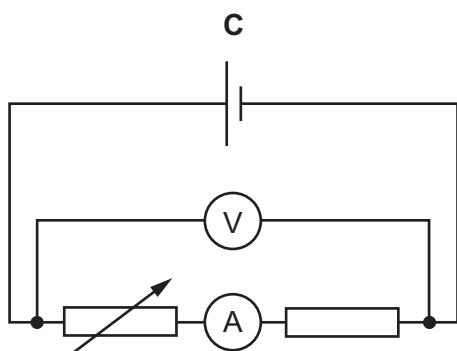
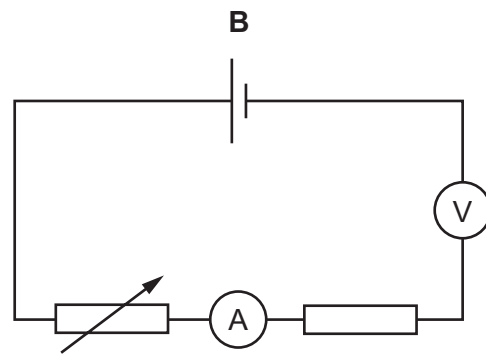
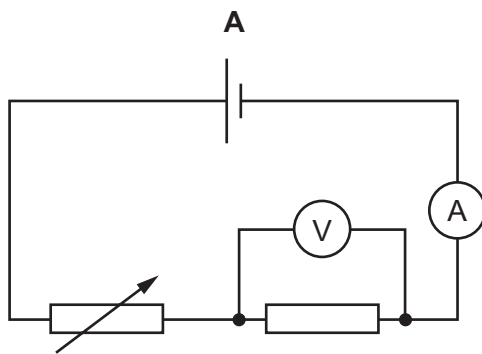
- 31** A positively charged plastic rod is held near but not touching an insulated and previously uncharged metal ball. No charge leaves the rod.



What are possible charges on the left-hand side and right-hand side of the ball?

	charge on left-hand side of ball	charge on right-hand side of ball
A	$-2.0 \times 10^{-9} \text{ A}$	$-1.0 \times 10^{-9} \text{ A}$
B	$-2.0 \times 10^{-9} \text{ A}$	$+2.0 \times 10^{-9} \text{ A}$
C	$-2.0 \times 10^{-9} \text{ C}$	$+2.0 \times 10^{-9} \text{ C}$
D	$+2.0 \times 10^{-9} \text{ C}$	$-2.0 \times 10^{-9} \text{ C}$

- 32** Which circuit is connected to measure the current in the fixed resistor and the potential difference (p.d.) across the same resistor?



33 Quantity X and quantity Y have the same unit Z.

X is the energy dissipated by a battery in driving unit charge round the complete circuit.

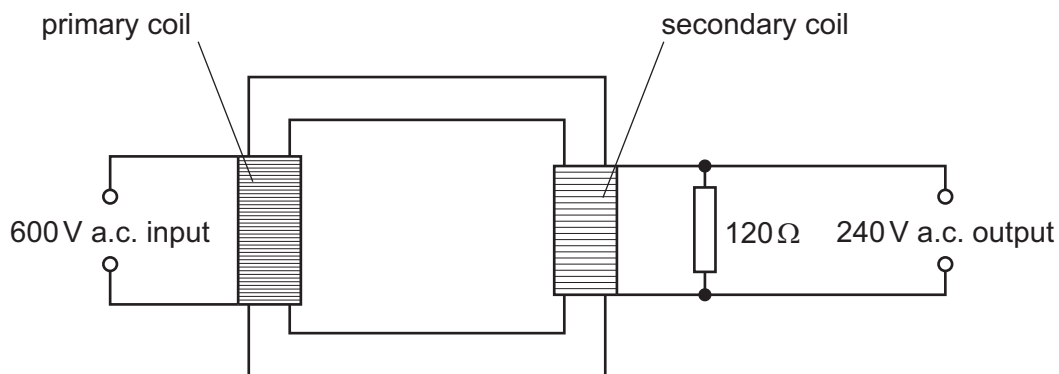
Y is the work done in a component in driving unit charge through the component.

What are X, Y and Z?

	X	Y	Z
A	electromotive force	potential difference	coulomb
B	electromotive force	potential difference	volt
C	potential difference	electromotive force	coulomb
D	potential difference	electromotive force	volt

34 A transformer with an efficiency of 100% has a primary voltage input of 600 V and a secondary voltage output of 240 V.

The secondary coil is attached to a resistor of resistance $120\ \Omega$.



What is the power dissipated in the resistor and what is the current in the primary coil?

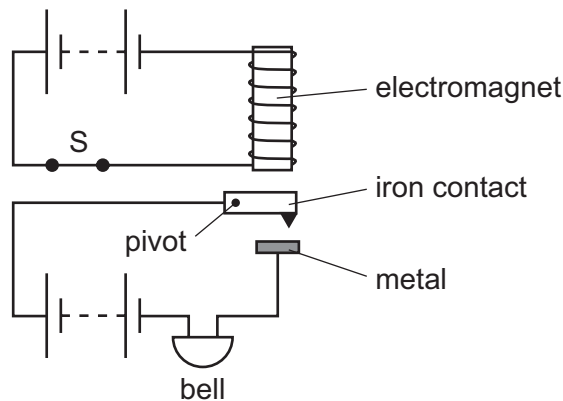
	power/W	current/A
A	120	0.20
B	120	5.0
C	480	0.80
D	480	1.3

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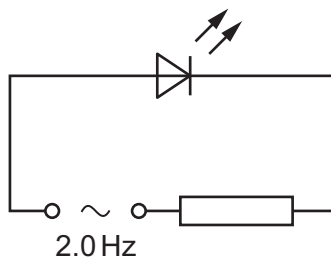
- 35** The diagram shows an alarm system in which the switch S is shown closed. The top circuit is arranged so that the electromagnet is positioned over the soft iron contact.



What happens when the switch S is opened?

	iron contact	bell
A	drops	rings
B	drops	stops ringing
C	moves up	rings
D	moves up	stops ringing

- 36** The diagram shows an LED connected in a circuit with a resistor and an a.c. power supply of frequency 2.0 Hz.



Which statement is correct?

- A** The LED does not light.
- B** The LED flashes on twice each second and off twice each second.
- C** The LED flashes on four times each second and off four times each second.
- D** The LED stays on all the time.

37 Which statement about the production of electricity in a nuclear power station is correct?

- A In the reactor, the main reaction occurs when protons hit uranium nuclei.
- B The process taking place in the reactor is called nuclear fusion.
- C The reactor produces energy to boil water and to produce steam.
- D Carbon dioxide is the major waste product from the reactor.

38 A radioactive sample contains an isotope that emits alpha particles.

Which quantity stays constant?

- A the half-life of the isotope
- B the mass of the sample
- C the number of neutrons in the sample
- D the rate of decay of the isotope

39 In the simple model of an atom, X orbits around Y.



What are X and Y?

	X	Y
A	electron	nucleus
B	neutron	electron
C	nucleus	proton
D	proton	neutron

40 How do the proton numbers (atomic numbers) and the nucleon numbers (mass numbers) of two different isotopes of the same element compare with each other?

	proton number	nucleon number
A	different	different
B	different	same
C	same	different
D	same	same



Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

October/November 2021

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
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INSTRUCTIONS

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INFORMATION

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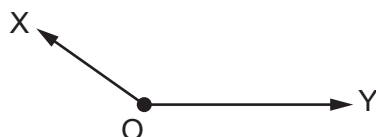
This document has **20** pages. Any blank pages are indicated.



1 Which quantity is a vector?

- A speed
- B force
- C mass
- D distance

2 Two forces, X and Y, act upon an object O. The arrows represent the magnitudes and directions of the forces.



Which arrow shows the direction of the resultant force?



3 Two identical objects begin to fall from rest.

One object falls from 200 m above the Earth's surface and the other falls from 200 m above the Moon's surface. One second after they have started to fall, both objects are still accelerating.

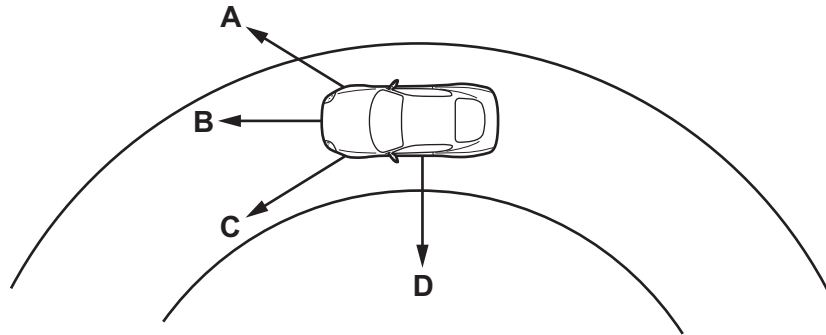
There is no atmosphere on the Moon and the weight of each object is constant.

Which row describes the motion of both objects at this time?

	acceleration of object falling above the Earth's surface	acceleration of object falling above the Moon's surface
A	decreasing	constant
B	constant	constant
C	increasing	decreasing
D	increasing	increasing

- 4 The diagram shows a car going around a circular track at constant speed.

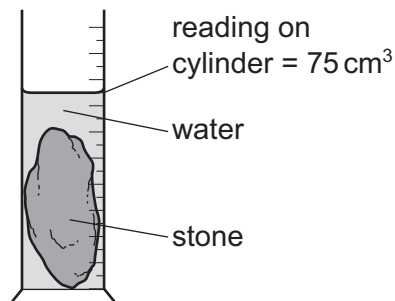
Which arrow shows the direction of the resultant force on the car?



- 5 An electron has mass.

Where does a stationary electron experience a force?

- A in an electric field only
 - B in a gravitational field only
 - C in a gravitational field and an electric field only
 - D in a gravitational, electric and magnetic field
- 6 A stone is placed in a measuring cylinder of water, as shown.



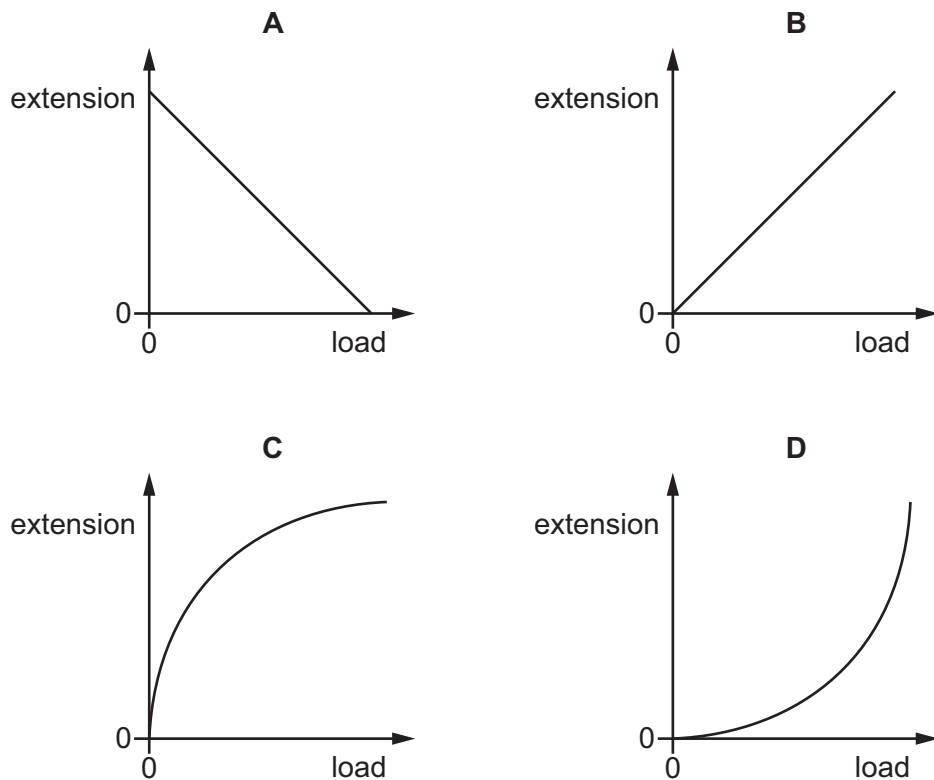
The volume of water in the cylinder before adding the stone is 15 cm³ and the stone has a mass of 90 g.

What is the density of the stone?

- A 1.0 g/cm³
- B 1.2 g/cm³
- C 1.5 g/cm³
- D 6.0 g/cm³

- 7 The extension of a spring is measured for various loads. The limit of proportionality is not reached.

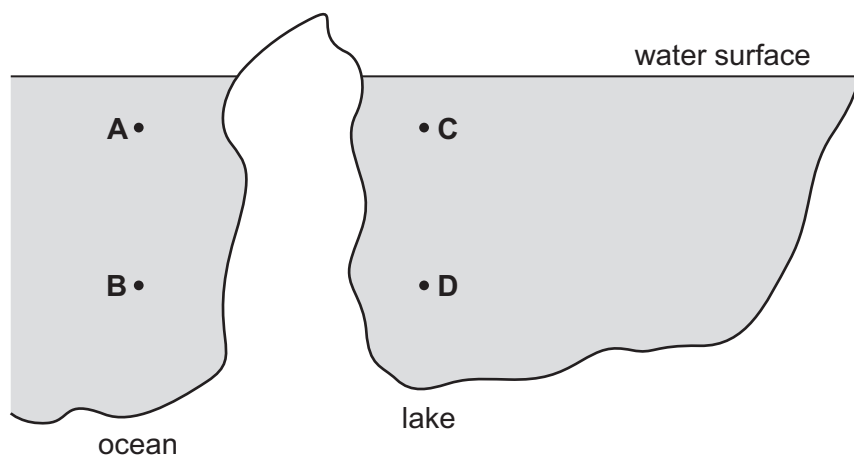
Which graph shows the extension–load graph for the spring?



- 8 An underwater diver moves from the ocean to a fresh water lake.

The density of water in the lake is less than in the ocean.

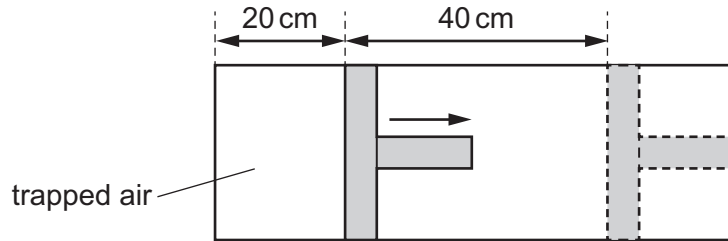
In which position does the diver experience the smallest pressure due to the water?



- 9 Air is trapped in a cylinder by a piston. The pressure of the air is p and the length of the air column is 20 cm.

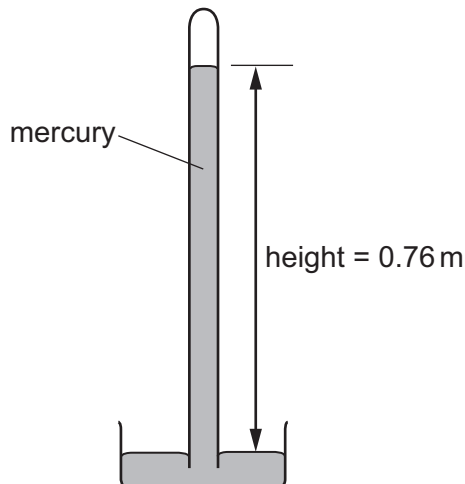
The piston is moved outwards. The length of the air column increases by 40 cm.

The temperature of the air remains constant.



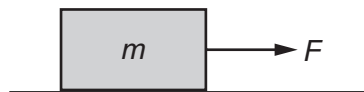
What is the new air pressure?

- A $\frac{p}{2}$ B $\frac{p}{3}$ C $2p$ D $3p$
- 10 When the pressure exerted by the atmosphere is 0.10 MPa, the height of the mercury in a barometer is 0.76 m.



What is the pressure exerted by the atmosphere when the height of the mercury in the simple barometer is 0.57 m?

- A 0.025 MPa B 0.075 MPa C 0.10 MPa D 0.13 MPa
- 11 The diagram shows a block of mass m pulled in a straight line along a horizontal surface by a force F .

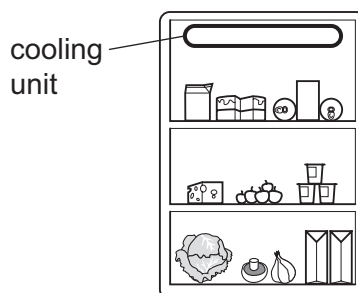


The block moves a distance d in time t . The average speed at which the block moves is v .

Which two quantities must be known to calculate the work done?

- A F and d B F and m C F and t D F and v

12 The diagram shows the inside of a refrigerator.



When the refrigerator is first switched on, what happens to the air near the cooling unit?

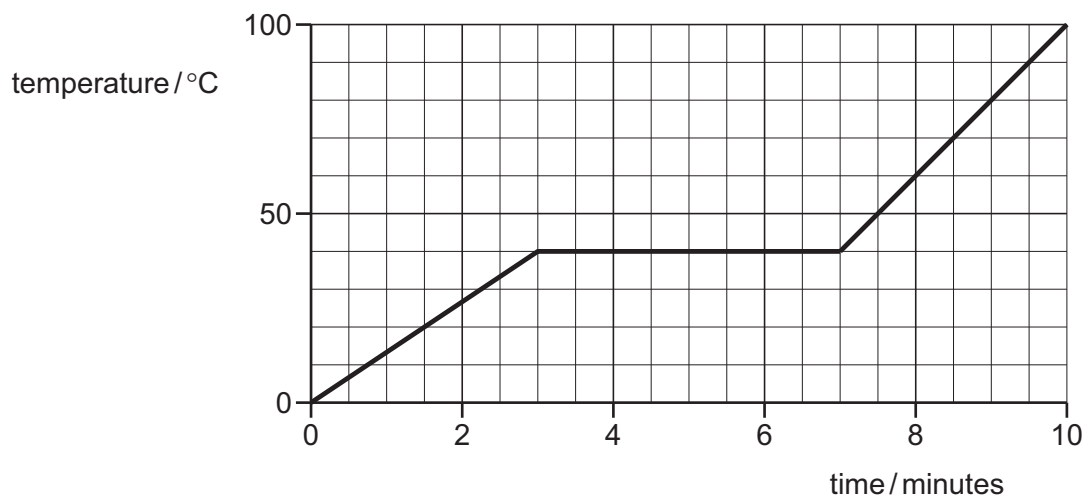
	the molecules of this air	the density of this air
A	become smaller	decreases
B	become smaller	increases
C	move closer together	decreases
D	move closer together	increases

13 Less thermal energy is needed to raise the temperature of 1.0 kg of copper by 1.0 °C than is needed to raise the temperature of 1.0 kg of water by 1.0 °C.

Which statement explains this?

- A** Copper has a higher melting point.
- B** Copper has a lower specific heat capacity.
- C** Copper has a smaller specific latent heat.
- D** Copper is a better conductor of thermal energy.

- 14 The graph is the temperature–time graph for a sample of wax that is heated so that it melts.



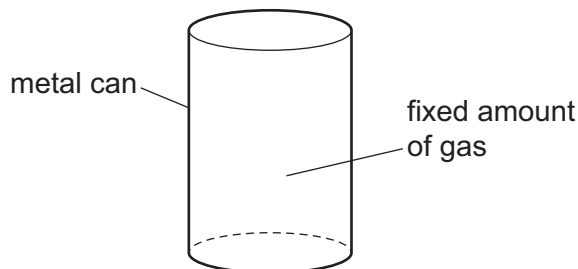
The mass of the wax is 200 g.

Thermal energy is supplied to the wax at a constant rate of 12 000 J/minute.

What is the specific latent heat of fusion of the wax?

- A** 180 J/g **B** 240 J/g **C** 480 J/g **D** 600 J/g

- 15 A fixed amount of gas is trapped inside a metal can.

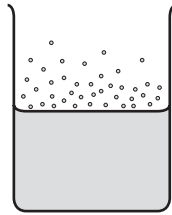


The temperature of the gas decreases but the volume does not change.

How does this change affect the kinetic energy of the gas molecules and the pressure inside the metal can?

	kinetic energy	pressure
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	stays the same	decreases

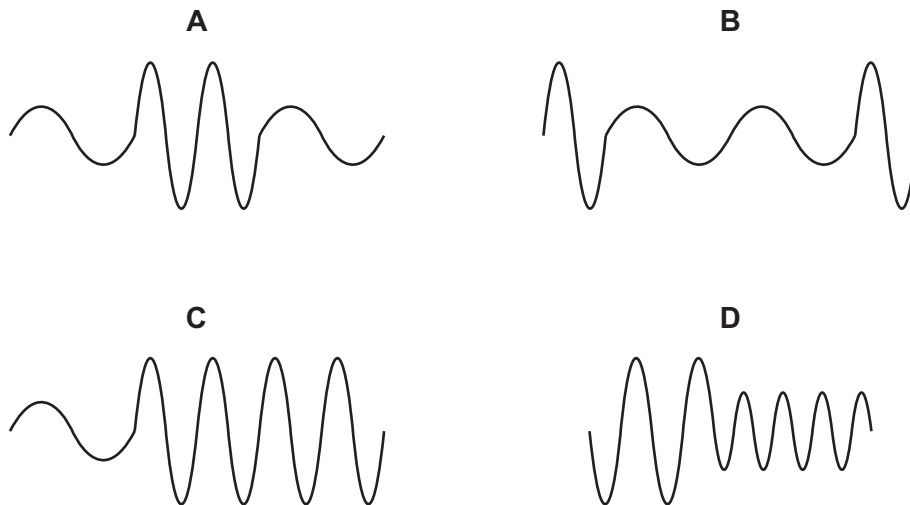
- 16 The diagram shows a room temperature liquid in a beaker.



What reduces the rate of loss of liquid by evaporation?

- A blowing air across the top of the beaker
 - B heating the liquid
 - C putting a lid on the beaker
 - D transferring the liquid to a beaker with a larger diameter
- 17 A water wave on a pond passes into a region where the wave travels more slowly. It then returns to a region where it travels at the original speed. Assume that the frequency of the wave stays constant.

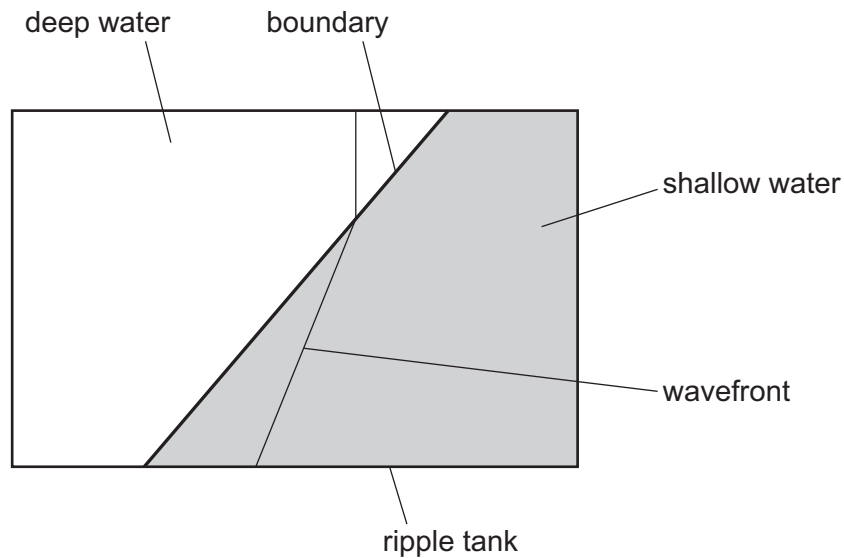
Which side-view of the wave is correct?



- 18 What is meant by the *amplitude* of a wave?

- A the distance between the extreme points of a particle's motion
- B the maximum distance a particle moves from its equilibrium position
- C the maximum energy carried by the wave
- D the maximum power of the vibrations carrying the wave

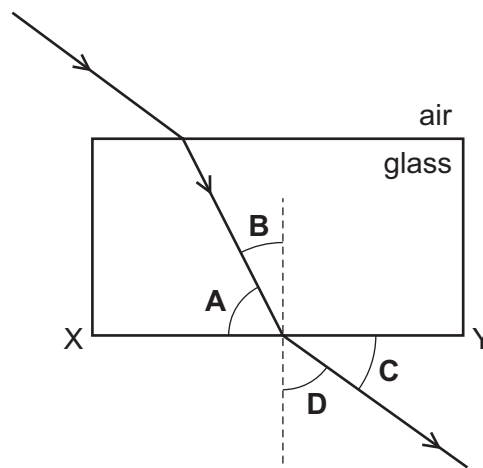
- 19 The diagram shows one wavefront of a wave as it travels from deep water to shallow water in a ripple tank.



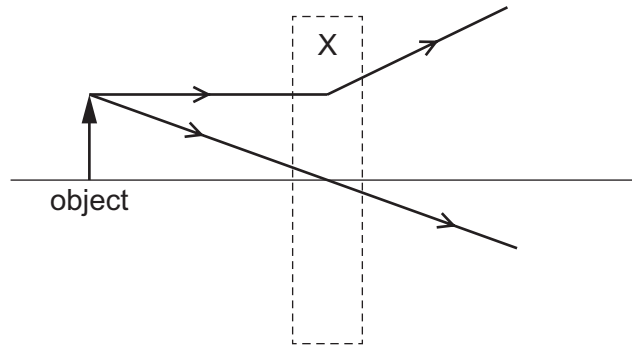
What happens as the wavefront moves into the shallow water?

- A The speed of the wavefront increases.
 - B The speed of the wavefront decreases.
 - C The wavelength of the wave remains constant.
 - D The wavelength of the wave increases.
- 20 A ray of light passes into a glass block. It travels through the glass block and then emerges into the air.

Which angle is the angle of refraction at the surface XY?



21 Two rays of light pass through a lens in region X.



Which type of lens is in region X and which type of image is formed?

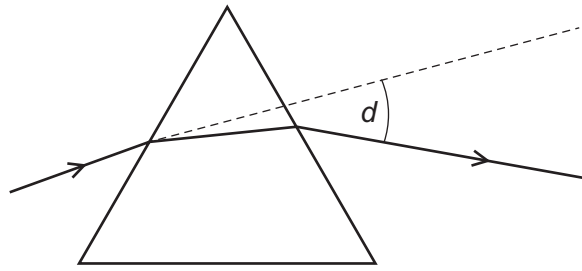
	type of lens	type of image
A	converging	real
B	converging	virtual
C	diverging	real
D	diverging	virtual

22 A girl is long-sighted.

Which statement is correct?

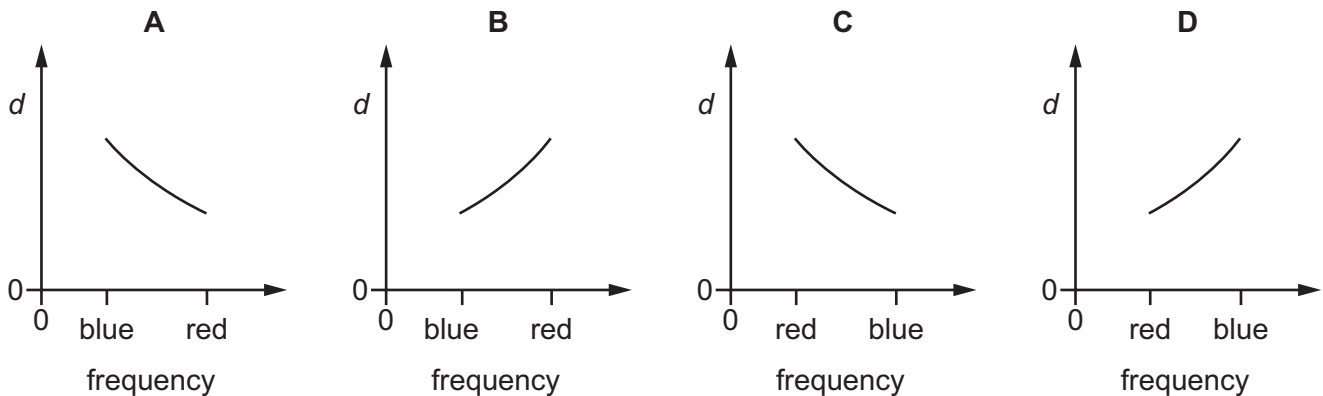
- A** She sees close objects less clearly than a person with normal vision.
- B** She sees distant objects more clearly than a person with normal vision.
- C** The fault is corrected with a diverging lens.
- D** The image of a close object is formed in front of her retina.

23 Light rays are deviated by a prism.



The deviation angle d is measured for light rays of different frequency, including blue light and red light.

Which graph of d against frequency is correct?



24 Which component of the electromagnetic spectrum is used for the remote control of a television?

- A** gamma rays
- B** infrared rays
- C** radio waves
- D** ultraviolet rays

25 A student stands a distance x in front of a large wall.

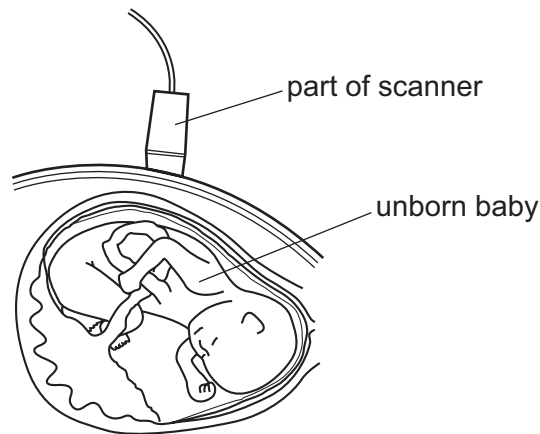
He claps his hands at a regular rate so that each clap coincides with the echo from the previous clap.

In a time t , he claps his hands N times.

Which expression is used to calculate the speed of sound in air?

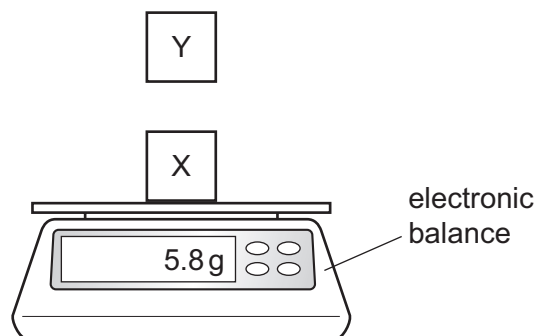
- A** $\frac{x}{Nt}$
- B** $\frac{2x}{Nt}$
- C** $\frac{Nx}{t}$
- D** $\frac{2Nx}{t}$

- 26 An ultrasound scanner produces an image of an unborn baby.



What does the scanner use to form an image of the baby?

- A ultrasound absorbed by the baby
 - B ultrasound emitted by the baby
 - C ultrasound reflected by the baby
 - D ultrasound refracted by the baby
- 27 Object X is on an electronic balance. Object Y is held above object X as shown.



The reading on the balance increases when Y is moved closer to X.

Why does this happen?

- A X and Y are both made of iron.
- B X and Y are both magnets.
- C Y is a magnet and X is made of iron.
- D Y is made of iron and X is a magnet.

28 A teacher wants to demagnetise a bar magnet and tries three different methods.

- 1 Heat the bar magnet to a high temperature and let it cool.
- 2 Place the bar magnet east to west and hammer it.
- 3 Place the bar magnet inside a coil that has a direct current in it and remove it from the coil slowly.

Which methods demagnetise the bar magnet?

- A** 1 only **B** 1 and 2 only **C** 2 and 3 only **D** 1, 2 and 3

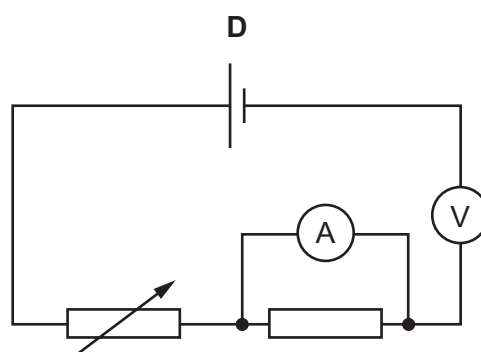
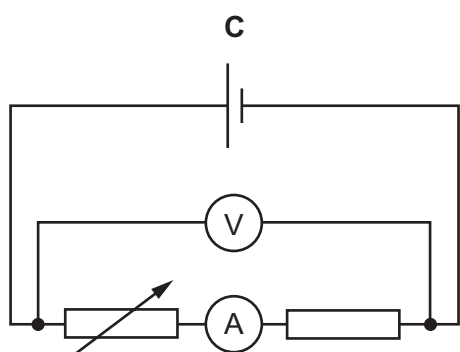
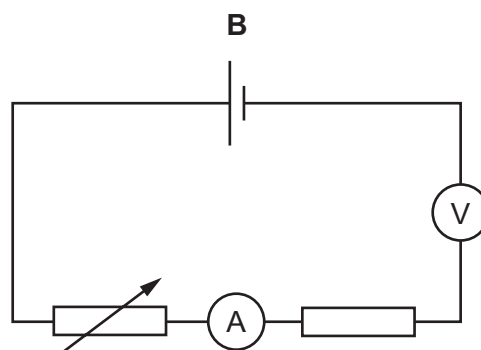
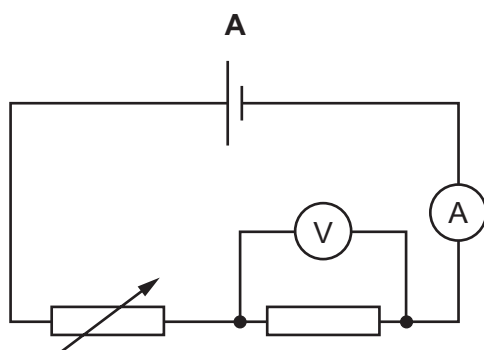
29 A piece of paper torn out of an exercise book is shown.

	is the work done (measured in joules)
	on unit charge (measured in coulombs) as it
	moves around the circuit.

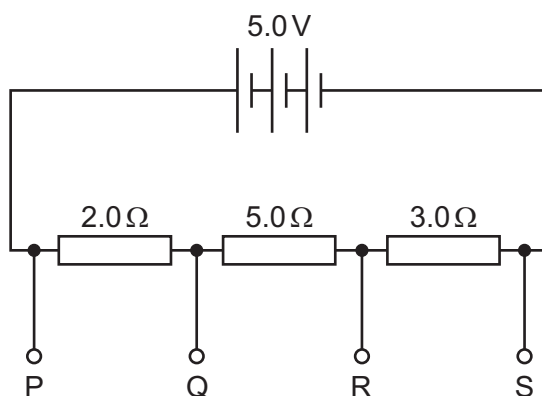
What is being defined?

- A** electrical resistance
B electric current
C electrical power
D electromotive force

- 30 Which circuit is connected to measure the current in the fixed resistor and the potential difference (p.d.) across the same resistor?



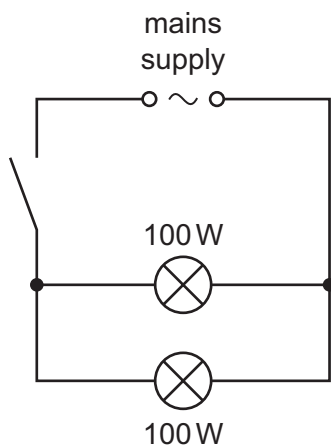
- 31 The diagram shows a circuit with a 5.0 V power supply, three resistors and four output terminals P, Q, R and S.



Between which pair of terminals is there a potential difference of 1.5 V?

- A** PQ **B** PR **C** QS **D** RS

- 32** Two 100 W lamps are connected in parallel to the mains supply, as shown.



How much electrical energy is supplied by the mains when the switch is closed for 36 minutes?

- A** 0.030 kWh **B** 0.060 kWh **C** 0.12 kWh **D** 7.2 kWh
- 33** The cost of electricity is 9.0 cents / kWh.

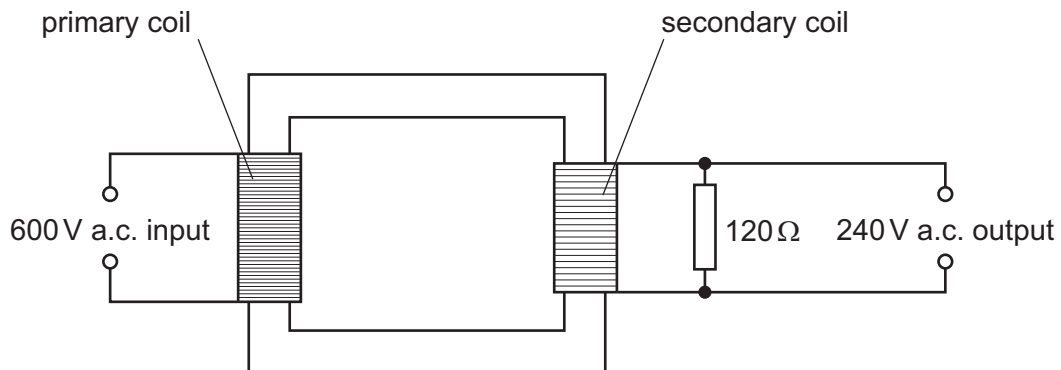
It takes 0.060 hours and costs 1.2 cents to boil the water in a kettle.

What is the power rating of the kettle?

- A** 0.45 kW **B** 0.65 kW **C** 2.2 kW **D** 130 kW

- 34** A transformer with an efficiency of 100% has a primary voltage input of 600 V and a secondary voltage output of 240 V.

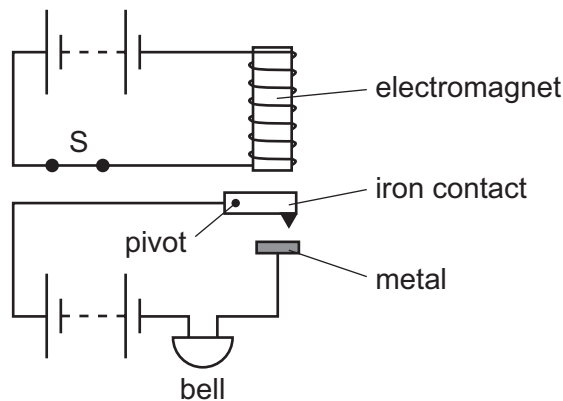
The secondary coil is attached to a resistor of resistance $120\ \Omega$.



What is the power dissipated in the resistor and what is the current in the primary coil?

	power / W	current / A
A	120	0.20
B	120	5.0
C	480	0.80
D	480	1.3

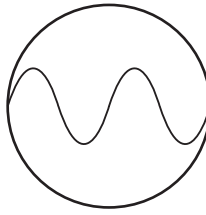
- 35** The diagram shows an alarm system in which the switch S is shown closed. The top circuit is arranged so that the electromagnet is positioned over the soft iron contact.



What happens when the switch S is opened?

	iron contact	bell
A	drops	rings
B	drops	stops ringing
C	moves up	rings
D	moves up	stops ringing

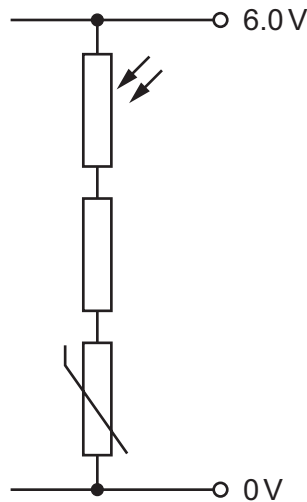
- 36** A microphone detects a musical sound and the signal is fed into an oscilloscope. The diagram shows the trace which appears on the screen.



The spot of light on the oscilloscope screen takes 1.2 ms to travel across the screen.

What is the frequency of the musical sound?

- A** 170 Hz **B** 330 Hz **C** 1700 Hz **D** 3300 Hz
- 37** A student connects two components whose resistances can change and a fixed resistor across a 6.0 V supply.



Under which conditions is the voltage across the fixed resistor the greatest?

- A** cold and dark
B cold and bright
C warm and dark
D warm and bright

- 38** Technetium-99m is a radioactive isotope used in medical scanning. It is injected into the body and its emissions are detected outside the body.

Which characteristics of technetium-99m make it suitable for use in medical scanning?

- A** It has a long half-life and emits alpha radiation.
 - B** It has a long half-life and emits gamma radiation.
 - C** It has a short half-life and emits alpha radiation.
 - D** It has a short half-life and emits gamma radiation.
- 39** Which statement about the production of electricity in a nuclear power station is correct?
- A** In the reactor, the main reaction occurs when protons hit uranium nuclei.
 - B** The process taking place in the reactor is called nuclear fusion.
 - C** The reactor produces energy to boil water and to produce steam.
 - D** Carbon dioxide is the major waste product from the reactor.
- 40** In the simple model of an atom, X orbits around Y.



What are X and Y?

	X	Y
A	electron	nucleus
B	neutron	electron
C	nucleus	proton
D	proton	neutron

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Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2021

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Question	Answer	Marks
1	C	1
2	B	1
3	B	1
4	D	1
5	D	1
6	C	1
7	A	1
8	D	1
9	B	1
10	D	1
11	B	1
12	C	1
13	A	1
14	D	1
15	B	1
16	B	1
17	B	1
18	D	1
19	B	1
20	D	1
21	B	1
22	B	1
23	C	1
24	D	1
25	A	1
26	D	1
27	C	1
28	B	1

Question	Answer	Marks
29	C	1
30	A	1
31	C	1
32	A	1
33	B	1
34	C	1
35	A	1
36	B	1
37	C	1
38	A	1
39	A	1
40	C	1



Cambridge O Level

PHYSICS

5054/11

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Question	Answer	Marks
1	B	1
2	A	1
3	A	1
4	D	1
5	C	1
6	C	1
7	B	1
8	C	1
9	B	1
10	B	1
11	A	1
12	D	1
13	B	1
14	B	1
15	A	1
16	C	1
17	A	1
18	B	1
19	B	1
20	D	1
21	D	1
22	A	1
23	D	1
24	B	1
25	D	1
26	C	1
27	B	1
28	B	1

Question	Answer	Marks
29	D	1
30	A	1
31	D	1
32	C	1
33	C	1
34	C	1
35	A	1
36	C	1
37	D	1
38	D	1
39	C	1
40	A	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2020

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

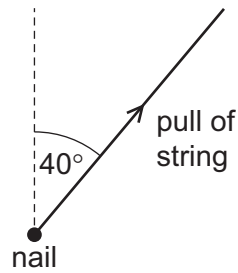
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



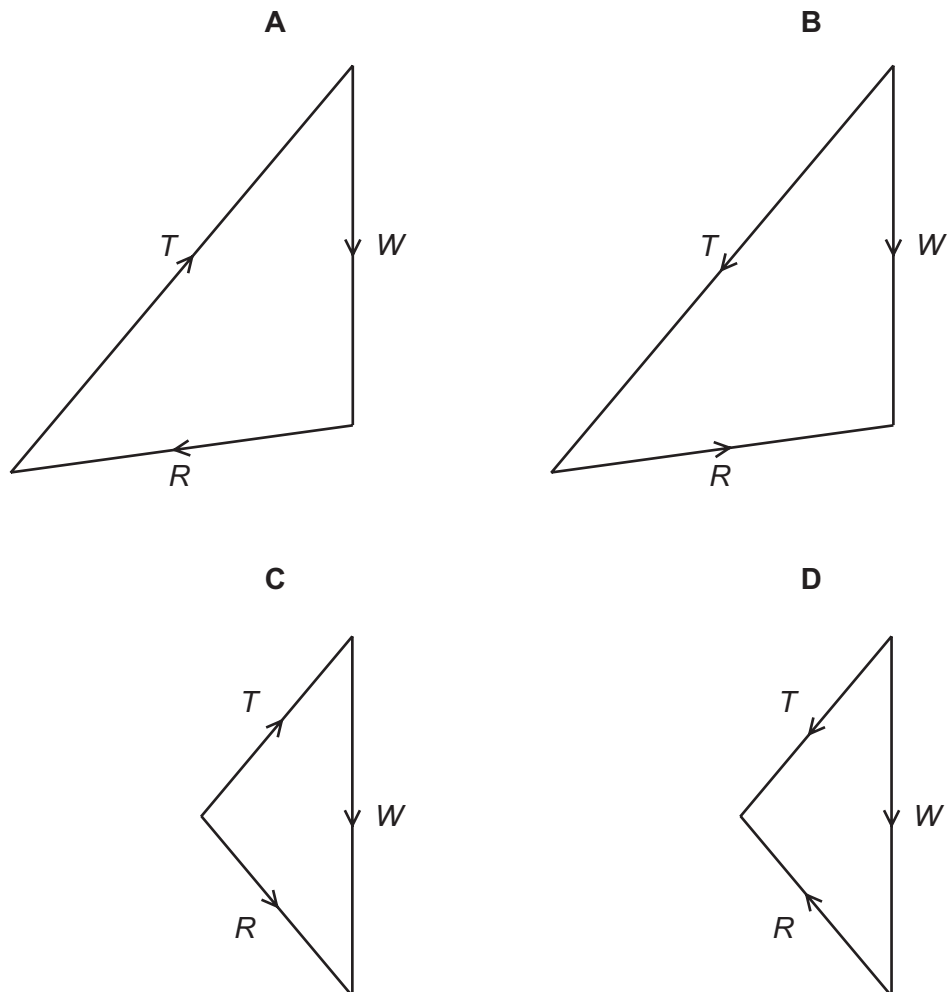
- 1 A heavy nail is fixed firmly to a wall. It is pulled by a string at 40° to the vertical. The nail does not move.



Three forces act on the nail:

- its weight W ,
- the tension T in the string,
- the force R exerted by the wall.

Which diagram, drawn to scale, represents the three forces?



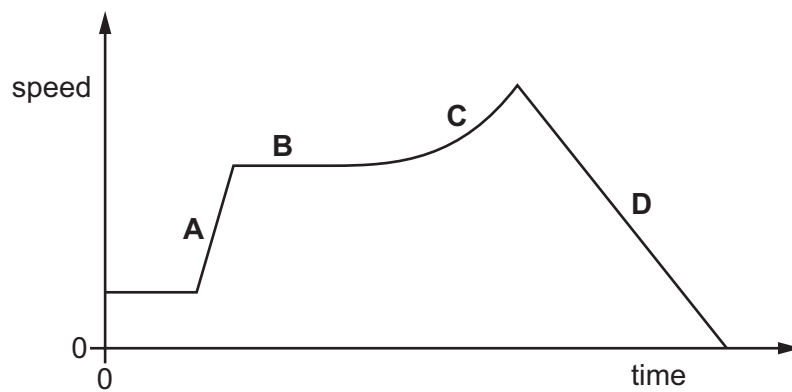
- 2 A car begins to move. It speeds up until it reaches a constant speed. It continues to travel at this constant speed for the rest of the journey.

What happens to the acceleration and what happens to the velocity of the car during the journey?

- A Both the acceleration and the velocity change.
- B Only the acceleration changes.
- C Only the velocity changes.
- D Neither the acceleration nor the velocity changes.

- 3 The speed–time graph for a car's journey is shown.

During which part of the journey is the car moving with non-uniform acceleration?



- 4 A car is travelling forwards at high speed. The brakes are applied and the car skids along the road surface.

In which direction and where does the friction force act?

- A backwards on the tyres
- B forwards on the tyres
- C upwards on the tyres
- D downwards on the road

- 5 A car of mass 800 kg has a forward acceleration of 2.5 m/s^2 . A frictional force of 1200 N opposes the motion of the car.

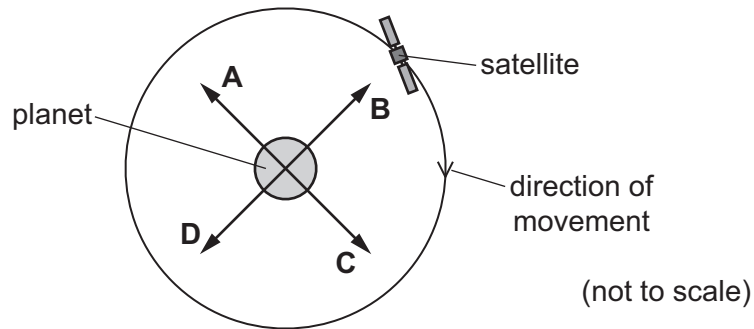
What is the driving force due to the engine of the car?

- A 800 N
- B 1200 N
- C 2000 N
- D 3200 N

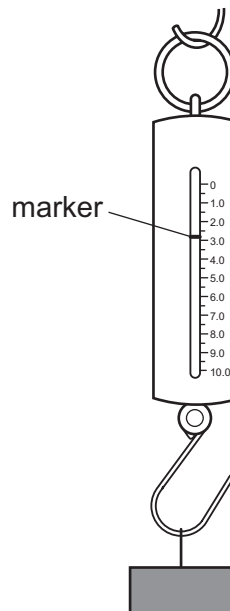
- 6 A satellite orbits a planet in a circular path as shown. It has constant speed.

There is a force on the planet due to the satellite.

In which direction is the force on the planet?



- 7 A block is hung on a spring balance. The marker inside the balance is pulled down by the block.



What can the position of the marker be used to determine?

- A the mass of the block in kg
- B the mass of the block in N
- C the moment caused by the block in N
- D the weight of the block in kg

- 8 There is an astronaut on a planet which has a gravitational field strength one fifth of the value on Earth.

He hangs samples of rock from a spring balance designed for use on Earth. Each sample has a mass of 0.1 kg.

The spring balance reads 1.0 kg.

How many samples of rock are there?

- A** 2 **B** 10 **C** 25 **D** 50

- 9 An electrically charged plastic ball is at rest.

Which types of field are caused by the ball?

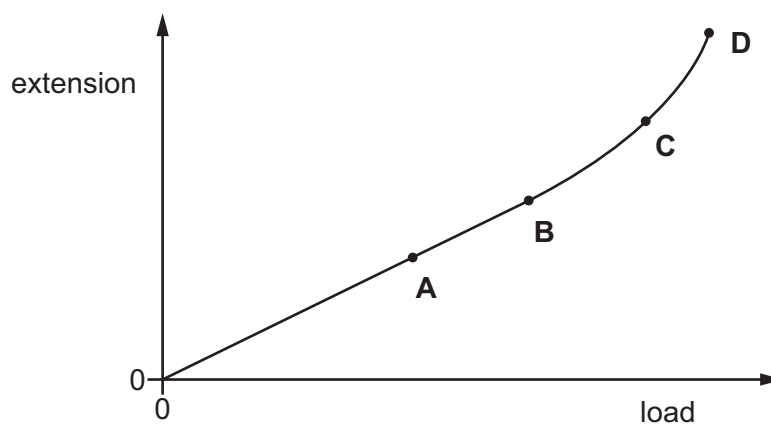
- A** electric, gravitational and magnetic
B electric and gravitational only
C electric and magnetic only
D gravitational and magnetic only

- 10 Which list contains only quantities that can be changed by a force?

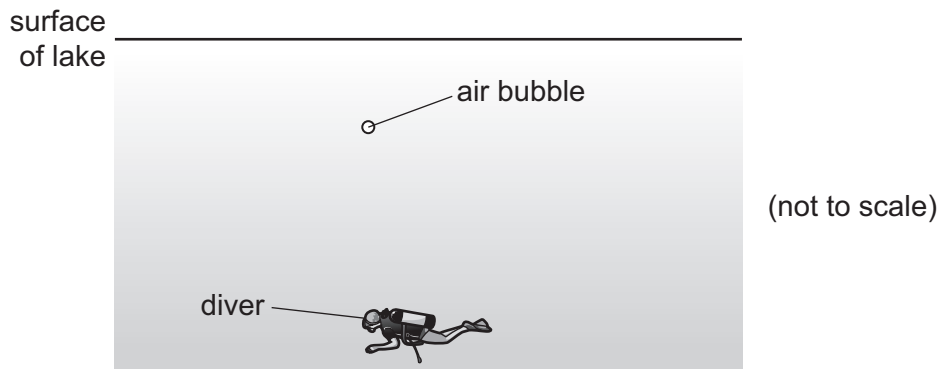
- A** mass, shape, velocity
B mass, shape, volume
C mass, velocity, volume
D shape, velocity, volume

- 11 An extension–load graph is plotted for a spring.

Which point marks the limit of proportionality for this spring?



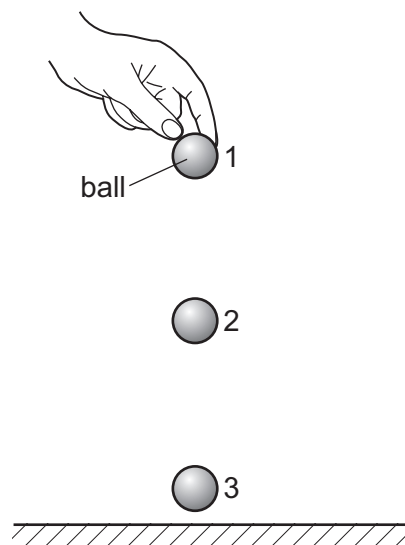
- 12 The diagram shows a diver 20 m below the surface of a lake. The total pressure at this depth is $3.0 \times 10^5 \text{ Pa}$.



An air bubble has a volume of 0.60 cm^3 as it leaves the diver. It rises to the surface of the lake where the pressure is $1.0 \times 10^5 \text{ Pa}$. The temperature of the air in the bubble remains constant.

What is the volume of the air bubble at the surface of the lake?

- A** 0.20 cm^3 **B** 0.60 cm^3 **C** 1.8 cm^3 **D** 2.4 cm^3
- 13 A ball is dropped from a height as shown.



Ignoring the effects of air resistance, which statement about the total energy of the ball is correct?

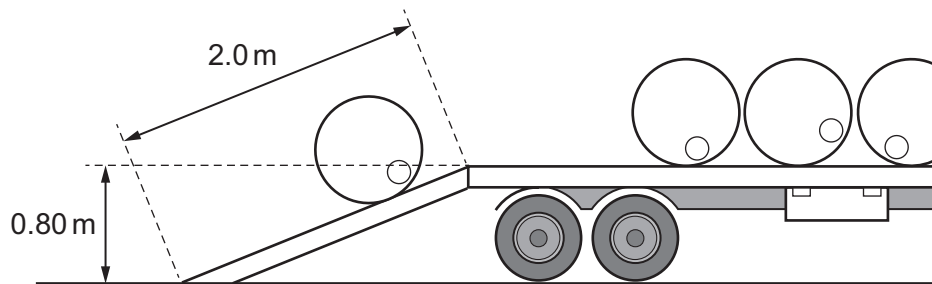
- A** It is the same at all points.
B It is greatest at point 1.
C It is greatest at point 2.
D It is greatest at point 3.

- 14 A student uses an average force of F to push a box a distance of d across a horizontal table. The mass of the box is m and the gravitational field strength is g .

How is the work done by the student pushing the box calculated?

- A $F \times d$ B $\frac{F \times d}{g}$ C $m \times d$ D $m \times g \times d$

- 15 A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

- A 1000 J B 1600 J C 2500 J D 4000 J

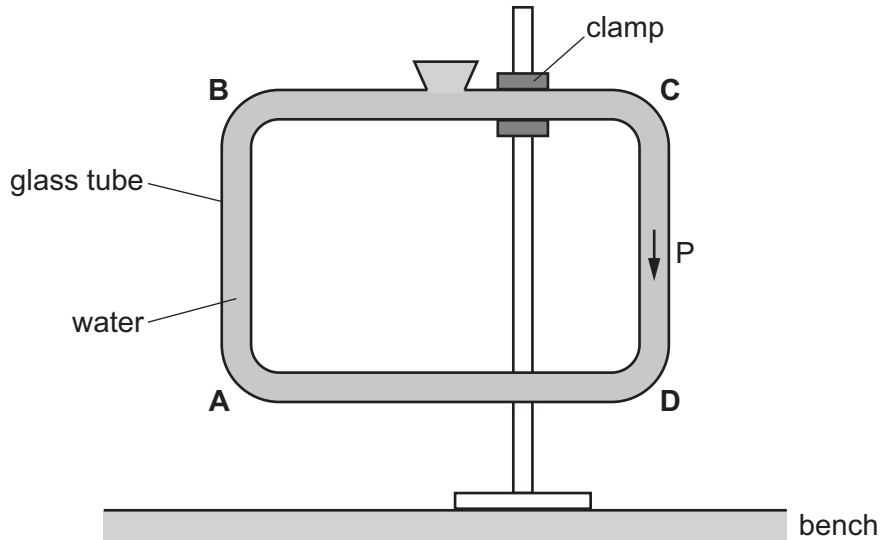
- 16 What is *power*?

- A force per second
B the maximum force an object exerts
C the total energy of an object
D work done per second

- 17 The diagram shows a glass tube filled with water and suspended above a bench. The water is free to circulate around the tube.

At point P, there is a convection current moving in a downwards direction.

At which point is the tube heated to cause this convection current?



- 18 Which physical property **cannot** be used for the measurement of temperature?
- A the length of a metal bar
 - B the mass of a plastic block
 - C the pressure of a gas
 - D the resistance of a wire
- 19 A student performs an experiment to find out how much energy is needed to change the temperature of a steel nail by 1.0°C .

Which quantity is the student determining?

- A the heat capacity of the nail
- B the latent heat of the nail
- C the specific heat capacity of steel
- D the specific latent heat of steel

- 20** A substance has a melting point of -17°C and a boiling point of 117°C .

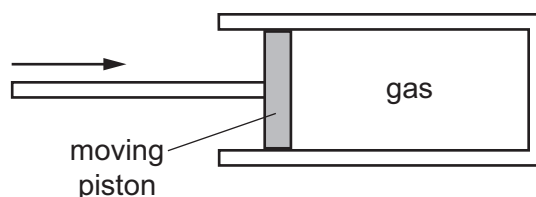
In which state does the substance exist at -10°C and at 110°C ?

	at -10°C	at 110°C
A	solid	liquid
B	solid	gas
C	liquid	liquid
D	liquid	gas

- 21** Which description of a liquid is correct?

- A** fixed shape, fixed volume
- B** fixed shape, no fixed volume
- C** no fixed shape, fixed volume
- D** no fixed shape, no fixed volume

- 22** A mass of gas is trapped inside a cylinder by a moving piston.



As the piston moves in, the volume of the gas decreases but the temperature stays the same.

What happens to the gas pressure, and why?

	pressure	reason
A	decreases	the molecules have less room to move in the cylinder
B	decreases	the molecules move more slowly between collisions
C	increases	the molecules collide more frequently with the piston
D	unchanged	the temperature of the gas is constant

- 23** A plane mirror on a vertical wall forms an image of an object placed in front of it.

Which characteristics describe the image?

- A** real, inverted and smaller than the object
- B** real, upright and the same size as the object
- C** virtual, upright and smaller than the object
- D** virtual, upright and the same size as the object

- 24** Which statement about light passing from air to glass is correct?

- A** The frequency of the light waves decreases.
- B** The speed of the light waves decreases.
- C** The wavelength of the light waves increases.
- D** The wavelength of the light waves remains unchanged.

- 25** Applications use different components of the electromagnetic spectrum.

Which shows correct applications for X-rays, ultraviolet light and microwaves?

	X-rays	ultraviolet light	microwaves
A	mobile phone	fluorescent tube	intruder alarm
B	killing cancerous cells	sterilising surgical instruments	satellite television
C	medical imaging	television controller	sunbed
D	sterilising surgical instruments	television controller	detecting cracks in metal

- 26** Where does sound travel the fastest?

- A** in a vacuum
- B** in gases
- C** in liquids
- D** in solids

- 27** A scientific research boat uses pulses of ultrasound to detect fish underneath it.

A pulse of ultrasound is bounced from a group of fish 6000 m directly below the boat.

The next pulse is sent out 2.0 s after the echo from the previous pulse has been received.

The speed of ultrasound in water is 1500 m/s.

What is the interval between the two pulses?

- A** 4.0 s **B** 6.0 s **C** 8.0 s **D** 10 s

- 28** Which row correctly describes ultrasound?

	type of wave	frequency
A	longitudinal	less than 20 kHz
B	longitudinal	greater than 20 kHz
C	transverse	less than 20 kHz
D	transverse	greater than 20 kHz

- 29** What is stored in a battery and what is its unit?

	quantity	unit
A	current	A
B	current	As
C	energy	J
D	energy	J/s

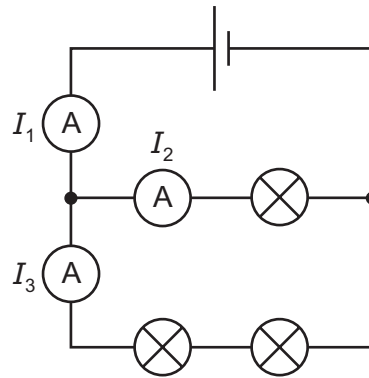
- 30** A car battery delivers 150 kC to a circuit before it needs recharging.

Two headlamps connected in parallel to the battery are switched on. There is a current of 4.0 A in each lamp.

How much time passes before the battery needs recharging?

- A** 2.6 hours **B** 5.2 hours **C** 7.8 hours **D** 10.4 hours

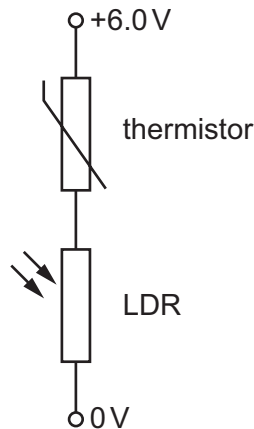
- 31 Three identical lamps and three ammeters are connected as shown.



The readings on the ammeters are I_1 , I_2 and I_3 .

How are the readings related?

- A $I_1 = I_2 = I_3$
 - B $I_1 > I_2$ and $I_2 = I_3$
 - C $I_1 > I_3 > I_2$
 - D $I_1 > I_2 > I_3$
- 32 A thermistor and a light-dependent resistor (LDR) are connected in series. A potential difference (p.d.) of 6.0 V is applied across them as shown.



The thermistor has a resistance of $6000\ \Omega$ in a cold room and $1000\ \Omega$ in a warm room. The LDR has a resistance of $2000\ \Omega$ in dim light and $500\ \Omega$ in bright light.

When is the p.d. across the LDR equal to 2.0 V?

- A in a cold room with bright light
- B in a cold room with dim light
- C in a warm room with bright light
- D in a warm room with dim light

- 33** A 240 V mains circuit contains eight 60 W lamps in parallel. At the time when the lamps are switched on, the filaments are cold and the current is four times as large as the final steady current in the circuit.

What is the initial current supplied by the mains?

- A** 0.25 A **B** 1.0 A **C** 2.0 A **D** 8.0 A

- 34** What is used with a magnet to create an induced electromotive force (e.m.f.) in a simple a.c. generator?

- A** a battery
B a coil of wire
C a voltmeter
D a relay

- 35** What changes kinetic (movement) energy into electrical energy?

- A** a car battery
B an electric fire
C an electric motor
D a wind turbine

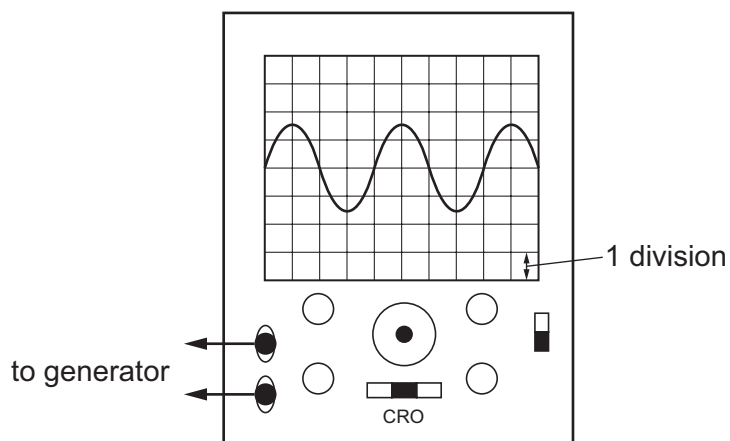
- 36** A transformer has 4800 turns on its primary coil and 480 turns on its secondary coil.

The primary coil is connected to a 240 V a.c. supply. The secondary coil is connected to a lamp.

How does the output current in the lamp compare with the input current?

- A** higher frequency a.c.
B lower frequency a.c.
C current in one direction only
D the same frequency a.c.

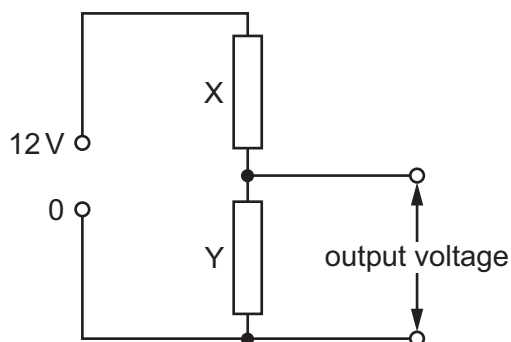
- 37 A cathode-ray oscilloscope with the Y-gain set at 2 V/division is connected to a generator.



Which row describes the signal from the generator and its amplitude?

	signal	amplitude / V
A	a.c.	3
B	a.c.	6
C	d.c.	3
D	d.c.	6

- 38 A potential divider uses a power supply of voltage 12 V. The resistors X and Y initially have equal resistances.



The resistance of X is halved.

What is the change in the output voltage?

- A** -3.0 V **B** -2.0 V **C** +2.0 V **D** +3.0 V

39 What is the definition of the half-life of a radioactive element?

- A** half of the time it takes for one atom to decay
- B** half of the time taken for all the atoms in a sample to decay
- C** the time taken for half of the atoms in a sample to decay
- D** the time taken for the nucleon number (mass number) to halve

40 Which row shows the atomic structure of a neutral atom with a nucleon number of 9?

	number of neutrons	number of protons	number of electrons
A	4	5	4
B	4	5	5
C	5	4	5
D	5	5	4

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

October/November 2020

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

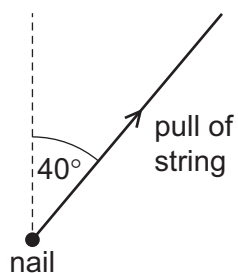
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



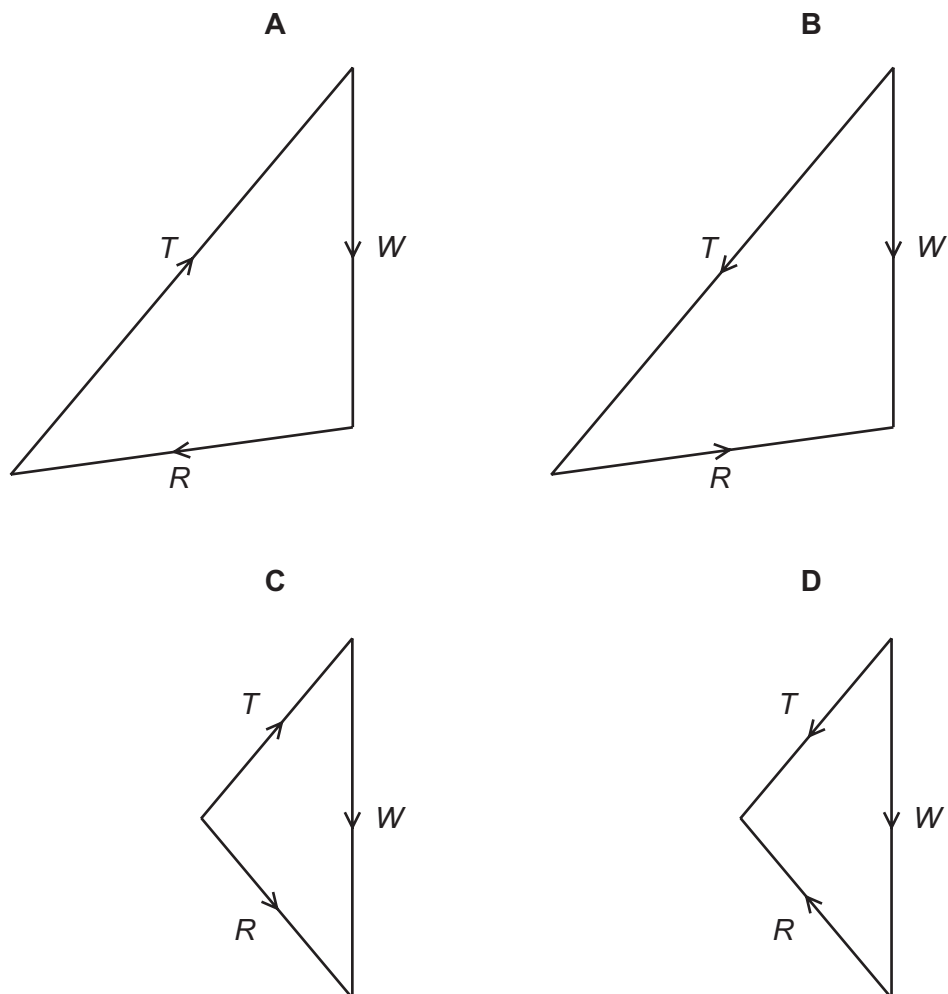
- 1 A heavy nail is fixed firmly to a wall. It is pulled by a string at 40° to the vertical. The nail does not move.



Three forces act on the nail:

- its weight W ,
- the tension T in the string,
- the force R exerted by the wall.

Which diagram, drawn to scale, represents the three forces?



- 2 A car begins to move. It speeds up until it reaches a constant speed. It continues to travel at this constant speed for the rest of the journey.

What happens to the acceleration and what happens to the velocity of the car during the journey?

- A Both the acceleration and the velocity change.
 B Only the acceleration changes.
 C Only the velocity changes.
 D Neither the acceleration nor the velocity changes.

- 3 A car of mass 800 kg has a forward acceleration of 2.5 m/s^2 . A frictional force of 1200 N opposes the motion of the car.

What is the driving force due to the engine of the car?

- A 800 N B 1200 N C 2000 N D 3200 N

- 4 A parachutist falling at a steady speed opens her parachute.

Which row is correct for the direction of the resultant force and for the direction of the acceleration of the parachutist just after her parachute opens?

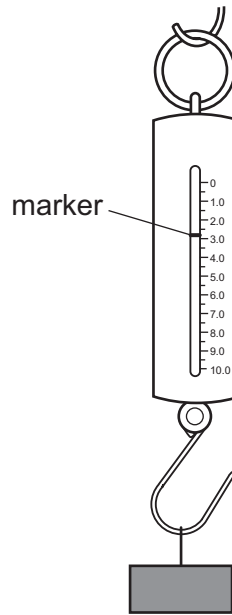
	resultant force direction	acceleration direction
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

- 5 Some gas trapped in a cylinder is compressed at constant temperature by a piston.

Which property of the gas does **not** change?

- A density
 B mass
 C pressure
 D volume

- 6 A block is hung on a spring balance. The marker inside the balance is pulled down by the block.



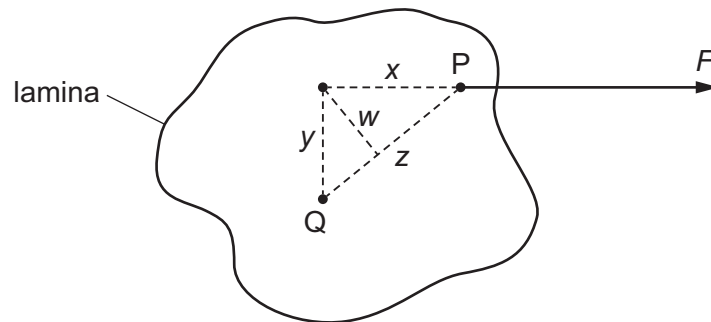
What can the position of the marker be used to determine?

- A the mass of the block in kg
 - B the mass of the block in N
 - C the moment caused by the block in N
 - D the weight of the block in kg
- 7 An electrically charged plastic ball is at rest.
- Which types of field are caused by the ball?
- A electric, gravitational and magnetic
 - B electric and gravitational only
 - C electric and magnetic only
 - D gravitational and magnetic only
- 8 Which statement about mass and weight is correct?
- A Mass is a scientific term that means the same as weight.
 - B The mass of an object on Earth is 10 times its weight.
 - C Weight is a scalar quantity, mass is a vector quantity.
 - D Weight is the force of gravity pulling on a mass.

9 Which expression is used to calculate density?

- A mass $\times$ volume
- B $\frac{\text{mass}}{\text{volume}}$
- C volume + mass
- D $\frac{\text{volume}}{\text{mass}}$

10 A length of thread is attached to a lamina at point P, as shown in the diagram.



The lamina is free to rotate about point Q.

The tension in the thread is F .

What is the moment of F about Q?

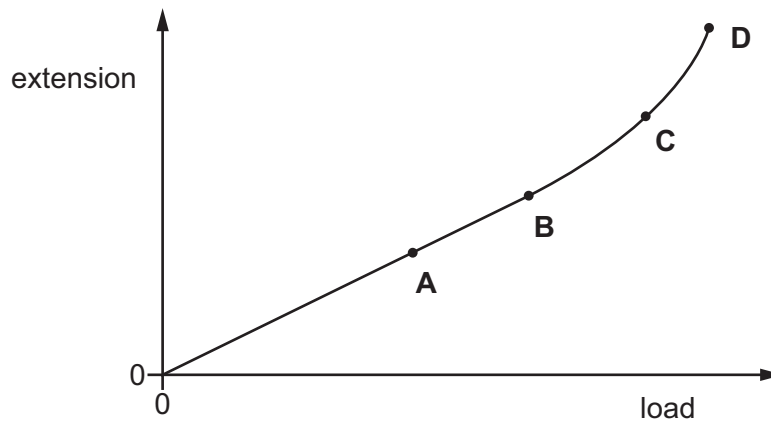
- A Fw
- B Fx
- C Fy
- D Fz

11 Which list contains only quantities that can be changed by a force?

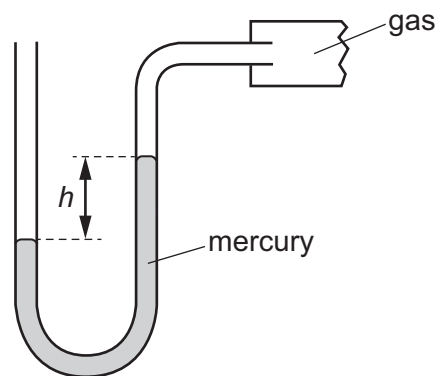
- A mass, shape, velocity
- B mass, shape, volume
- C mass, velocity, volume
- D shape, velocity, volume

- 12 An extension–load graph is plotted for a spring.

Which point marks the limit of proportionality for this spring?



- 13 A mercury manometer is used to measure the pressure of a gas.

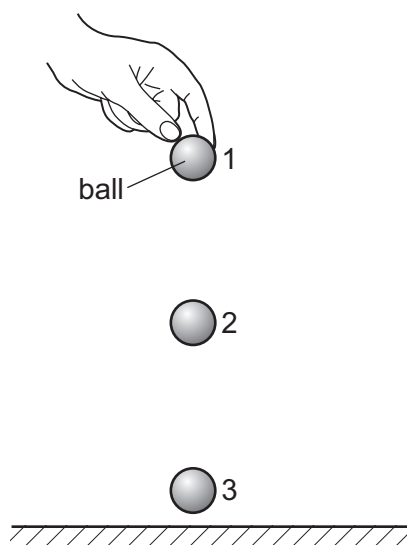


The pressure of the atmosphere is p_0 and the density of mercury is ρ .

What is the pressure of the gas?

- A** $p_0 - h\rho g$ **B** $p_0 + h\rho g$ **C** p_0 **D** $h\rho g$
- 14 There is a current in a resistor.
- Which energy transfer takes place?
- A** electrical $\rightarrow$ chemical
B electrical $\rightarrow$ internal
C kinetic $\rightarrow$ electrical
D potential $\rightarrow$ electrical

15 A ball is dropped from a height as shown.



Ignoring the effects of air resistance, which statement about the total energy of the ball is correct?

- A It is the same at all points.
- B It is greatest at point 1.
- C It is greatest at point 2.
- D It is greatest at point 3.

16 Which source of electrical energy is **not** renewable?

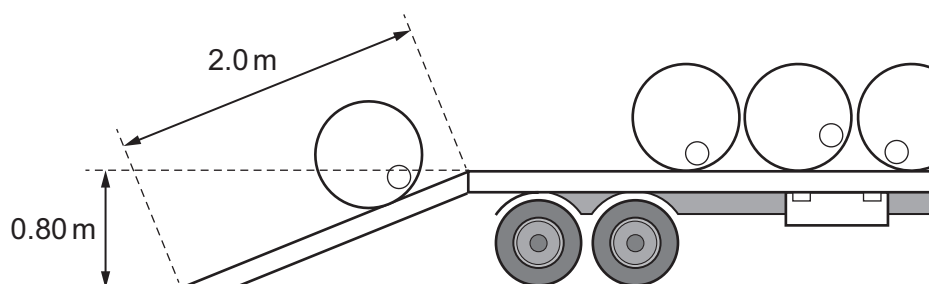
- A solar cells
- B hydroelectric generators
- C wind turbines
- D nuclear reactors

17 The table shows some data from a high-jump competition.

Which athlete jumps the highest?

	weight / N	increase in gravitational potential energy / J
A	600	1320
B	700	1610
C	800	1760
D	900	1800

- 18** A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

- A** 1000 J **B** 1600 J **C** 2500 J **D** 4000 J

- 19** Data for three types of electricity generator are shown.

	input energy /MJ	wasted energy /MJ
oil	500	300
nuclear	200	160
hydroelectric	10	1.0

Which is the least efficient generator and which is the most efficient?

	least efficient	most efficient
A	hydroelectric	nuclear
B	hydroelectric	oil
C	nuclear	hydroelectric
D	oil	hydroelectric

- 20** An instruction in a physics book states 'divide the length between two marks on the glass tube into 100 equal parts'.

What is being described?

- A** how to calculate weight using the extension of a spring
B how to calibrate a thermometer
C how to determine pressure with a manometer
D how to measure volume with a measuring cylinder

- 21** What is a unit for heat capacity?

- A** J **B** J/g **C** J/°C **D** J/(g °C)

22 A block of aluminium is heated.

What happens to the kinetic energy and to the potential energy of the molecules?

	kinetic energy	potential energy
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

23 Which statement about the melting of ice and the boiling of water is correct?

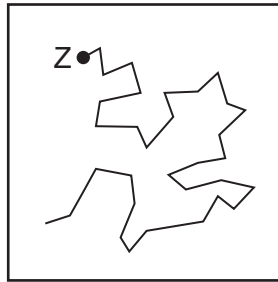
- A** Both processes are accompanied by a decrease in volume.
- B** Both processes involve energy transfer and a temperature change.
- C** Both processes involve the absorption of latent heat.
- D** Both processes result in the increase of intermolecular forces.

24 A substance has a melting point of -17°C and a boiling point of 117°C .

In which state does the substance exist at -10°C and at 110°C ?

	at -10°C	at 110°C
A	solid	liquid
B	solid	gas
C	liquid	liquid
D	liquid	gas

- 25 A smoke particle Z is seen to move randomly when suspended in air as illustrated.

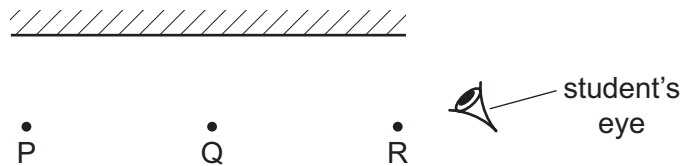


Which statement explains why Z moves randomly?

- A** Air molecules are much larger than Z.
B Air molecules are smaller than Z.
C Air molecules hit Z from different directions.
D Air molecules vibrate about a fixed position.
- 26 An electromagnetic wave has a speed of 3.0×10^8 m/s and a wavelength of 10 cm.

What is the frequency of the wave?

- A** 3.3×10^{-10} Hz
B 3.3×10^{-8} Hz
C 3.0×10^7 Hz
D 3.0×10^9 Hz
- 27 Three objects P, Q and R are placed in front of a plane mirror.



The student's eye is positioned as shown.

Which of the images of P, Q and R can the student see in the mirror?

	P	Q	R
A	✓	✓	✓
B	✓	✓	x
C	✓	x	x
D	x	x	x

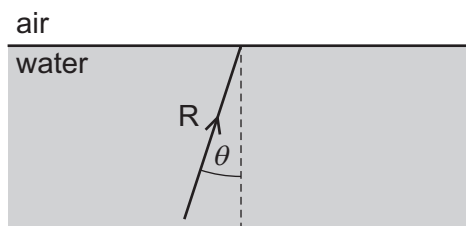
key

✓ = can see

x = cannot see

- 28** A ray of light R is incident on a water-to-air surface with an angle of incidence θ .

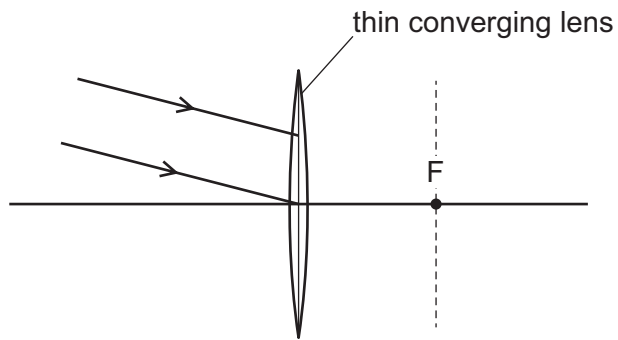
The angle θ is less than the critical angle c .



Which statement describes the subsequent path of R?

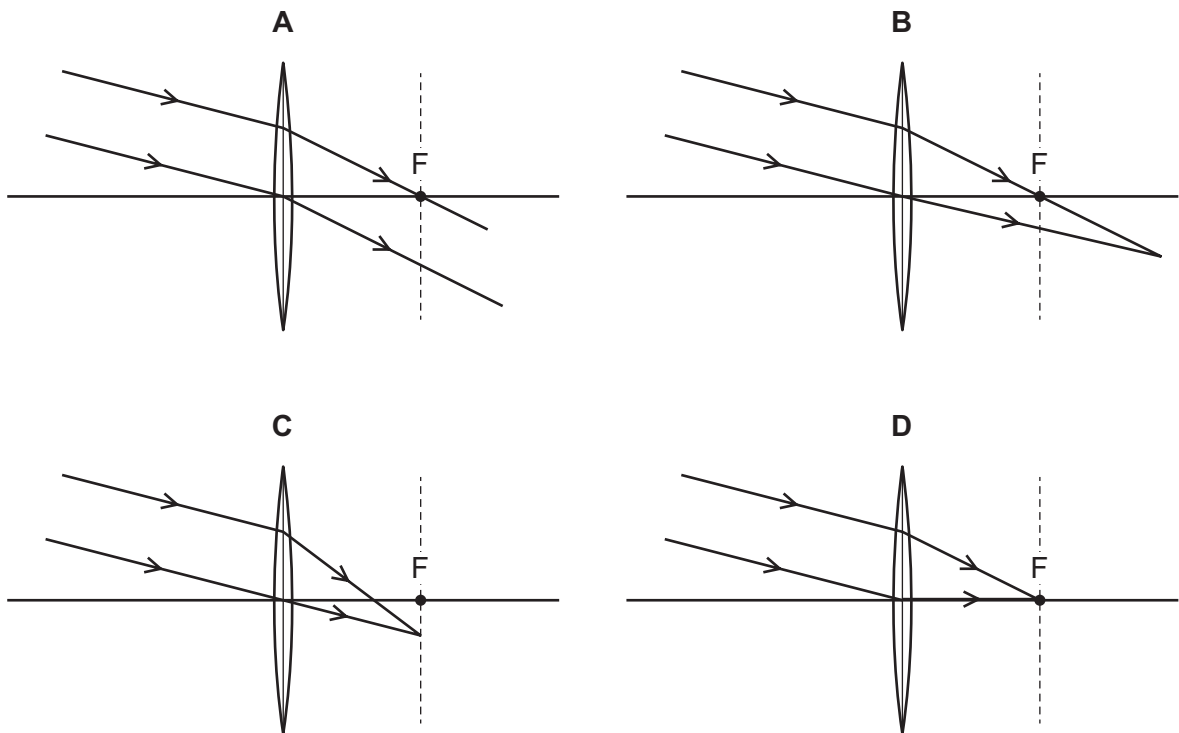
- A** It travels back into the water with an angle of reflection equal to c .
- B** It travels back into the water with an angle of reflection greater than c .
- C** It travels into air with an angle of refraction greater than θ .
- D** It travels into air with an angle of refraction smaller than θ .

- 29** A parallel beam of light is incident on a thin converging lens.



F is one focal point of the lens.

Which ray diagram shows the light after it has passed through the lens?



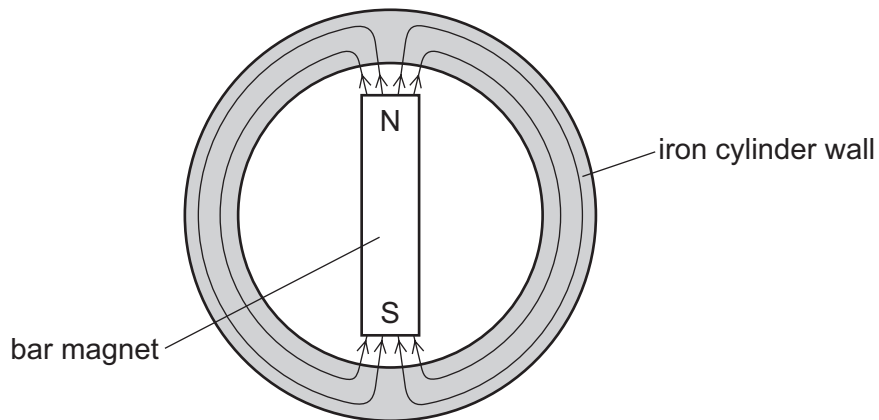
- 30** Which description of a sound wave is correct?

- A** a longitudinal electromagnetic wave
- B** a transverse electromagnetic wave
- C** an oscillation of particles parallel to the direction of travel of the wave energy
- D** an oscillation of particles perpendicular to the direction of travel of the wave energy

31 Where does sound travel the fastest?

- A in a vacuum
- B in gases
- C in liquids
- D in solids

32 A bar magnet is placed in a hollow iron cylinder. The diagram shows the magnetic field pattern produced.

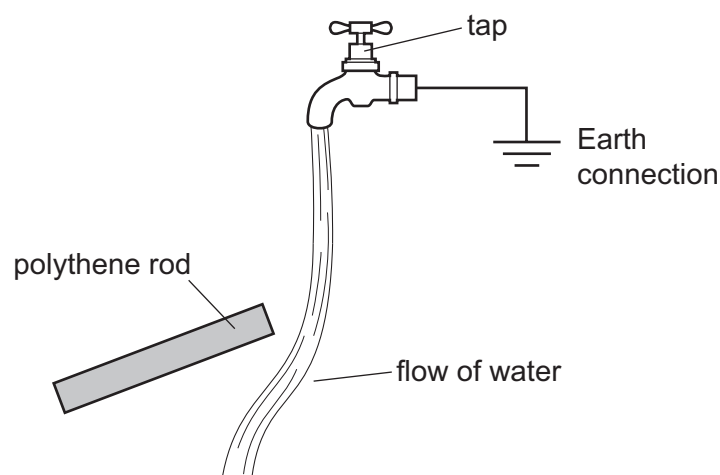


What is the cause of the field inside the cylinder wall?

- A electromagnetic generation
- B electromagnetic induction
- C electrostatic induction
- D magnetic induction

- 33** A polythene rod is brought near to a stream of water from a metal tap which is earthed. Tap water is an electrical conductor.

The water moves towards the rod, as shown.



Which description of the rod and water is correct?

	rod	water
A	charged	charged
B	charged	uncharged
C	uncharged	charged
D	uncharged	uncharged

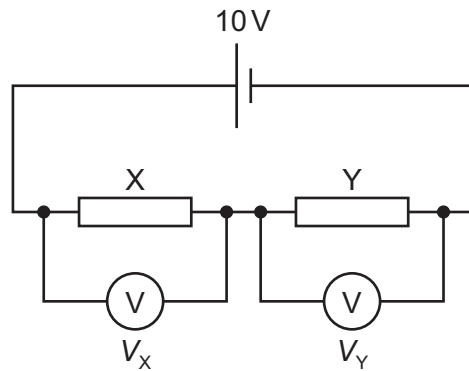
- 34** A small negative charge is placed at a point where an electric field is vertically upwards.

There is a force on the charge due to the field.

In which direction does it act?

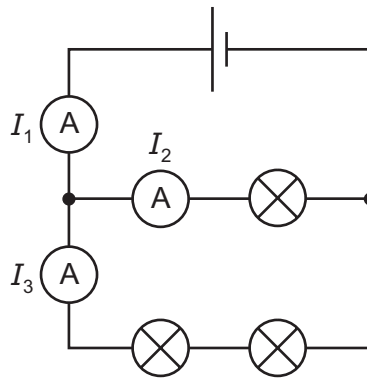
- A** vertically upwards
- B** vertically downwards
- C** horizontally to the right
- D** horizontally to the left

- 35 In the circuit shown, the resistance of resistor Y is four times greater than the resistance of resistor X.



What is the difference $V_Y - V_X$ of the voltages shown on the voltmeters?

- A** 2.0V **B** 5.0V **C** 6.0V **D** 8.0V
- 36 Two identical resistors connected in parallel have a total resistance of 4.0Ω .
What is the total resistance when the same two resistors are connected in series?
- A** 1.0Ω **B** 4.0Ω **C** 8.0Ω **D** 16Ω
- 37 Three identical lamps and three ammeters are connected as shown.

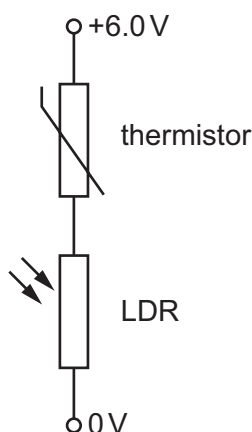


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How are the readings related?

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The thermistor has a resistance of $6000\ \Omega$ in a cold room and $1000\ \Omega$ in a warm room. The LDR has a resistance of $2000\ \Omega$ in dim light and $500\ \Omega$ in bright light.

When is the p.d. across the LDR equal to 2.0 V?

- A in a cold room with bright light
 - B in a cold room with dim light
 - C in a warm room with bright light
 - D in a warm room with dim light
- 39 A student calculates the amount of energy used by an electric heater.
- What is the equation for calculating the energy E in kWh?
- A $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (second)}$
 - B $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (hour)}$
 - C $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (second)} \div 1000$
 - D $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (hour)} \div 1000$
- 40 An isotope P is radioactive and has a half-life of 7.0 years. A sample initially contains 0.016 kg of P.

After how long will the sample contain 0.0020 kg of P?

- A 7.0 years
- B 14 years
- C 21 years
- D 28 years

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Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2020

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	A	1
3	C	1
4	A	1
5	D	1
6	B	1
7	A	1
8	D	1
9	B	1
10	D	1
11	B	1
12	C	1
13	A	1
14	A	1
15	B	1
16	D	1
17	A	1
18	B	1
19	A	1
20	C	1
21	C	1
22	C	1
23	D	1
24	B	1
25	B	1
26	D	1
27	D	1

Question	Answer	Marks
28	B	1
29	C	1
30	B	1
31	D	1
32	C	1
33	D	1
34	B	1
35	D	1
36	D	1
37	A	1
38	C	1
39	C	1
40	B	1



Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

October/November 2020

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	A	1
3	D	1
4	D	1
5	B	1
6	A	1
7	B	1
8	D	1
9	B	1
10	C	1
11	D	1
12	B	1
13	A	1
14	B	1
15	A	1
16	D	1
17	B	1
18	B	1
19	C	1
20	B	1
21	C	1
22	A	1
23	C	1
24	C	1
25	C	1
26	D	1
27	B	1

Question	Answer	Marks
28	C	1
29	C	1
30	C	1
31	D	1
32	D	1
33	A	1
34	B	1
35	C	1
36	D	1
37	D	1
38	C	1
39	D	1
40	C	1



PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2019

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

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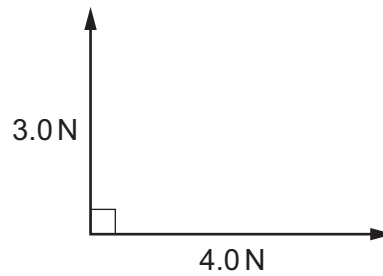
This document consists of **16** printed pages.



1 Which word is the name of a vector quantity?

- A density
- B displacement
- C energy
- D speed

2 What is the size of the resultant of the two forces shown in the diagram?

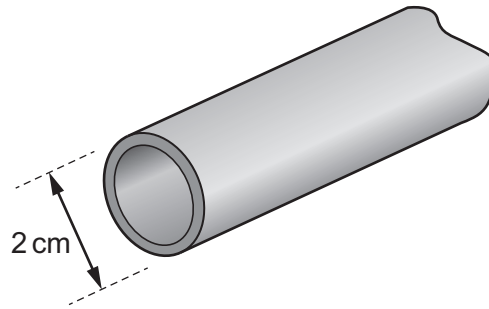


- A 1.0 N B 3.5 N C 5.0 N D 7.0 N

3 What is the correct unit for the quantity shown?

	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg/m ³
D	weight	kg

- 4 A length of copper pipe, of uniform cross-section and several metres long, carries water to a tap.



Measurements are taken to determine accurately the volume of copper in the pipe.

Which instruments are used?

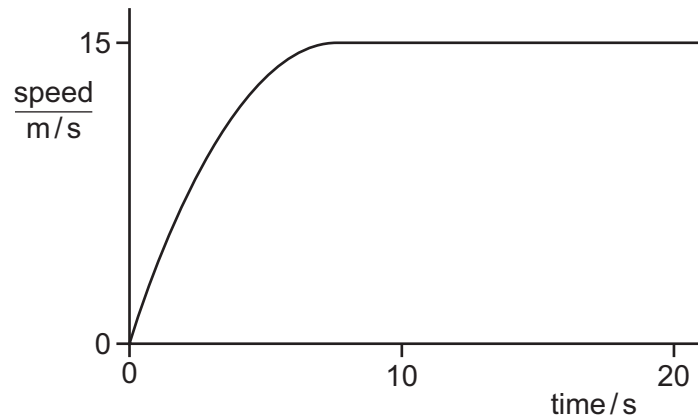
- A micrometer and rule
 - B micrometer and calipers
 - C rule and tape
 - D tape and calipers
- 5 Part of a speed–time graph for two runners is shown. They are running along the same track.



Which statement **must** be correct?

- A Both runners started at the same moment.
- B Runner 1 has a greater acceleration than runner 2.
- C Runner 1 is moving faster than runner 2.
- D The distance between the two runners stays constant.

- 6 The graph shows how the speed of a car varies with time.

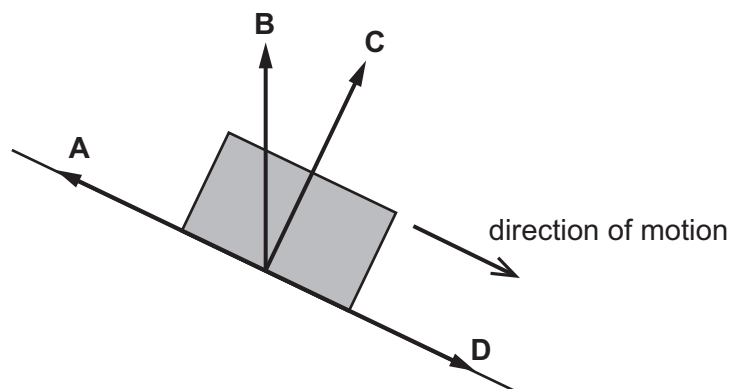


Which statement about the acceleration of the car between the times 10 s and 20 s is correct?

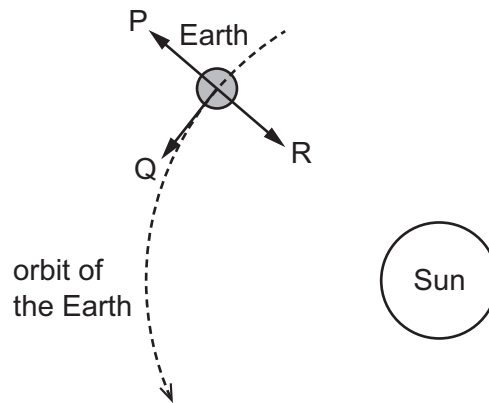
- A The acceleration decreases.
 - B The acceleration increases.
 - C The acceleration is constant, but not zero.
 - D The acceleration is zero.
- 7 In which descent is the acceleration equal to the acceleration of free fall g at all times?
- A a bungee jumper leaping from a bridge
 - B a feather falling in a vertical tube that contains a vacuum
 - C a hailstone travelling to Earth at terminal velocity
 - D a sky-diver dropping from an aircraft towards the ground
- 8 A body slides down a frictionless slope, as shown.

As the body presses on the surface, the surface pushes back on the body.

In which direction does the surface push back on the body?



- 9 The Earth travels at constant speed in a circular orbit around the Sun.

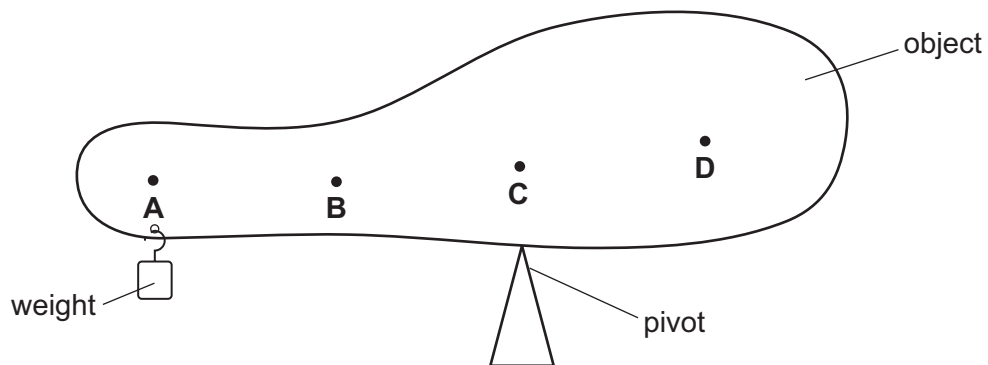


Which arrows show the direction of the acceleration of the Earth and the direction of the velocity of the Earth?

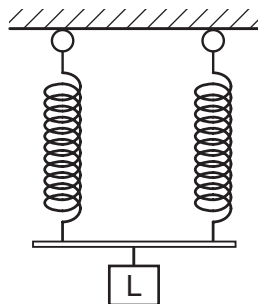
	direction of acceleration	direction of velocity
A	P	Q
B	Q	P
C	Q	R
D	R	Q

- 10 A student balances a non-uniform object on a pivot. To do this, a weight is suspended near the left-hand end of the object, as shown.

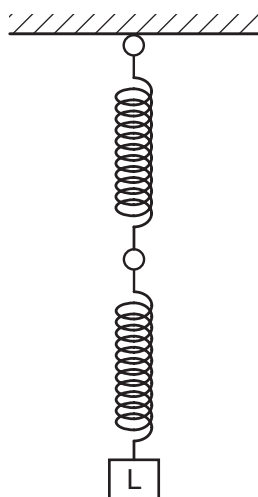
Where is the centre of mass of the object?



- 11 A load L is suspended from two springs that are in parallel. The extension of each spring is x .



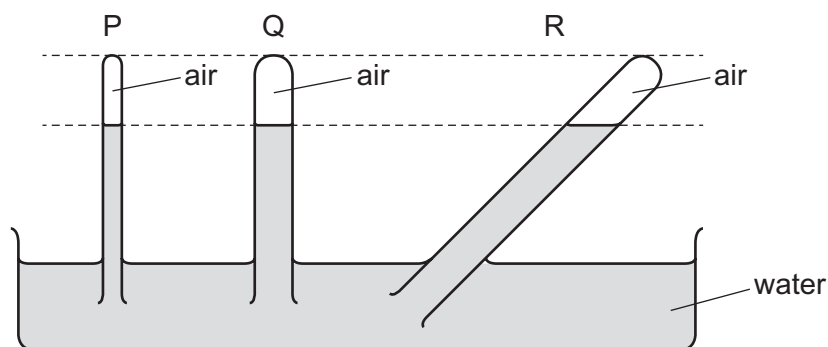
The springs are then arranged to hang vertically, one below the other.



In this new arrangement, what is the total extension of the two springs?

- A** $\frac{1}{2}x$ **B** x **C** $2x$ **D** $4x$

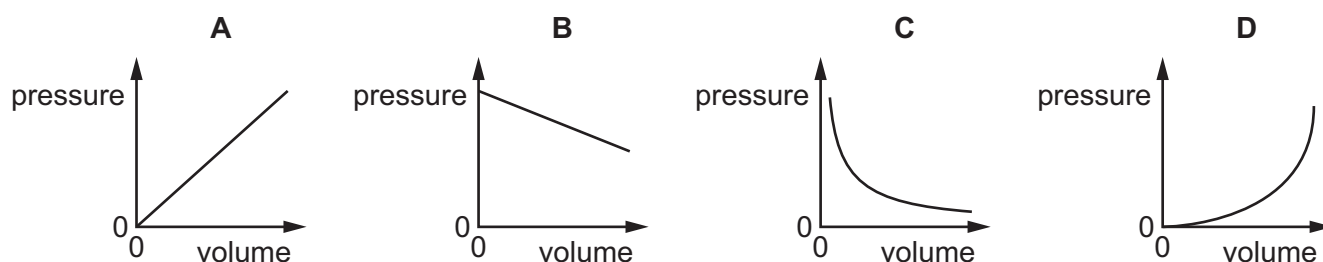
- 12 The diagram shows three tubes P, Q and R. Each tube contains air trapped by a water column.



Which statement is correct?

- A** The pressure of the trapped air is equal in tubes P, Q and R.
B The pressure of the trapped air is greatest in tube Q.
C The pressure of the trapped air is greatest in tube R.
D The pressure of the trapped air is greatest in tube P.

- 13 Which graph shows the relationship between the pressure and the volume of a fixed mass of gas at constant temperature?



- 14 An object X has mass m and velocity v .

A second object Y has mass $2m$ and velocity $2v$.

What is the value of the ratio $\frac{\text{kinetic energy of Y}}{\text{kinetic energy of X}}$?

- A** 1 **B** 2 **C** 4 **D** 8

- 15 Where is energy released by the fusion of hydrogen nuclei to form helium?

- A** in a nuclear power station
B in a radioactive isotope emitting alpha-particles
C in the core of the Earth
D in the core of the Sun

- 16 A student uses a newton meter to pull an object a distance d horizontally along a flat surface in a time of t . The newton meter reading is F .

What is given by the expression $(F \times d)/t$?

- A** efficiency
B energy
C power
D work

- 17** A student is given a thermometer that reads 1°C in pure melting ice and 101°C in pure boiling water. She uses it to measure the temperature of some water before and after it is heated.

How does the rise in temperature of the water calculated from her results compare with the correct value?

- A** It is 1°C lower.
- B** It is the same.
- C** It is 1°C higher.
- D** It is 2°C higher.

- 18** A student makes four statements about evaporation and boiling.

- 1 Both involve an increase in molecular separation.
- 2 Both occur only at a fixed temperature.
- 3 Both involve the absorption of thermal energy.
- 4 Both occur throughout the liquid.

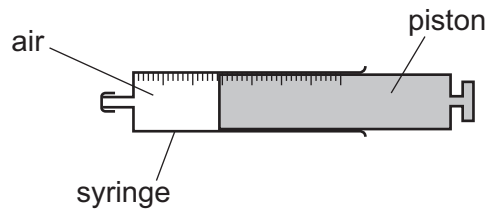
Which two statements are correct?

- A** 1 and 4 **B** 2 and 4 **C** 1 and 3 **D** 2 and 3

- 19** How is *specific latent heat* defined?

- A** the heat absorbed or emitted by unit mass during a change of a state without a temperature change
- B** the heat absorbed or emitted by unit mass during unit temperature change
- C** the heat absorbed or emitted during a change of a state without a temperature change
- D** the heat absorbed or emitted during unit temperature change

- 20** A syringe containing air is sealed at one end. The piston is free to move.



The air is heated.

What happens to the average distance between air molecules and the density of the air?

	average distance between air molecules	density of the air
A	increases	decreases
B	increases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

- 21** A metal disc is heated to 600°C and then lowered into a beaker of water.

What happens to the mass of the metal disc and to the mass of the water in the beaker?

	mass of metal disc	mass of water in beaker
A	increases	decreases
B	increases	no change
C	no change	decreases
D	no change	increases

- 22** Which description defines the amplitude of a water wave in a ripple tank?

- A** half of the vertical distance between a trough and a peak
- B** the horizontal distance between adjacent peaks
- C** the horizontal distance between a trough and an adjacent peak
- D** the vertical distance between a trough and a peak

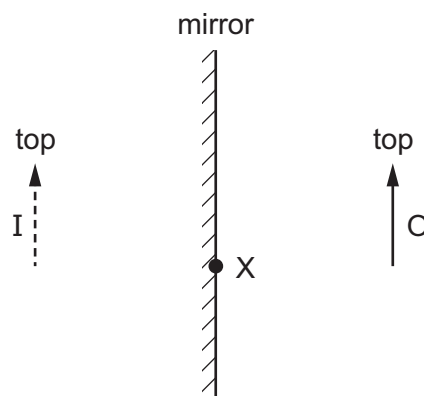
23 Here are four statements about wavefronts.

- 1 Wavefronts from a point source are straight lines.
- 2 Wavefronts from a point source are circular.
- 3 Wavefronts are parallel to the direction of energy movement.
- 4 Wavefronts are perpendicular to the direction of energy movement.

Which two statements are correct?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

24 An object O is placed in front of a plane mirror. I is the image formed.

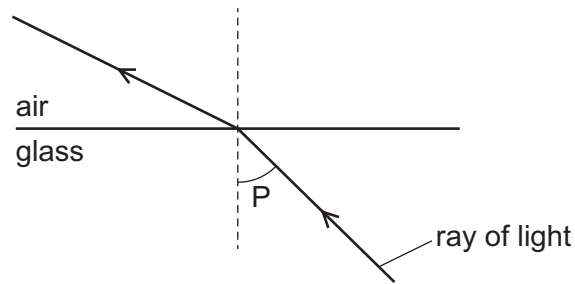


A ray from the top of the object is incident on the mirror at X.

What happens to this ray?

- A** It reflects and passes through the bottom of O.
B It reflects and passes through the top of O.
C It reflects as though it came from the bottom of I.
D It reflects as though it came from the top of I.

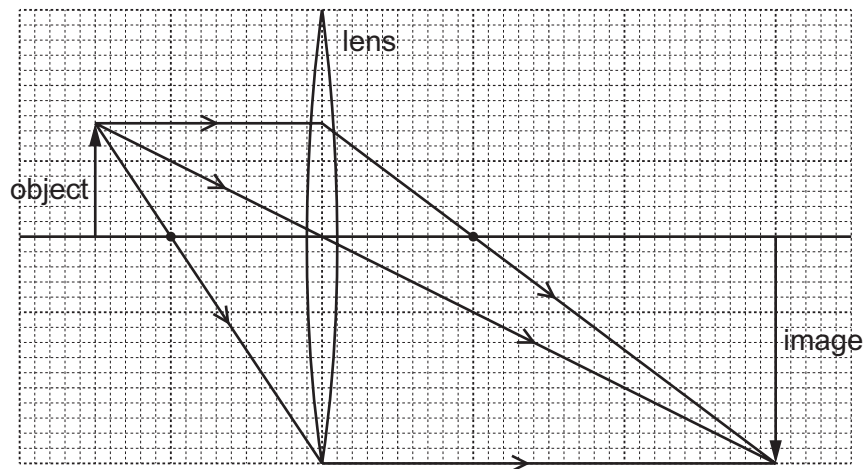
- 25 The diagram shows light passing from glass into air.



What is the name of angle P?

- A the angle of incidence
 - B the angle of reflection
 - C the angle of refraction
 - D the critical angle
- 26 An object is placed in front of a converging lens of focal length 4.0 cm. The height of the image is 6.0 cm.

The arrangement is shown on the scale diagram.



What is the linear magnification produced by the lens?

- A 0.50
- B 1.5
- C 2.0
- D 6.0

- 27 Which row shows the speed of sound in air (c_a), liquid (c_L) and solid (c_s) in order from slowest to fastest?

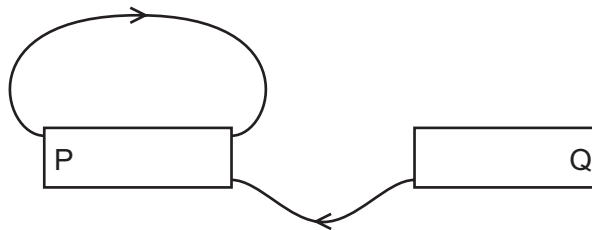
	slowest	→	fastest
A	c_L		c_s
B	c_L		c_s
C	c_a		c_s
D	c_s		c_a

- 28 A man stands 80 m in front of a cliff face. He makes a loud bang and listens for the echo.

He makes a loud bang once every second. He hears an echo exactly halfway between the bang that caused it and the next bang.

What is the speed of sound?

- A** 40 m/s **B** 80 m/s **C** 160 m/s **D** 320 m/s
- 29 The diagram shows two magnetic field lines of the magnetic field around two bar magnets.

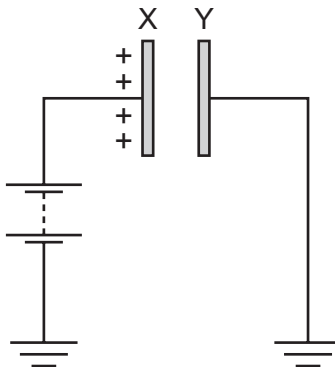


Which row shows the magnetic poles at end P and end Q of the two magnets?

	P	Q
A	N	N
B	N	S
C	S	N
D	S	S

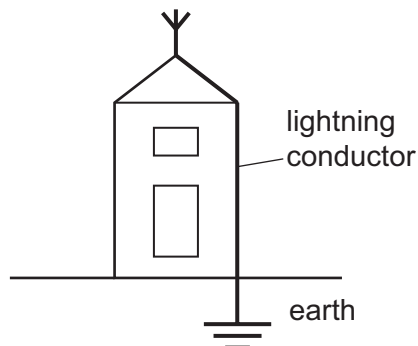
- 30** A metal plate X is attached to earth by a battery. The plate has a positive charge.

A second metal plate Y is initially uncharged and connected to earth. This plate is positioned parallel to X.



What happens as Y is brought closer to X?

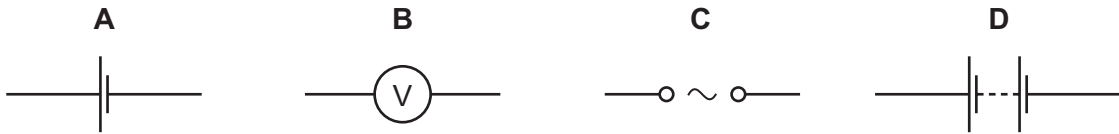
- A** Electrons flow from Y to earth only.
 - B** Electrons flow to Y from earth only.
 - C** Protons flow from Y to earth and electrons flow to Y from earth.
 - D** Protons flow to Y from earth and electrons flow from Y to earth.
- 31** The diagram shows a metal lightning conductor attached to a building.



Why is the lightning conductor made of copper?

- A** Both positive and negative charges can flow through copper.
- B** Copper can be electrostatically charged.
- C** So the lightning conductor has a low electrical resistance.
- D** So the heat is conducted away quickly.

32 Which symbol does **not** represent a power source?

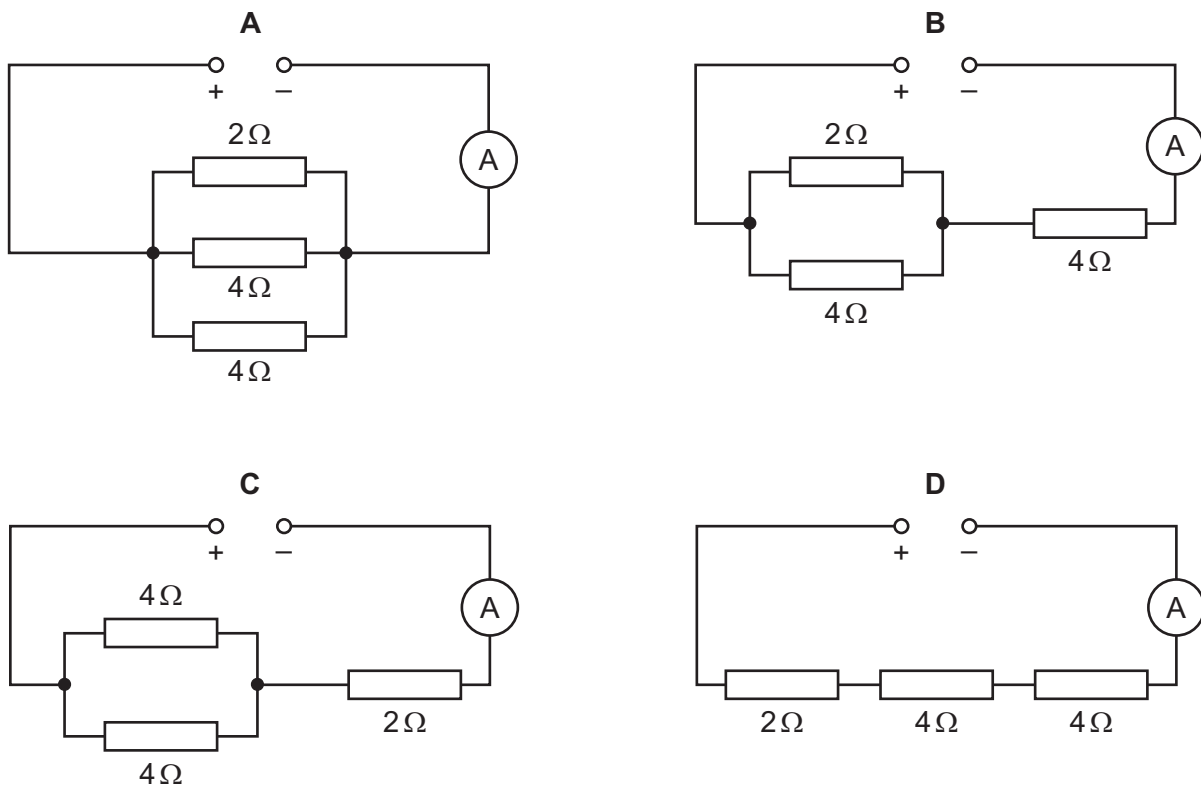


33 Resistors of different resistances are connected to a power supply in either a parallel circuit or a series circuit. Which statement is correct?

- A The current is the same in all resistors connected in parallel.
- B The current is the same in all resistors connected in series.
- C The voltage across each resistor is different for all resistors connected in parallel.
- D The voltage across each resistor is the same for all resistors connected in series.

34 An ammeter is connected to three resistors and a power supply.

Which arrangement of resistors gives the greatest ammeter reading?



- 35** A hotplate on an electric cooker contains two identical resistors.

The switch has three positions.

- position 1 The two resistors are connected in series to the mains supply.
 position 2 The two resistors are connected in parallel to the mains supply.
 position 3 Just one resistor is connected to the mains supply.

Which positions correspond to the low, the medium and the high power settings?

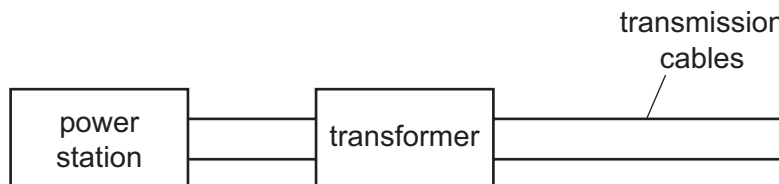
	low power	medium power	high power
A	1	2	3
B	1	3	2
C	3	1	2
D	3	2	1

- 36** A mains electric circuit is fitted with a circuit breaker rather than a fuse.

Where is the circuit breaker connected and what happens when the current is too large?

	A circuit breaker is connected in...	When the current is too large...
A	the live wire.	a thin wire melts and breaks the circuit.
B	the live wire.	an electromagnet opens a switch.
C	the neutral wire.	a thin wire melts and breaks the circuit.
D	the neutral wire.	an electromagnet opens a switch.

- 37** Transformers are used to transmit electrical energy between power stations and transmission cables, as shown.

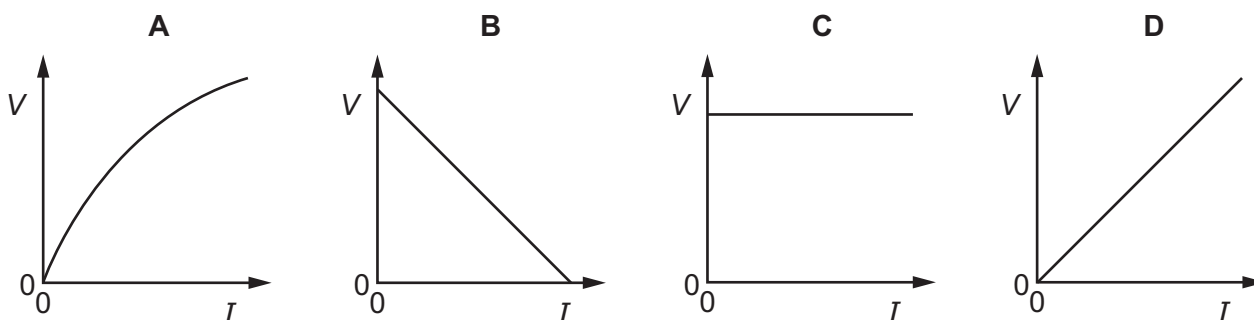


What is the purpose of the transformer in the diagram?

- A** to decrease the current and the potential difference from the power station
B to decrease the current and increase the potential difference from the power station
C to increase the current and the potential difference from the power station
D to increase the current and decrease the potential difference from the power station

- 38 A thermistor warms up as the current in it increases.

Which graph shows how the voltage V across the thermistor changes as the current I through it is increased?



- 39 Radioactive substances are used as tracers in medical examinations.

A liquid tracer is injected into a patient. A detector 20 cm above the patient shows where the radioactive substance is in the body.

What is the best radioactive substance to use?

- A an alpha-particle emitter with a half-life of a few hours
 - B an alpha-particle emitter with a half-life of a few years
 - C a gamma ray emitter with a half-life of a few hours
 - D a gamma ray emitter with a half-life of a few years
- 40 An isotope of lead, $^{212}_{82}\text{Pb}$, is radioactive. After three decays, two beta-particles and an alpha-particle have been emitted. Another radioactive isotope of lead is formed.

What is the mass number (nucleon number) of the new isotope of lead?

- A 200
- B 208
- C 212
- D 216

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PHYSICS

5054/11

Paper 1 Multiple Choice

October/November 2019

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

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Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

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There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

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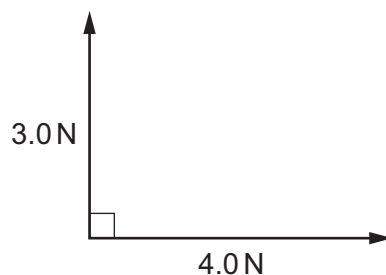
This document consists of **19** printed pages and **1** blank page.



1 Which word is the name of a scalar quantity?

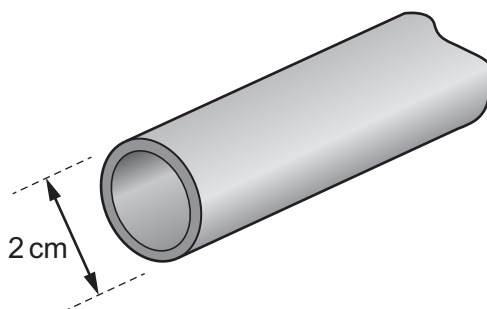
- A acceleration
- B force
- C mass
- D velocity

2 What is the size of the resultant of the two forces shown in the diagram?



- A 1.0 N B 3.5 N C 5.0 N D 7.0 N

3 A length of copper pipe, of uniform cross-section and several metres long, carries water to a tap.



Measurements are taken to determine accurately the volume of copper in the pipe.

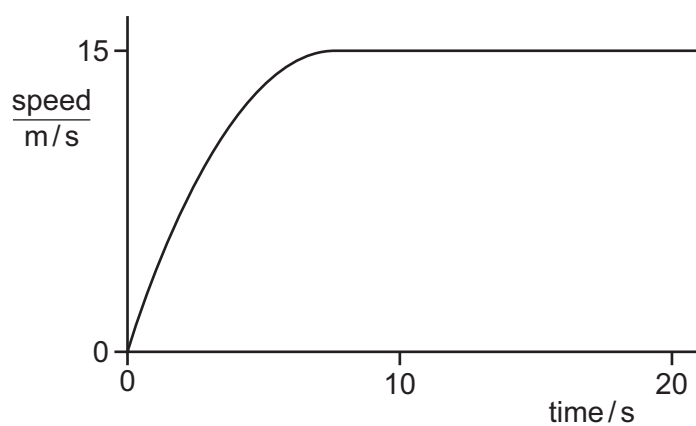
Which instruments are used?

- A micrometer and rule
- B micrometer and calipers
- C rule and tape
- D tape and calipers

4 What is the correct unit for the quantity shown?

	quantity	unit
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B	latent heat	J
C	pressure	kg/m^3
D	weight	kg

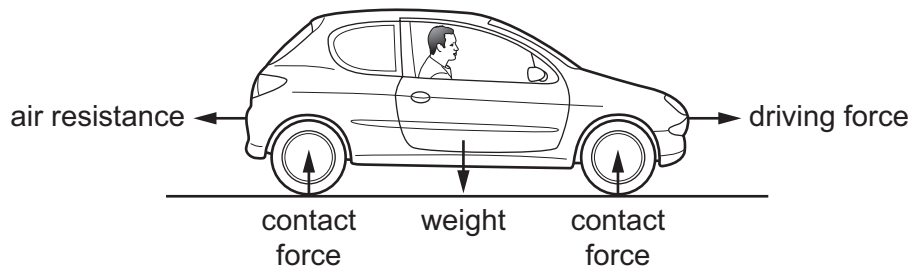
5 The graph shows how the speed of a car varies with time.



Which statement about the acceleration of the car between the times 10 s and 20 s is correct?

- A** The acceleration decreases.
- B** The acceleration increases.
- C** The acceleration is constant, but not zero.
- D** The acceleration is zero.

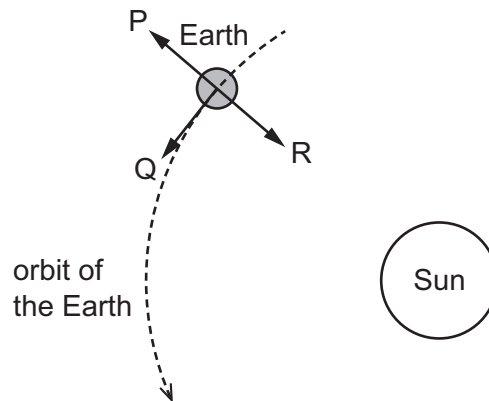
- 6 A car is accelerating along a straight, horizontal road. The diagram shows forces acting on the car.



Which forces are balanced?

- A contact forces and air resistance
 - B contact forces and weight
 - C driving force and air resistance
 - D driving force and weight
- 7 A single force is applied to a body.
- What **cannot** happen?
- A a change in the direction of the moving body
 - B a change in the mass of the body initially at rest
 - C a change in the position of the body initially at rest
 - D a change in the speed of the moving body

- 8 The Earth travels at constant speed in a circular orbit around the Sun.



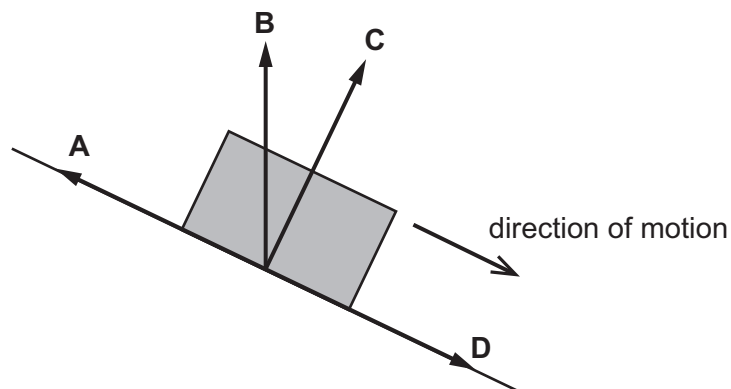
Which arrows show the direction of the acceleration of the Earth and the direction of the velocity of the Earth?

	direction of acceleration	direction of velocity
A	P	Q
B	Q	P
C	Q	R
D	R	Q

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As the body presses on the surface, the surface pushes back on the body.

In which direction does the surface push back on the body?



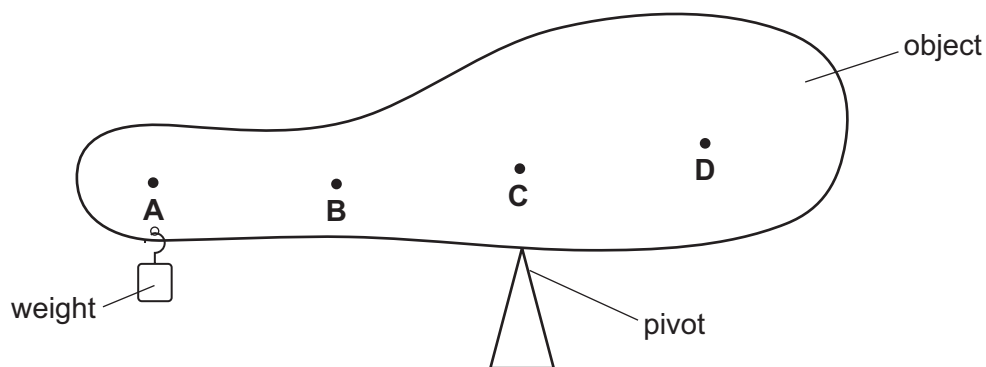
- 10 There is no atmosphere on the Moon.

Two metal spheres of identical volume but different mass are at rest at the same height above the ground on the Moon. They are dropped at the same time.

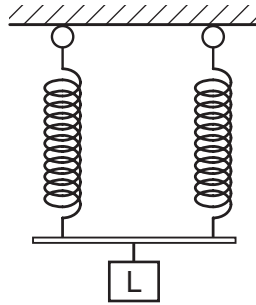
How do the spheres move after release?

- A The sphere with the larger mass has a smaller acceleration.
 - B The sphere with the larger mass hits the ground first.
 - C The sphere with the smaller mass has a smaller acceleration.
 - D The two spheres hit the ground together.
- 11 What is an expression for the moment of a force about a pivot?
- A distance the force moves from the pivot $\div$ force
 - B force $\times$ distance the force moves from the pivot
 - C force $\times$ perpendicular distance of the force from the pivot
 - D force $\div$ perpendicular distance of the force from the pivot
- 12 A student balances a non-uniform object on a pivot. To do this, a weight is suspended near the left-hand end of the object, as shown.

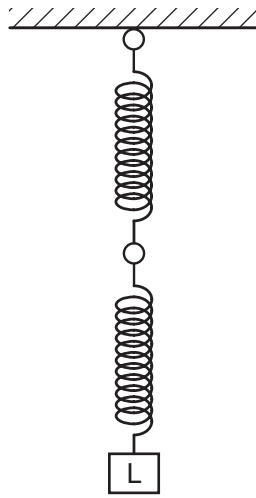
Where is the centre of mass of the object?



- 13 A load L is suspended from two springs that are in parallel. The extension of each spring is x .



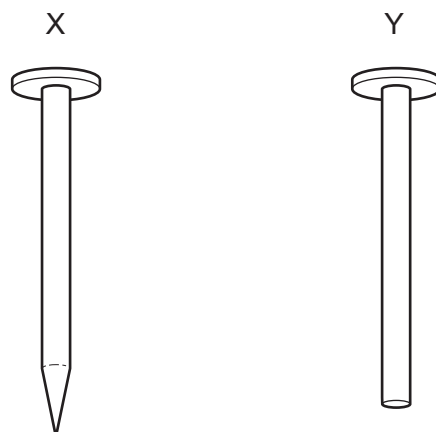
The springs are then arranged to hang vertically, one below the other.



In this new arrangement, what is the total extension of the two springs?

- A** $\frac{1}{2}x$ **B** x **C** $2x$ **D** $4x$

- 14 Nail X is tapped into wood. Nail Y is tapped into the same wood using the same force.



Which nail moves furthest into the wood and why?

	nail	why
A	X	X produces greater pressure on the wood
B	X	X produces smaller pressure on the wood
C	Y	Y produces greater pressure on the wood
D	Y	Y produces smaller pressure on the wood

- 15 The list gives the symbols for some physical quantities.

acceleration of free-fall g

atmospheric pressure P

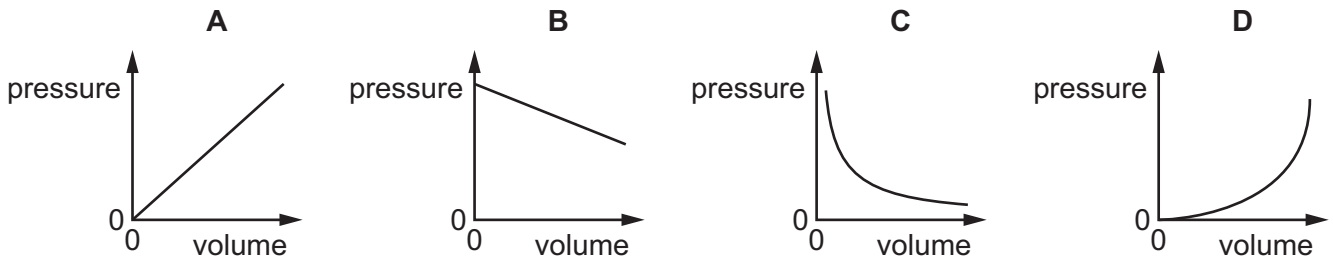
density of water ρ

depth of water h

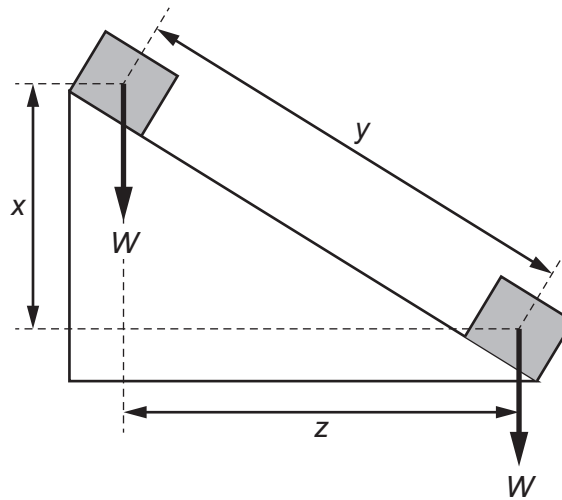
Which quantities from the list are used to calculate the total pressure at the bottom of a lake?

- A** g , h , P and ρ
- B** g , h and P only
- C** h , P and ρ only
- D** g , h and ρ only

- 16 Which graph shows the relationship between the pressure and the volume of a fixed mass of gas at constant temperature?



- 17 A heavy box of weight W slides down a frictionless slope.



What is the work done by the weight W in moving from top to bottom?

- A** Wx **B** Wy **C** $W(y - z)$ **D** $W(x + z)$
- 18 Where is energy released by the fusion of hydrogen nuclei to form helium?
- A** in a nuclear power station
 - B** in a radioactive isotope emitting alpha-particles
 - C** in the core of the Earth
 - D** in the core of the Sun
- 19 Which change, on its own, increases the range of a liquid-in-glass thermometer?
- A** decrease the length of the capillary
 - B** decrease the thickness of glass around the bulb
 - C** increase the diameter of the capillary
 - D** increase the volume of liquid in the bulb

- 20** A student is given a thermometer that reads 1°C in pure melting ice and 101°C in pure boiling water. She uses it to measure the temperature of some water before and after it is heated.

How does the rise in temperature of the water calculated from her results compare with the correct value?

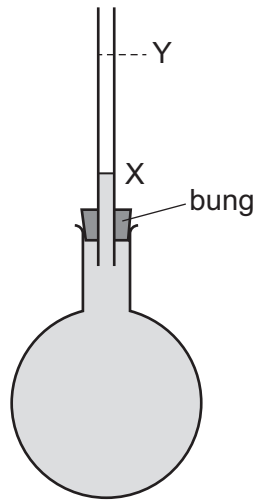
- A** It is 1°C lower.
 - B** It is the same.
 - C** It is 1°C higher.
 - D** It is 2°C higher.
- 21** What is the name of the change of state that occurs when a substance changes from a solid to a liquid?
- A** boiling
 - B** condensation
 - C** evaporation
 - D** melting

- 22** A metal rod is heated.

What happens to the rod?

- A** Its density decreases and its mass decreases.
- B** Its density decreases and its volume increases.
- C** Its density increases and its mass decreases.
- D** Its density increases and its volume increases.

- 23** The diagram shows a glass flask full of water at 10°C . A long glass tube passes through the bung into the water. The water level in the tube is at X.



The flask is placed in hot water and the water level in the tube eventually rises to Y.

Why does the water level rise?

- A** The flask contracts and the water expands.
 - B** The flask contracts more than the water contracts.
 - C** The flask expands and the water contracts.
 - D** The flask expands less than the water expands.
- 24** A glass beaker contains water in a warm room.

The water is initially at the same temperature as the room.

Which change does **not** affect the initial rate of evaporation of the water?

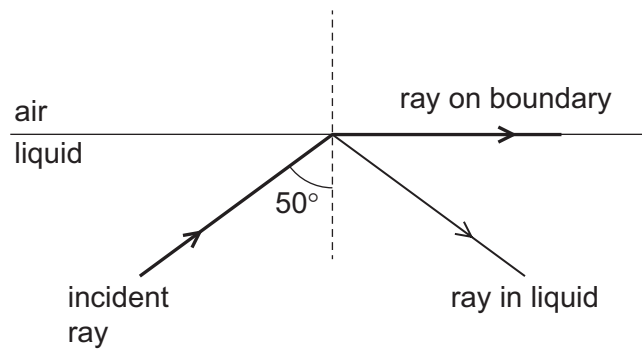
- A** changing the temperature of the water
- B** using a beaker made of thicker glass
- C** using a beaker with a larger cross-sectional area
- D** using a fan to blow air over the beaker

- 25** The frequency of microwaves used in an experiment is 3.0×10^9 Hz. Their speed in air is 3.0×10^8 m/s.

What is a possible frequency and a possible speed in air of a radio wave?

	frequency / Hz	<u>speed in air</u> m/s
A	3.0×10^5	1500
B	3.0×10^5	3.0×10^8
C	3.0×10^{14}	1500
D	3.0×10^{14}	3.0×10^8

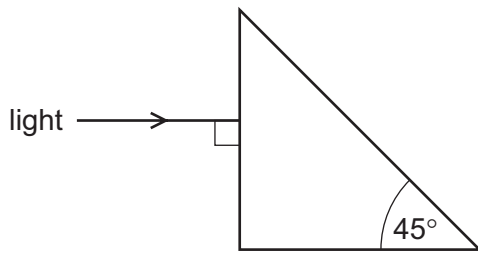
- 26** The diagram shows a ray of light in liquid incident on the boundary with air. Two other rays are observed. One is in the liquid and the other is in the air on the boundary.



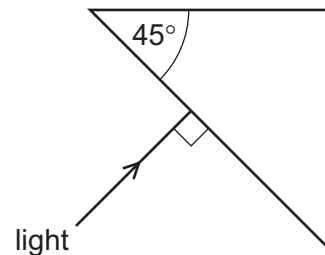
What is the angle of refraction?

- A** 0° **B** 40° **C** 50° **D** 90°

- 27 Light is incident at 90° on the surfaces of two glass prisms P and Q.



glass prism P



glass prism Q

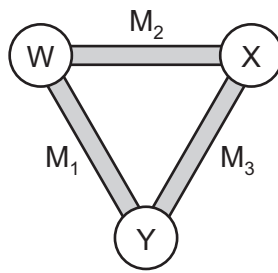
The critical angle for light travelling from glass into air is 42° .

Where does total internal reflection occur?

- A in P and in Q
 - B in P only
 - C in Q only
 - D in neither P nor Q
- 28 A person hears an echo after a sound hits a solid cliff face.
- What causes the echo?
- A absorption
 - B dispersion
 - C reflection
 - D refraction
- 29 What is a typical value for the speed of sound in air?

- A $3.3 \times 10^4 \text{ cm/s}$
- B $3.3 \times 10^4 \text{ m/s}$
- C $3.3 \times 10^2 \text{ km/s}$
- D $3.3 \times 10^8 \text{ m/s}$

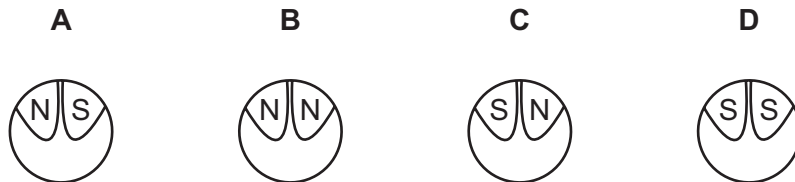
30 Magnets M_1 , M_2 and M_3 are joined by iron balls W, X and Y as shown.



Magnetic poles are induced in the iron balls W and X as shown.

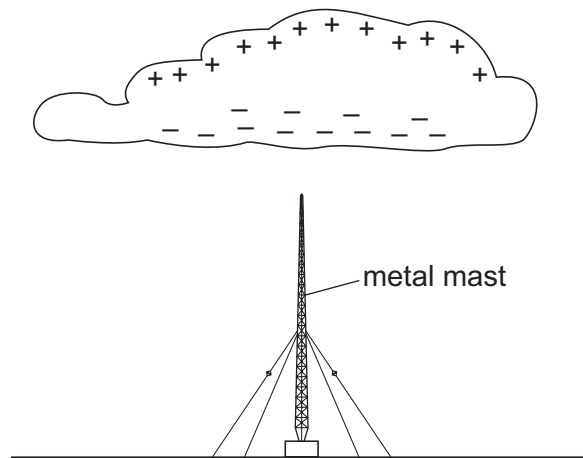


Which diagram shows the poles induced in ball Y?



- 31** The base of a cloud is negatively charged.

The cloud is over a metal mast as shown.



Which row describes the charge induced at the top of the mast and how it is produced?

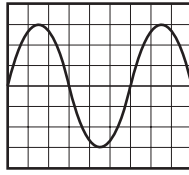
	charge at the top of the mast	charge at the top of mast is produced by
A	negative	electrons moving to the top of the mast
B	negative	protons moving to the bottom of the mast
C	positive	electrons moving to the bottom of the mast
D	positive	protons moving to the top of the mast

- 32** An aircraft flying through the air becomes electrically charged.

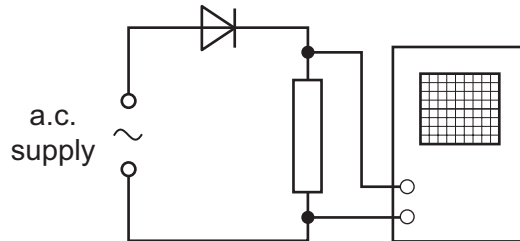
What causes this to happen?

- A** background radiation in the aircraft
- B** friction of the aircraft with the air
- C** heating of the aircraft surface
- D** radio waves emitted by the aircraft

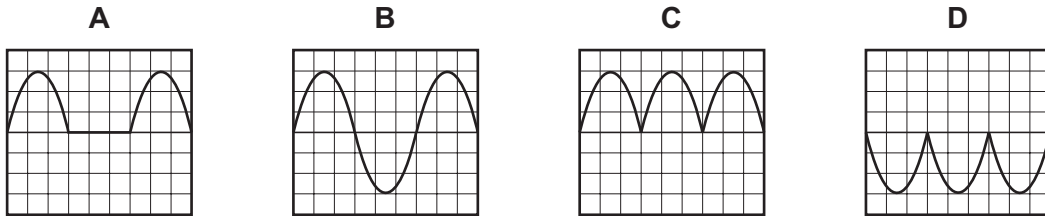
- 33 An a.c. supply connected to an oscilloscope produces the trace shown.



A diode is now connected in series to the circuit as shown.



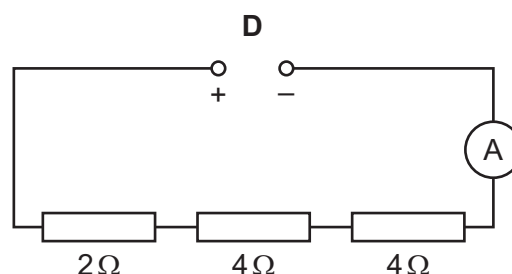
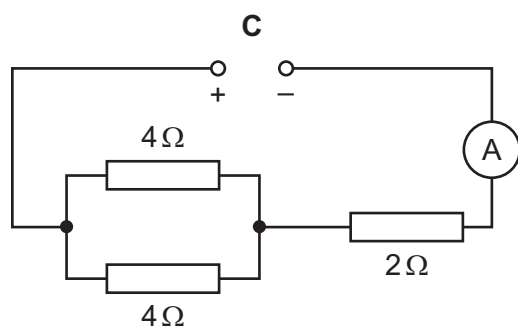
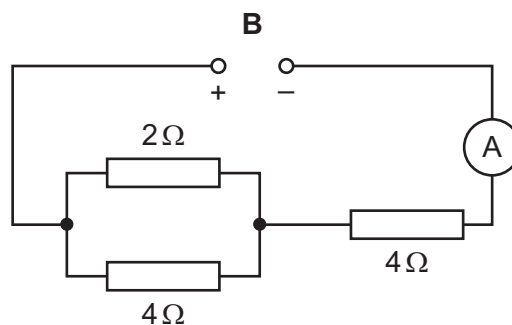
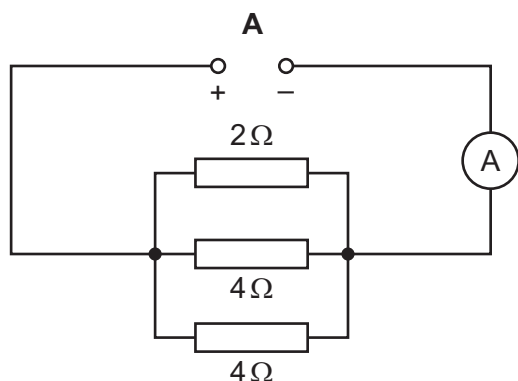
What is the new trace on the screen?



- 34 Resistors of different resistances are connected to a power supply in either a parallel circuit or a series circuit. Which statement is correct?
- A The current is the same in all resistors connected in parallel.
 - B The current is the same in all resistors connected in series.
 - C The voltage across each resistor is different for all resistors connected in parallel.
 - D The voltage across each resistor is the same for all resistors connected in series.

35 An ammeter is connected to three resistors and a power supply.

Which arrangement of resistors gives the greatest ammeter reading?



36 A lamp is connected to a mains plug.

Where are the switch and the fuse connected so that the lamp is safe to use?

	switch	fuse
A	live wire	live wire
B	live wire	neutral wire
C	neutral wire	live wire
D	neutral wire	neutral wire

37 The input voltage to a transformer is 24 V a.c. and the output voltage is 6.0 V.

The input coil has 720 turns.

How many turns are on the output coil?

- A** 5 **B** 180 **C** 2900 **D** 100 000

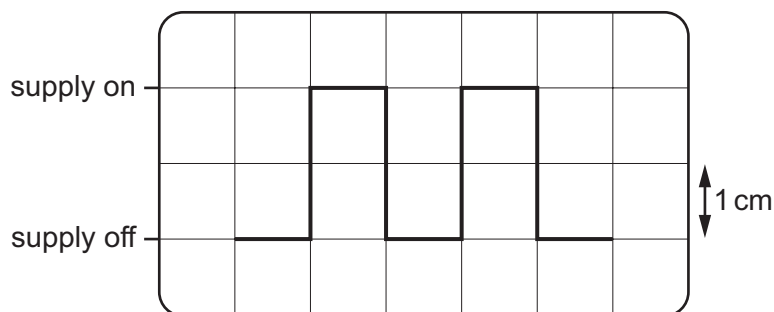
38 A student suggests four conditions for the production of a continuous flow of electrons from a metal filament.

- 1 The filament is hot.
- 2 A high-pressure gas is present.
- 3 A high positive electrical potential is present.
- 4 A strong magnetic field is present.

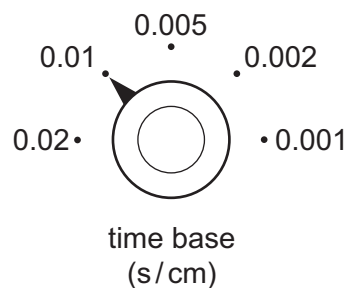
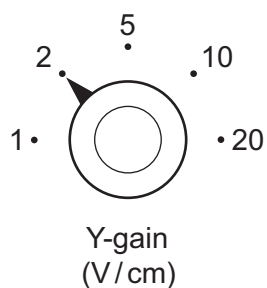
Which of the four conditions are needed?

- A** 1 and 4 **B** 2 and 3 **C** 1 and 3 **D** 2 and 4

39 The diagram shows the trace on an oscilloscope screen when connected to a d.c. supply which is being switched on and off.



The settings of the Y-gain control and the time base control are shown.



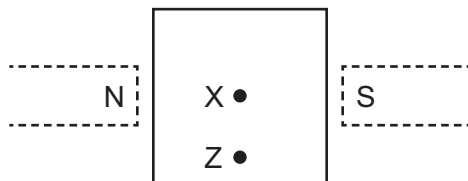
The voltage of the supply is v .

The supply is switched on f times per second.

What are the values of v and of f ?

	v/V	f
A	2	50
B	2	100
C	4	50
D	4	100

- 40 A narrow beam of radiation from a radioactive source travels out of the page through the point X.



The poles of a magnet are placed on either side of the beam. Radiation is now detected both at X and at Z.

Which types of radiation are present in the beam?

- A alpha and beta radiation
- B alpha and gamma radiation
- C alpha, beta and gamma radiation
- D beta and gamma radiation

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PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2019

MARK SCHEME

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Question	Answer	Marks
1	B	1
2	C	1
3	B	1
4	D	1
5	C	1
6	D	1
7	B	1
8	C	1
9	D	1
10	D	1
11	D	1
12	A	1
13	C	1
14	D	1
15	D	1
16	C	1
17	B	1
18	C	1
19	A	1
20	A	1
21	C	1
22	A	1
23	D	1
24	D	1
25	A	1
26	C	1
27	C	1
28	D	1

Question	Answer	Marks
29	B	1
30	B	1
31	C	1
32	B	1
33	B	1
34	A	1
35	B	1
36	B	1
37	B	1
38	A	1
39	C	1
40	B	1



Cambridge Assessment International Education
Cambridge Ordinary Level

PHYSICS

5054/11

Paper 1 Multiple Choice

October/November 2019

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Question	Answer	Mark
1	C	1
2	C	1
3	D	1
4	B	1
5	D	1
6	B	1
7	B	1
8	D	1
9	C	1
10	D	1
11	C	1
12	D	1
13	D	1
14	A	1
15	A	1
16	C	1
17	A	1
18	D	1
19	C	1
20	B	1
21	D	1
22	B	1
23	D	1
24	B	1
25	B	1
26	D	1
27	A	1
28	C	1

Question	Answer	Mark
29	A	1
30	B	1
31	C	1
32	B	1
33	A	1
34	B	1
35	A	1
36	A	1
37	B	1
38	C	1
39	C	1
40	D	1



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/12

October/November 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

This document consists of **15** printed pages and **1** blank page.



- 1 An apple falls from a tree.

Which row describes the forces acting on the apple and the motion of the apple as it falls?

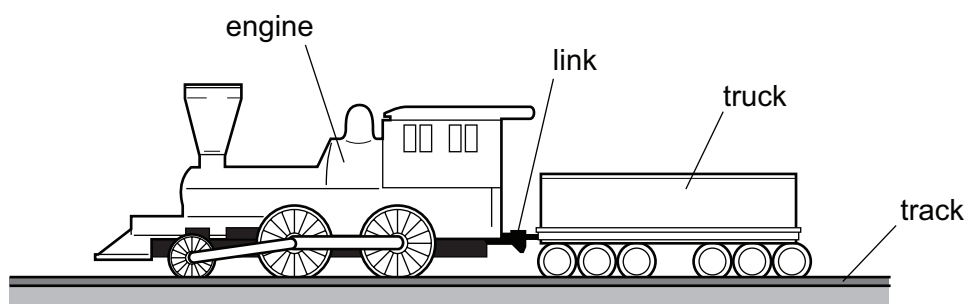
	forces acting on the apple	motion of the apple
A	balanced	acceleration
B	balanced	constant speed
C	unbalanced	acceleration
D	unbalanced	constant speed

- 2 The table shows how the speeds of four bodies, **A**, **B**, **C** and **D**, change with time.

Which body has an acceleration that is **not** constant?

time/s	<u>speed of A</u> m/s	<u>speed of B</u> m/s	<u>speed of C</u> m/s	<u>speed of D</u> m/s
0	0	0	0	5.5
1	1.0	2.0	3.0	6.5
2	3.0	4.0	6.0	7.5
3	6.0	6.0	9.0	8.5

- 3 An engine pulls a truck at constant speed on a level track.

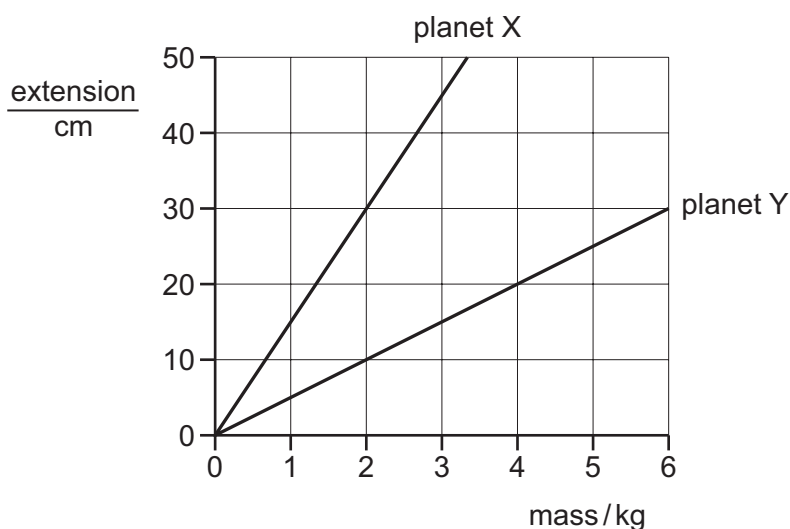


The link between the engine and the truck breaks. The driving force on the engine remains constant.

What effect does this have on the engine and on the truck?

	engine	truck
A	speed stays constant	slows down
B	speeds up	slows down
C	speed stays constant	stops immediately
D	speeds up	stops immediately

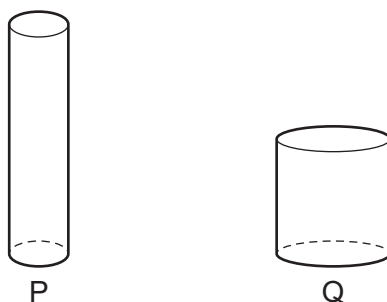
- 4 On which car is there a resultant force?
- A a car moving along a straight horizontal road at constant speed
 - B a car moving around a bend at constant speed
 - C a car moving uphill at constant velocity
 - D a car that is stationary
- 5 Which piece of apparatus may be used to compare the masses of two objects?
- A balance
 - B manometer
 - C measuring cylinder
 - D micrometer
- 6 The graph shows how the extension of a spring changes with the masses suspended from it when the spring is on planet X and when the spring is on planet Y.



Which conclusion can be drawn from these graphs?

- A It is not possible to compare the gravitational field strengths on planets X and Y.
- B The gravitational field strength on planet X is equal to the gravitational field strength on planet Y.
- C The gravitational field strength on planet X is one third of the gravitational field strength on planet Y.
- D The gravitational field strength on planet X is three times the gravitational field strength on planet Y.

- 7 Two cylinders P and Q are made of copper.



The height of P is twice the height of Q. The diameter of P is half the diameter of Q.

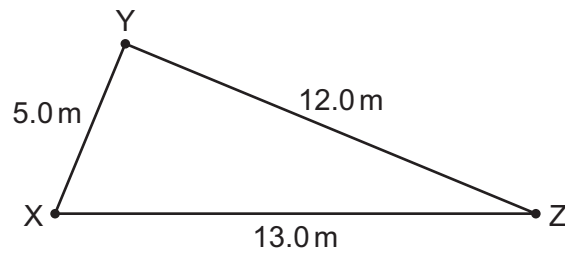
Which statement is correct?

- A** The density of cylinder P is four times that of cylinder Q.
- B** The density of cylinder P is twice that of cylinder Q.
- C** The density of cylinder P is equal to that of cylinder Q.
- D** The density of cylinder P is half that of cylinder Q.
- 8 A tennis ball is compressed. Work is done and energy is transferred to the molecules of air inside the ball, and also to the rubber from which the ball is made.

Which row is correct?

	kinetic energy of air molecules inside the ball	temperature of ball
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 9 Paths are laid as shown between points X, Y and Z.

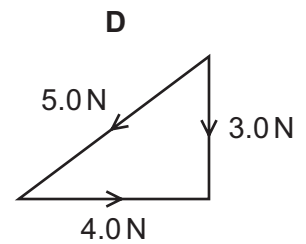
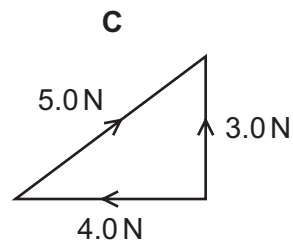
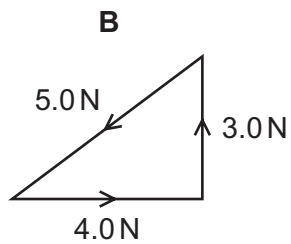
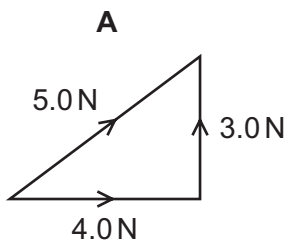


A person walks along the paths from X to Y to Z and then back to X.

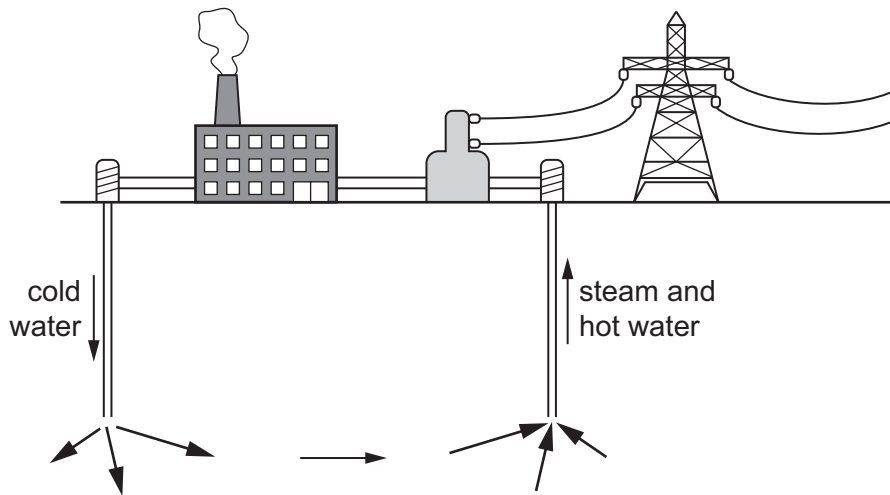
What is the value of the total displacement and of the total distance travelled?

	total displacement /m	total distance travelled /m
A	0	0
B	0	30
C	30	0
D	30	30

- 10 Which diagram shows the addition of the 4.0 N and the 3.0 N forces?



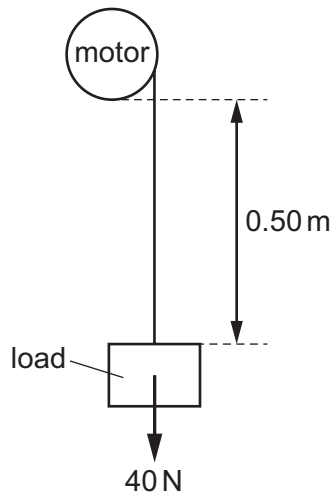
11 The diagram represents a geothermal power station.



Which useful energy transformation is taking place?

- A electrical energy $\rightarrow$ potential energy
- B electrical energy $\rightarrow$ thermal energy
- C potential energy $\rightarrow$ electrical energy
- D thermal energy $\rightarrow$ electrical energy

12 A motor is used to lift a load 0.50 m vertically, as shown.



The load weighs 40 N. The power of the motor is 20 W and the system is 25% efficient.

How long does it take to raise the load?

- A 0.040 s
- B 0.25 s
- C 4.0 s
- D 40 s

- 13 Different liquids are poured into four different containers.

In which container does the liquid produce the greatest pressure at the bottom of the container?

	area of base /cm ²	density of liquid g/cm ³	depth of liquid /cm
A	10	1.3	50
B	20	0.80	80
C	40	1.0	60
D	50	0.92	75

- 14 A rectangular block of metal has weight 6.0 N and measures 3.0 cm × 4.0 cm × 5.0 cm.

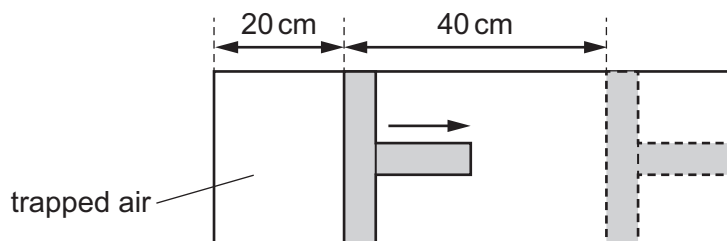
What is the smallest pressure that it can exert when resting on a horizontal surface?

- A** 0.10 N/cm² **B** 0.30 N/cm² **C** 0.40 N/cm² **D** 0.50 N/cm²

- 15 Air is trapped in a cylinder by a piston. The pressure of the air is p and the length of the air column is 20 cm.

The piston is moved outwards until the length of the air column has increased by 40 cm.

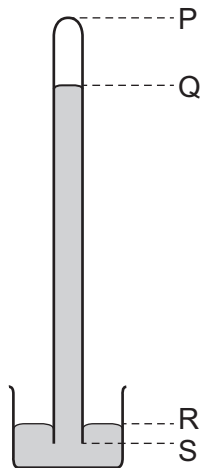
The temperature of the air remains constant.



What is the new air pressure?

- A** $\frac{p}{2}$ **B** $\frac{p}{3}$ **C** $2p$ **D** $3p$

- 16** A long tube, full of mercury, is inverted in a small dish of mercury.



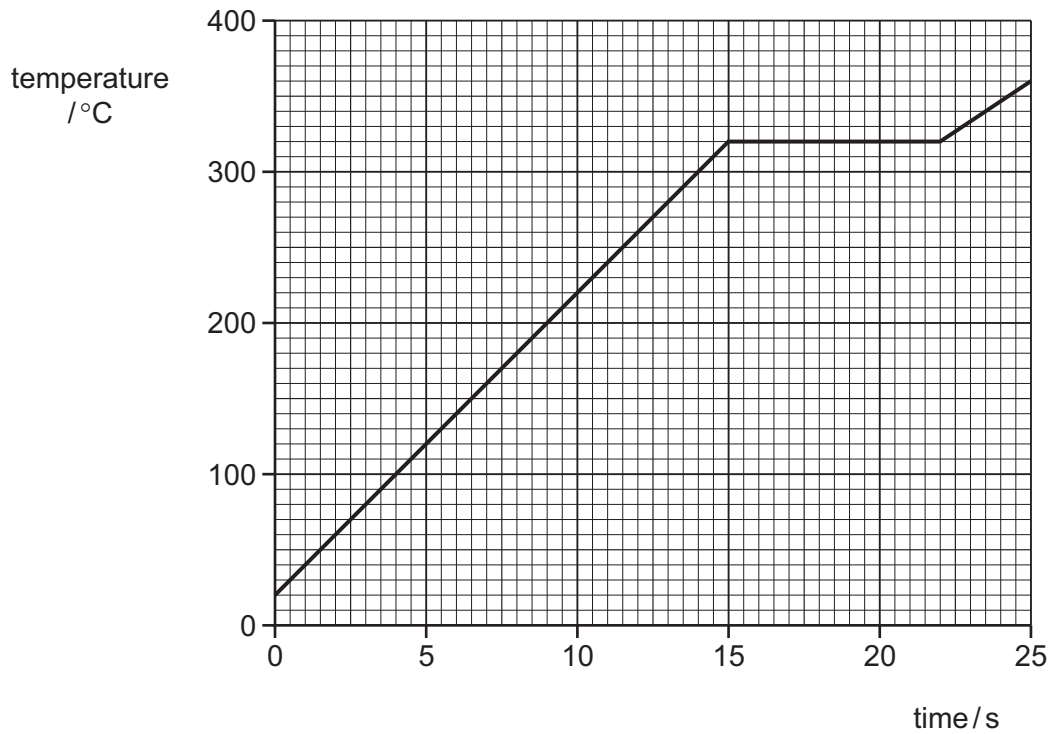
The mercury level in the tube falls, leaving a vacuum at the top.

When the atmospheric pressure decreases, which length decreases?

- A** PQ **B** PS **C** QR **D** RS
- 17** A cold bottle containing a drink is placed on a table on a warm day. Drops of water form on the outside of the bottle.
- Which process causes the drops to form?
- A** condensation
B conduction
C convection
D evaporation
- 18** Some gas is trapped in a closed container. The gas is cooled and the volume of the container is kept constant.
- What happens to the gas molecules?
- A** They collide with the walls more often.
B They contract.
C They get closer together.
D They move more slowly.

- 19 A 100 g piece of solid lead at room temperature is heated. After 22 s, it has all become liquid.

The graph shows how its temperature varies with time.



The power of the heater is 320 W.

Which expression gives the specific latent heat, in J/kg, of the lead?

- A** $\frac{7.0 \times 320}{0.10 \times 300}$
 B $\frac{22 \times 320}{0.10 \times 300}$
 C $\frac{7.0 \times 320}{0.10}$
 D $\frac{22 \times 320}{0.10}$

- 20 What increases when a liquid becomes a gas at its boiling point?

- A** the average kinetic energy of the molecules
B the molecular size
C the molecular spacing
D the total number of molecules

- 21** Two liquid-in-glass thermometers are almost identical. They contain the same quantity of the same liquid and the diameters of their capillary tubes are the same.

The only difference is that one thermometer is longer than the other.

Which row shows how the ranges and the sensitivities of the thermometers compare?

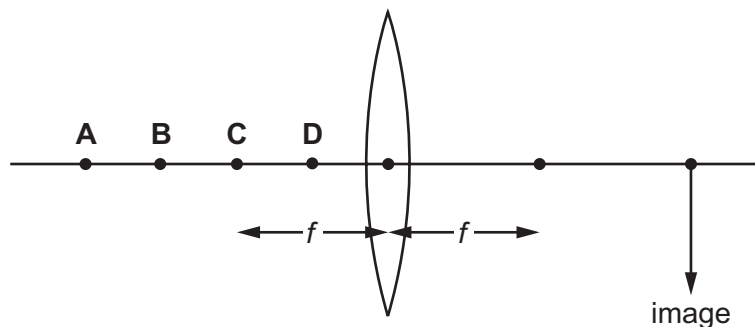
	ranges	sensitivities
A	different	different
B	different	same
C	same	different
D	same	same

- 22** Which statement about a water wave is correct?

- A** The amplitude is the vertical distance between a trough and a peak.
- B** The frequency is the number of troughs passing a point in one second added to the number of peaks passing a point in one second.
- C** The speed is the horizontal distance travelled per second by a peak.
- D** The wavelength is the horizontal distance between a trough and a peak.

- 23** The diagram shows a thin converging lens of focal length f .

Where must an object be placed to produce a real image in the position shown?



- 24** White light is dispersed by a prism. Compared with blue light, the **red** light is

- A** slowed down less and refracted less.
- B** slowed down less and refracted more.
- C** slowed down more and refracted less.
- D** slowed down more and refracted more.

25 A student reads the following in her physics book.

‘The incident angle is greater than 42° which is the critical angle for glass in air.’

What is the student reading about?

- A focal length of a glass lens
- B reflection in a plane mirror
- C refraction as light enters glass
- D total internal reflection

26 Which statement about radio waves is correct?

- A Radio waves are sound waves.
- B Radio waves are used to kill cancerous cells.
- C Radio waves are used in television communications.
- D Radio waves have frequencies higher than those of visible light.

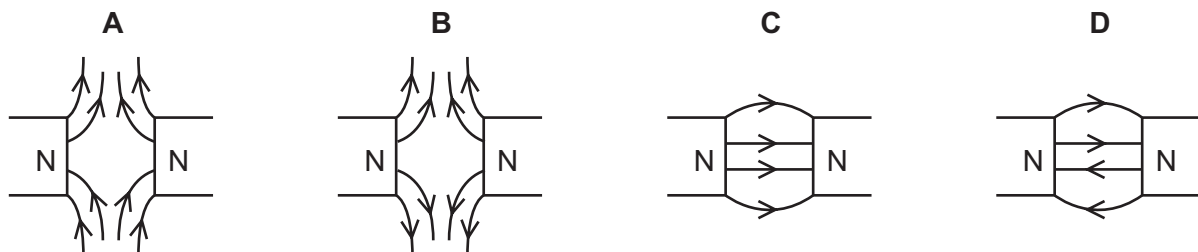
27 Ultrasound is used to map the ocean floor.

During one survey, the depth of water is 1200 m. An ultrasound pulse is sent from the surface and when it returns to the ship, another pulse is sent immediately. In any period of 8.0 s, five pulses are sent down from the surface.

What is the speed of the ultrasound in water?

- A 150 m/s
- B 300 m/s
- C 750 m/s
- D 1500 m/s

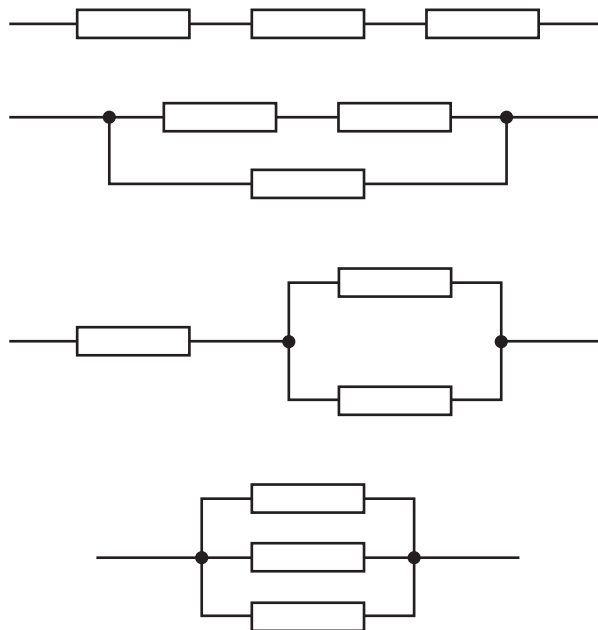
28 Which diagram shows the magnetic field pattern in the region between the N-poles of the two bar magnets?



- 29** Students are asked to describe an experiment to measure the resistance of a metallic conductor.
Which description is correct?

- A** Connect an ammeter in parallel and a voltmeter in series with the conductor then use $R = \frac{I}{V}$.
B Connect an ammeter in parallel and a voltmeter in series with the conductor then use $R = \frac{V}{I}$.
C Connect a voltmeter in parallel and an ammeter in series with the conductor then use $R = \frac{I}{V}$.
D Connect a voltmeter in parallel and an ammeter in series with the conductor then use $R = \frac{V}{I}$.

- 30** The diagram shows three $6.0\ \Omega$ resistors connected to give various total resistances.



What is the difference between the smallest total resistance and the largest total resistance?

- A** $7.0\ \Omega$ **B** $9.0\ \Omega$ **C** $12\ \Omega$ **D** $16\ \Omega$
- 31** A wire of length $0.50\ \text{m}$ and cross-sectional area $1.0 \times 10^{-6}\ \text{m}^2$ has a resistance of $0.75\ \Omega$.

Another wire of the same material has a length of $2.0\ \text{m}$ and a cross-sectional area of $0.50 \times 10^{-6}\ \text{m}^2$.

What is the resistance of the longer wire?

- A** $0.094\ \Omega$ **B** $0.38\ \Omega$ **C** $1.5\ \Omega$ **D** $6.0\ \Omega$

- 32** A defibrillator is a device that is used to give an electric shock to a patient's heart.

It supplies an electric shock with energy 240 J at an average voltage of 2000 V for 10 ms.

What is the average current it supplies?

- A** 0.012 A **B** 1.2 A **C** 12 A **D** 120 A

- 33** In a lighting circuit, the switch is placed in the live wire.

Why is this?

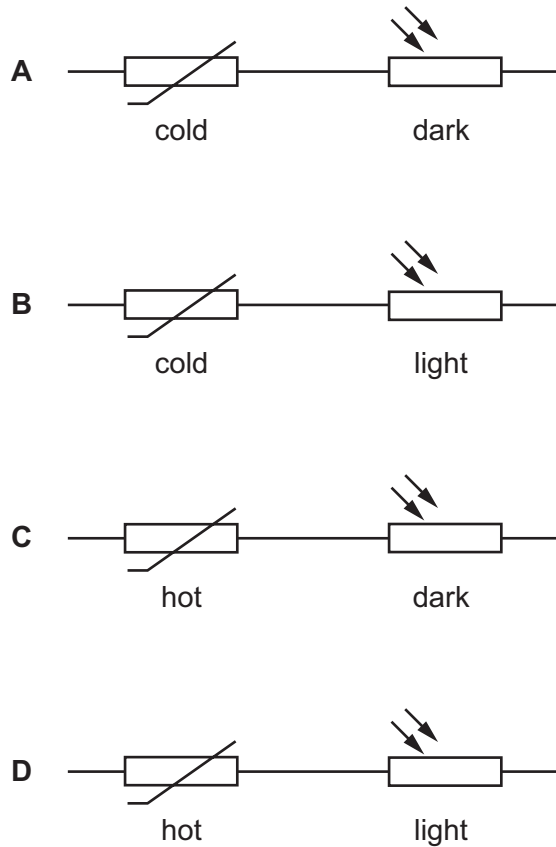
- A** A lamp in the circuit can be isolated from the supply.
B The fuse is in the neutral wire.
C The switch does not work in the neutral wire.
D Too much current flows in the earth wire.

- 34** Which cable transmits electrical energy at the highest voltage?

- A** a cable from a bicycle dynamo to a lamp
B a household lighting cable
C a power cable between towns
D an electric cooker cable

35 A thermistor and a light-dependent resistor are connected in series.

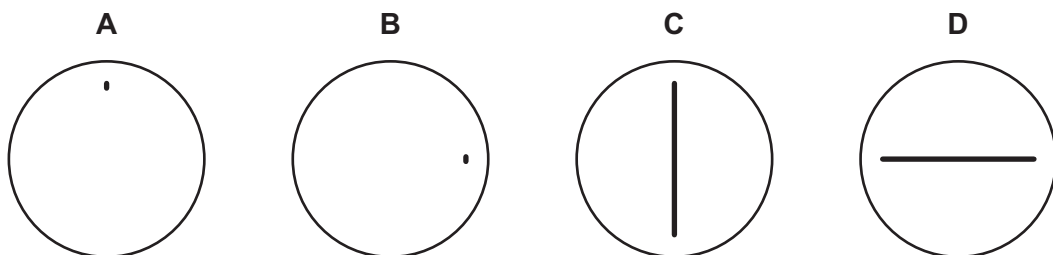
Which conditions give the least resistance?



36 An oscilloscope is connected to a d.c. power supply.

The time-base on the oscilloscope is switched off.

What is seen on the screen of the oscilloscope?



37 What does the alpha-radiation given off by radioactive nuclei consist of?

- A** fast-moving protons
- B** helium nuclei
- C** microwaves
- D** radio waves

- 38** Which statement about the reactor in a nuclear power station is correct?
- A** In the reactor, the main reaction occurs when protons hit uranium nuclei.
 - B** The process taking place in the reactor is called nuclear fusion.
 - C** The reactor produces energy to boil water and to produce steam.
 - D** Carbon dioxide is the major waste product from the reactor.
- 39** The results of the alpha-particle scattering experiment gave evidence for which of the following?
- A** nuclear fusion
 - B** radioactive decay
 - C** the existence of isotopes
 - D** the nuclear atom
- 40** Which statement about two different isotopes of the same element is correct?
- A** Both isotopes have the same number of electrons in a nucleus.
 - B** Both isotopes have the same number of protons in a nucleus.
 - C** Both isotopes have the same number of neutrons in a nucleus.
 - D** Both isotopes have the same number of nucleons in a nucleus.

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/11

October/November 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

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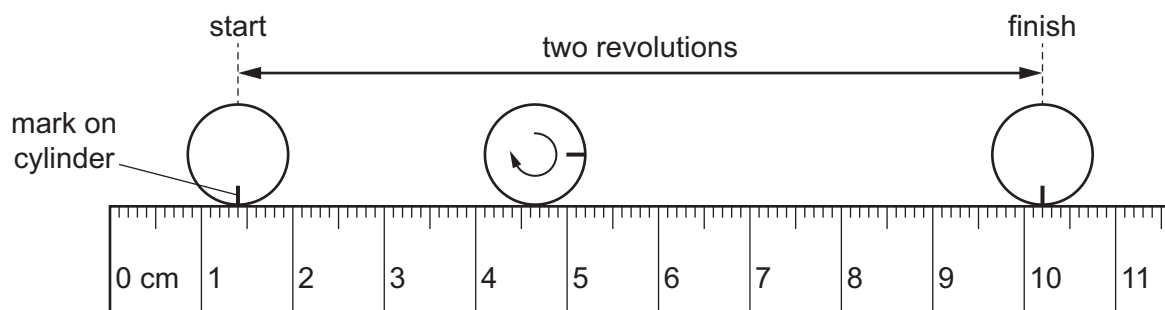
Electronic calculators may be used.

This document consists of **14** printed pages and **2** blank pages.

1 What is a possible mass for a normal adult person?

- A** 7.5 kg **B** 75 kg **C** 750 kg **D** 7500 kg

2 A small cylinder is rolled along a ruler and completes two revolutions.



The circumference is the distance around the outside of a circle.

What is the circumference of the cylinder?

- A** 4.4 cm **B** 5.2 cm **C** 8.8 cm **D** 10.2 cm

3 A skydiver is falling at terminal velocity.

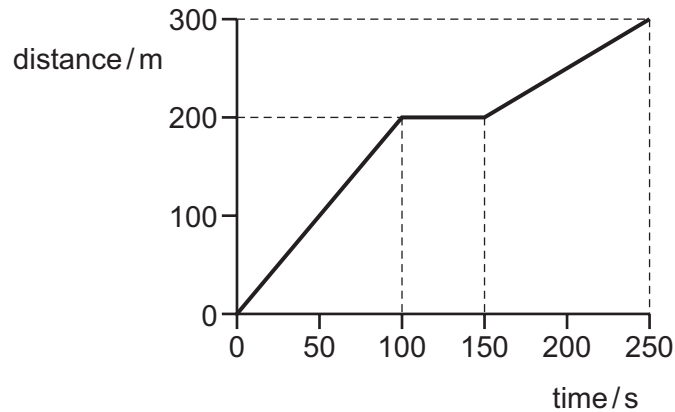


Which row describes the acceleration of the skydiver and the velocity of the skydiver?

	acceleration of the skydiver	velocity of the skydiver
A	downwards	constant
B	downwards	zero
C	zero	constant
D	zero	zero

- 4 A cyclist takes a ride lasting 250 s.

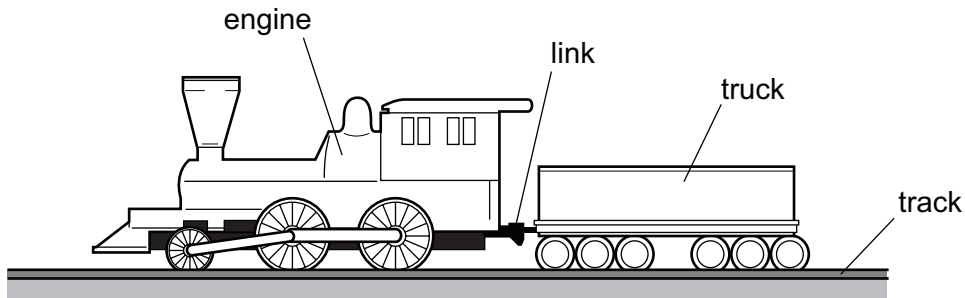
The graph shows how the distance from the starting position varies with time.



What is his average speed for the whole journey?

- A** 1.0 m/s **B** 1.2 m/s **C** 1.5 m/s **D** 2.0 m/s

- 5 An engine pulls a truck at constant speed on a level track.

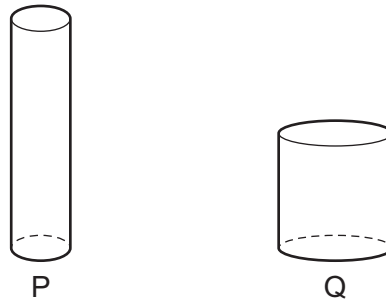


The link between the engine and the truck breaks. The driving force on the engine remains constant.

What effect does this have on the engine and on the truck?

	engine	truck
A	speed stays constant	slows down
B	speeds up	slows down
C	speed stays constant	stops immediately
D	speeds up	stops immediately

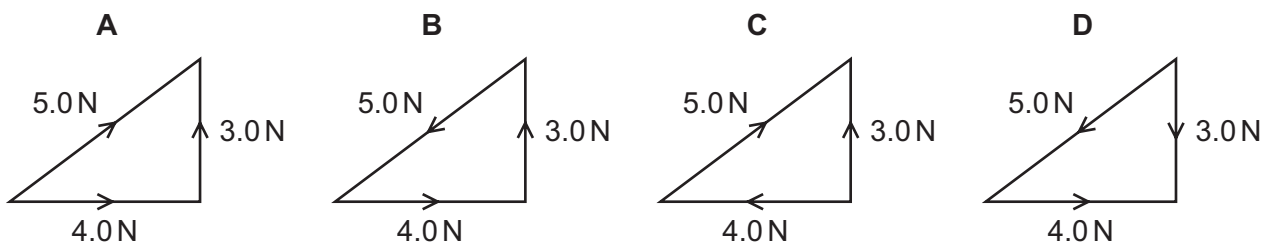
- 6 What happens to an object when it is moved to a location where the gravitational field strength is slightly greater?
- A Its density decreases.
 - B Its mass decreases.
 - C Its weight increases.
 - D Its volume increases.
- 7 Two cylinders P and Q are made of copper.



The height of P is twice the height of Q. The diameter of P is half the diameter of Q.

Which statement is correct?

- A The density of cylinder P is four times that of cylinder Q.
 - B The density of cylinder P is twice that of cylinder Q.
 - C The density of cylinder P is equal to that of cylinder Q.
 - D The density of cylinder P is half that of cylinder Q.
- 8 Which diagram shows the addition of the 4.0 N and the 3.0 N forces?



- 9 An electric motor, connected to the mains, is used to lift bricks to the top of a building.

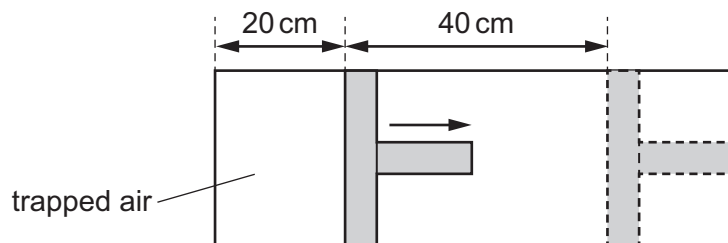
There is a fuse in the plug.

How can the efficiency of the motor be increased?

- A** Increase the friction in the motor.
B Reduce the energy losses in the motor.
C Use a fuse with a higher current rating.
D Use a fuse with a lower current rating.
- 10 Air is trapped in a cylinder by a piston. The pressure of the air is p and the length of the air column is 20 cm.

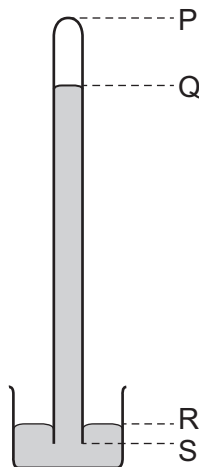
The piston is moved outwards until the length of the air column has increased by 40 cm.

The temperature of the air remains constant.



What is the new air pressure?

- A** $\frac{p}{2}$ **B** $\frac{p}{3}$ **C** $2p$ **D** $3p$
- 11 A long tube, full of mercury, is inverted in a small dish of mercury.



The mercury level in the tube falls, leaving a vacuum at the top.

When the atmospheric pressure decreases, which length decreases?

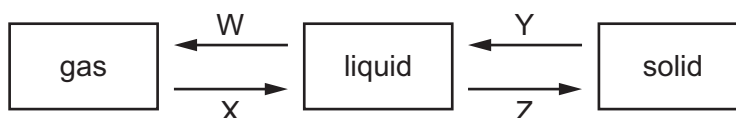
- A** PQ **B** PS **C** QR **D** RS

- 12** Some gas is trapped in a closed container. The gas is cooled and the volume of the container is kept constant.

What happens to the gas molecules?

- A** They collide with the walls more often.
- B** They contract.
- C** They get closer together.
- D** They move more slowly.

- 13** The diagram shows four changes of state W, X, Y and Z.



Which row gives the names of the four changes of state?

	W	X	Y	Z
A	condensation	boiling	freezing	melting
B	condensation	boiling	melting	freezing
C	boiling	condensation	freezing	melting
D	boiling	condensation	melting	freezing

- 14** What increases when a liquid becomes a gas at its boiling point?

- A** the average kinetic energy of the molecules
- B** the molecular size
- C** the molecular spacing
- D** the total number of molecules

- 15** A metal lid fits tightly on a glass jar.

Which process makes it easier to remove the lid from the jar?

- A** cool the lid only
- B** put the jar and lid in a refrigerator
- C** warm the jar only
- D** warm the lid only

- 16** The specific heat capacity of lead is $130 \text{ J/(kg } ^\circ\text{C)}$. Lead of mass 0.50 kg is heated from 10°C to 35°C .

Which calculation gives the amount of thermal energy, in J, absorbed by the lead?

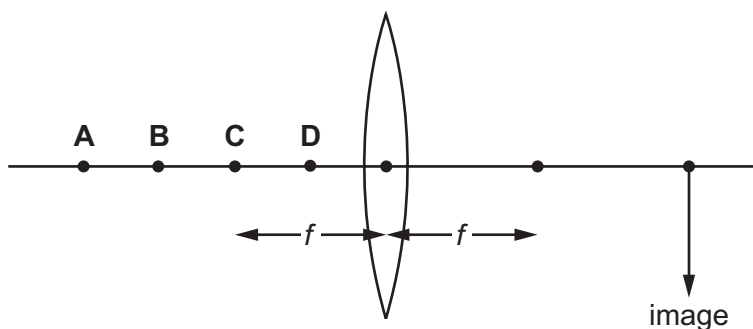
- A** $0.50 \times 130 \times 25$
- B** $0.50 \times 130 \times 35$
- C** $\frac{0.50 \times 130}{25}$
- D** $\frac{0.50 \times 130}{35}$
- 17** A cold solid is placed on top of a hot solid. Thermal energy flows from the hot solid to the cold one.

What is the explanation for this?

- A** A hot solid expands, so its molecules will move further apart.
- B** Energy is passed from one molecule to the next.
- C** Heat always rises.
- D** Molecules are free to move randomly through the solids.
- 18** Which term is the number of crests of a wave passing a point per second?
- A** amplitude
- B** frequency
- C** speed
- D** wavelength

- 19** The diagram shows a thin converging lens of focal length f .

Where must an object be placed to produce a real image in the position shown?



20 White light is dispersed by a prism. Compared with blue light, the **red** light is

- A slowed down less and refracted less.
- B slowed down less and refracted more.
- C slowed down more and refracted less.
- D slowed down more and refracted more.

21 What is the approximate value for

$$\frac{\text{speed of light in air}}{\text{speed of sound in air}} ?$$

- A 10^2 B 10^4 C 10^6 D 10^8

22 What are the lowest frequency and the highest frequency of the sound that a human being with normal hearing can hear?

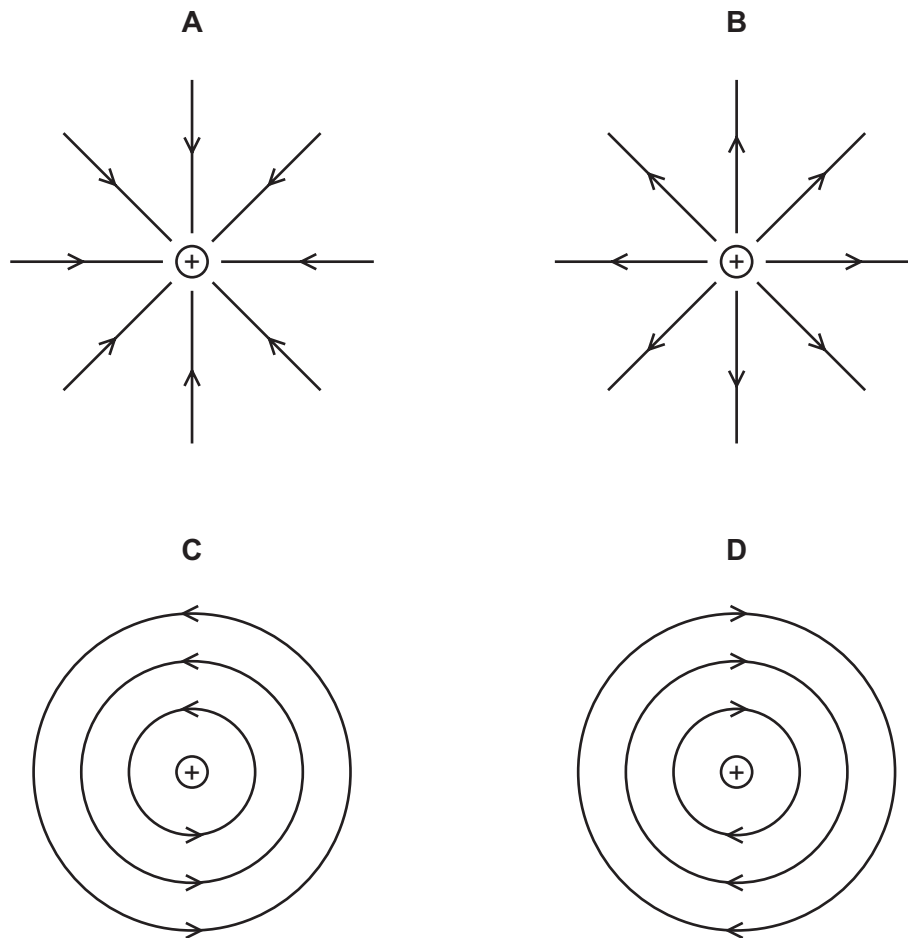
	lowest frequency	highest frequency
A	2.0 Hz	2.0 kHz
B	2.0 Hz	20 000 Hz
C	20 Hz	20 kHz
D	20 kHz	20 000 kHz

23 Ultrasound has a number of uses in engineering and medicine.

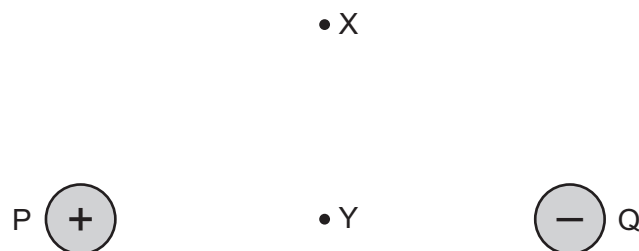
What is ultrasound **not** used for?

- A cleaning
- B pre-natal scanning
- C quality control
- D sterilising medical equipment

24 Which diagram shows the electric field around an isolated point positive charge?



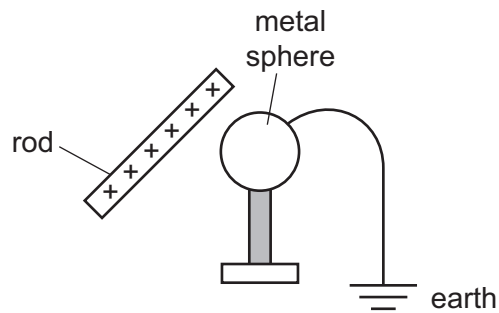
25 An electric field is formed by two isolated, equal and opposite charges P and Q.



How does the field at X compare with the field at Y?

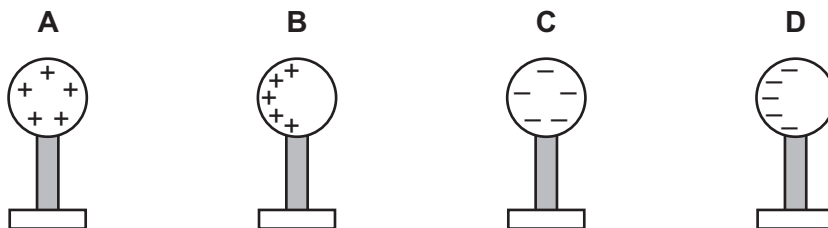
- A** It is stronger at X than at Y and is in the opposite direction.
- B** It is stronger at X than at Y and is in the same direction.
- C** It is weaker at X than at Y and is in the opposite direction.
- D** It is weaker at X than at Y and is in the same direction.

- 26** A positively charged rod is held close to an insulated metal sphere. The sphere is earthed as shown.



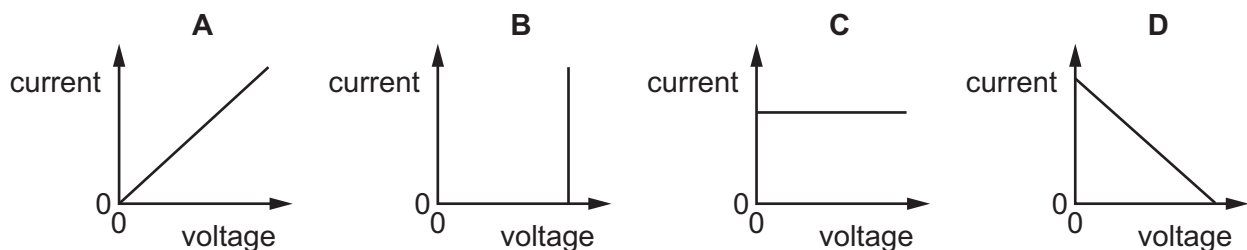
The earth connection is removed and then the rod is removed.

Which diagram shows the charges on the sphere after the rod is removed?



- 27** The voltage across a resistor of fixed resistance changes.

Which graph shows how the current in the resistor changes with the voltage?



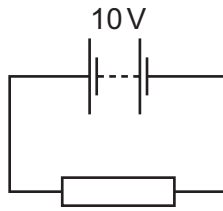
- 28** A wire of length 0.50 m and cross-sectional area $1.0 \times 10^{-6} \text{ m}^2$ has a resistance of 0.75Ω .

Another wire of the same material has a length of 2.0 m and a cross-sectional area of $0.50 \times 10^{-6} \text{ m}^2$.

What is the resistance of the longer wire?

- A** 0.094Ω **B** 0.38Ω **C** 1.5Ω **D** 6.0Ω

- 29 A battery of electromotive force (e.m.f.) 10 V is connected to a resistor.

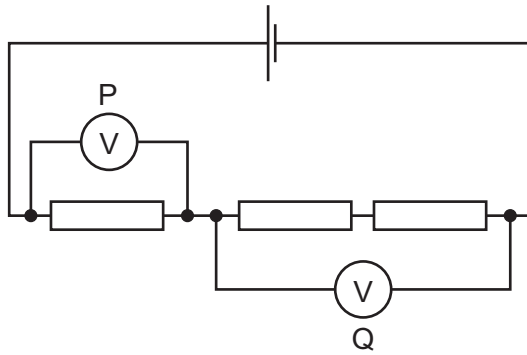


A charge of 2.0 coulombs passes through the resistor.

How much work is done as the charge passes through the resistor?

- A** 0.20 J **B** 2.0 J **C** 5.0 J **D** 20 J

- 30 Three identical resistors are joined in series to a cell.



Voltmeter Q reads 8.0 V.

What is the reading on voltmeter P and what is the e.m.f. of the cell?

	reading on P / V	e.m.f. of cell / V
A	4.0	8.0
B	4.0	12
C	8.0	8.0
D	8.0	12

- 31 The power of an electrical heater is 3.0 kW.

The heater is switched on for a time of 2 hours and 30 minutes.

The cost of using the heater for this time is 120 c.

What is the cost of 1.0 kWh of electrical energy?

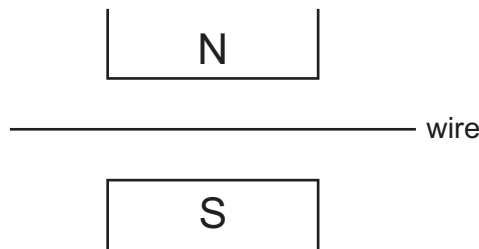
- A** 16 c **B** 17 c **C** 40 c **D** 48 c

- 32** In a lighting circuit, the switch is placed in the live wire.

Why is this?

- A** A lamp in the circuit can be isolated from the supply.
- B** The fuse is in the neutral wire.
- C** The switch does not work in the neutral wire.
- D** Too much current flows in the earth wire.

- 33** A wire is placed in the magnetic field between the poles of a magnet, as shown.



When there is a current in the wire, it experiences a force.

Which change increases the size of this force?

- A** reverse both the direction of the current and the poles of the magnet
 - B** reverse just the direction of the current
 - C** use a stronger magnet
 - D** use a thicker wire with the same current
- 34** A 12V lamp glows at normal brightness when connected to the secondary coil of a mains transformer. The mains voltage is 240 V.

The transformer is 100% efficient and the primary coil has 200 turns.

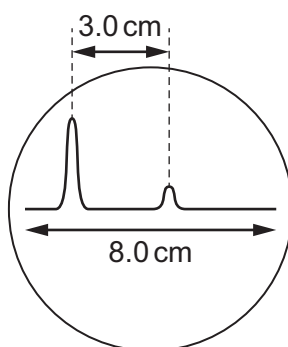
How many turns are there on the secondary coil?

- A** 10
- B** 14
- C** 4000
- D** 580 000

- 35** Which row shows how the electrical energy produced by a power station is transmitted to distant towns?

	current	voltage
A	alternating	low
B	alternating	very high
C	direct	low
D	direct	very high

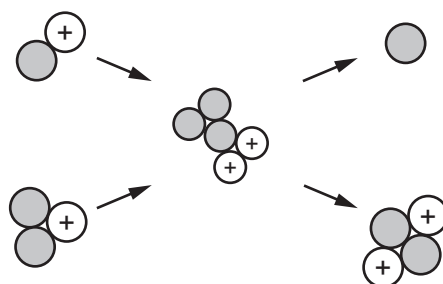
- 36 The diagram shows the trace on an oscilloscope screen. The first peak occurs when a pulse of sound is emitted. The second, smaller peak is the echo received from a wall.



The trace moves across the screen in a time of 24 ms.

How long does it take for the sound to travel to the wall?

- A** 4.5 ms **B** 9.0 ms **C** 12 ms **D** 32 ms
- 37 The diagram shows a nuclear reaction.



The shaded particles are uncharged.

What are the products shown in this reaction?

- A** a helium atom and a neutron
B a helium atom and a proton
C a helium nucleus and a neutron
D a helium nucleus and a proton
- 38 The results of the alpha-particle scattering experiment gave evidence for which of the following?
- A** nuclear fusion
B radioactive decay
C the existence of isotopes
D the nuclear atom

39 What does the alpha-radiation given off by radioactive nuclei consist of?

- A** fast-moving protons
- B** helium nuclei
- C** microwaves
- D** radio waves

40 One nucleus of a uranium isotope contains 238 nucleons of which 146 are neutrons.

Which is the nucleus of another isotope of uranium?

	number of nucleons in the nucleus	number of neutrons in the nucleus
A	235	143
B	237	147
C	238	144
D	241	146

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PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2018

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the October/November 2018 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	A	1
3	B	1
4	B	1
5	A	1
6	D	1
7	C	1
8	D	1
9	B	1
10	A	1
11	D	1
12	C	1
13	D	1
14	B	1
15	B	1
16	C	1
17	A	1
18	D	1
19	C	1
20	C	1
21	B	1
22	C	1
23	A	1
24	A	1
25	D	1
26	C	1
27	D	1
28	B	1

Question	Answer	Marks
29	D	1
30	D	1
31	D	1
32	C	1
33	A	1
34	C	1
35	D	1
36	A	1
37	B	1
38	C	1
39	D	1
40	B	1



PHYSICS

5054/11

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October/November 2018

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Question	Answer	Marks
1	B	1
2	A	1
3	C	1
4	B	1
5	B	1
6	C	1
7	C	1
8	A	1
9	B	1
10	B	1
11	C	1
12	D	1
13	D	1
14	C	1
15	D	1
16	A	1
17	B	1
18	B	1
19	A	1
20	A	1
21	C	1
22	C	1
23	D	1
24	B	1
25	D	1
26	C	1
27	A	1
28	D	1

Question	Answer	Marks
29	D	1
30	B	1
31	A	1
32	A	1
33	C	1
34	A	1
35	B	1
36	A	1
37	C	1
38	D	1
39	B	1
40	A	1



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/12

October/November 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

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Electronic calculators may be used.

This document consists of **16** printed pages.

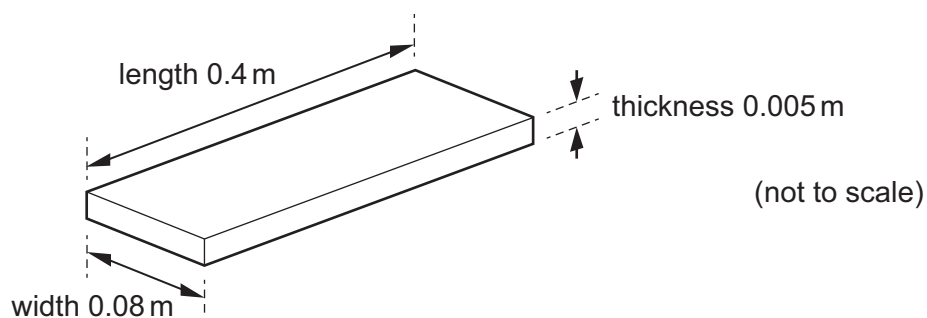


1 Which list contains only scalar quantities?

- A acceleration, displacement, velocity
- B distance, force, speed
- C force, length, time
- D length, mass, speed

2 A manufacturer measures the three dimensions of a wooden floor tile using three different instruments.

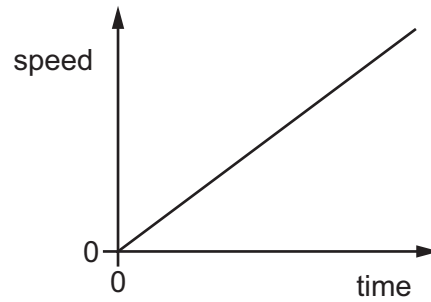
The approximate dimensions of the tile are shown.



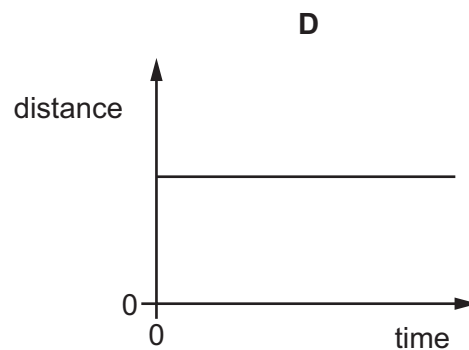
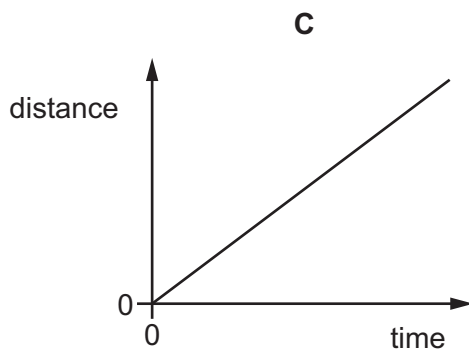
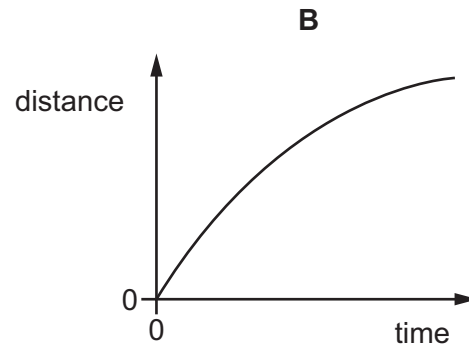
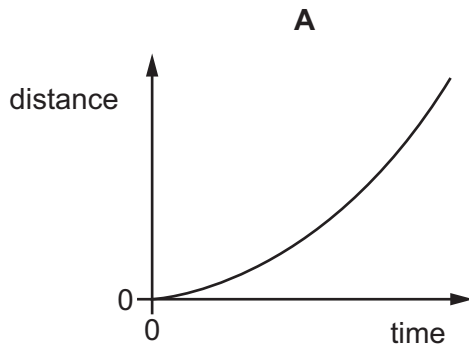
Which instruments are used to measure accurately each of these dimensions?

	length	thickness	width
A	metre rule	micrometer	calipers
B	metre rule	calipers	micrometer
C	micrometer	metre rule	calipers
D	calipers	micrometer	metre rule

- 3 The speed-time graph represents a short journey.



Which distance-time graph represents the same journey?

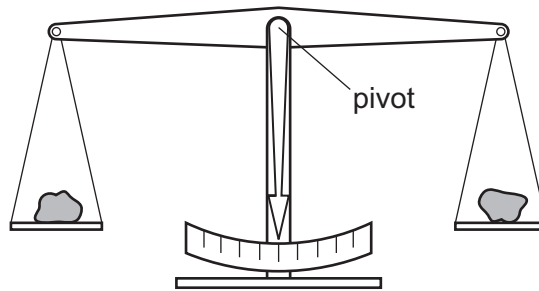


- 4 A free-fall skydiver jumps from a plane. As he falls there is a force acting upwards and a force acting downwards on his body. These produce a resultant force.

Before he reaches terminal velocity, how do the sizes of the forces change?

	downward force	upward force	resultant force
A	decreases	decreases	stays the same
B	increases	stays the same	decreases
C	stays the same	increases	decreases
D	stays the same	increases	increases

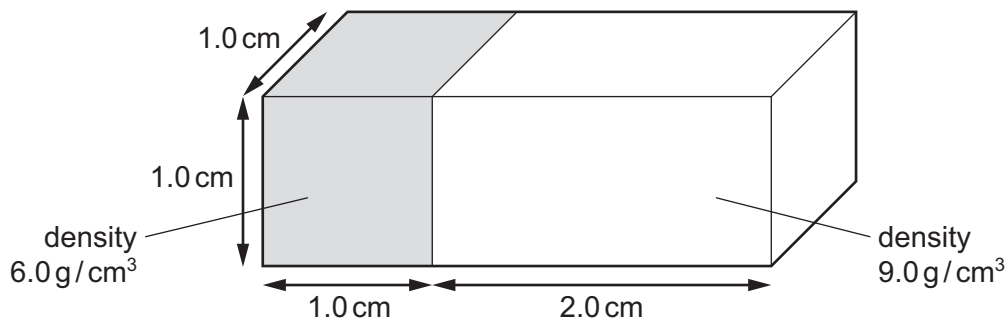
- 5 The diagram shows two objects on a beam balance.



The beam balance is in equilibrium.

Which quantities may be different?

- A** the masses of the two objects
 - B** the moments about the pivot of the two objects
 - C** the volumes of the two objects
 - D** the weights of the two objects
- 6 Two blocks are joined together.

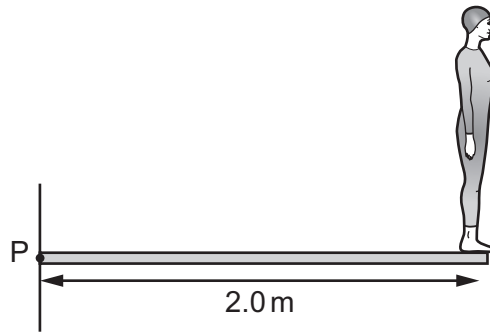


One block has a density of 6.0 g/cm^3 and the other has a density of 9.0 g/cm^3 .

What is the overall density of the two blocks joined together?

- A** 7.0 g/cm^3
- B** 7.5 g/cm^3
- C** 8.0 g/cm^3
- D** 15 g/cm^3

- 7 A diver of weight 500 N stands at the end of a springboard that is 2.0 m long and is fixed at point P.



The springboard has a weight of 500 N. The centre of mass of the springboard is in the centre of the board.

What is the total moment about point P of the diver and the board?

- A** 500 N m **B** 750 N m **C** 1000 N m **D** 1500 N m

- 8 A tennis ball of mass 56 g is travelling at 1500 metres/minute.

Which expression is equal to the kinetic energy, in joules, of the tennis ball?

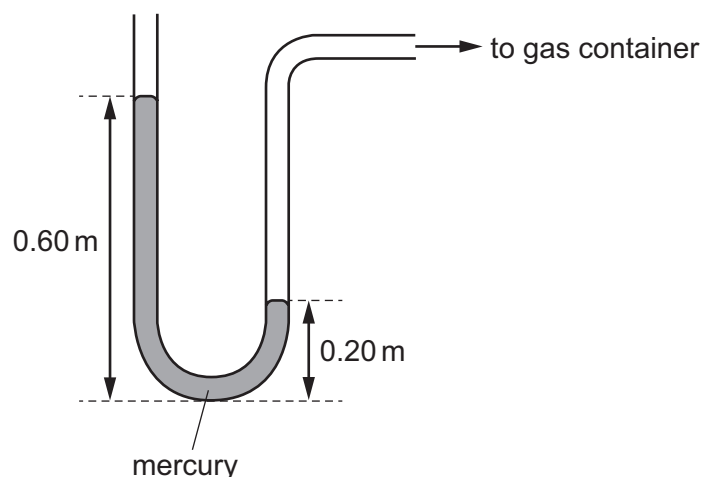
- A** $\frac{1}{2} \times 0.056 \times (25)^2$
B $\frac{1}{2} \times 0.056 \times (1500)^2$
C $\frac{1}{2} \times 56 \times (25)^2$
D $\frac{1}{2} \times 56 \times (1500)^2$

- 9 In a hydroelectric power station, 4.2×10^5 kg of water passes through the turbines every second. The turbines are at a height of 50 m below the surface of the reservoir. The gravitational field strength g is 10 N/kg.

Assuming there are no energy losses, what is the power output of the power station?

- A** 8.4×10^3 W **B** 8.4×10^4 W **C** 2.1×10^7 W **D** 2.1×10^8 W

- 10 The diagram shows a mercury manometer connected to a gas container.



The density of mercury is $14\,000\text{ kg/m}^3$. The gravitational field strength g is 10 N/kg .

What is the pressure difference between the gas in the container and the atmosphere?

- A** 28 000 Pa **B** 42 000 Pa **C** 56 000 Pa **D** 84 000 Pa
- 11 A gas syringe contains a fixed mass of air. The volume of the air is 240 cm^3 and it exerts a pressure of $5.0 \times 10^4\text{ Pa}$. The air is slowly compressed, keeping the temperature constant, until the pressure is $1.5 \times 10^5\text{ Pa}$.

What is the final volume of air?

- A** 72 cm^3 **B** 80 cm^3 **C** 720 cm^3 **D** 800 cm^3
- 12 An extremely large increase in pressure is needed to compress a liquid. A gas can be compressed by a much smaller increase in pressure.

Which statement explains this?

- A** Molecules repel each other very strongly when very close.
B The attractive forces between molecules are small at large distances.
C The molecules in the gas collide with the walls of their container and rebound.
D The molecules of a liquid are constantly moving at random.

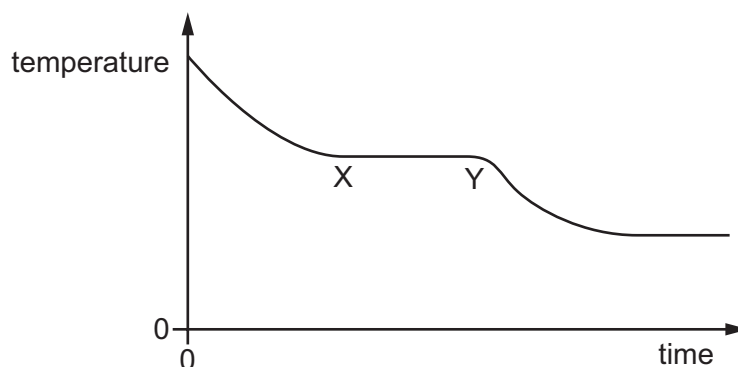
- 13** A dish of liquid is left on a laboratory bench. Some of the liquid evaporates.

What happens and why?

- A** The liquid cools because liquid molecules have more potential energy than gas molecules.
- B** The liquid cools because faster-moving molecules escape.
- C** The liquid warms because liquid molecules have less potential energy than gas molecules.
- D** The liquid warms because slower-moving molecules are left behind.

- 14** Solid wax is melted in a boiling tube and then allowed to cool.

The graph shows the cooling curve.



What is happening between points X and Y?

- A** The liquid is at its boiling point.
 - B** The liquid is turning to solid.
 - C** The solid has reached room temperature.
 - D** The solid is turning to liquid.
- 15** What is meant by the *heat capacity* of a solid object?
- A** the maximum thermal energy it can store per unit mass
 - B** the maximum thermal energy it can store per unit volume
 - C** the thermal energy required to cause a unit temperature rise
 - D** the thermal energy required to raise its temperature to its melting point
- 16** The distance between the ice point and the steam point in a liquid-in-glass thermometer is 20 cm.
- The top of the liquid thread is 12 cm above the ice point.

What is the temperature?

- A** 40 °C
- B** 60 °C
- C** 80 °C
- D** 88 °C

- 17** A thermocouple is a type of thermometer.

Which statement applies to a thermocouple?

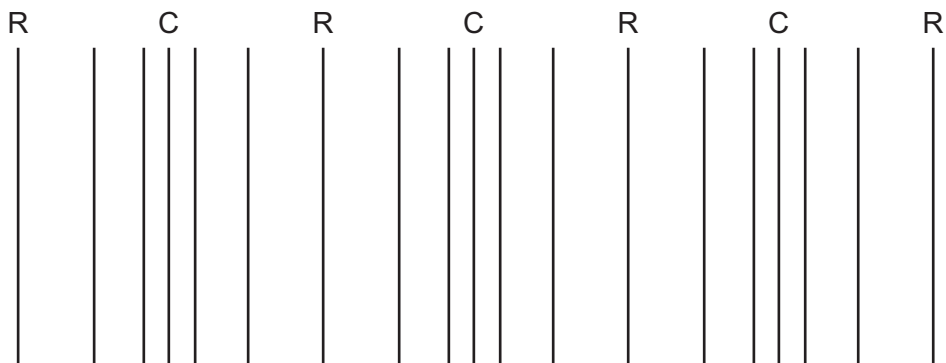
- A** It can measure rapidly varying temperatures.
- B** It must be made from three different metals.
- C** Its resistance decreases when one junction is heated.
- D** The distance between the two junctions increases as the temperature increases.

- 18** A heater is designed to radiate thermal energy.

Which change to the design decreases the thermal energy emitted by radiation?

- A** a darker coloured surface
- B** a higher surface temperature
- C** a larger surface area
- D** a shinier surface

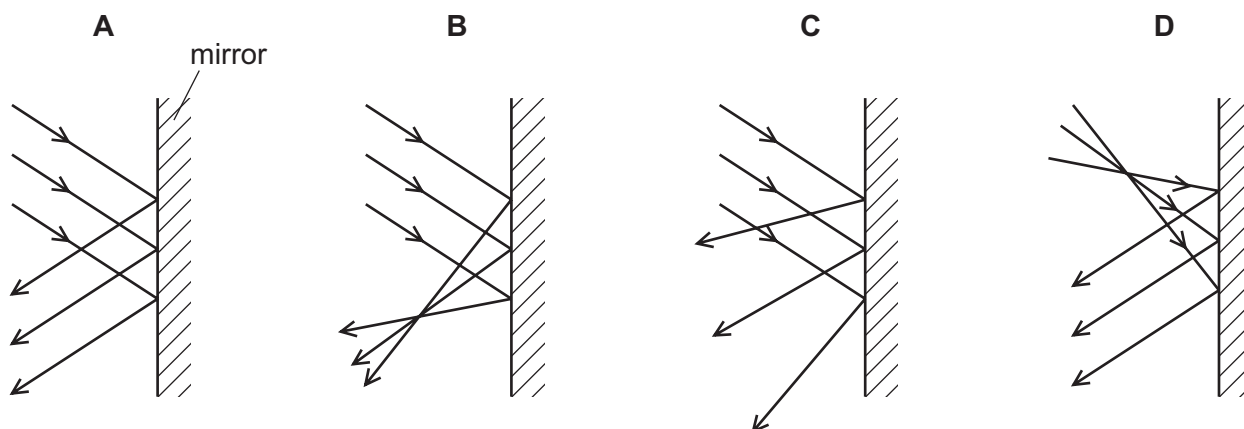
- 19** A sound wave in a solid is represented by a series of lines. The diagram shows compressions C and rarefactions R at one instant in time.



Which statement is correct?

- A** All particles on one line move in the same direction at the same time.
- B** The direction of travel of the sound wave is parallel to the lines.
- C** The distance between a compression and its adjacent rarefaction is a wavelength.
- D** The pattern of lines represents a transverse wave.

20 Which diagram shows reflection by a plane mirror?



21 Light is incident on a plastic block of refractive index 1.5.

The angle of incidence is 50° .

What is the angle of refraction?

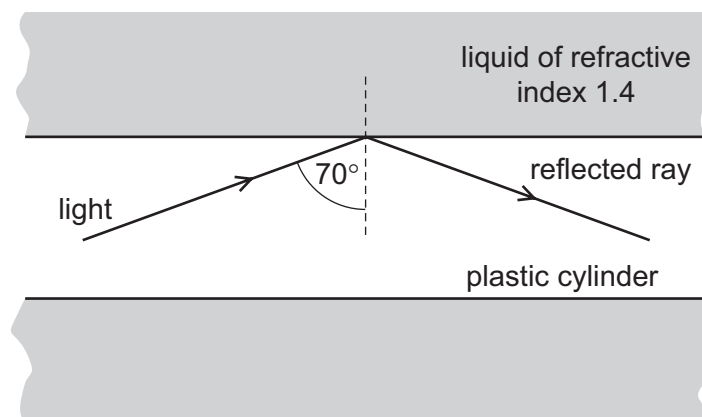
A 31°

B 33°

C 40°

D 75°

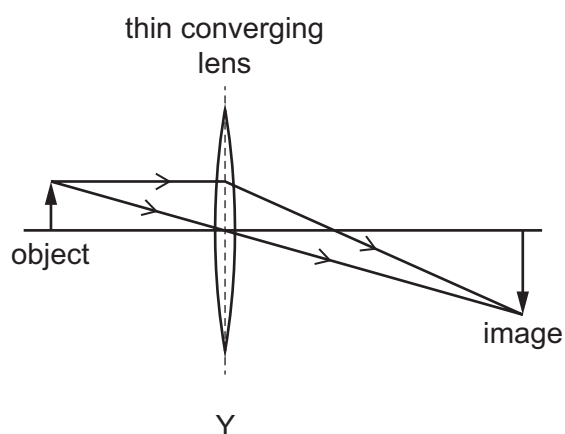
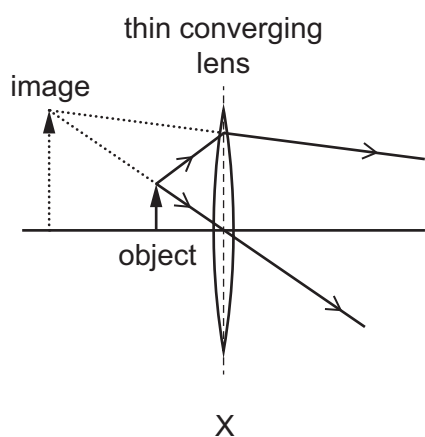
22 A solid plastic cylinder is immersed in a liquid of refractive index 1.4. Light travelling in the plastic cylinder strikes the inside surface at an angle of incidence of 70° . The light undergoes total internal reflection.



What are the values of the critical angle in the plastic and the refractive index of the plastic?

	critical angle in plastic	refractive index of plastic
A	greater than 70°	greater than 1.4
B	greater than 70°	less than 1.4
C	less than 70°	greater than 1.4
D	less than 70°	less than 1.4

- 23 The ray diagrams, X and Y, show two ways in which a thin converging lens produces an image that is larger than the object.



Which devices use a lens as shown in diagram X and in diagram Y?

	X	Y
A	camera	magnifying glass
B	magnifying glass	projector
C	photographic enlarger	camera
D	photographic enlarger	projector

- 24 The diagram shows the electromagnetic spectrum with three components named. The spectrum is in order from long wavelength to short wavelength.

Which component of the spectrum is used in a sunbed to produce a suntan?

long wavelength

short wavelength

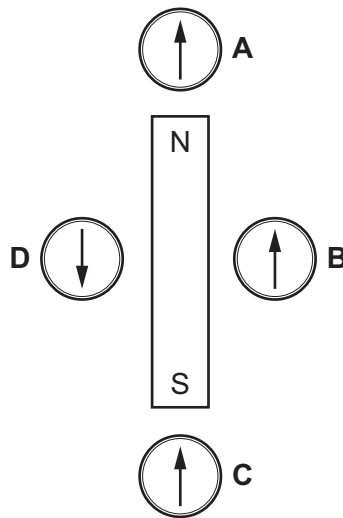
radio waves	A	B	visible light	C	D	gamma rays
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- 25 Which row gives the speed of sound in air, in water and in steel?

	<u>speed in air</u> m/s	<u>speed in water</u> m/s	<u>speed in steel</u> m/s
A	330	1500	6000
B	330	6000	1500
C	6000	330	1500
D	6000	1500	330

- 26 A compass needle is moved to different places around a strong bar magnet.

Which position shows **incorrectly** the direction of the compass needle?

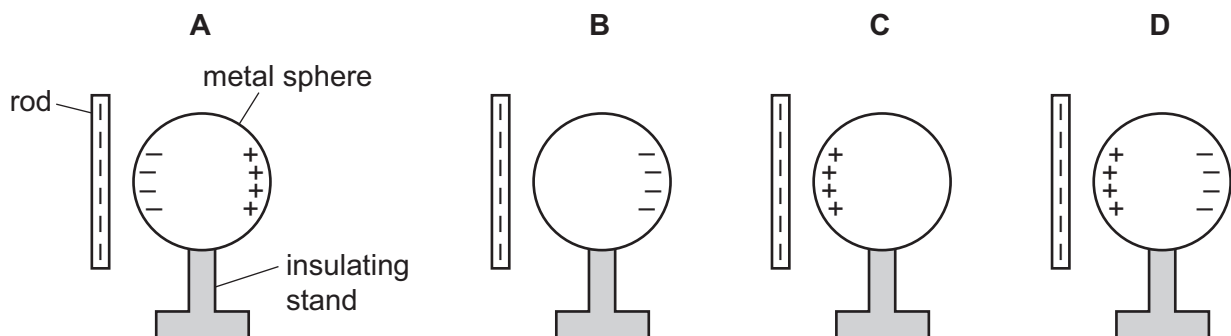


- 27 A plastic block is rubbed with a silk handkerchief and the block becomes charged.

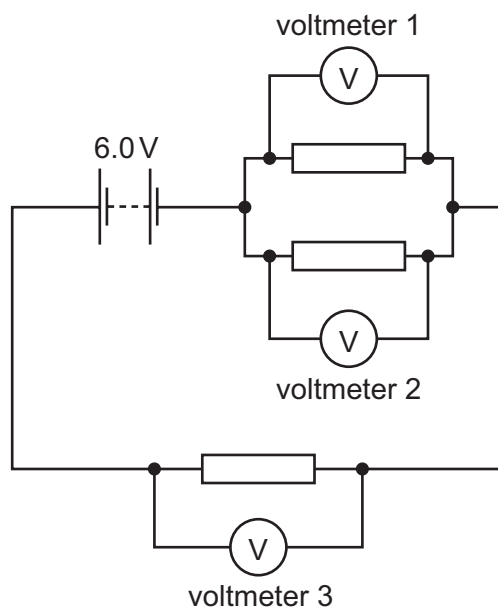
What causes the block to become charged?

- A Negative electrons flow from the handkerchief and positive electrons flow from the block.
 - B Negative electrons flow from the handkerchief to the block.
 - C Positive electrons flow from the block to the handkerchief.
 - D Positive electrons flow to the handkerchief and negative electrons flow to the block.
- 28 A negatively-charged rod is brought near to an uncharged metal sphere on an insulating stand.

Which diagram shows the charge distribution on the sphere?



- 29 A 6.0 V battery, three resistors and three voltmeters are connected in the circuit shown.



The reading on voltmeter 1 is 2.0 V.

What is the reading on voltmeter 2 and what is the reading on voltmeter 3?

	reading on voltmeter 2 /V	reading on voltmeter 3 /V
A	2.0	4.0
B	2.0	8.0
C	4.0	4.0
D	4.0	8.0

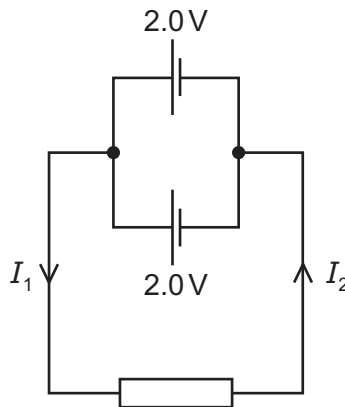
- 30 A piece of wire has a resistance of $16\ \Omega$.

The wire is 20 cm long and has a cross-sectional area of $2.0\ \text{mm}^2$.

Which wire of the same material has a resistance of $8.0\ \Omega$?

	length / cm	cross-sectional area / mm^2
A	10	1.0
B	10	4.0
C	20	1.0
D	20	4.0

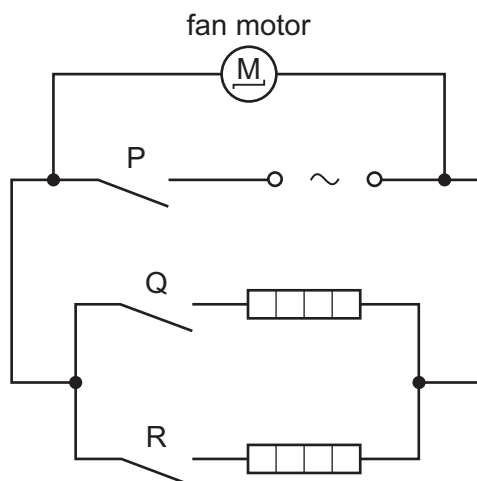
- 31 The battery in a circuit contains two identical cells connected in parallel.



Which row shows the relationship between currents I_1 and I_2 and gives the electromotive force (e.m.f.) of the battery?

	relationship	e.m.f./V
A	$I_1 > I_2$	2.0
B	$I_1 > I_2$	4.0
C	$I_1 = I_2$	2.0
D	$I_1 = I_2$	4.0

- 32 The diagram shows the circuit for a hair-dryer.



The fan motor has a power rating of 0.10 kW and the heaters each have a rating of 0.40 kW. The cost of electricity is 8.0 cents / kWh.

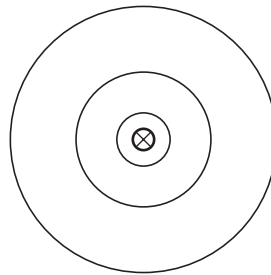
What is the cost of running the hair-dryer for two hours with switches P and Q closed and switch R open?

- A** 1.6 cents **B** 4.0 cents **C** 6.4 cents **D** 8.0 cents

33 Why is the coil of an electric motor wound on a soft-iron cylinder?

- A** to decrease the electric current
- B** to increase the electric current
- C** to reverse the magnetic field
- D** to strengthen the magnetic field

34 A straight wire carries a current into the paper. The diagram shows three magnetic field lines around the wire.



key

⊗ current into paper

The current in the wire increases.

What is the direction of the field lines and which change occurs to the field lines as the current increases?

	direction	change
A	clockwise	the lines move closer together
B	clockwise	the lines move further apart
C	anticlockwise	the lines move closer together
D	anticlockwise	the lines move further apart

35 Which particles are emitted by a hot metal filament?

- A** alpha-particles
- B** electrons
- C** neutrons
- D** protons

- 36** A circuit includes a thermistor and a light-dependent resistor (LDR).

The resistance of the thermistor and the resistance of the LDR both increase.

Which changes cause this?

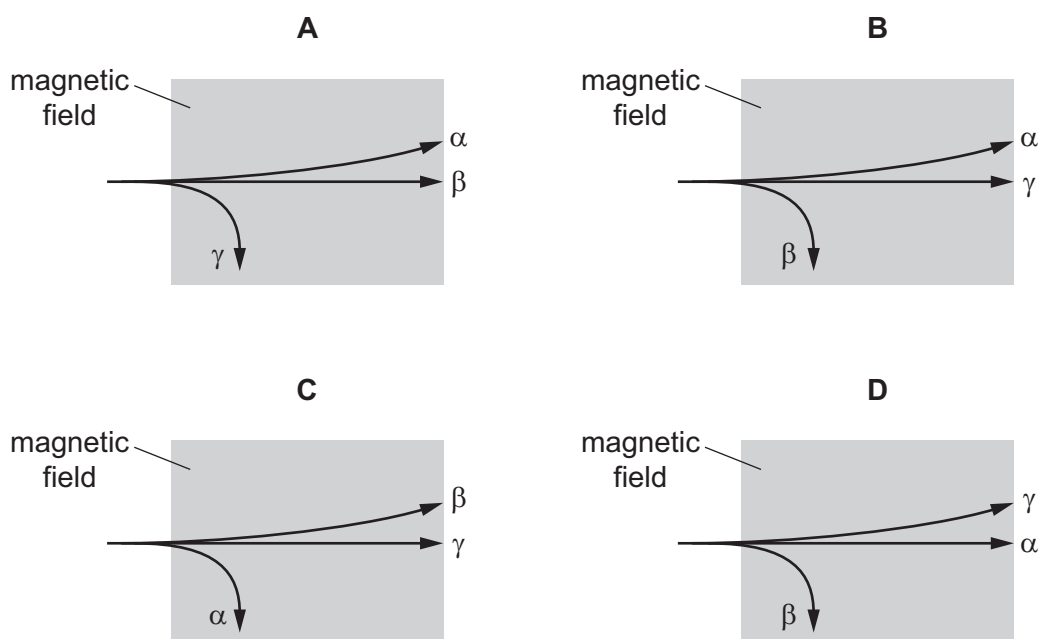
	change to thermistor	change to LDR
A	decrease in temperature	decrease in brightness
B	decrease in temperature	increase in brightness
C	increase in temperature	decrease in brightness
D	increase in temperature	increase in brightness

- 37** A sample contains atoms of an isotope that has a half-life of 7.2 years.

Which quantity halves every 7.2 years?

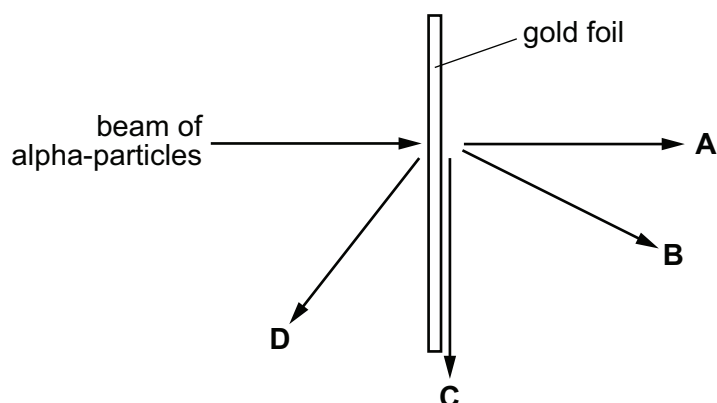
- A** the mass of the sample
B the nucleon (mass) number
C the number of atoms of the isotope
D the proton (atomic) number
- 38** A beam, consisting of alpha-particles (α), beta-particles (β), and gamma rays (γ), passes into a magnetic field.

Which diagram shows their paths in the magnetic field?



39 A narrow beam of alpha-particles is fired at a thin piece of gold foil.

What is the final direction of the largest number of alpha-particles?



40 A nucleus of phosphorus $^{32}_{15}\text{P}$ emits a beta-particle to form a new nucleus.

What is the nucleon number and what is the proton number of the new nucleus?

	nucleon number (mass number)	proton number (atomic number)
A	28	13
B	31	14
C	31	15
D	32	16

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/11

October/November 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

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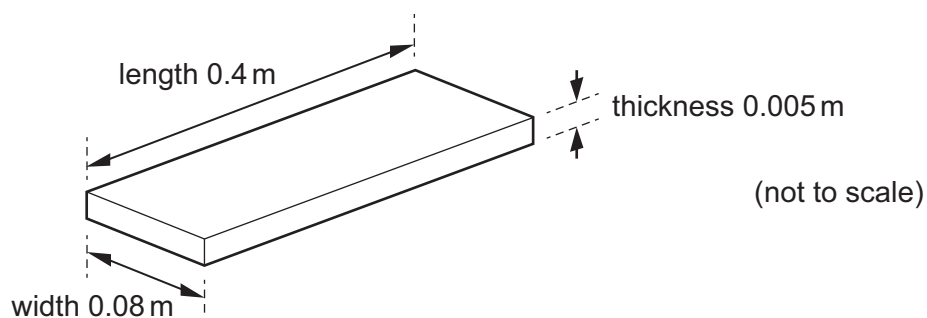


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- B distance, force, speed
- C force, length, time
- D length, mass, speed

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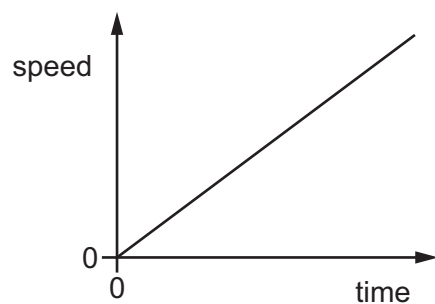
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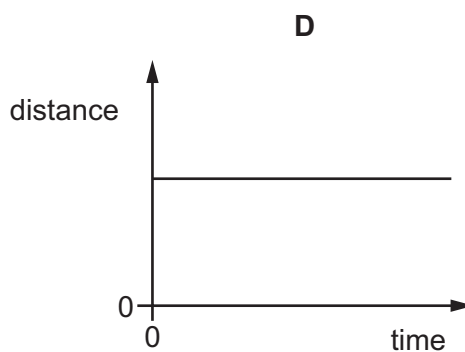
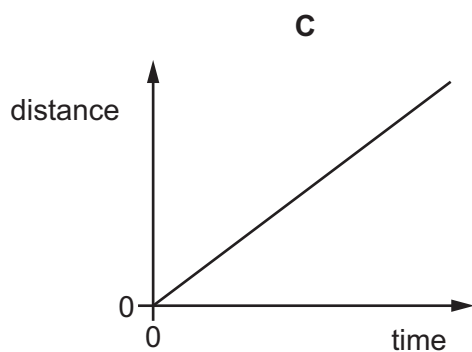
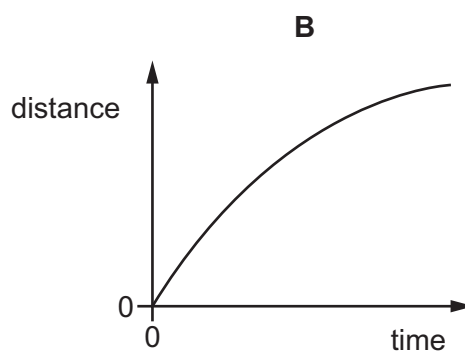
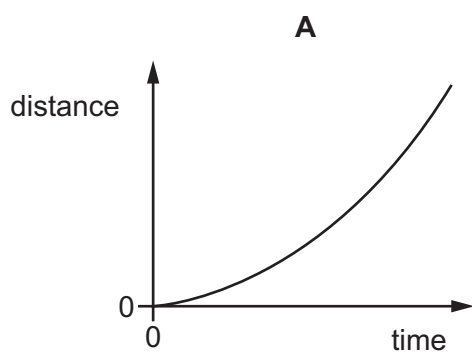
Which instruments are used to measure accurately each of these dimensions?

	length	thickness	width
A	metre rule	micrometer	calipers
B	metre rule	calipers	micrometer
C	micrometer	metre rule	calipers
D	calipers	micrometer	metre rule

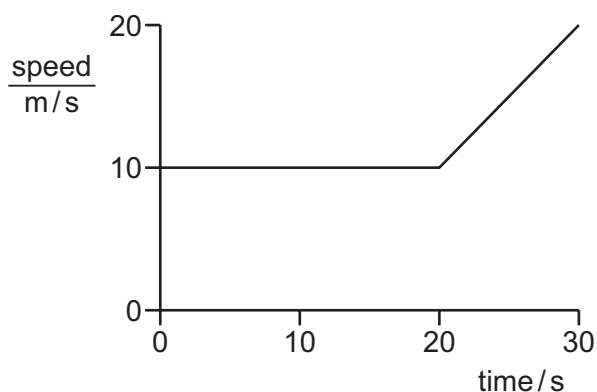
- 3 The speed-time graph represents a short journey.



Which distance-time graph represents the same journey?



- 4 An object travels for 20 s with a constant speed of 10 m/s. For the next 10 s, it accelerates uniformly to 20 m/s.



What is the total distance travelled by the object in the 30 s?

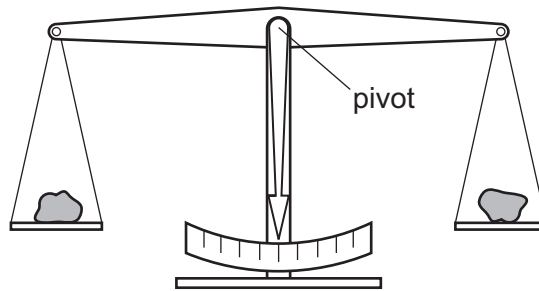
- A** 300 m **B** 350 m **C** 400 m **D** 450 m

- 5 A skydiver falls at terminal velocity. He then opens his parachute.

Which row gives the direction of the resultant force on the skydiver and the direction of the acceleration of the skydiver immediately after the parachute opens?

	resultant force	acceleration
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

- 6 The diagram shows two objects on a beam balance.



The beam balance is in equilibrium.

Which quantities may be different?

- A the masses of the two objects
 - B the moments about the pivot of the two objects
 - C the volumes of the two objects
 - D the weights of the two objects
- 7 A metal wire is suspended from a support and loads of different mass are attached to the lower end.

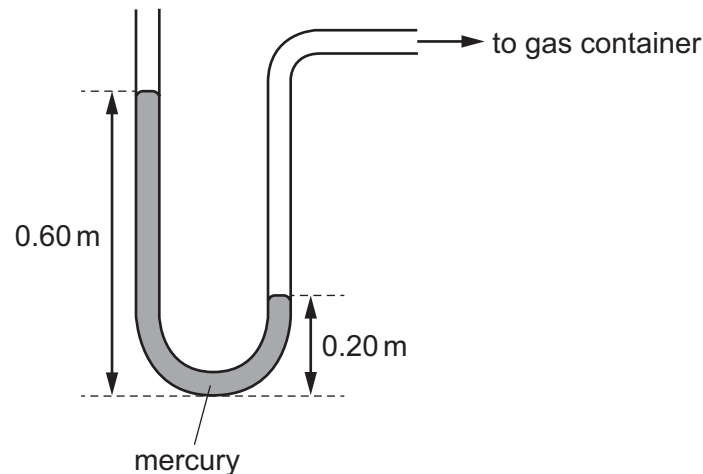
The table shows how the extension of the wire depends on the mass of the load.

mass/kg	0	2.0	4.0	6.0	8.0	10.0
extension/mm	0	0.35	0.70	1.05	1.50	2.05

Between which values of mass does the limit of proportionality lie?

- A 0 and 2.0 kg
- B 4.0 kg and 6.0 kg
- C 6.0 kg and 8.0 kg
- D 8.0 kg and 10.0 kg

- 8 The diagram shows a mercury manometer connected to a gas container.



The density of mercury is $14\,000\text{ kg/m}^3$. The gravitational field strength g is 10 N/kg .

What is the pressure difference between the gas in the container and the atmosphere?

- A** 28 000 Pa **B** 42 000 Pa **C** 56 000 Pa **D** 84 000 Pa
- 9 An astronaut on the Moon drops a tool box of mass 3.0 kg . It falls from rest and its kinetic energy as it hits the surface is 0.96 J .
- At which speed does the tool box hit the surface of the Moon?
- A** 0.46 m/s **B** 0.57 m/s **C** 0.64 m/s **D** 0.80 m/s
- 10 A car travels a distance of 200 m in 20 s . The engine of the car provides a driving force of 1000 N .
- What is the power output of the engine?
- A** 0.25 W **B** 4.0 W **C** 100 W **D** $10\,000\text{ W}$
- 11 A gas is heated in a closed container of constant volume.
- What happens to the molecules of the gas?
- A** They collide with the walls with less force.
B They expand.
C They move faster.
D They move further apart.

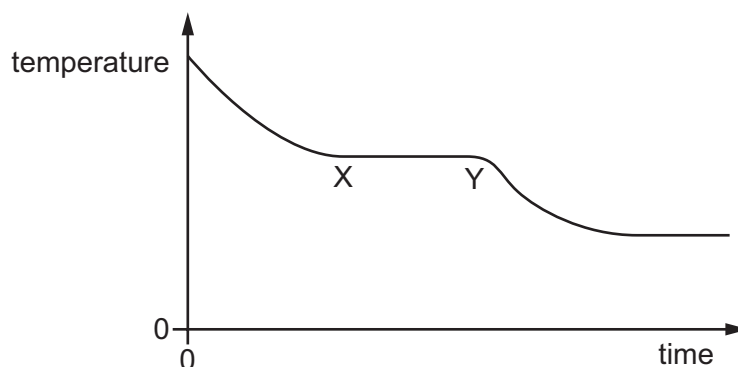
- 12** Scientists believe that some lakes are shrinking because of evaporation.

What increases the rate of evaporation?

- A** a decrease in the surface area
- B** a fall in water temperature
- C** an increase in the depth of the lake
- D** wind blowing across the surface

- 13** Solid wax is melted in a boiling tube and then allowed to cool.

The graph shows the cooling curve.



What is happening between points X and Y?

- A** The liquid is at its boiling point.
 - B** The liquid is turning to solid.
 - C** The solid has reached room temperature.
 - D** The solid is turning to liquid.
- 14** During boiling and during melting, a substance either takes in energy or gives out energy.

Which row gives the energy transfer corresponding to the change of state?

	during boiling	during melting
A	gives out energy	gives out energy
B	gives out energy	takes in energy
C	takes in energy	gives out energy
D	takes in energy	takes in energy

- 15** The temperature of a 50 g mass of a metal is raised by 40 °C. The specific heat capacity of the metal is 0.40 J/(g °C).

How much thermal energy is supplied?

- A** 0.32 J **B** 0.50 J **C** 500 J **D** 800 J

- 16** In which substance is the conduction of thermal energy mainly due to the movement of electrons?

- A** air
B ice
C iron
D water

- 17** Four similar metal plates are the same distance from a heater that emits infra-red radiation.

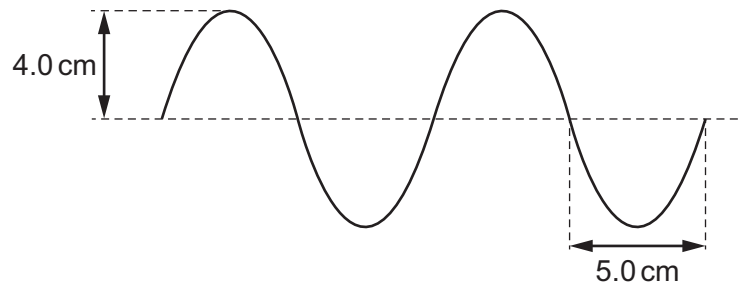
The plates are painted dull black, dull white, shiny black and shiny white.

Which plate absorbs the most radiation and which plate reflects the most radiation?

	absorbs most radiation	reflects most radiation
A	dull black	dull white
B	dull black	shiny white
C	shiny black	dull black
D	shiny white	dull black

- 18** A wave is travelling in water in a glass-sided tank.

The diagram shows a side view of the wave at one instant.



What are the amplitude and the wavelength of the wave?

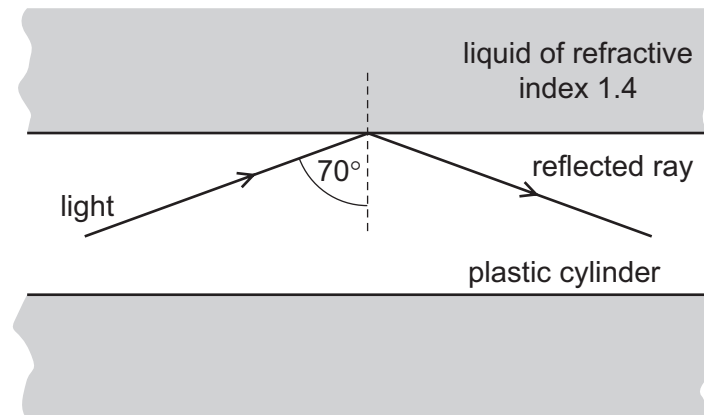
	amplitude / cm	wavelength / cm
A	4.0	5.0
B	4.0	10
C	8.0	5.0
D	8.0	10

- 19** An earthquake wave travels through the solid surface of the Earth from east to west. The solid surface vibrates in a north-south direction.

How can the earthquake wave be described?

- A** electromagnetic
- B** longitudinal
- C** sound
- D** transverse

- 20** A solid plastic cylinder is immersed in a liquid of refractive index 1.4. Light travelling in the plastic cylinder strikes the inside surface at an angle of incidence of 70° . The light undergoes total internal reflection.



What are the values of the critical angle in the plastic and the refractive index of the plastic?

	critical angle in plastic	refractive index of plastic
A	greater than 70°	greater than 1.4
B	greater than 70°	less than 1.4
C	less than 70°	greater than 1.4
D	less than 70°	less than 1.4

- 21** What is the name and shape of the lens used to correct short sight?

	name of lens	shape of lens
A	converging	
B	converging	
C	diverging	
D	diverging	

- 22** An object is placed in front of a converging lens. The lens forms a magnified image of the object on a screen.

Which statement is correct?

- A** The distance between the object and the lens is greater than the focal length.
- B** The image formed is a virtual image.
- C** The image is the right way up.
- D** The lens is acting as a magnifying glass.

- 23** A type of electromagnetic radiation possesses the following properties.

- It is ionising.
- Its frequency is higher than the frequency of microwaves.
- It is not detected by the human eye.

What is this radiation?

- A** gamma rays
- B** infra-red
- C** light
- D** radio waves

- 24** Which row gives the speed of sound in air, in water and in steel?

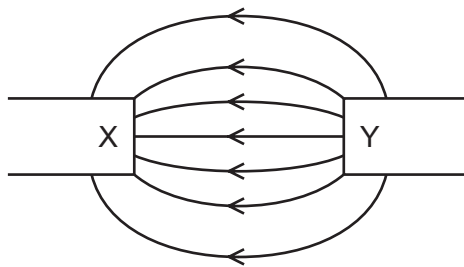
	<u>speed in air</u> m/s	<u>speed in water</u> m/s	<u>speed in steel</u> m/s
A	330	1500	6000
B	330	6000	1500
C	6000	330	1500
D	6000	1500	330

- 25** As a sound wave travels from one medium to another, its wavelength increases.

What happens to the frequency and to the speed of the sound?

	frequency	speed
A	decreases	decreases
B	decreases	increases
C	stays constant	decreases
D	stays constant	increases

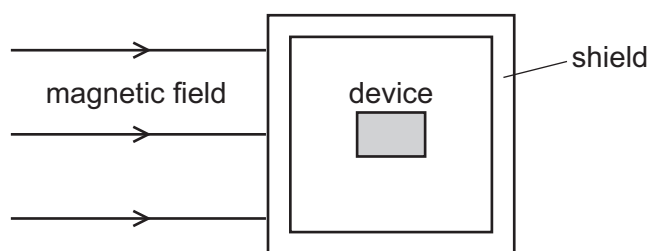
- 26** The diagram shows the pattern and the direction of the magnetic field between two magnetic poles X and Y.



Which types of pole are X and Y?

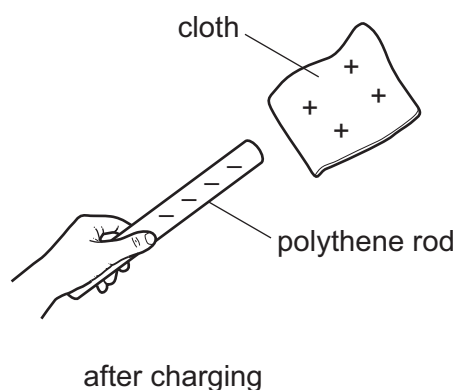
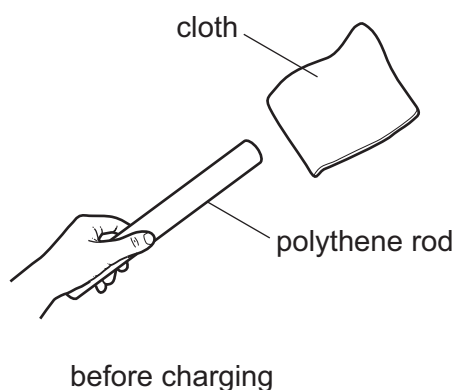
	X	Y
A	N-pole	N-pole
B	N-pole	S-pole
C	S-pole	N-pole
D	S-pole	S-pole

- 27 The diagram shows a sensitive device surrounded by a shield that prevents the magnetic field from reaching the device.



How does the shield protect the device?

- A It channels the magnetic field around the shield until it emerges on the right-hand side.
 - B Its left-hand side absorbs the magnetic field.
 - C Its left-hand side reflects the magnetic field.
 - D Its left-hand side scatters the magnetic field in all directions.
- 28 A polythene rod is charged by rubbing it with a cloth.

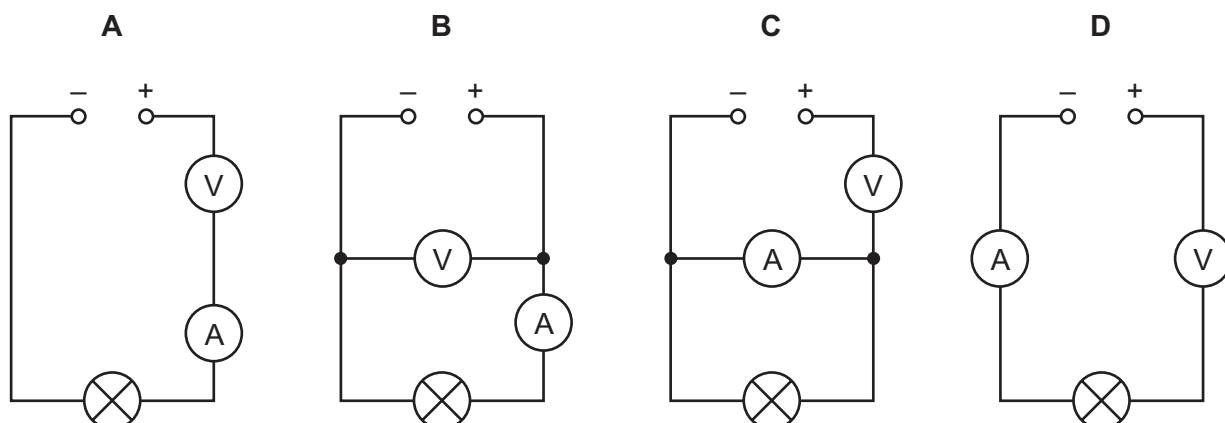


Which statement explains why the cloth is charged?

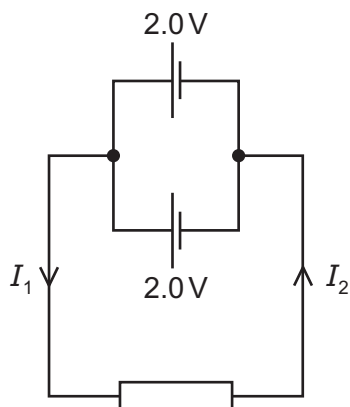
- A Electrons move from the cloth to the rod.
 - B Electrons move from the rod to the cloth.
 - C Protons move from the cloth to the rod.
 - D Protons move from the rod to the cloth.
- 29 What is *electromotive force* (*e.m.f.*)?
- A $\frac{\text{charge}}{\text{work done}}$
 - B $\frac{\text{charge}}{\text{time}}$
 - C $\frac{\text{work done}}{\text{charge}}$
 - D $\frac{\text{work done}}{\text{time}}$

- 30 In a circuit, a voltmeter is used to measure the potential difference across a lamp. An ammeter is used to measure the current in the lamp.

Which diagram shows the circuit?



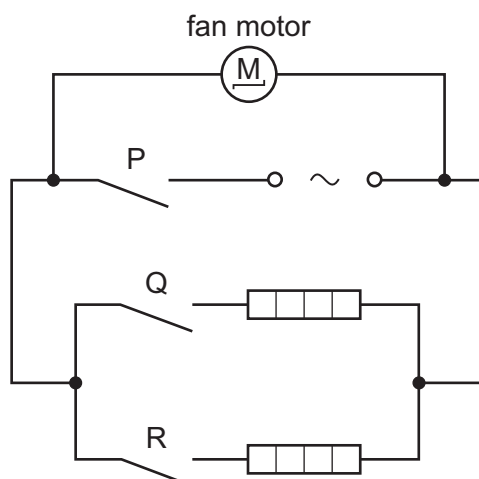
- 31 The battery in a circuit contains two identical cells connected in parallel.



Which row shows the relationship between currents I_1 and I_2 and gives the electromotive force (e.m.f.) of the battery?

	relationship	e.m.f./V
A	$I_1 > I_2$	2.0
B	$I_1 > I_2$	4.0
C	$I_1 = I_2$	2.0
D	$I_1 = I_2$	4.0

32 The diagram shows the circuit for a hair-dryer.



The fan motor has a power rating of 0.10 kW and the heaters each have a rating of 0.40 kW. The cost of electricity is 8.0 cents / kWh.

What is the cost of running the hair-dryer for two hours with switches P and Q closed and switch R open?

- A** 1.6 cents **B** 4.0 cents **C** 6.4 cents **D** 8.0 cents

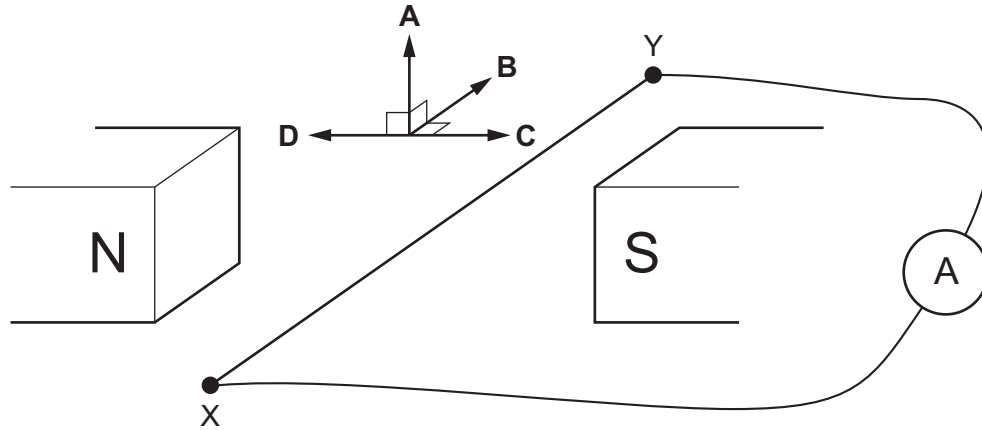
33 Why are the metal casings of electrical appliances earthed?

- A** to complete the circuit
B to ensure the casing is not at a dangerous voltage
C to ensure the fuse blows when the current in the appliance is too large
D to protect the appliance from overheating

- 34** The diagram shows a wire XY lying between the poles of a magnet.

The ends of the wire are connected to a sensitive ammeter. The wire is moved and a reading is registered.

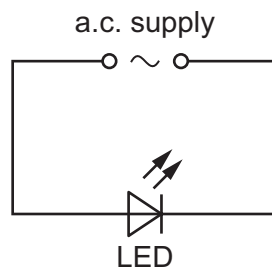
In which direction is the wire moved?



- 35** Which particles are emitted by a hot metal filament?

- A** alpha-particles
- B** electrons
- C** neutrons
- D** protons

- 36** A light-emitting diode (LED) is connected to an alternating current (a.c.) supply.



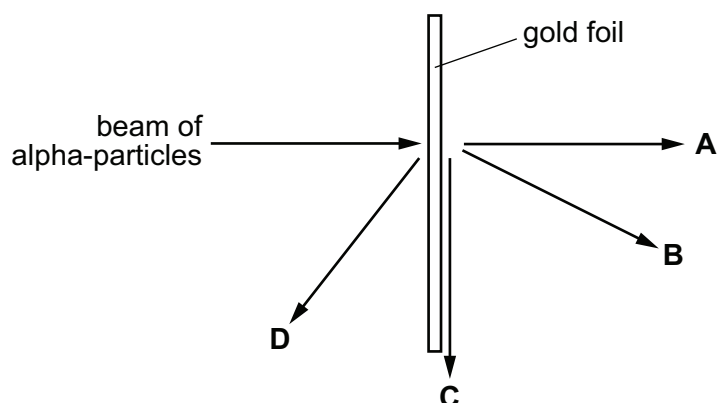
The LED flashes on and off.

What is the direction of the current in the LED when it emits light?

- A** towards the left only
- B** towards the right only
- C** towards the left or towards the right
- D** there is no current in the LED

37 A narrow beam of alpha-particles is fired at a thin piece of gold foil.

What is the final direction of the largest number of alpha-particles?



38 A nucleus of phosphorus $^{32}_{15}\text{P}$ emits a beta-particle to form a new nucleus.

What is the nucleon number and what is the proton number of the new nucleus?

	nucleon number (mass number)	proton number (atomic number)
A	28	13
B	31	14
C	31	15
D	32	16

39 How does a nucleus of the isotope chlorine-37 differ from a nucleus of the isotope chlorine-35?

- A** It contains one more neutron and one more electron.
- B** It contains one more proton and one more electron.
- C** It contains two more neutrons.
- D** It contains two more protons.

40 Which types of radiation may be emitted by radioactive decay?

- A** alpha-particles and gamma rays
- B** microwaves and infra-red
- C** radio waves and microwaves
- D** ultra-violet and X-rays

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Cambridge Ordinary Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2017

MARK SCHEME

Maximum Mark: 40

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International Education

Question	Answer	Marks
1	D	1
2	A	1
3	A	1
4	C	1
5	C	1
6	C	1
7	D	1
8	A	1
9	D	1
10	C	1
11	B	1
12	A	1
13	B	1
14	B	1
15	C	1
16	B	1
17	A	1
18	D	1
19	A	1
20	A	1
21	A	1
22	C	1
23	B	1
24	C	1
25	A	1
26	B	1
27	B	1
28	D	1

Question	Answer	Marks
29	A	1
30	D	1
31	C	1
32	D	1
33	D	1
34	A	1
35	B	1
36	A	1
37	C	1
38	B	1
39	A	1
40	D	1



Cambridge Assessment International Education
Cambridge Ordinary Level

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5054/11

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Cambridge Assessment
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Question	Answer	Marks
1	D	1
2	A	1
3	A	1
4	B	1
5	D	1
6	C	1
7	C	1
8	C	1
9	D	1
10	D	1
11	C	1
12	D	1
13	B	1
14	D	1
15	D	1
16	C	1
17	B	1
18	B	1
19	D	1
20	C	1
21	C	1
22	A	1
23	A	1
24	A	1
25	D	1
26	C	1
27	A	1
28	A	1

Question	Answer	Marks
29	C	1
30	B	1
31	C	1
32	D	1
33	B	1
34	A	1
35	B	1
36	B	1
37	A	1
38	D	1
39	C	1
40	A	1



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/12

October/November 2016

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

This document consists of **15** printed pages and **1** blank page.

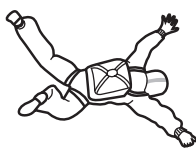


- 1 A length of copper wire is labelled 'length 30 m' and 'diameter 0.50 mm'.

Which instruments are most suitable to measure accurately the length and the diameter of the wire?

	length	diameter
A	rule	calipers
B	rule	micrometer
C	tape	calipers
D	tape	micrometer

- 2 A skydiver falls from rest through the air and reaches terminal velocity.



What is the acceleration of the skydiver during his fall?

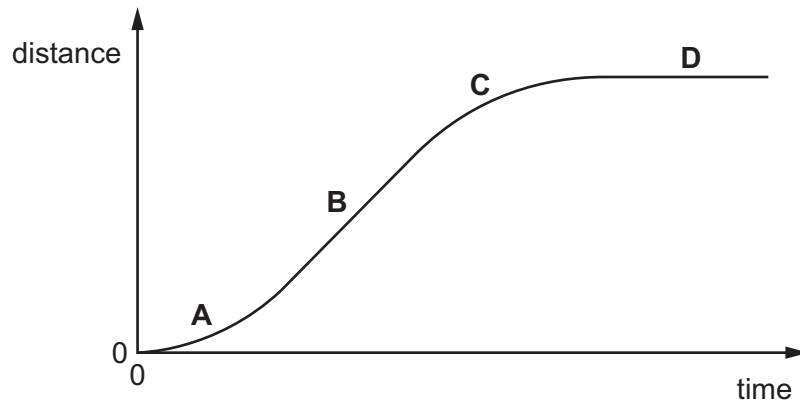
- A** constant at 0 m/s^2
B constant at 10 m/s^2
C starting at 0 m/s^2 and increasing to 10 m/s^2
D starting at 10 m/s^2 and decreasing to 0 m/s^2
- 3 Two aluminium spheres P and Q, of masses 0.50 kg and 1.0 kg respectively, are released at the same time from the same height.

Why do they have the same acceleration?

- A** The density of P is equal to the density of Q.
B The force on P is equal to the force on Q.
C The force on Q is double the force on P.
D The gravitational field strength for Q is double that for P.

- 4 The graph shows a distance-time graph for a car travelling in a straight line.

In which region is the car decelerating?



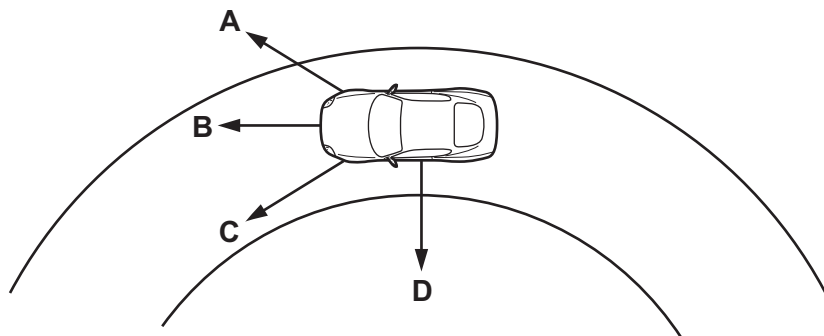
- 5 A force of 20 N acts on an object of mass 5.0 kg in the forward direction. A second force also acts on the object. The forward acceleration of the object is 3.0 m/s^2 .

What is the size and direction of the second force?

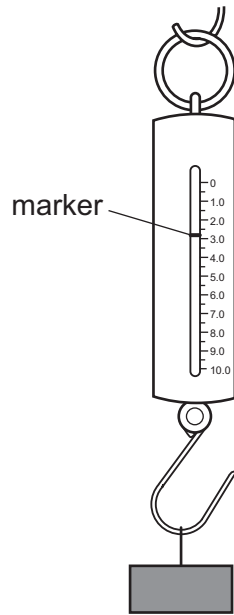
	size of force / N	direction of force
A	5.0	backwards
B	15	backwards
C	15	forwards
D	35	forwards

- 6 The diagram shows a car going around a circular track at constant speed.

Which arrow shows the direction of the resultant force on the car?

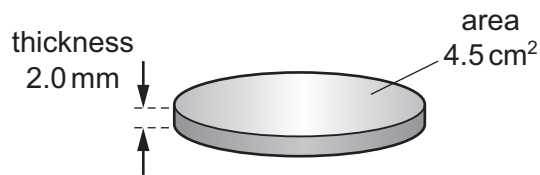


- 7 A block is hung on a spring balance. The marker inside the balance is pulled down by the block.



What can the position of the marker be used to determine?

- A the mass of the block in kg
 - B the mass of the block in N
 - C the moment caused by the block in N m
 - D the weight of the block in kg
- 8 The density of gold is 19 g/cm^3 .
- A gold coin is 2.0 mm thick. Each face of the coin has an area of 4.5 cm^2 .



What is the mass of the coin?

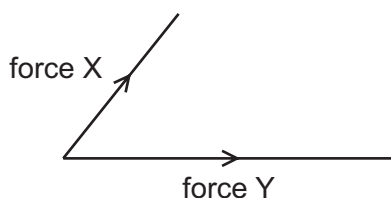
- A 2.1 g
- B 17 g
- C 21 g
- D 170 g

- 9 A man with an open parachute is falling to Earth at constant speed. The following forces are acting.

- P the upward force of the parachute on the man.
- Q the upward force of the man on the Earth.
- R the downward force of the Earth on the parachute.
- S the downward force of the man on the parachute.

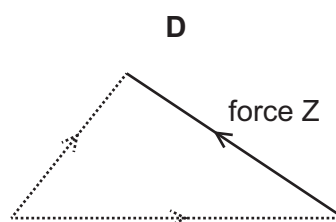
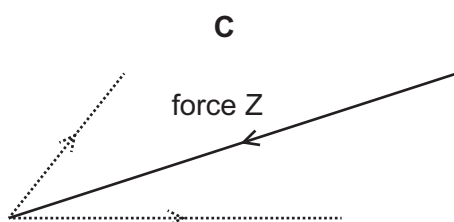
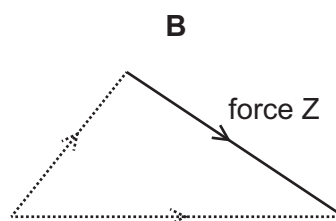
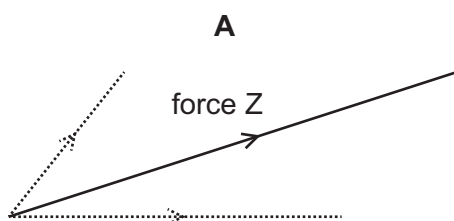
Which two forces are a Newton's third law pair?

- A** P and Q **B** P and R **C** P and S **D** Q and R
- 10 Two forces, X and Y, act on an object and produce a resultant force. The scale diagram represents the sizes and directions of forces X and Y.



Force Z balances the resultant force due to X and Y and keeps the object stationary.

Which arrow represents force Z?

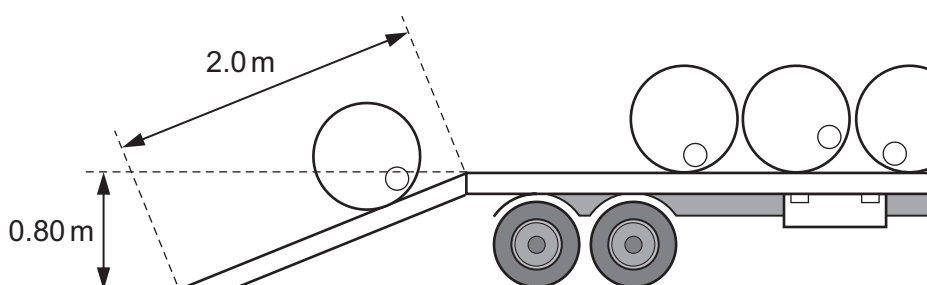


- 11 Three objects P, Q and R have different masses and different speeds, as shown in the table.

	$\frac{\text{mass}}{\text{kg}}$	$\frac{\text{speed}}{\text{m/s}}$
P	1.0	3.0
Q	2.0	2.0
R	5.0	1.0

What is the order of increasing kinetic energy (smallest first) of the objects?

- A** P → Q → R
- B** P → R → Q
- C** R → P → Q
- D** R → Q → P
- 12 Which power stations all rely on renewable sources of energy?
- A** hydroelectric, oil-fired, solar power stations
- B** hydroelectric, solar, wind power stations
- C** hydroelectric, oil-fired, wind power stations
- D** oil-fired, wind, solar power stations
- 13 Which force does no work?
- A** the force from a car's engine that causes the car to accelerate
- B** the force that causes a box to slide over a rough horizontal surface
- C** the force that causes a satellite to move in a circular orbit
- D** the force that causes a spring to stretch
- 14 A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

- A** 1000 J **B** 1600 J **C** 2500 J **D** 4000 J

- 15 An electric motor is used to operate a lift (elevator). The motor does 140 000 J of work in 70 s.

What is the output power of the motor?

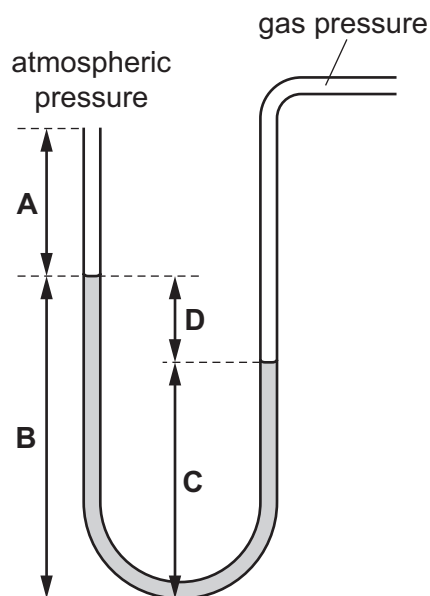
- A 0.50 milliwatts
- B 2.0 watts
- C 2.0 kilowatts
- D 9.8 megawatts

- 16 Which expression gives the efficiency of an a.c. generator?

- A $\frac{\text{electrical energy output}}{\text{total energy input}}$
- B $\frac{\text{electrical energy output}}{\text{thermal energy output}}$
- C $\frac{\text{total energy output}}{\text{electrical energy input}}$
- D $\frac{\text{thermal energy output}}{\text{total energy input}}$

- 17 The pressure of the gas supply in the laboratory is measured using a manometer.

Which distance gives the pressure of the gas above atmospheric pressure?

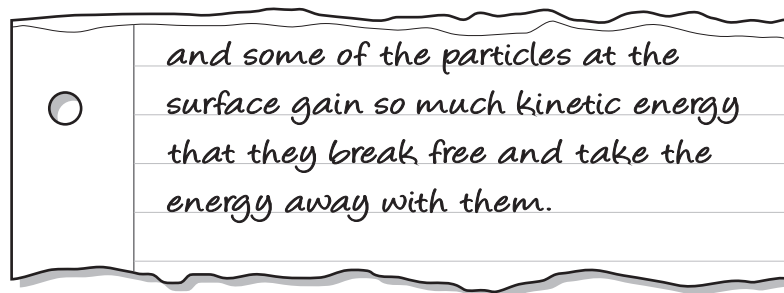


- 18 A gas expands slowly and its temperature remains constant.

What happens to the gas molecules?

- A They move further apart and their average speed decreases.
- B They move further apart and their average speed increases.
- C They move further apart and their average speed remains unchanged.
- D They stay the same distance apart and their average speed decreases.

- 19 A piece of paper torn out of an exercise book is shown.

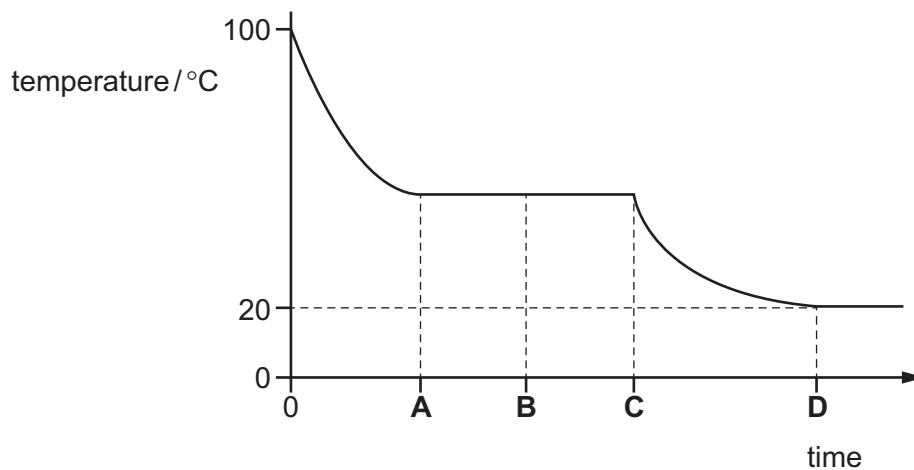


Which process is being described?

- A conduction
 - B convection
 - C evaporation
 - D radiation
- 20 A sample of liquid at 100°C is allowed to cool to a room temperature of 20°C .

The diagram shows how the temperature changes with time during the cooling process.

At which time does the sample become completely solid?



- 21** One kilogram of water needs more thermal energy to raise its temperature by 1°C than one kilogram of copper.

Why is this?

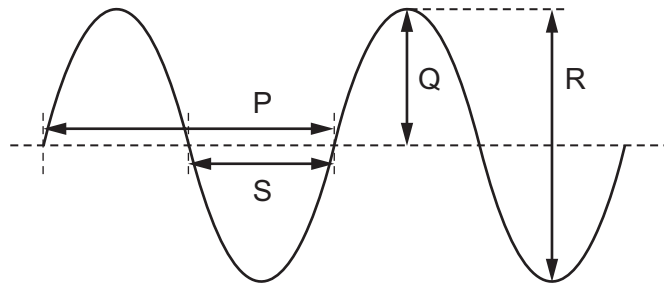
- A** copper has a higher specific heat capacity than water
 - B** copper has a lower specific heat capacity than water
 - C** copper is a better conductor of heat than water
 - D** copper is a poorer conductor of heat than water
- 22** The thermal energy produced by an electric heater in three minutes is used to melt wax. The wax is initially at its melting point of 60°C .

The specific latent heat of the wax is 220 J/g . The heater supplies 7700 J of thermal energy to the wax.

How much wax melts?

- A** 0.58 g **B** 1.7 g **C** 12 g **D** 35 g
- 23** What makes a clinical thermometer suitable for measuring small changes in body temperature?
- A** The amount of liquid in the bulb is small.
 - B** The bore of the capillary tube is narrow.
 - C** The capillary tube is long.
 - D** The glass bulb has a thin wall.
- 24** In which states of matter does convection occur?
- A** gases, liquids and solids
 - B** gases and liquids only
 - C** gases and solids only
 - D** liquids and solids only
- 25** Which statement about waves is correct?
- A** All electromagnetic waves are transverse waves.
 - B** All transverse waves travel at $300\,000\,000\text{ m/s}$.
 - C** Longitudinal waves cannot be refracted.
 - D** Water waves are longitudinal waves.

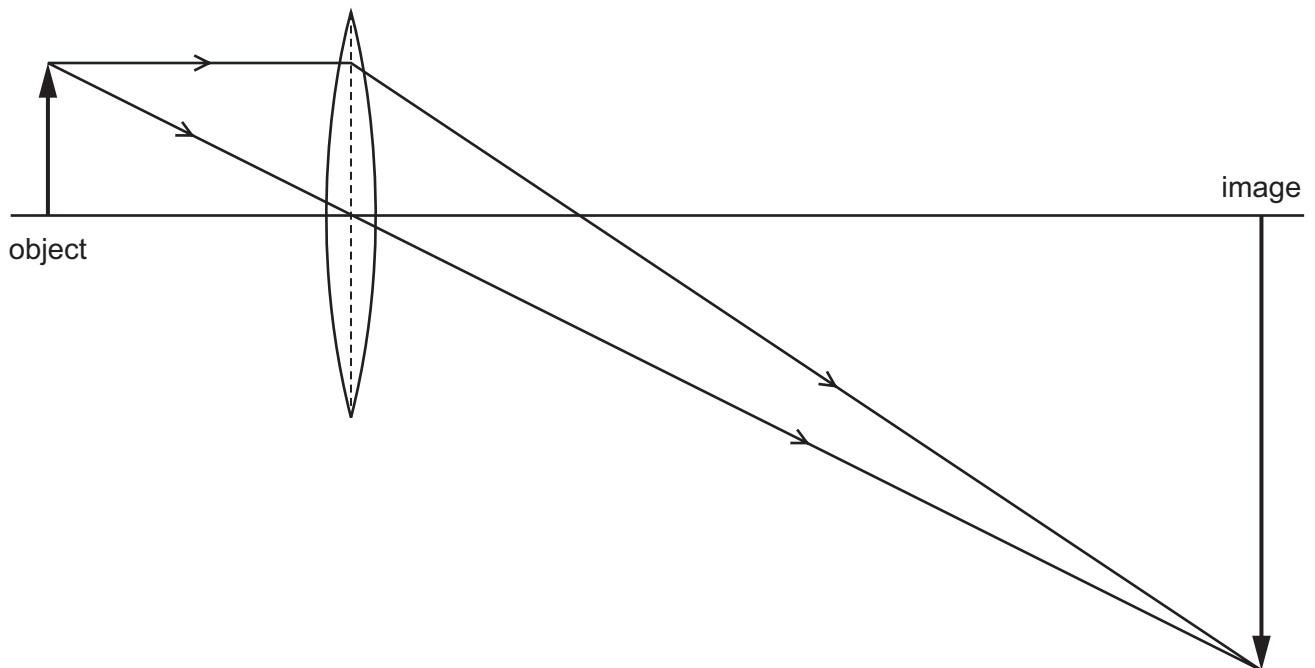
- 26 The diagram shows the surface of water as a wave passes across a ripple tank.



Which lengths represent the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	Q	P
B	Q	S
C	R	P
D	R	S

- 27 A lens is used to produce a magnified image, as shown in the scale diagram.



What is the linear magnification produced by the lens?

- A** 0.33 **B** 3.0 **C** 4.0 **D** 6.0

- 28 A girl is long-sighted.

Which statement is correct?

- A She sees close objects less clearly than a person with normal vision.
- B She sees distant objects more clearly than a person with normal vision.
- C The fault is corrected with a diverging lens.
- D The image of a close object is formed in front of her retina.

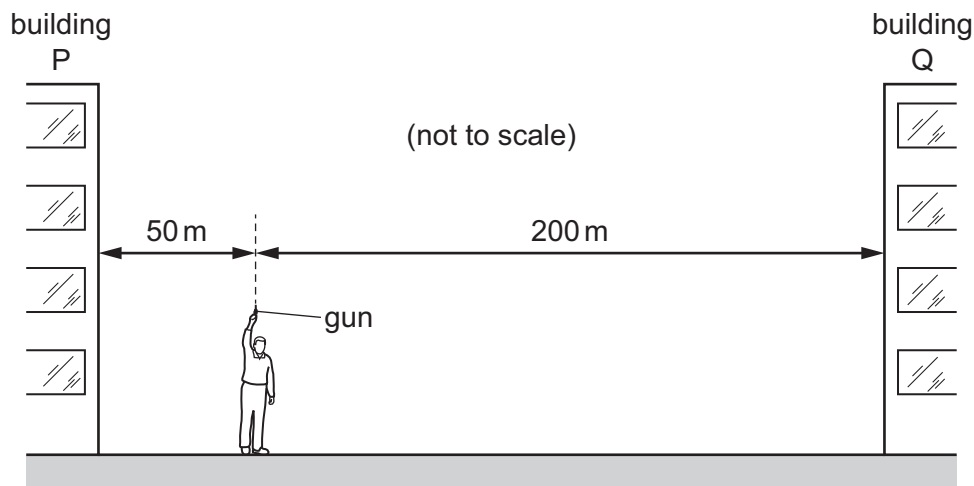
- 29 Sound travels at different speeds in solids, liquids and gases.

Which list shows these states of matter in order of increasing speed of sound (slowest to fastest)?

- A gas → liquid → solid
- B gas → solid → liquid
- C liquid → solid → gas
- D solid → liquid → gas

- 30 A man stands between two tall buildings, P and Q. The diagram is not drawn to scale.

He is 50 m from P and 200 m from Q.



He fires a gun and the first two echoes he hears are 1.0 second apart.

What is the speed of sound calculated from this observation?

- A 250 m/s
- B 300 m/s
- C 330 m/s
- D 500 m/s

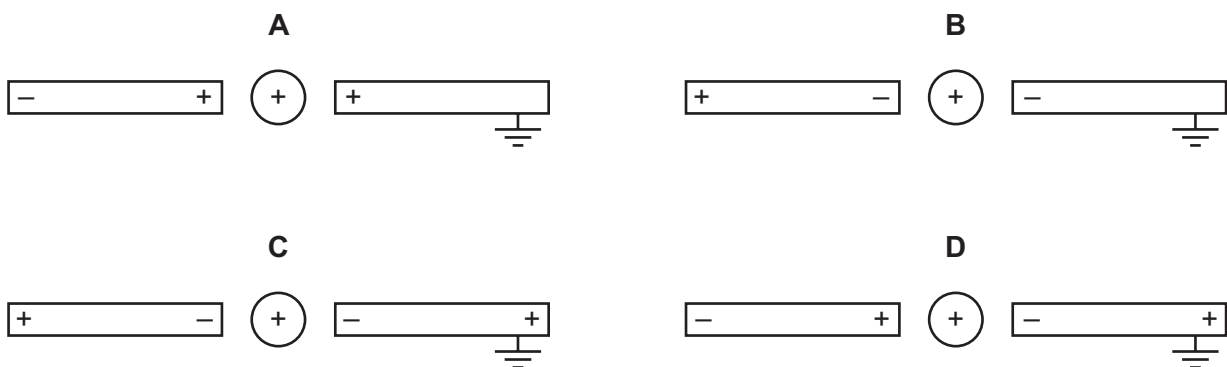
- 31** Delicate instruments are often placed in a box to screen them from stray magnetic fields.

What is the material used for the box and why is it chosen?

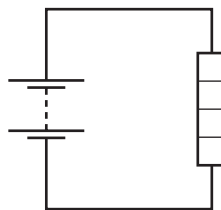
- A** Aluminium is used because it is a non-magnetic material.
- B** Copper is used because it has a low electrical resistance.
- C** Polythene is used because it is a good electrical insulator.
- D** Soft iron is used because it is a magnetic material.

- 32** A positively charged metal sphere is placed midway between two previously uncharged metal rods, one of which is connected to earth.

Which diagram shows the charges on the rods?



- 33** An ammeter and a voltmeter are connected into the circuit shown when determining the power produced by a heater.



How are these meters connected, in relation to the heater?

	ammeter	voltmeter
A	in parallel	in parallel
B	in parallel	in series
C	in series	in parallel
D	in series	in series

34 Resistors of different resistances are connected to a power supply in either a parallel circuit or a series circuit. Which statement is correct?

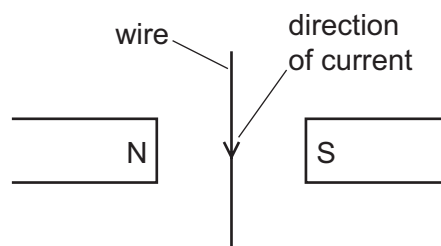
- A** The current is the same in all resistors connected in parallel.
- B** The current is the same in all resistors connected in series.
- C** The voltage across each resistor is different for all resistors connected in parallel.
- D** The voltage across each resistor is the same for all resistors connected in series.

35 The headlamp circuit in a car includes a 7A fuse.

What is the purpose of the fuse?

- A** It increases the resistance of the circuit.
- B** It makes it easier for the current to flow.
- C** It protects the wiring from overheating.
- D** It reduces the voltage across the lamp.

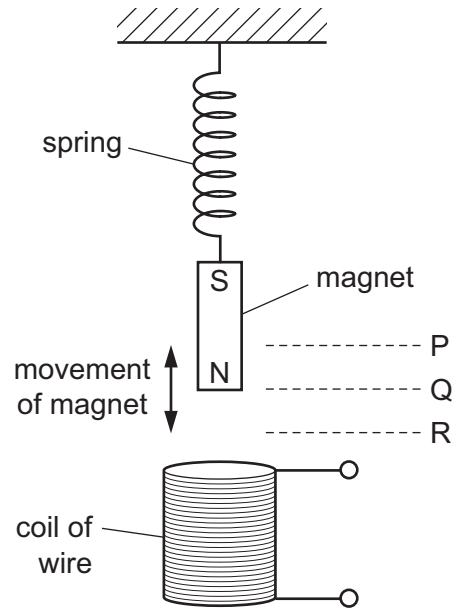
36 A current-carrying wire lies between the poles of two magnets, as shown.



What is the direction of the force on the wire?

- A** into the plane of the paper
- B** out of the plane of the paper
- C** towards the left
- D** towards the right

37 A magnet moves up and down above a coil of wire.

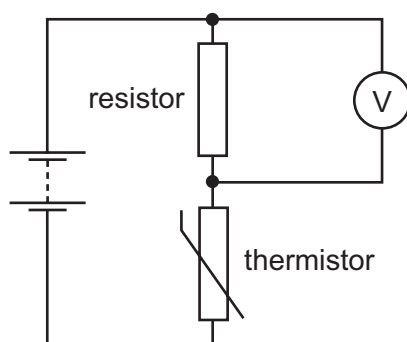


The bottom of the magnet moves up and down between P and R.

Where is the bottom of the magnet when there is no induced electromotive force (e.m.f.) in the coil?

- A** at P and at Q
- B** at P and at R
- C** at Q only
- D** at R only

- 38 A resistor of constant resistance and a thermistor are connected in series with a battery. A voltmeter is connected in parallel with the resistor.



The temperature falls.

What are the correct changes?

	resistance of thermistor	voltmeter reading
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 39 A freshly made sample of radioactive material gives a count rate of 8000 counts per minute. After 20 days, it gives a count rate of 500 counts per minute.

What is the half-life of the material?

- A** 4.0 days
- B** 5.0 days
- C** 20 days
- D** 80 days

- 40 A nucleus of uranium ${}_{92}^{238}\text{U}$ decays to thorium by emitting an alpha-particle.

What is the resulting thorium nucleus?

- A** ${}_{90}^{234}\text{Th}$
- B** ${}_{90}^{236}\text{Th}$
- C** ${}_{93}^{238}\text{Th}$
- D** ${}_{92}^{239}\text{Th}$

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/11

October/November 2016

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

This document consists of **17** printed pages and **3** blank pages.

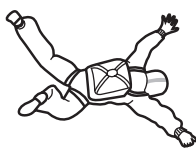


- 1 A length of copper wire is labelled 'length 30 m' and 'diameter 0.50 mm'.

Which instruments are most suitable to measure accurately the length and the diameter of the wire?

	length	diameter
A	rule	calipers
B	rule	micrometer
C	tape	calipers
D	tape	micrometer

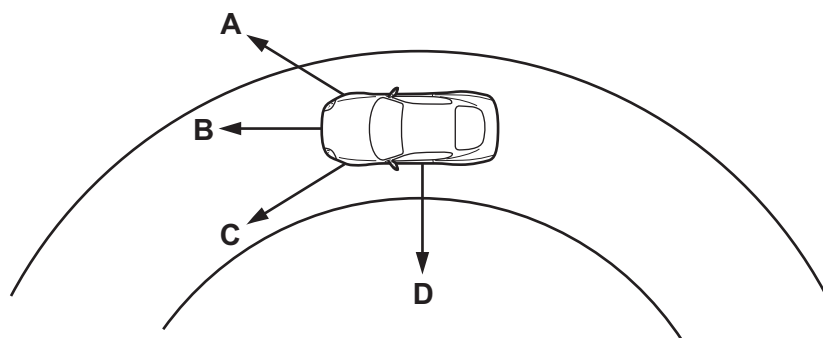
- 2 A skydiver falls from rest through the air and reaches terminal velocity.



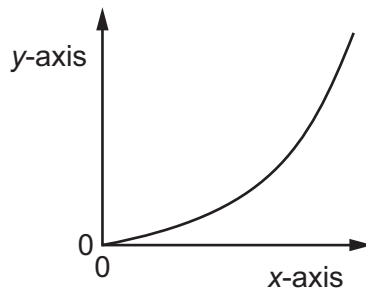
What is the acceleration of the skydiver during his fall?

- A** constant at 0 m/s^2
B constant at 10 m/s^2
C starting at 0 m/s^2 and increasing to 10 m/s^2
D starting at 10 m/s^2 and decreasing to 0 m/s^2
- 3 The diagram shows a car going around a circular track at constant speed.

Which arrow shows the direction of the resultant force on the car?



- 4 A car has stopped at a red light. When the light changes to green, the car starts moving with a constant acceleration. The graph represents this motion.



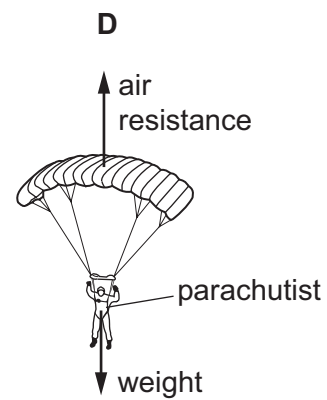
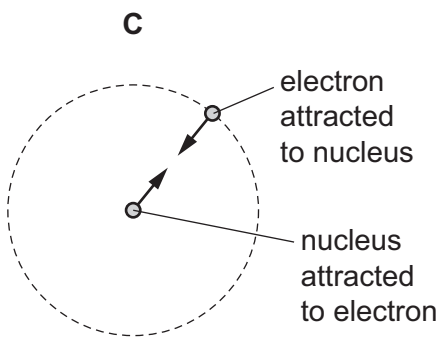
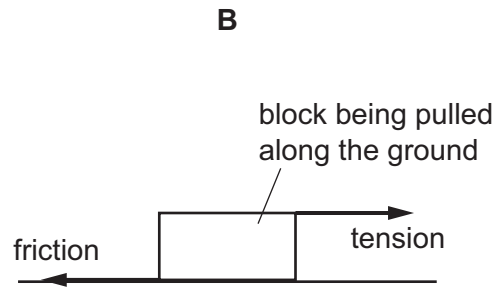
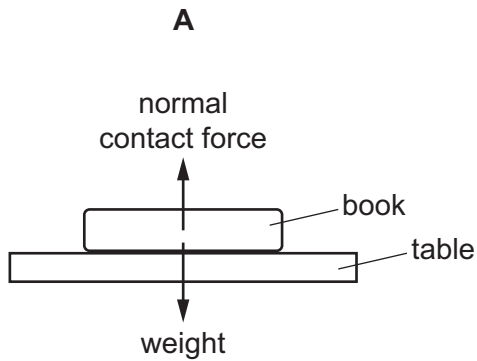
Which quantity is plotted on the x-axis and which quantity is plotted on the y-axis?

	x-axis	y-axis
A	distance	time
B	speed	time
C	time	distance
D	time	speed

- 5 Which statement about mass and weight is correct?
- A** A mass experiences a weight due to a gravitational field.
 - B** Mass and weight are different types of force.
 - C** Mass and weight have the same unit.
 - D** When an object expands, its mass changes but its weight stays the same.

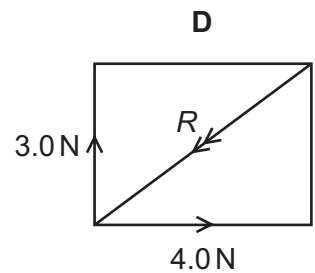
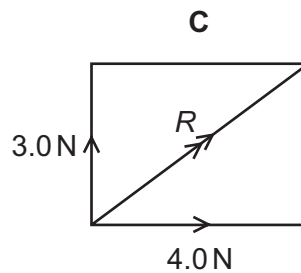
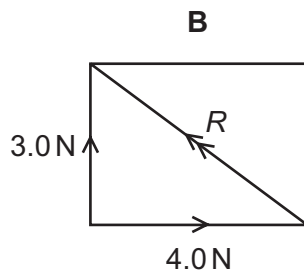
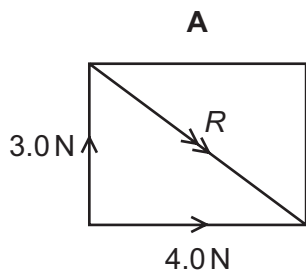
- 6 Each diagram shows two forces that are equal in magnitude.

Which diagram shows two forces that are a Newton's third law pair of forces?

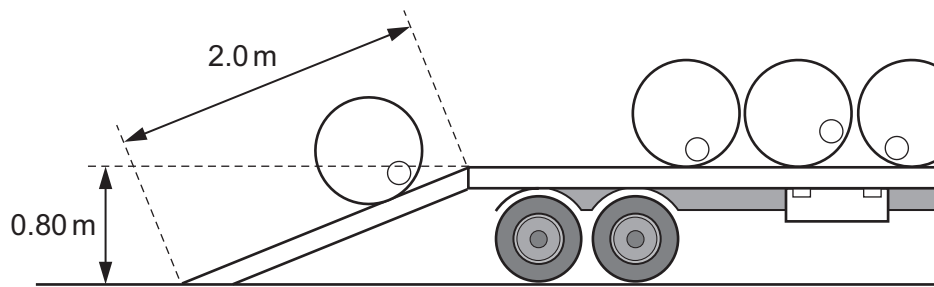


- 7 Two forces of 3.0 N and 4.0 N act at right-angles to each other.

Which diagram shows the resultant R of these forces?



- 8 A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

- A** 1000 J **B** 1600 J **C** 2500 J **D** 4000 J
- 9 Three objects P, Q and R have different masses and different speeds, as shown in the table.

	$\frac{\text{mass}}{\text{kg}}$	$\frac{\text{speed}}{\text{m/s}}$
P	1.0	3.0
Q	2.0	2.0
R	5.0	1.0

What is the order of increasing kinetic energy (smallest first) of the objects?

- A** $P \rightarrow Q \rightarrow R$
B $P \rightarrow R \rightarrow Q$
C $R \rightarrow P \rightarrow Q$
D $R \rightarrow Q \rightarrow P$

- 10 A weightlifter lifts some masses through a height of 1.8 m.



The gravitational field strength g is 10 N/kg .

During the lift, the gravitational potential energy of the masses increases by 4500 J .

What is the total mass lifted?

- A 25 kg B 250 kg C 450 kg D 810 kg
- 11 Which expression is used to calculate power?

- A $\frac{\text{distance}}{\text{work done}}$
- B work done $\times$ distance
- C $\frac{\text{work done}}{\text{time taken}}$
- D work done $\times$ time taken

- 12 The pressure of the atmosphere is $100\,000 \text{ Pa}$.

What force does the atmosphere exert on the upper surface of a pond of surface area 20 m^2 ?

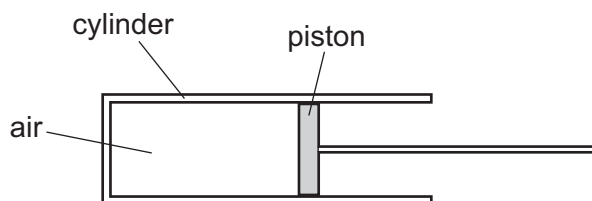
- A 5000 N B 100 000 N C 100 020 N D 2 000 000 N
- 13 Oil of density $8.5 \times 10^2 \text{ kg/m}^3$ is stored in a large tank.

The gravitational field strength g is 10 N/kg .

What is the pressure due to the oil 6.0 m below its surface?

- A 51 Pa B 510 Pa C 5100 Pa D 51 000 Pa

- 14 Air is trapped in a cylinder by a piston.



The piston is pushed further into the cylinder.

How do the pressure and the volume of the air in the syringe change?

	pressure	volume
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

- 15 Which statement explains how a pressure is exerted by a gas on a container?

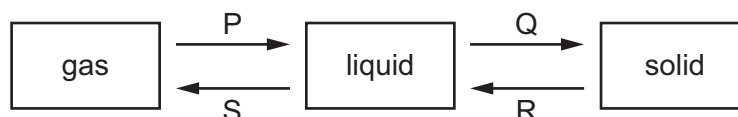
- A** Gas molecules collide with other gas molecules in the container.
- B** Gas molecules collide with the walls of the container.
- C** Gas molecules exert strong attractive forces on each other.
- D** Gas molecules exert strong repulsive forces on each other.

- 16 When a liquid evaporates, molecules escape from its surface.

Which molecules escape, and what happens to the average kinetic energy of the molecules remaining in the liquid?

- A** The less energetic molecules escape and the average kinetic energy decreases.
- B** The less energetic molecules escape and the average kinetic energy increases.
- C** The more energetic molecules escape and the average kinetic energy decreases.
- D** The more energetic molecules escape and the average kinetic energy increases.

17 The diagram shows changes of state.



What are the names of the changes of state P, Q, R and S?

	P	Q	R	S
A	condensation	melting	solidification	boiling
B	condensation	solidification	melting	boiling
C	boiling	melting	solidification	condensation
D	boiling	solidification	melting	condensation

18 A boiling liquid absorbs thermal energy (heat) at a rate of 450 W.

The specific latent heat of vaporisation is $2.7 \times 10^6 \text{ J/kg}$.

How much liquid is vaporised in 9.0 minutes?

- A** 1.5 g **B** 11 g **C** 90 g **D** 5400 g

19 What makes a clinical thermometer suitable for measuring small changes in body temperature?

- A** The amount of liquid in the bulb is small.
B The bore of the capillary tube is narrow.
C The capillary tube is long.
D The glass bulb has a thin wall.

20 Which statement about the transfer of thermal energy is correct?

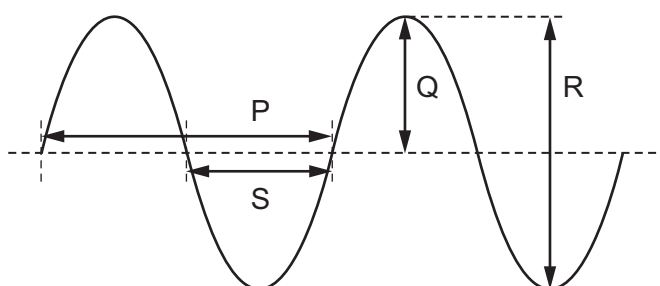
- A** Transfer by radiation does not require a medium.
B Transfer is from a region of lower temperature to one of higher temperature.
C Transfer upwards in fluids is mainly through the vibrations of neighbouring particles.
D Transfer in solids is by means of density changes in the material.

21 A phone rings and lights up.

Which types of wave are associated with the phone's operation?

	light	sound
A	longitudinal	longitudinal
B	longitudinal	transverse
C	transverse	longitudinal
D	transverse	transverse

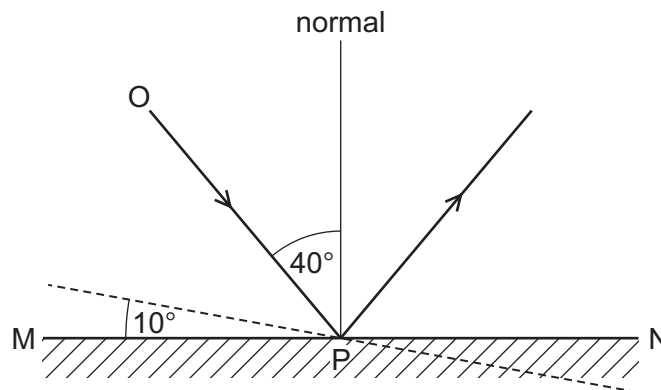
22 The diagram shows the surface of water as a wave passes across a ripple tank.



Which lengths represent the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	Q	P
B	Q	S
C	R	P
D	R	S

- 23 The angle of incidence of ray OP on the plane mirror MN is 40° .

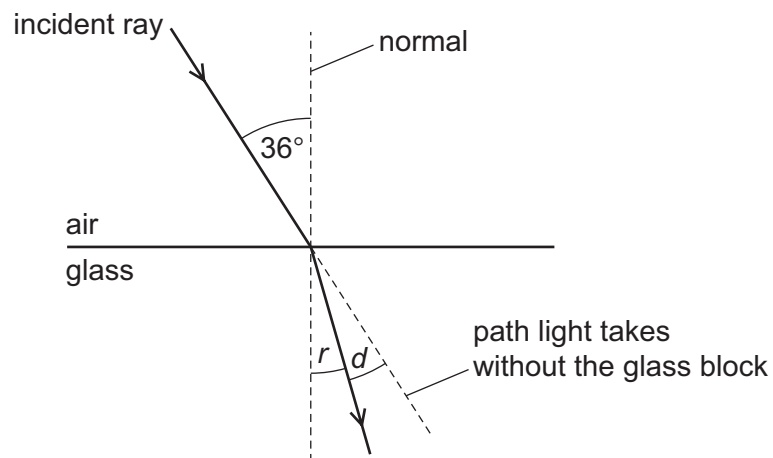


The mirror is rotated through 10° , as shown by the dashed line. The direction of the incident ray OP does not change.

What is the new angle of incidence?

- A** 30° **B** 40° **C** 50° **D** 60°
- 24 A ray of light is incident on the surface of a glass block. The diagram is not drawn to scale.

The refractive index of the glass is 1.5.

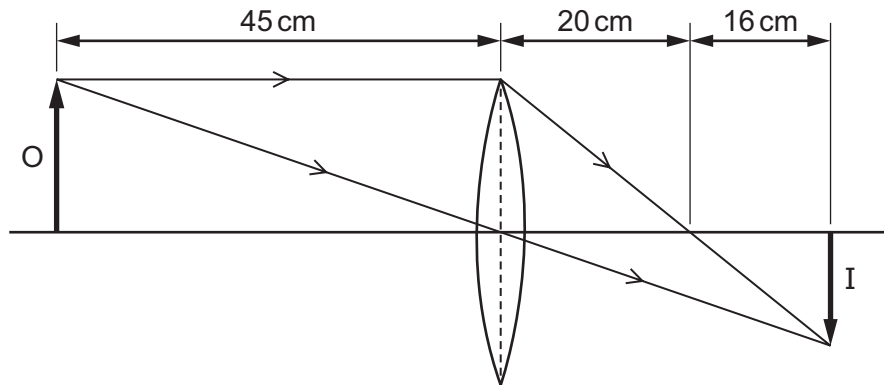


The angle of refraction is r . The angle between the refracted ray and the path the light takes without the glass block is d .

What are r and d ?

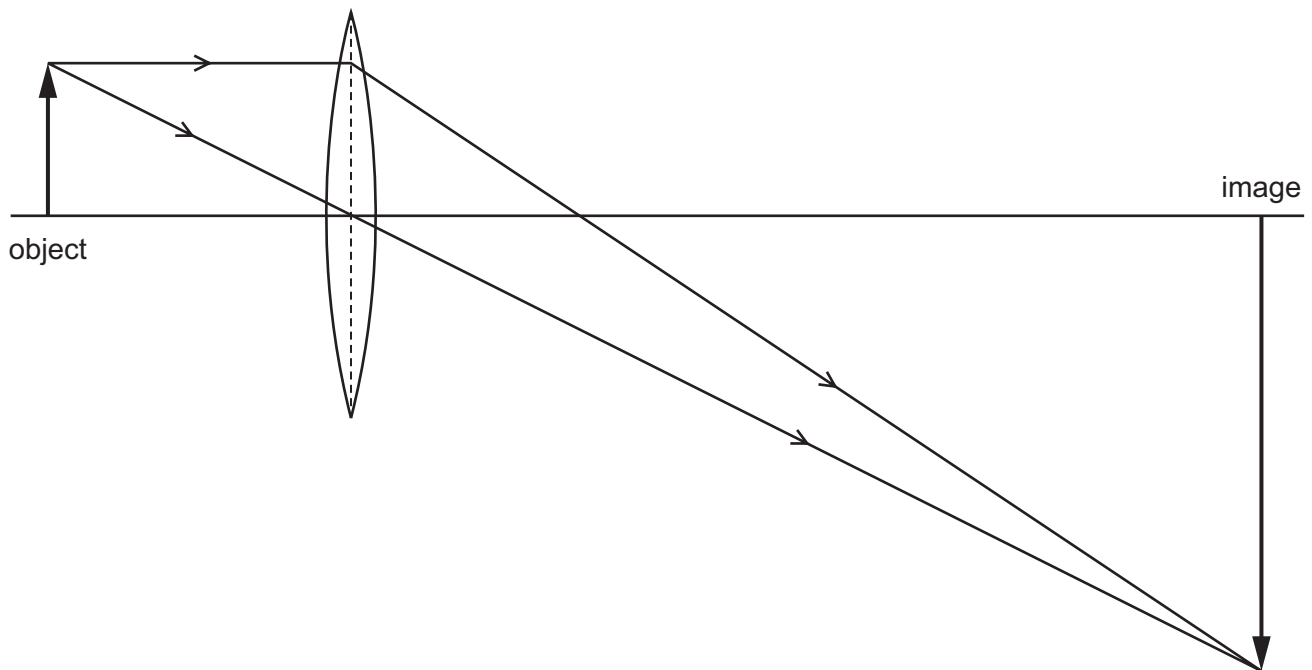
	$r/^\circ$	$d/^\circ$
A	23	12
B	24	12
C	23	13
D	24	13

- 25** In the diagram, a convex lens forms an image I of an object O. The diagram is not drawn to scale.



What is the focal length of the lens?

- A** 16 cm **B** 20 cm **C** 36 cm **D** 45 cm
- 26** A lens is used to produce a magnified image, as shown in the scale diagram.



What is the linear magnification produced by the lens?

- A** 0.33 **B** 3.0 **C** 4.0 **D** 6.0

- 27** Blue and yellow are colours in the visible spectrum.

Which colour has the lower frequency and which colour has the longer wavelength?

	lower frequency	longer wavelength
A	blue	blue
B	blue	yellow
C	yellow	blue
D	yellow	yellow

- 28** Which two types of electromagnetic radiation are both used to kill cancerous cells and are both used to detect cracks in metals?

- A** gamma-rays and microwaves
- B** gamma-rays and X-rays
- C** microwaves and ultra-violet radiation
- D** ultra-violet radiation and X-rays

- 29** Which property of a sound wave increases as the loudness of the sound increases?

- A** amplitude
- B** frequency
- C** speed
- D** wavelength

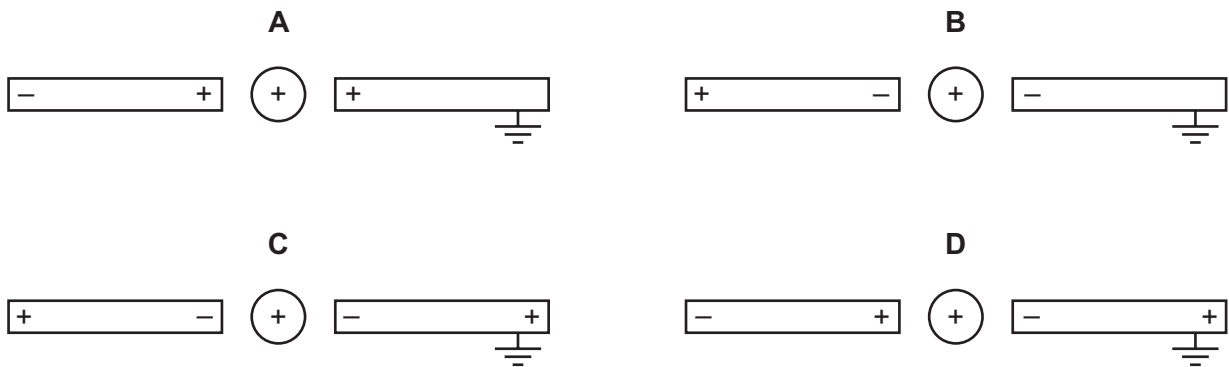
- 30** Delicate instruments are often placed in a box to screen them from stray magnetic fields.

What is the material used for the box and why is it chosen?

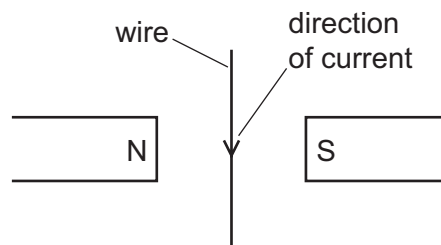
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- B** Copper is used because it has a low electrical resistance.
- C** Polythene is used because it is a good electrical insulator.
- D** Soft iron is used because it is a magnetic material.

- 31** A positively charged metal sphere is placed midway between two previously uncharged metal rods, one of which is connected to earth.

Which diagram shows the charges on the rods?



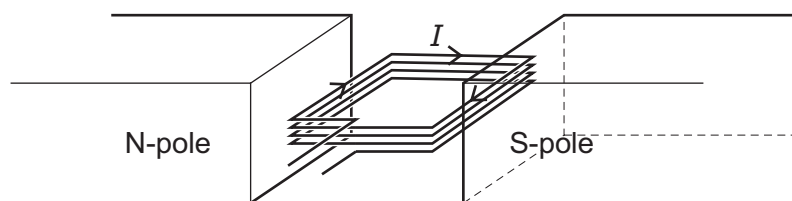
- 32** A current-carrying wire lies between the poles of two magnets, as shown.



What is the direction of the force on the wire?

- A** into the plane of the paper
- B** out of the plane of the paper
- C** towards the left
- D** towards the right

- 33** A coil P of N turns is made from a length L of wire. The coil carries a current I when between two magnetic poles.

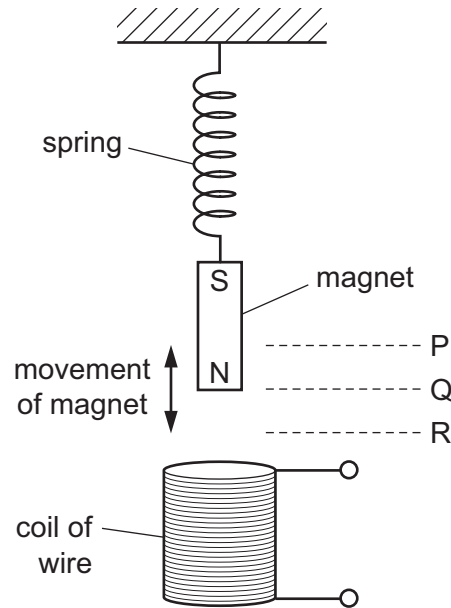


A similar coil Q of $2N$ turns is made from a length $2L$ of identical wire. It also carries a current I when between the two magnetic poles.

Which coil has the greater resistance and which coil experiences the greater turning effect?

	greater resistance	greater turning effect
A	P	P
B	P	Q
C	Q	P
D	Q	Q

- 34 A magnet moves up and down above a coil of wire.



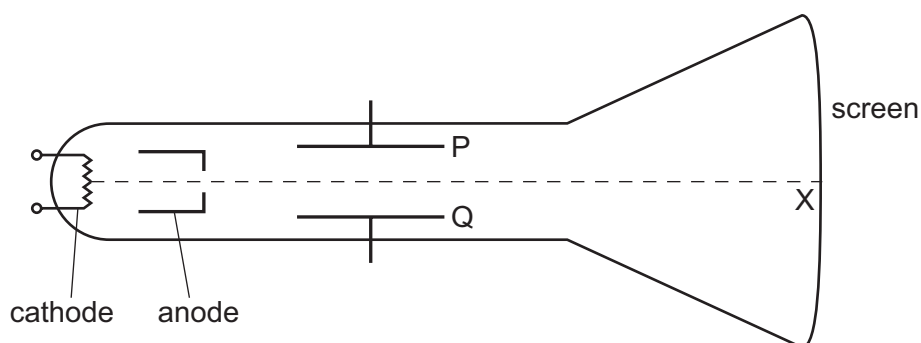
The bottom of the magnet moves up and down between P and R.

Where is the bottom of the magnet when there is no induced electromotive force (e.m.f.) in the coil?

- A at P and at Q
 - B at P and at R
 - C at Q only
 - D at R only
- 35 Which material is most suitable for the core of a transformer and which material is most suitable for the coils in the transformer?

	material for core	material for coils
A	iron	copper
B	iron	steel
C	steel	copper
D	steel	iron

- 36** The diagram shows a cathode-ray tube. The dotted line shows the path of a beam of electrons. The beam hits the screen at point X and produces a bright spot. P and Q are horizontal plates on each side of the beam.

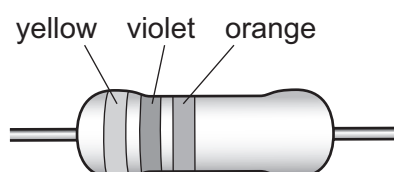


Which change will move the bright spot to a different position on the screen?

- A** Apply a voltage between metal plates P and Q.
 - B** Decrease the positive voltage of the anode.
 - C** Increase the positive voltage of the anode.
 - D** Increase the voltage between the terminals of the cathode.
- 37** The table gives the colour code used for marking resistors.

black	brown	red	orange	yellow	green	blue	violet
0	1	2	3	4	5	6	7

What is the resistance of the resistor in the diagram?



- A** $4.73\ \Omega$
 - B** $473\ \Omega$
 - C** $4700\ \Omega$
 - D** $47\ 000\ \Omega$
- 38** Which types of radiation may be emitted by radioactive nuclei?
- A** beta and gamma
 - B** microwaves and infra-red
 - C** radio waves and microwaves
 - D** ultra-violet and X-rays

- 39** A freshly made sample of radioactive material gives a count rate of 8000 counts per minute. After 20 days, it gives a count rate of 500 counts per minute.

What is the half-life of the material?

- A** 4.0 days
 - B** 5.0 days
 - C** 20 days
 - D** 80 days
- 40** Which number is always equal to the proton number of a neutral atom of a radioactive isotope?
- A** the number of electrons in the atom
 - B** the number of neutrons in the atom
 - C** the number of nucleons in the atom
 - D** the number of particles in the atom

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

5054/12

Paper 1 Multiple Choice

October/November 2016

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2016	5054	12

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	D	21	B
2	D	22	D
3	C	23	B
4	C	24	B
5	A	25	A
6	D	26	A
7	A	27	B
8	B	28	A
9	C	29	A
10	C	30	B
11	D	31	D
12	B	32	B
13	C	33	C
14	B	34	B
15	C	35	C
16	A	36	B
17	D	37	B
18	C	38	C
19	C	39	B
20	C	40	A



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

5054/11

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Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2016	5054	11

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	D	21	C
2	D	22	A
3	D	23	C
4	C	24	C
5	A	25	B
6	C	26	B
7	C	27	D
8	B	28	B
9	D	29	A
10	B	30	D
11	C	31	B
12	D	32	B
13	D	33	D
14	C	34	B
15	B	35	A
16	C	36	A
17	B	37	D
18	C	38	A
19	B	39	B
20	A	40	A



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/12

October/November 2015

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



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Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

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There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

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Any rough working should be done in this booklet.

Electronic calculators may be used.

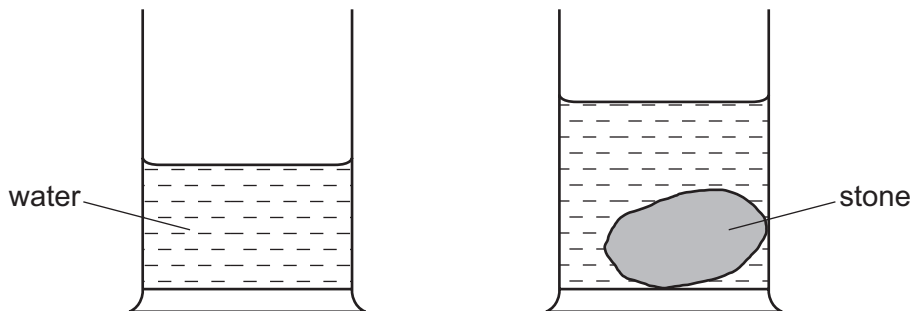
This document consists of **15** printed pages and **1** blank page.



1 Which is a scalar quantity?

- A mass
- B force
- C velocity
- D weight

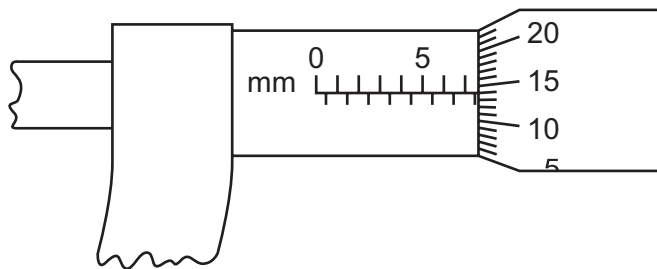
2 During an experiment to find the density of a stone, the stone is lowered into a measuring cylinder partly filled with water.



Which statement is correct?

- A The difference between the readings gives the density of the stone.
- B The difference between the readings gives the volume of the stone.
- C The final reading gives the density of the stone.
- D The final reading gives the volume of the stone.

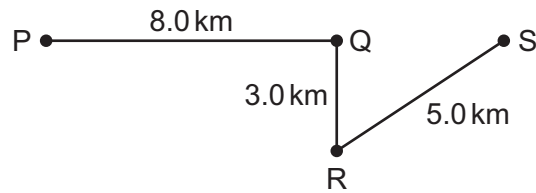
3 The diagram shows a micrometer scale.



Which reading is shown?

- A 5.64 mm
- B 7.14 mm
- C 7.16 mm
- D 7.64 mm

- 4 A lorry takes 15 minutes to travel along the path PQRS.



What is the average speed of the lorry?

- A** 4.0 km/h **B** 22 km/h **C** 48 km/h **D** 64 km/h
- 5 An apple of mass 0.15 kg and weight 1.5 N falls from a tree. At one point during its fall, the air resistance on the apple is 0.60 N upwards.
- What is the acceleration of the apple at this point?
- A** 4.0 m/s² **B** 6.0 m/s² **C** 10 m/s² **D** 14 m/s²
- 6 At time t_1 , a stone is dropped from a stationary balloon. The stone reaches terminal velocity at time t_2 .

Which row gives the acceleration of the stone at time t_1 and at time t_2 ?

	<u>acceleration at t_1</u> m/s ²	<u>acceleration at t_2</u> m/s ²
A	0	0
B	10	0
C	0	10
D	10	10

- 7 Four of the gravitational forces that act between bodies in the Solar System are described below.

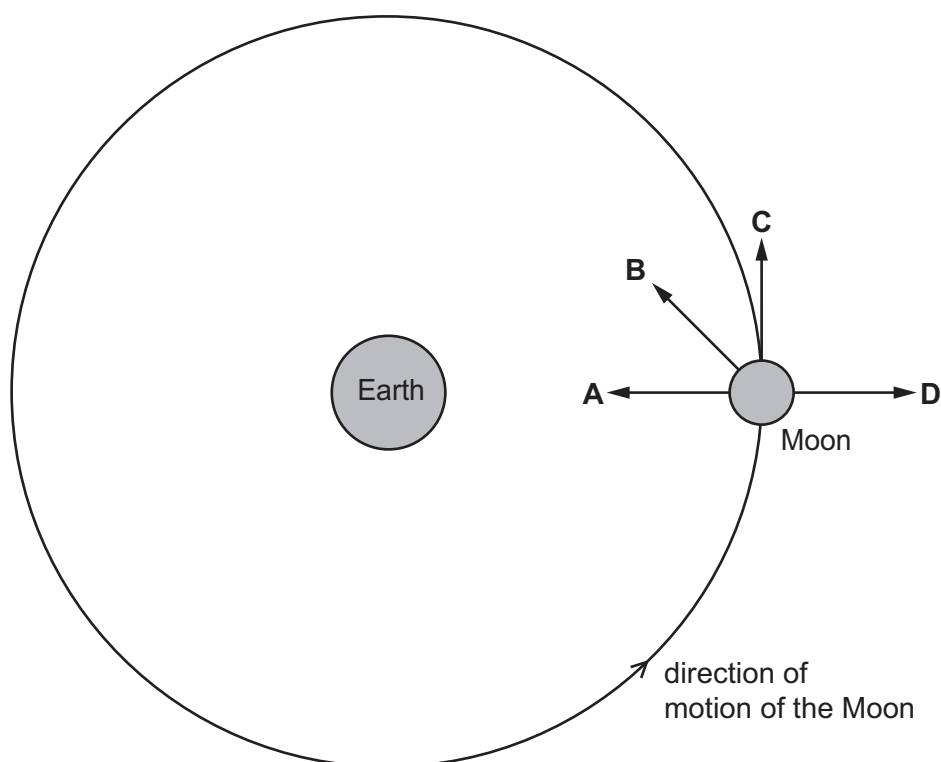
- P the force on the Moon due to the Earth
 Q the force on the Earth due to the Sun
 R the force on the Earth due to the Moon
 S the force on the Moon due to the Sun

Which two forces are a Newton's third law pair (action and reaction)?

- A** P and Q **B** P and R **C** Q and S **D** R and S

- 8 The diagram represents the Moon in its orbit around the Earth.

Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?

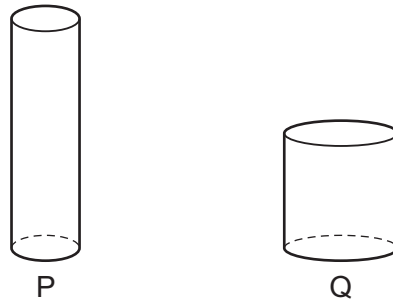


- 9 A student produces some revision notes on gravity, mass and weight.

Which statement is **not** correct?

- A A gravitational field is a region in which a mass experiences a gravitational force.
- B Gravitational field strength has the unit of N/kg .
- C Mass is a measure of the amount of matter.
- D Mass is equal to weight multiplied by gravitational field strength.

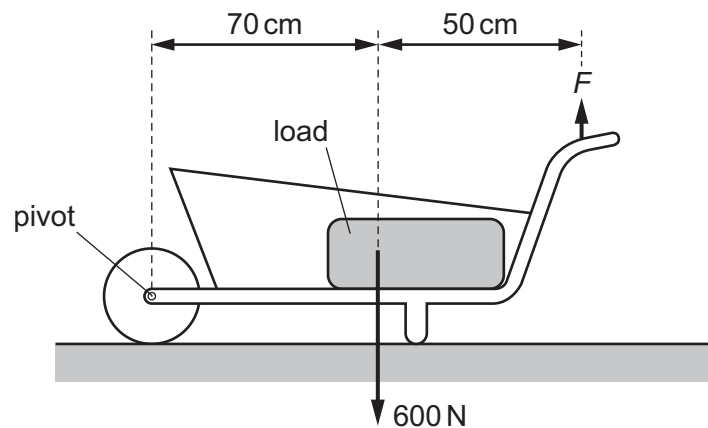
- 10 Two cylinders P and Q are made of copper.



The height of P is twice the height of Q. The diameter of P is half the diameter of Q.

Which statement is correct?

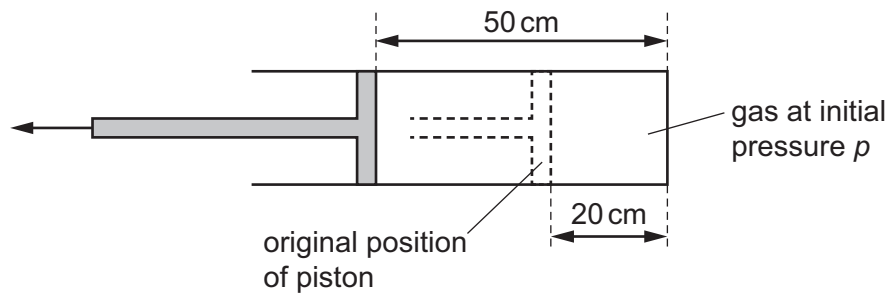
- A The density of cylinder P is four times that of cylinder Q.
 - B The density of cylinder P is twice that of cylinder Q.
 - C The density of cylinder P is equal to that of cylinder Q.
 - D The density of cylinder P is half that of cylinder Q.
- 11 The total weight of the load and the wheelbarrow shown is 600 N.



What is the size of force F needed just to lift the loaded wheelbarrow?

- A 350 N
- B 430 N
- C 600 N
- D 840 N

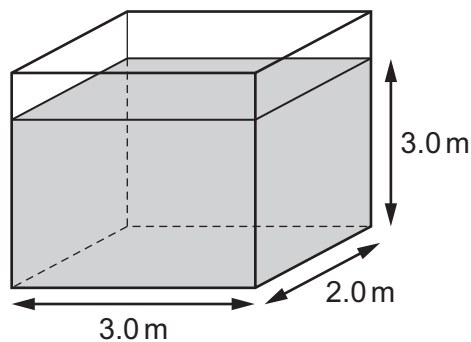
- 12 A gas is trapped inside a cylinder by a movable piston. The length of the gas column is 20 cm and the pressure inside the cylinder is p .



The piston is pulled out a distance of 30 cm, so that the length of the gas column is now 50 cm. The temperature of the gas does not change.

What is the new pressure of the gas?

- A** $0.40p$ **B** $0.60p$ **C** $1.5p$ **D** $2.5p$
- 13 The base of a rectangular storage tank is 2.0 m by 3.0 m. The tank is filled with paraffin to a depth of 3.0 m.



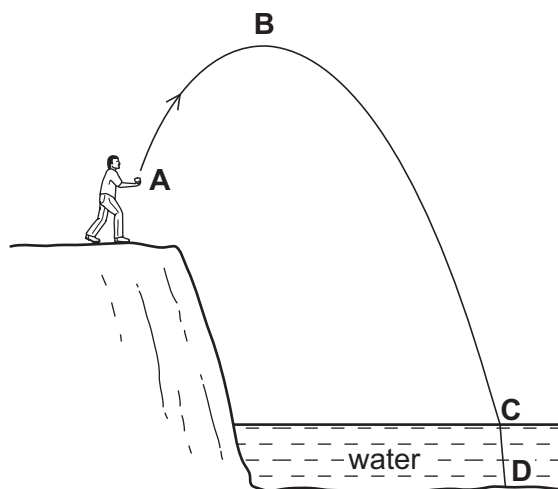
The density of paraffin is 800 kg/m^3 and the gravitational field strength is 10 N/kg .

What is the pressure at the bottom of the tank due to the paraffin?

- A** 2400 Pa **B** 14 400 Pa **C** 24 000 Pa **D** 144 000 Pa

- 14 A person throws a stone so that it follows the path shown in the diagram.

In which position does the stone have the most gravitational potential energy?



- 15 A piston of area 10 cm^2 is pushed slowly into a very large cylinder containing gas at a pressure of 10 N/cm^2 . The pressure of the gas remains constant as the piston moves a distance of 0.10 m .

What is the force of the gas on the piston and what is the work done by the piston on the gas?

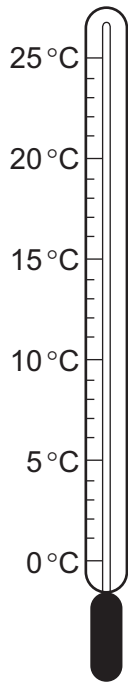
	force / N	work done / J
A	1.0	0.1
B	1.0	10
C	100	10
D	100	1000

- 16 A gas is in a sealed container of constant volume. The gas is heated and the pressure of the gas on the walls of the container increases.

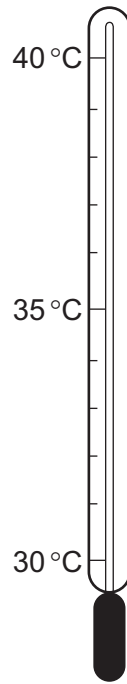
How do the particles of the gas cause this increase in pressure?

- A** They expand.
- B** They hit each other more frequently.
- C** They hit the container more frequently.
- D** They vibrate faster.

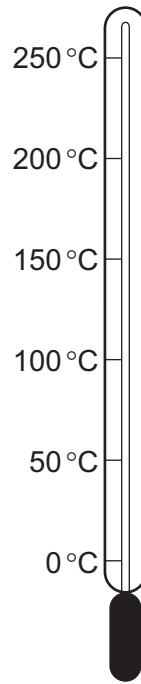
17 The diagrams represent four thermometers.



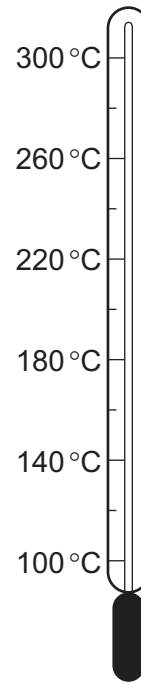
P



Q



R

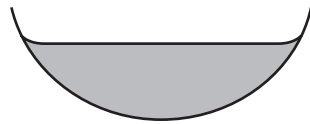


S

Which thermometer has the greatest sensitivity and which thermometer has the greatest range?

	greatest sensitivity	greatest range
A	P	R
B	P	S
C	Q	R
D	Q	S

- 18 Some of the liquid in a dish evaporates, as shown in the diagrams.



before evaporation



after evaporation

Which molecules leave the liquid and which molecules in the liquid have greater average kinetic energy?

	molecules that leave have	molecules in the liquid have greater average kinetic energy
A	high energy	before evaporation
B	high energy	after evaporation
C	low energy	before evaporation
D	low energy	after evaporation

- 19 An insulated beaker contains 300 g of water, initially at 30 °C. Water at 100 °C is added until the temperature of the mixture reaches 50 °C.

The specific heat capacity of water is 4.2 J/(g °C).

How much water is added?

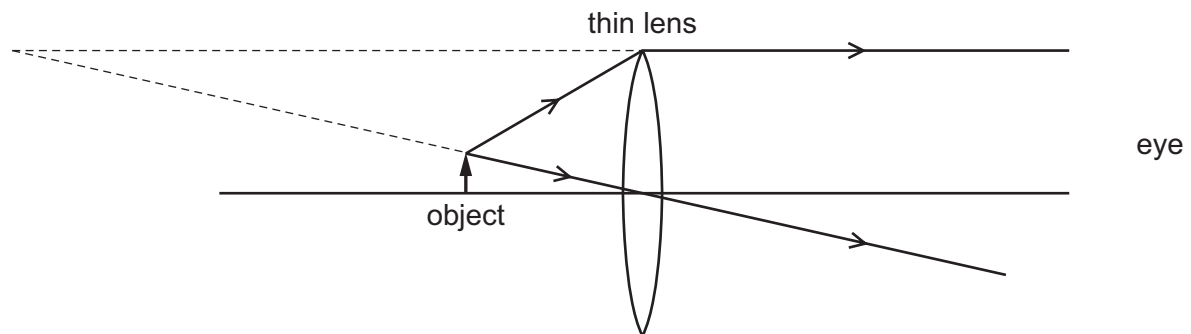
- A** 60 g **B** 120 g **C** 180 g **D** 750 g
- 20 Which statement is correct?
- A** Infra-red radiation cannot travel in a vacuum.
- B** Infra-red radiation cannot travel in solids or in gases.
- C** Infra-red radiation can only travel in a vacuum.
- D** Infra-red radiation can travel in a vacuum and in gases.
- 21 An echo sounder produces ultrasound of frequency 24 kHz. The ultrasound travels in water at a speed of 1.5 km/s.

What is the wavelength in water of ultrasound of this frequency?

- A** 0.063 m **B** 16 m **C** 36 m **D** 63 m

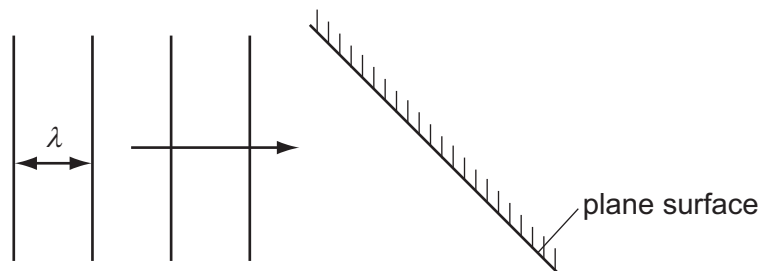
- 22 An object is viewed through a converging lens.

The diagram shows the paths of two rays from the top of the object to an eye.



How does the image compare with the object?

- A It is larger and inverted.
 - B It is larger and upright.
 - C It is smaller and inverted.
 - D It is smaller and upright.
- 23 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	larger	shorter
B	smaller	shorter
C	the same	longer
D	the same	the same

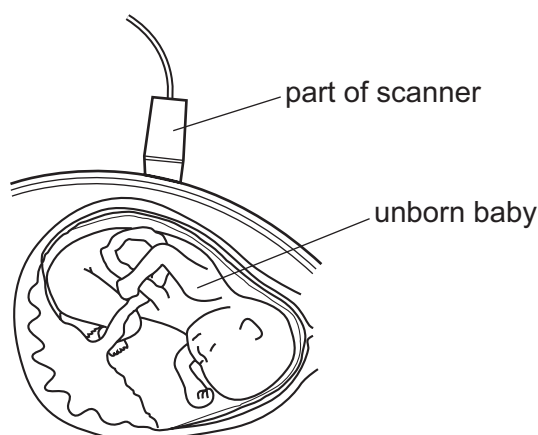
- 24 Which statement about red light and blue light is correct?

- A Red light has a higher frequency than blue light.
- B Red light has a longer wavelength than blue light.
- C Red light has the same speed in glass as blue light.
- D Red light is refracted by a glass prism more than blue light.

- 25 A loudspeaker produces a sound wave of frequency 50 Hz. The amplitude of the sound wave is increased.

What is heard?

- A a louder sound of a higher pitch
 - B a louder sound of the same pitch
 - C a sound of higher pitch but the same loudness
 - D a sound of the same pitch and loudness as before
- 26 An ultrasound scanner produces an image of an unborn baby.



How does the scanner form an image?

- A from ultrasound absorbed by the baby
 - B from ultrasound emitted by the baby
 - C from ultrasound reflected by the baby
 - D from ultrasound refracted by the baby
- 27 The sound from the siren of a ship is reflected by a cliff. An echo is heard by a sailor on the deck of the ship, 4.0 s after the siren is sounded. The speed of sound in air is 320 m/s.

How far from the cliff is the ship?

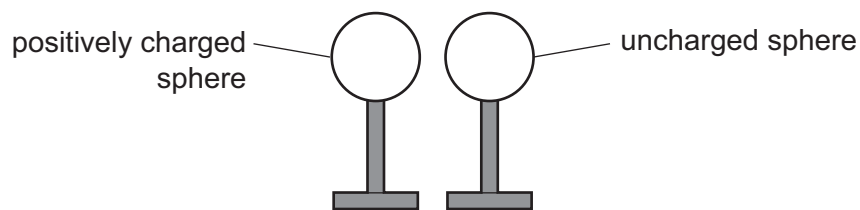
- A 80 m
 - B 160 m
 - C 640 m
 - D 1280 m
- 28 Which material can be picked up by a magnet?

- A aluminium
- B copper
- C iron
- D plastic

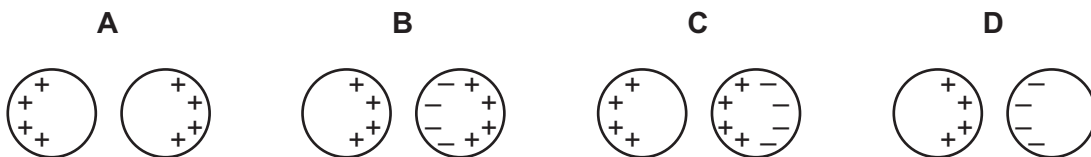
29 Which row correctly names an electrical conductor and an insulator?

	electrical conductor	insulator
A	aluminium	iron
B	iron	rubber
C	plastic	aluminium
D	rubber	plastic

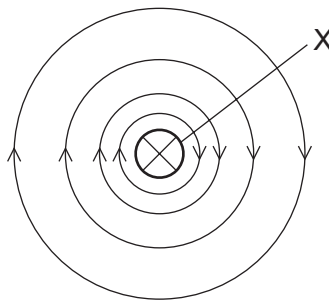
30 A positively charged insulated metal sphere is brought close to an uncharged insulated metal sphere.



Which diagram shows the charge distribution on the spheres?



31 The diagram shows the magnetic field around wire X which carries a current into the paper.

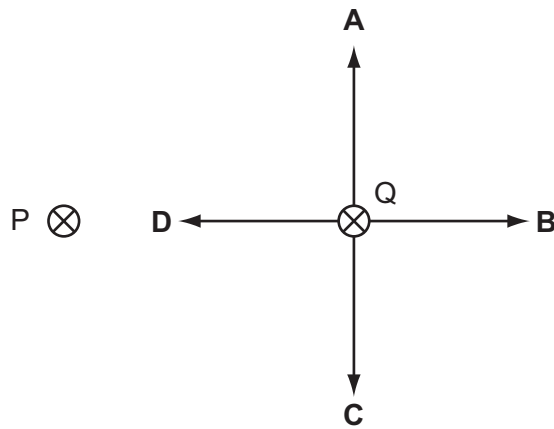


The arrows on the field lines show the direction of the force on

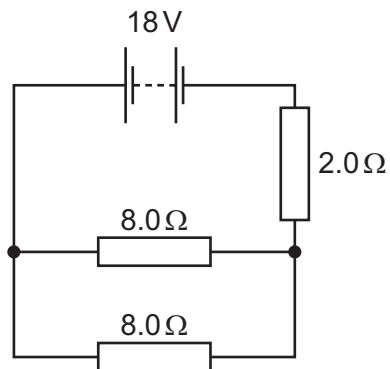
- A** a N-pole.
- B** a S-pole.
- C** a small negative charge.
- D** a small positive charge.

- 32 P and Q represent two, parallel, straight wires carrying currents into the plane of the paper. P and Q exert a force on each other.

Which arrow shows the force on Q?



- 33 A power supply of 18 V is connected to three resistors, as shown.



What is the potential difference across the $2.0\ \Omega$ resistor?

- A** 2.0 V **B** 3.6 V **C** 6.0 V **D** 12 V
- 34 An immersion heater is labelled 24 V, 120 W.

What is the current in the heater when it is connected to a 24 V supply?

- A** 0.20 A **B** 5.0 A **C** 24 A **D** 120 A

- 35** Electric power cables transmit electrical energy over large distances using high-voltage, alternating current.

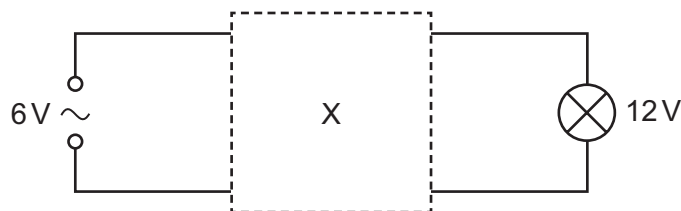
What are the advantages of using a high voltage and of using an alternating current?

	advantage of using a high voltage	advantage of using an alternating current
A	high current is produced in the cable	the resistance of the cable is reduced
B	high current is produced in the cable	the voltage can be changed using a transformer
C	less energy is wasted in the cable	the resistance of the cable is reduced
D	less energy is wasted in the cable	the voltage can be changed using a transformer

- 36** Which particles are emitted by the hot filament of a cathode-ray oscilloscope (c.r.o.) and which type of field is used to accelerate them?

	particles	type of field
A	electrons	electric
B	electrons	magnetic
C	protons	electric
D	protons	magnetic

- 37** The diagram shows an electrical device X connected between a 6 V a.c. supply and a 12 V lamp

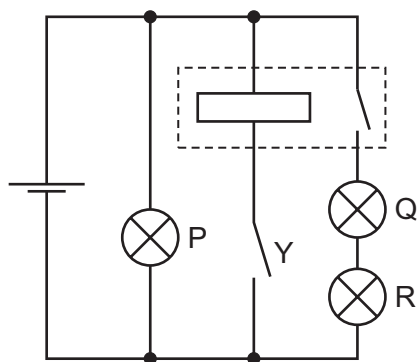


The lamp is seen to glow with normal brightness.

What is X?

- A** a capacitor
- B** a potential divider
- C** a relay
- D** a transformer

- 38 In the circuit shown, all lamps are identical. Lamp P lights with normal brightness.



Switch Y is closed and lamps come on.

Which row indicates the brightness of the lamps?

	P	Q	R
A	dim	dim	dim
B	normal	dim	dim
C	normal	off	off
D	off	normal	normal

- 39 A radioactive isotope ^{14}C emits beta-particles.

A sample of this isotope is left for a period of time that is equal to its half-life.

Which two quantities decrease to **half** of their initial value during this time?

- A** the decay rate and the number of protons
 - B** the mass of the sample and the number of ^{14}C nuclei
 - C** the number of ^{14}C nuclei and the decay rate
 - D** the number of protons and the mass of the sample
- 40 In the Geiger-Marsden experiment, a beam of alpha-particles is fired at a very thin sheet of gold foil, in a vacuum.
- What is deduced from this experiment?
- A** Alpha-particles are repelled by electrons.
 - B** Atoms contain air.
 - C** Electrons are found in atomic nuclei.
 - D** Nuclei are much smaller than atoms.

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/11

October/November 2015

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

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Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

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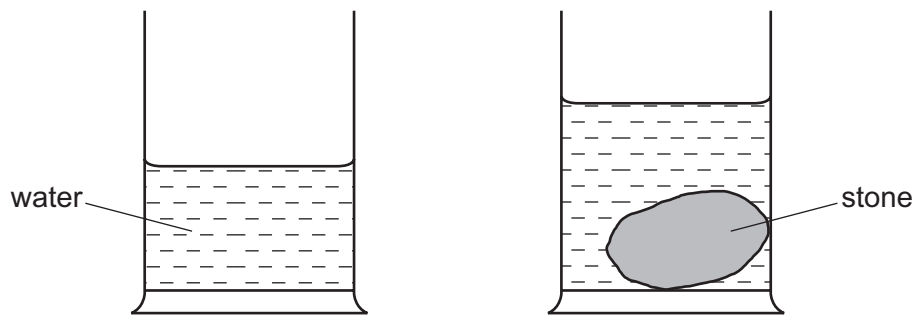
Any rough working should be done in this booklet.

Electronic calculators may be used.

This document consists of **16** printed pages.



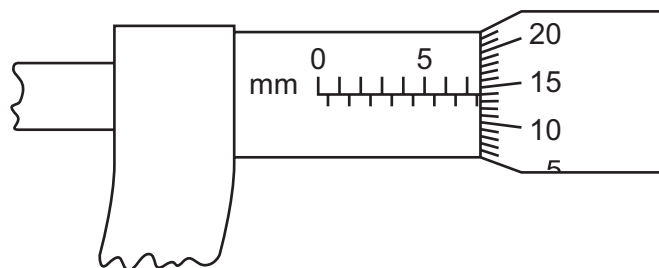
- 1 During an experiment to find the density of a stone, the stone is lowered into a measuring cylinder partly filled with water.



Which statement is correct?

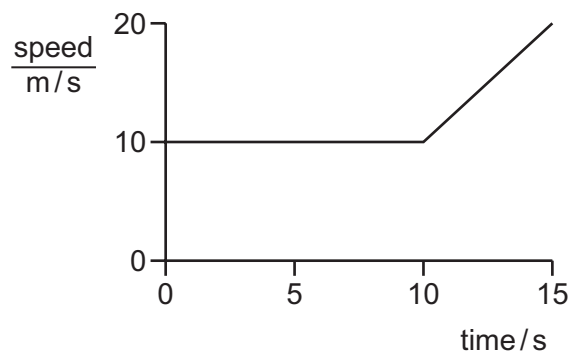
- A The difference between the readings gives the density of the stone.
- B The difference between the readings gives the volume of the stone.
- C The final reading gives the density of the stone.
- D The final reading gives the volume of the stone.

- 2 The diagram shows a micrometer scale.



Which reading is shown?

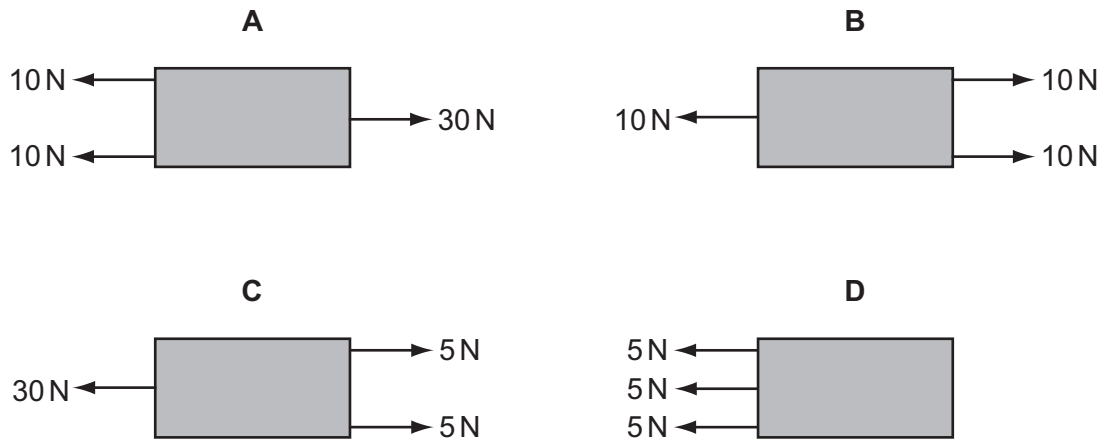
- A 5.64 mm
 - B 7.14 mm
 - C 7.16 mm
 - D 7.64 mm
- 3 An object travels at a constant speed of 10 m/s for 10 s. During the next 5 s, it accelerates uniformly to 20 m/s.



What is the total distance travelled by the object?

- A 150 m
- B 175 m
- C 200 m
- D 300 m

4 Which object has the largest resultant force?

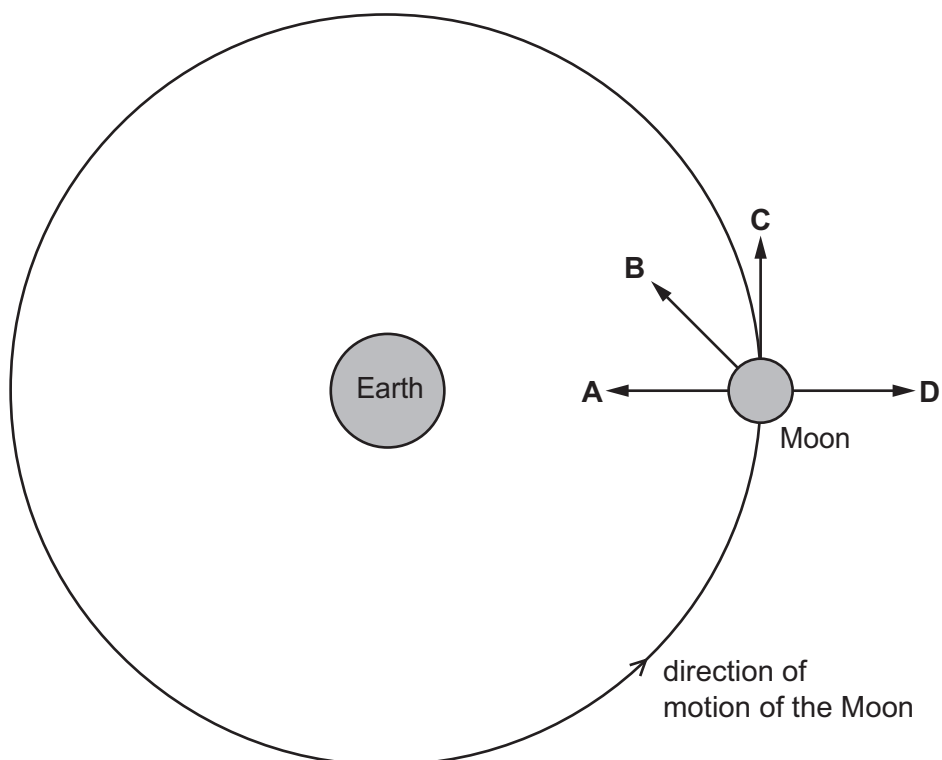


5 In which example does friction act in the direction of forward motion of the object on which it acts?

- A on a suitcase accelerating on a conveyor belt
- B on sandpaper rubbing a wooden block
- C on the skis of a skier accelerating downhill on smooth snow
- D on the tyres of a braking car

6 The diagram represents the Moon in its orbit around the Earth.

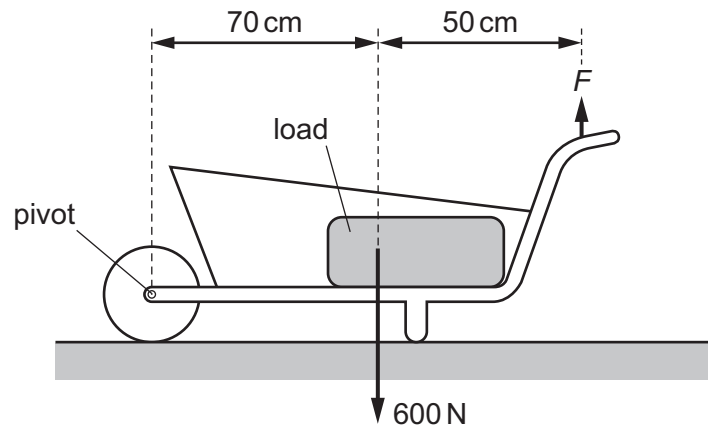
Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?



7 Which statement about mass is correct?

- A Mass is density divided by volume.
- B Mass is the amount of space occupied by a body.
- C Mass is weight multiplied by the gravitational field strength.
- D Mass resists a change in motion.

8 The total weight of the load and the wheelbarrow shown is 600 N.



What is the size of force F needed just to lift the loaded wheelbarrow?

- A 350 N
- B 430 N
- C 600 N
- D 840 N

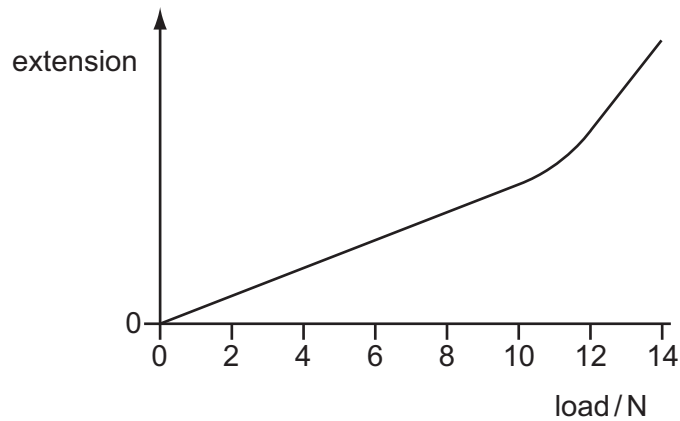
9 A force acts on a body.

Which properties of the body may all be changed by the force?

- A mass, shape and size
- B mass, shape and velocity
- C mass, size and velocity
- D shape, size and velocity

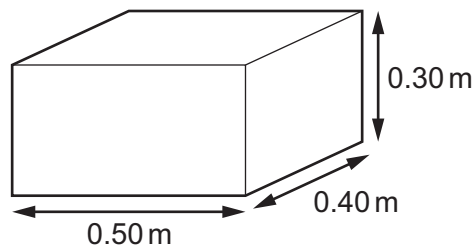
- 10** Some students plan to use a spring to make a spring balance with a linear scale.

The graph shows how the extension of their spring changes with the load on it.



What is the total range of a balance with a linear scale using this spring?

- A** 0 N to 10 N **B** 0 N to 12 N **C** 10 N to 12 N **D** 10 N to 14 N
- 11** A block of weight 900 N has rectangular faces. The diagram shows the lengths of the sides.

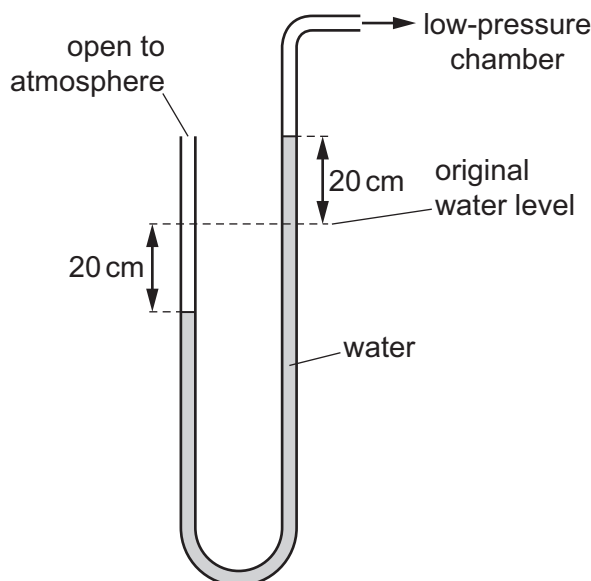


The block can rest on any of its faces.

What is the minimum pressure that the block can exert on the ground when resting on one of its faces?

- A** 900 Pa **B** 4500 Pa **C** 6000 Pa **D** 7500 Pa

- 12 A U-tube containing water is used as a manometer.



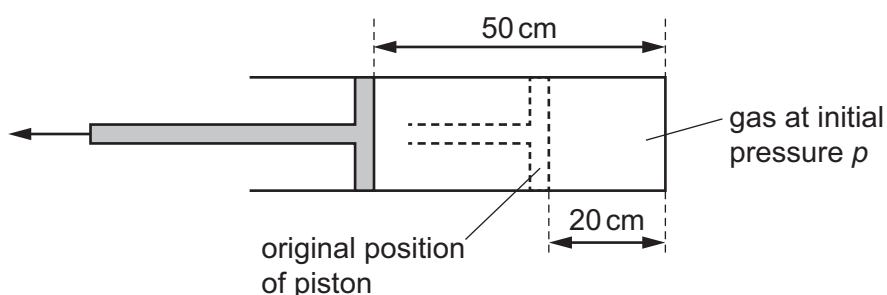
When one end of the manometer is connected to a low-pressure chamber, both water levels in the manometer change by 20 cm. The gravitational field strength g is 10 N/kg .

The density of water is 1000 kg/m^3 .

How far below atmospheric pressure is the pressure in this chamber?

- A** 2000 Pa **B** 4000 Pa **C** 200 000 Pa **D** 400 000 Pa

- 13 A gas is trapped inside a cylinder by a movable piston. The length of the gas column is 20 cm and the pressure inside the cylinder is p .



The piston is pulled out a distance of 30 cm, so that the length of the gas column is now 50 cm. The temperature of the gas does not change.

What is the new pressure of the gas?

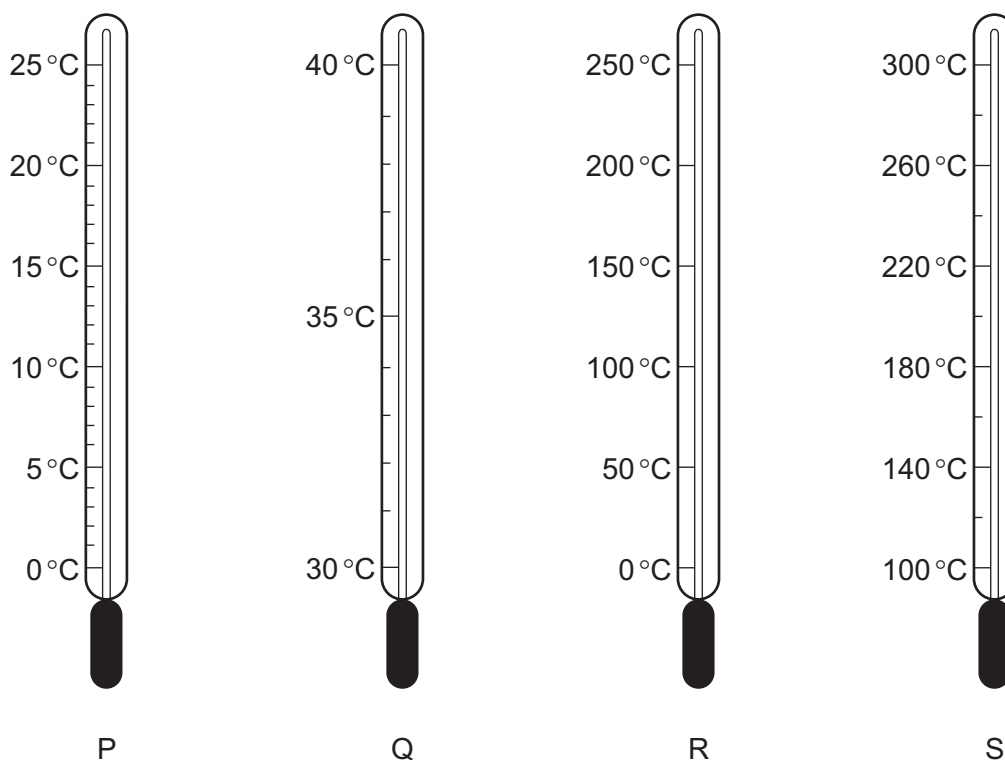
- A** $0.40p$ **B** $0.60p$ **C** $1.5p$ **D** $2.5p$

- 14 The input power to a lamp is 6.0 W . The lamp wastes 2.7 J of energy in 3.0 s .

What is the efficiency of the lamp?

- A** 0.15 **B** 0.45 **C** 0.55 **D** 0.85

15 The diagrams represent four thermometers.



Which thermometer has the greatest sensitivity and which thermometer has the greatest range?

	greatest sensitivity	greatest range
A	P	R
B	P	S
C	Q	R
D	Q	S

16 Which type of thermometer is used for measuring temperatures above 1000°C and which type is used for measuring temperatures that change rapidly?

	measures temperatures above 1000°C	measures temperatures that change rapidly
A	liquid-in-glass	liquid-in-glass
B	liquid-in-glass	thermocouple
C	thermocouple	liquid-in-glass
D	thermocouple	thermocouple

- 17 An electric shower takes in cold water at 17°C . The shower gives 6000 J of energy every second to the cold water and heats it to 37°C . The specific heat capacity of water is $4200\text{ J}/(\text{kg }^{\circ}\text{C})$.

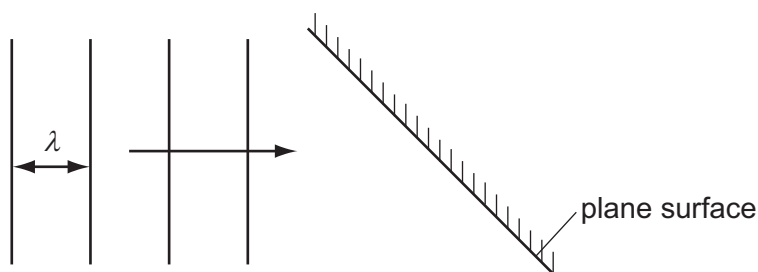
What is the mass of hot water supplied by the shower in one second?

- A** 0.035 kg **B** 0.039 kg **C** 0.071 kg **D** 0.084 kg

- 18 What is latent heat of vaporisation?

- A** the energy required to make molecules expand
B the energy required to make molecules expand and move apart
C the energy required to make molecules move apart
D the energy required to make molecules move faster

- 19 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



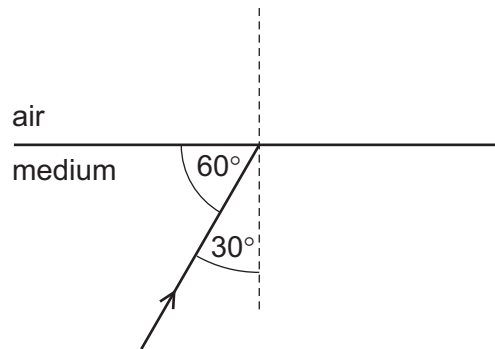
Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	larger	shorter
B	smaller	shorter
C	the same	longer
D	the same	the same

- 20 Which statement about red light and blue light is correct?

- A** Red light has a higher frequency than blue light.
B Red light has a longer wavelength than blue light.
C Red light has the same speed in glass as blue light.
D Red light is refracted by a glass prism more than blue light.

- 21** A ray of light in a transparent medium of refractive index 1.8 is incident on the surface as shown. The light enters air.



What is the angle between the refracted ray and the normal in air?

- A** 29° **B** 33° **C** 54° **D** 64°
- 22** A musical note is produced by two sources.
- The traces produced by each source on the screen of a cathode-ray oscilloscope (c.r.o.) are shown below.



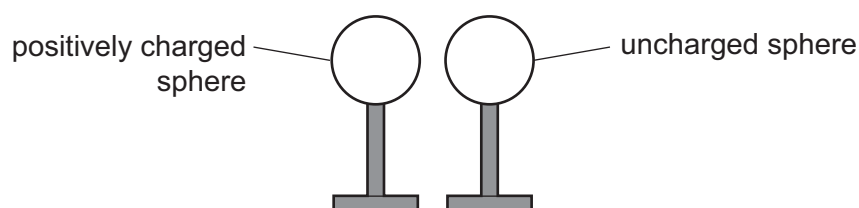
What is the difference between the sounds?

- A** the amplitude
B the frequency
C the pitch
D the quality
- 23** What is a suitable metal for the core of an electromagnet?
- A** aluminium
B copper
C iron
D steel

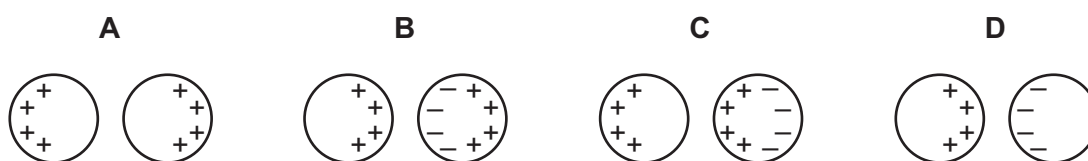
- 24** A polythene rod becomes charged when rubbed with a dry woollen cloth.

The charge caused is a result of

- A** friction producing a movement of electrons.
 - B** friction producing a movement of protons.
 - C** magnetism producing a movement of electrons.
 - D** magnetism producing a movement of protons.
- 25** A positively charged insulated metal sphere is brought close to an uncharged insulated metal sphere.



Which diagram shows the charge distribution on the spheres?

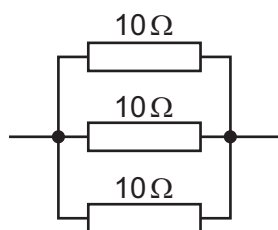


- 26** The flash on a camera operates when a capacitor discharges through a flash-tube. In 3.2 ms, a charge of 6.0 C passes through the flash-tube.

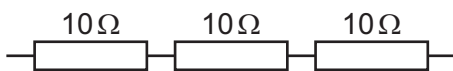
What is the average current in the flash-tube?

- A** 0.019 A
 - B** 1.9 A
 - C** 19 A
 - D** 1900 A
- 27** Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d.)?
- A** current
 - B** electromotive force (e.m.f.)
 - C** energy
 - D** power

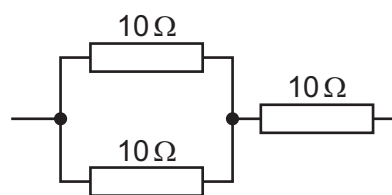
28 Three 10Ω resistors are connected together to form networks X, Y and Z.



network X



network Y

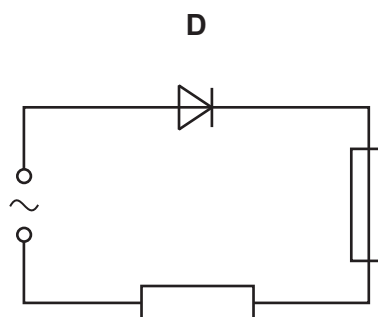
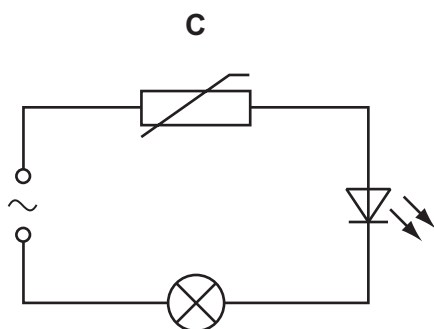
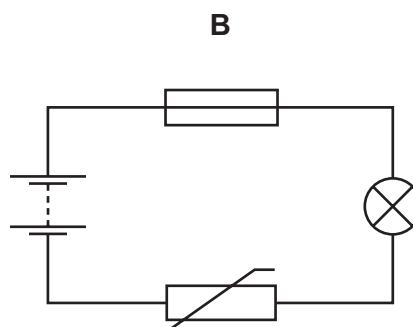
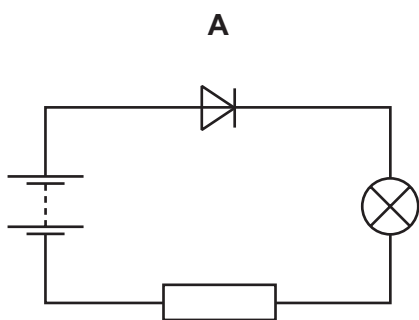


network Z

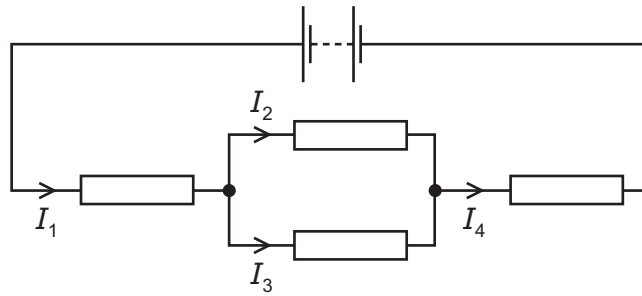
What is the order of resistance of the networks going from the smallest total resistance to the largest total resistance?

- A** $X \rightarrow Y \rightarrow Z$ **B** $X \rightarrow Z \rightarrow Y$ **C** $Z \rightarrow X \rightarrow Y$ **D** $Z \rightarrow Y \rightarrow X$

29 Which circuit contains a fuse and a diode?



30 Four resistors of unequal resistances are connected to a power supply as shown.

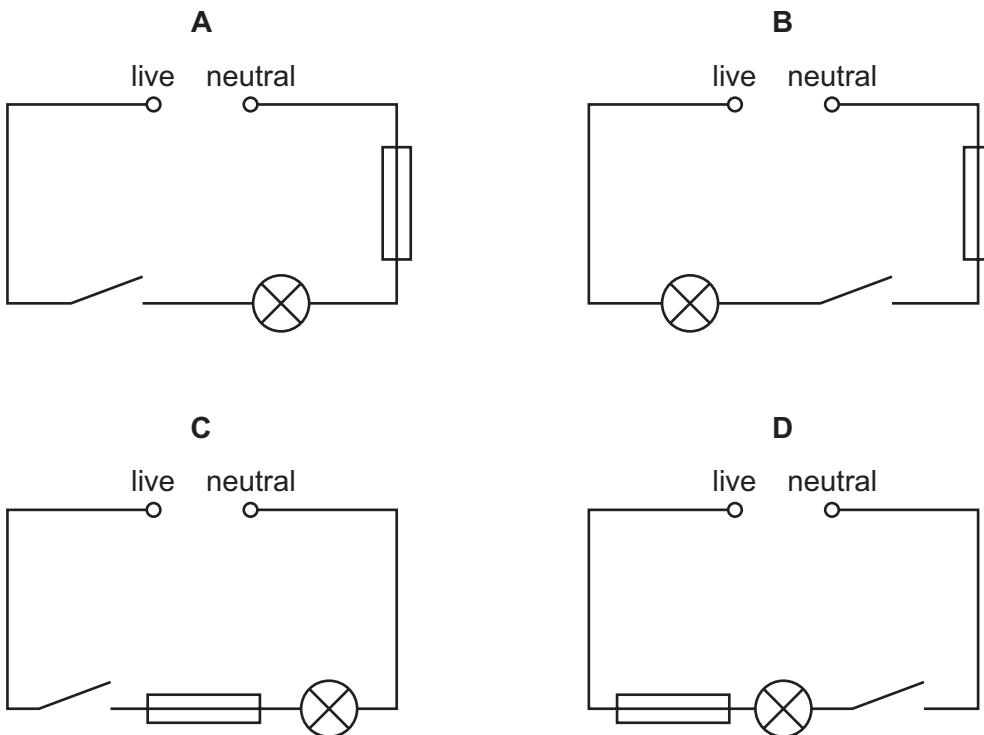


The currents in the four resistors are I_1 , I_2 , I_3 and I_4 .

Which equation is correct?

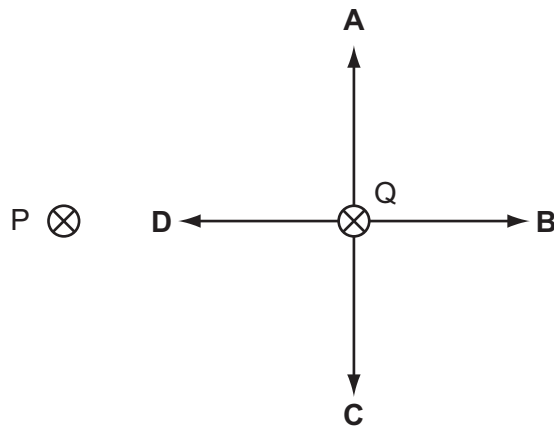
- A** $I_1 = I_2 - I_3$
- B** $I_2 = I_1 + I_4$
- C** $I_3 = I_4 - I_1$
- D** $I_4 = I_2 + I_3$

31 Which diagram shows a lamp wired correctly to the mains supply in a house?

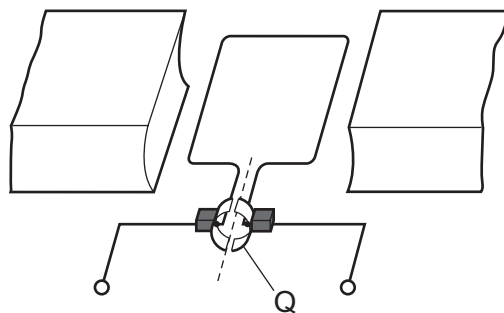


- 32 P and Q represent two, parallel, straight wires carrying currents into the plane of the paper. P and Q exert a force on each other.

Which arrow shows the force on Q?



- 33 The diagram shows a simple d.c. motor.

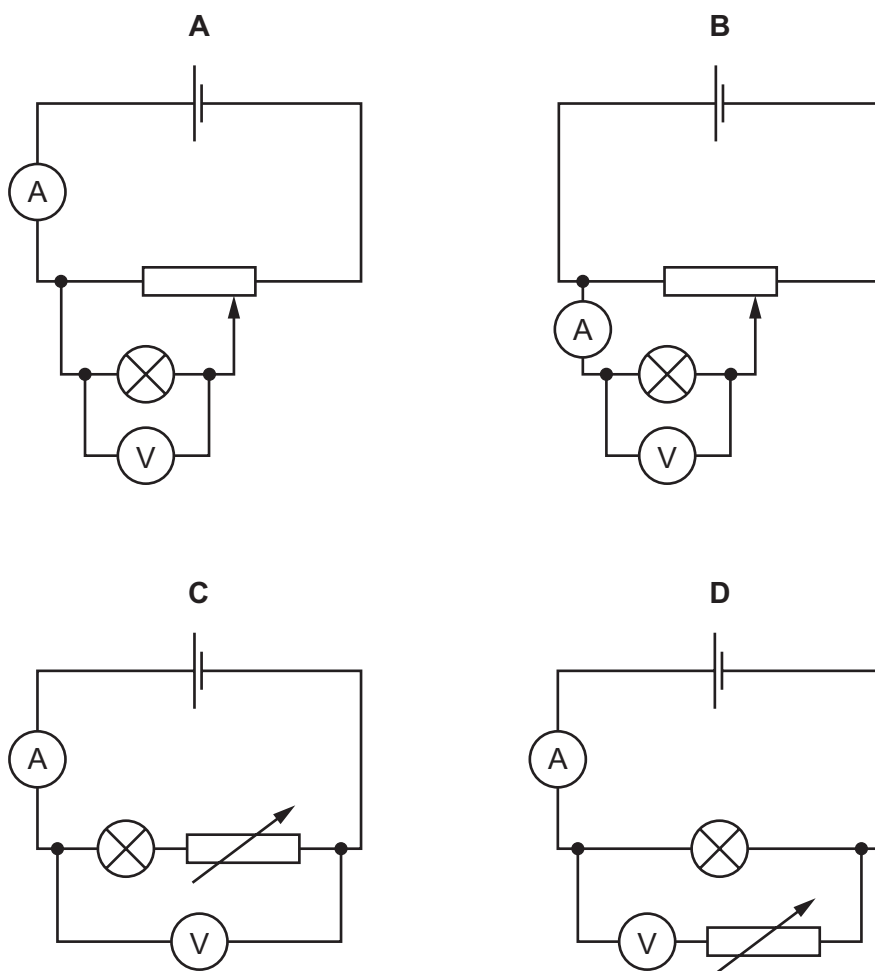


What is the part labelled Q?

- A a coil
- B a commutator
- C a magnet
- D a slip ring

- 34** An experiment is set up to investigate how the current in a filament lamp changes with the potential difference across it.

Which circuit is correct?

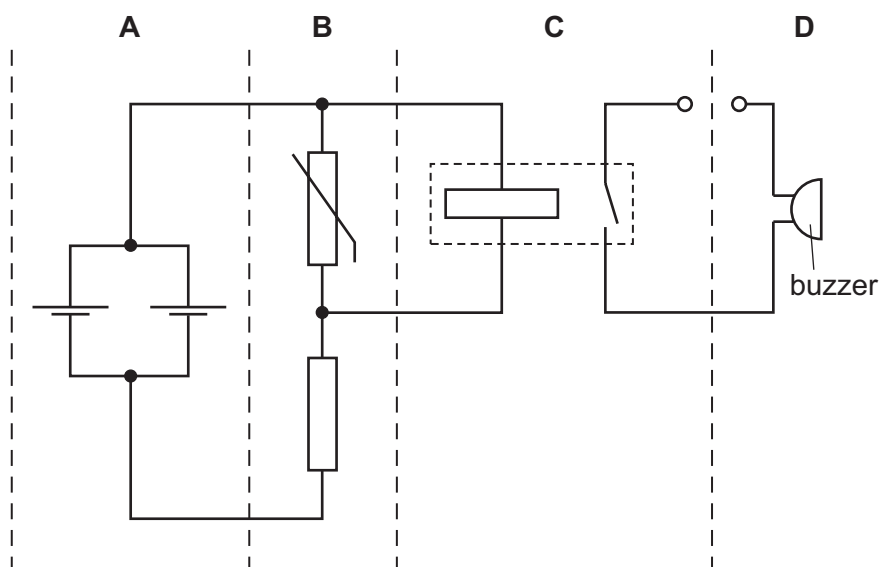


- 35** Electric power cables transmit electrical energy over large distances using high-voltage, alternating current.

What are the advantages of using a high voltage and of using an alternating current?

	advantage of using a high voltage	advantage of using an alternating current
A	high current is produced in the cable	the resistance of the cable is reduced
B	high current is produced in the cable	the voltage can be changed using a transformer
C	less energy is wasted in the cable	the resistance of the cable is reduced
D	less energy is wasted in the cable	the voltage can be changed using a transformer

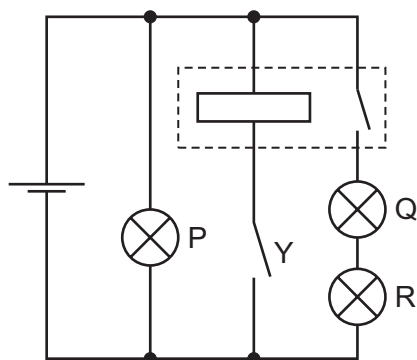
36 Which section of the circuit contains a potential divider?



37 Which particles are emitted by the hot filament of a cathode-ray oscilloscope (c.r.o.) and which type of field is used to accelerate them?

	particles	type of field
A	electrons	electric
B	electrons	magnetic
C	protons	electric
D	protons	magnetic

38 In the circuit shown, all lamps are identical. Lamp P lights with normal brightness.



Switch Y is closed and lamps come on.

Which row indicates the brightness of the lamps?

	P	Q	R
A	dim	dim	dim
B	normal	dim	dim
C	normal	off	off
D	off	normal	normal

39 Which type of electromagnetic radiation is produced during radioactive decay?

- A** alpha-particles
- B** beta-particles
- C** gamma-rays
- D** X-rays

40 Which row states where nuclear fusion occurs and what nuclear fusion is?

	nuclear fusion occurs in	nuclear fusion is
A	a power station	the joining of small nuclei
B	a power station	the splitting of large nuclei
C	a star	the joining of small nuclei
D	a star	the splitting of large nuclei

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Cambridge Ordinary Level

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5054 PHYSICS

5054/12

Paper 1 (Multiple Choice), maximum raw mark 40

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Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2015	5054	12

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	A	21	A
2	B	22	B
3	D	23	D
4	D	24	B
5	B	25	B
6	B	26	C
7	B	27	C
8	A	28	C
9	D	29	B
10	C	30	B
11	A	31	A
12	A	32	D
13	C	33	C
14	B	34	B
15	C	35	D
16	C	36	A
17	C	37	D
18	A	38	B
19	B	39	C
20	D	40	D

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5054 PHYSICS

5054/11

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Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2015	5054	11

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	B	21	D
2	D	22	D
3	B	23	C
4	C	24	A
5	A	25	B
6	A	26	D
7	D	27	C
8	A	28	B
9	D	29	D
10	A	30	D
11	B	31	C
12	B	32	D
13	A	33	B
14	D	34	B
15	C	35	D
16	D	36	B
17	C	37	A
18	C	38	B
19	D	39	C
20	B	40	C



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2025

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

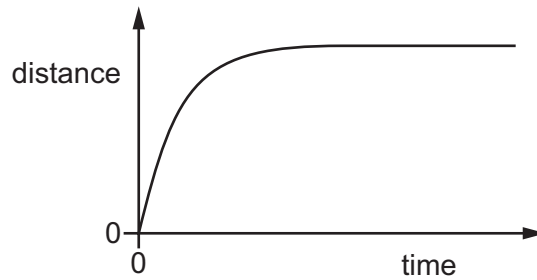
This document has **16** pages.



1 Which list contains **only** vector quantities?

- A mass, distance, weight
- B momentum, gravitational field strength, displacement
- C energy, force, velocity
- D acceleration, time, speed

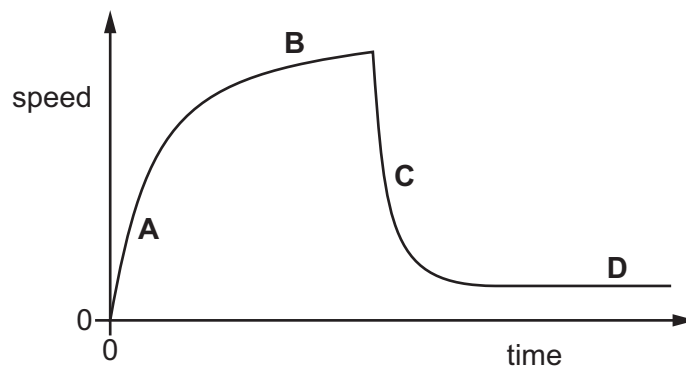
2 The diagram shows the distance–time graph for a moving object.



What is a description of the moving object?

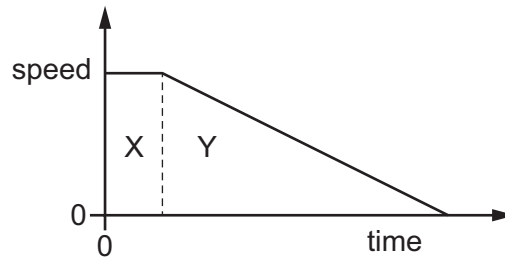
- A a ball thrown vertically upwards and falling back to the thrower
 - B a car starting from rest and speeding up as it moves away from a traffic signal
 - C a rock dropped from a high cliff and falling into the sea below
 - D a train braking to a halt as it stops at a station
- 3 The speed–time graph for a falling skydiver is shown. As he falls, the skydiver spreads out his arms and legs and then opens his parachute.

Which part of the graph shows the skydiver falling with terminal velocity?



- 4 A car driver travelling at constant speed in a straight line sees a hazard in the road ahead at time = 0.

The speed–time graph shows how the speed of the car changes from time = 0.



Three students suggest what area X and area Y represent.

student 1 Area X represents the thinking distance.

student 2 Area Y represents the stopping distance.

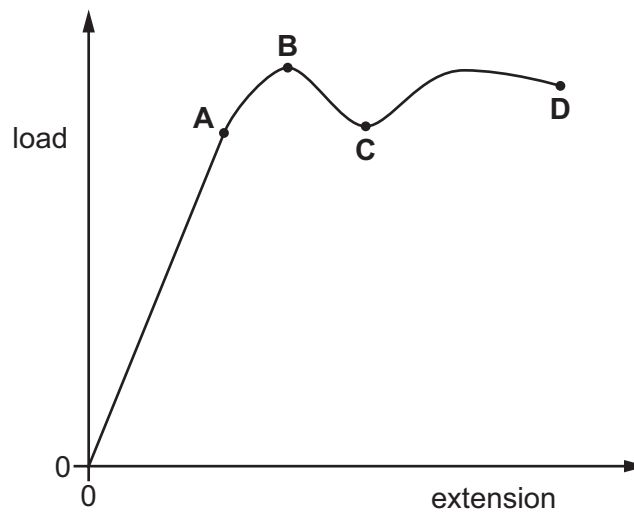
student 3 Area X + area Y represents the stopping distance.

Which students are correct?

- A** 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 only

- 5 The diagram shows a graph of load against extension for a material.

At which point is the limit of proportionality?



- 6 There is no atmosphere on the Moon.

Two metal spheres of identical volume but different mass are at rest at the same height above the ground on the Moon. The two spheres are dropped at the same time.

How do the spheres move after they are dropped?

- A The sphere with the larger mass has a smaller acceleration.
- B The sphere with the larger mass hits the ground first.
- C The sphere with the smaller mass has a smaller acceleration.
- D The two spheres hit the ground together.

- 7 A student determines the density of a small rock.

The weight of the small rock is 1.2 N.

She pours 100 cm^3 of water into a measuring cylinder.

She carefully submerges the rock in the 100 cm^3 of water. The new reading on the measuring cylinder is 147 cm^3 .

What is the density of the rock?

- A 0.026 N/cm^3 B 0.83 g/cm^3 C 1.8 N/cm^3 D 2.6 g/cm^3

- 8 A rock of mass 20 kg is travelling in space at a speed of 6.0 m/s.

What is its kinetic energy?

- A 60 J B 120 J C 360 J D 720 J

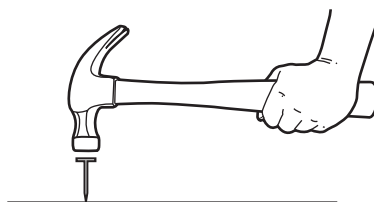
- 9 Which row shows an advantage and a disadvantage of using nuclear fuel to generate electrical power?

	advantage	disadvantage
A	does not produce carbon dioxide	produces radioactive waste
B	does not produce carbon dioxide	only works in some weather conditions
C	energy source is renewable	produces radioactive waste
D	energy source is renewable	only works in some weather conditions

10 Which row gives the definition of work done and the unit of work done?

	definition of work done	unit of work done
A	the energy transferred per unit time	J
B	the energy transferred per unit time	N/m
C	the force multiplied by the distance moved in the direction of the force	J
D	the force multiplied by the distance moved in the direction of the force	N/m

11 A hammer hits a nail into a piece of wood.



The weight of the nail is negligible.

The hammer exerts a force F_H and a pressure p_H on the nail.

The nail exerts a force F_W and a pressure p_W on the wood.

How do these forces and pressures compare?

	forces	pressures
A	$F_W = F_H$	$p_W > p_H$
B	$F_W = F_H$	$p_W < p_H$
C	$F_W > F_H$	$p_W > p_H$
D	$F_W > F_H$	$p_W < p_H$

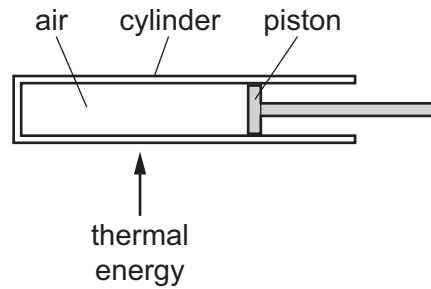
12 The pressure at a point beneath the surface of a liquid varies with the depth of the point and with the density of the liquid.

Which changes both increase the pressure beneath the surface of a liquid?

	change in depth	change in density
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

- 13 The diagram shows a cylinder containing air.

The piston is free to move inside the cylinder and atmospheric pressure acts outside the cylinder.



As the air inside the cylinder is heated, the piston moves to the right.

Which quantity decreases?

- A** the average force exerted on the piston each time that one air particle hits the piston
- B** the average speed of the air particles
- C** the mass of air inside the cylinder
- D** the number of collisions made by air particles with the piston in one second

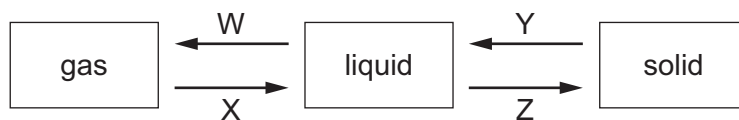
- 14 A fixed mass of air has a volume of 150 cm^3 and a pressure of 120 kPa .

The volume of the air is now changed to 100 cm^3 . The temperature of the air remains constant.

What is the new pressure of the air?

- A** 80 kPa **B** 180 kPa **C** $80\,000 \text{ kPa}$ **D** $180\,000 \text{ kPa}$

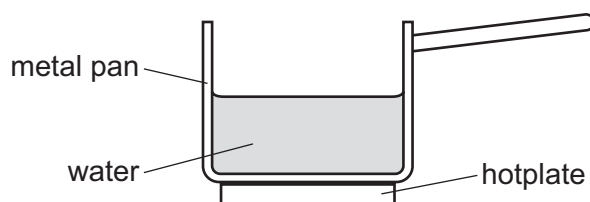
- 15 The diagram shows four changes of state, W, X, Y and Z.



Which row gives the names of the four changes of state?

	W	X	Y	Z
A	condensation	boiling	freezing	melting
B	condensation	boiling	melting	freezing
C	boiling	condensation	freezing	melting
D	boiling	condensation	melting	freezing

- 16 The diagram shows a metal pan on a hotplate.

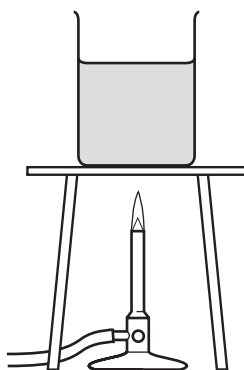


The pan is heated by the hotplate.

How is the thermal energy transferred from the hotplate to the metal pan?

- A** conduction
- B** convection
- C** diffraction
- D** refraction

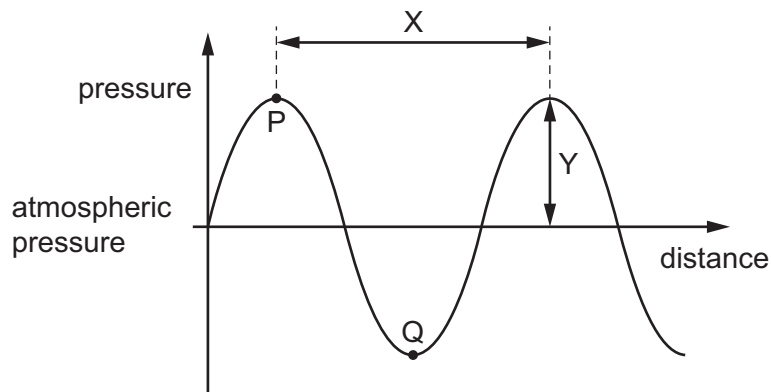
- 17 A beaker of cold water is heated at its base.



What happens to the heated water?

	density of heated water	movement of heated water
A	decreases	no movement
B	decreases	upwards
C	increases	no movement
D	increases	upwards

- 18 The graph shows, at one instant, the pressure variation along a sound wave.

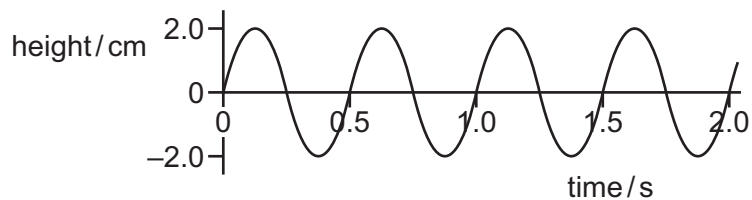


Which point on the diagram represents a rarefaction and what is the wavelength of the sound wave?

	rarefaction	wavelength
A	P	X
B	P	Y
C	Q	X
D	Q	Y

- 19 A cork on the surface of water moves up and down as a wave passes across the surface.

The diagram shows how the height of the cork above the average level of the surface varies with time.



The numerical value of the wavelength of this wave is equal to the numerical value of the amplitude of this wave.

What is the speed of the wave?

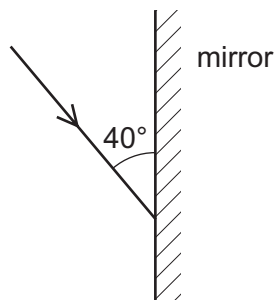
- A** 1.0 cm/s **B** 2.0 cm/s **C** 4.0 cm/s **D** 8.0 cm/s

- 20** A water wave travels from deep water to shallow water in a ripple tank.

What happens to the speed and to the wavelength of the wave?

	speed of wave	wavelength
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

- 21** The diagram shows a ray of light directed at a plane mirror.

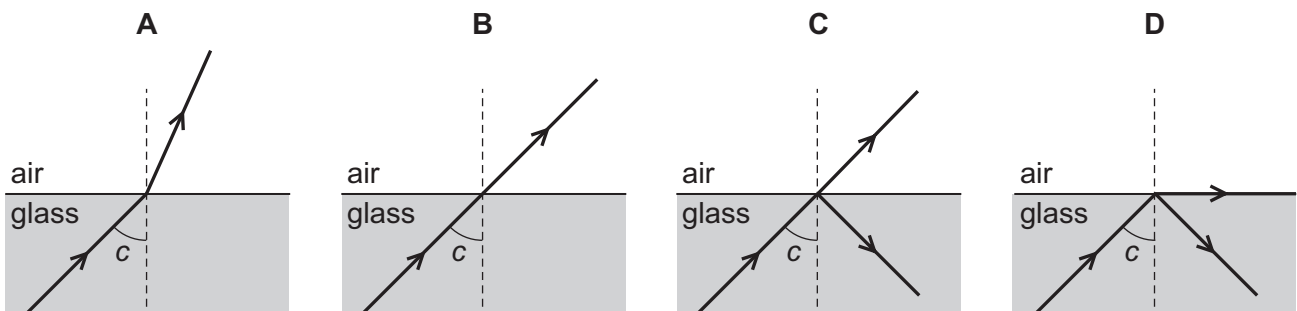


What are the angle of incidence and the angle of reflection?

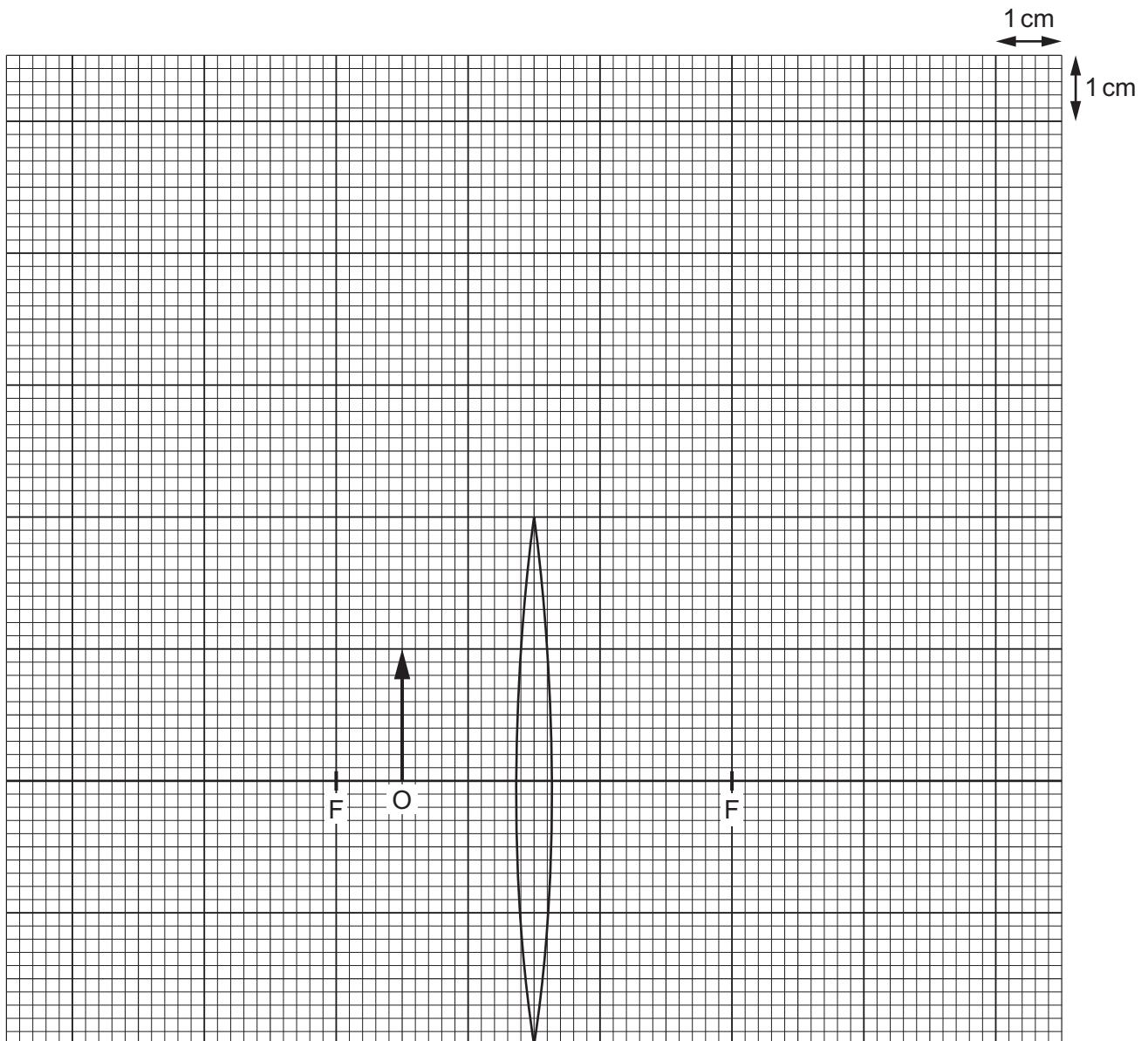
	angle of incidence / °	angle of reflection / °
A	40	40
B	40	50
C	50	40
D	50	50

- 22** A ray of light in glass is incident on the surface at an angle c . The angle c is the critical angle.

Which diagram shows what happens to the light?



- 23 An object O is placed between a converging lens and one principal focus F of the lens.



What is the magnification of the image formed?

- A** 1.0 **B** 2.0 **C** 3.0 **D** 6.0
- 24 Which statement about red light and violet light is correct?
- A** A prism deviates red light more than it deviates violet light.
- B** Red light has a lower frequency than violet light.
- C** Red light has a shorter wavelength than violet light.
- D** The speed of red light in a vacuum is less than the speed of violet light in a vacuum.

- 25** Ultrasound is used to clean jewellery in a liquid.

What is another use of ultrasound?

- A** optical fibre communication
- B** prenatal scanning
- C** photography
- D** telephone communications

- 26** A ship that is stationary on the surface of the sea sends pulses of sound vertically downwards towards the sea bed.

Each pulse that reflects from the sea bed is received 1.0 s after it is sent out.

A whale swims under the boat and a pulse is received 0.60 s after it is sent out.

The speed of sound in sea water is 1500 m/s.

What is the distance of the whale above the sea bed?

- A** 300 m **B** 450 m **C** 600 m **D** 750 m

- 27** Which row describes ultrasound waves?

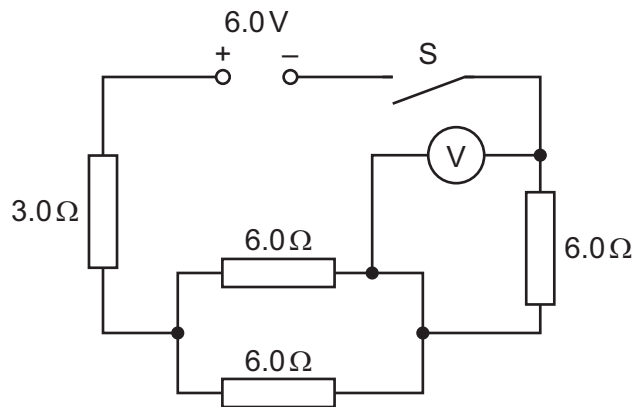
	type of wave	frequency
A	longitudinal	smaller than 20 kHz
B	longitudinal	greater than 20 kHz
C	transverse	smaller than 20 kHz
D	transverse	greater than 20 kHz

- 28** A quantity is defined as the work done by a source in moving unit charge around a complete circuit.

What is the quantity and what is its unit?

	quantity	unit
A	electromotive force	N
B	electromotive force	V
C	potential difference	N
D	potential difference	V

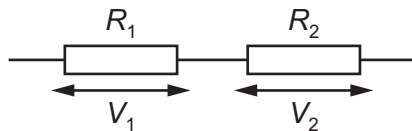
- 29 A student connects the circuit shown.



The switch S is closed.

What is the reading on the voltmeter?

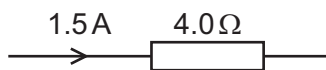
- A** 1.0V **B** 3.0V **C** 4.5V **D** 6.0V
- 30 Two resistors connected in series act as a potential divider as shown.



The resistors have resistances R_1 and R_2 and the potential differences across them are V_1 and V_2 respectively.

Which expression is equal to $\frac{V_1}{V_2}$?

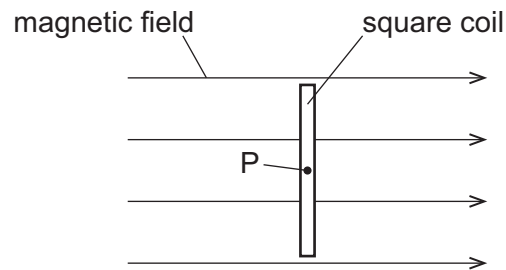
- A** $\frac{R_1}{R_2}$ **B** $\frac{R_2}{R_1}$ **C** $\left(\frac{R_1}{R_2}\right)^2$ **D** $\left(\frac{R_2}{R_1}\right)^2$
- 31 There is an electric current of 1.5 A in a 4.0Ω resistor.



How much energy is transferred electrically to thermal energy in 1.0 minute?

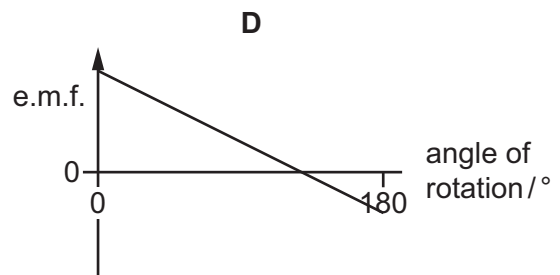
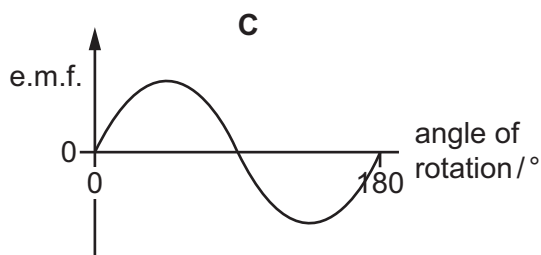
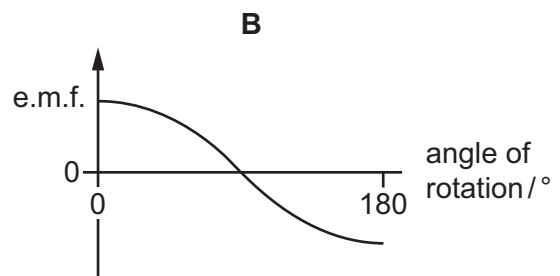
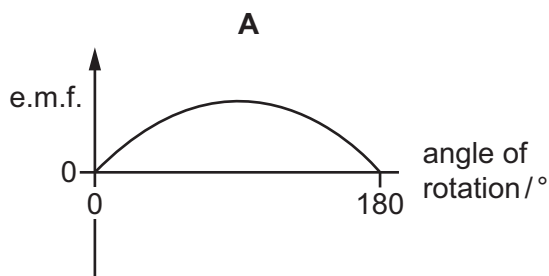
- A** 6.0 J **B** 360 J **C** 540 J **D** 1440 J

- 32** A square coil is placed in a horizontal uniform magnetic field. The plane of the coil is perpendicular to the magnetic field.



The coil is now rotated about an axis through P at a steady rate. The coil rotates through 180° .

Which diagram shows how the electromotive force (e.m.f.) induced in the coil varies as it rotates through 180° ?



- 33** Three pieces of equipment made of magnetic materials are listed.

- 1 the core of a transformer
- 2 the needle of a compass
- 3 iron filings used to show a magnetic field pattern

Which pieces of equipment act as temporary magnets when in use?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 34** Students are asked to suggest factors that affect the speed of rotation of a direct current (d.c.) motor.

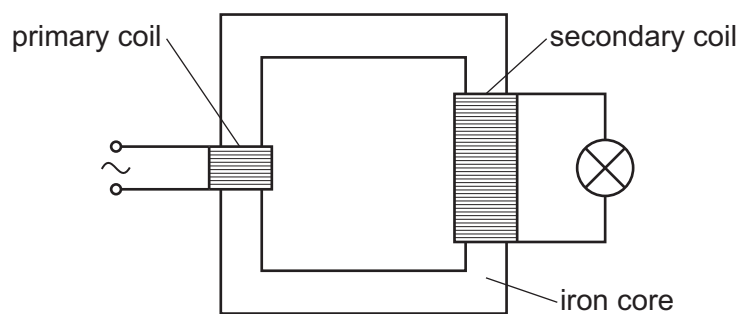
Three of their suggestions are listed.

- 1 the strength of the magnetic field
- 2 the current
- 3 the number of turns on the coil

Which suggestions are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 35** A transformer is used to power a lamp.



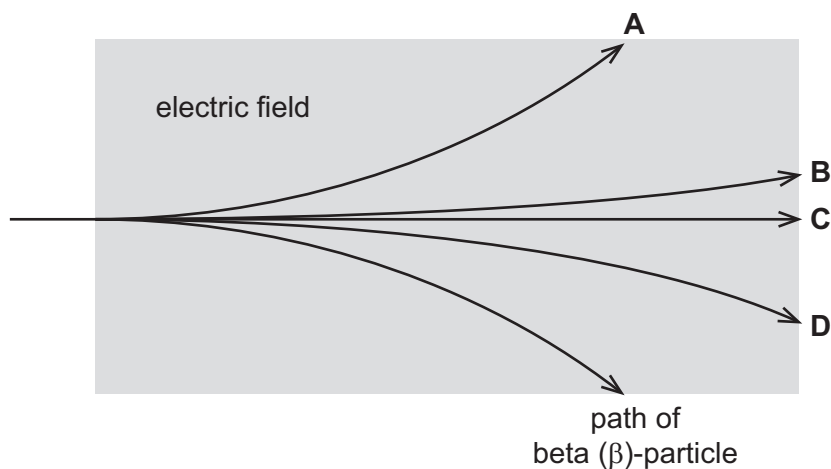
There are more turns on the secondary coil than on the primary coil.

Which statement is correct?

- A** There is an alternating current (a.c.) in the primary coil and a direct current (d.c.) in the secondary coil.
- B** There is a larger current in the secondary coil than in the primary coil.
- C** There is a changing magnetic field in the iron core.
- D** In the iron core there is an alternating current from the primary coil to the secondary coil.

- 36 In an electric field, a beta (β)-particle follows the path shown.

Which path does an alpha (α)-particle travelling at the same speed follow?



- 37 Which row gives the sign of the charge on the nucleus of an atom and the sign of the charge of an electron?

	charge on nucleus	charge on electron
A	negative	positive
B	neutral	negative
C	neutral	positive
D	positive	negative

- 38 An isotope has a half-life of 6000 years.

How much time passes before the count rate due to emissions from a sample of the isotope decreases to $\frac{1}{16}$ of the initial value?

- A** 6000 years
B 18 000 years
C 24 000 years
D 96 000 years
- 39 What is the approximate time taken by light from the Sun to travel to the Earth?
- A** 0.50 s **B** 5.0 s **C** 50 s **D** 500 s

40 Where in the Solar System is the asteroid belt?

- A** between the Sun and Mercury
- B** between Venus and the Earth
- C** between Mars and Jupiter
- D** between Uranus and Neptune

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

May/June 2025

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

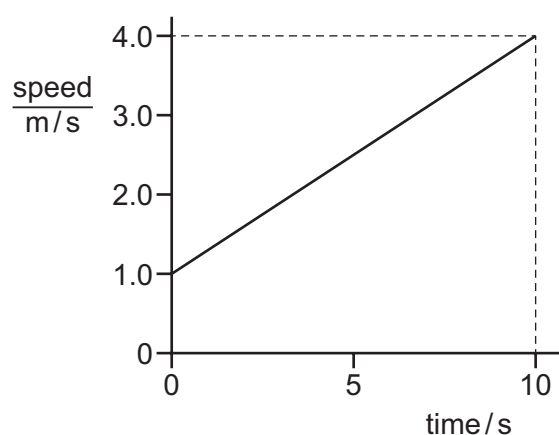
This document has **20** pages. Any blank pages are indicated.



- 1 Which quantity has both magnitude and direction, and is it a scalar quantity or a vector quantity?

	quantity with magnitude and direction	scalar quantity or vector quantity
A	mass	scalar
B	mass	vector
C	weight	scalar
D	weight	vector

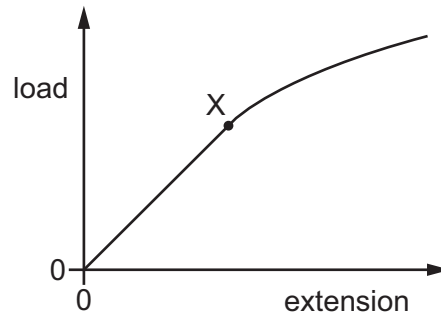
- 2 The speed–time graph shows part of the journey of a bicycle for a time of 10 s.



What is the distance that the bicycle travels in the 10 s?

- A** 20 m **B** 25 m **C** 30 m **D** 40 m
- 3 Which statement explains why a heavy coin that falls through a short distance towards the ground does **not** reach terminal velocity?
- A** The coin has **not** hit the ground.
- B** The weight of the coin equals the air resistance.
- C** The weight of the coin increases as the air resistance increases.
- D** The weight of the coin is always more than the air resistance.

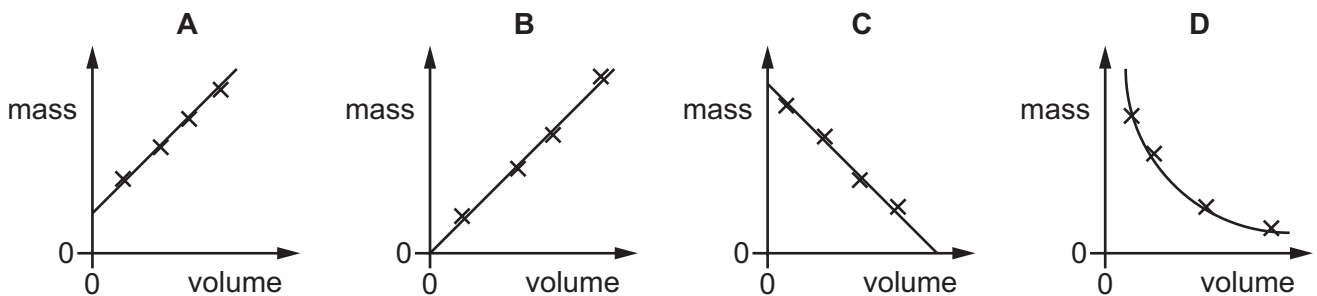
- 4 The diagram shows the load–extension graph for a spring.



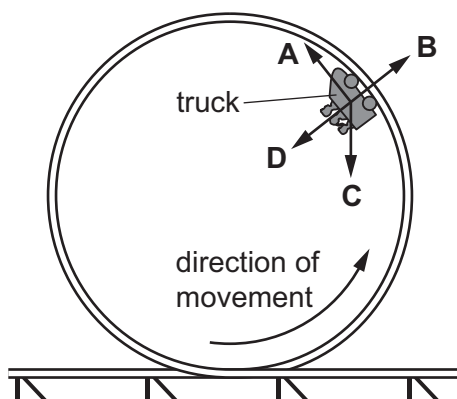
What is the name given to point X on the graph?

- A** limit of force
 - B** limit of proportionality
 - C** limit of the extension
 - D** limit of the spring
- 5 Some students measure the masses and the volumes of differently sized samples of a type of wood.

Which graph shows their results?



- 6 The diagram shows a section of a roller coaster ride that is a circle arranged vertically. The truck is moving at constant speed at the position shown. Which arrow shows the direction of the resultant force acting on the truck?



- 7 What is a unit for momentum?

A kg m/s **B** kg m/s^2 **C** Nm **D** N/m^2

- 8 In a safety test, a car collides with a concrete block.

The impulse exerted by the block on the car is $14\,000\text{ N s}$.

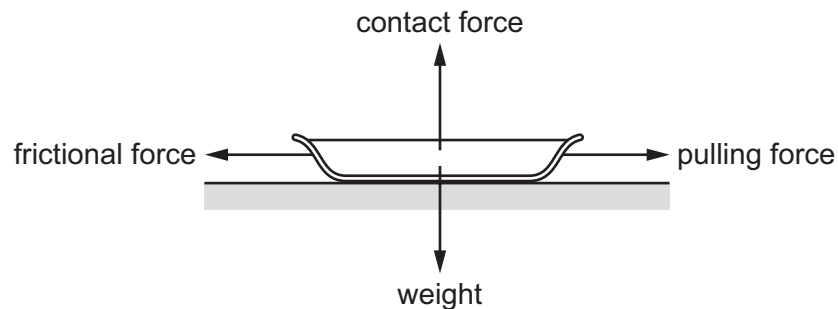
The collision with the block lasts for 70 ms .

What is the force exerted on the car?

A 200 N **B** $14\,000\text{ N}$ **C** $200\,000\text{ N}$ **D** $980\,000\text{ N}$

- 9 An object pulled along horizontal ground accelerates to the right.

The diagram shows the directions of the forces acting on the object.



Which force is multiplied by the distance moved to calculate the work done to increase the kinetic energy store?

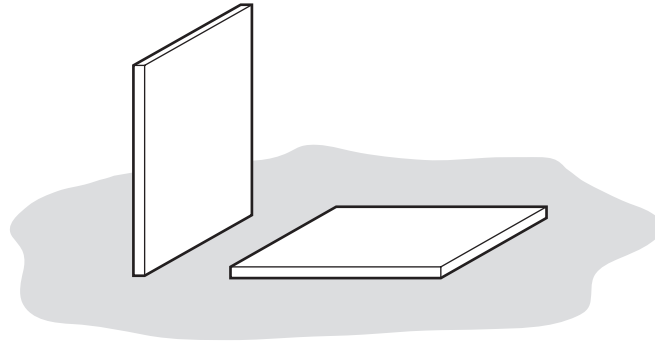
- A contact force
 - B frictional force
 - C pulling force
 - D weight
- 10 The mass of object P is greater than the mass of object Q.

The objects contain different amounts of matter and have a different resistance to change of motion.

Which row is correct?

	greater amount of matter	greater resistance to change of motion
A	P	P
B	P	Q
C	Q	P
D	Q	Q

- 11 A builder leaves two identical, heavy, stone tiles resting on soft earth. One is vertical and the other is horizontal.



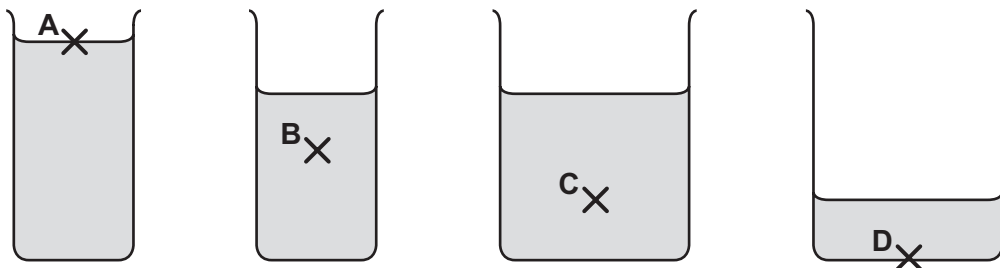
After a few hours, the vertical tile has started to sink into the soft earth. The horizontal one has **not** started to sink.

Which row correctly compares the forces and the pressures that the tiles exert on the earth?

	forces	pressures
A	different	different
B	different	same
C	same	different
D	same	same

- 12 Four beakers contain the same liquid.

At which point is the pressure the greatest?



- 13 The strength of the forces between the particles of a solid determines how much the solid expands when heated.

Which statement about these forces is correct?

- A** The forces are strong so solids expand less than liquids and gases.
- B** The forces are strong so solids expand more than liquids and gases.
- C** The forces are weak so solids expand less than liquids and gases.
- D** The forces are weak so solids expand more than liquids and gases.

- 14** Air is heated in a sealed container with constant volume.

Why does the air pressure increase when the temperature increases?

- A** The air molecules expand.
- B** The air molecules bounce off each other more frequently.
- C** The air molecules bounce off the walls more frequently.
- D** The number of air molecules increases.

- 15** A student determines the specific heat capacity of a metal by carrying out an experiment.

The list shows the apparatus that the student has available.

- 1 electronic balance
- 2 ruler
- 3 thermometer
- 4 heater operating at a power of 50 W
- 5 stopwatch
- 6 ammeter
- 7 voltmeter

Which apparatus does the student use?

- A** 1, 3, 4 and 5
- B** 1, 3, 6 and 7
- C** 2, 3, 4 and 6
- D** 3, 4, 5, 6 and 7

- 16** When a liquid evaporates, molecules escape from its surface.

Which molecules escape, and what happens to the average kinetic energy of the molecules remaining in the liquid?

- A** The less energetic molecules escape and the average kinetic energy decreases.
- B** The less energetic molecules escape and the average kinetic energy increases.
- C** The more energetic molecules escape and the average kinetic energy decreases.
- D** The more energetic molecules escape and the average kinetic energy increases.

- 17** A sample of water in a beaker is at a temperature T .

The surface area of the water is A and an electric fan blows air across the top of the beaker at a speed v .

For which values of T , A and v does the water evaporate the most rapidly?

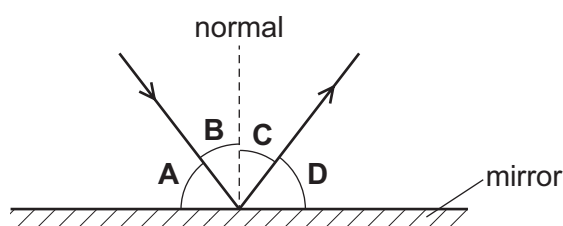
	$T/^{\circ}\text{C}$	A/cm^2	$\frac{v}{\text{m/s}}$
A	18	45	0.1
B	24	50	1.9
C	38	70	5.0
D	38	45	4.7

- 18** Thermal energy passes quickly from hot regions of a metal to cold regions of the metal.

Why are metals good conductors of thermal energy?

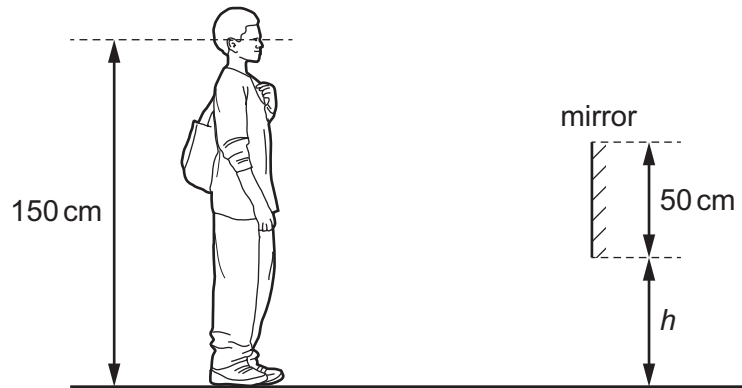
- A** Electrons in the metal vibrate about fixed points and quickly pass energy from one electron to the next.
 - B** Electrons move quickly from ions in hot regions to ions in cold regions.
 - C** Protons in the metal are close together and move quickly from hot to cold regions.
 - D** Small ions of the metal move quickly from hot to cold regions through spaces in the metal lattice.
- 19** The diagram shows light striking a plane mirror and reflecting.

Which angle is the angle of incidence?



- 20** A shoe shop puts a mirror on the wall so that customers can look at their shoes.

The length of the mirror is 50 cm. A customer has eyes 150 cm above ground level.

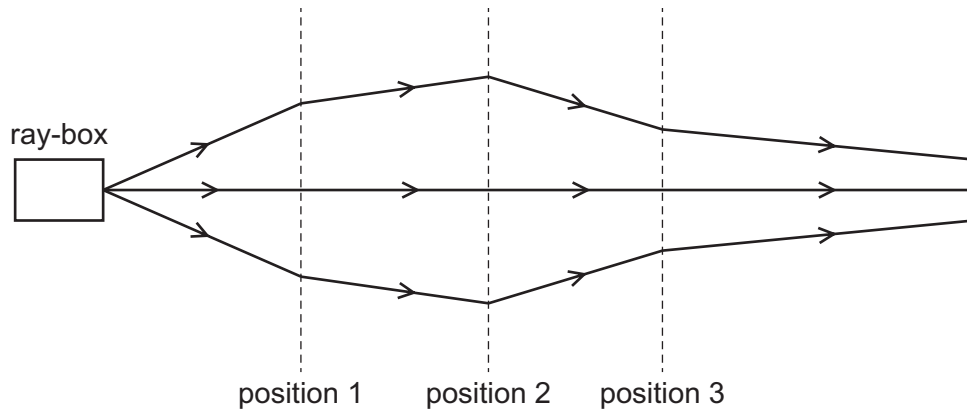


The bottom of the mirror is at height h above the ground.

What is the smallest value of h that allows the customer to see an image of his shoes in the mirror?

- A** 0 **B** 25 cm **C** 50 cm **D** 75 cm
- 21** Which description of a dull black surface is correct?
- A** good emitter, good absorber and good reflector of radiation
- B** good emitter, poor absorber and poor reflector of radiation
- C** good emitter, good absorber and poor reflector of radiation
- D** poor emitter, poor absorber and poor reflector of radiation

22 The rays of light from a ray-box pass through three lenses placed at positions 1, 2 and 3.



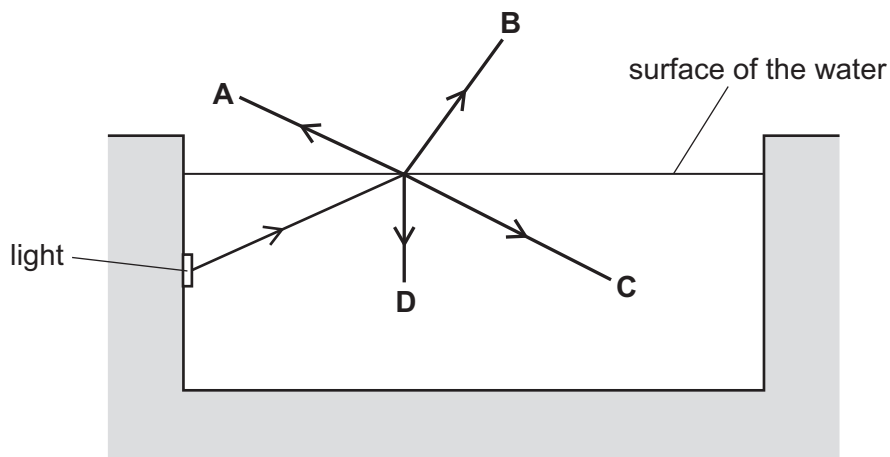
Which type of lens is used at each position?

	position 1	position 2	position 3
A	converging	converging	converging
B	converging	converging	diverging
C	diverging	converging	diverging
D	diverging	diverging	converging

23 A swimming pool is lit by an underwater light.

A ray of light is incident on the surface of the water.

What is the path for the ray of light?



24 When a converging lens is used as a magnifying glass, what is the nature of the image?

- A** real and inverted
- B** real and upright
- C** virtual and inverted
- D** virtual and upright

25 A sound wave consists of compressions and rarefactions.

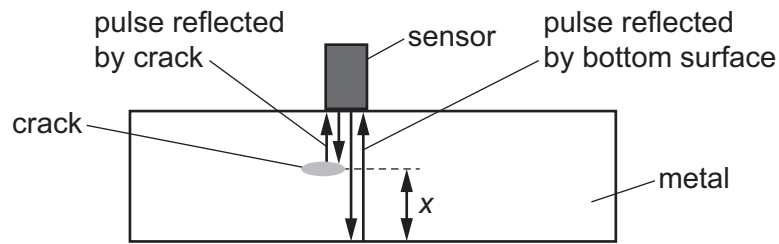
Which type of wave is sound and how does the pressure in a rarefaction compare with the pressure in a compression?

	type of wave	pressure in a rarefaction
A	longitudinal	smaller
B	longitudinal	greater
C	transverse	smaller
D	transverse	greater

26 What happens to a sound wave when the note heard gets louder?

- A** Its amplitude increases.
- B** Its frequency increases.
- C** Its speed increases.
- D** Its wavelength increases.

- 27** A pulse of ultrasound from a sensor is reflected back to the sensor by a crack in a piece of metal and by the bottom surface of the metal.



The reflection from the crack is received back at the sensor 1.0×10^{-5} s after the pulse is sent out.

The reflection from the bottom surface is received back at the sensor 1.5×10^{-5} s after the pulse is sent out.

The speed of ultrasound in the metal is 5200 m/s.

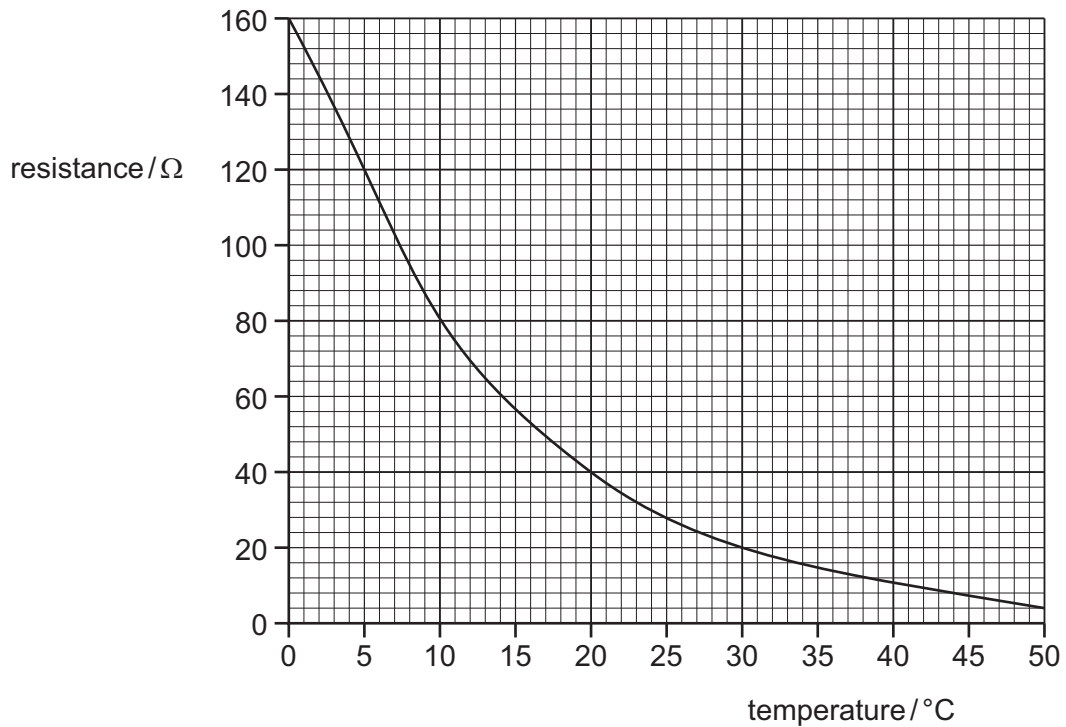
What is the distance x between the crack and the bottom surface of the metal?

- A** 0.013 m **B** 0.026 m **C** 0.052 m **D** 0.078 m
- 28** A 12 V battery is connected to a resistor of resistance 100Ω .
- How much charge passes through the resistor in 30 minutes?
- A** 3.6 C **B** 220 C **C** 250 C **D** 15 000 C
- 29** A circuit consists of a resistor connected between the positive terminal and the negative terminal of a power supply.

Which row describes the direction of conventional current and the direction of flow of free electrons in the resistor?

	direction of conventional current	direction of flow of free electrons
A	from negative to positive	from negative to positive
B	from negative to positive	from positive to negative
C	from positive to negative	from negative to positive
D	from positive to negative	from positive to negative

30 The graph shows how the resistance of a thermistor varies with temperature.

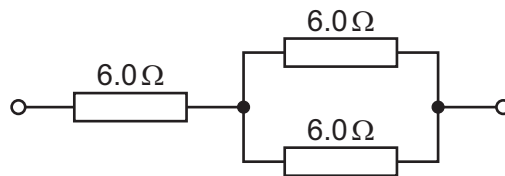


The thermistor is at a temperature of 10°C and is connected in series with a fixed resistor of resistance $20\,\Omega$.

What is the combined resistance of both components?

- A** $16\,\Omega$ **B** $20\,\Omega$ **C** $80\,\Omega$ **D** $100\,\Omega$

31 Each of the resistors in the circuit shown has a resistance of $6.0\,\Omega$.

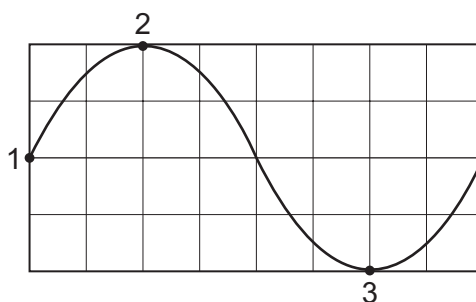


What is the total resistance of the circuit?

- A** $2.0\,\Omega$ **B** $4.0\,\Omega$ **C** $9.0\,\Omega$ **D** $18\,\Omega$

- 32 An alternating current (a.c.) generator is connected to an oscilloscope.

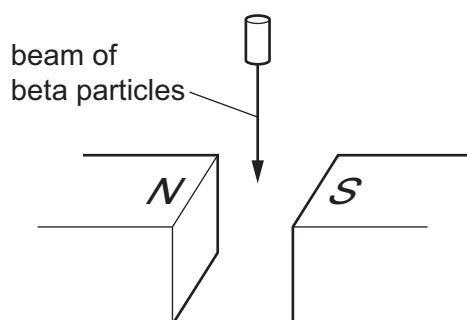
The diagram shows the trace produced on the oscilloscope screen.



What are possible angles between the plane of the coil and the magnetic field direction at the times represented by points 1, 2 and 3?

	1	2	3
A	0°	90°	180°
B	0°	90°	270°
C	90°	180°	0°
D	90°	270°	0°

- 33 A beam of beta particles travelling vertically downwards enters the magnetic field between two magnetic poles.



The beta particles experience a force due to the magnetic field.

What is the direction of the force on a beta particle as it enters the magnetic field?

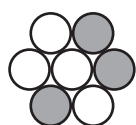
- A** towards the top of the page
- B** towards the bottom of the page
- C** into the page
- D** out of the page

- 34** A step-up transformer is connected between a power station and a long-distance electricity transmission cable.

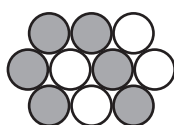
What is the purpose of the step-up transformer, and does it operate using alternating current (a.c.) or using direct current (d.c.)?

	purpose of a step-up transformer	current used
A	to decrease the current in the cable	d.c.
B	to decrease the current in the cable	a.c.
C	to increase the current in the cable	d.c.
D	to increase the current in the cable	a.c.

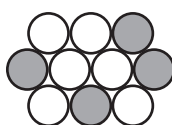
- 35** Each diagram shows the nucleus of an atom.



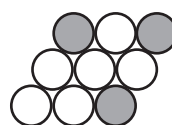
1



2



3



4

key

proton

neutron

Which diagrams show isotopes of the same element?

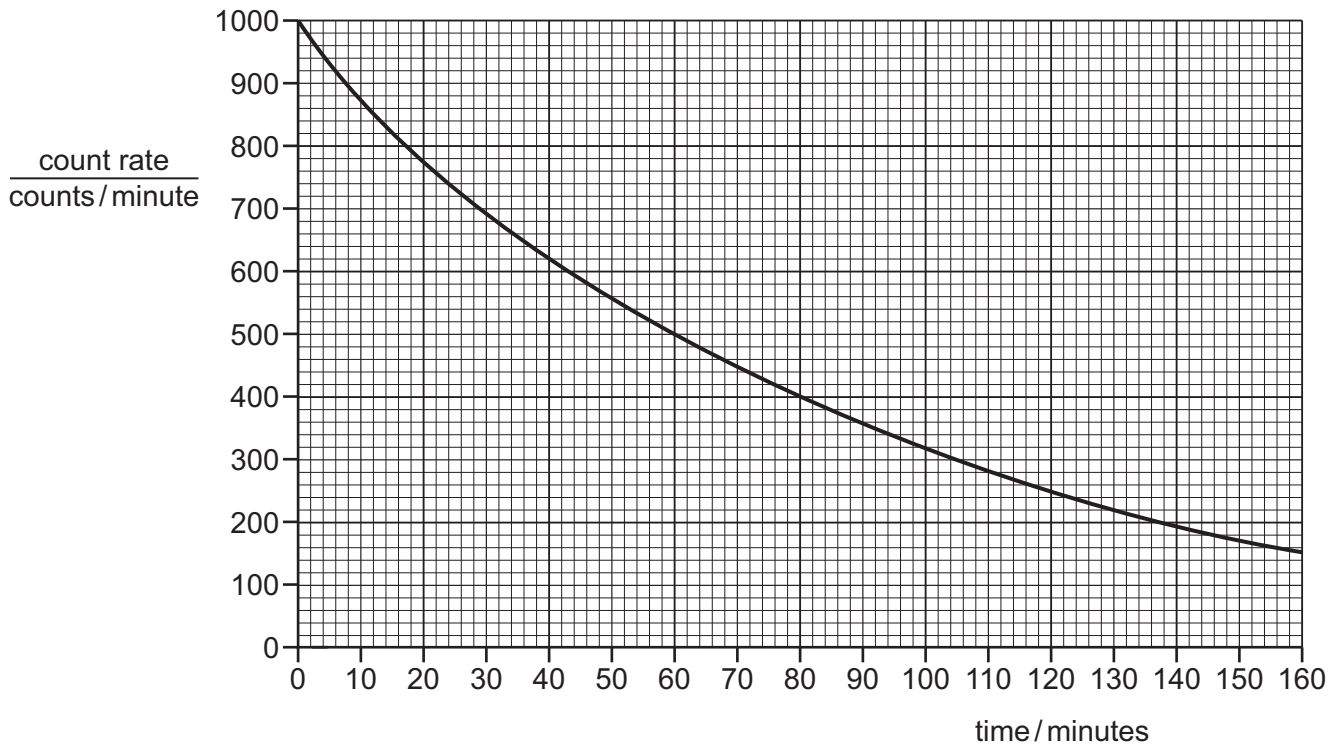
- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4
- 36** A radioactive nucleus X decays to another radioactive nucleus. After a number of decays, a stable nucleus Q is produced.

The proton number (atomic number) of X is equal to that of Q. The nucleon number (mass number) of X is four more than that of Q.

Which radiations are emitted to produce Q?

- A** two alpha-particles
B one alpha-particle and one beta-particle
C two alpha-particles and one beta-particle
D one alpha-particle and two beta-particles

37 The graph shows the results of an experiment to measure the count rate of a radioactive isotope.



What is the half-life of the isotope?

- A** 60 minutes
- B** 80 minutes
- C** 100 minutes
- D** 160 minutes

38 During fission, a nucleus of uranium-235 (U-235) absorbs one neutron to produce two daughter nuclei and three neutrons.

Krypton-92 (Kr-92) is one of the daughter nuclei and it contains 92 nucleons.

How many nucleons does the other daughter nucleus contain?

- A** 92
- B** 141
- C** 144
- D** 327

- 39** The orbital speed of a planet around the Sun is v and its orbital period is T .

Jupiter is further from the Sun than the Earth.

Which planet has the greater value for v and which planet has the greater value for T ?

	greater orbital speed v	greater orbital period T
A	Earth	Earth
B	Earth	Jupiter
C	Jupiter	Jupiter
D	Jupiter	Earth

- 40** What is the Earth's position in the order of planets from the Sun?

- A** closest planet to the Sun
- B** second closest planet to the Sun
- C** third closest planet to the Sun
- D** fourth closest planet to the Sun

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Cambridge O Level

PHYSICS**5054/12**

Paper 1 Multiple Choice

May/June 2025

MARK SCHEME

Maximum Mark: 40

Published

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This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	D	1
3	D	1
4	B	1
5	A	1
6	D	1
7	D	1
8	C	1
9	A	1
10	C	1
11	A	1
12	D	1
13	D	1
14	B	1
15	D	1
16	A	1
17	B	1
18	C	1
19	C	1
20	D	1
21	D	1
22	D	1
23	C	1
24	B	1
25	B	1
26	A	1
27	B	1
28	B	1

Question	Answer	Marks
29	B	1
30	A	1
31	C	1
32	A	1
33	C	1
34	A	1
35	C	1
36	B	1
37	D	1
38	C	1
39	D	1
40	C	1

Cambridge O Level

PHYSICS**5054/11**

Paper 1 Multiple Choice

May/June 2025

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Question	Answer	Marks
1	D	1
2	B	1
3	D	1
4	B	1
5	B	1
6	D	1
7	A	1
8	C	1
9	C	1
10	A	1
11	C	1
12	C	1
13	A	1
14	C	1
15	A	1
16	C	1
17	C	1
18	B	1
19	B	1
20	B	1
21	C	1
22	B	1
23	C	1
24	D	1
25	A	1
26	A	1
27	A	1
28	B	1

Question	Answer	Marks
29	C	1
30	D	1
31	C	1
32	C	1
33	C	1
34	B	1
35	B	1
36	D	1
37	A	1
38	B	1
39	B	1
40	C	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2024

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

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- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

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- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

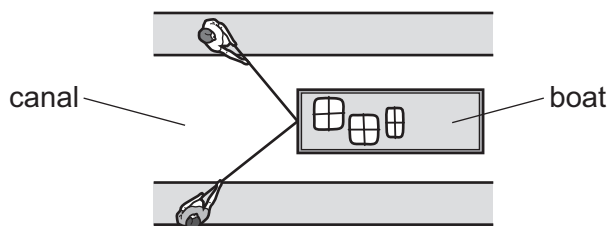
This document has **16** pages. Any blank pages are indicated.



1 What is measured using a micrometer?

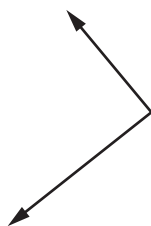
- A area
- B current
- C length
- D mass

2 Two people pull on ropes to move a boat along a canal.

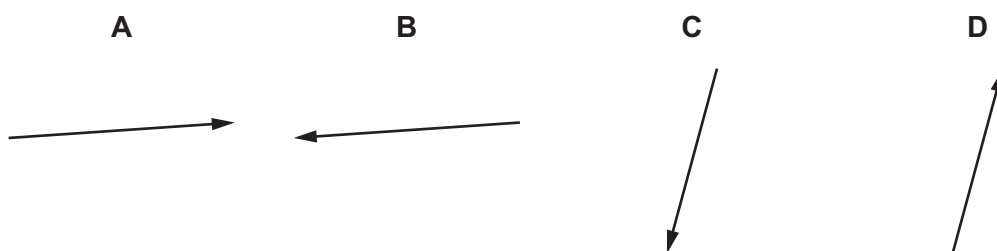


The boat moves at a constant velocity.

The vector diagram for the tension in the ropes is shown.

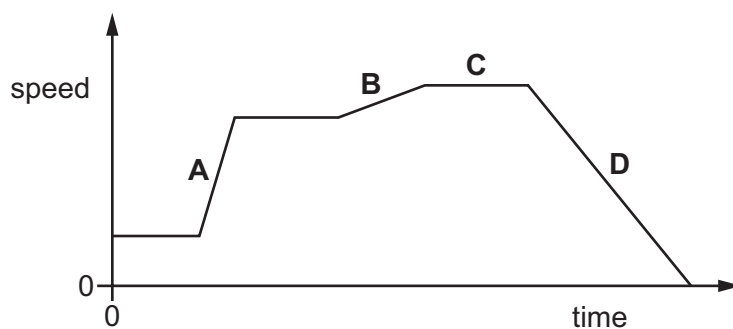


Which vector is the drag acting on the boat?



3 The graph shows how the speed of a car travelling in a straight line changes with time.

Which section shows the largest acceleration?



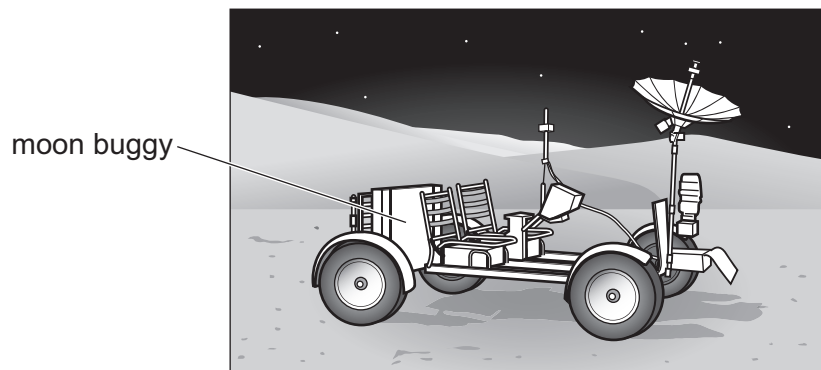
- 4 A man walks along a path from X to Y. The diagram shows the path from above.



The man measures the distance he walks and the time taken.

Which quantity can be calculated using this data only?

- A acceleration
 - B average speed
 - C average velocity
 - D power
- 5 Which property of an object determines its resistance to a change from its state of rest or motion?
- A its mass
 - B its shape
 - C its surface area
 - D its volume
- 6 The diagram shows a moon buggy used by astronauts.



The mass of the moon buggy on the Earth is 210 kg. The gravitational field strength on the Moon is $\frac{1}{6}$ of that on the Earth.

What is the weight of the moon buggy on the Moon?

- A zero B 35 N C 210 N D 340 N

- 7 A car travels along a road. The force on the car due to the engine is 800 N.

The motion of the car depends on the value of the total resistive force R .



Which row shows the motion of the car for the given value of R ?

	value of resistive force R/N	motion
A	500	deceleration
B	800	acceleration
C	900	deceleration
D	1000	acceleration

- 8 Four of the gravitational forces that act between objects in the Solar System are listed.

P the force on the Moon due to the Earth

Q the force on the Earth due to the Sun

R the force on the Earth due to the Moon

S the force on the Moon due to the Sun

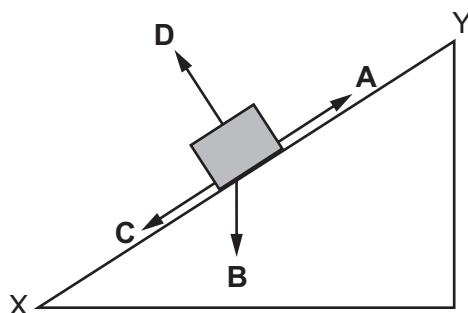
Which two forces are a Newton's third law pair?

- A** P and Q **B** P and R **C** Q and S **D** R and S

- 9 A box is moved up a rough slope from X to Y.

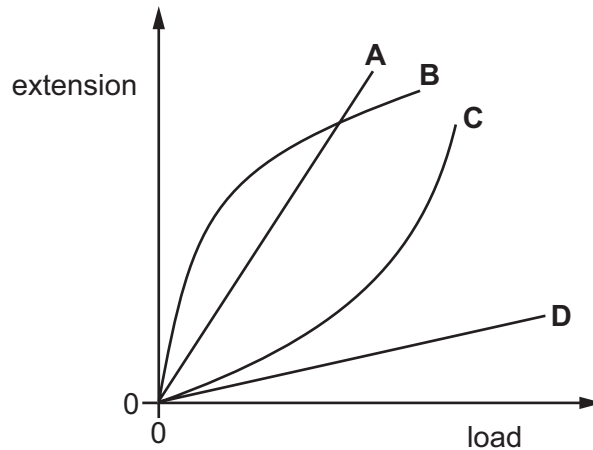
The diagram shows four forces acting on the box.

Which force is the force due to friction on the box?



- 10 The graph shows how the extension of four different threads depends on the load attached.

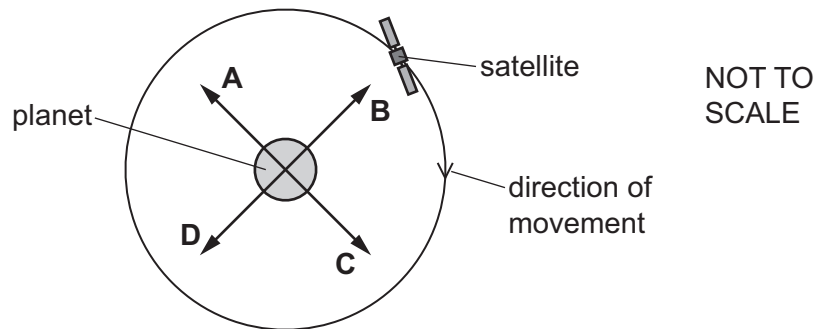
Which thread is the most difficult to stretch over the range of loads shown?



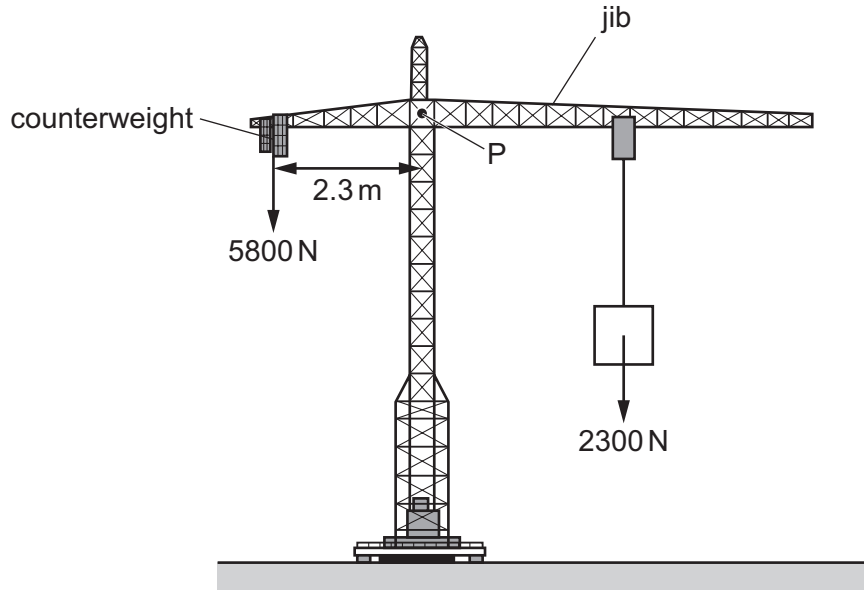
- 11 A satellite orbits a planet in a circular path as shown. It has constant speed.

There is a force on the satellite due to the planet.

In which direction is the force on the satellite when it is in the position shown?



- 12** A crane has a 5800 N counterweight positioned 2.3 m from the tower along a horizontal jib. The centre of gravity P of the crane jib is marked.



What is the horizontal distance between the 2300 N load and P so that there is no moment about P?

- A** 0.91 m **B** 3.5 m **C** 5.8 m **D** 8.1 m
- 13** A car of mass 750 kg travels 400 m at 25 m/s. It then accelerates to 35 m/s and travels a further 400 m.
- What is the change in the momentum of the car due to acceleration?
- A** 7500 kg m/s
B 24 000 kg m/s
C 45 000 kg m/s
D 75 000 kg m/s
- 14** A ball is dropped from rest at the top of a building. Air resistance is negligible.
- The velocity of the ball is 14 m/s when it hits the ground.
- What is the height of the building?

- A** 2.9 m **B** 10 m **C** 20 m **D** 40 m

15 Which energy source is available constantly over a 24-hour period?

- A natural gas
- B solar cells
- C tidal
- D wind

16 A 15W lamp is turned on for 30 minutes. It wastes 7000 J of energy.

What is the efficiency of the lamp?

- A 0.26 B 0.35 C 0.59 D 0.74

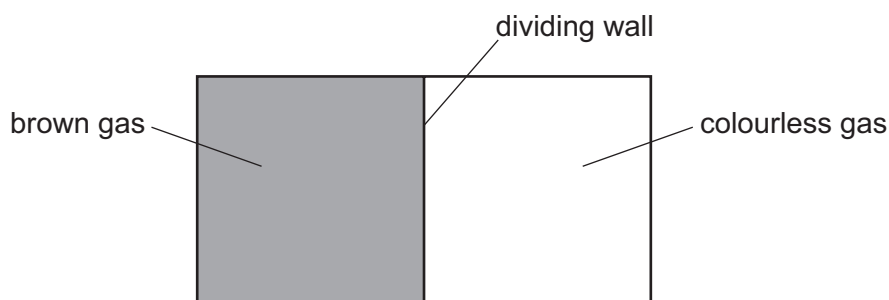
17 Which quantity is **not** measured in joules (J)?

- A gravitational potential energy
- B latent heat
- C power
- D work

18 Which description of a liquid is correct?

- A fixed shape, fixed volume
- B fixed shape, variable volume
- C variable shape, fixed volume
- D variable shape, variable volume

- 19** A transparent box has a dividing wall in its middle. It contains two different gases, one in each half, as shown.



The dividing wall is removed. The box is left for a long time. The gases do not react.

What is then seen in the box?

- A** brown gas on the right and colourless gas on the left
 - B** pale brown gas throughout
 - C** several distinct clouds of colourless and brown gas throughout
 - D** colourless gas on the right and brown gas on the left
- 20** A bottle containing a cold liquid is placed on a table on a warm day. Drops of water form on the outside of the bottle.

Which process causes the drops to form?

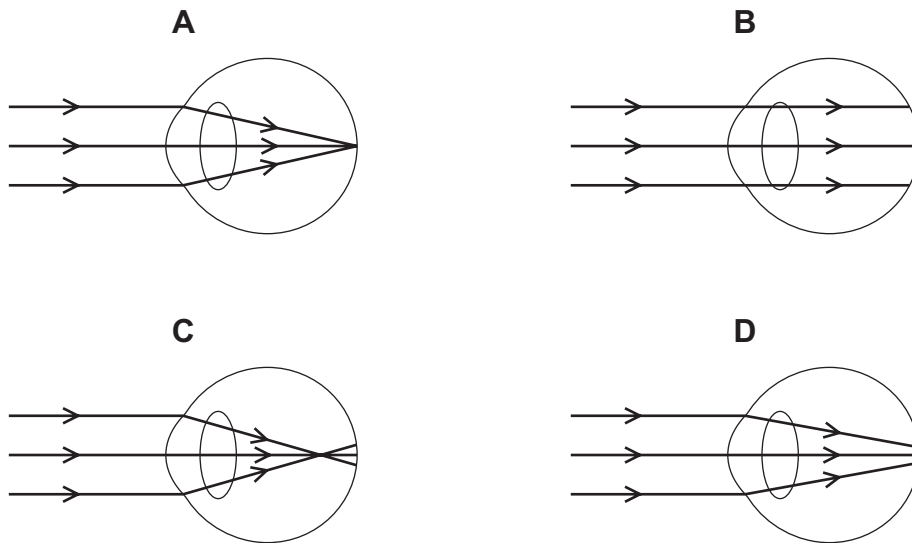
- A** condensation
 - B** conduction
 - C** convection
 - D** evaporation
- 21** What is the specific heat capacity of a liquid?
- A** the difference between the boiling temperature and the melting temperature of the liquid
 - B** the energy required to change the state of 1 kg of the liquid
 - C** the energy required to heat 1 kg of the liquid through 1 °C
 - D** the increase in temperature of the liquid when it is heated
- 22** Which statement about infrared radiation is correct?
- A** In a vacuum, infrared radiation travels at the speed of light.
 - B** Infrared radiation is a longitudinal wave.
 - C** Infrared radiation has a higher frequency than ultraviolet radiation.
 - D** White surfaces are better emitters of infrared radiation than black surfaces.

23 The speed of sound in air is 330 m/s.

Which sound is classed as ultrasound?

- A** a sound with a wavelength of 250 cm
- B** a sound with a wavelength of 25 cm
- C** a sound with a wavelength of 2.5 cm
- D** none of the above

24 Which diagram shows how light from a distant object forms an image in a normal eye?



25 The colour of visible light is related to the wavelength of the light.

Which list of colours is in order of increasing wavelength?

- A** blue → green → yellow → red
- B** blue → green → red → yellow
- C** green → red → yellow → blue
- D** red → yellow → green → blue

26 A thin converging lens is used as a magnifying glass.

Which row gives the nature of the image produced and an expression that is the linear magnification?

	nature of image	expression for linear magnification
A	real	$\frac{\text{image length}}{\text{object length}}$
B	real	$\frac{\text{object length}}{\text{image length}}$
C	virtual	$\frac{\text{image length}}{\text{object length}}$
D	virtual	$\frac{\text{object length}}{\text{image length}}$

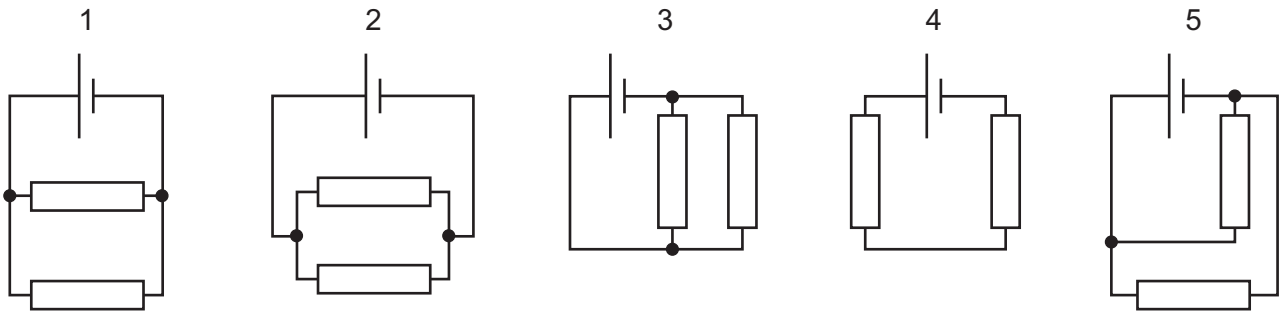
27 Which application is **not** a use for ultrasound?

- A** cleaning jewellery
- B** scanning an unborn baby
- C** sonar
- D** sterilising water

28 Which equation is correct for potential difference (p.d.)?

- A** p.d. = voltage $\times$ current
- B** p.d. = $\frac{\text{energy}}{\text{time}}$
- C** p.d. = $\frac{\text{work done}}{\text{charge}}$
- D** p.d. = $\frac{\text{current}}{\text{resistance}}$

29 The diagrams show five electrical circuits. All of the resistors shown are identical.



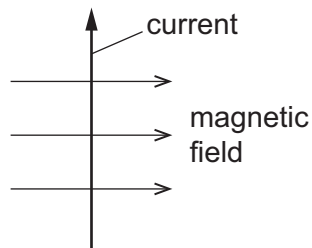
Which circuits have equal resistance?

- A** 1, 2, 3 and 5
B 1, 2, 4 and 5
C 1, 3, 4 and 5
D 2, 3, 4 and 5
- 30 Which electrical appliance uses the heating effect of electricity?
- A** a cell phone (mobile phone)
B a fan
C a hairdryer
D a lawnmower
- 31 How many kilowatt-hours of energy are used by a 1000 W heater connected to a 230 V supply for 30 minutes?
- A** 0.30 kWh **B** 0.50 kWh **C** 30 kWh **D** 120 kWh
- 32 Which safety precautions must be taken when wiring an electrical kettle that has a stainless-steel outer casing?
- A** It must be earthed and have a fuse in the live wire.
B It must be earthed and have a fuse in the neutral wire.
C It needs a fuse in the live wire but does **not** need to be earthed.
D It needs a fuse in the neutral wire but does **not** need to be earthed.

33 What is the purpose of the earth wire in a plug connected to an appliance?

- A** to complete the circuit so that the appliance works
- B** to conduct thermal energy so that the appliance does not get too hot
- C** to prevent a person getting a shock
- D** to protect the appliance from a current that is too large

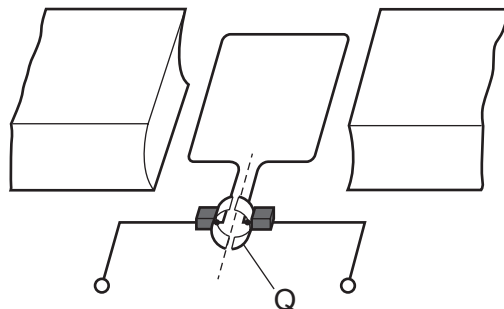
34 The diagram shows a wire carrying a current in a magnetic field.



What is the direction of the force on the wire?

- A** left to right
- B** right to left
- C** into the page
- D** out of the page

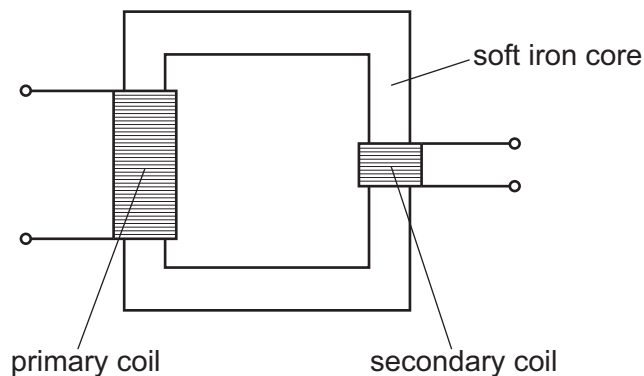
35 The diagram shows a simple d.c. motor.



What is the part labelled Q?

- A** a coil
- B** a magnet
- C** a slip ring
- D** a split-ring commutator

36 The diagram shows a transformer.



A student writes four statements about how the transformer works.

- 1 An alternating voltage across the primary coil induces an unchanging voltage across the secondary coil.
- 2 An alternating voltage across the primary coil produces a changing magnetic field in the iron core.
- 3 A changing magnetic field in the iron core induces an alternating voltage across the secondary coil.
- 4 An unchanging voltage across the primary coil produces a changing magnetic field across the secondary coil.

Which statements explain how the transformer works?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

37 Which particle has the smallest mass?

- A** alpha particle
B electron
C neutron
D proton

38 Which two atoms are isotopes of the same element?

atom	number of neutrons	number of protons
1	22	12
2	22	14
3	25	13
4	24	14

- A** 1 and 2 **B** 1 and 4 **C** 2 and 4 **D** 3 and 4

39 Four types of ionising radiation are listed.

- alpha particles
- beta particles
- X-rays
- gamma rays

Which types of radiation can be emitted from the unstable nuclei of a radioactive material?

- A** alpha particles, beta particles and gamma rays
- B** alpha particles and beta particles only
- C** gamma rays only
- D** X-rays and gamma rays

40 What is the nuclear reaction that powers the Sun?

- A** the fission of hydrogen into helium
- B** the fission of helium into hydrogen
- C** the fusion of hydrogen into helium
- D** the fusion of helium into hydrogen

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

May/June 2024

1 hour

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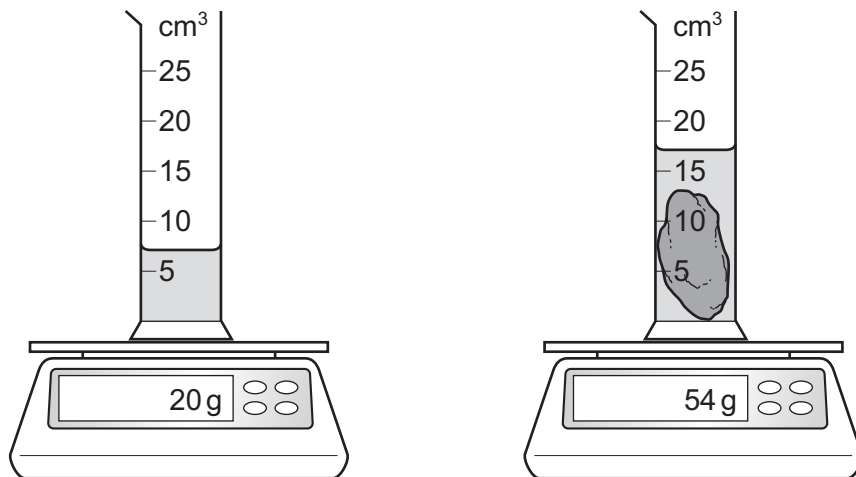
INFORMATION

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- 1 A measuring cylinder containing water is placed on a top-pan balance. An object is placed into the measuring cylinder.



What is the mass and what is the volume of the object?

	mass /g	volume /cm ³
A	10	7
B	10	34
C	34	10
D	34	17

- 2 Which quantity is a vector?

- A** distance
- B** force
- C** mass
- D** speed

- 3 The table shows the distance travelled by a car in each 2 s time interval during a 12 s period of its journey.

time interval / s	0–2	2–4	4–6	6–8	8–10	10–12
distance travelled in time interval / m	10	10	11	13	16	23

Which statement describes the motion of the car?

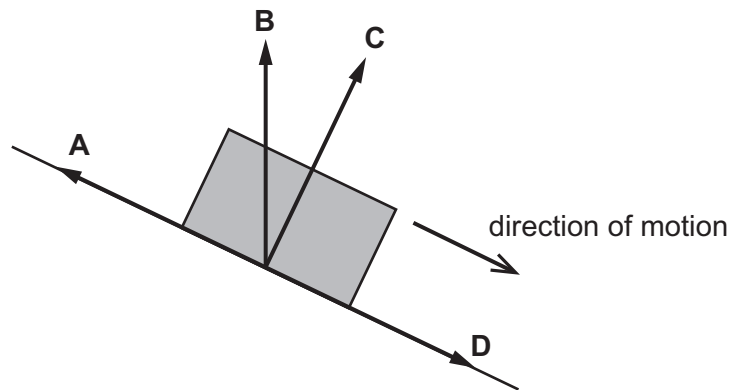
- A** moving with constant speed from 0–4 s, followed by non-uniform acceleration
B moving with constant speed from 0–4 s, then uniform acceleration
C at rest from 0–4 s, then non-uniform acceleration
D at rest from 0–4 s, then uniform acceleration
- 4 Two cubes, X and Y, have the same mass, but the length of a side of X is twice that of Y.

What is the value of the ratio $\frac{\text{density of X}}{\text{density of Y}}$?

- A** 0.125 **B** 1.0 **C** 4.0 **D** 8.0
- 5 An object slides down a frictionless slope as shown.

As the object presses on the surface, the surface pushes back on the object.

In which direction does the surface push back on the object?



- 6 A car, travelling on a straight road, brakes and comes to a stop.

Which row gives the expression for calculating the thinking distance and gives a variable that affects thinking distance?

	expression for calculating thinking distance	variable that affects thinking distance
A	braking distance + stopping distance	load carried
B	braking distance + stopping distance	speed of car
C	stopping distance – braking distance	load carried
D	stopping distance – braking distance	speed of car

- 7 An unstretched spring of length 15.0 cm stretches by 4.0 cm when a mass of weight 6.0 N is suspended from it. The spring does not exceed the limit of proportionality.

What is the total length of the spring when the weight of the suspended mass is 3.0 N?

- A** 2.0 cm **B** 7.5 cm **C** 9.5 cm **D** 17.0 cm

- 8 Which statements describing the moment of a force about a pivot are correct?

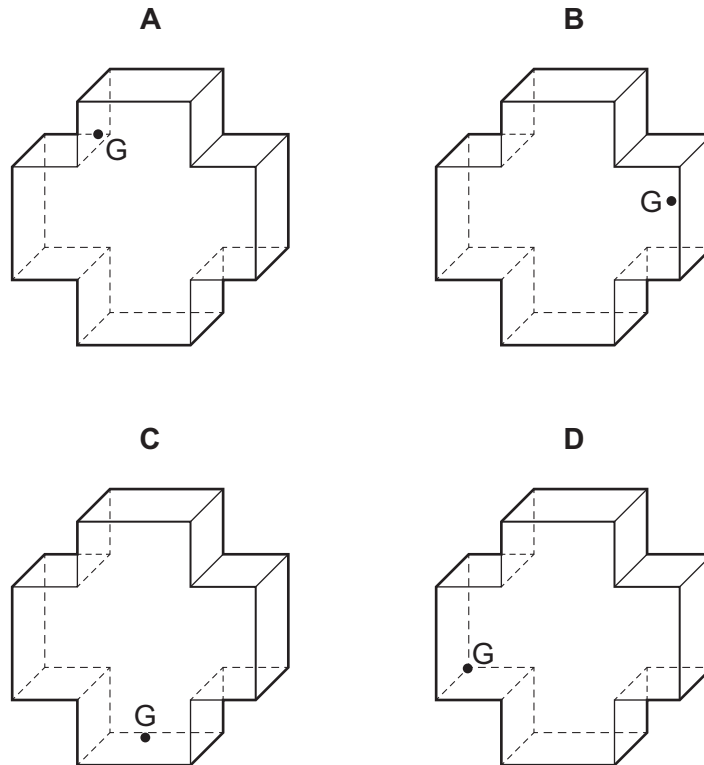
- 1 The moment of a force is a measure of the turning effect of the force.
- 2 The moment of a force is equal to the force $\times$ perpendicular distance from the pivot.
- 3 The moment of a force is equal to the force $\times$ pressure on the pivot.

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

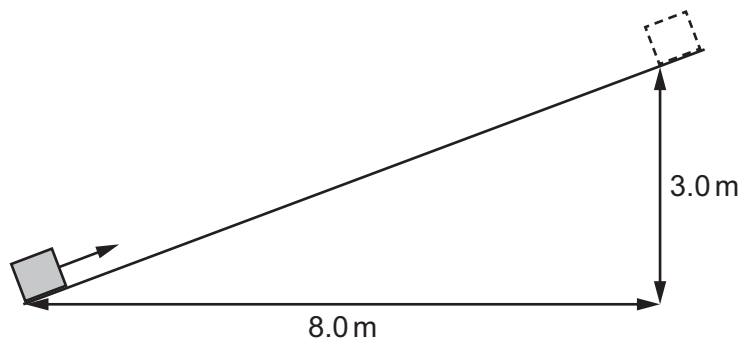
- 9 Four objects have identical shapes but different distributions of mass.

The centre of gravity of each object is shown by the dot marked G.

Which object is the most stable?



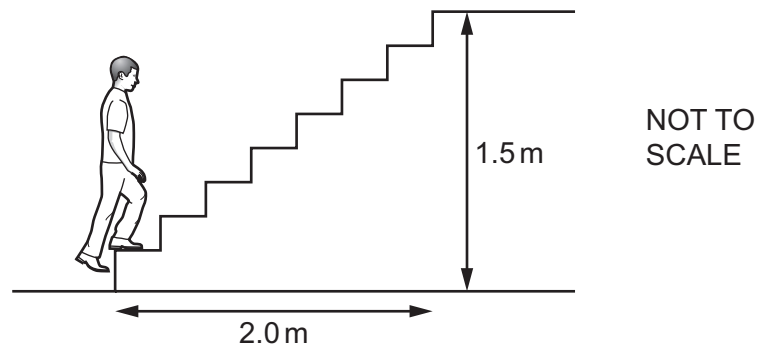
- 10 An object of mass 5.0 kg is pushed up a slope at a steady speed of 2.0 m/s.



Which calculation gives the change in gravitational potential energy from the bottom to the top?

- A $0.50 \times 5.0 \times 2.0$
 B $5.0 \times 9.8 \times 3.0$
 C $0.50 \times 5.0 \times 2.0^2$
 D $5.0 \times 9.8 \times 8.0$

- 11 A student of mass 60 kg climbs some steps. He travels a horizontal distance of 2.0 m and a vertical distance of 1.5 m.



What is the work done against the force of gravity?

- A** 88 J **B** 118 J **C** 880 J **D** 1200 J
- 12 Which method of producing electricity does **not** involve a turbine?
- A** hydroelectric power station
B nuclear power station
C solar cells
D wind generator
- 13 The input power to a lamp is 6.0 W. The lamp wastes 2.7 J of energy in 3.0 s.
- What is the efficiency of the lamp?
- A** 0.15 **B** 0.45 **C** 0.55 **D** 0.85
- 14 A camera is taken under water and left at a depth of 8.0 m.
- atmospheric pressure = 100 000 Pa
 density of water = 1000 kg/m³
- What is the total pressure acting on the camera?
- A** 22 000 Pa **B** 78 000 Pa **C** 110 000 Pa **D** 178 000 Pa
- 15 In which example are the particles in fixed positions?
- A** ice in an ice cube
B water in a lake
C air in an air bubble
D water vapour in the atmosphere

- 16** A gas expands slowly and its temperature remains constant.

What happens to the gas particles?

- A** They move further apart and their average speed decreases.
 - B** They move further apart and their average speed increases.
 - C** They move further apart and their average speed remains unchanged.
 - D** They stay the same distance apart and their average speed decreases.
- 17** The air in each of four syringes is slowly compressed so that the temperature of the air stays constant. The volumes before and after compression for each syringe are given in the table.

The air in all four syringes is initially at the same pressure.

Which syringe shows the smallest pressure change?

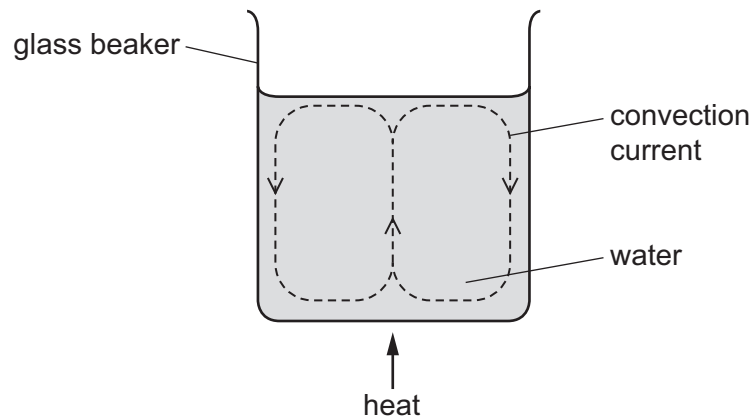
	volume before compression / cm ³	volume after compression / cm ³
A	50	10
B	100	50
C	400	25
D	400	100

- 18** The temperature of a substance is measured with a liquid-in-glass thermometer.

Which physical property changes so that the temperature can be measured?

- A** length of the thermometer
 - B** mass of the liquid
 - C** specific heat capacity of the liquid
 - D** volume of the liquid
- 19** Boiling and evaporation are different processes.
- Which statement is correct?
- A** Boiling only occurs at the surface of a liquid.
 - B** Evaporation only occurs when the temperature of the liquid is high enough.
 - C** Evaporation does not change the temperature of a liquid.
 - D** When a boiling liquid is heated its temperature remains constant.

- 20** A glass beaker contains water. When the centre of the base of the beaker is heated, a convection current is set up.



Which statement explains this?

- A** The evaporation of water causes water molecules to rise to the surface.
 - B** The expansion of water molecules causes them to rise to the surface.
 - C** The water above the heat source rises because it becomes less dense.
 - D** The water at the side falls because it becomes less dense.
- 21** A hot meal is supplied in a container made of shiny metal foil.
- Why does the container help to keep the food hot?
- A** The container traps air which increases convection.
 - B** The foil is a poor thermal conductor.
 - C** Shiny metal foil is a good absorber of infrared radiation.
 - D** Shiny metal foil is a poor emitter of infrared radiation.
- 22** Which quantity is defined as the number of wavelengths passing a point per second?
- A** amplitude
 - B** frequency
 - C** speed
 - D** wavelength

- 23** A plane mirror forms the image of an object.

What are characteristics of the image formed?

	size	type
A	same as object	real
B	same as object	virtual
C	smaller than object	real
D	smaller than object	virtual

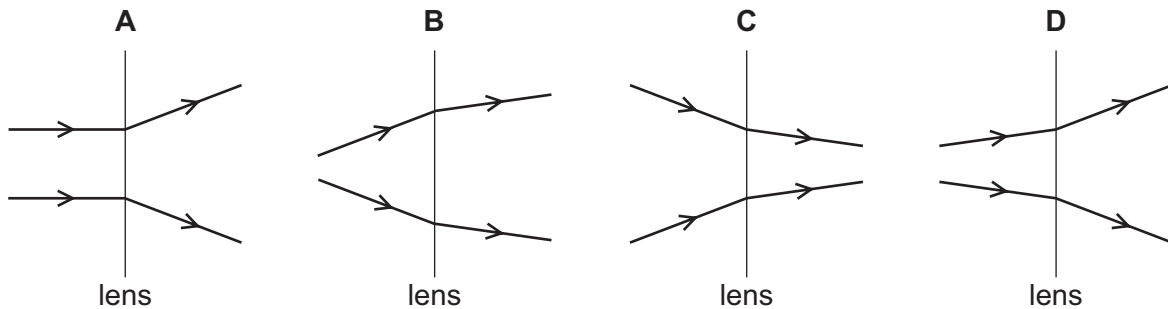
- 24** A ray of light strikes the surface of a glass block at an angle of incidence of 45° .

The refractive index of the glass is 1.8.

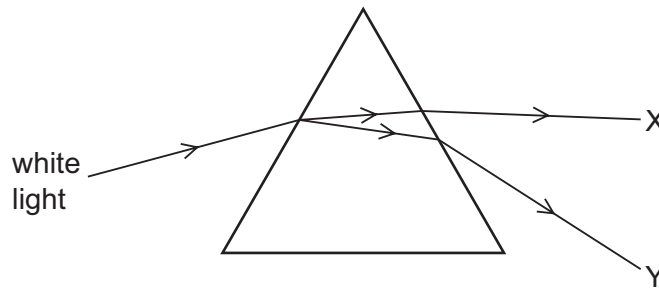
What is the angle of refraction inside the block?

- A** 23° **B** 25° **C** 45° **D** 81°

- 25** Which diagram shows rays of light passing through a converging lens?



- 26 A beam of white light is incident on a glass prism.



X and Y are the two ends of the visible spectrum.

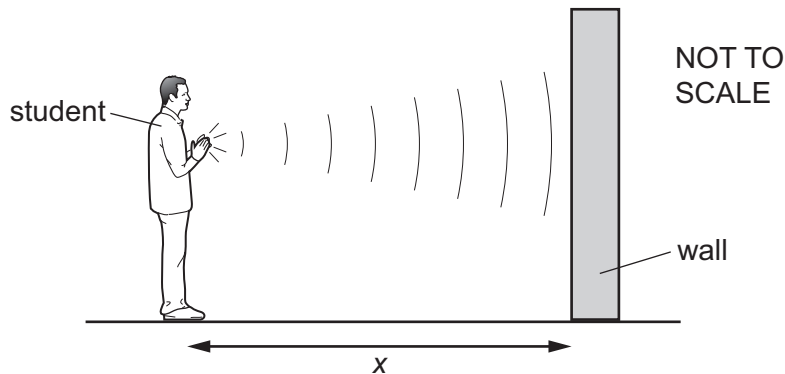
Which statement is correct?

- A All the light travels at the same speed in the prism.
 - B The light at X has a lower frequency than the light at Y.
 - C The light at X refracts more as it leaves the prism than the light at Y.
 - D The white light refracts away from the normal as it enters the prism.
- 27 Which components of the electromagnetic spectrum are used to communicate from a remote controller to a television set and from the Earth to a satellite?

	remote controller to TV set	the Earth to a satellite
A	infrared	microwave
B	infrared	ultraviolet
C	light	microwave
D	light	ultraviolet

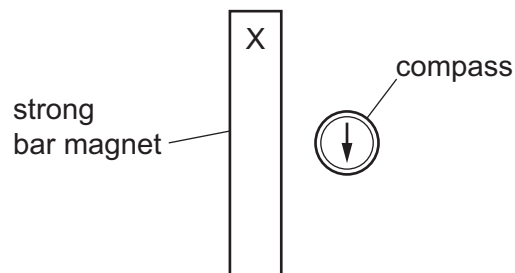
- 28** A student claps his hands in front of a wall and hears the echoes. He claps at the same moment as he hears the echo of the previous clap.

Another student starts a stop-watch on the first clap and stops it on the eleventh clap. The reading on the stop-watch is 9.4 s and the speed of sound is 320 m/s.



What is the distance x between the student and the wall?

- A** 140 m **B** 150 m **C** 270 m **D** 300 m
- 29** A compass is placed close to a strong bar magnet and pivots to the position shown.



Which type of magnet is a compass needle and what is pole X?

- A** The needle is an induced magnet and X is a north magnetic pole.
B The needle is an induced magnet and X is a south magnetic pole.
C The needle is a permanent magnet and X is a north magnetic pole.
D The needle is a permanent magnet and X is a south magnetic pole.

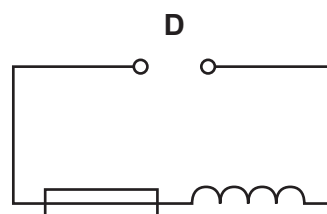
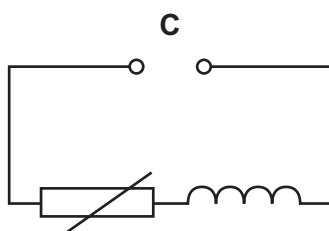
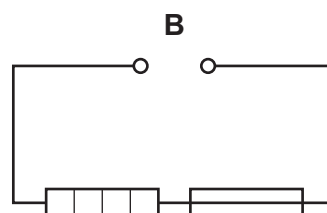
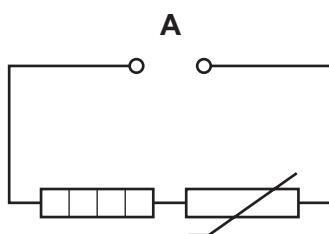
30 Plastic and wool are insulating materials.

Samples of plastic and wool are rubbed together. This causes a transfer of electrons.

Which row shows the direction of electron transfer and the final charge on both materials?

	plastic	wool	final charge on plastic	final charge on wool
A	gains electrons	loses electrons	neutral	positive
B	gains electrons	loses electrons	negative	positive
C	loses electrons	gains electrons	negative	positive
D	loses electrons	gains electrons	positive	neutral

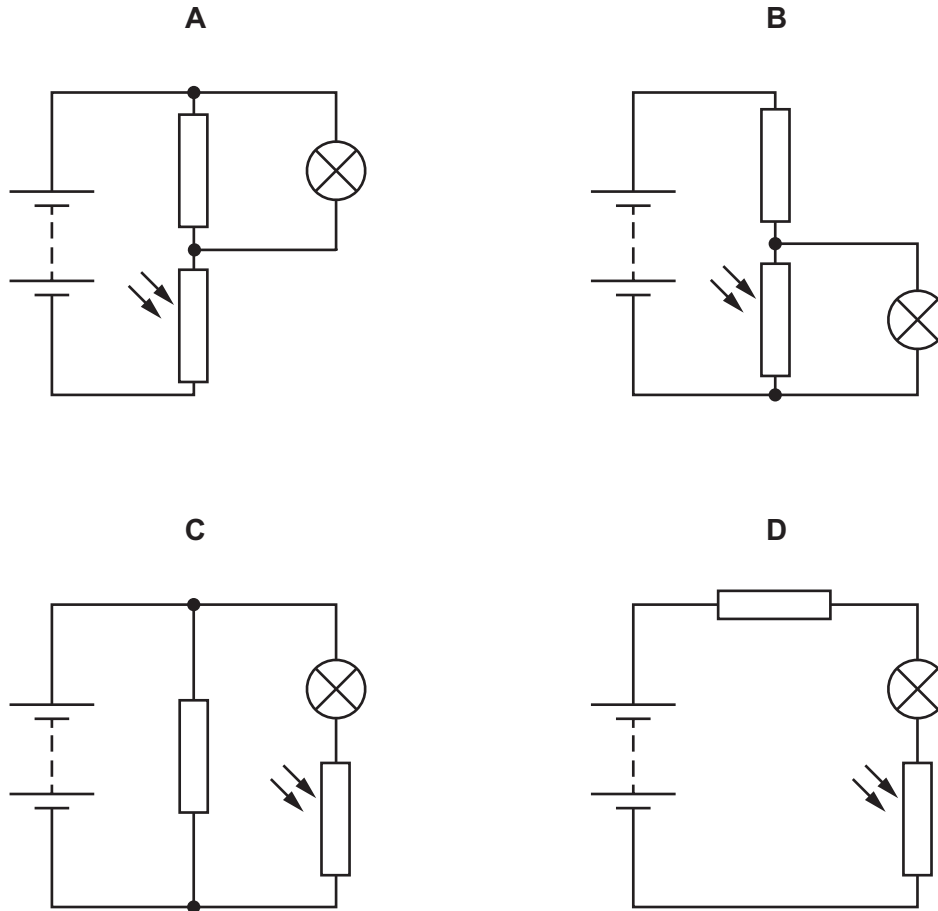
31 Which circuit contains a heater and a fuse?



- 32** A light-dependent resistor (LDR) is to be used as an input sensor for an intruder alarm circuit. When there is no light on the LDR, the lamp switches on.

The resistance of the fixed resistor is less than that of the lamp.

Which diagram shows the circuit?

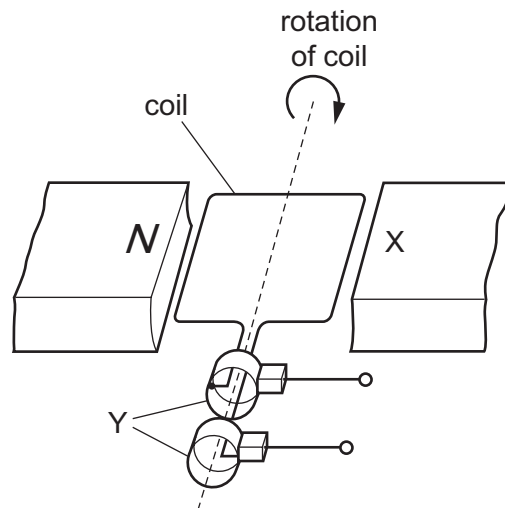


- 33** A potential difference of 4.0 V produces a current of 3.0 A in a resistor.

How much energy is transferred to thermal energy during 10 s ?

- A** 12 J **B** 30 J **C** 40 J **D** 120 J

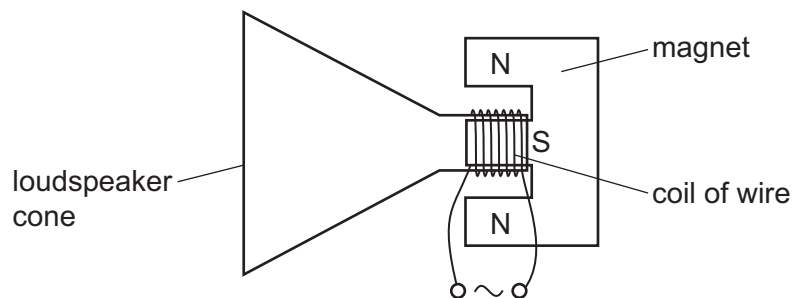
34 The diagram shows a generator.



What are X and Y?

	X	Y
A	N pole	split-ring commutator
B	N pole	slip rings
C	S pole	split-ring commutator
D	S pole	slip rings

35 The diagram shows a loudspeaker that is producing a sound.



A student writes four sentences about the loudspeaker.

- 1 There is a direct current in the coil of wire.
- 2 The current in the coil produces a changing magnetic field.
- 3 A force is produced on the coil of wire attached to the cone.
- 4 The cone spins around due to the force.

Which sentences are correct?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 36 A step-down transformer changes 240 V a.c. to 12 V a.c. There are 600 turns on the primary coil.

How many turns are on the secondary coil?

- A 20 B 30 C 600 D 12 000

- 37 Alpha particles are directed at a thin gold foil.

Only a very small proportion of the alpha particles are deflected through very large angles.

Which fact about the nuclei of the gold atoms is **not** a conclusion of this experiment?

- A They are surrounded by mostly empty space.
B They are very small.
C They contain most of the mass of the atom.
D They contain protons and neutrons.

- 38 A radioactive source emitting only gamma radiation is placed in front of a radiation detector which measures the count rate.

What must cause the count rate to increase?

- A a thick sheet of lead placed between the source and the detector
B a strong magnetic field placed between the source and the detector
C the radioactive source moved closer to the detector
D the detector moved further away from the radioactive source

- 39 Nuclear fusion is one source of energy.

What is nuclear fusion?

- A the decay of a radioactive nucleus
B the joining together of two nuclei to make a larger nucleus
C the melting of an unstable nucleus
D the splitting of a nucleus into two lighter nuclei

40 Which statement is correct?

A The orbital speed of the Earth around the Sun is:

$$\frac{2\pi \times \text{average diameter of the orbit}}{\text{orbital period}}.$$

B The orbit of the Earth around the Sun is an ellipse.

C The Moon takes approximately one month to orbit the Sun.

D Light from the Sun takes 5.0 minutes to reach the Earth.

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Cambridge O Level

PHYSICS**5054/12**

Paper 1 Multiple Choice

May/June 2024

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	A	1
3	A	1
4	B	1
5	A	1
6	D	1
7	C	1
8	B	1
9	C	1
10	D	1
11	D	1
12	C	1
13	A	1
14	B	1
15	A	1
16	D	1
17	C	1
18	C	1
19	B	1
20	A	1
21	C	1
22	A	1
23	D	1
24	A	1
25	A	1
26	C	1
27	D	1
28	C	1

Question	Answer	Marks
29	A	1
30	C	1
31	B	1
32	A	1
33	C	1
34	C	1
35	D	1
36	C	1
37	B	1
38	C	1
39	A	1
40	C	1

Cambridge O Level

PHYSICS**5054/11**

Paper 1 Multiple Choice

May/June 2024

MARK SCHEME

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	B	1
3	A	1
4	A	1
5	C	1
6	D	1
7	D	1
8	A	1
9	C	1
10	B	1
11	C	1
12	C	1
13	D	1
14	D	1
15	A	1
16	C	1
17	B	1
18	D	1
19	D	1
20	C	1
21	D	1
22	B	1
23	B	1
24	A	1
25	B	1
26	B	1
27	A	1
28	B	1

Question	Answer	Marks
29	C	1
30	B	1
31	B	1
32	B	1
33	D	1
34	D	1
35	C	1
36	B	1
37	D	1
38	C	1
39	B	1
40	B	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2023

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

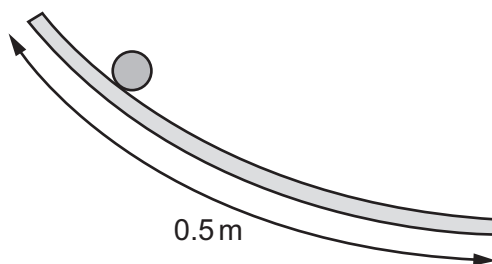
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 In an experiment, a ball is rolled down a curved track that is about half a metre long.

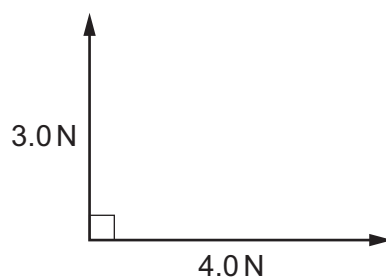


Which measuring device is used to measure the length accurately?

- A** metre rule
B micrometer
C stop-watch
D tape measure
- 2 Which row describes acceleration, displacement, distance and speed?

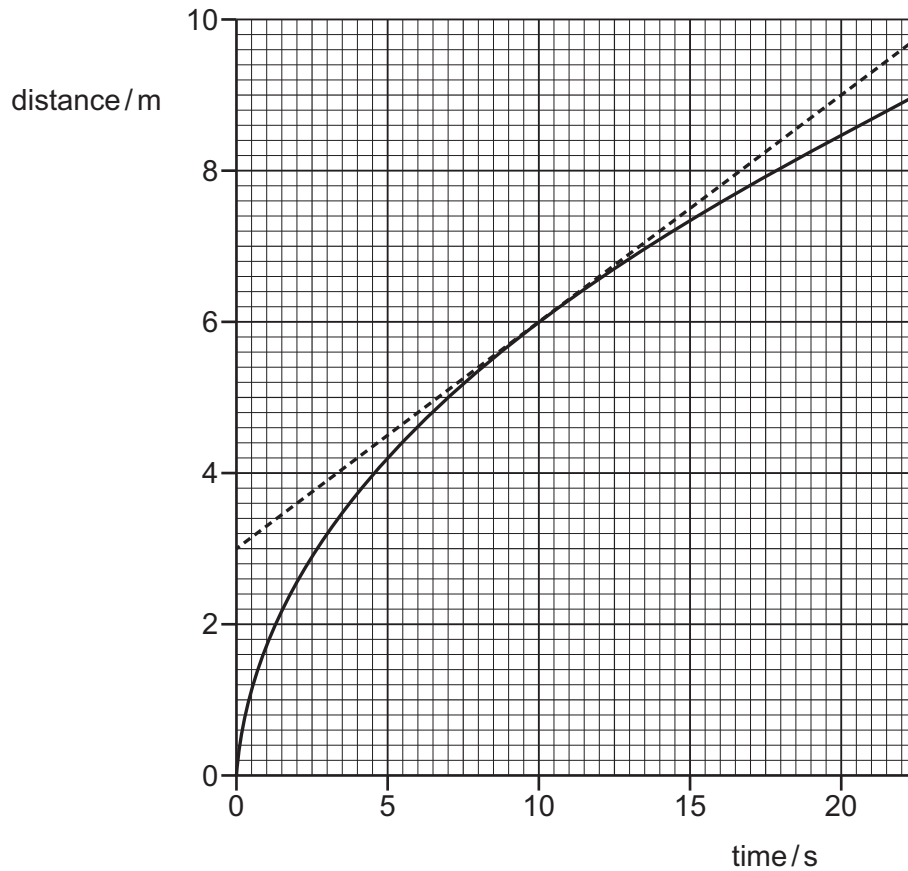
	acceleration	displacement	distance	speed
A	scalar	scalar	vector	scalar
B	scalar	vector	scalar	vector
C	vector	scalar	vector	vector
D	vector	vector	scalar	scalar

- 3 What is the size of the resultant of the two forces shown?



- A** 1.0 N **B** 3.5 N **C** 5.0 N **D** 7.0 N

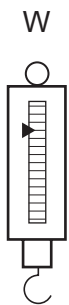
- 4 The curved line on the distance–time graph shows the motion of a toy car. The straight line is the tangent to the curve at 10 s.



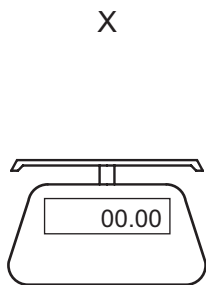
What is the speed of the toy car at 10 s?

- A** 0.3 m/s **B** 0.6 m/s **C** 1.7 m/s **D** 3.3 m/s

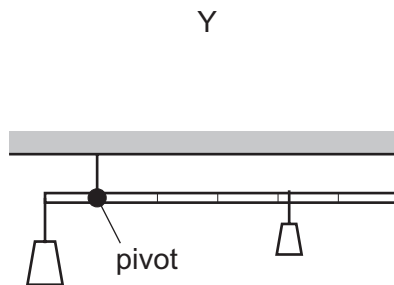
- 5 The diagrams show four balances, W, X, Y and Z.



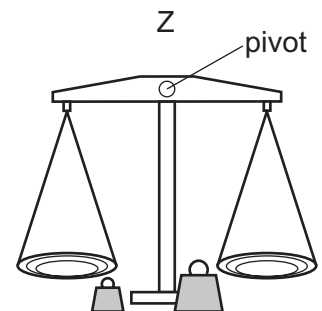
spring balance



top-pan balance



lever-arm balance



two-pan balance

The scales of all the balances are calibrated on the Earth to measure mass.

Which balances also measure mass correctly when used on the Moon?

- A** W and X **B** W and Z **C** X and Y **D** Y and Z

- 6 A spacecraft carries a probe to Mars where the gravitational field strength is 3.7 N/kg . The weight of the probe on Mars is 370 N .

The gravitational field strength on Earth is 9.8 N/kg .

What are the mass and the weight of the probe on Earth?

	mass / kg	weight / N
A	100	370
B	100	980
C	370	98
D	370	3700

- 7 A student determines the density of a liquid.

Which equipment does the student use?

- A** a balance and a ruler
- B** a balance and a measuring cylinder
- C** a beaker and a measuring cylinder
- D** a ruler and a beaker

- 8 Newton's third law describes a pair of forces.

Which row shows whether the two forces are of the same type and whether they act on the same object?

	the types of forces in the pair	the objects on which they act
A	different	different
B	different	same
C	same	different
D	same	same

- 9 The stopping distance of a car is the sum of the thinking distance and the braking distance.

Which factors affect the braking distance?

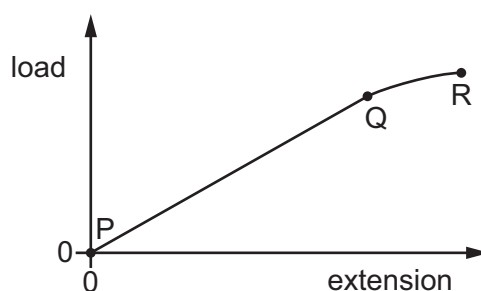
	speed of car	tiredness of driver	condition of road
A	✓	✗	✗
B	✓	✓	✓
C	✓	✗	✓
D	✗	✓	✓

key

✓ = affects braking distance

✗ = does not affect braking distance

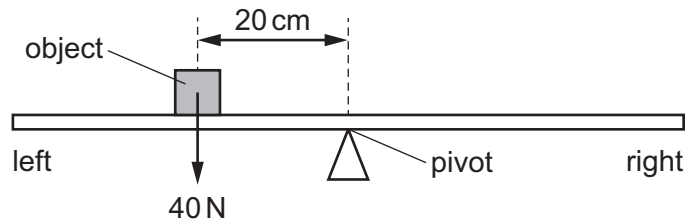
- 10 A load–extension graph for a spring is shown.



Which point is the limit of proportionality for the spring and what is the spring constant of the spring equal to?

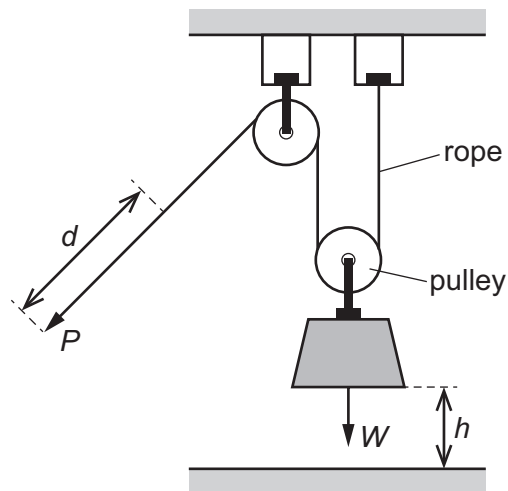
	limit of proportionality	spring constant is equal to
A	Q	the gradient of PQ
B	Q	the gradient of QR
C	R	the gradient of PQ
D	R	the gradient of QR

- 11 A uniform beam is pivoted at its midpoint. An object weighing 40 N is placed on the beam 20 cm to the left of the pivot as shown.



Which force and position balances the system?

- A 20 N acting downwards, 40 cm to the right of the pivot
 B 20 N acting upwards, 40 cm to the right of the pivot
 C 50 N acting downwards, 10 cm to the left of the pivot
 D 50 N acting upwards, 10 cm to the left of the pivot
- 12 A worker pulls a rope with a force P through a distance d in the direction of the force as shown. This causes a block of weight W to move vertically upwards through a height h .



How much work is done by the worker?

- A Ph B Pd C $(P + W)h$ D $(P + W)d$
- 13 A battery is connected in a circuit to light a lamp.

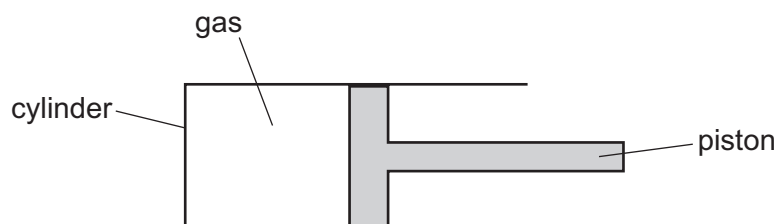
What is the energy store in the battery?

- A chemical
 B electrostatic
 C gravitational
 D kinetic

14 What is a property of both liquids and gases?

- A They always fill their containers.
- B They are incompressible.
- C They can flow.
- D They have particles in fixed positions.

15 In an experiment, the volume of a gas in a cylinder is measured as the pressure of the gas is increased.



The values obtained are shown.

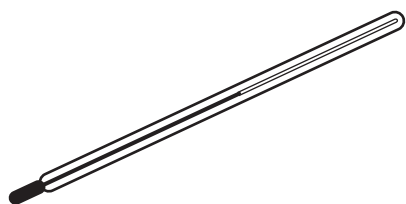
pressure P/Pa	1.0×10^5	2.0×10^5	3.0×10^5	4.0×10^5
volume V/m^3	20×10^{-5}	10×10^{-5}	6.0×10^{-5}	4.0×10^{-5}

What is a possible explanation for these results?

- A After the pressure is doubled, gas starts leaking into the cylinder.
- B After the pressure is doubled, gas starts leaking out of the cylinder.
- C The temperature of the gas is constant.
- D The temperature of the gas is increasing.

16 The diagrams show examples of thermal expansion.

P



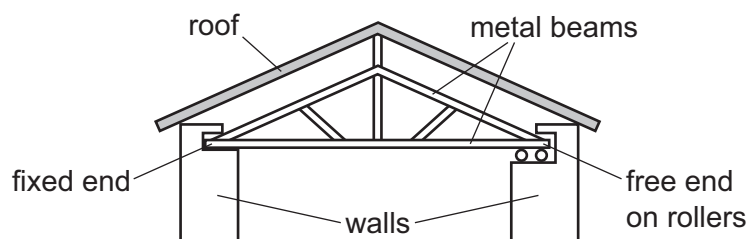
expansion of liquid in a thermometer

Q



expansion of a tight lid on a glass container

R



expansion of the long metal beams supporting a roof

In which diagrams is the thermal expansion useful?

- A** P, Q and R **B** P and Q only **C** P only **D** Q and R only

17 Which statement about copper explains why it is a better thermal conductor than glass?

- A** Atomic vibration is passed on to neighbouring copper atoms slowly.
- B** Atoms move through the copper and pass on kinetic energy.
- C** There are density changes within the copper.
- D** There are free electrons in the copper.

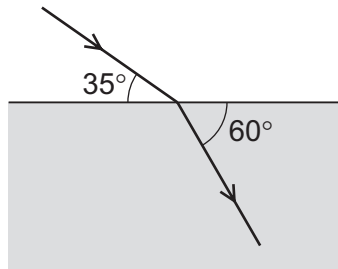
18 Which statement describes the transfer of thermal energy in a liquid by convection?

- A** A decrease in its density causes the heated liquid to rise.
- B** Free electrons carry energy large distances through the liquid.
- C** Infrared radiation passes through the liquid and transfers energy.
- D** Particles of the liquid vibrate and pass energy to neighbouring particles.

19 Which characteristics describe the image formed by a vertical plane mirror?

- A real and upside down
- B virtual and upright
- C real and larger than the object
- D virtual and smaller than the object

20 A ray of light passes into a block of transparent material as shown.

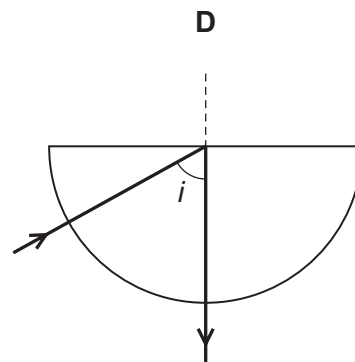
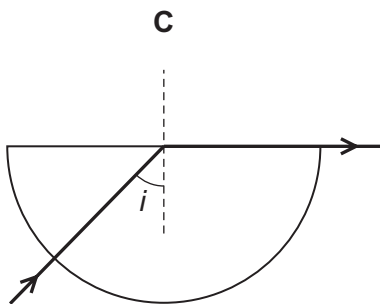
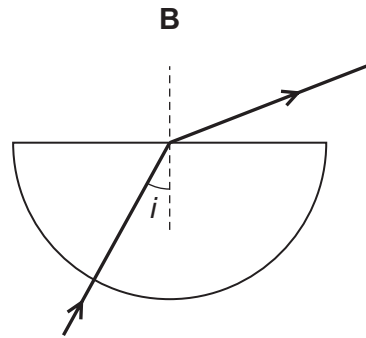
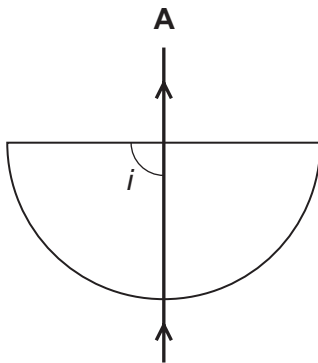


What is the refractive index of the transparent material?

- A 0.66
- B 1.15
- C 1.64
- D 1.83

21 The diagrams show light incident on the straight edge of a semi-circular glass block after passing through it.

Which diagram shows refraction and an angle i equal to the critical angle of the glass?



22 Which radioactive isotope can be used for measuring cracks or flaws in metal components?

- A americium-241, an alpha emitter with a half-life of 432 years
- B cobalt-60, a gamma emitter with a half-life of 5 years
- C radium-223, an alpha emitter with a half-life of 11 days
- D barium-137, a gamma emitter with a half-life of 3 minutes

23 Ultrasound is a type of sound wave that is not audible to a healthy human ear.

Which range of frequencies contains **only** ultrasound?

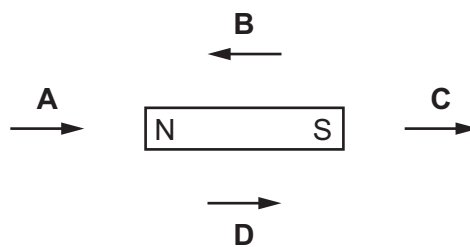
- A 30 Hz–3.0 kHz
- B 300 Hz–30 kHz
- C 3.0 kHz–3.0 MHz
- D 30 kHz–30 MHz

24 What proves that a metal bar is a permanent magnet?

- A It attracts both ends of a compass needle.
- B It attracts one end of another magnet.
- C It conducts electricity.
- D It repels one end of another magnet.

25 The diagram shows a bar magnet and four labelled arrows.

Which arrow shows the direction of the magnetic field at the point where the arrow is shown?



26 A shoe becomes positively charged by friction when it rubs against a carpet.

What happens as the shoe becomes charged?

- A Negative electrons are transferred to the carpet.
- B Negative electrons are transferred to the shoe.
- C Positive electrons are transferred to the carpet.
- D Positive electrons are transferred to the shoe.

- 27** A metal conductor is connected between the positive and negative terminals of a battery.

Which row gives the direction of movement of the particles that flow in the conductor and their name?

	direction of movement	name of particles
A	from positive terminal to negative terminal	protons
B	from positive terminal to negative terminal	electrons
C	from negative terminal to positive terminal	protons
D	from negative terminal to positive terminal	electrons

- 28** Three identical cells are connected in parallel to a resistor.

What is the advantage of using three cells in parallel rather than using a single cell?

- A** Each cell produces more energy.
- B** Each cell supplies more charge.
- C** Each cell takes longer to discharge.
- D** The e.m.f. is greater than that of a single cell.

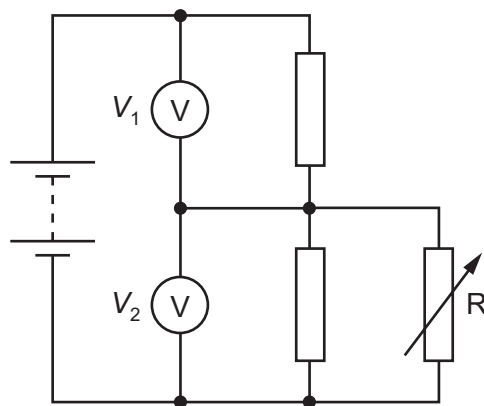
- 29** A student has three 15Ω resistors.

She connects the resistors together in different arrangements.

What is the minimum resistance and what is the maximum resistance of the arrangements that the student can produce?

	minimum resistance/ Ω	maximum resistance/ Ω
A	5.0	15
B	5.0	45
C	15	30
D	15	45

- 30** The circuit diagram shows a variable resistor R connected in parallel to the lower section of a potential divider.



The resistance of R increases.

What happens to the two voltmeter readings V_1 and V_2 ?

	V_1	V_2
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

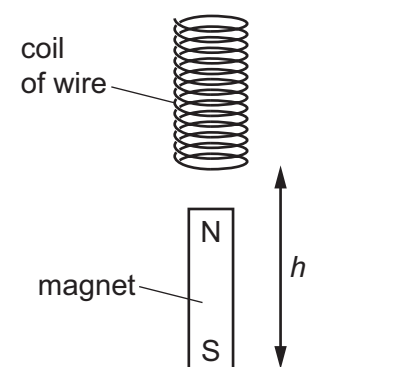
- 31** The power consumption of a microwave oven is 1500 W.

The cost of 1.0 kWh of electrical energy is 24 cents.

What is the cost of using the microwave oven for 10 minutes?

- A** 6 cents
- B** 36 cents
- C** 60 cents
- D** 360 cents

- 32 A coil of wire above a magnet is dropped. As the coil falls over the magnet, an e.m.f. is induced across the coil.



What does **not** affect the magnitude of the e.m.f. induced across the coil?

- A changing the number of turns of wire in the coil
 - B changing the height h from which the coil is dropped
 - C changing the direction of the magnet, so the north pole is at the bottom
 - D changing the strength of the magnetic field from the magnet
- 33 The results of the alpha-particle scattering experiment give evidence for which of the following?
- A nuclear fusion
 - B radioactive decay
 - C the existence of isotopes
 - D the size of the nucleus
- 34 A nucleus of an isotope of cobalt is represented by the symbol ${}^{59}_{27}\text{Co}$.

What is the composition of this nucleus?

	number of protons	number of neutrons
A	27	32
B	27	59
C	59	27
D	59	32

- 35 The only stable isotope of gold, Au, has 79 protons and 118 neutrons in a nucleus of an atom.

How many electrons are there in a neutral atom of this isotope?

- A 39
- B 79
- C 118
- D 197

- 36** A sample of a radioactive isotope with a very long half-life is placed next to a detector. Sheets of different materials are inserted, one at a time, between the radioactive sample and the detector.

The background count rate is subtracted from the readings on the detector and the corrected count rates are recorded in the table.

type of material	<u>corrected count rate</u> counts / s
no sheet inserted	450
thin aluminium sheet	381
thin lead sheet	285
thin paper sheet	452

Which types of radiation are emitted by the radioactive sample?

- A** alpha-particles and beta-particles only
 - B** alpha-particles only
 - C** beta-particles and gamma radiation only
 - D** gamma radiation only
- 37** In a fission reactor, which particle causes a uranium-235 nucleus to split?
- A** alpha-particle
 - B** gamma ray
 - C** neutron
 - D** proton
- 38** What is the purpose of moderators in nuclear reactors?
- A** to absorb all the neutrons in order to stop the chain reaction
 - B** to produce more neutrons in order to create a chain reaction
 - C** to remove thermal energy in order to control the chain reaction
 - D** to slow down neutrons in order to continue the chain reaction
- 39** In which region of the electromagnetic spectrum does the Sun radiate the most energy?
- A** infrared region
 - B** microwave region
 - C** radio wave region
 - D** X-ray region

40 Four of the stages in the life cycle of a star, until it becomes a red giant, are shown.

- W Inward force of gravitational attraction is balanced by an outward force from its centre.
- X Internal gravitational collapse produces an increase in temperature.
- Y It expands.
- Z Most of the hydrogen has been converted to helium.

In which order do these stages occur, starting with the earliest?

- A** $W \rightarrow X \rightarrow Y \rightarrow Z$
- B** $W \rightarrow X \rightarrow Z \rightarrow Y$
- C** $X \rightarrow W \rightarrow Y \rightarrow Z$
- D** $X \rightarrow W \rightarrow Z \rightarrow Y$

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

May/June 2023

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



- 1 A student wishes to measure directly the circumference of a football.

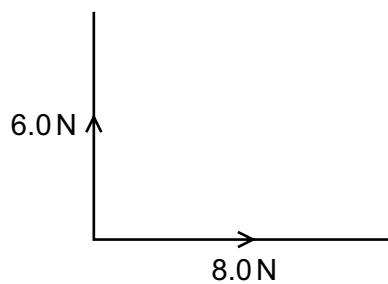
What is the most suitable instrument to use?

- A a clock only
- B a measuring tape only
- C a micrometer only
- D a ruler only

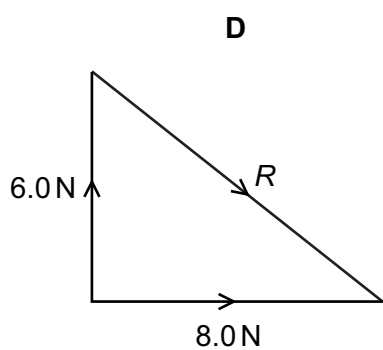
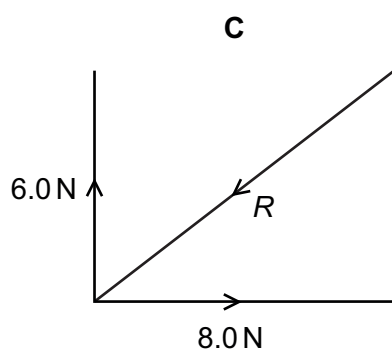
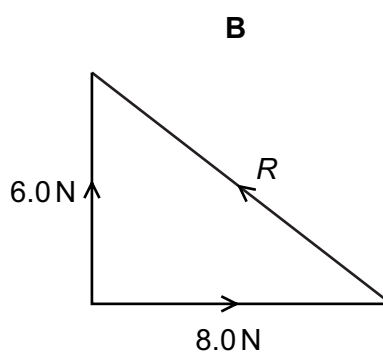
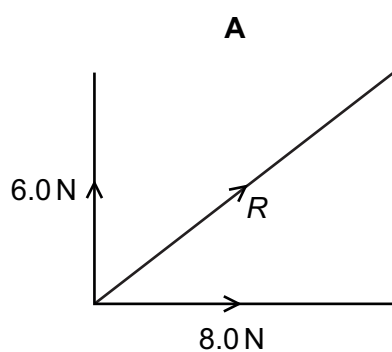
- 2 Which equation contains two vector quantities?

- A $\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$
- B $\text{average speed} = \frac{\text{distance travelled}}{\text{time taken}}$
- C $\text{density} = \frac{\text{mass}}{\text{volume}}$
- D $\text{volume} = \text{length} \times \text{width} \times \text{height}$

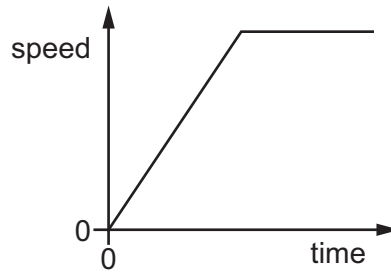
3 Forces of 6.0 N and 8.0 N act as shown.



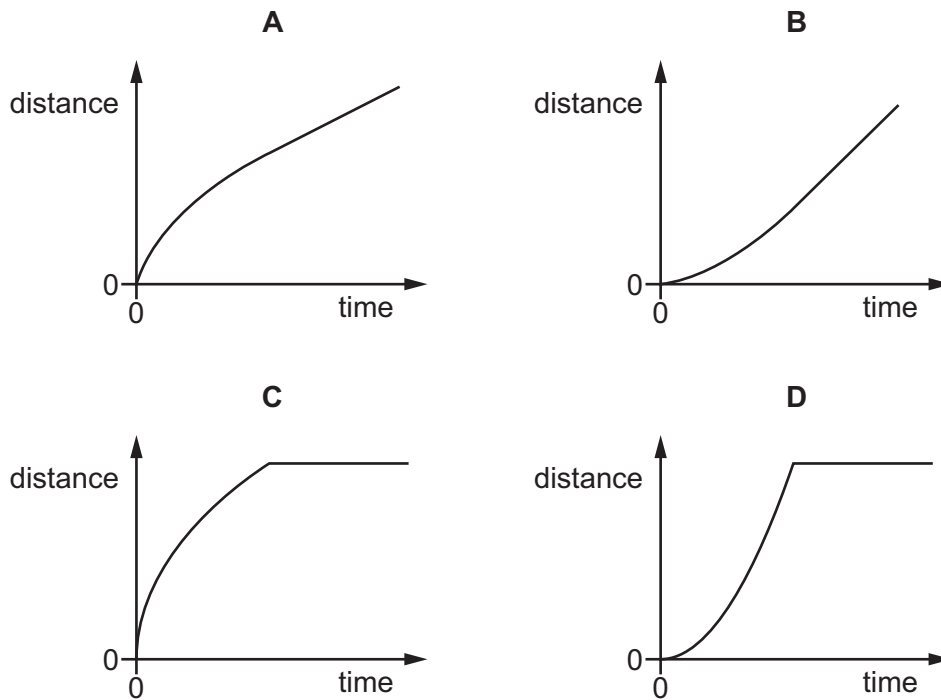
Which diagram shows the resultant R of these two forces?



- 4 A speed–time graph for the journey of a car is shown.



What is the distance–time graph for the journey?



- 5 The gravitational field strength on Earth is 9.8 N/kg .

The gravitational field strength on Mars is 3.7 N/kg .

The difference between the weight of an object on Earth and the weight of the same object on Mars is 25 N .

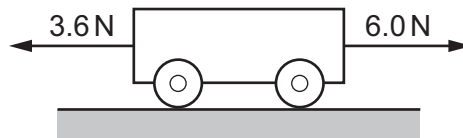
What is the mass of the object?

- A** 1.9 kg **B** 2.6 kg **C** 4.1 kg **D** 6.8 kg

- 6 A student determines the density of a solid metal cube.

Which pieces of equipment does the student use?

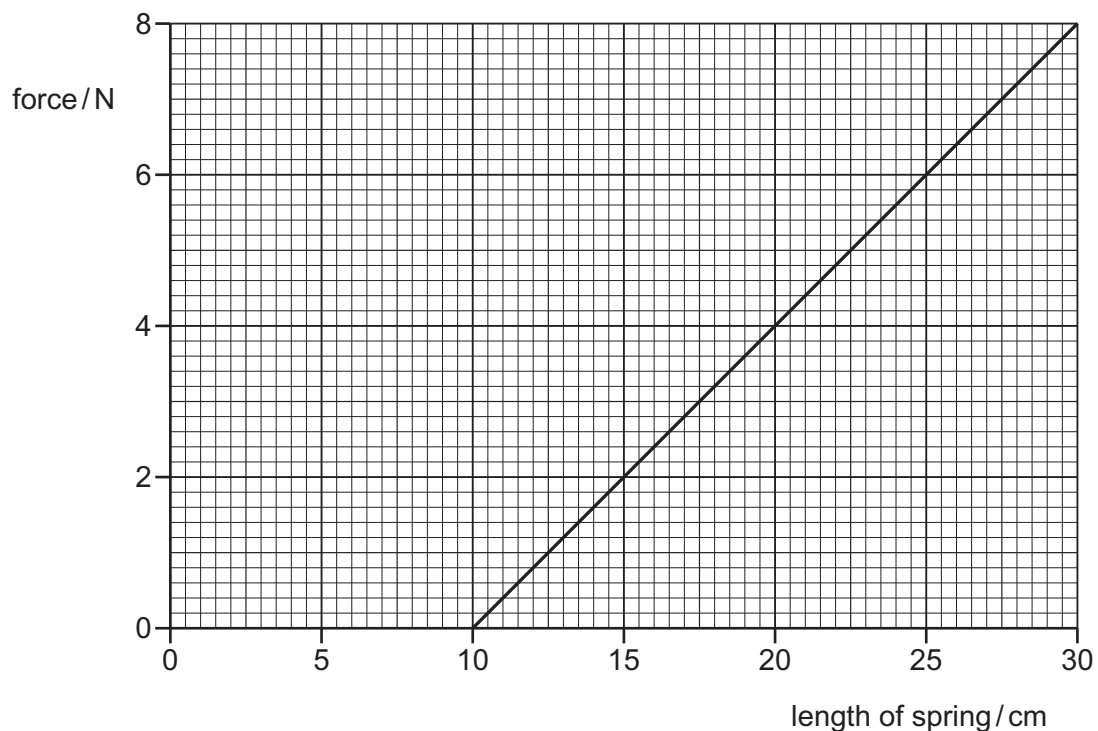
- A** a balance and a force meter
B a balance and a ruler
C a micrometer and a measuring cylinder
D a ruler and a measuring cylinder
- 7 The diagram shows the only two horizontal forces acting on a trolley that is accelerating at 0.12 m/s^2 .



What is the mass of the trolley?

- A** 0.050 kg **B** 0.13 kg **C** 20 kg **D** 80 kg
- 8 A spring is stretched to different lengths by a variable force.

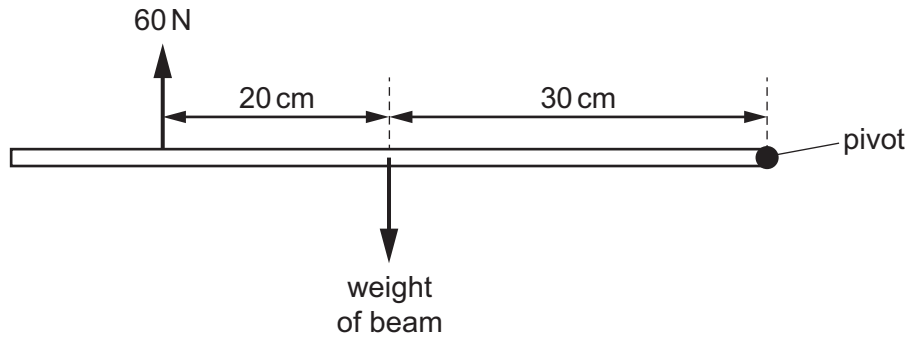
The graph shows how the size of the force depends on the length to which the spring is stretched.



What is the spring constant k of the spring?

- A** 0.27 N/cm **B** 0.40 N/cm **C** 2.5 N/cm **D** 3.8 N/cm

- 9 A uniform horizontal beam, pivoted at its right-hand end, is in equilibrium. A force of 60 N acts vertically upwards on the beam as shown.

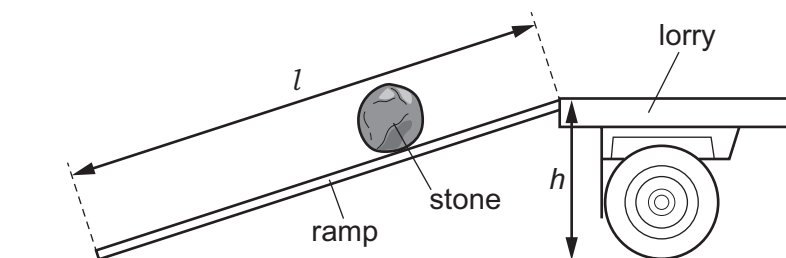


What is the weight of the beam?

- A** 36 N **B** 40 N **C** 90 N **D** 100 N
- 10 A constant frictional force acts on a moving object for a short amount of time causing the momentum of the object to decrease.

Which expression is used to calculate the time for which the force acts?

- A** $\text{time} = \frac{\text{change in momentum}}{\text{force}}$
- B** $\text{time} = \text{change in momentum} \times \text{force}$
- C** $\text{time} = \frac{\text{momentum}}{\text{force}}$
- D** $\text{time} = \text{momentum} \times \text{force}$
- 11 A large stone is rolled onto a lorry using a ramp.



The length of the ramp is l . The stone is lifted a height of h .

The mass of the stone is m . The weight of the stone is W .

Which expression is equal to the useful work done on the stone?

- A** $m \times h$ **B** $W \times h$ **C** $m \times l$ **D** $W \times l$

- 12 The diagram shows a burning candle.



Which energy store is associated with the burning candle?

- A** chemical
B elastic
C electrostatic
D nuclear
- 13 An African elephant has a mass of 6000 kg and each of its four feet has an area of 0.20 m^2 . The gravitational field strength g is 9.8 N/kg .

What is the average pressure due to the elephant when it is standing stationary on all four feet on level ground?

- A** 7.5 kPa **B** 30 kPa **C** 74 kPa **D** 290 kPa
- 14 Which row describes the shape and the volume of a liquid at constant temperature?

	shape	volume
A	fixed	fixed
B	fixed	not fixed
C	not fixed	fixed
D	not fixed	not fixed

- 15 Which row shows how the forces and the distances between the particles in a solid generally compare with the forces and distances in a liquid?

	forces between particles in a solid	distances between particles in a solid
A	stronger	greater
B	stronger	smaller
C	weaker	greater
D	weaker	smaller

- 16** An incomplete statement about latent heat is given.

Latent heat is the1..... required to overcome the bonds between molecules and change the2..... of a substance.

Which words are inserted into gap 1 and gap 2 to correctly complete the sentence?

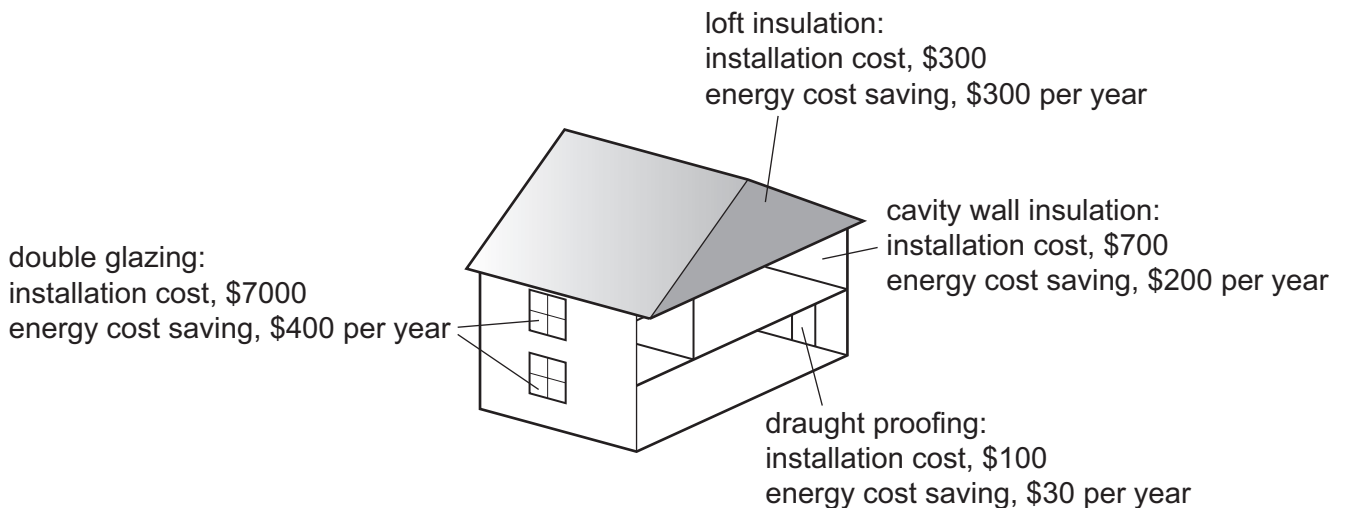
	gap 1	gap 2
A	energy	state
B	force	state
C	energy	temperature
D	force	temperature

- 17** A cold solid is placed on top of a hot solid. Thermal energy is transferred from the hot solid to the cold one.

What is the explanation for this?

- A** A hot solid expands, so its particles will move further apart.
- B** Energy is transferred from one particle to the next.
- C** Heat always rises.
- D** Molecules are free to move randomly through the solids.

- 18** A house is to be thermally insulated to reduce thermal energy loss and reduce fuel costs.

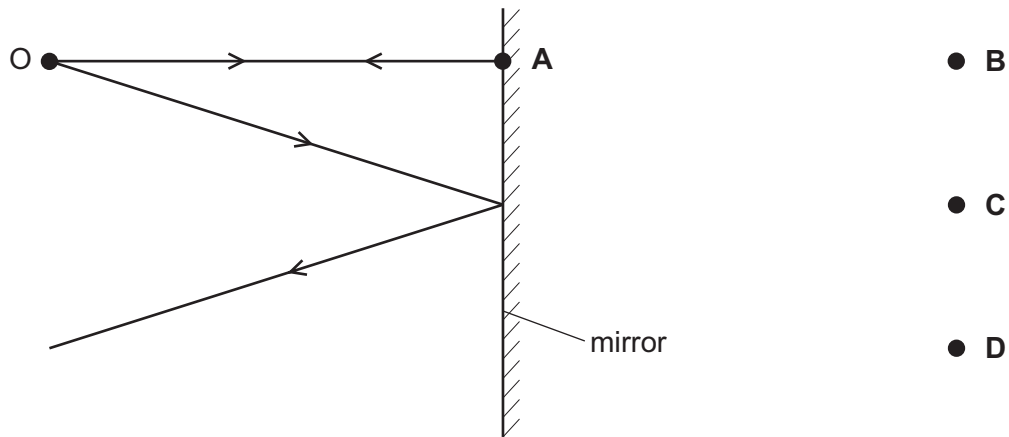


Which type of thermal insulation pays back the cost of installation in the shortest time?

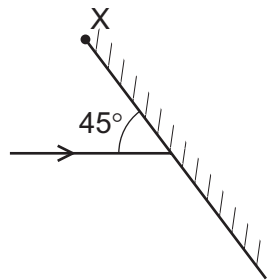
- A** cavity wall insulation
- B** double glazing
- C** draught proofing
- D** loft insulation

- 19 The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

At which position is the image formed?



- 20 Light is incident on a mirror at an angle of 45° as shown. The mirror can be rotated about an axis into the page through point X.



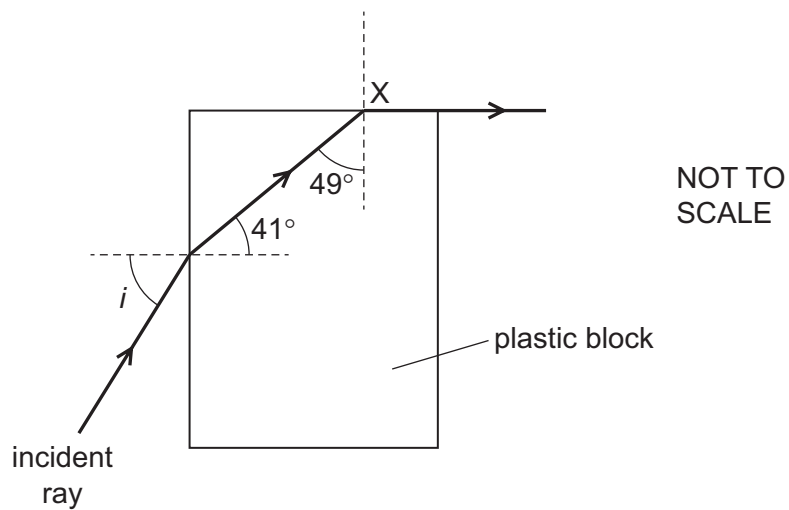
The mirror is rotated until the light is reflected back along its original path.

Through which angle is the mirror rotated?

- A 22.5° clockwise
- B 22.5° anticlockwise
- C 45° clockwise
- D 45° anticlockwise

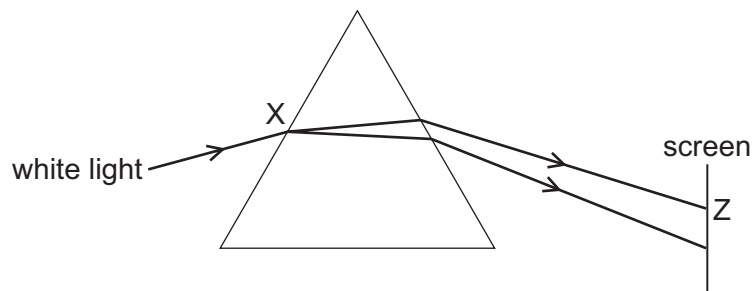
- 21** A ray of light is incident on a plastic block in air, at an angle of incidence i . The refractive index of the plastic is n .

The light ray refracts along the plastic-air boundary when it reaches X.



Which equation is correct?

- A** $\sin i = \frac{\sin 41^\circ}{n}$
- B** $\sin i = \frac{n}{\sin 49^\circ}$
- C** $\sin i = n \times \sin 41^\circ$
- D** $\sin i = n \times \sin 49^\circ$
- 22** The diagram shows white light entering a prism.



What happens at point X and what is the colour of the light that strikes the screen at point Z?

	at point X	at point Z
A	dispersion only	blue light
B	dispersion and refraction	blue light
C	dispersion and refraction	red light
D	refraction only	red light

- 23** Which row names a region of the electromagnetic spectrum with a wavelength longer than that of visible light and a region with a frequency that is greater than that of visible light?

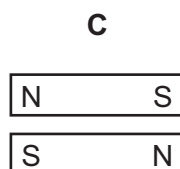
	wavelength longer than visible light	frequency greater than visible light
A	infrared	ultraviolet
B	microwave	radio
C	radio	microwave
D	ultraviolet	infrared

- 24** Sound travels at different speeds in different states of matter.

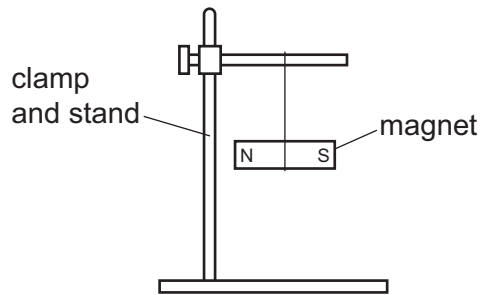
Which list gives the three states of matter arranged in order of the speed of sound from slowest to fastest?

- A** gas → liquid → solid
B liquid → solid → gas
C solid → gas → liquid
D solid → liquid → gas
- 25** Bar magnets and various non-magnetic and demagnetised metal bars are placed in the different arrangements shown.

In which arrangement do the bars repel?

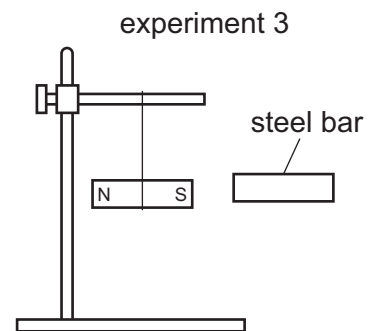
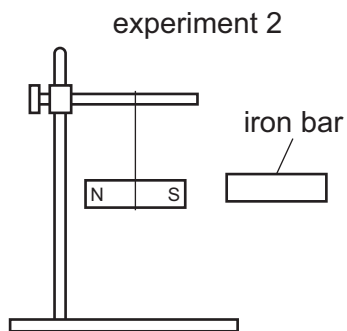
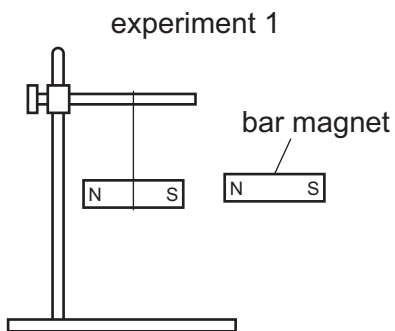


26 A student uses a wooden clamp and a stand to hold a freely suspended bar magnet.



The diagrams show that:

- in experiment 1, the student holds a bar magnet next to the suspended magnet
- in experiment 2, the student holds an iron bar next to the suspended magnet
- in experiment 3, the student holds a steel bar next to the suspended magnet.



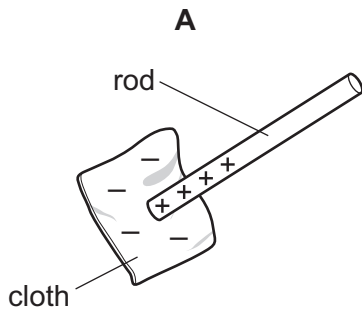
In which experiments does the suspended magnet move?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

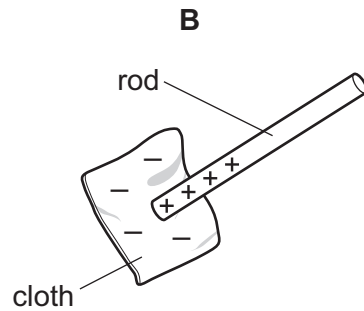
27 In an electrostatics experiment, a plastic rod is rubbed with a cloth.

The **cloth** becomes negatively charged.

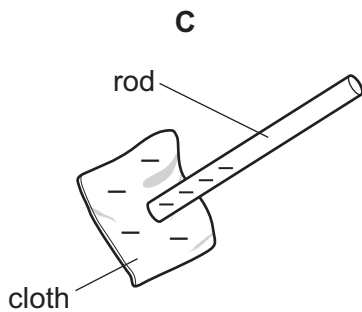
Which diagram shows the charge on the rod and describes the movement of charge?



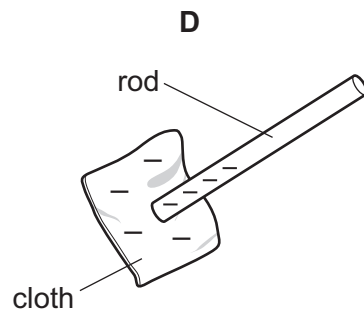
electrons move from the rod on to the cloth



protons move from the cloth on to the rod



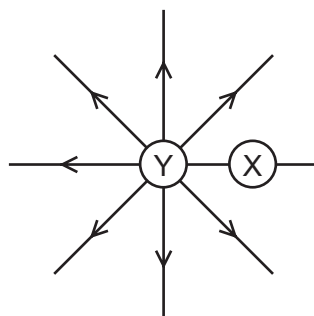
electrons move from the cloth on to the rod



protons move from the rod on to the cloth

28 Object X is stationary and positively charged. It experiences a force due to the field produced by object Y.

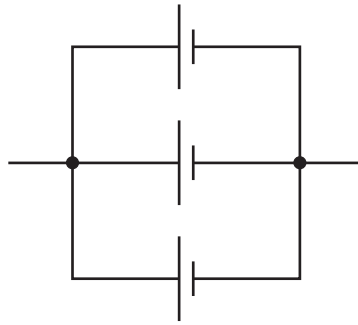
The arrows show the direction of the field produced by Y.



Which statement about the direction of the force on X is correct?

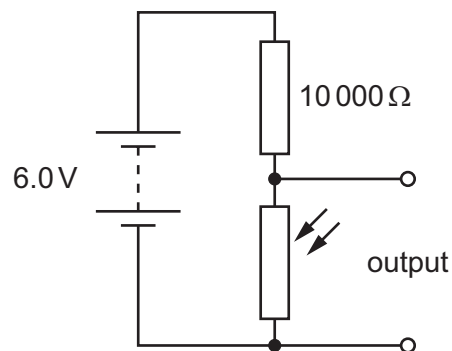
- A** It is towards the right because it is in an electric field.
- B** It is towards the left because it is in an electric field.
- C** It is towards the right because it is in a magnetic field.
- D** It is towards the left because it is in a magnetic field.

- 29 The diagram shows a battery of three 1.5 V cells.



What is an advantage of this arrangement of cells?

- A The battery can supply a current for a longer time than a single 1.5 V cell.
 - B The battery can supply any e.m.f. between 0 V and 4.5 V.
 - C The battery supplies more energy to each coulomb of charge than a single 1.5 V cell.
 - D The e.m.f. of the battery is 4.5 V.
- 30 A potential divider consists of a light-dependent resistor (LDR) and a resistor connected to a 6.0 V battery.



The output is 3.0 V.

What is the resistance of the LDR?

- A 0
- B between 0 and 10 000 Ω
- C 10 000 Ω
- D more than 10 000 Ω

- 31 Three students each suggest a consequence of wiring lamps in parallel rather than in series.

student 1 If one lamp breaks, the other lamps stay lit.

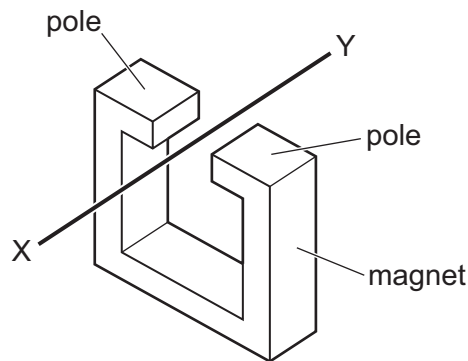
student 2 The lamps can be switched on and off separately.

student 3 The p.d. across each lamp is equal to the p.d. across the supply.

Which students are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

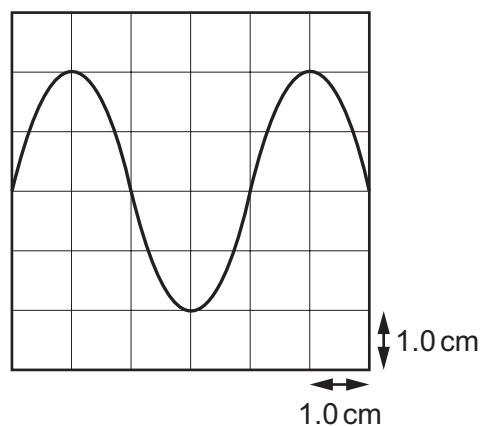
- 32 When wire XY is moved downwards between the poles of a stationary magnet, an e.m.f. is produced across X and Y.



Which action produces an e.m.f. across X and Y in the opposite direction?

- A** Move both the wire and the magnet upwards at the same speed.
B The wire is kept stationary and the magnet is moved upwards.
C The wire is moved downwards and the magnet is moved upwards.
D The wire is moved upwards and the magnet is kept stationary.

- 33 An a.c. voltage is displayed on an oscilloscope screen. The Y-gain is set at 2.0 V/cm .



What is the maximum value of the voltage?

- A** 2.0 V **B** 4.0 V **C** 8.0 V **D** 12 V

- 34** The results of a scattering experiment show that a very small nucleus contains most of the matter in an atom.

What is scattered in this experiment?

- A** alpha-particles
- B** beta-particles
- C** gamma radiation
- D** gold nuclei

- 35** Atoms P and Q are isotopes of the same element.

How does the composition of an atom of P compare with the composition of an atom of Q?

	number of protons	number of neutrons
A	different	different
B	different	same
C	same	different
D	same	same

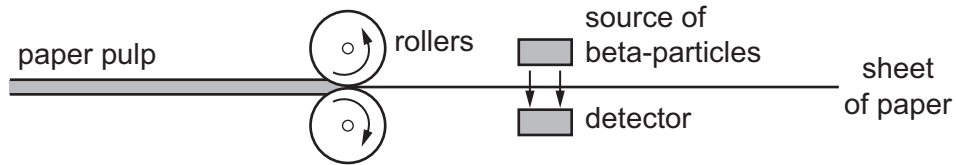
- 36** The nucleus of an atom emits an alpha-particle.

How do the proton number and nucleon number change?

	proton number	nucleon number
A	decreases by 2	decreases by 4
B	decreases by 4	decreases by 2
C	increases by 2	increases by 4
D	increases by 4	increases by 2

- 37** The diagram shows how the thickness of paper is measured during manufacture.

If the sheet is too thick, fewer beta-particles can reach the detector.



Why are beta-particles used for this purpose?

- A** Alpha-particles are all stopped by the paper.
- B** Alpha-particles make the paper radioactive.
- C** Gamma radiation is all stopped by the paper.
- D** Gamma radiation is only produced by sources with a short half-life.

- 38** A sample of tritium has a mass of 32 mg.

Tritium has a half-life of 12 years.

How much of the sample of tritium decays in 36 years?

- A** 4.0 mg **B** 8.0 mg **C** 24 mg **D** 28 mg
- 39** It takes 27 days for the Moon to orbit the Earth.

The average value of the radius of the Moon's orbit is 3.8×10^8 m.

What is the average orbital speed of the Moon?

- A** 160 m/s
 - B** 1000 m/s
 - C** 1.4×10^8 m/s
 - D** 8.8×10^8 m/s
- 40** What is the nuclear reaction that takes place in stable stars?
- A** fission of helium to produce hydrogen
 - B** fission of hydrogen to produce helium
 - C** fusion of helium into hydrogen
 - D** fusion of hydrogen into helium

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Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	D	1
3	C	1
4	A	1
5	D	1
6	B	1
7	B	1
8	C	1
9	C	1
10	A	1
11	A	1
12	B	1
13	A	1
14	C	1
15	B	1
16	B	1
17	D	1
18	A	1
19	B	1
20	C	1
21	C	1
22	B	1
23	D	1
24	D	1
25	D	1
26	A	1
27	D	1
28	C	1

Question	Answer	Marks
29	B	1
30	B	1
31	A	1
32	C	1
33	D	1
34	A	1
35	B	1
36	C	1
37	C	1
38	D	1
39	A	1
40	D	1



Cambridge O Level

PHYSICS

5054/11

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Question	Answer	Marks
1	B	1
2	A	1
3	A	1
4	B	1
5	C	1
6	B	1
7	C	1
8	B	1
9	D	1
10	A	1
11	B	1
12	A	1
13	C	1
14	C	1
15	B	1
16	A	1
17	B	1
18	D	1
19	B	1
20	C	1
21	C	1
22	C	1
23	A	1
24	A	1
25	D	1
26	A	1
27	A	1
28	A	1

Question	Answer	Marks
29	A	1
30	C	1
31	A	1
32	D	1
33	B	1
34	A	1
35	C	1
36	A	1
37	A	1
38	D	1
39	B	1
40	D	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2022

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

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- 1 A force of 3.0 N and a force of 4.0 N act on an object.

What is the maximum possible resultant of these two forces?

- A** 1.0 N **B** 5.0 N **C** 7.0 N **D** 12 N

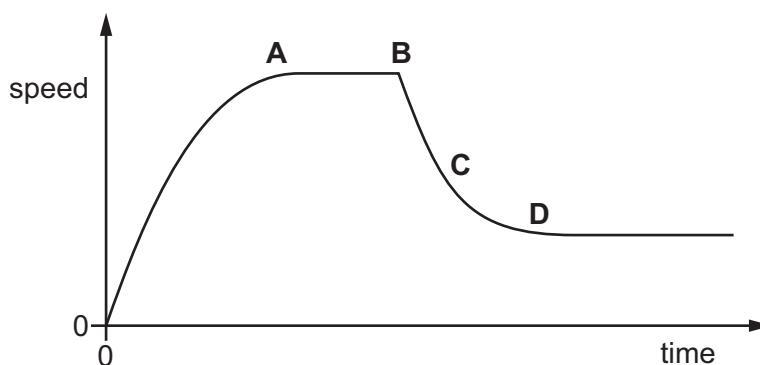
- 2 A length of copper wire is labelled: length 0.50 m and diameter 0.50 mm.

Which instruments are most suitable to measure accurately the length and the diameter of the wire?

	length	diameter
A	metre rule	metre rule
B	metre rule	micrometer
C	calipers	metre rule
D	calipers	micrometer

- 3 The graph shows the speed–time graph for a parachutist who jumps from a plane but does not open his parachute immediately.

At which point does he open his parachute?



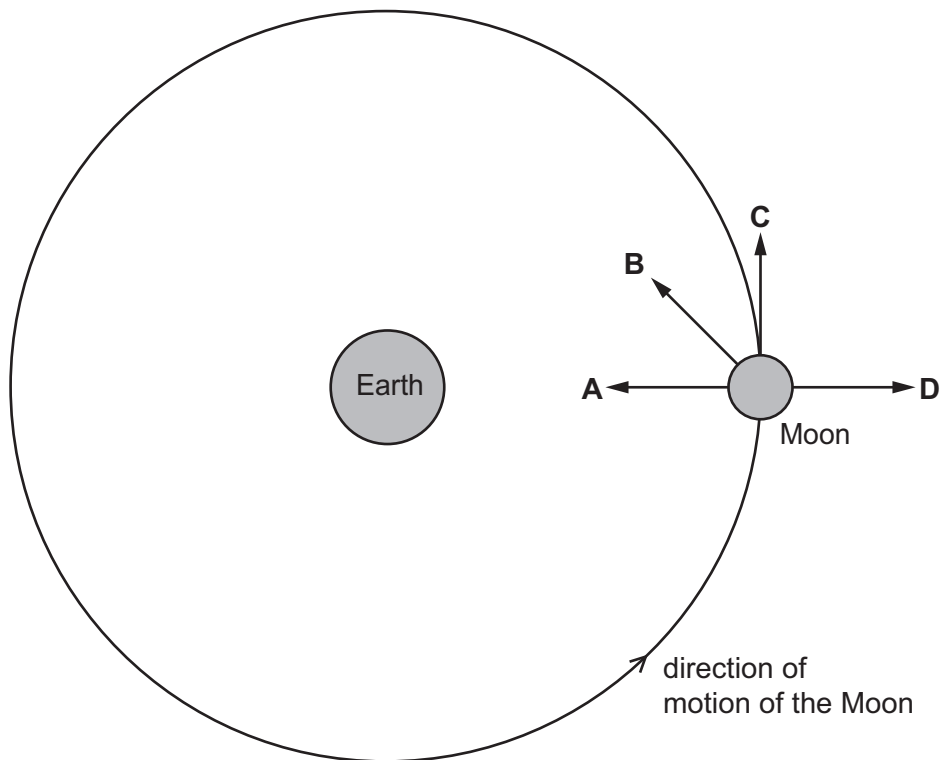
- 4 A 60 kg passenger enters a stationary lift. The gravitational field strength g is 10 N/kg.

How much force does the floor of the lift exert on the passenger when the lift accelerates upwards at 2.0 m/s^2 ?

- A** 120 N **B** 480 N **C** 600 N **D** 720 N

- 5 The diagram represents the Moon in its orbit around the Earth.

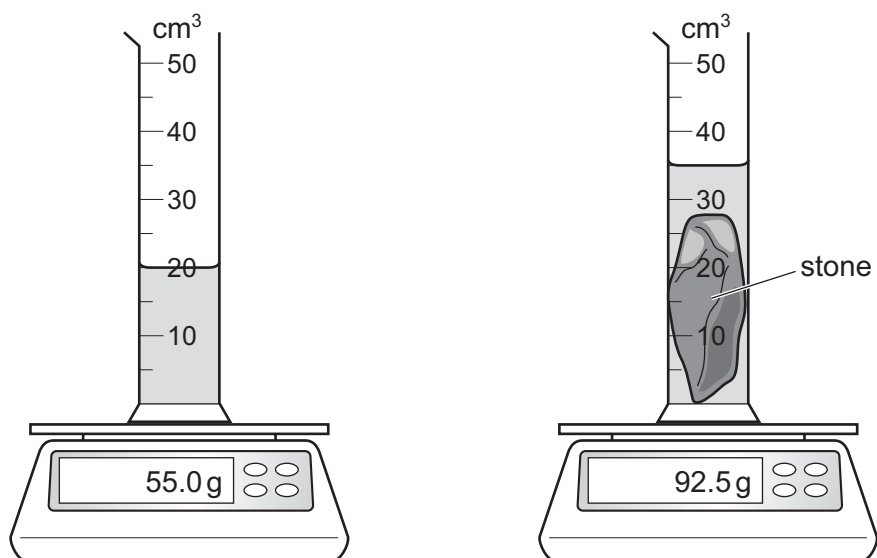
Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?



- 6 A measuring cylinder containing water is placed on a balance.

A stone is placed into the water.

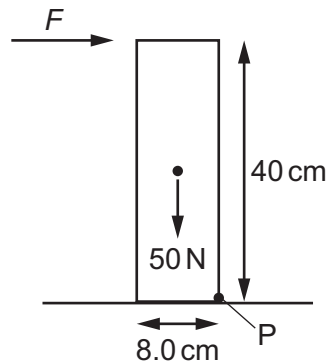
The diagram shows the readings on the balance and on the measuring cylinder.



What is the density of the stone?

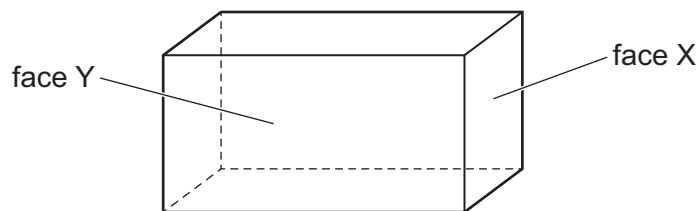
- A** 1.1 g/cm^3 **B** 1.5 g/cm^3 **C** 2.5 g/cm^3 **D** 2.6 g/cm^3

- 7 The diagram shows a uniform solid rectangular block of weight 50 N that is pivoted about point P. The height of the block is 40 cm. The base of the block is 8.0 cm wide.



Which horizontal force F just makes the block start to rotate about P?

- A** 2.5 N **B** 5.0 N **C** 10 N **D** 160 N
- 8 The centre of mass of a solid rectangular block is at its centre. A small heavy weight is available.

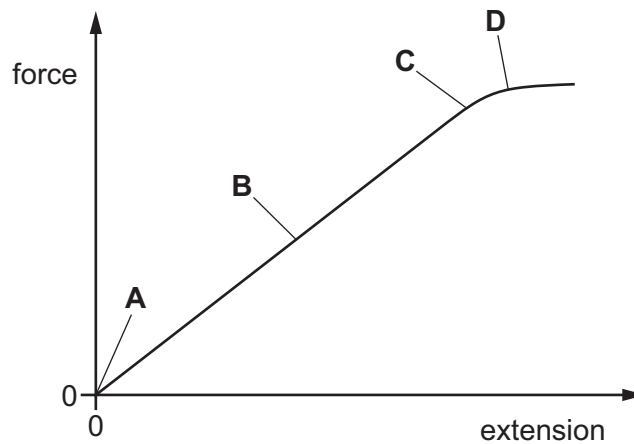


In which arrangement is the centre of mass the lowest?

- A** with face X on a table
B with face Y on a table
C with face X on a table and the heavy weight attached centrally on top of the block
D with face Y on a table and the heavy weight attached centrally on top of the block

- 9 The graph shows how the extension of a spring depends on the force applied.

Which point is the limit of proportionality?



- 10 An elastic spring has an unstretched length of 30 cm.

A load of 6.0 N is hung from the spring and the length of the spring is now 66 cm.

The 6.0 N load is removed and the spring returns to its original length. A load of 2.0 N is now hung from the spring.

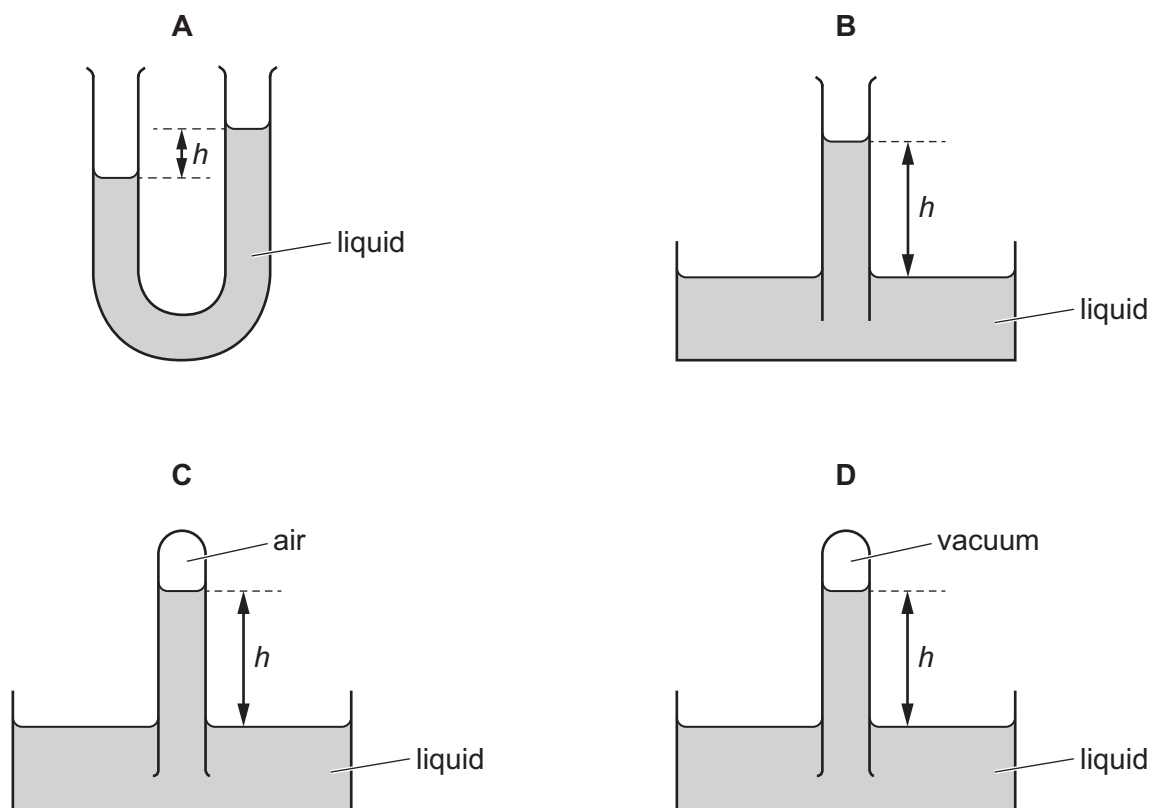
What is the new length of the spring?

- A 22 cm B 40 cm C 42 cm D 52 cm

11 A barometer is an instrument used to measure atmospheric pressure.

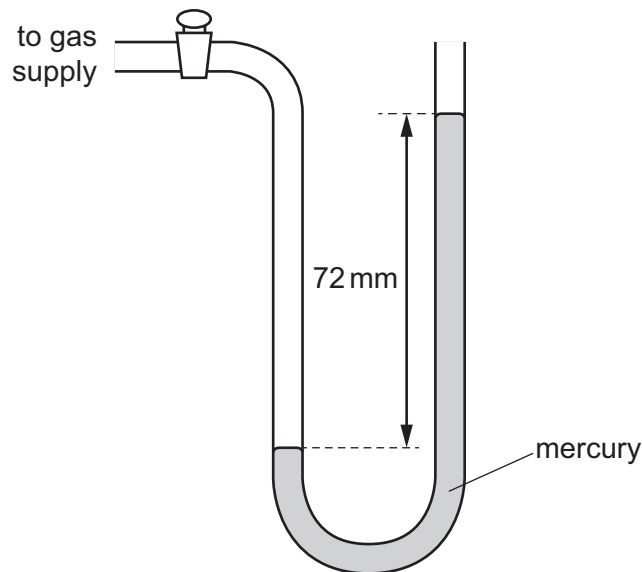
In one type of barometer the height of a liquid in a tube is measured.

In which diagram does the height h represent atmospheric pressure?



- 12** A manometer contains mercury of density $14\,000\text{ kg/m}^3$. The manometer is connected to a gas supply and the difference in the height of the mercury levels is 72 mm . The atmospheric pressure is 100 kPa .

The gravitational field strength g is 10 N/kg .



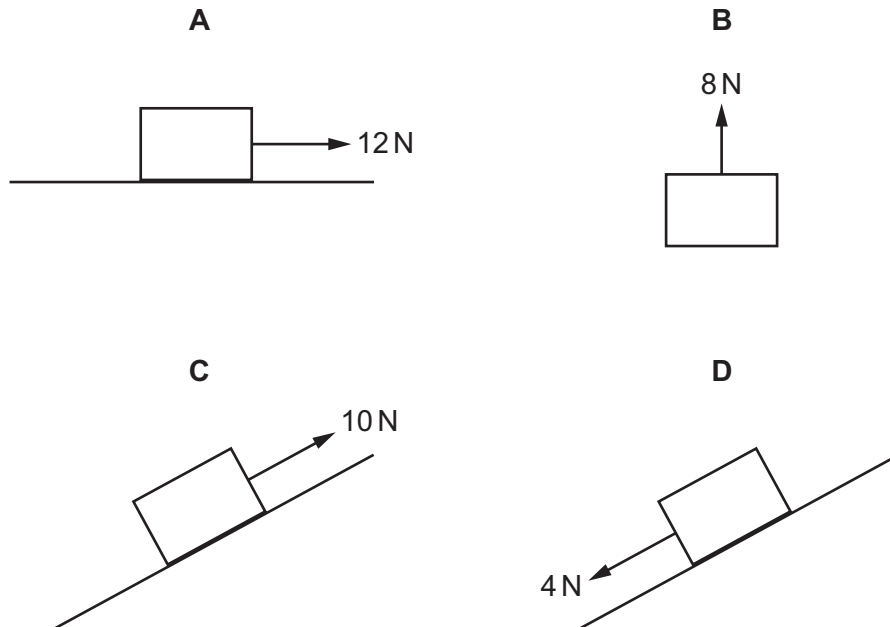
What is the pressure of the gas supply?

- A** 90 kPa **B** 99 kPa **C** 110 kPa **D** 200 kPa
- 13** Which device provides a continuous, steady energy output for the longest time?
- A** a nuclear reactor
B a solar panel
C a 1.5 V cell
D a wind generator

- 14** A load is pulled by a rope attached to a motor. The resultant force exerted by the rope on the load is shown in the diagrams.

In each diagram, the load moves in the direction of the force shown and takes 10 s to travel 1.0 m.

In which diagram does the motor work with the greatest power?



- 15** The electrical energy supplied to a kettle is 600 kJ.

Of this energy, 45 kJ is transferred to the surroundings and 15 kJ is used to heat the casing of the kettle.

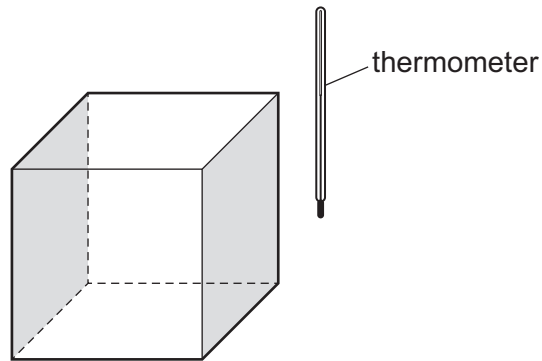
The remaining energy is used to heat the water.

What is the efficiency of the kettle?

- A** 0.10 **B** 0.90 **C** 0.925 **D** 0.975
- 16** How is heat energy transferred through the vacuum of space from the Sun to the Moon's surface?
- A** conduction only
B convection only
C radiation only
D conduction, convection and radiation

- 17 A metal cube contains boiling water.

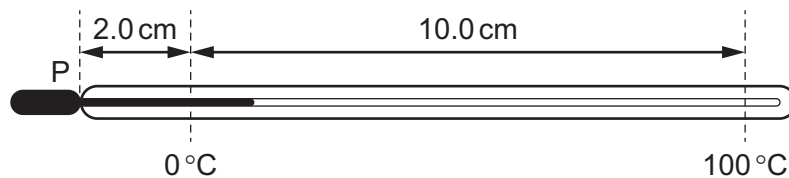
Each vertical face of the cube is painted a different colour or has a different texture.



Identical thermometers are held at equal distances from each vertical face.

Near which face does a thermometer read the greatest temperature?

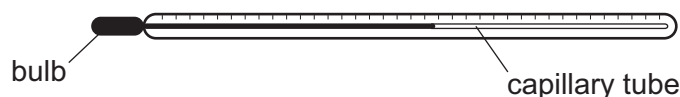
- A dull black
 - B dull white
 - C shiny black
 - D shiny white
- 18 A liquid column in a liquid-in-glass thermometer is 2.0 cm long at 0°C . The column expands by 10.0 cm when heated to 100°C .



Measuring from P, how long is the liquid column at 30°C ?

- A 2.3 cm
- B 3.0 cm
- C 5.0 cm
- D 7.0 cm

- 19 A liquid-in-glass thermometer consists of a bulb containing a liquid which expands into a thin capillary tube.



The liquid in the thermometer is replaced by the same volume of a different liquid that expands more for the same temperature rise. The length of the capillary tube remains the same.

How does the new thermometer compare with the old thermometer?

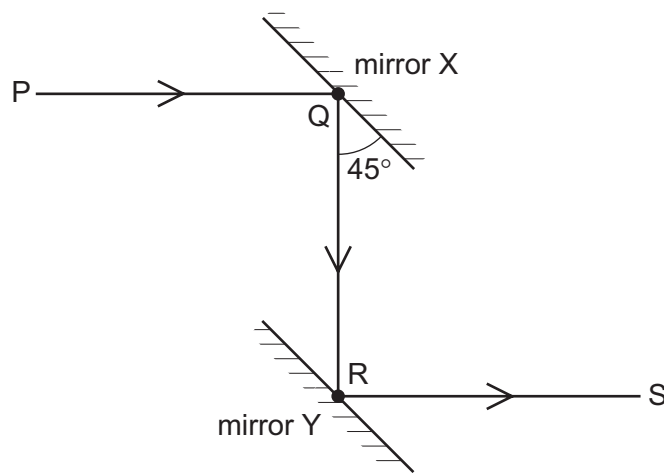
- A It has greater sensitivity and a greater range.
 - B It has greater sensitivity but a smaller range.
 - C It has the same sensitivity and the same range.
 - D It has the same sensitivity but a greater range.
- 20 Steam at 110°C condenses on a surface to form water droplets at 100°C .
- What happens after the steam comes into contact with the surface?
- A The molecules slow down and absorb energy from the surroundings.
 - B The molecules slow down and emit energy to the surroundings.
 - C The molecules stay at the same speed and absorb energy from the surroundings.
 - D The molecules stay at the same speed and emit energy to the surroundings.
- 21 Which list shows the states of matter in order of expansion from smallest to greatest for the same temperature rise?
- A gas $\rightarrow$ liquid $\rightarrow$ solid
 - B liquid $\rightarrow$ solid $\rightarrow$ gas
 - C solid $\rightarrow$ gas $\rightarrow$ liquid
 - D solid $\rightarrow$ liquid $\rightarrow$ gas
- 22 Which row describes the boiling and the evaporation of a liquid?

	boiling	evaporation
A	bubbles form throughout the liquid	only occurs at one temperature
B	bubbles form throughout the liquid	produces cooling
C	occurs at any temperature	only occurs at one temperature
D	occurs at any temperature	produces cooling

- 23** A hairdryer is used to blow air across the surface of water in a beaker so that the water evaporates.

What increases the rate of evaporation of the water?

- A** decreasing the speed of the air from the hairdryer
 - B** decreasing the mass of the water in the beaker
 - C** increasing the surface area of the water by using a wider beaker
 - D** increasing the volume of the water in the beaker
- 24** The diagram shows a ray PQ reflected by mirror X to a parallel mirror Y. The reflected ray along RS is parallel to PQ.

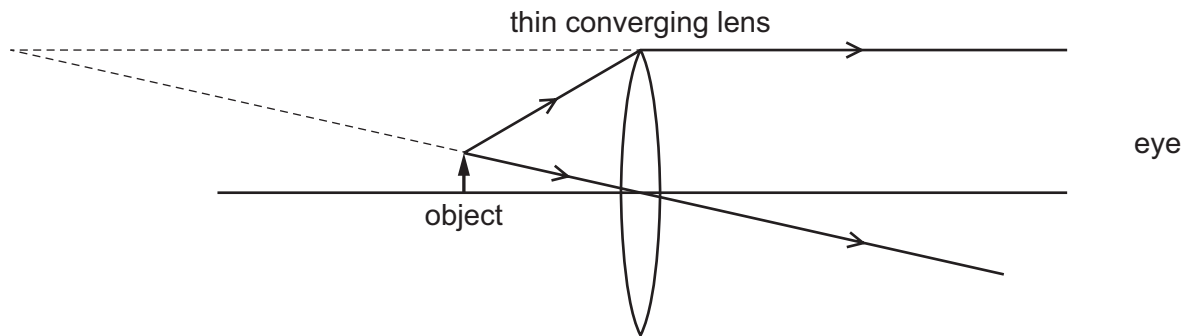


Which statement is correct?

- A** The angle between PQ and QR is 45° .
- B** The angle between QR and RS is 180° .
- C** The angle of incidence of PQ on mirror X is 60° .
- D** The angle of incidence of QR on mirror Y is 45° .

- 25** An object is viewed through a thin converging lens.

The diagram shows the paths of two rays from the top of the object to an eye.



How does the image compare with the object?

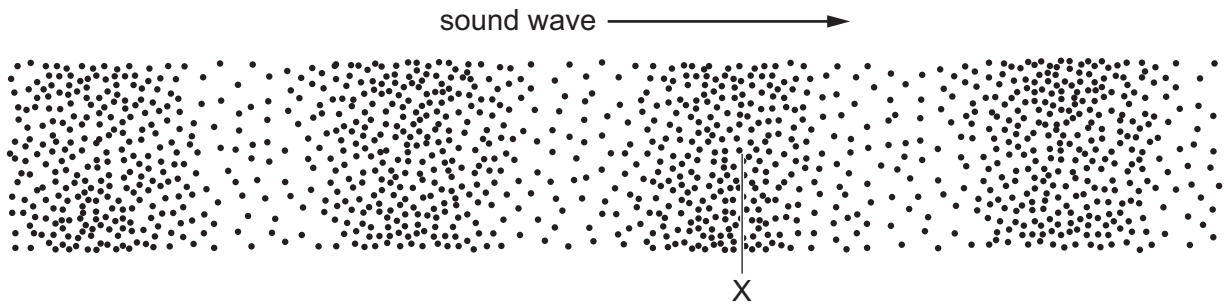
- A** It is larger and inverted.
 - B** It is larger and upright.
 - C** It is smaller and inverted.
 - D** It is smaller and upright.
- 26** Violet and indigo light have the shortest wavelengths in the spectrum of visible light.

Which three colours, in order of increasing wavelength, immediately follow indigo?

- A** blue → green → orange
- B** blue → green → yellow
- C** green → blue → yellow
- D** yellow → green → orange

- 27 A sound wave passes through a substance.

The diagram shows the position of molecules of the substance at one point in time.

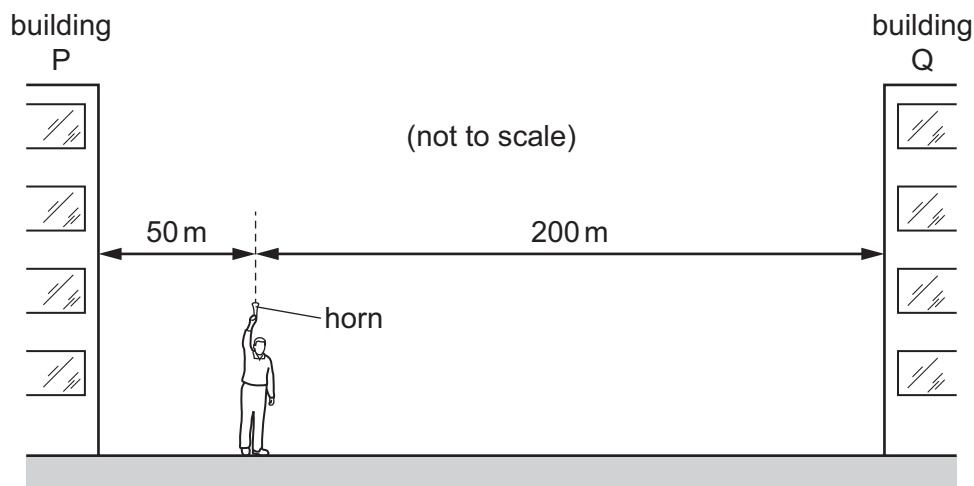


Which type of wave is a sound wave and what is at X?

	type of wave	at X
A	longitudinal	compression
B	longitudinal	rarefaction
C	transverse	compression
D	transverse	rarefaction

- 28 A man stands between two tall buildings, P and Q.

He is 50 m from P and 200 m from Q.



He sounds a horn.

He hears the first echo from building P and 1.0 s later he hears the first echo from building Q.

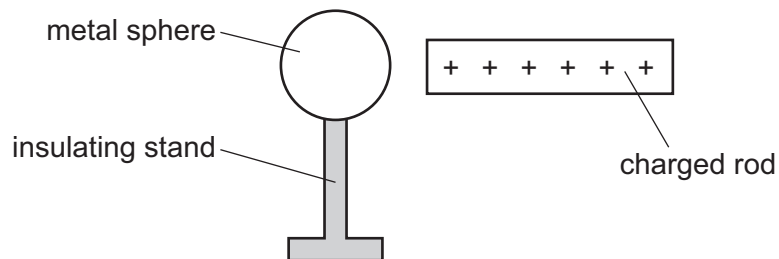
What is the speed of sound calculated using this information?

- A** 250 m/s **B** 300 m/s **C** 330 m/s **D** 500 m/s

29 What is one of the uses of ultrasound?

- A cleaning jewellery
- B fluorescent tubes
- C optical fibres
- D sunbeds

30 A positively charged rod is held near an uncharged metal sphere on an insulating stand.

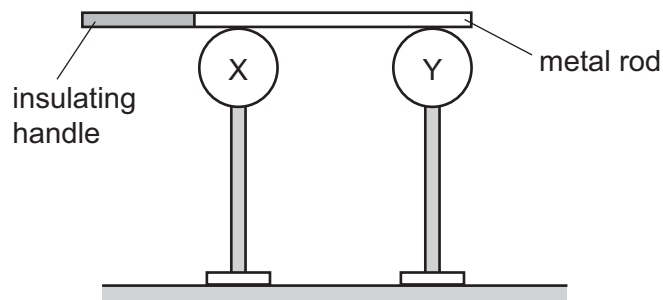


What is the distribution of charge on the sphere?



31 Two metal spheres are mounted on insulating stands. Sphere X is initially uncharged and sphere Y is initially positively charged.

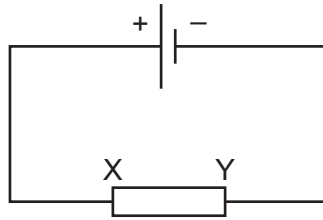
A metal rod, held by an insulating handle, is placed in contact with X and Y as shown.



What are the charges on X and on Y after the rod is placed in contact with them?

	charge on X	charge on Y
A	positive	positive
B	positive	uncharged
C	uncharged	positive
D	uncharged	uncharged

32 The diagram shows a simple electric circuit.



Which row describes the charge on an electron and the direction of electron flow through the resistor?

	charge on an electron	direction of electron flow
A	negative	X to Y
B	negative	Y to X
C	positive	X to Y
D	positive	Y to X

33 Charge of 0.40 C passes through a resistor in 1.0 s.

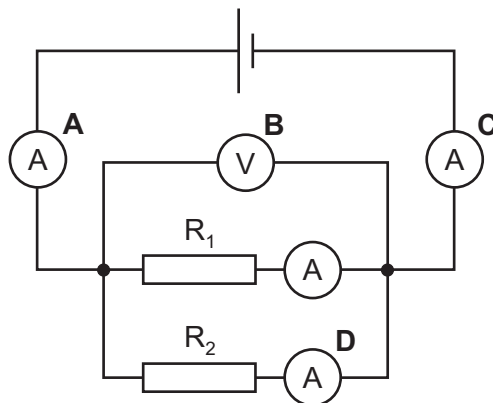
In 2.0 s, 20 J of thermal energy is produced in the resistor.

What is the potential difference across the resistor?

- A** 0.020 V **B** 0.040 V **C** 25 V **D** 50 V

34 The diagram shows a simple electric circuit.

Which meter measures the current in resistor R_2 ?



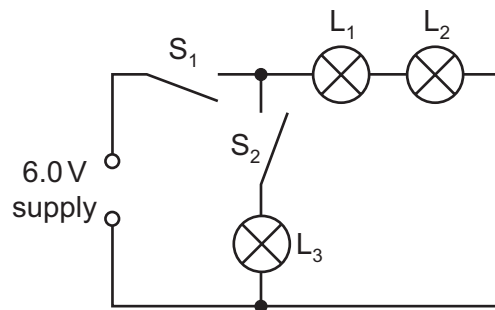
35 When the flash on a camera is used, a charge of 1.5 C flows for 0.0030 s through the flash lamp.

The average voltage across the flash lamp is 3600 V.

What is the electrical energy supplied to the flash lamp and what is the average power supplied?

	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

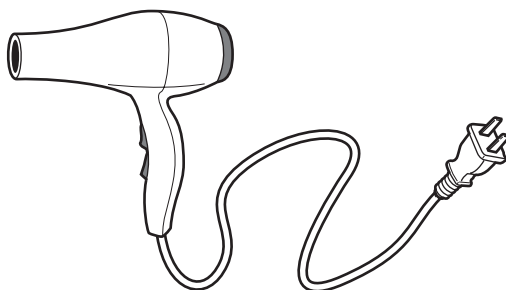
36 The diagram shows an electric circuit.



Which lamps light when switch S_1 is closed and S_2 is open?

- A** L_1 only
- B** L_3 only
- C** L_1 and L_2 only
- D** L_1 , L_2 and L_3

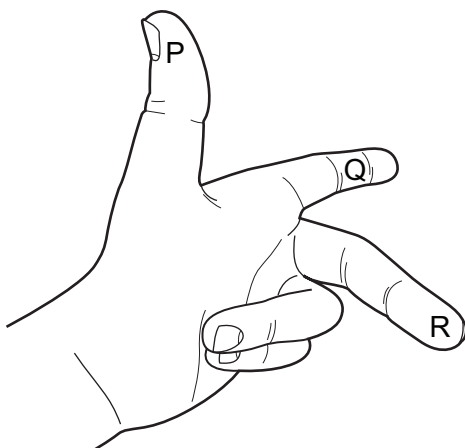
37 The diagram shows a hairdryer.



Which statement explains why the hairdryer does **not** need an earth wire?

- A It has a live wire.
- B It has a neutral wire.
- C It has double insulation.
- D The cable is insulated.

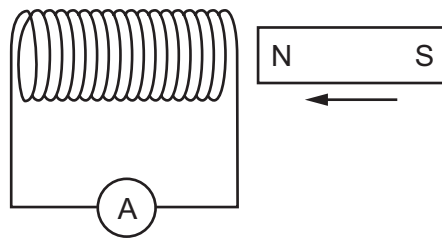
38 A left hand can be used to determine the direction of the force when a current-carrying conductor is perpendicular to a magnetic field.



Which quantities are represented by the direction of fingers P, Q and R?

	P	Q	R
A	current	field	force
B	field	force	current
C	force	current	field
D	force	field	current

- 39** As a magnet is moved into the coil of wire as shown, there is a small positive reading on the sensitive ammeter.



Which change must increase the size of the reading?

- A** moving the opposite pole into the coil
 - B** pulling the magnet out of the coil
 - C** pushing the magnet in faster
 - D** unwinding some of the turns of wire
- 40** Electrical power is transmitted by cables over long distances at very high voltages.

What are the effects of using a high voltage transmission system?

	power loss in the cables	current in the cables
A	high	high
B	high	low
C	low	high
D	low	low

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

May/June 2022

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 A length of copper wire is labelled: length 0.50 m and diameter 0.50 mm.

Which instruments are most suitable to measure accurately the length and the diameter of the wire?

	length	diameter
A	metre rule	metre rule
B	metre rule	micrometer
C	calipers	metre rule
D	calipers	micrometer

- 2 A driver drives a car at uniform velocity along a road.

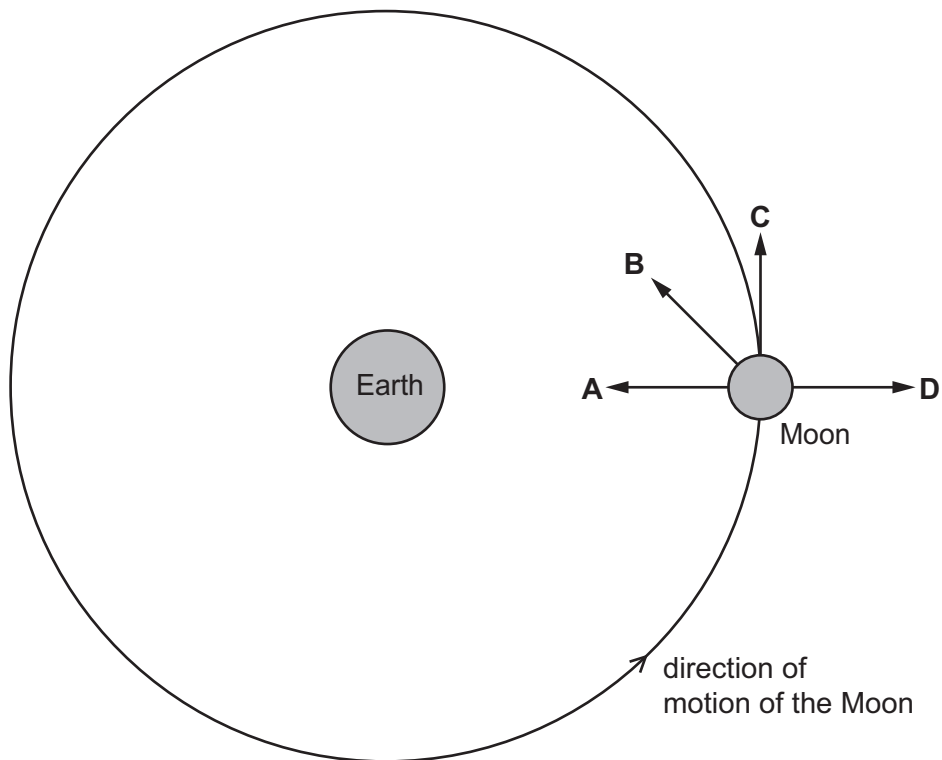
The driver sees a hazard and applies the brakes.

What is the car's motion as it moves through the thinking distance and as it moves through the braking distance?

	motion through the thinking distance	motion through the braking distance
A	deceleration	deceleration
B	deceleration	uniform velocity
C	uniform velocity	deceleration
D	uniform velocity	uniform velocity

- 3 The diagram represents the Moon in its orbit around the Earth.

Which arrow represents the direction of the resultant force acting on the Moon at the instant shown?



- 4 The mass m of an object is related to its weight W and the gravitational field strength g .

Which equation shows the relationship?

- A** $m = W \times g$
 B $m = \frac{W}{g}$
 C $m = \frac{g}{W}$
 D $m = W - g$

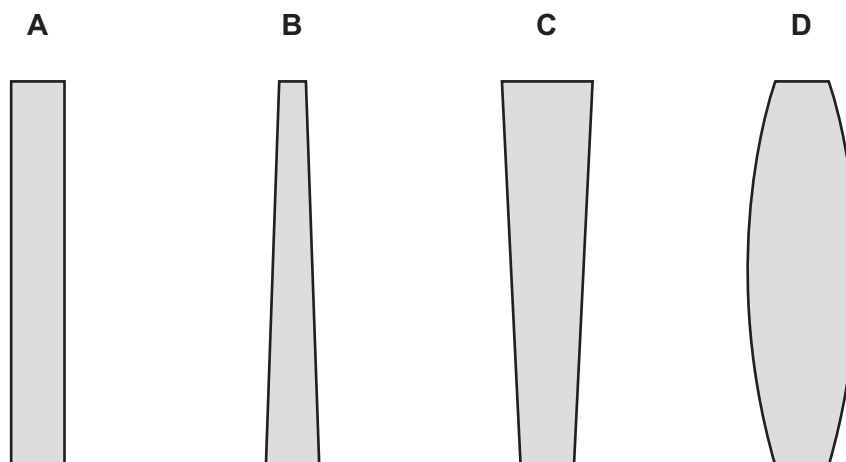
- 5 A plastic cube, of density ρ , is compressed so that each of its sides becomes one-half of its original length.

What is the new density of the material of the cube?

- A** 0.125ρ
 B 0.50ρ
 C 2.0ρ
 D 8.0ρ

- 6 Four glass objects have square bases of equal area.

Which object is the least stable?



- 7 The initial pressure of a volume V_1 of a fixed mass of gas is p_1 . The gas expands at a constant temperature until it has volume V_2 and pressure p_2 .

Which statement about the pressure and volume of the gas is **not** correct?

- A Pressure is measured in N/m^3 .
 - B Pressure multiplied by volume is a constant.
 - C Pressure is directly proportional to $\frac{1}{\text{volume}}$.
 - D The ratio of V_1 to V_2 is equal to the ratio of p_2 to p_1 .
- 8 A stone falls at terminal velocity in air.

What is equal to the change in gravitational potential energy of the stone?

- A the change in kinetic energy
- B the change in temperature of the stone
- C the friction between the stone and the air
- D the thermal energy produced

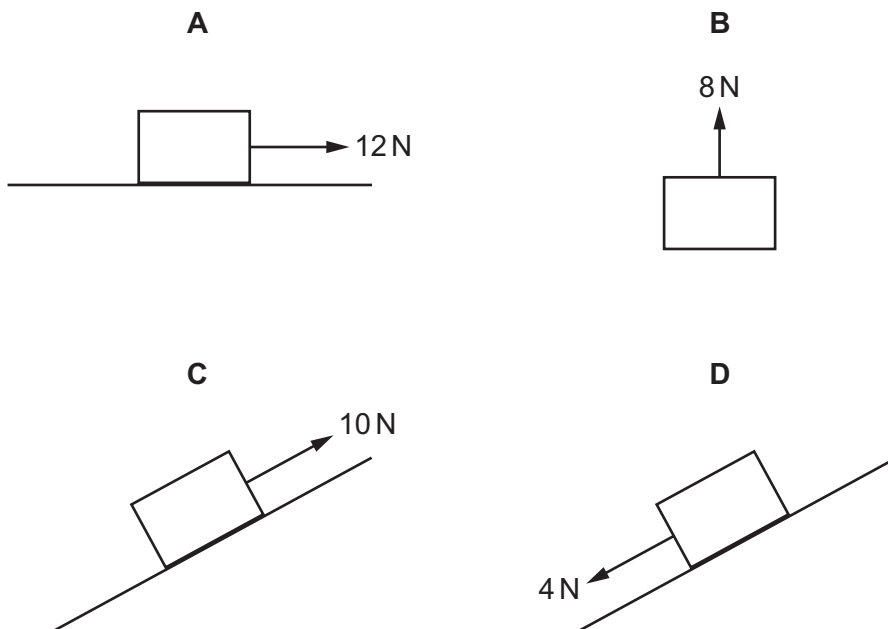
- 9 Water in a river is used to turn a turbine as the water flows downstream.

How is this type of power source described?

- A hydroelectric
 - B tidal
 - C geothermal
 - D nuclear
- 10 A load is pulled by a rope attached to a motor. The resultant force exerted by the rope on the load is shown in the diagrams.

In each diagram, the load moves in the direction of the force shown and takes 10 s to travel 1.0 m.

In which diagram does the motor work with the greatest power?



- 11** The total energy incident in 1.0 s on a group of solar panels is 120 J/m^2 .

The group of solar panels converts 720 J of light energy to 120 J of useful electrical energy in 1.0 s.

What is the total surface area of the panels and what is the efficiency of the system?

	total surface area of solar panels / m^2	efficiency / %
A	6.0	17
B	0.17	6.0
C	6.0	6.0
D	0.17	17

- 12** A 1000 W, 240 V electrical device is switched on.

How much energy does it use in 5.0 minutes?

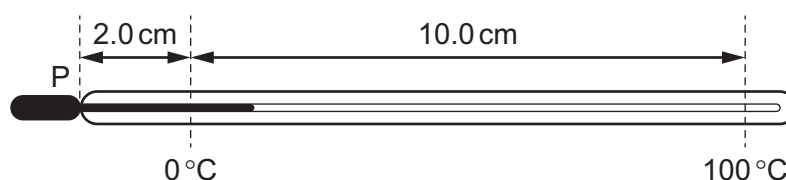
- A** 5000 J **B** 12000 J **C** 72000 J **D** 300000 J

- 13** A building has walls made from wood and steel.

What describes the way in which thermal energy is transferred through the walls?

- A** conduction through the wood by movement of molecules through the wood only
B conduction through the metal by both the vibration of particles and movement of electrons through the steel
C conduction through the wood by both the vibration and movement of molecules through the wood
D convection through the metal by movement of electrons through the steel only

- 14** A liquid column in a liquid-in-glass thermometer is 2.0 cm long at 0°C . The column expands by 10.0 cm when heated to 100°C .



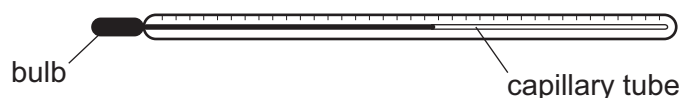
Measuring from P, how long is the liquid column at 30°C ?

- A** 2.3 cm **B** 3.0 cm **C** 5.0 cm **D** 7.0 cm

- 15 In many applications, a thermocouple is used to measure temperature instead of a liquid-in-glass thermometer.

Which property of a thermocouple is a major advantage?

- A its greater heat capacity
 - B its smaller temperature range
 - C its quicker response time
 - D it has a non-linear output
- 16 A liquid-in-glass thermometer consists of a bulb containing a liquid which expands into a thin capillary tube.



The liquid in the thermometer is replaced by the same volume of a different liquid that expands more for the same temperature rise. The length of the capillary tube remains the same.

How does the new thermometer compare with the old thermometer?

- A It has greater sensitivity and a greater range.
 - B It has greater sensitivity but a smaller range.
 - C It has the same sensitivity and the same range.
 - D It has the same sensitivity but a greater range.
- 17 What is the usual symbol and what is the unit for the heat capacity of an object?

	symbol	unit
A	C	$\text{J}/^{\circ}\text{C}$
B	C	$\text{J}/(\text{kg } ^{\circ}\text{C})$
C	H	$\text{J}/^{\circ}\text{C}$
D	H	$\text{J}/(\text{kg } ^{\circ}\text{C})$

- 18 Which row describes the arrangement of particles and the forces between particles in a solid?

	arrangement of particles	forces between particles
A	regular	strong
B	regular	weak
C	irregular	strong
D	irregular	weak

- 19** A wave in the sea collides with a cliff.

A crest of the wave hits the cliff once every 6.0 s.

The horizontal distance between a crest and the adjacent trough of the wave is 4.5 m.

What is the speed of the wave?

- A** 0.67 m/s **B** 0.75 m/s **C** 1.3 m/s **D** 1.5 m/s

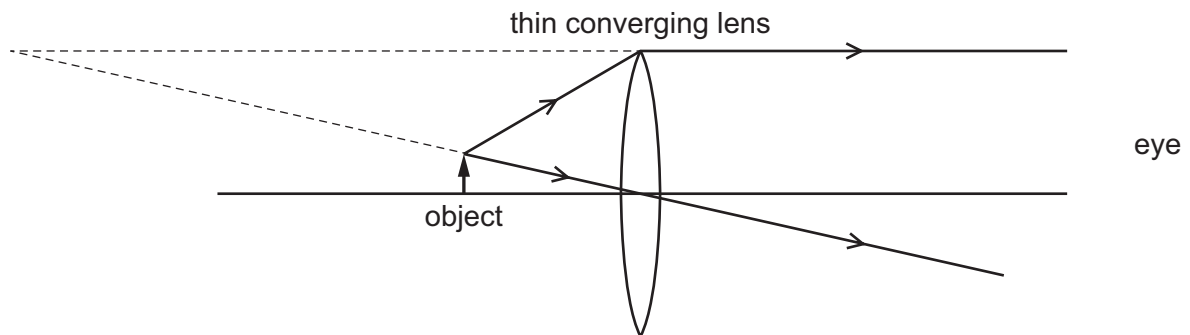
- 20** A sound wave refracts as it passes from air to water.

Which quantities change?

- A** speed and frequency only
B speed and wavelength only
C wavelength and frequency only
D speed, frequency and wavelength

- 21** An object is viewed through a thin converging lens.

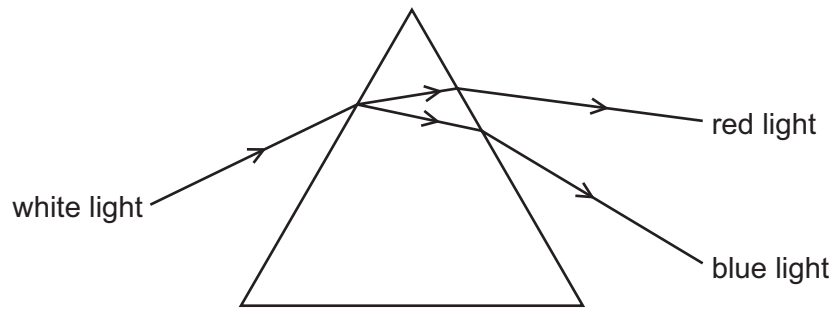
The diagram shows the paths of two rays from the top of the object to an eye.



How does the image compare with the object?

- A** It is larger and inverted.
B It is larger and upright.
C It is smaller and inverted.
D It is smaller and upright.

22 A ray of white light enters a prism as shown.



Which row is correct?

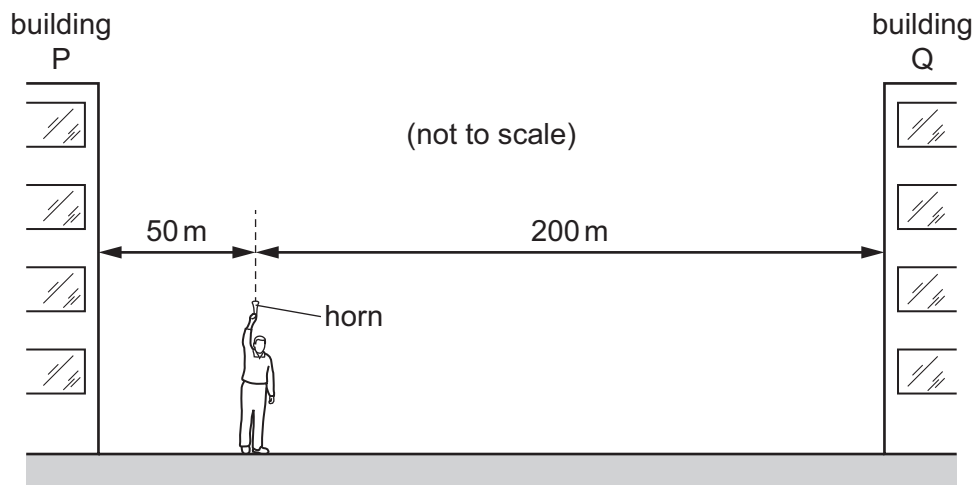
	wave properties observed	frequency of red light compared with frequency of blue light
A	dispersion only	smaller
B	refraction only	greater
C	dispersion and refraction	smaller
D	dispersion and refraction	greater

23 Which waves are **not** part of the electromagnetic spectrum?

- A** radio waves
- B** television signals
- C** ultrasound
- D** ultraviolet

- 24** A man stands between two tall buildings, P and Q.

He is 50 m from P and 200 m from Q.



He sounds a horn.

He hears the first echo from building P and 1.0 s later he hears the first echo from building Q.

What is the speed of sound calculated using this information?

- A** 250 m/s **B** 300 m/s **C** 330 m/s **D** 500 m/s

- 25** Sound travels at different speeds in each of the three states of matter, solid, liquid and gas.

Which row shows where sound travels fastest and where sound travels slowest?

	state where sound travels fastest	state where sound travels slowest
A	gas	liquid
B	gas	solid
C	liquid	gas
D	solid	gas

- 26** What is one of the uses of ultrasound?

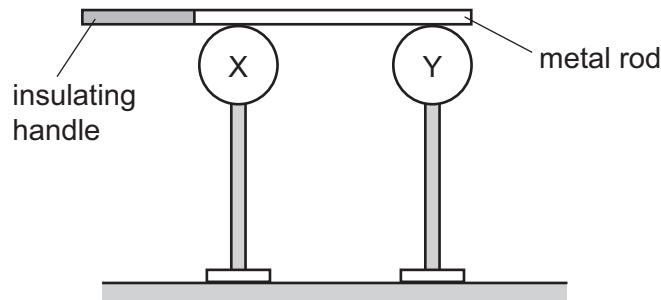
- A** cleaning jewellery
B fluorescent tubes
C optical fibres
D sunbeds

27 Which material is used for the core of an electromagnet?

- A aluminium
- B copper
- C iron
- D steel

28 Two metal spheres are mounted on insulating stands. Sphere X is initially uncharged and sphere Y is initially positively charged.

A metal rod, held by an insulating handle, is placed in contact with X and Y as shown.



What are the charges on X and on Y after the rod is placed in contact with them?

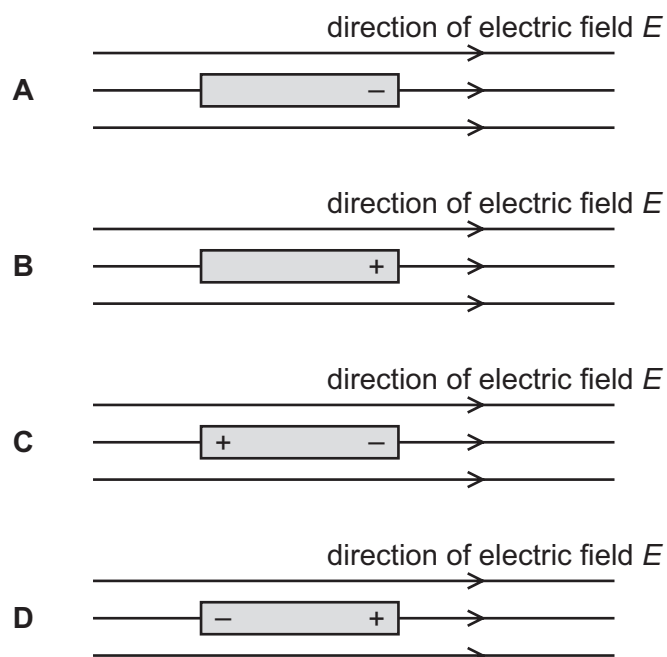
	charge on X	charge on Y
A	positive	positive
B	positive	uncharged
C	uncharged	positive
D	uncharged	uncharged

29 Which device involves the use of static electrical charges?

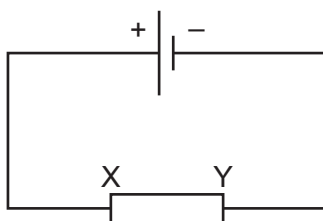
- A a computer hard disk drive
- B a motor
- C a photocopier
- D a transformer

- 30 An initially uncharged copper rod is placed in a uniform electric field E . The rod is parallel to the field.

Which diagram shows the charges induced on the rod?



- 31 The diagram shows a simple electric circuit.



Which row describes the charge on an electron and the direction of electron flow through the resistor?

	charge on an electron	direction of electron flow
A	negative	X to Y
B	negative	Y to X
C	positive	X to Y
D	positive	Y to X

- 32** When the flash on a camera is used, a charge of 1.5 C flows for 0.0030 s through the flash lamp.

The average voltage across the flash lamp is 3600 V.

What is the electrical energy supplied to the flash lamp and what is the average power supplied?

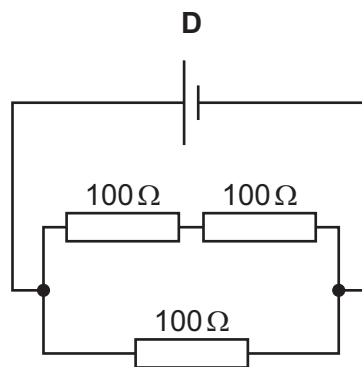
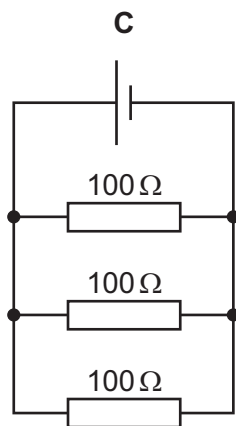
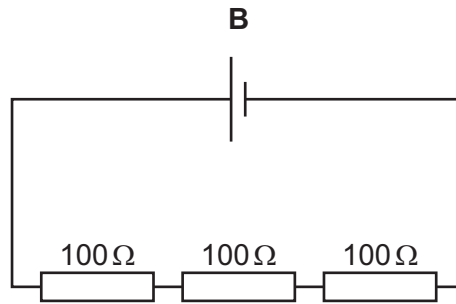
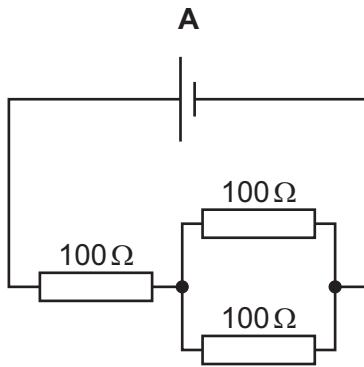
	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

- 33** What is the unit of potential difference (p.d.) and which measuring instrument is used to measure p.d.?

	unit	instrument
A	ampere	ammeter
B	ampere	voltmeter
C	volt	ammeter
D	volt	voltmeter

34 The diagrams show four circuits. Each circuit contains three $100\ \Omega$ resistors.

Which circuit has the smallest total resistance?



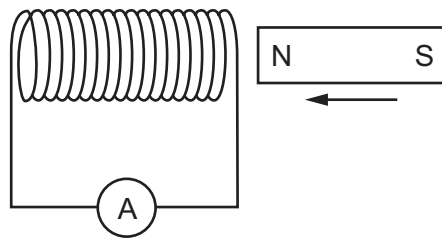
35 A metal kettle is plugged into a mains socket. The plug contains a suitable fuse.

The kettle's cable is damaged and the fuse blows.

Why does the fuse blow?

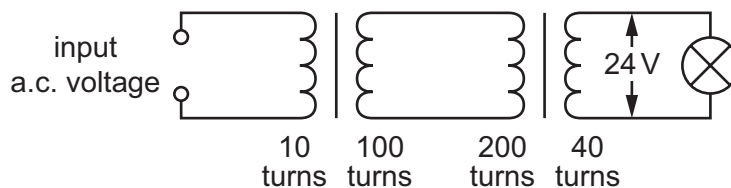
- A** The bare earth wire in the cable touches the bare live wire.
- B** The metal casing of the kettle touches the bare earth wire.
- C** The bare neutral wire in the cable touches the bare earth wire.
- D** The live wire breaks without touching any other wire.

- 36 As a magnet is moved into the coil of wire as shown, there is a small positive reading on the sensitive ammeter.



Which change **must** increase the size of the reading?

- A moving the opposite pole into the coil
 - B pulling the magnet out of the coil
 - C pushing the magnet in faster
 - D unwinding some of the turns of wire
- 37 The diagram shows the circuit for a model power line with two transformers and a lamp at the output.



What is the input voltage?

- A 0.12 V
 - B 6.0 V
 - C 12 V
 - D 96 V
- 38 Electrical power is transmitted by cables over long distances at very high voltages.

What are the effects of using a high voltage transmission system?

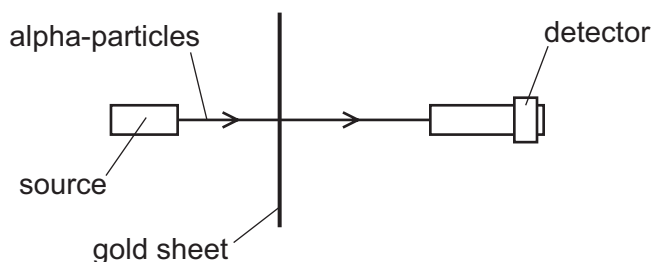
	power loss in the cables	current in the cables
A	high	high
B	high	low
C	low	high
D	low	low

- 39 The diagram shows a beam of electrons about to enter a magnetic field. The magnetic field is directed into the page.



What is the direction of the deflection of the electrons as they enter the magnetic field?

- A into the page
 - B out of the page
 - C up the page
 - D down the page
- 40 In the Geiger-Marsden experiment, alpha-particles are fired at a thin gold sheet.



Most alpha-particles pass straight through the thin gold sheet.

A few are deflected.

What can be deduced from this experiment?

- A The nucleus is very small.
- B The nucleus has no charge.
- C Electrons surround the nucleus.
- D Electrons have a negative charge.

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Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	B	1
3	B	1
4	D	1
5	A	1
6	C	1
7	B	1
8	B	1
9	C	1
10	C	1
11	D	1
12	C	1
13	A	1
14	A	1
15	B	1
16	C	1
17	A	1
18	C	1
19	B	1
20	B	1
21	D	1
22	B	1
23	C	1
24	D	1
25	B	1
26	B	1
27	A	1
28	B	1

Question	Answer	Marks
29	A	1
30	C	1
31	A	1
32	B	1
33	C	1
34	D	1
35	D	1
36	C	1
37	C	1
38	D	1
39	C	1
40	D	1



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4	B	1
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6	C	1
7	A	1
8	D	1
9	A	1
10	A	1
11	A	1
12	D	1
13	B	1
14	C	1
15	C	1
16	B	1
17	A	1
18	A	1
19	D	1
20	B	1
21	B	1
22	C	1
23	C	1
24	B	1
25	D	1
26	A	1
27	C	1
28	A	1

Question	Answer	Marks
29	C	1
30	D	1
31	B	1
32	D	1
33	D	1
34	C	1
35	A	1
36	C	1
37	C	1
38	D	1
39	D	1
40	A	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2021

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
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- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



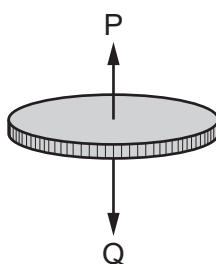
- 1 A student determines the circumference of a football.

Which instrument gives a reading that is the circumference of the football?

- A calipers
- B micrometer
- C rule
- D tape

- 2 A coin falls from rest through the air and eventually reaches a constant speed.

There is a resultant force acting on the coin due to the two forces P and Q shown in the diagram.

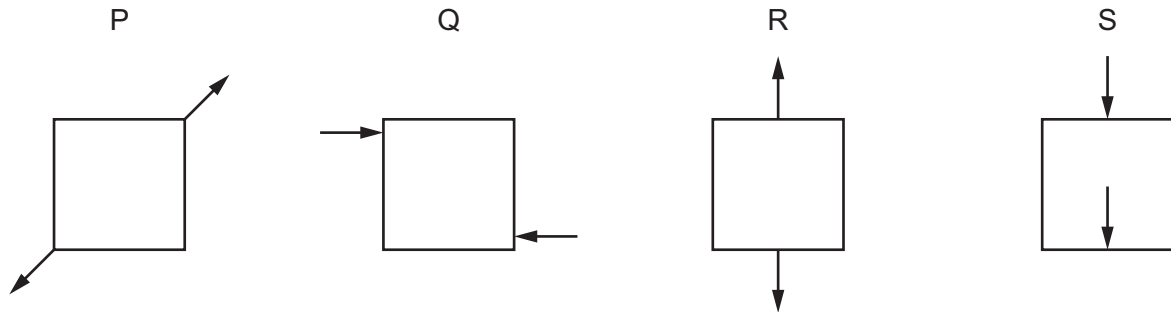


What happens to force P and what happens to the resultant force before the coin reaches constant speed?

	force P	resultant force
A	decreases	increases
B	decreases	decreases
C	increases	decreases
D	increases	increases

3 The diagrams show the forces acting on four identical solid blocks.

Each arrow represents a force of 20 N.



Which blocks are in equilibrium?

- A** P and R only
- B** Q and S only
- C** P, Q and R only
- D** P, Q, R and S

4 A student lists three changes that affect the stopping distance of a car.

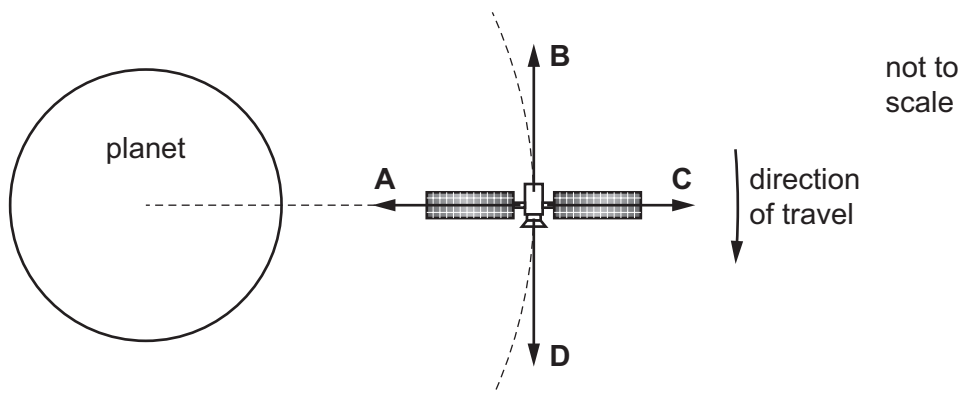
- 1 increasing the braking force
- 2 increasing the friction between the tyres and the road
- 3 increasing the speed of the car

Which change or changes increase the stopping distance?

- A** 1 and 2
- B** 1 and 3
- C** 1 only
- D** 3 only

5 The diagram shows a satellite orbiting a planet at a steady speed.

In which direction does the resultant force act on the satellite?



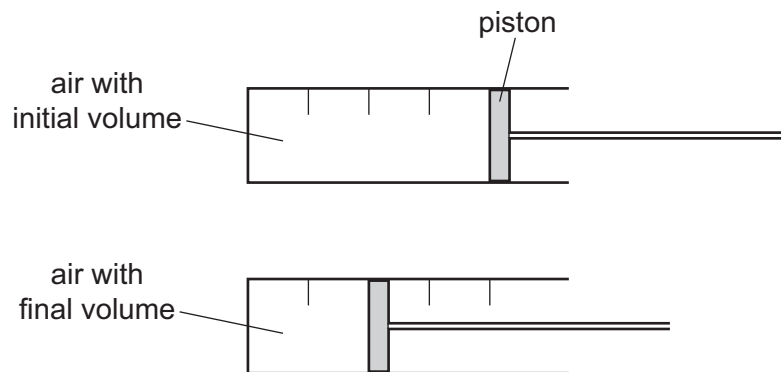
- 6 The gravitational field strength in space is smaller than on the Earth's surface.

A rocket is used to launch a satellite from the Earth's surface into space.

How are the mass and the weight of the satellite affected as the satellite moves away from the surface of the Earth and into space?

- A Both the mass and the weight are unaffected.
 - B The mass decreases and the weight decreases.
 - C The mass increases and the weight is unaffected.
 - D The mass is unaffected and the weight decreases.
- 7 Air is trapped in a cylinder by a piston.

The piston is moved inwards so that the volume of the air reduces.



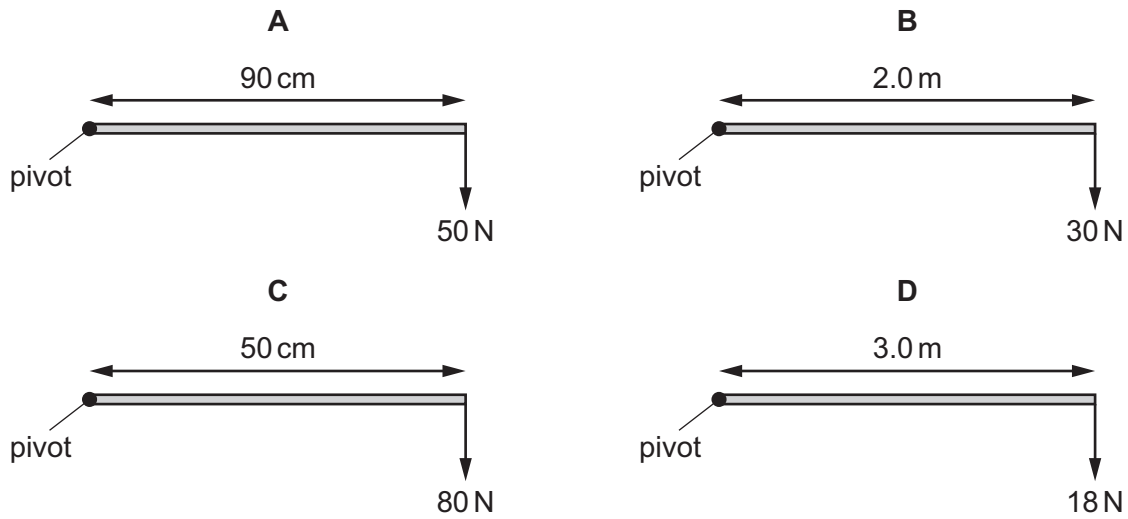
The density of the air in the syringe at the initial volume is 0.0012 g/cm^3 .

What is the density of the air in the syringe at the final volume?

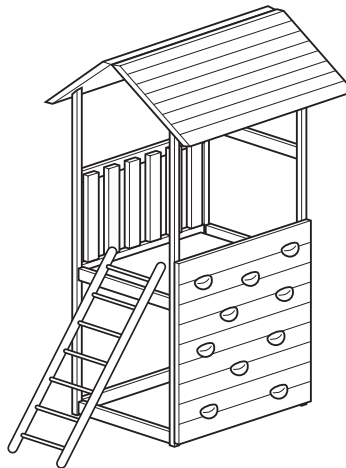
- A 0.0006 g/cm^3
- B 0.0012 g/cm^3
- C 0.0024 g/cm^3
- D 0.0048 g/cm^3

- 8 Some forces are applied at different distances from a pivot.

Which diagram shows the force that produces the largest moment about the pivot?



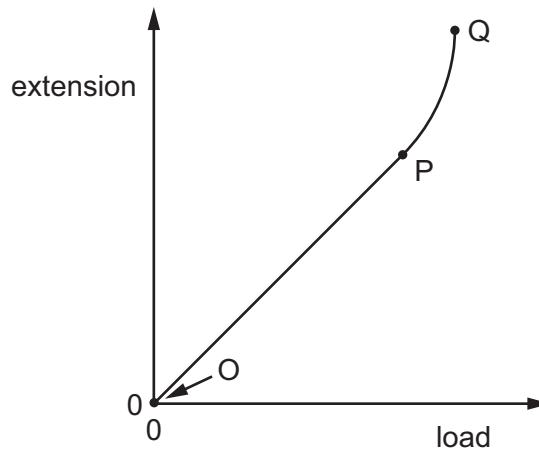
- 9 The diagram shows a children's wooden play tower.



Which change to the tower makes it more stable?

- A making the tower narrower
- B making the tower taller
- C lowering the centre of mass
- D raising the centre of mass

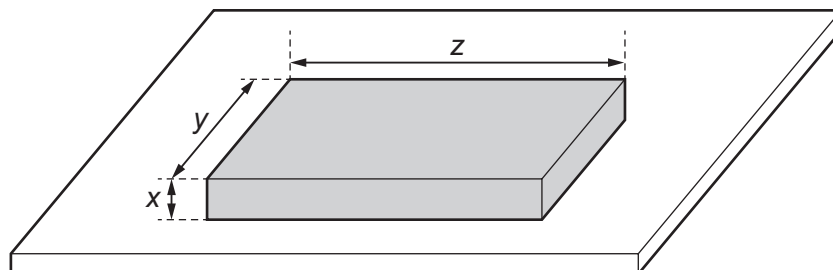
- 10 Where on the graph is the limit of proportionality for an elastic solid?



- A between O and P
 B at P
 C between P and Q
 D at Q
- 11 A weight of 2.0 N is hung from a spring. The extension produced is 6.0 cm. The 2.0 N weight is removed and an 8.0 N weight is hung from the spring. The spring does not pass its limit of proportionality.

What is the new extension of the spring?

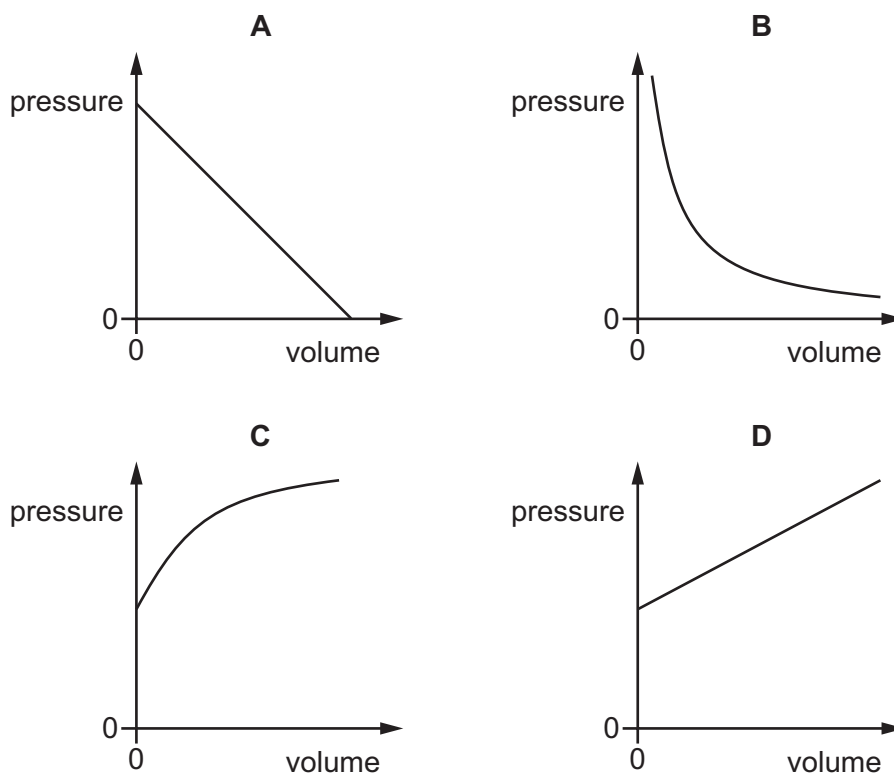
- A 6.0 cm B 24 cm C 36 cm D 48 cm
- 12 An object is placed at a depth d below the surface of a liquid of density ρ .
 On which expression does the pressure on the object depend?
- A $d + \rho$ B $d \times \rho$ C $d \div \rho$ D $\rho \div d$
- 13 The diagram shows the dimensions of a box of mass M and weight W at rest on a table.



What is the pressure on the table due to the box?

- A $\frac{M}{xyz}$ B $\frac{M}{yz}$ C $\frac{W}{xyz}$ D $\frac{W}{yz}$

- 14 Which graph shows the relationship between the pressure and the volume of a fixed mass of gas at constant temperature?



- 15 The work done by a force F on a body is calculated by multiplying F by a quantity q .

What is q ?

- A** the distance travelled in the direction of the force
 - B** the distance travelled perpendicular to the direction of the force
 - C** the velocity in the direction of the force
 - D** the velocity in the direction perpendicular to the force
- 16 A girl lifts an object of mass 1.0 kg from the floor and puts it on a ledge 2.0 m above the floor.
- How much potential energy is gained by the object?
- [gravitational field strength $g = 10 \text{ N/kg}$]
- A** 2.0 J
 - B** 5.0 J
 - C** 10 J
 - D** 20 J

17 A copper rod is heated at one end.

Which statement describes how heat transfer occurs in the copper?

- A Energetic copper molecules move from the cooler end to the hotter end.
- B Energetic copper molecules move from the hotter end to the cooler end.
- C Energetic free electrons move from the cooler end to the hotter end.
- D Energetic free electrons move from the hotter end to the cooler end.

18 Which heat transfer processes do **not** require a medium?

- A conduction only
- B convection only
- C radiation only
- D conduction and radiation

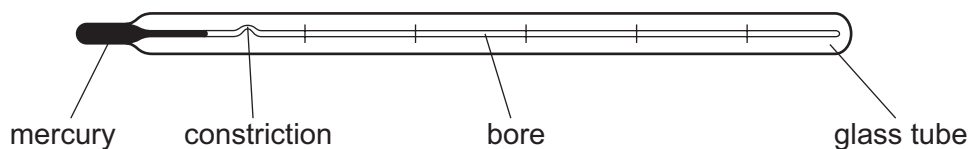
19 The list shows four physical properties.

mass resistance voltage volume

How many of these properties can be used to measure temperature?

- A 1 B 2 C 3 D 4

20 The diagram shows a clinical thermometer.



Which factor affects the sensitivity of the thermometer?

- A the constriction
- B the diameter of the bore
- C the length of the glass tube
- D the thickness of the glass tube

21 Which statement about the thermal expansion of solids, liquids and gases is correct?

- A Liquids do not expand.
- B Liquids expand more than gases but less than solids.
- C Liquids expand more than solids but less than gases.
- D Liquids expand the same amount as solids but less than gases.

22 Which row describes the shape of a liquid and the arrangement of its molecules?

	shape	arrangement of molecules
A	fixed	not regular
B	fixed	regular
C	takes shape of container	not regular
D	takes shape of container	regular

23 Air in a sealed container of fixed volume is heated.

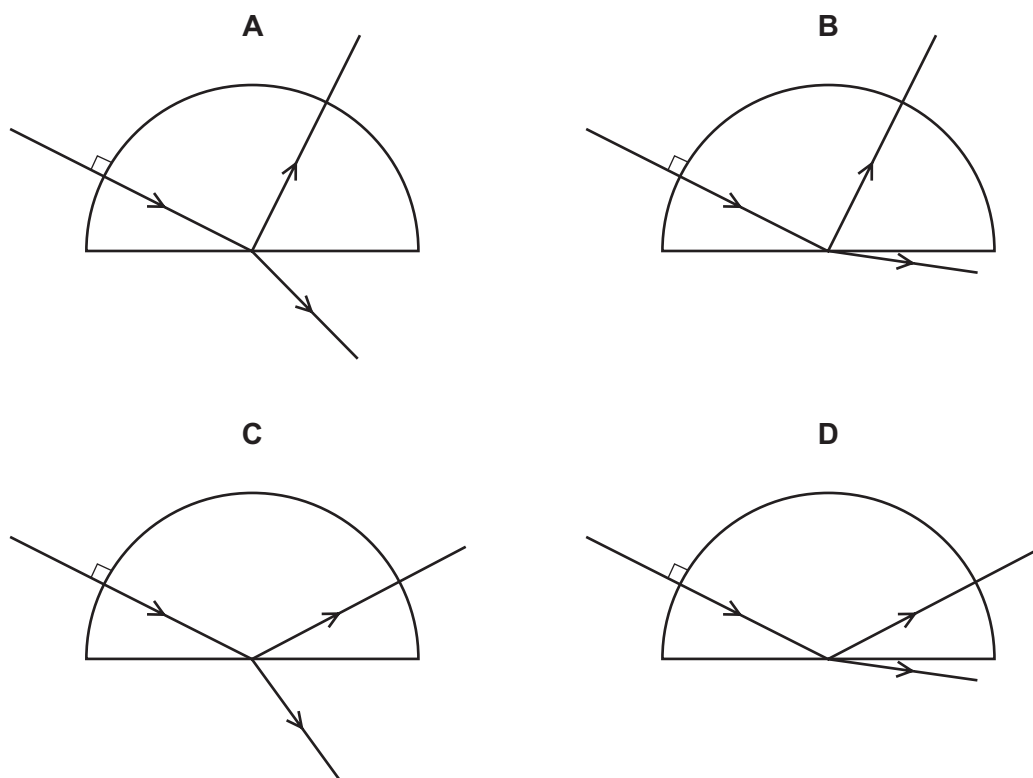
The pressure of the air increases.

How do the molecules of air cause this increase in pressure?

- A** The molecules expand and push on the container with a greater force.
- B** The molecules move at the same speed and hit the container with a greater force.
- C** The molecules move faster and hit the container more often.
- D** The molecules move further apart and hit the container more often.

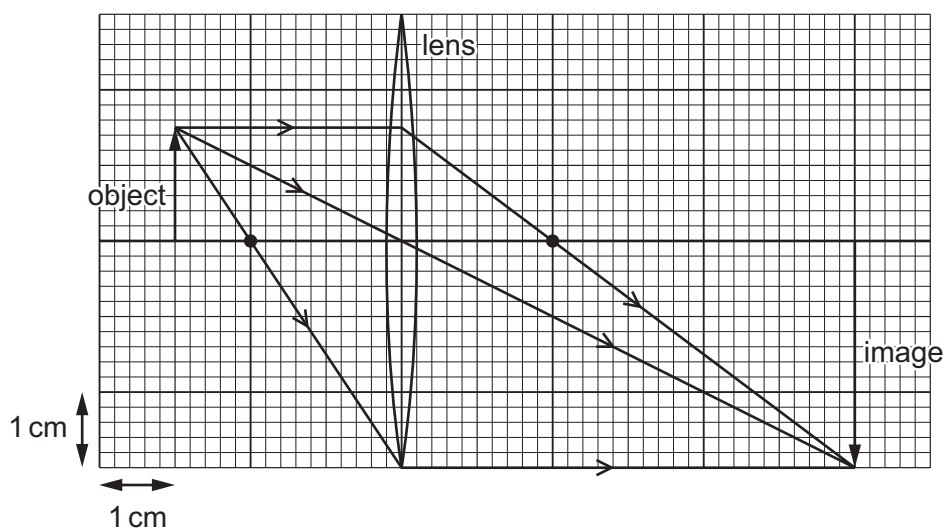
24 A ray of red light in air enters a semi-circular block.

Which diagram shows the partial reflection and the refraction of the ray?



- 25** An object of height 1.5 cm is placed in front of a converging lens of focal length 2.0 cm.

The arrangement is shown on the full-scale ray diagram.



What is the linear magnification produced by the lens?

- A** 2.0 **B** 3.0 **C** 4.0 **D** 6.0
- 26** In which optical instrument is the distance between the object and the lens less than the focal length of the lens?
- A** camera
B magnifying glass
C photographic enlarger
D projector
- 27** Which two waves are components of the electromagnetic spectrum?
- A** light and sound
B water waves and infrared
C ultrasound and ultraviolet
D X-rays and microwaves
- 28** The sound from a ship is reflected by a cliff. An echo is heard by a sailor on the ship 4.0 s after the sound is made. The speed of sound in air is 320 m/s.
- How far from the cliff is the ship?
- A** 160 m **B** 640 m **C** 1280 m **D** 2560 m

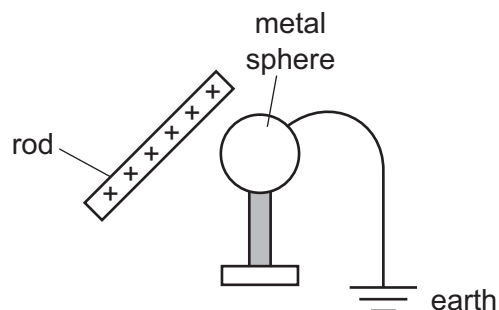
- 29 Students are asked for the uses of ultrasound.

Three suggested uses for ultrasound are:

- P cleaning jewellery
- Q tanning on a sunbed
- R obtaining an image of an unborn baby.

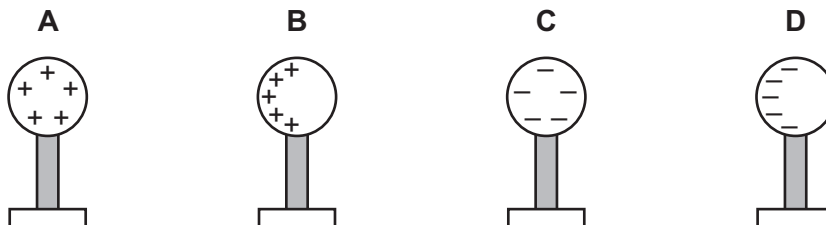
Which suggestions are correct?

- A** P and Q only **B** P and R only **C** Q and R only **D** P, Q and R
- 30 End X of a metal rod attracts the North pole of a compass needle.
- Which statement about the rod is correct?
- A** It is made of copper that is not initially magnetised.
 - B** It is made of copper with a South pole at X.
 - C** It is made of steel that is not initially magnetised.
 - D** It is made of steel with a North pole at X.
- 31 A positively charged rod is held close to an insulated metal sphere. The sphere is earthed as shown.

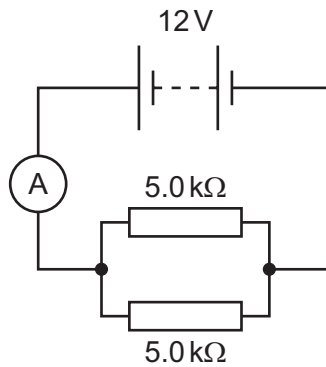


The earth connection is removed and then the rod is removed.

Which diagram shows the charges on the sphere after the rod is removed?



32 The diagram shows a circuit.



The ammeter has four different range settings.

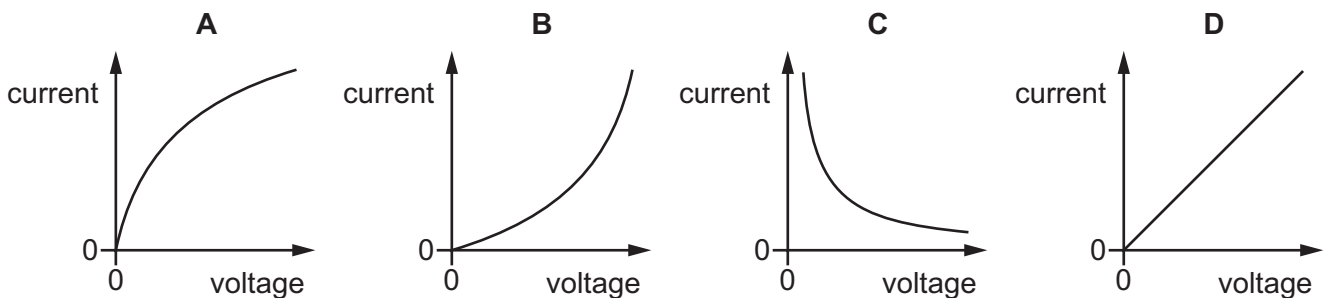
Which range setting gives a reading near the mid-point of the range?

- A** 0–1 mA **B** 0–10 mA **C** 0–1.0 A **D** 0–10 A

33 In which example is the current 1.0 A?

- A** a charge of 0.10 C passes through a resistor in 100 ms
B a 10 V lamp with input power 0.10 W
C a resistor of resistance $0.50\ \Omega$ with a potential difference (p.d.) of 2.0 V across it
D two resistors with a current of 0.50 A connected in series

34 Which graph shows the relationship between current and voltage for a filament lamp?



35 A 100 W lamp is switched on for five hours each day for three weeks.

The cost of one unit of electricity is \$0.24.

How much does it cost to run the lamp for this time?

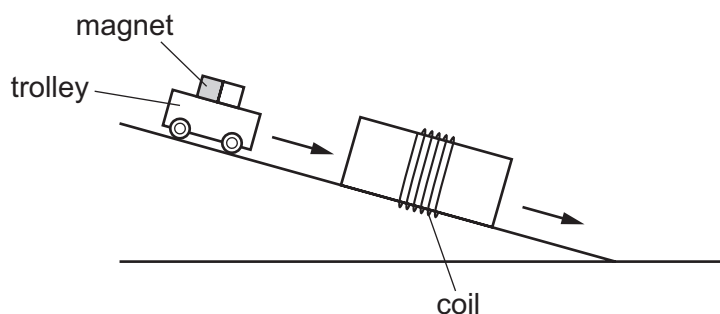
- A** \$0.36 **B** \$0.84 **C** \$2.52 **D** \$25.20

- 36** A three-pin mains plug contains a fuse. A device with double insulation has a cable connected to the plug.

Which part of the plug is **not** connected to the cable?

- A** earth pin
- B** fuse
- C** live pin
- D** neutral pin

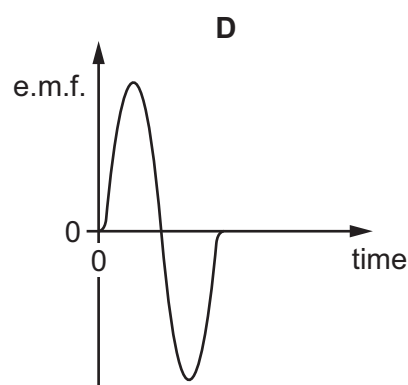
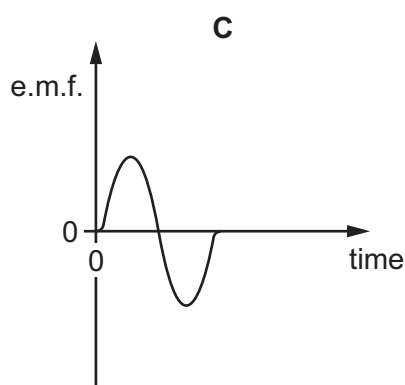
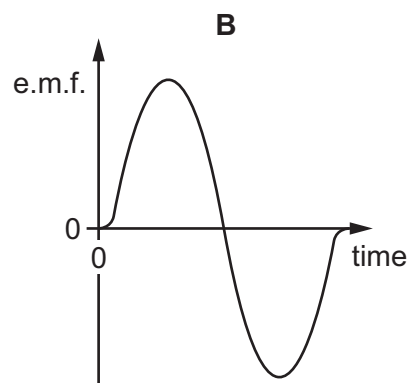
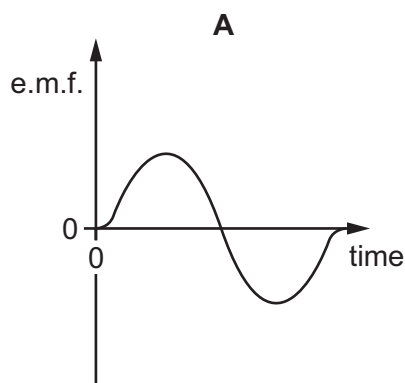
- 37** A trolley carrying a strong magnet rolls down a ramp at constant speed. It passes through a coil as shown.



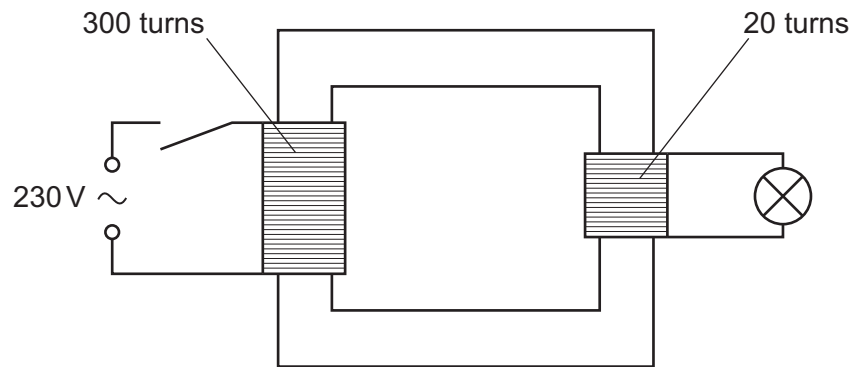
An electromotive force (e.m.f.) is induced in the coil. A graph of the e.m.f. against time is plotted.

The experiment is repeated with different coils and with a steeper ramp. The trolley moves at a greater constant speed on the steeper ramp.

Which graph is produced using the coil with the least number of turns and the steepest ramp? All graphs are drawn to the same scale.



- 38** A student uses a transformer to light a filament lamp using a 230 V a.c. supply. The lamp has a maximum voltage rating of 6.0 V.



What happens when the circuit is switched on?

- A** The lamp does not light at all.
 - B** The lamp lights at normal brightness.
 - C** The lamp lights dimly.
 - D** The lamp lights up brightly and then goes out.
- 39** What does an alpha-particle consist of?
- A** two electrons and four neutrons only
 - B** two protons and two neutrons only
 - C** two protons and four neutrons only
 - D** two protons, two electrons and two neutrons
- 40** The count rate from a radioactive source falls from 4000 counts per minute to 500 counts per minute in 72 minutes.

What is the half-life of the source?

- A** 8 minutes
- B** 9 minutes
- C** 18 minutes
- D** 24 minutes

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

May/June 2021

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
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Soft pencil (type B or HB is recommended)

INSTRUCTIONS

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INFORMATION

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- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 A list of various quantities is shown.

acceleration
displacement
force
length
mass
velocity

How many of these quantities are vectors?

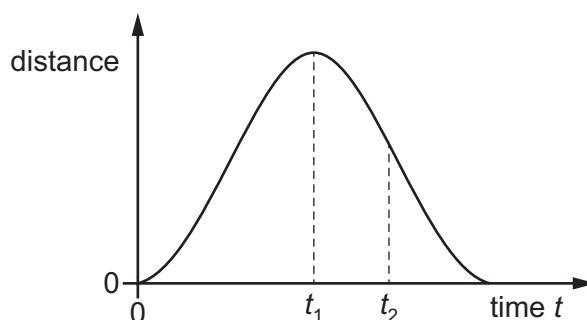
- A** 2 **B** 3 **C** 4 **D** 5

- 2 A student determines the circumference of a football.

Which instrument gives a reading that is the circumference of the football?

- A** calipers
B micrometer
C rule
D tape

- 3 A train sets off from a station at time $t = 0$. The graph shows how the distance between the train and the station varies with time.

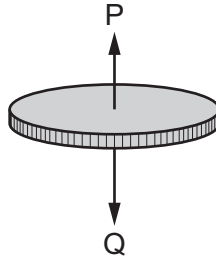


Which statement about the movement of the train between time t_1 and t_2 is correct?

- A** Its speed is decreasing and it is moving away from the station.
B Its speed is decreasing and it is moving towards the station.
C Its speed is increasing and it is moving away from the station.
D Its speed is increasing and it is moving towards the station.

- 4 A coin falls from rest through the air and eventually reaches a constant speed.

There is a resultant force acting on the coin due to the two forces P and Q shown in the diagram.

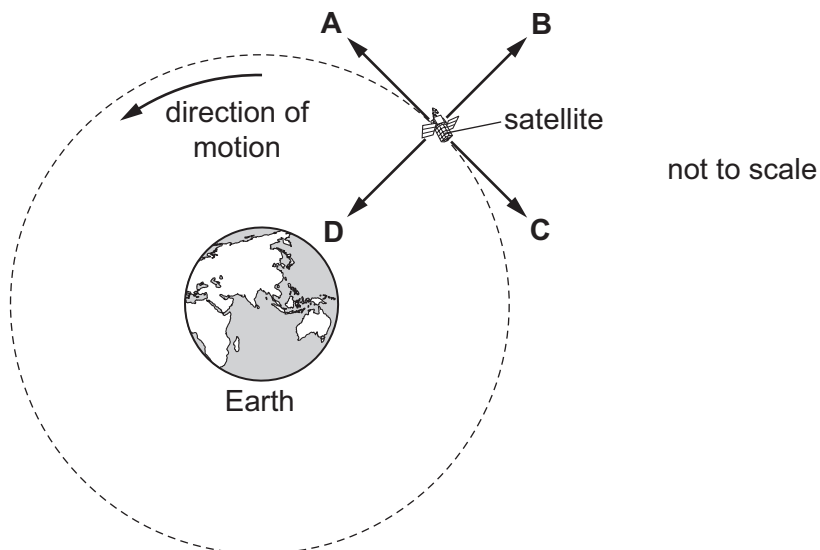


What happens to force P and what happens to the resultant force before the coin reaches constant speed?

	force P	resultant force
A	decreases	increases
B	decreases	decreases
C	increases	decreases
D	increases	increases

- 5 A satellite is shown moving around the Earth in a circular path at a constant speed.

Which arrow shows the direction of the force on the satellite?



- 6 Which row shows the mass and the weight of an object on the Earth's surface?

[gravitational field strength $g = 10 \text{ N/kg}$]

	mass/kg	weight/N
A	2	0.20
B	2	10
C	5	5.0
D	5	50

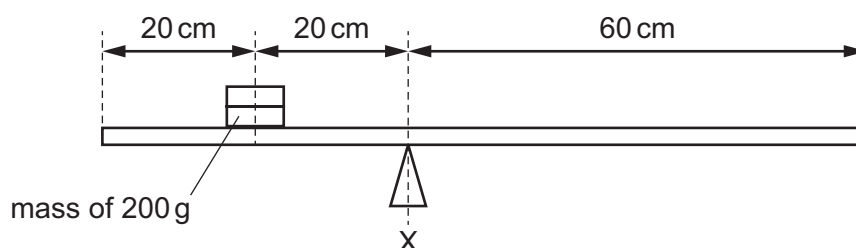
- 7 Water is added to a measuring cylinder containing 100 cm^3 of liquid paraffin.

(The density of paraffin is 0.80 g/cm^3 and that of the water is 1.0 g/cm^3 .)

As the water is added, the level of the paraffin rises to 150 cm^3 . The paraffin and water do not mix.

What finally is the total mass of liquid in the measuring cylinder?

- A** 130 g **B** 140 g **C** 167 g **D** 175 g
- 8 A horizontal beam is pivoted at X. A mass of 200 g rests on the beam as shown. The centre of mass of the beam is 50 cm from the right-hand end of the beam.

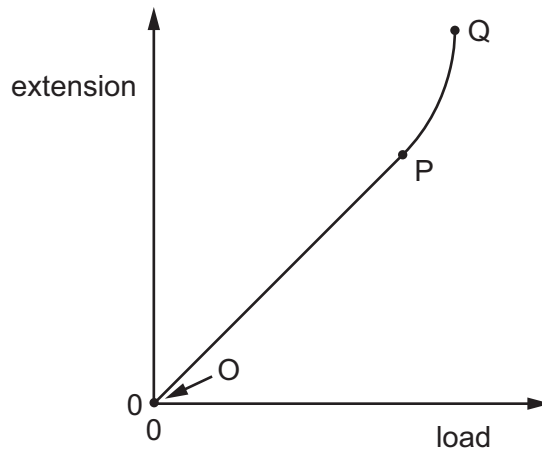


The beam is balanced.

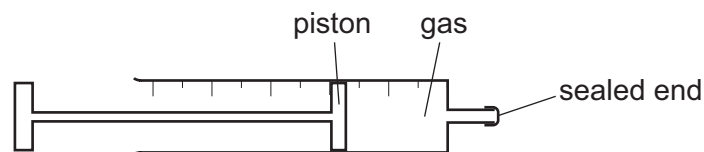
What is the mass of the beam?

- A** 80 g **B** 100 g **C** 400 g **D** 800 g

- 9 Where on the graph is the limit of proportionality for an elastic solid?



- A between O and P
 - B at P
 - C between P and Q
 - D at Q
- 10 A sealed gas syringe contains a fixed mass of gas.



The piston is moved and the volume of the gas doubles. The temperature of the gas does not change.

What happens to the pressure of the gas?

- A halves
 - B no change
 - C doubles
 - D triples
- 11 Which expression for pressure is correct?

- A force $\times$ area
- B force $\div$ area
- C mass $\times$ area
- D mass $\div$ area

- 12** At a depth d in sea-water, the total pressure experienced by a diver is $2P$, where P is atmospheric pressure.

At which depth is the pressure $4P$?

- A** $1.5d$ **B** $2d$ **C** $3d$ **D** $4d$

- 13** The work done by a force F on a body is calculated by multiplying F by a quantity q .

What is q ?

- A** the distance travelled in the direction of the force
B the distance travelled perpendicular to the direction of the force
C the velocity in the direction of the force
D the velocity in the direction perpendicular to the force

- 14** Some solar panels have a total area of 12 m^2 .

Each 1.0 m^2 of the panels receives 0.85 kJ of energy from the Sun in 1.0 s .

The efficiency of the panels is 16% .

How much power do they produce?

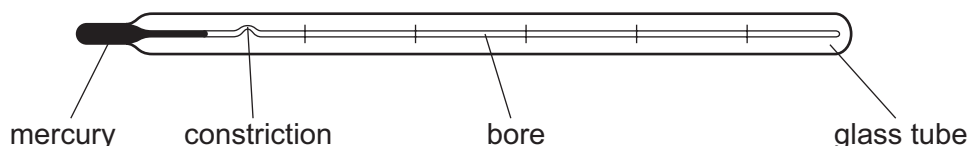
- A** 1.6 kW **B** 2.2 kW **C** 64 kW **D** 160 kW

- 15** A copper rod is heated at one end.

Which statement describes how heat transfer occurs in the copper?

- A** Energetic copper molecules move from the cooler end to the hotter end.
B Energetic copper molecules move from the hotter end to the cooler end.
C Energetic free electrons move from the cooler end to the hotter end.
D Energetic free electrons move from the hotter end to the cooler end.

16 The diagram shows a clinical thermometer.



Which factor affects the sensitivity of the thermometer?

- A the constriction
- B the diameter of the bore
- C the length of the glass tube
- D the thickness of the glass tube

17 Which row is correct for a thermocouple thermometer?

	measures very high temperatures	responds quickly to change in temperature
A	no	no
B	no	yes
C	yes	no
D	yes	yes

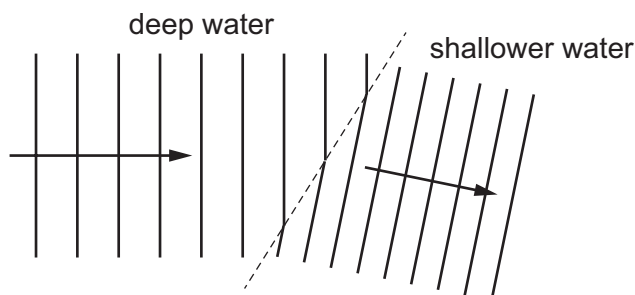
18 What is the *heat capacity* of a body?

- A the amount of thermal energy that the body can absorb without melting
- B the amount of thermal energy required to raise the temperature of the body by 1.0°C
- C the amount of thermal energy required to raise the temperature of 1.0 kg of the body by 1.0°C
- D the amount of thermal energy required to raise the temperature of 1.0 m^3 of the body by 1.0°C

19 Which statement about water is correct?

- A At the boiling point, water vapour molecules have the same kinetic energy as liquid water molecules.
- B Evaporation occurs only at the boiling point.
- C Water molecules become heavier when water freezes.
- D Water molecules lose all of their kinetic energy when water freezes.

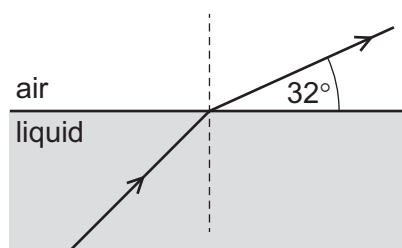
- 20 A water wave in a ripple tank refracts as it moves from deep water into shallower water.



What happens to the speed and to the frequency of the wave as it moves into shallower water?

	speed	frequency
A	decreases	stays constant
B	increases	stays constant
C	stays constant	decreases
D	stays constant	increases

- 21 Light refracts from a liquid into air as shown.



not to scale

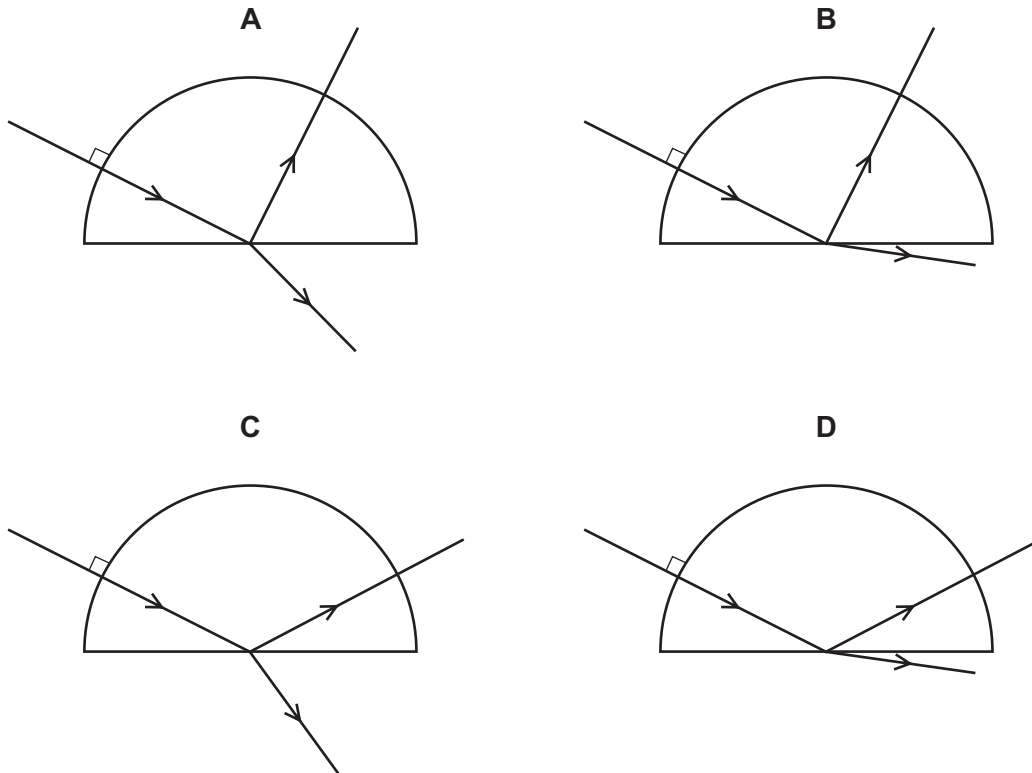
The refractive index for light moving from air to the liquid is 1.4.

What is the angle of incidence in the liquid?

- A** 22° **B** 37° **C** 41° **D** 45°

22 A ray of red light in air enters a semi-circular block.

Which diagram shows the partial reflection and the refraction of the ray?



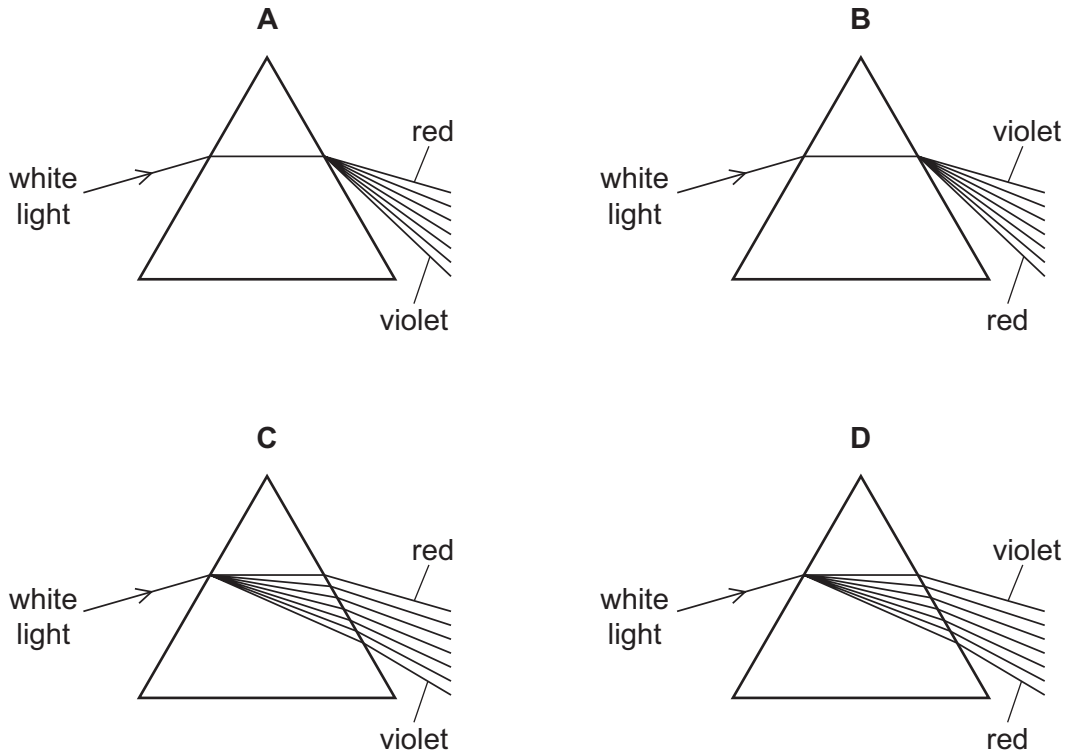
23 Which statement about human vision is correct?

- A** In a normal eye, the image on the retina is magnified and upright.
- B** In a long-sighted eye, distant objects form images in front of the retina.
- C** Short-sighted eyes produce only virtual images.
- D** Short-sight is corrected by the use of a diverging lens.

- 24** White light enters a prism and forms a spectrum.

The rays in the air are labelled.

Which diagram shows how the white light is dispersed by the prism?



- 25** The sound from a ship is reflected by a cliff. An echo is heard by a sailor on the ship 4.0 s after the sound is made. The speed of sound in air is 320 m/s.

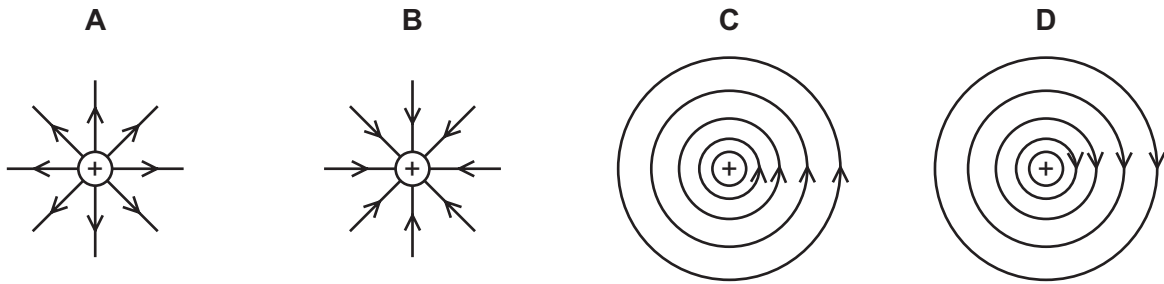
How far from the cliff is the ship?

- A** 160 m **B** 640 m **C** 1280 m **D** 2560 m
- 26** End X of a metal rod attracts the North pole of a compass needle.

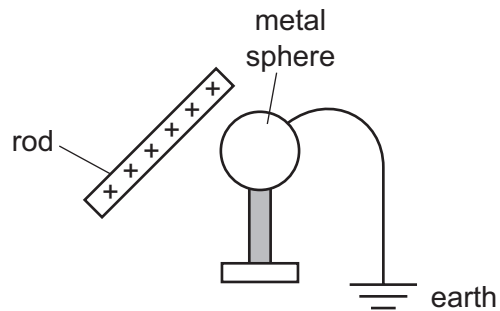
Which statement about the rod is correct?

- A** It is made of copper that is not initially magnetised.
B It is made of copper with a South pole at X.
C It is made of steel that is not initially magnetised.
D It is made of steel with a North pole at X.

27 Which diagram shows the pattern and direction of the electric field lines near a positive charge?

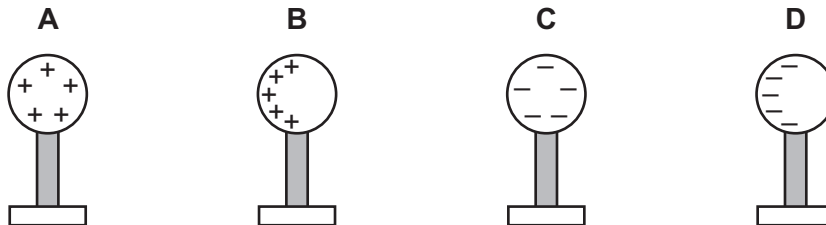


28 A positively charged rod is held close to an insulated metal sphere. The sphere is earthed as shown.

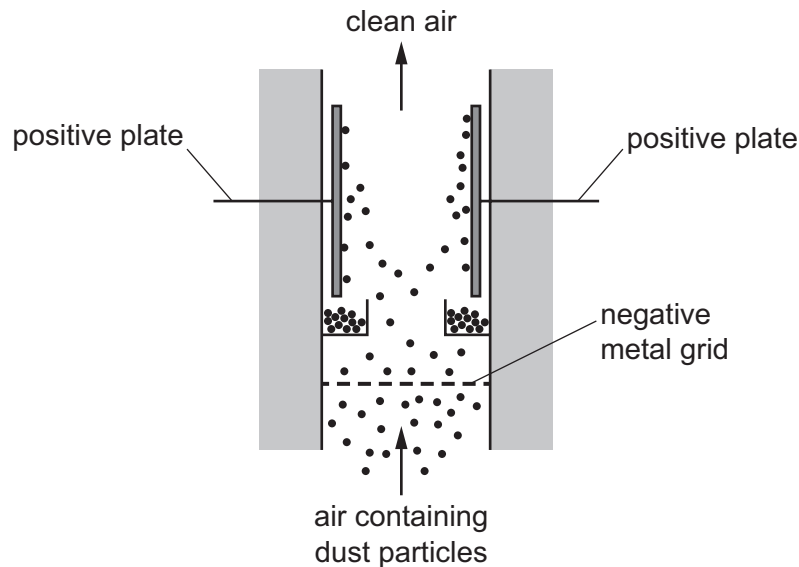


The earth connection is removed and then the rod is removed.

Which diagram shows the charges on the sphere after the rod is removed?



- 29 The diagram shows an electrostatic precipitator. It can be used to remove dust from air.



What happens at the negative grid?

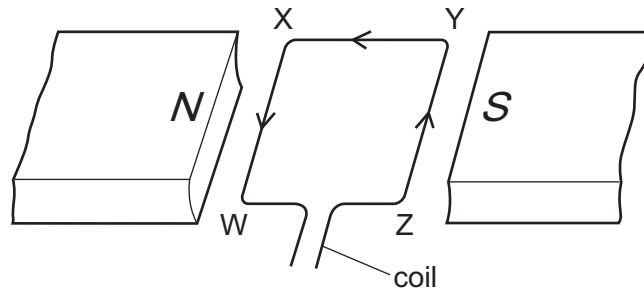
- A Dust particles gain electrons.
 - B Dust particles gain protons.
 - C Dust particles lose electrons.
 - D Dust particles lose protons.
- 30 A 100 W lamp is switched on for five hours each day for three weeks.
- The cost of one unit of electricity is \$0.24.
- How much does it cost to run the lamp for this time?
- A \$0.36
 - B \$0.84
 - C \$2.52
 - D \$25.20
- 31 The information on the back of an electric room heater is shown.

rating 220–240 V
~50 Hz
4.2 A

What is a suitable fuse rating for this room heater?

- A 4.0 A
- B 4.2 A
- C 5.0 A
- D 13.0 A

- 32 The diagram shows a horizontal rectangular wire coil WXYZ between the poles of a magnet.

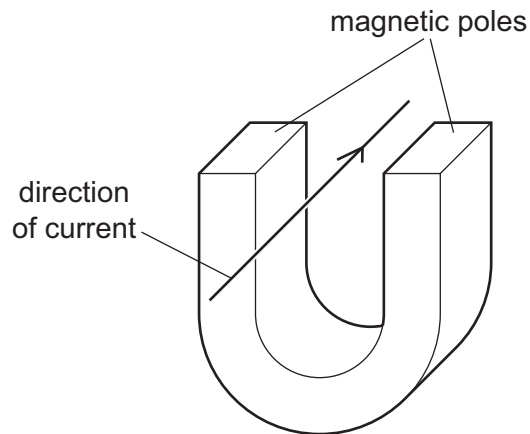


There is a current in the coil in the direction shown.

Which statement is correct?

- A The side WX experiences an upward force.
 - B The side XY experiences an outward force.
 - C The side YZ experiences an inward force.
 - D The side ZW experiences a downward force.
- 33 Which energy transfer takes place in an electric kettle?
- A chemical to electrical
 - B electrical to heat
 - C electrical to chemical
 - D heat to electrical

- 34** The diagram shows a current-carrying conductor between the poles of a magnet. The force on the wire acts downwards.



Four changes are possible.

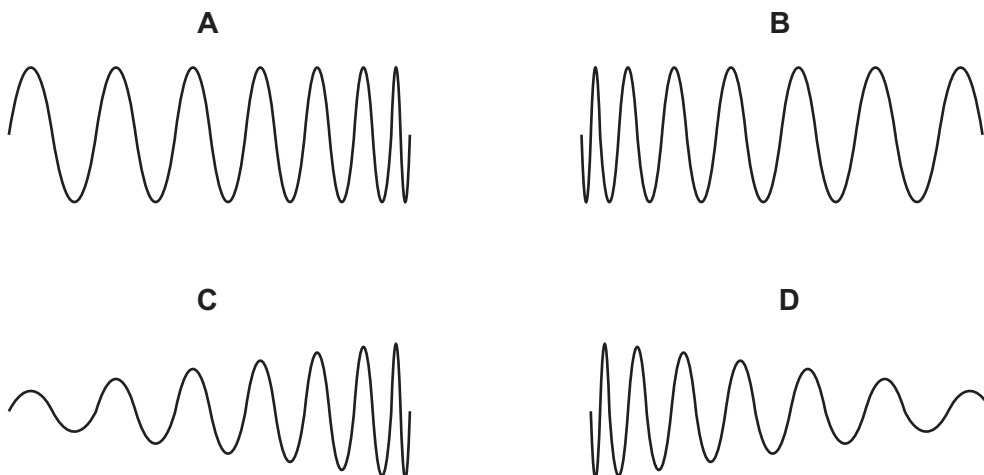
- 1 The current is increased.
- 2 A stronger magnet is used.
- 3 The current is reversed.
- 4 The poles exchange positions.

Which two changes made together keep the force acting downwards?

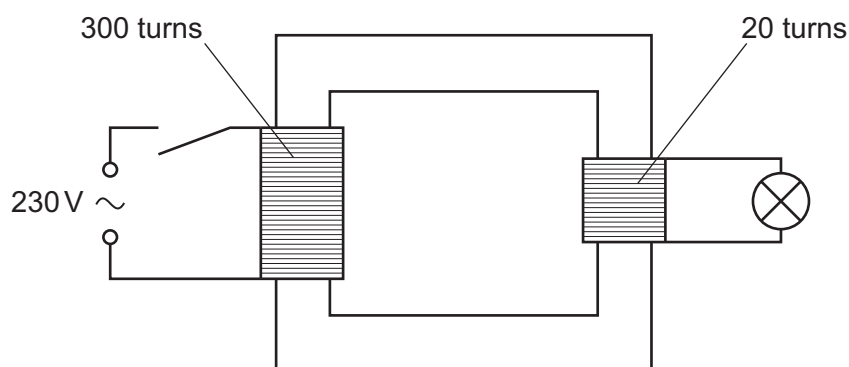
- A** 1 and 3 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

- 35** In an alternating current (a.c.) generator, a magnet rotates near a coil of wire. The induced electromotive force (e.m.f.) in the coil is displayed on an oscilloscope screen.

Which trace is produced as the magnet slows down?



- 36 A student uses a transformer to light a filament lamp using a 230 V a.c. supply. The lamp has a maximum voltage rating of 6.0 V.



What happens when the circuit is switched on?

- A The lamp does not light at all.
 - B The lamp lights at normal brightness.
 - C The lamp lights dimly.
 - D The lamp lights up brightly and then goes out.
- 37 Which statement about nuclear fusion is correct?
- A Nuclear fusion occurs at low temperatures.
 - B Nuclear fusion occurs only between heavy nuclei.
 - C Nuclear fusion occurs in the formation of many stars.
 - D Nuclear fusion powers most electricity-generating stations.
- 38 In one radioactive decay, radium-226 decays to radon-222 as shown.



Which particles are also produced?

- A both an alpha-particle and a beta-particle
- B an alpha-particle only
- C a beta-particle only
- D a neutron

- 39** The count rate from a radioactive source falls from 4000 counts per minute to 500 counts per minute in 72 minutes.

What is the half-life of the source?

- A** 8 minutes
 - B** 9 minutes
 - C** 18 minutes
 - D** 24 minutes
- 40** Which particles are found inside the nucleus of an atom?
- A** neutrons and electrons
 - B** electrons and protons
 - C** neutrons only
 - D** neutrons and protons

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Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2021

MARK SCHEME

Maximum Mark: 40

Published

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This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	C	1
3	A	1
4	D	1
5	A	1
6	D	1
7	C	1
8	B	1
9	C	1
10	B	1
11	B	1
12	B	1
13	D	1
14	B	1
15	A	1
16	D	1
17	D	1
18	C	1
19	C	1
20	B	1
21	C	1
22	C	1
23	C	1
24	D	1
25	A	1
26	B	1
27	D	1
28	B	1

Question	Answer	Marks
29	B	1
30	C	1
31	C	1
32	B	1
33	A	1
34	A	1
35	C	1
36	A	1
37	C	1
38	D	1
39	B	1
40	D	1



Cambridge O Level

PHYSICS

5054/11

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Question	Answer	Marks
1	C	1
2	D	1
3	D	1
4	C	1
5	D	1
6	D	1
7	A	1
8	C	1
9	B	1
10	A	1
11	B	1
12	C	1
13	A	1
14	A	1
15	D	1
16	B	1
17	D	1
18	B	1
19	A	1
20	A	1
21	B	1
22	D	1
23	D	1
24	C	1
25	B	1
26	C	1
27	A	1
28	C	1

Question	Answer	Marks
29	A	1
30	C	1
31	C	1
32	A	1
33	B	1
34	D	1
35	D	1
36	D	1
37	C	1
38	B	1
39	D	1
40	D	1



Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2020

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

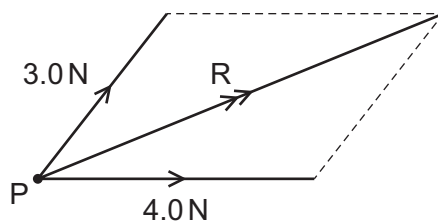
This document has **16** pages. Blank pages are indicated.



1 Which quantity is a vector?

- A acceleration
- B distance
- C speed
- D time

2 The diagram shows the resultant R of a 3.0 N force and a 4.0 N force that act at a point P .



The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90° .

Which value of R is **not** possible?

- A 4.0 N
- B 5.0 N
- C 6.0 N
- D 7.0 N

3 What is measured using a micrometer?

- A area
- B current
- C length
- D mass

4 Stop-watches are used to time the runners in a race.

The stop-watches show the times recorded for the winner and another runner.



winner



other runner

What is the difference in time between the winner and the other runner?

- A 0.4608 s
- B 6.08 s
- C 46.08 s
- D 608 s

- 5 Two cameras are a known distance apart. The exact time that a vehicle passes each of the cameras is recorded.

What can be obtained from the information?

- A average acceleration of the vehicle
- B average speed of the vehicle
- C maximum acceleration of the vehicle
- D maximum speed of the vehicle

- 6 A car of weight 11 000 N moves with constant velocity along a horizontal road. A driving force of 5000 N acts on the car.

What is the force opposing the motion of the car?

- A 5000 N B 6000 N C 11 000 N D 16 000 N

- 7 A man with an open parachute falls to Earth at constant speed. The following forces act:

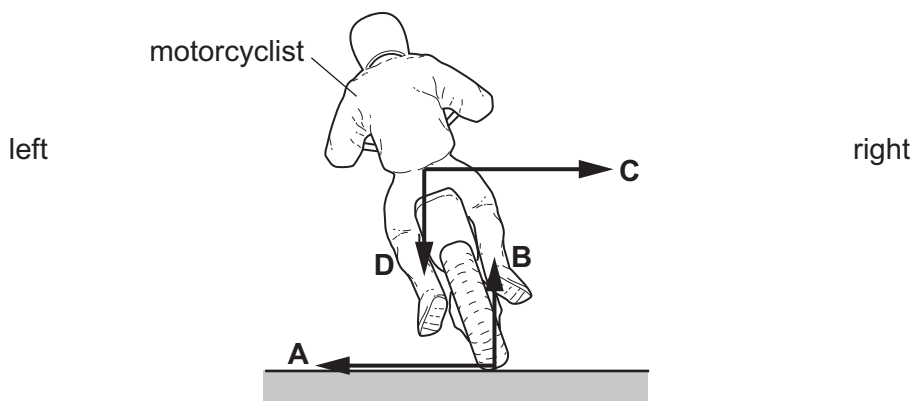
- P the upward force of the parachute on the man
- Q the upward force of the man on the Earth
- R the downward force of the Earth on the parachute
- S the downward force of the man on the parachute

Which two forces are a Newton's third law pair?

- A P and Q B P and R C P and S D Q and R

- 8 The diagram shows a motorcyclist leaning over in order to turn the corner to the left.

Which force causes him to turn?

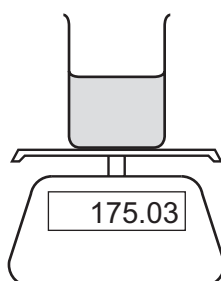


- 9 An object weighs 40 N on Earth, where the gravitational field strength is 10 N/kg. The object is taken to a planet where the gravitational field strength is 4.0 N/kg.

Which row is correct?

	mass of object on the planet / kg	weight of object on the planet / N
A	4.0	1.0
B	4.0	16
C	400	100
D	400	1600

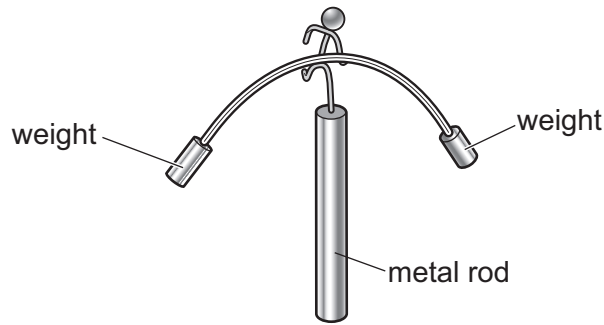
- 10 The diagram shows the equipment used to determine the density of a liquid.



Which equation is used to calculate the density of the liquid?

- A**
$$\frac{\text{mass of beaker and liquid} - \text{mass of beaker}}{\text{volume of liquid}}$$
- B**
$$\frac{\text{mass of beaker and liquid} - \text{mass of liquid}}{\text{volume of liquid}}$$
- C**
$$\frac{\text{volume of liquid}}{\text{mass of beaker and liquid} - \text{mass of beaker}}$$
- D**
$$\frac{\text{volume of liquid}}{\text{mass of beaker and liquid} - \text{mass of liquid}}$$
- 11 Which single item can be used to find the centre of mass of a plane lamina of irregular shape?
- A** a balance
- B** a measuring tape
- C** a micrometer
- D** a vertical pin

- 12 A student makes a model toy which balances on the end of a metal rod.



Why are the weights needed?

- A** They increase the pressure on the rod.
B They increase the weight of the model.
C They lower the centre of mass below the top of the rod.
D They raise the centre of mass above the top of the rod.
- 13 An unstretched elastic cord of length 15.0 cm stretches by 4.0 cm when a weight of 6.0 N is suspended from it. The cord does not exceed the limit of proportionality.

What is the total length of the cord when the weight is 3.0 N?

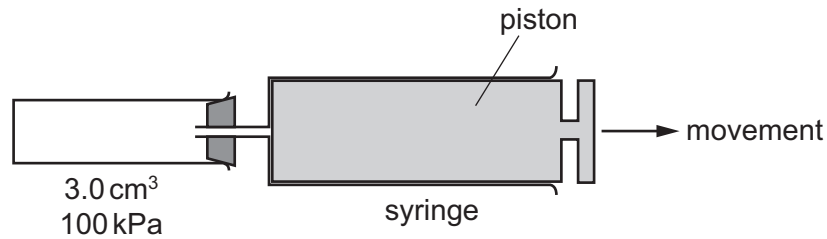
- A** 2.0 cm **B** 7.5 cm **C** 9.5 cm **D** 17.0 cm
- 14 An object is placed at different depths in liquids of different densities.

In which liquid and at what depth is the pressure on the object the greatest?

	<u>density of liquid</u> kg/m ³	depth / m
A	1000	4.0
B	1000	10
C	1200	4.0
D	1200	10

- 15** A small vessel of volume 3.0 cm^3 contains air at a pressure of 100 kPa .

The small vessel is connected to a syringe. The piston is fully inserted into the syringe.



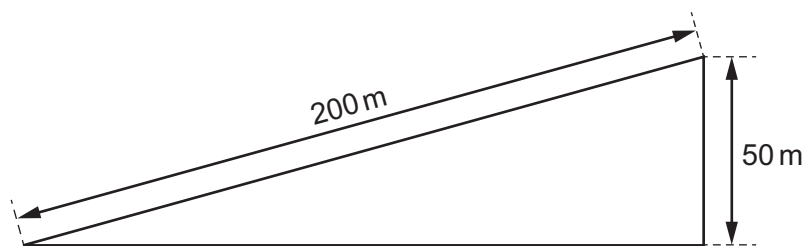
The piston is moved slowly to the right so that the air in the syringe has a volume of 12 cm^3 . The temperature of the air does not change.

What is the pressure of the air in the syringe?

- A** 20 kPa **B** 25 kPa **C** 80 kPa **D** 100 kPa
- 16** A rocket of mass M when empty carries a mass M of fuel. The rocket and fuel travel at speed v . The engine of the rocket is fired and all of the fuel is expelled. The speed of the rocket increases to $2v$.

What happens to the kinetic energy of the rocket?

- A** It doubles.
B It halves.
C It increases by a factor of four.
D It stays the same.
- 17** A car of mass 1000 kg is driven 200 m up an incline so that it rises 50 m vertically.



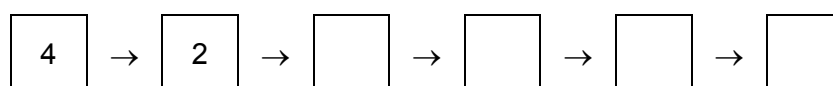
The acceleration of free fall g is 10 m/s^2 .

What is the gain in gravitational potential energy?

- A** 5000 J **B** 200 000 J **C** 500 000 J **D** 2 000 000 J

- 18 In which situation is energy being released by the fusion of hydrogen nuclei to form helium?
- A in the decay of ^{14}C used to date an object
 - B in a radioactive isotope emitting alpha-particles
 - C in the centre of the Earth
 - D in the centre of the Sun
- 19 The main stages in the operation of a coal-fired power station are listed. They are **not** in the correct order.
- 1 The turbine turns a generator.
 - 2 Water in the boiler becomes hot.
 - 3 Steam turns a turbine.
 - 4 Burning coal produces thermal energy.
 - 5 Electromagnetic induction produces electrical energy.
 - 6 Steam is produced.

The flow chart shows the first two stages.



What is the correct order for the remaining stages?

- A $6 \rightarrow 3 \rightarrow 1 \rightarrow 5$
 - B $6 \rightarrow 1 \rightarrow 3 \rightarrow 5$
 - C $1 \rightarrow 6 \rightarrow 3 \rightarrow 5$
 - D $6 \rightarrow 3 \rightarrow 5 \rightarrow 1$
- 20 In which situation is there **no** transfer of energy?
- A a car battery turning the starter motor
 - B a car moving along a horizontal road at constant speed
 - C a solar panel warming water
 - D a spacecraft orbiting the earth at constant height

- 21** The 200 V battery of an electric car is charged with a current of 33 A for 4.0 hours. The efficiency of the charging process is 90%.

How much useable energy is transferred to the battery?

- A** 21 MJ **B** 86 MJ **C** 95 MJ **D** 106 MJ

- 22** In an experiment to measure the power output of a small steam engine, a known load is lifted by the engine.

Which two measuring instruments are also required?

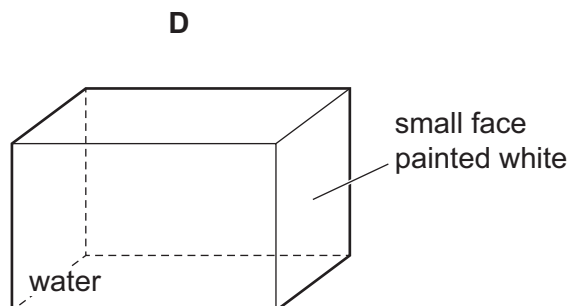
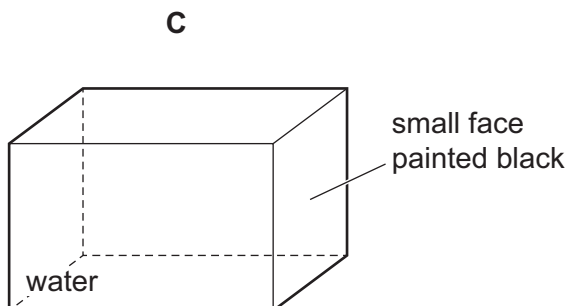
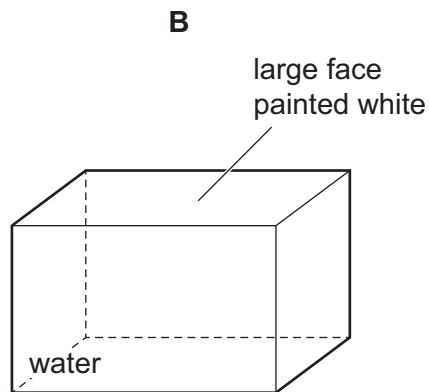
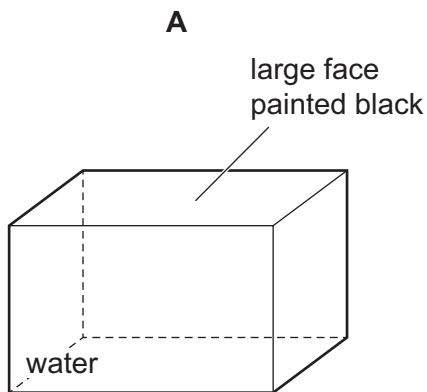
- A** measuring cylinder and thermometer
B measuring cylinder and metre rule
C metre rule and stop-watch
D stop-watch and thermometer

- 23** Four metal containers, with identical dimensions, are filled with water at 90 °C.

All the faces, except one, of each container are covered in a very good insulator.

The one exposed face on each container is painted either black or white.

In which container does the water cool the fastest?



24 Thermal energy is used to turn water at 100 °C to steam at 100 °C.

Which of the following occurs?

- 1 The forces between the molecules decrease.
- 2 The speed of the molecules increases.
- 3 The separation of the molecules increases.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

25 The thermal energy produced by an electric heater in three minutes is used to melt wax. The solid wax is initially at its melting point of 60 °C.

The specific latent heat of the wax is 220 J/g. The heater supplies 7700 J of thermal energy to the wax. Some of the wax melts.

How much wax melts?

A 0.58 g **B** 1.7 g **C** 12 g **D** 35 g

26 What always happens when the temperature increases?

- A** A gas at constant pressure expands.
- B** A metal rod becomes longer, but its volume remains constant.
- C** A liquid contracts and then expands.
- D** The hole in the centre of a metal disc becomes smaller.

27 Which row explains why a liquid has a fixed volume but does **not** have a fixed shape?

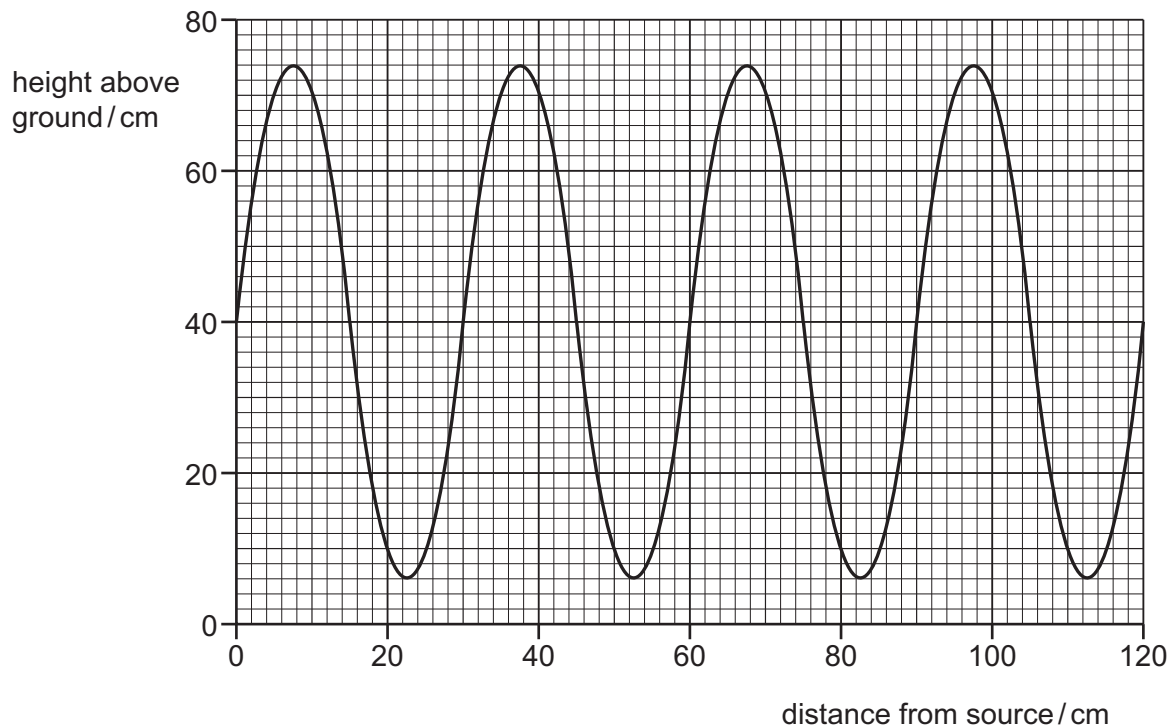
	force between molecules in the liquid	movement of molecules in the liquid
A	large	free to move within the liquid
B	large	vibrate at fixed positions
C	small	free to move within the liquid
D	small	vibrate at fixed positions

28 The air in a sealed can is heated.

Which statement explains the increase in pressure inside the can?

- A** The molecules collide with the walls of the can more frequently when the air is heated.
- B** The mass of the molecules is greater when the air is heated.
- C** There are more molecules per unit volume when the air is heated.
- D** The separation of molecules becomes greater when the air is heated.

- 29 The diagram shows how the height above the ground of a rope transmitting a wave varies with distance from the source of vibrations at one instant in time.



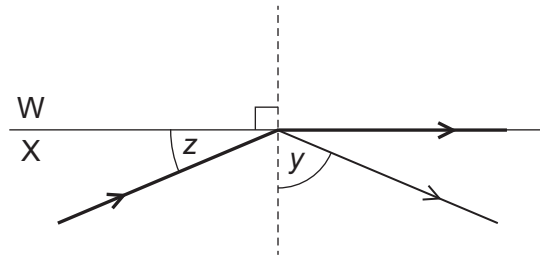
What is the amplitude of the wave shown?

- A** 30 cm **B** 34 cm **C** 72 cm **D** 74 cm
- 30 A vibrator produces 12 wavelengths on the surface of water in 10 s. The spacing between the first crest and the third crest is 60 cm.

What is the speed of the wave?

- A** 24 cm/s **B** 25 cm/s **C** 36 cm/s **D** 72 cm/s

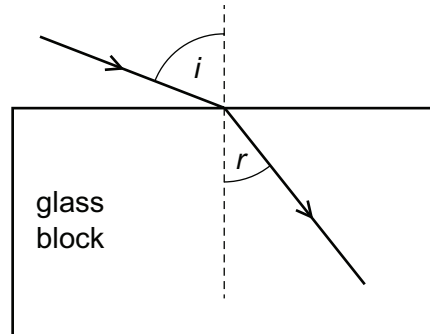
- 31 The diagram shows a ray of light incident on the boundary between two mediums W and X. The mediums have different refractive indexes.



Some light is reflected and some passes along the surface between the two mediums. Angle y is greater than angle z .

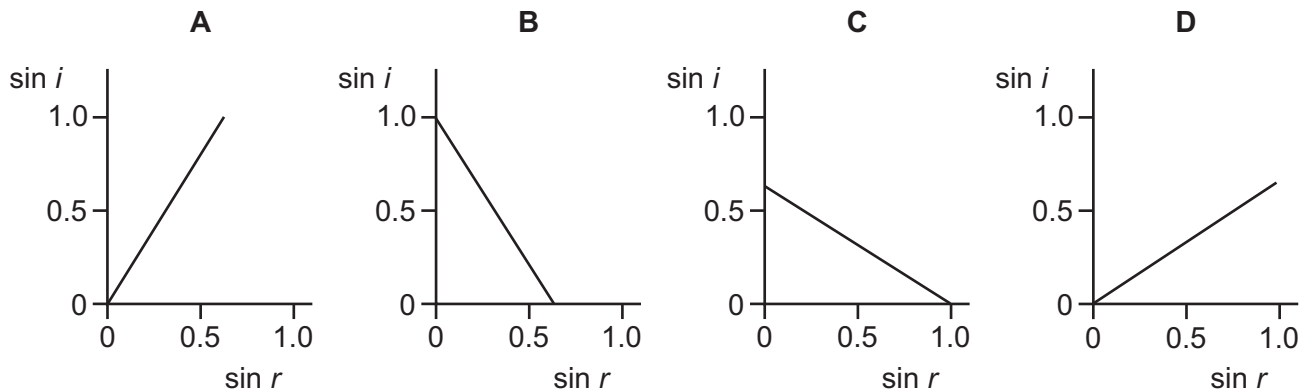
Which statement is correct?

- A W has a greater refractive index than X and angle y is equal to the critical angle.
 - B W has a greater refractive index than X and angle z is equal to the critical angle.
 - C X has a greater refractive index than W and angle y is equal to the critical angle.
 - D X has a greater refractive index than W and angle z is equal to the critical angle.
- 32 Light enters a glass block at an angle of incidence i and it produces an angle of refraction r in the glass.



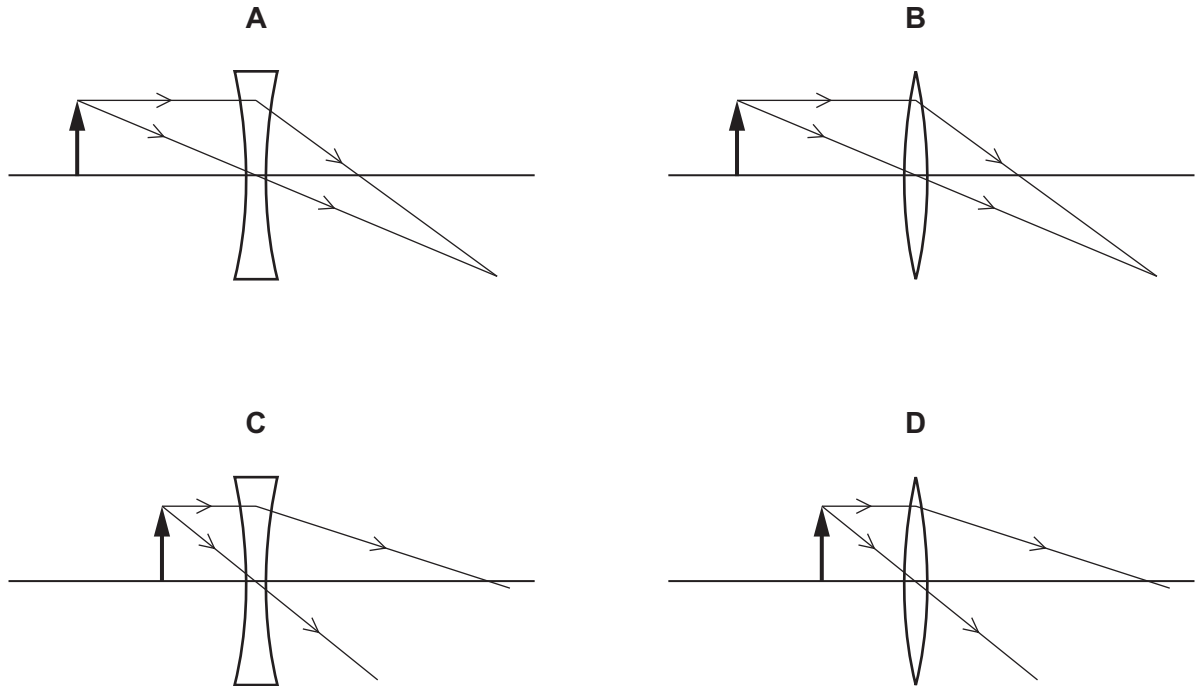
Several different values of i and r are measured, and a graph is drawn of $\sin i$ against $\sin r$.

Which graph is correct?



33 A converging glass lens is used to produce a virtual, magnified image.

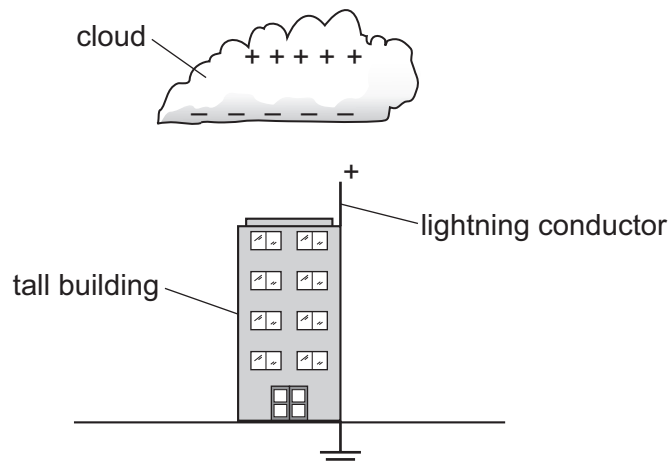
Which ray diagram shows the rays passing through the converging lens?



34 A metal lightning conductor is placed on the top of a tall building.

The conductor is connected to Earth.

When a charged cloud passes over the building, the metal lightning conductor becomes positively charged.



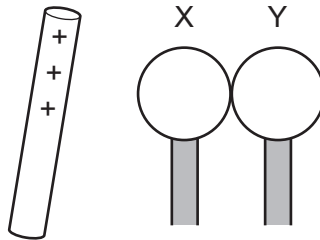
What happens to the lightning conductor to produce a positive charge at its top?

- A** Electrons move downwards and protons move upwards in the conductor.
- B** Only electrons move in the conductor.
- C** Only protons move in the conductor.
- D** Protons in the air attach themselves to the conductor.

- 35** Two insulated and uncharged metal spheres X and Y are touching.

A positively charged rod is held near X and then the spheres are moved apart.

X now has a negative charge.



What is the charge on Y?

- A** negative and smaller than that on X
 - B** negative and the same size as that on X
 - C** positive and smaller than that on X
 - D** positive and the same size as that on X
- 36** A stationary negative charge in an electric field experiences an electric force in the direction shown.



What is the direction of the electric field?

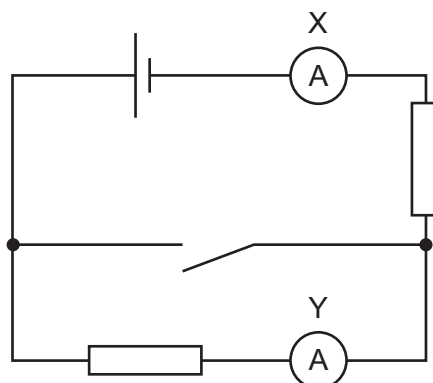
- A** to the left
 - B** to the right
 - C** down the page
 - D** up the page
- 37** A metal wire of length l and cross-sectional area A has resistance R .

A second wire is made from the same metal. It has a length $2l$ and a cross-sectional area $4A$.

What is the resistance of the second wire?

- A** $8R$
- B** $2R$
- C** $\frac{R}{2}$
- D** $\frac{R}{8}$

38 The diagram shows a circuit.



What effect does closing the switch have on the readings of ammeters X and Y?

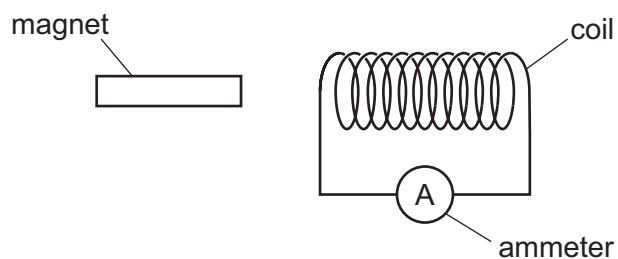
	reading on ammeter X	reading on ammeter Y
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

39 A rectangular current-carrying coil is pivoted between the poles of an electromagnet.

Which action does **not**, on its own, increase the size of the turning effect exerted on the coil?

- A** increasing the current in the coils of the electromagnet
- B** increasing the current in the rectangular coil
- C** reversing the current in the electromagnet
- D** increasing the number of turns on the rectangular coil

- 40 The diagram shows a coil connected to a very sensitive ammeter. A magnet is next to the coil.



Which action results in a zero reading on the ammeter?

- A** moving the coil and the magnet at the same speed in opposite directions
- B** moving the coil and the magnet at the same speed in the same direction
- C** moving the coil away from the stationary magnet
- D** moving the magnet towards the stationary coil

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Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

May/June 2020

1 hour

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

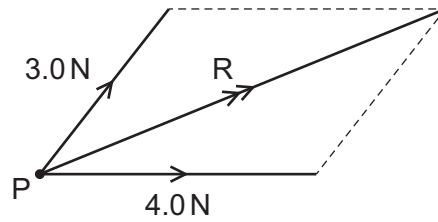
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



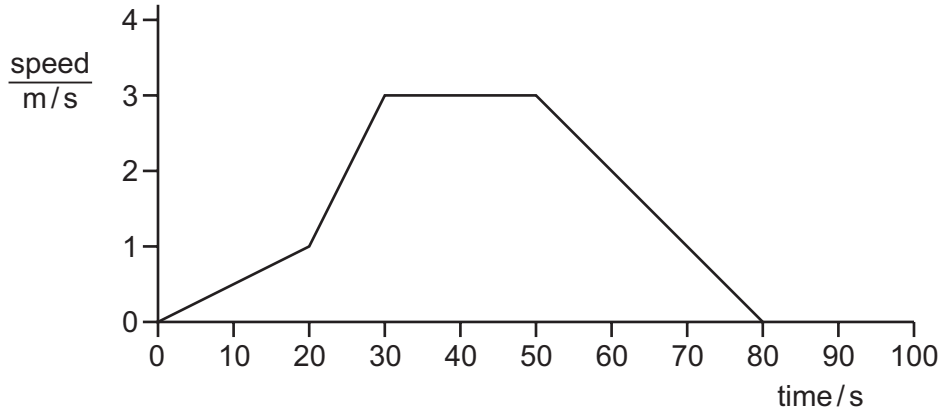
- 1 The diagram shows the resultant R of a 3.0 N force and a 4.0 N force that act at a point P .



The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90° .

Which value of R is **not** possible?

- A** 4.0 N **B** 5.0 N **C** 6.0 N **D** 7.0 N
- 2 What is measured using a micrometer?
- A** area
B current
C length
D mass
- 3 The graph shows a short journey. It is a speed–time graph.



What is the greatest speed reached?

- A** 1 m/s **B** 2 m/s **C** 3 m/s **D** 4 m/s
- 4 A car of weight $11\,000\text{ N}$ moves with constant velocity along a horizontal road. A driving force of 5000 N acts on the car.

What is the force opposing the motion of the car?

- A** 5000 N **B** 6000 N **C** $11\,000\text{ N}$ **D** $16\,000\text{ N}$

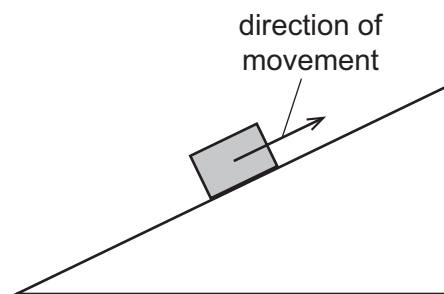
- 5 A man with an open parachute falls to Earth at constant speed. The following forces act:

- P the upward force of the parachute on the man
- Q the upward force of the man on the Earth
- R the downward force of the Earth on the parachute
- S the downward force of the man on the parachute

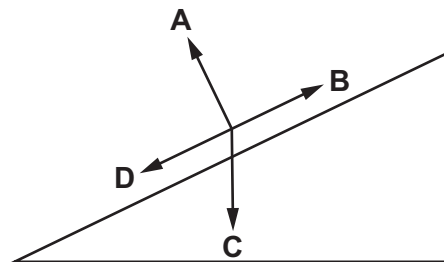
Which two forces are a Newton's third law pair?

- A** P and Q **B** P and R **C** P and S **D** Q and R

- 6 A box is pulled up a rough slope, as shown.

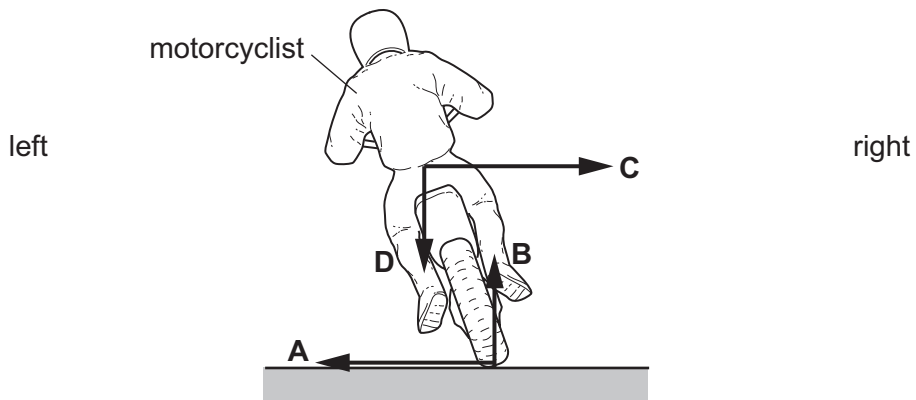


In which direction does friction act on the box?



- 7 The diagram shows a motorcyclist leaning over in order to turn the corner to the left.

Which force causes him to turn?



8 Which type of force causes the Earth to orbit the Sun?

- A elastic
- B electrostatic
- C gravitational
- D magnetic

9 Four objects of different masses are on different planets.

The weight of each object on its planet is determined.

Which object is on the planet with the smallest gravitational field strength?

	mass / kg	weight / N
A	5	125
B	15	150
C	20	220
D	25	225

10 Which physical quantity does **not** change when a piece of copper is heated?

- A density
- B mass
- C temperature
- D volume

11 A single metal bolt has a mass of 34 g.

Three of the bolts are immersed in a measuring cylinder that contains 30 cm^3 of water.

The reading on the measuring cylinder rises to 42 cm^3 .

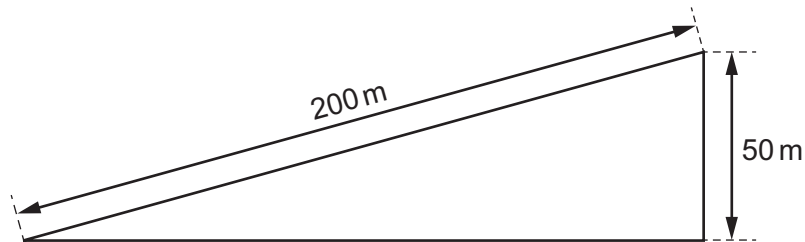
What is the density of the metal?

- A 0.81 g/cm^3 B 2.8 g/cm^3 C 7.5 g/cm^3 D 8.5 g/cm^3

- 12** A rocket of mass M when empty carries a mass M of fuel. The rocket and fuel travel at speed v . The engine of the rocket is fired and all of the fuel is expelled. The speed of the rocket increases to $2v$.

What happens to the kinetic energy of the rocket?

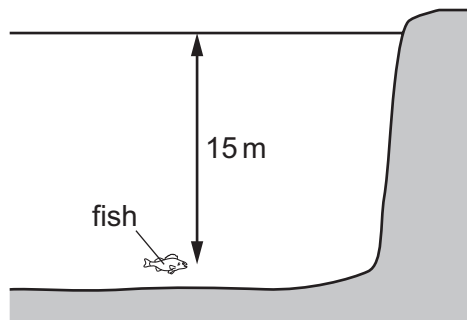
- A** It doubles.
B It halves.
C It increases by a factor of four.
D It stays the same.
- 13** A car of mass 1000 kg is driven 200 m up an incline so that it rises 50 m vertically.



The acceleration of free fall g is 10 m/s^2 .

What is the gain in gravitational potential energy?

- A** 5000 J **B** 200 000 J **C** 500 000 J **D** 2 000 000 J
- 14** A fish is swimming 15 m below the surface of a lake, as shown.



The density of the water is 1000 kg/m^3 .

Atmospheric pressure is 100 000 Pa.

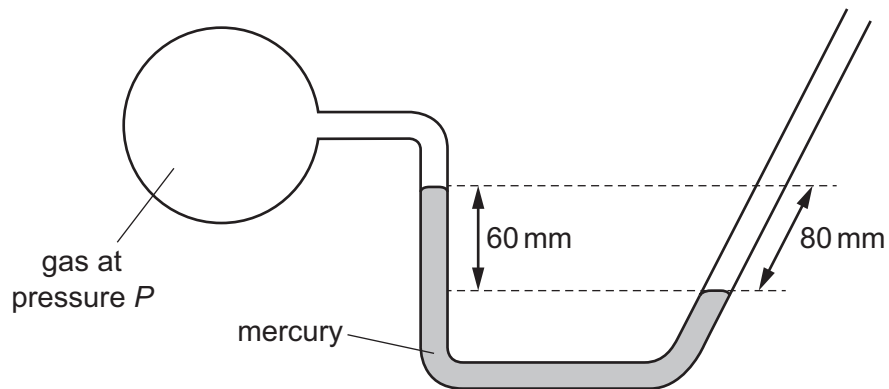
The acceleration of free fall g is 10 m/s^2 .

What is the total pressure on the fish?

- A** 50 000 Pa **B** 120 000 Pa **C** 150 000 Pa **D** 250 000 Pa

- 15** The diagram shows a mercury manometer. The tube is open to the atmosphere on the right-hand side.

The left-hand side is connected to a container containing a gas at pressure P .

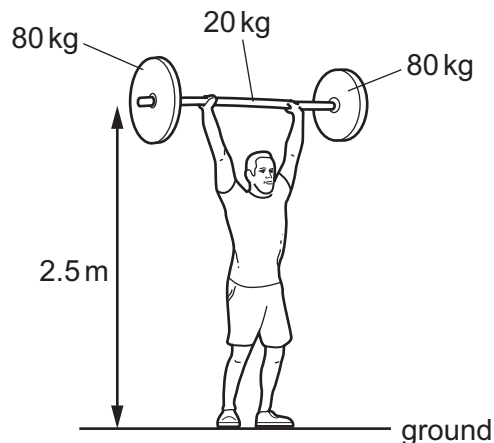


Atmospheric pressure on its own supports a column of mercury of height 756 mm.

Which height of column does pressure P on its own support?

- A** 676 mm of mercury
B 696 mm of mercury
C 816 mm of mercury
D 836 mm of mercury
- 16** In a weightlifting contest, an athlete lifts a metal bar of mass 20 kg fitted with a mass of 80 kg on each end.

The lift from ground to a height of 2.5 m takes 0.50 s.

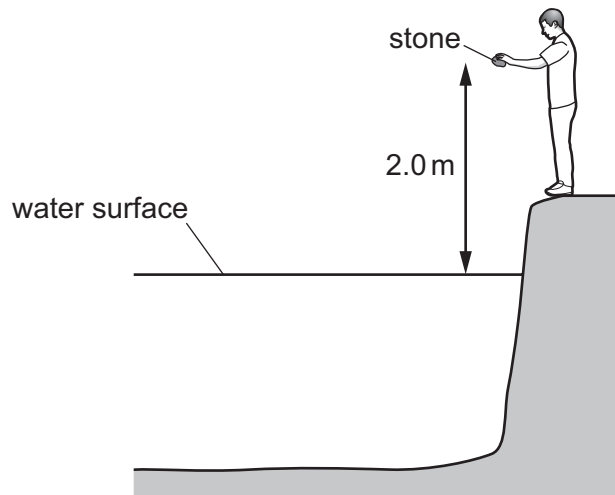


The gravitational field strength g is 10 N/kg.

What average power does the weightlifter exert in providing the gravitational potential energy during this lift?

- A** 0.90 kW **B** 5.0 kW **C** 8.0 kW **D** 9.0 kW

- 17 A boy is standing by the side of a lake.

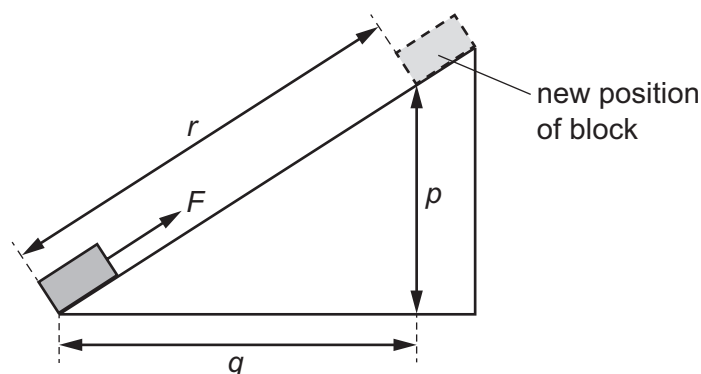


The boy drops a heavy stone from a height of 2.0 m above the water surface.

The acceleration due to gravity is 10 m/s^2 .

What is the speed of the stone when it hits the surface of the water?

- A** 4.5 m/s **B** 6.3 m/s **C** 20 m/s **D** 40 m/s
- 18 In which situation is energy being released by the fusion of hydrogen nuclei to form helium?
- A** in the decay of ^{14}C used to date an object
- B** in a radioactive isotope emitting alpha-particles
- C** in the centre of the Earth
- D** in the centre of the Sun
- 19 The diagram shows a block being pulled up a slope by a force F .



The block reaches the new position at the top of the slope.

What is the work done by force F in moving the block to its new position?

- A** $F \times r$ **B** $F \times p$ **C** $F(q + p)$ **D** $F(r + p)$

20 Which statement about a mercury-in-glass clinical thermometer is correct?

- A It covers a much larger range than an ordinary laboratory thermometer.
- B It is more sensitive than an ordinary laboratory thermometer.
- C Its scale is not linear.
- D Its sensitivity is affected by the constriction in the capillary tube.

21 Which description of a dull black surface is correct?

- A good emitter, good absorber and good reflector of radiation
- B good emitter, poor absorber and poor reflector of radiation
- C good emitter, good absorber and poor reflector of radiation
- D poor emitter, poor absorber and poor reflector of radiation

22 The temperature of a body increases by 1°C .

Which other quantity also increases?

- A heat capacity
- B internal energy
- C specific heat capacity
- D specific latent heat

23 The thermal energy produced by an electric heater in three minutes is used to melt wax. The solid wax is initially at its melting point of 60°C .

The specific latent heat of the wax is 220 J/g . The heater supplies 7700 J of thermal energy to the wax. Some of the wax melts.

How much wax melts?

- A 0.58 g B 1.7 g C 12 g D 35 g

24 Which row explains why a liquid has a fixed volume but does **not** have a fixed shape?

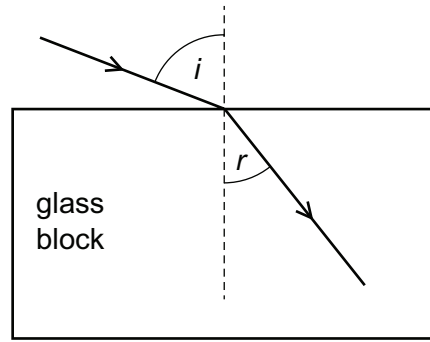
	force between molecules in the liquid	movement of molecules in the liquid
A	large	free to move within the liquid
B	large	vibrate at fixed positions
C	small	free to move within the liquid
D	small	vibrate at fixed positions

- 25** A fixed mass of gas is enclosed in a cylinder by a piston which is free to move.

Which combination of changes to the pressure and to the temperature **must** increase the density of the gas?

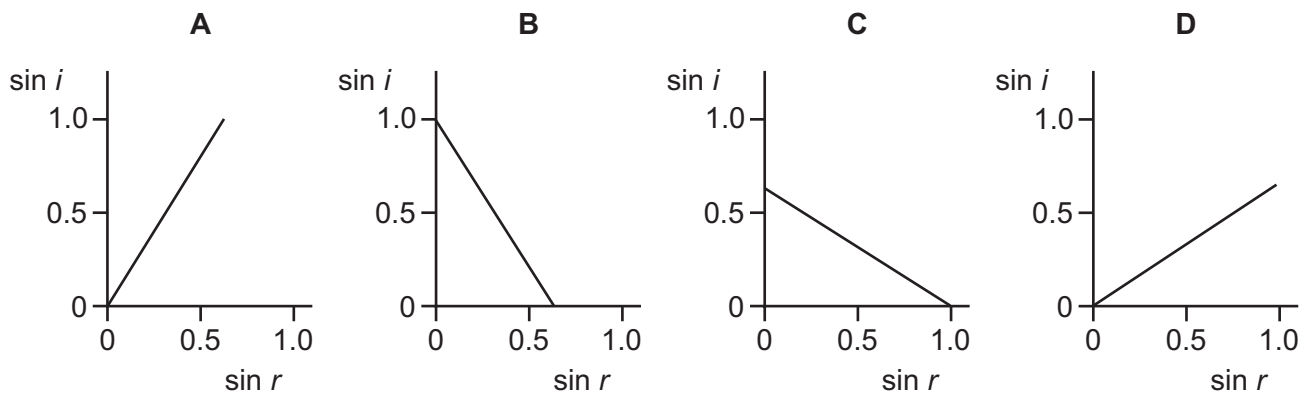
	pressure	temperature
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

- 26** Light enters a glass block at an angle of incidence i and it produces an angle of refraction r in the glass.



Several different values of i and r are measured, and a graph is drawn of $\sin i$ against $\sin r$.

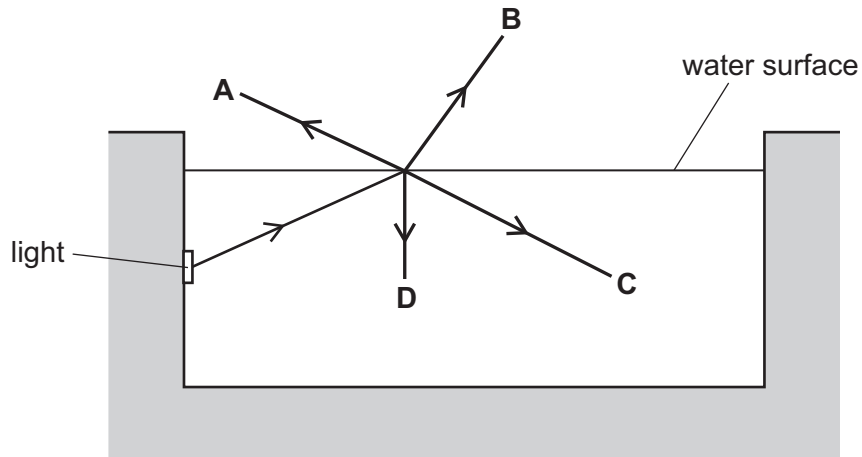
Which graph is correct?



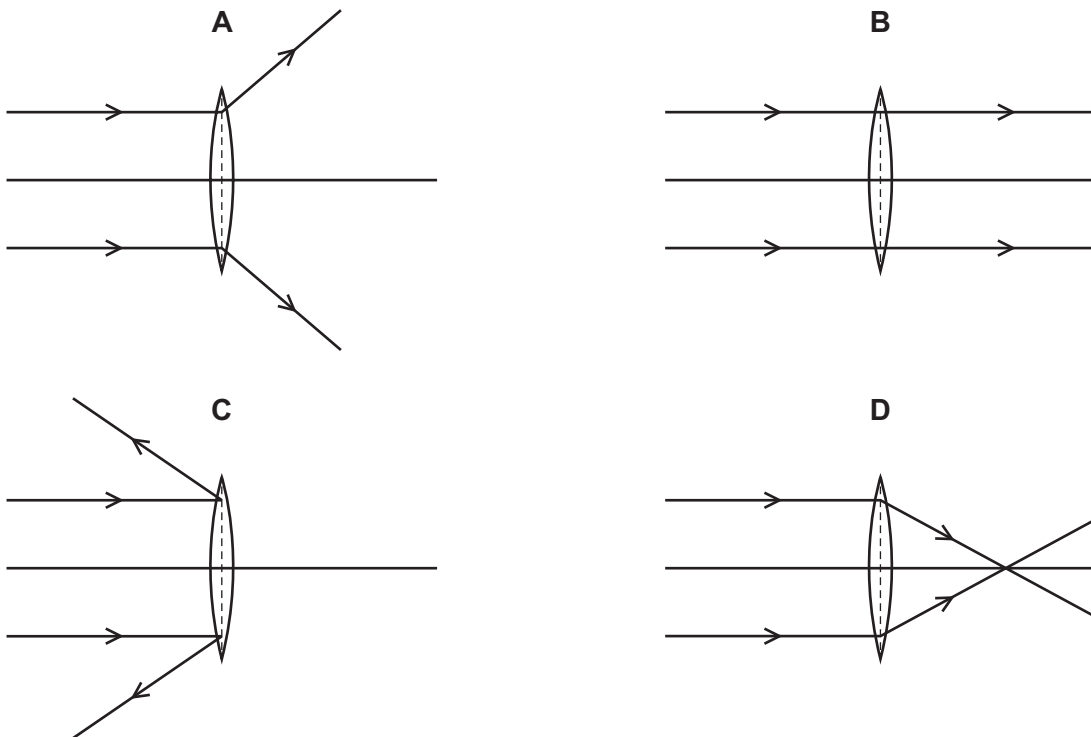
27 A swimming pool is lit by an underwater light.

A ray of light is incident on the water surface.

What is the correct path for the ray of light?



28 Which diagram shows the action of a converging lens on a parallel beam of light?



29 What is a feature of red light compared with that of violet light?

- A** A prism deviates red light more.
- B** Red light has a lower frequency.
- C** Red light has a shorter wavelength.
- D** The speed of red light in a vacuum is smaller.

30 What is the ratio of the longest sound wavelength audible to a healthy human ear to the shortest?

- A** 20:1 **B** 1000:1 **C** 10 000:1 **D** 20 000:1

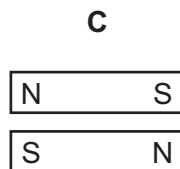
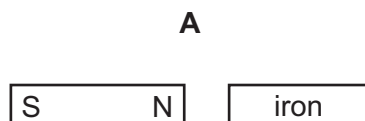
31 Ultrasound is used to clean jewellery in a liquid.

What is another use of ultrasound?

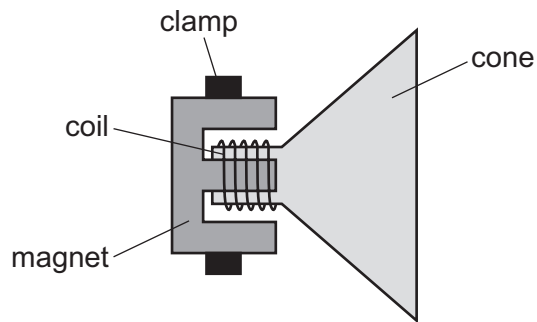
- A** optical fibre communication
B pre-natal scanning
C sunbeds
D telephone communications

32 Bar magnets and various non-magnetic and demagnetised metal bars are placed in the different arrangements shown.

In which arrangement do the bars repel?

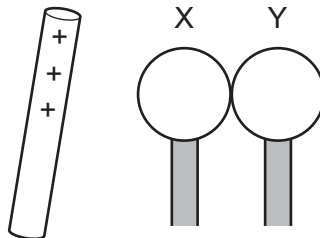


- 33** There is varying current in the coil of the loudspeaker shown. The loudspeaker is producing a sound. The magnet is clamped.



What is vibrating to produce the sound?

- A** coil only
 - B** cone only
 - C** magnet only
 - D** coil and cone
- 34** Two insulated and uncharged metal spheres X and Y are touching.
- A positively charged rod is held near X and then the spheres are moved apart.
- X now has a negative charge.



What is the charge on Y?

- A** negative and smaller than that on X
- B** negative and the same size as that on X
- C** positive and smaller than that on X
- D** positive and the same size as that on X

- 35 A stationary negative charge in an electric field experiences an electric force in the direction shown.



What is the direction of the electric field?

- A** to the left
B to the right
C down the page
D up the page
- 36 A battery consists of three identical cells in parallel.

What is the unit of electromotive force (e.m.f.) and to what is the e.m.f. of the battery equal?

	unit	e.m.f. of the battery is equal to
A	J/C	the sum of the e.m.f.s of the three cells
B	J/C	the e.m.f. of one of the cells
C	N/V	the sum of the e.m.f.s of the three cells
D	N/V	the e.m.f. of one of the cells

- 37 The nuclide notation for an isotope of X is A_ZX .

How many neutrons are there in a nucleus?

- A** A **B** A – Z **C** Z – A **D** Z
- 38 Radioactivity is used in several activities that take place in hospitals.

Which activity **never** uses radioactivity?

- A** cooking meals
B diagnosing illnesses
C sterilising equipment
D treating illnesses

- 39** $^{14}_6\text{C}$ is a natural radioactive isotope of carbon and is present in all living things. It has a half-life of 5700 years.

How old is a bone that is found to have only $\frac{1}{4}$ of the natural proportion of $^{14}_6\text{C}$?

- A** 5700 years
 - B** 11 400 years
 - C** 17 100 years
 - D** 22 800 years
- 40** A kettle is connected to the 240 V mains supply using a plug containing a 13 A fuse.

The kettle contains water.

When it is switched on, the fuse blows. This happens again after a new fuse is fitted.

Someone replaces the fuse with a nail, and the kettle works.

What else might happen as a result of replacing the fuse with a nail?

- A** A very large current overloads the wiring, causing a fire.
- B** The kettle boils the water less quickly.
- C** The kettle uses more energy to boil the water.
- D** The water boils at a higher temperature.

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Cambridge O Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

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Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	A	1
3	C	1
4	B	1
5	B	1
6	A	1
7	C	1
8	A	1
9	B	1
10	A	1
11	D	1
12	C	1
13	D	1
14	D	1
15	A	1
16	C	1
17	C	1
18	D	1
19	A	1
20	D	1
21	B	1
22	C	1
23	A	1
24	C	1
25	D	1
26	A	1
27	A	1
28	A	1

Question	Answer	Marks
29	B	1
30	C	1
31	C	1
32	A	1
33	D	1
34	B	1
35	D	1
36	A	1
37	C	1
38	C	1
39	C	1
40	B	1



Cambridge O Level

PHYSICS

5054/11

Paper 1 Multiple Choice

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	A	1
2	C	1
3	C	1
4	A	1
5	C	1
6	D	1
7	A	1
8	C	1
9	D	1
10	B	1
11	D	1
12	C	1
13	C	1
14	D	1
15	B	1
16	D	1
17	B	1
18	D	1
19	A	1
20	B	1
21	C	1
22	B	1
23	D	1
24	A	1
25	C	1
26	A	1
27	C	1
28	D	1

Question	Answer	Marks
29	B	1
30	B	1
31	B	1
32	D	1
33	D	1
34	D	1
35	A	1
36	B	1
37	B	1
38	A	1
39	B	1
40	A	1



Cambridge Assessment International Education
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/12

May/June 2019

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

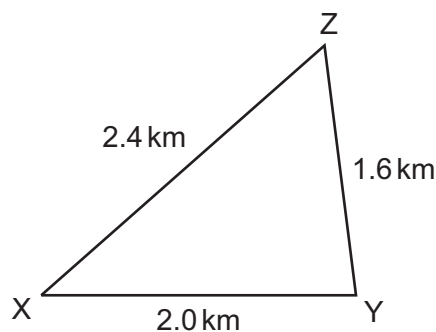
Any rough working should be done in this booklet.

Electronic calculators may be used.

This document consists of **16** printed pages.



- 1 Which reading is given to one tenth of a millimetre?
- A** 3.3 cm **B** 3.31 cm **C** 3.310 cm **D** 3.312 cm
- 2 A woman runs 2.0 km from X to Y in 20 minutes and then rests at Y for 10 minutes. She then takes 10 minutes to run 1.6 km from Y to Z.



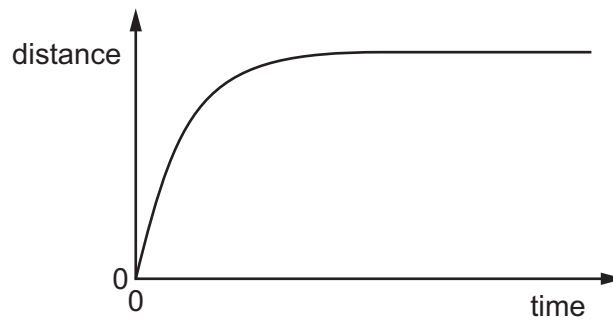
What is the size of her average velocity for the journey from X to Z?

- A** 0.060 km/minute
B 0.080 km/minute
C 0.090 km/minute
D 0.130 km/minute
- 3 A cyclist travelling in a straight line at 8.0 m/s accelerates to 12 m/s in a time of 6.0 s.

Which expression gives the cyclist's acceleration?

- A** $\frac{6}{12-8}$ **B** $\frac{8}{12-6}$ **C** $\frac{12-6}{8}$ **D** $\frac{12-8}{6}$

- 4 The diagram shows the distance–time graph for a moving object.



What is the moving object?

- A** a ball thrown vertically upwards and falling back to the thrower
 - B** a car, starting from rest, speeding up as it moves away from a traffic signal
 - C** a rock dropped from a high cliff, falling into the sea below
 - D** a train braking to a halt as it stops at a station
- 5 Four of the gravitational forces that act between bodies in the Solar System are listed below.

- P the force on the Moon due to the Earth
- Q the force on the Earth due to the Sun
- R the force on the Earth due to the Moon
- S the force on the Moon due to the Sun

Which two forces are a Newton's third law pair (action and reaction)?

- A** P and Q **B** P and R **C** Q and S **D** R and S
- 6 A parachutist of mass 60 kg falls at a steady speed of 10 m/s. The gravitational field strength g is 10 N/kg.

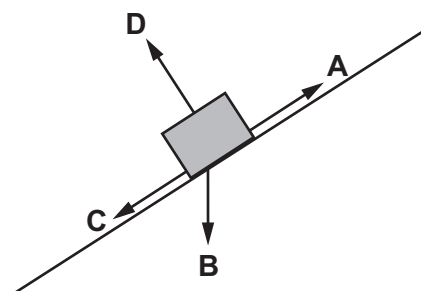
What is the upward force acting on the parachutist?

- A** 0 **B** 6.0 N **C** 60 N **D** 600 N

- 7 A force moves a box up a rough slope.

The diagram shows four forces acting on the box.

Which is the force due to friction on the box?



- 8 The mass of object P is greater than the mass of object Q.

The objects contain different amounts of matter and have a different resistance to change of motion.

Which row is correct?

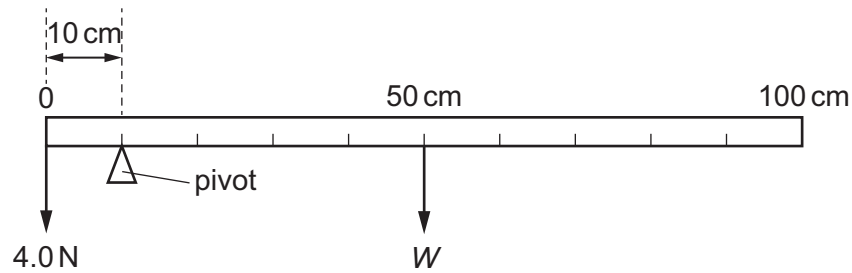
	greater amount of matter	greater resistance to change of motion
A	P	P
B	P	Q
C	Q	P
D	Q	Q

- 9 Four objects of different masses are situated in places with different gravitational field strengths.

Which object has the greatest weight?

	mass / kg	<u>gravitational field strength</u> N/kg
A	3.0	10.4
B	3.5	9.5
C	4.0	10.2
D	4.5	9.0

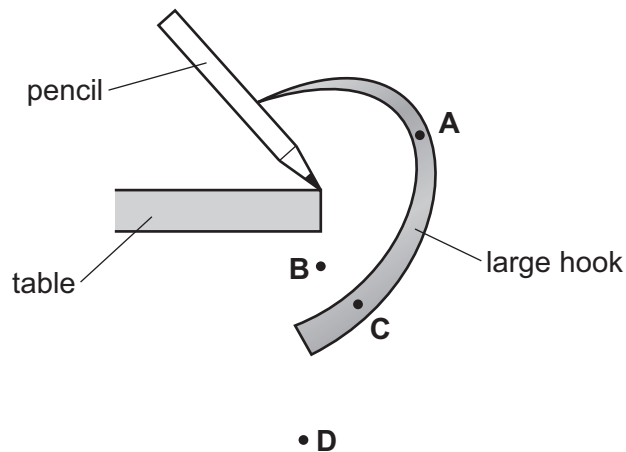
- 10 A uniform metre rule is balanced by a 4.0 N weight as shown.



What is the weight W of the metre rule?

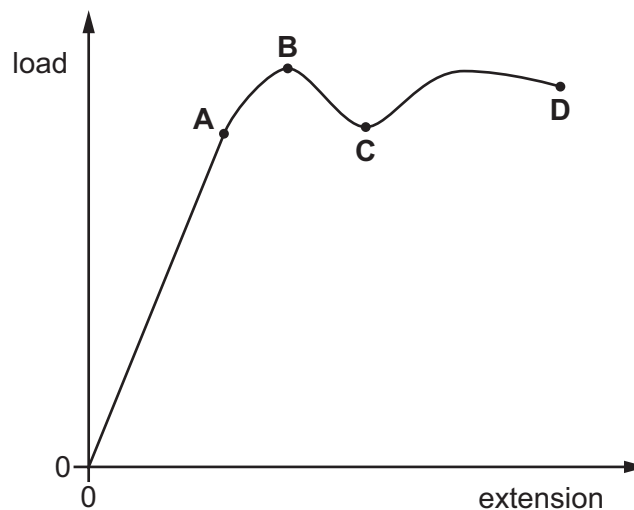
- A** 1.0 N **B** 4.0 N **C** 16 N **D** 40 N
- 11 A large hook sticks into a pencil. The hook and pencil are balanced on the edge of a table, as shown.

Where is the centre of mass of the hook and pencil?



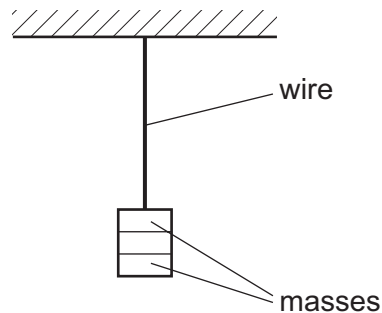
- 12 The diagram shows a graph of load against extension for a material.

At which point is the limit of proportionality?



- 13** A length of metal wire is made from a radioactive isotope of the metal.

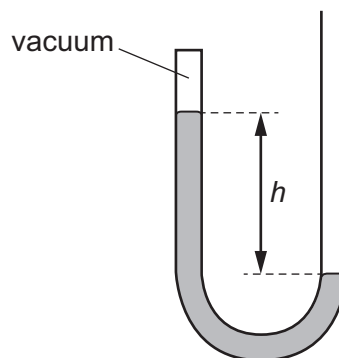
The diagram shows masses suspended from the end of the wire.



The masses are removed from the wire.

Which property changes?

- A** half-life of the isotope
 - B** length of the wire
 - C** mass of the wire
 - D** melting point of the metal
- 14** The diagram shows a manometer containing mercury that is sealed at one end.



What happens to the distance h when the manometer is taken higher up a mountain?

- A** It decreases, because atmospheric pressure decreases with height.
- B** It decreases, because atmospheric pressure increases with height.
- C** It increases, because atmospheric pressure decreases with height.
- D** It increases, because atmospheric pressure increases with height.

- 15 The pressure on the gas contained in a sealed syringe is slowly doubled. The volume of the gas halves.

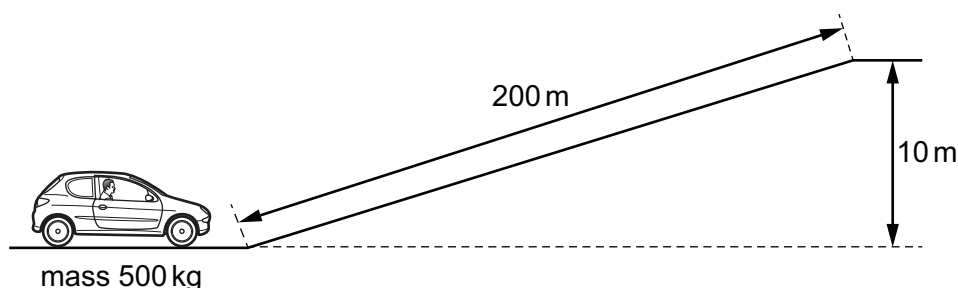
What **must** be true about the density and the temperature of the gas?

	density	temperature
A	decreases	increases
B	increases	decreases
C	increases	remains constant
D	remains constant	increases

- 16 An object is lifted.

Which expression gives the increase in gravitational potential energy of the object?

- A** mass multiplied by its velocity
B mass multiplied by its acceleration
C the rate at which work is done
D weight multiplied by the increase in height
- 17 The diagram shows a small car of mass 500 kg approaching a hill. It moves up the hill with constant speed.



The gravitational field strength g is 10 N/kg .

Ignoring friction, how much work is done in moving the car up the hill?

- A** $5.0 \times 10^3 \text{ J}$ **B** $5.0 \times 10^4 \text{ J}$ **C** $1.0 \times 10^5 \text{ J}$ **D** $1.0 \times 10^6 \text{ J}$
- 18 Which method of producing electricity does **not** use an a.c. generator?
- A** geothermal power station
B solar panels
C wave-energy generators
D wind turbine

- 19 Three solar panels have a total area of 7.2 m^2 . Together, they produce 1.0 kW of power.

An area of 1.0 m^2 of each panel receives 0.80 kJ of energy from the Sun in 1.0 s .

What is the efficiency of the solar panels?

- A** 11% **B** 17% **C** 52% **D** 80%

- 20 A student suggests three different definitions for power.

- 1 energy transferred per second
- 2 force $\times$ distance moved
- 3 rate of doing work

Which definitions are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 3 only

- 21 Which statement is correct?

- A** Infrared radiation cannot travel in a vacuum.
B Infrared radiation cannot travel in solids or in gases.
C Infrared radiation can only travel in a vacuum.
D Infrared radiation can travel in a vacuum and in gases.

- 22 How is *latent heat* defined?

- A** the thermal energy absorbed or emitted by unit mass during a change of state without a temperature change
B the thermal energy absorbed or emitted by unit mass during unit temperature change
C the thermal energy absorbed or emitted during a change of state without a temperature change
D the thermal energy absorbed or emitted during unit temperature change

- 23 In low-pressure hydrogen, there are no forces between the molecules except during collisions with each other.

Which statement describes the motion of a molecule between one collision and the next?

- A** It moves with changing speed and continuously changes direction.
B It moves with changing speed in a straight line.
C It moves with constant speed and continuously changes direction.
D It moves with constant speed in a straight line.

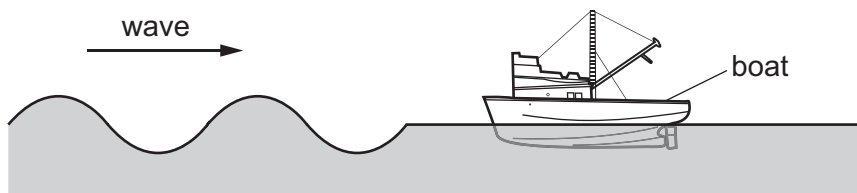
- 24** A gas is enclosed in a container of fixed volume.

The gas gains heat energy from an external source.

What happens to the molecules of the gas?

- A** They expand.
- B** They move faster inside the container.
- C** They move further apart.
- D** They vibrate with greater frequency.

- 25** A wave on water travels towards a stationary boat.

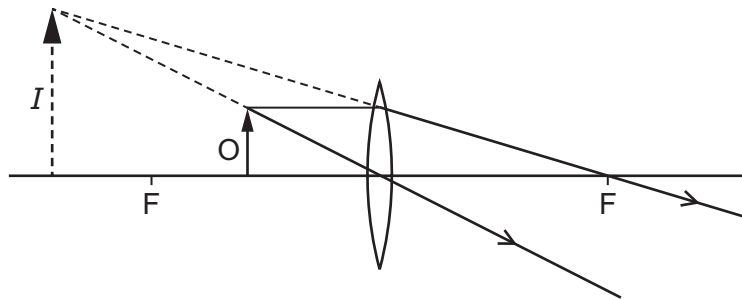


As the wave passes, the boat starts to move up and down and gains kinetic energy from the wave.

Which property of the wave changes as the wave passes the boat?

- A** amplitude
 - B** frequency
 - C** speed
 - D** wavelength
- 26** Which waves are longitudinal?
- A** gamma rays
 - B** light waves
 - C** ultrasound waves
 - D** X-rays
- 27** An object is placed at a distance from a converging lens that is equal to twice the focal length of the lens.
- Which statement about the image is correct?
- A** It is enlarged.
 - B** It is inverted.
 - C** It is on the same side of the lens as the object.
 - D** It is virtual.

- 28 The lens in the diagram produces an image I of the object O .



Why is this **not** the ray diagram for a photographic enlarger?

- A The image is magnified.
 - B The image is virtual.
 - C The lens is a converging lens.
 - D The lens is too thin.
- 29 What is the approximate range of audible frequencies for a healthy human ear?
- A 2.0 Hz to 2.0 kHz
 - B 20 Hz to 2.0 kHz
 - C 20 Hz to 20 kHz
 - D 200 Hz to 200 kHz
- 30 The speed of sound in air is c_a , the speed of sound in water is c_w and the speed of sound in ice is c_i .

Which relationship is correct?

- A $c_a > c_i$
- B $c_a > c_w$
- C $c_i > c_w$
- D $c_w = c_a$

- 31 One end of each of two bars is marked with an X or a Y, as shown.



bar 1



bar 2

End X of bar 1 is brought close to the N-pole and then to the S-pole of a magnet.

End Y of bar 2 is brought close to the N-pole and then to the S-pole of the magnet.

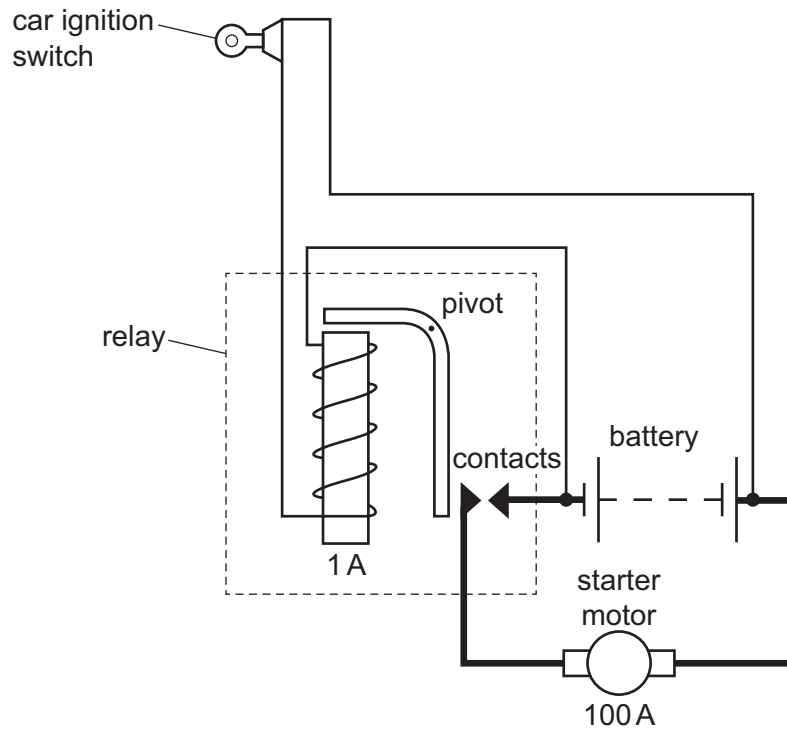
The table shows what happens.

	end X of bar 1	end Y of bar 2
N-pole of magnet	attracted	attracted
S-pole of magnet	attracted	repelled

What materials are bar 1 and bar 2 made from?

	bar 1	bar 2
A	copper	iron
B	iron	steel
C	steel	copper
D	steel	iron

32 The diagram shows a car ignition switch and starter motor.

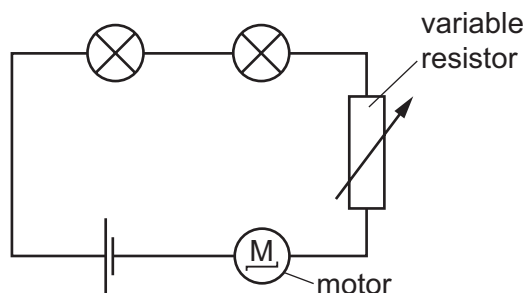


The ignition switch is in a circuit with long, thin wires. The starter motor is in a circuit with short, thick wires.

What is the explanation for the choice of wires?

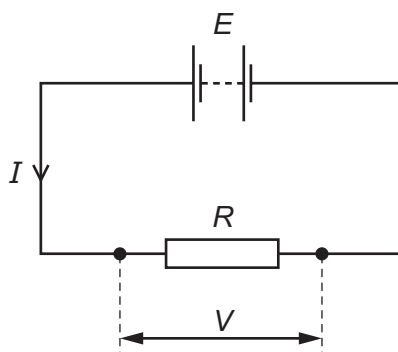
- A Each circuit needs to contain the same total mass of wire.
- B Thicker wires heat up more quickly when the relay is switched on.
- C Thin wires have lower resistances.
- D The ignition switch circuit carries a smaller current than the starter motor circuit.

- 33 The diagram shows a simple circuit.



What happens when the resistance of the variable resistor is increased?

- A The lamps are dimmer and the motor rotates more quickly.
 - B The lamps are dimmer and the motor rotates more slowly.
 - C The lamps have the same brightness and the motor rotates more quickly.
 - D The lamps have the same brightness and the motor rotates more slowly.
- 34 The diagram shows a simple electrical circuit.



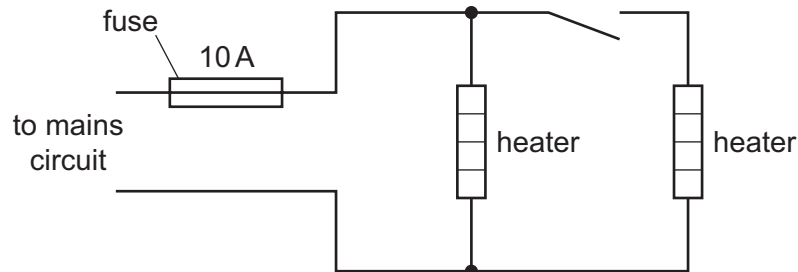
Which statement **cannot** be correct?

- A The current I is 2 A.
 - B The e.m.f. E is 12 J.
 - C The p.d. V (across the resistor) is 10 V.
 - D The resistance R is $5\ \Omega$.
- 35 The cable to an electric cooker contains a live wire, a neutral wire and an earth wire.
- When the cooker is working correctly, in which wires are the currents equal?
- A the live, the neutral and the earth
 - B the live and the earth only
 - C the neutral and the earth only
 - D the neutral and the live only

- 36** The manufacturer of an electrical heater states that only one heater must be connected to the mains wiring through any fuse.

The current in the circuit with only one heater is 9 A. The fuse rating is 10 A.

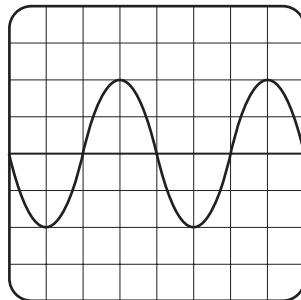
Despite the warning, a second identical heater is added, as shown. When the switch is closed the fuse blows.



The fuse is replaced with a 20 A fuse.

What happens when the switch is closed?

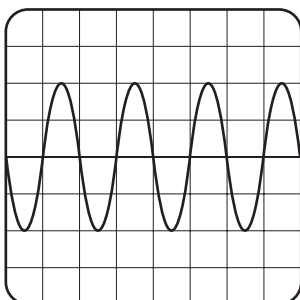
- A** Each heater only receives half the correct current.
 - B** Each heater receives twice the correct current.
 - C** The circuit wiring connecting the heaters to the mains may overheat.
 - D** The fuse melts when the heaters are switched on.
- 37** A coil is rotating in a magnetic field. The coil is connected to an oscilloscope. The diagram shows the trace on the screen of the oscilloscope.



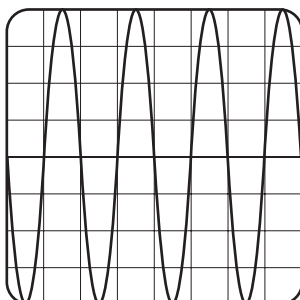
The coil now speeds up and rotates twice as fast.

Which diagram shows the new trace?

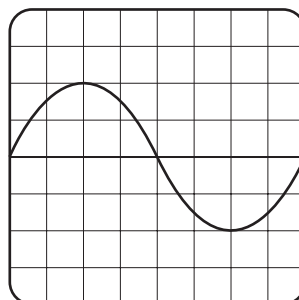
A



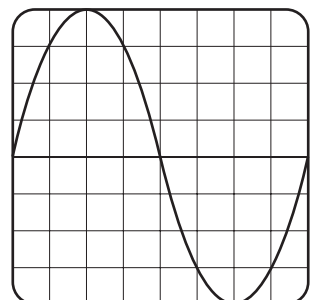
B



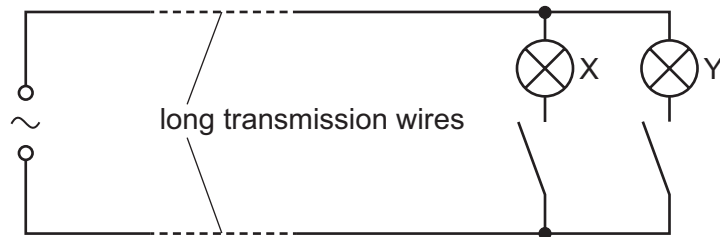
C



D



38 A teacher uses the circuit shown.



The identical lamps X and Y are connected to a low voltage a.c. power supply by high resistance transmission wires. Both lamps are switched on.

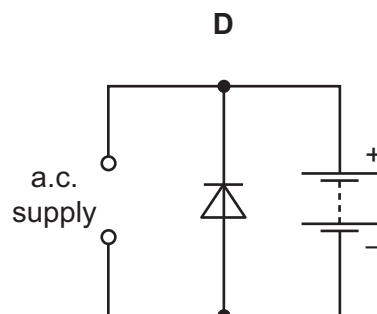
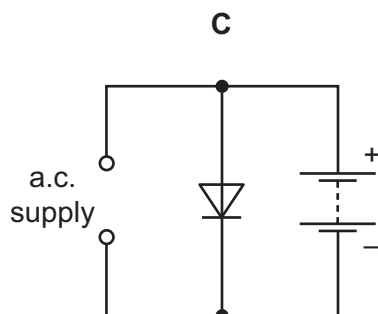
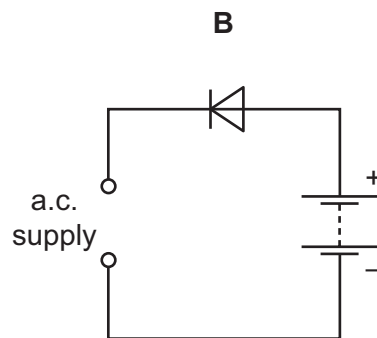
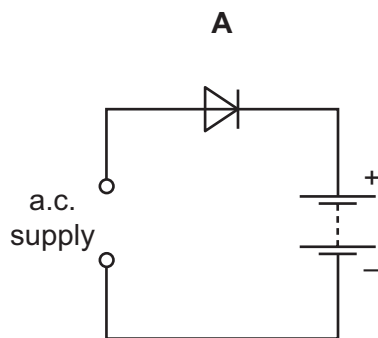
Lamp X is then switched off. Lamp Y stays switched on.

What happens to the voltage and the power supplied to lamp Y?

	the voltage supplied to lamp Y	the power supplied to lamp Y
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

39 A car battery supplies a current in one direction. A current in the opposite direction recharges the battery.

Which circuit recharges the battery, using an alternating current (a.c.) supply and a diode?



40 A proton has charge q and mass m .

What is the charge and what is the mass of an alpha-particle?

	charge	mass
A	q	$2m$
B	$2q$	$2m$
C	$2q$	$4m$
D	$4q$	$2m$

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PHYSICS

Paper 1 Multiple Choice

5054/11

May/June 2019

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



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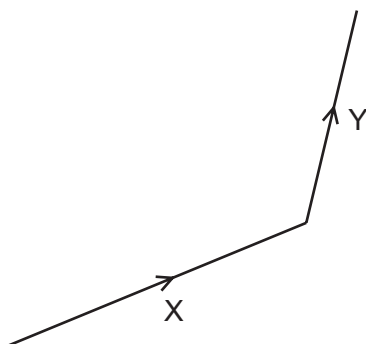
This document consists of **18** printed pages and **2** blank pages.



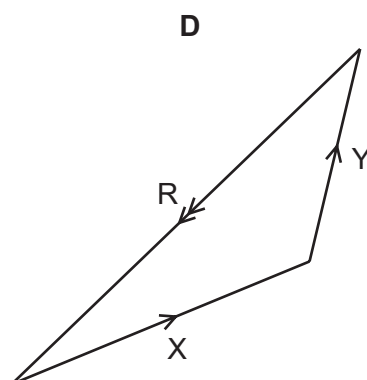
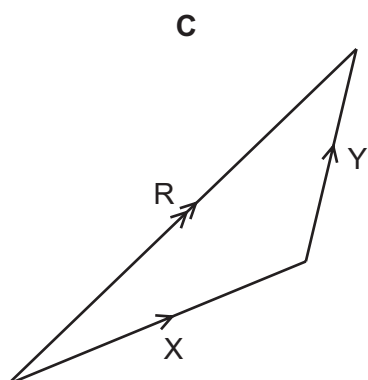
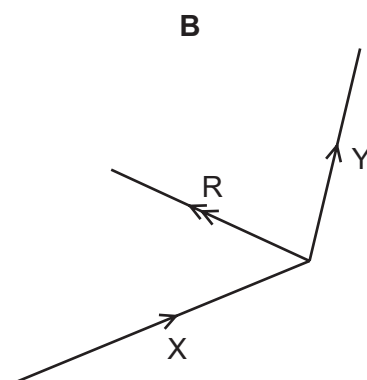
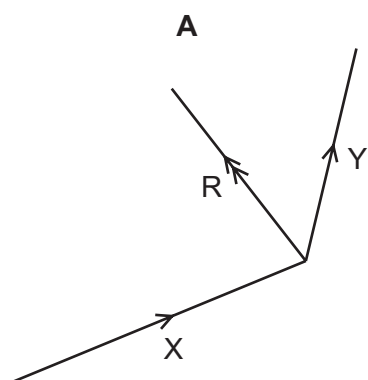
1 Which quantities are both vectors?

- A acceleration and length
- B distance and velocity
- C length and distance
- D velocity and acceleration

2 Two forces X and Y act as shown.



Which diagram shows the resultant force R of X and Y?



3 Which reading is given to one tenth of a millimetre?

- A 3.3 cm
- B 3.31 cm
- C 3.310 cm
- D 3.312 cm

- 4 The magnitudes of three different electric charges are given below.

What is the correct order of size, from largest to smallest?

- A $1 \text{ mC} \rightarrow 1 \text{ MC} \rightarrow 1 \text{ kC}$
- B $1 \text{ MC} \rightarrow 1 \text{ mC} \rightarrow 1 \text{ kC}$
- C $1 \text{ MC} \rightarrow 1 \text{ kC} \rightarrow 1 \text{ mC}$
- D $1 \text{ kC} \rightarrow 1 \text{ mC} \rightarrow 1 \text{ MC}$

- 5 Four of the gravitational forces that act between bodies in the Solar System are listed below.

P the force on the Moon due to the Earth

Q the force on the Earth due to the Sun

R the force on the Earth due to the Moon

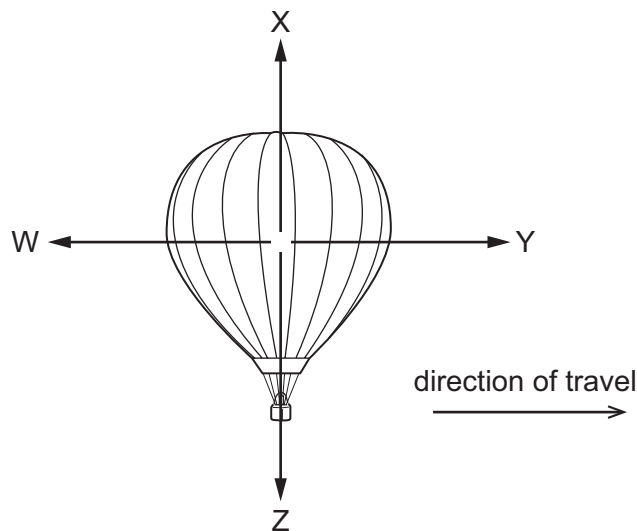
S the force on the Moon due to the Sun

Which two forces are a Newton's third law pair (action and reaction)?

- A P and Q
- B P and R
- C Q and S
- D R and S

- 6 A hot-air balloon is travelling at constant velocity and is at a constant height above the ground.

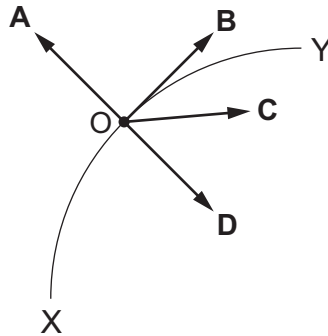
The diagram shows the only four forces acting on the balloon.



Which statement is correct?

- A W and Y are equal, X and Z are equal.
- B W and Y are equal, X is greater than Z.
- C X and Z are equal, Y is greater than W.
- D Y is greater than W, X is greater than Z.

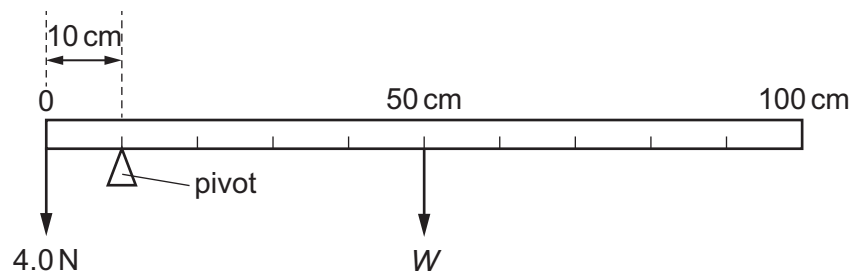
- 7 The diagram represents an object O moving from X to Y along a circular path at constant speed. What is the direction of the resultant force on O in the position shown?



- 8 Four objects of different masses are situated in places with different gravitational field strengths. Which object has the greatest weight?

	mass / kg	<u>gravitational field strength</u> N/kg
A	3.0	10.4
B	3.5	9.5
C	4.0	10.2
D	4.5	9.0

- 9 A uniform metre rule is balanced by a 4.0 N weight as shown.



What is the weight W of the metre rule?

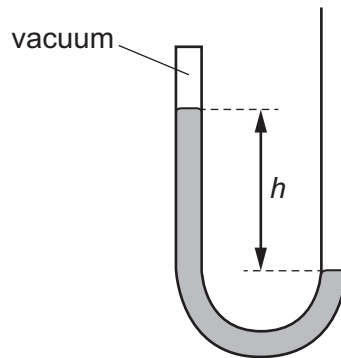
- A** 1.0 N **B** 4.0 N **C** 16 N **D** 40 N
- 10 Which statement about centre of mass is correct?
- A** Objects with a centre of mass at the same height are less stable when the base is larger.
- B** Objects with a centre of mass at the same height are more stable when the base is larger.
- C** Objects with higher centres of mass and smaller bases are more stable.
- D** Objects with identical bases are more stable when the centre of mass is higher.

- 11 A garden table weighs 60 N and has a top surface of area 2.0 m^2 . It is raining and the rain produces a pressure of 4.0 N/m^2 on the table.

Ignoring the pressure of the atmosphere, what is the force exerted by the table on the ground?

- A** 52 N **B** 58 N **C** 62 N **D** 68 N

- 12 The diagram shows a manometer containing mercury that is sealed at one end.



What happens to the distance h when the manometer is taken higher up a mountain?

- A** It decreases, because atmospheric pressure decreases with height.
B It decreases, because atmospheric pressure increases with height.
C It increases, because atmospheric pressure decreases with height.
D It increases, because atmospheric pressure increases with height.
- 13 Some doctors measure blood pressure by using a mercury manometer. Blood pressure varies by 5.6 kPa as a heart beats.

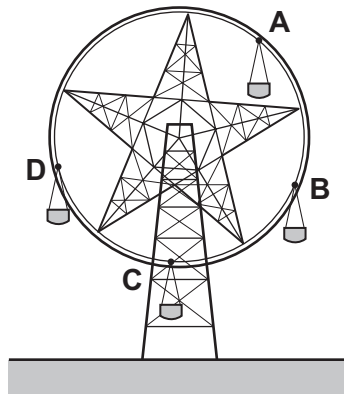
The density of mercury is $14\,000 \text{ kg/m}^3$ and the gravitational field strength g is 10 N/kg .

What is the change in the height difference between the levels in the manometer during a heartbeat?

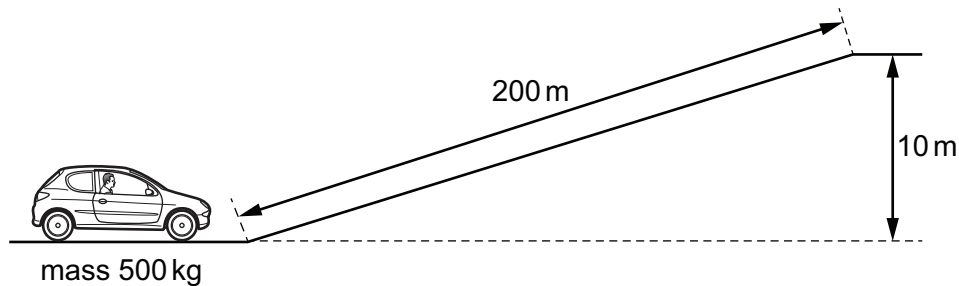
- A** 40 mm **B** 80 mm **C** 400 mm **D** 800 mm

- 14 The diagram shows a stationary fairground ride with four chairs of equal mass.

Which chair has the most gravitational potential energy?



- 15 The diagram shows a small car of mass 500 kg approaching a hill. It moves up the hill with constant speed.



The gravitational field strength g is 10 N/kg .

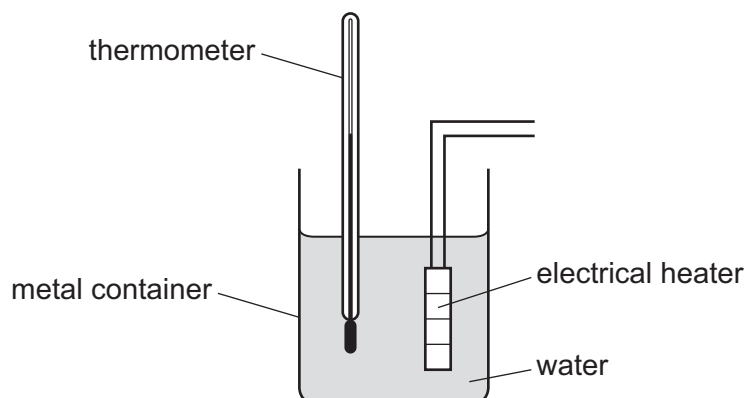
Ignoring friction, how much work is done in moving the car up the hill?

- A** $5.0 \times 10^3 \text{ J}$ **B** $5.0 \times 10^4 \text{ J}$ **C** $1.0 \times 10^5 \text{ J}$ **D** $1.0 \times 10^6 \text{ J}$

- 16 Which statement is correct?

- A** Infrared radiation cannot travel in a vacuum.
B Infrared radiation cannot travel in solids or in gases.
C Infrared radiation can only travel in a vacuum.
D Infrared radiation can travel in a vacuum and in gases.

- 17 The diagram shows a set of apparatus used to determine the specific heat capacity of water.



What does **not** affect the rate at which energy is lost to the surroundings?

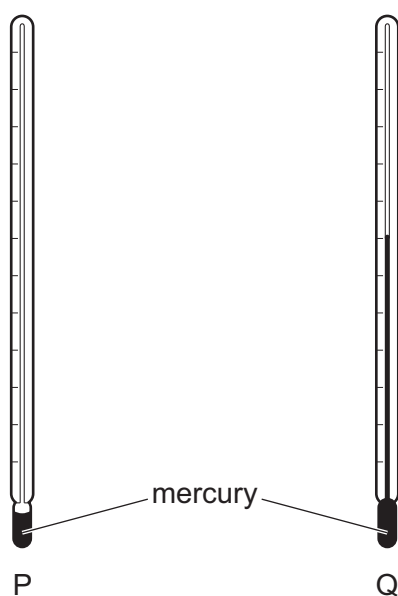
- A insulating the container
 - B placing a lid on the container
 - C polishing the outer surface of the container
 - D moving the thermometer closer to the heater
- 18 Two thermometers X and Y contain different liquids but are otherwise identical.

The thermometers are heated through the same temperature rise. The volume of liquid in thermometer X increases by more than the volume of liquid in thermometer Y.

Which is the more sensitive thermometer and which is the thermometer with the larger range?

	more sensitive thermometer	thermometer with the larger range
A	X	X
B	X	Y
C	Y	X
D	Y	Y

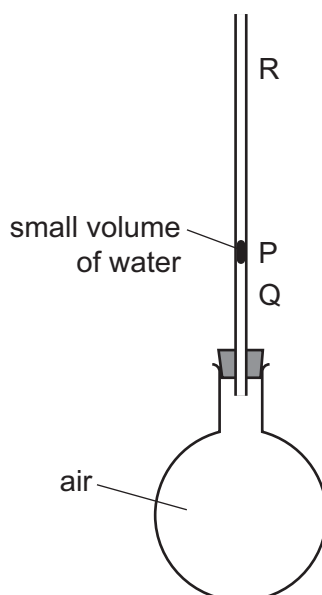
- 19 The diagram shows two thermometers P and Q that are identical except that P contains less mercury.



Which statement is correct?

- A The lowest measureable temperature is the same on P and on Q.
- B The range of P is equal to the range of Q.
- C The scales on P and Q are both linear.
- D The sensitivity of P is equal to the sensitivity of Q.

- 20 The diagram shows air trapped in a flask by a small volume of water in a thin tube.



When the flask is held in a student's hands, the small volume of water first moves down from P to Q, and then up to R.

Why does the small volume of water move like this?

- A The flask contracts and then the air expands.
 - B The flask expands and then the air contracts.
 - C The flask expands and then the air expands less than the flask.
 - D The flask expands and then the air expands more than the flask.
- 21 Which statement about particles in a liquid is correct?
- A They all have the same speed.
 - B They are stationary.
 - C They move at random.
 - D They vibrate about a fixed point.

- 22 A gas is enclosed in a container of fixed volume.

The gas gains heat energy from an external source.

What happens to the molecules of the gas?

- A They expand.
- B They move faster inside the container.
- C They move further apart.
- D They vibrate with greater frequency.

23 Which waves are longitudinal?

- A** gamma rays
- B** light waves
- C** ultrasound waves
- D** X-rays

24 A ray of light strikes a plane mirror at an angle of incidence of 20° .

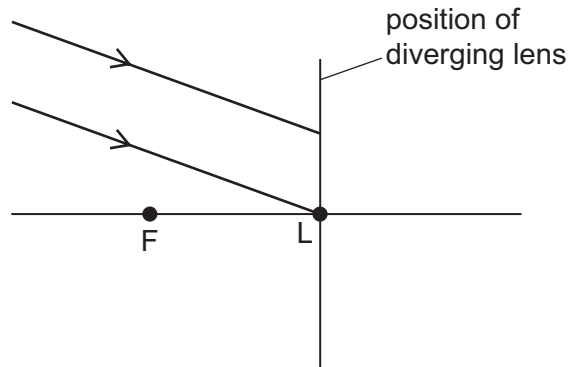
The angle of incidence is then increased by 5° .

What is the new angle between the incident ray and the reflected ray?

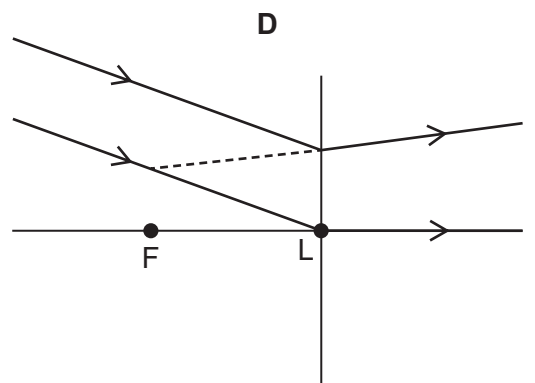
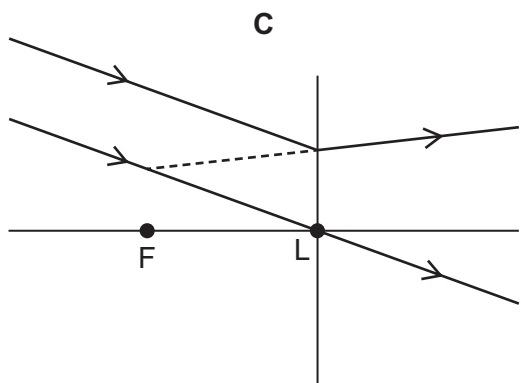
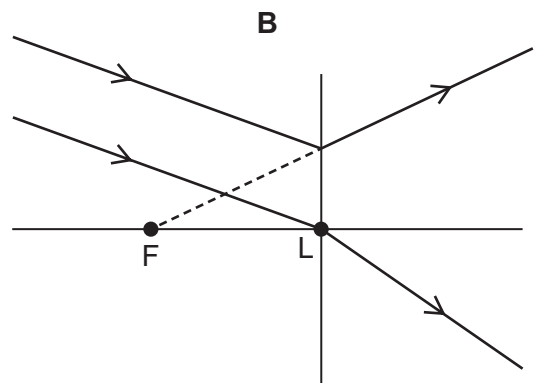
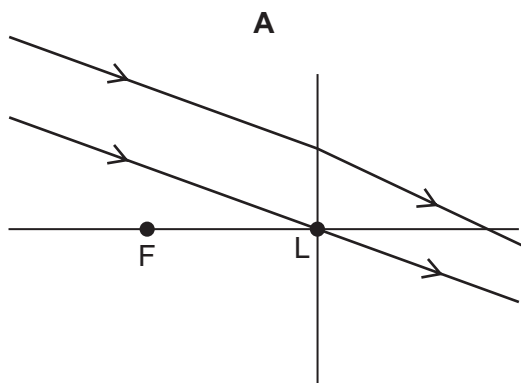
- A** 10° **B** 25° **C** 45° **D** 50°

- 25** A parallel beam of light is incident on a thin diverging lens.

The focal length of the lens is FL , as shown in the diagram.



Which ray diagram shows the beam after it has passed through the lens?



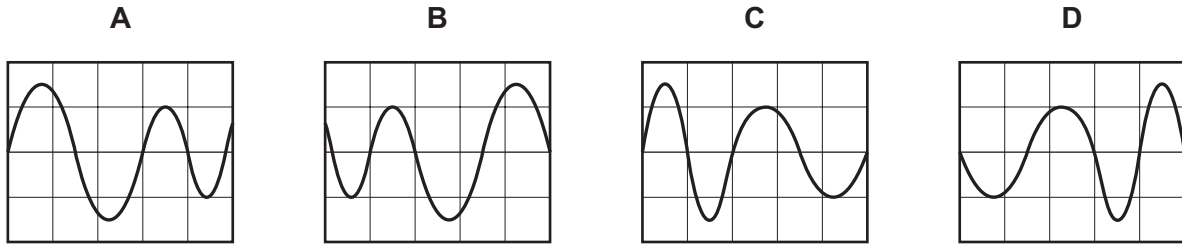
- 26** The following lists show colours of the spectrum.

Which list shows these colours in order of increasing frequency?

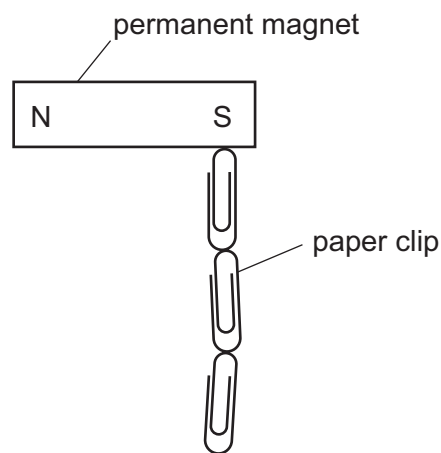
- A** blue, violet, red, orange, yellow, green
- B** green, blue, violet, red, orange, yellow
- C** red, orange, violet, yellow, green, blue
- D** red, orange, yellow, green, blue, violet

27 Four sound waves are displayed on the screen of a cathode-ray oscilloscope.

Which sound wave gets louder and has a pitch that decreases?



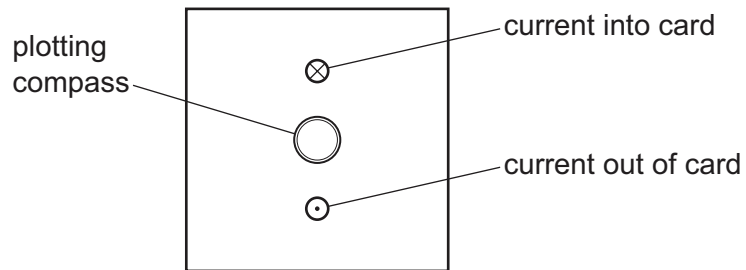
28 A student investigates a permanent magnet by suspending paper clips from the magnet as shown.



Which statement is correct?

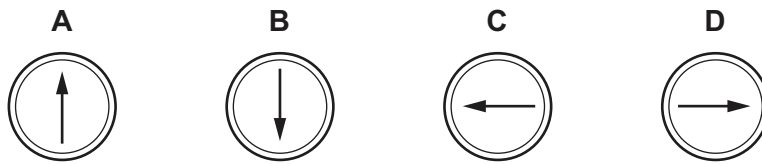
- A Each paper clip has a S-pole at the top and a N-pole at the bottom when suspended from the permanent magnet.
- B If the paper clips are made of soft iron they become demagnetised when the permanent magnet is removed.
- C If the paper clips are made of steel they become demagnetised when the permanent magnet is removed.
- D More paper clips can be suspended from the N-pole.

- 29 Two vertical wires pass at right-angles through a piece of card. There is a large current in each wire in the direction shown.

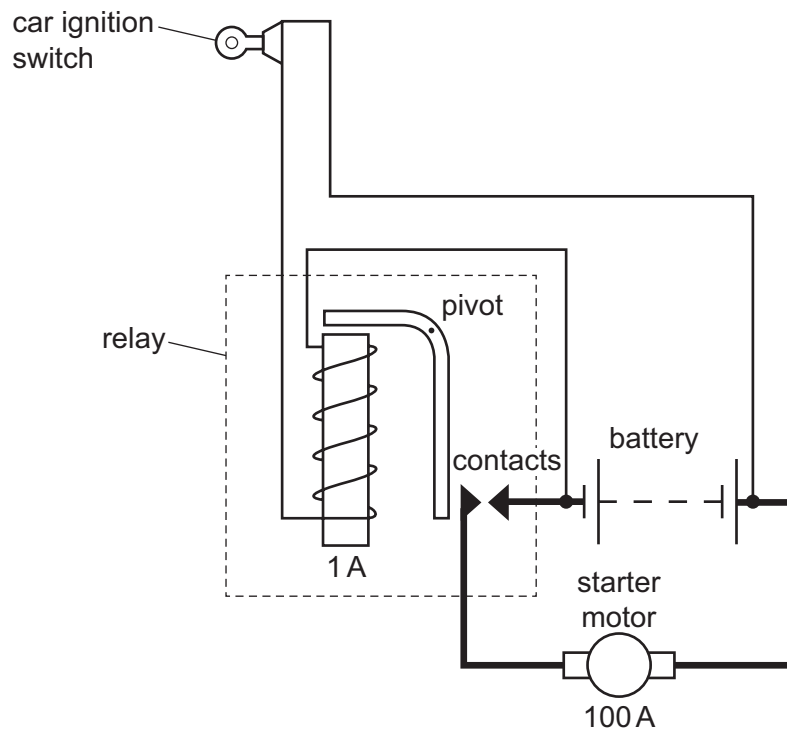


A plotting compass is placed on the card.

Which diagram shows the direction in which the needle of the plotting compass points?



30 The diagram shows a car ignition switch and starter motor.

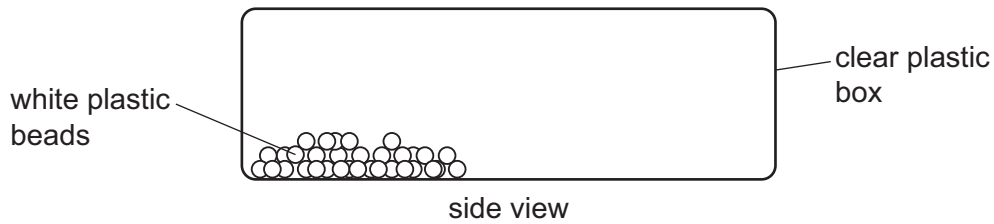


The ignition switch is in a circuit with long, thin wires. The starter motor is in a circuit with short, thick wires.

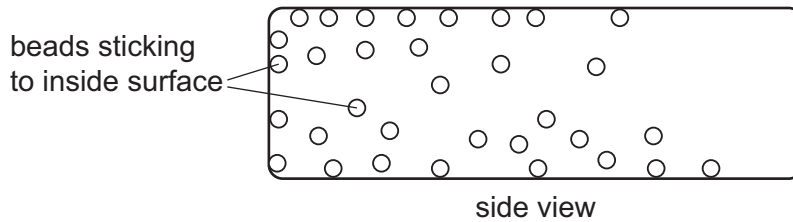
What is the explanation for the choice of wires?

- A Each circuit needs to contain the same total mass of wire.
- B Thicker wires heat up more quickly when the relay is switched on.
- C Thin wires have lower resistances.
- D The ignition switch circuit carries a smaller current than the starter motor circuit.

31 The diagram shows some white plastic beads in a clear plastic box.



The box is shaken and the beads rub against the box. The beads stick to the inside surface of the box.



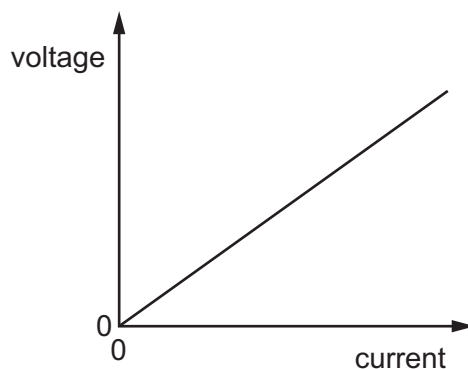
Which row is a possible explanation for this?

	box	beads
A	gains electrons	no change of electrons
B	loses electrons	gain electrons
C	no change of electrons	lose electrons
D	no change of electrons	gain electrons

32 Which unit is the same as a volt?

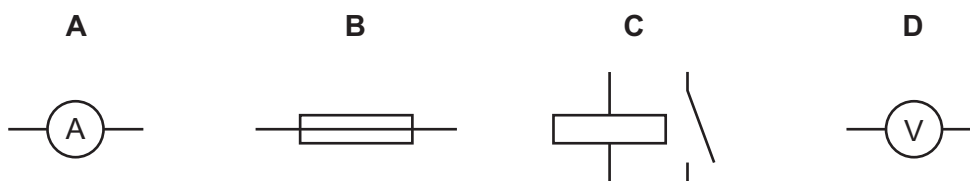
- A** ampere/ohm
- B** ampere/watt
- C** ohm/ampere
- D** watt/ampere

- 33 The voltage/current graph for a metal wire is shown.



What does the gradient of this graph represent?

- A the charge passing through the wire
 - B the e.m.f. of the battery connected to the wire
 - C the energy produced in the wire
 - D the resistance of the wire
- 34 What is the symbol for a device that measures current?



- 35 The cable to an electric cooker contains a live wire, a neutral wire and an earth wire.

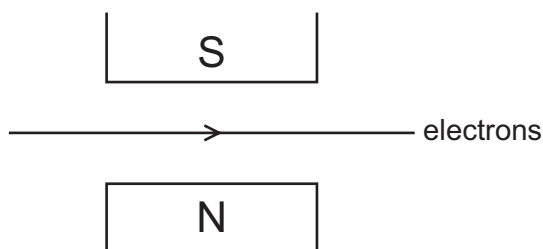
When the cooker is working correctly, in which wires are the currents equal?

- A the live, the neutral and the earth
 - B the live and the earth only
 - C the neutral and the earth only
 - D the neutral and the live only
- 36 The coil of a simple motor lies between the poles of a permanent magnet. The coil rotates about its axis when there is a current in it.

What decreases the frequency of rotation of the coil?

- A increasing the number of turns in the coil
- B reversing the current
- C using a lower voltage supply
- D using a stronger magnet

- 37 A beam of electrons travels through a vacuum. The beam passes between the poles of a magnet as shown.



What is the direction of the conventional current and what is the direction of the magnetic field?

	direction of the conventional current	direction of the magnetic field
A	→	↓
B	→	↑
C	←	↓
D	←	↑

- 38 A teacher uses the circuit shown.



The identical lamps X and Y are connected to a low voltage a.c. power supply by high resistance transmission wires. Both lamps are switched on.

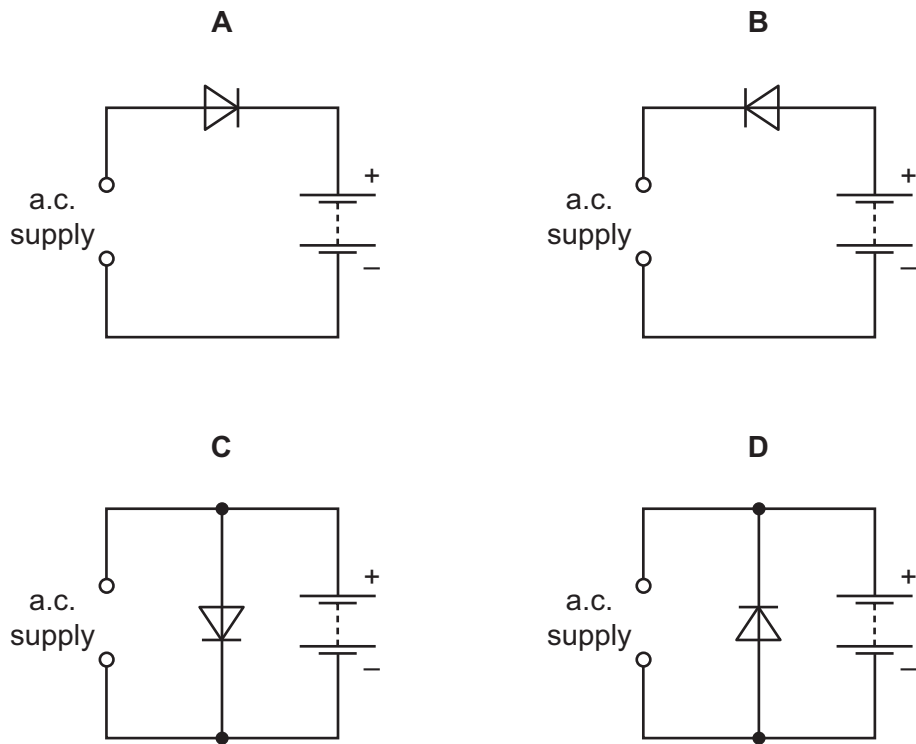
Lamp X is then switched off. Lamp Y stays switched on.

What happens to the voltage and the power supplied to lamp Y?

	the voltage supplied to lamp Y	the power supplied to lamp Y
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

- 39** A car battery supplies a current in one direction. A current in the opposite direction recharges the battery.

Which circuit recharges the battery, using an alternating current (a.c.) supply and a diode?



- 40** A neutral atom of boron contains 5 electrons and 5 protons.

The nucleon number (mass number) of the atom is 11.

How many neutrons are there in the atom?

- A** 1 **B** 5 **C** 6 **D** 11

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PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2019

MARK SCHEME

Maximum Mark: 40

Published

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This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	A	1
3	D	1
4	D	1
5	B	1
6	D	1
7	C	1
8	A	1
9	C	1
10	A	1
11	B	1
12	A	1
13	B	1
14	A	1
15	C	1
16	D	1
17	B	1
18	B	1
19	B	1
20	C	1
21	D	1
22	C	1
23	D	1
24	B	1
25	A	1
26	C	1
27	B	1
28	B	1

Question	Answer	Marks
29	C	1
30	C	1
31	B	1
32	D	1
33	B	1
34	B	1
35	D	1
36	C	1
37	B	1
38	C	1
39	A	1
40	C	1



PHYSICS

5054/11

Paper 1 Multiple Choice

May/June 2019

MARK SCHEME

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Published

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Question	Answer	Marks
1	D	1
2	C	1
3	B	1
4	C	1
5	B	1
6	A	1
7	D	1
8	C	1
9	A	1
10	B	1
11	D	1
12	A	1
13	A	1
14	A	1
15	B	1
16	D	1
17	D	1
18	B	1
19	C	1
20	D	1
21	C	1
22	B	1
23	C	1
24	D	1
25	C	1
26	D	1
27	B	1
28	B	1

Question	Answer	Marks
29	C	1
30	D	1
31	B	1
32	D	1
33	D	1
34	A	1
35	D	1
36	C	1
37	D	1
38	C	1
39	A	1
40	C	1



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/12

May/June 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

This document consists of **15** printed pages and **1** blank page.

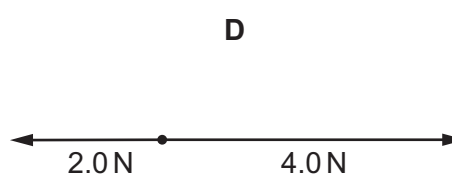
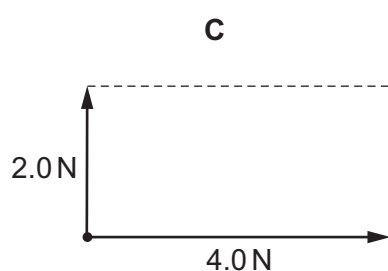
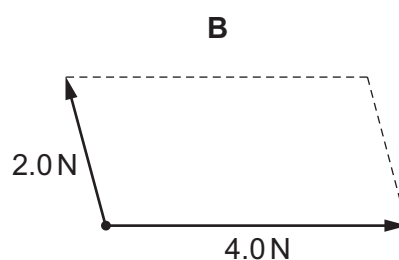
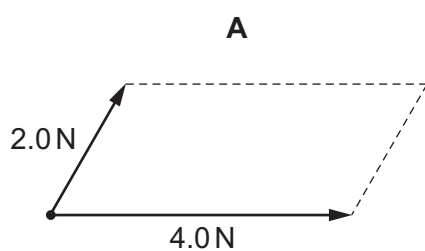


1 Which is a vector quantity?

- A a mass of 2.0 kg
- B a temperature of -10°C
- C a weight of 15 N
- D an average speed of 20 m/s

2 Forces of 4.0 N and 2.0 N act at a point.

Which scale diagram shows the forces that have a resultant of 4.0 N?



3 What is the name and value of the unit of power written as mW?

	name	value
A	megawatt	10^{-3} W
B	megawatt	10^6 W
C	milliwatt	10^{-3} W
D	milliwatt	10^6 W

- 4 Micrometers, metre rules, tapes and calipers are used for measuring lengths.

Which row identifies the most suitable device for accurately measuring the stated length?

	length	measuring device
A	0.15 mm	micrometer
B	0.50 mm	metre rule
C	0.15 m	tape
D	0.50 m	calipers

- 5 The planets in the Solar System orbit the Sun.

Which statement is correct?

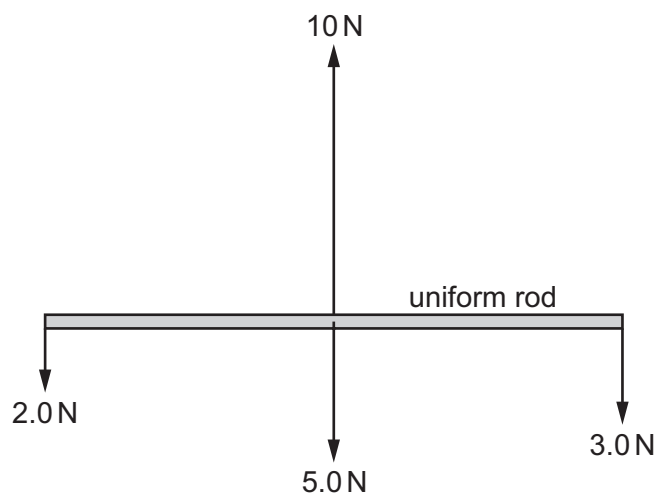
- A** There is a force on each planet away from the Sun.
- B** There is a force on each planet in the direction in which it travels.
- C** There is a force on each planet opposite to the direction in which it travels.
- D** There is a force on each planet towards the Sun.

- 6 Which forces act on a skydiver who is falling at terminal velocity?

- A** air resistance and weight
- B** air resistance only
- C** weight only
- D** no forces act

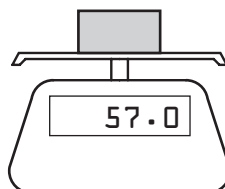
- 7 A uniform rod of weight 5.0 N is held initially at rest.

The diagram shows the forces acting on the rod when it is released.



What happens to the rod when it is released?

- A It does not move.
 - B It moves to the right.
 - C It moves upwards.
 - D It starts to rotate.
- 8 A block of metal is placed on an electronic balance to record its mass.



What is the unit of the reading on the electronic balance and what is the unit of weight?

	unit on electronic balance	unit of weight
A	g	g
B	g	N
C	N	g
D	N	N

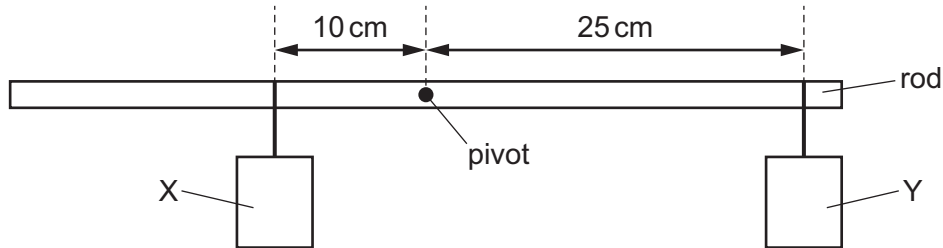
- 9 A body of mass 10 kg falling freely in the gravitational field close to the Moon's surface has an acceleration of 1.6 m/s^2 .

What is the gravitational field strength on the Moon?

- A 0 N/kg B 1.6 N/kg C 10 N/kg D 16 N/kg

- 10 Two objects X and Y are suspended from a uniform rod, pivoted at its centre.

The rod is in equilibrium.



Which statement about X and Y is correct?

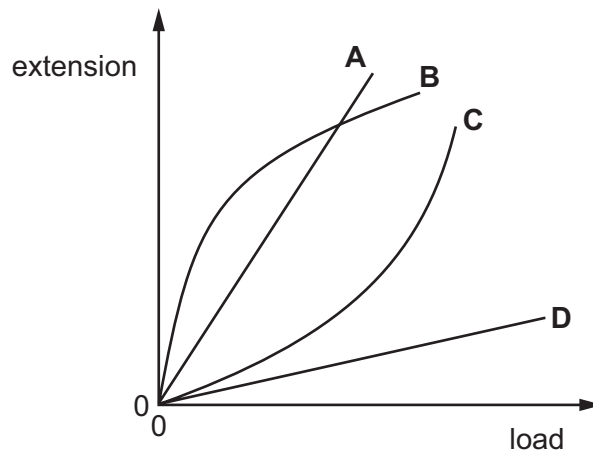
- A The mass of X is 0.4 times the mass of Y.
 B The mass of X is 2.5 times the mass of Y.
 C The mass of X is 3.5 times the mass of Y.
 D The mass of X is equal to the mass of Y.
- 11 A force of 1600 N accelerates a car, of mass 800 kg, from rest.

What is the car's acceleration and its velocity after 4.0 s?

	<u>acceleration</u> m/s^2	<u>velocity</u> m/s
A	0.50	0.13
B	0.50	0.50
C	2.0	2.0
D	2.0	8.0

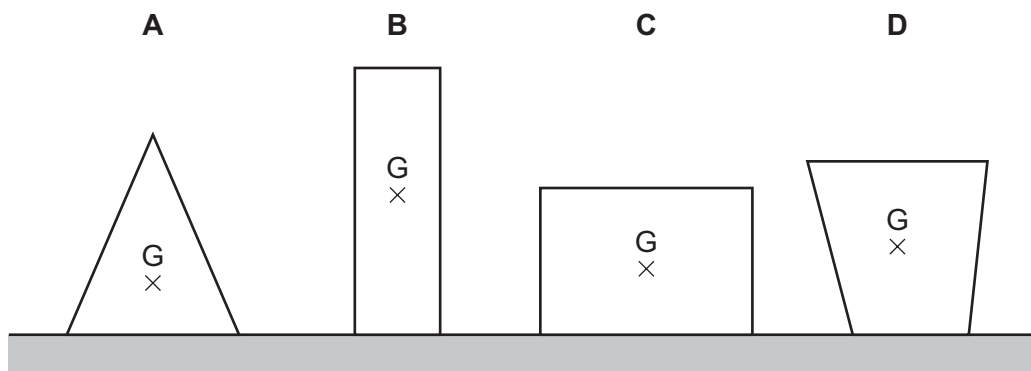
12 The graph shows extension-load curves for four fibres.

Which fibre is the most difficult to stretch over the range of loads shown?



13 Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

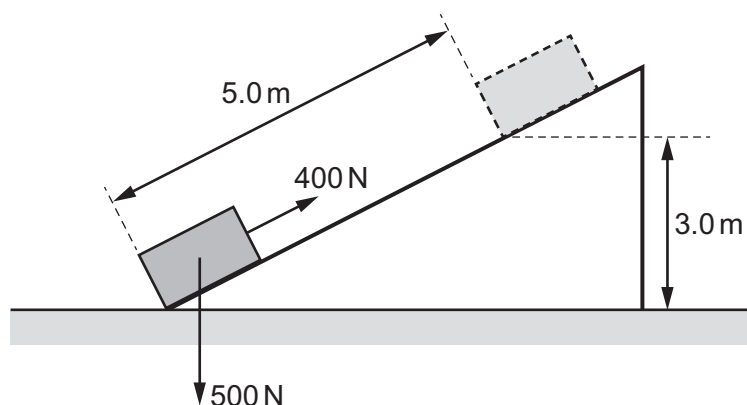
Which object is the least stable?



14 Which power station produces carbon dioxide when operating?

- A** gas-fired power station
- B** geothermal power station
- C** nuclear power station
- D** wind power station

- 15** Work is done when a force of 400 N pulls a crate of weight 500 N at a constant speed along a ramp, as shown.

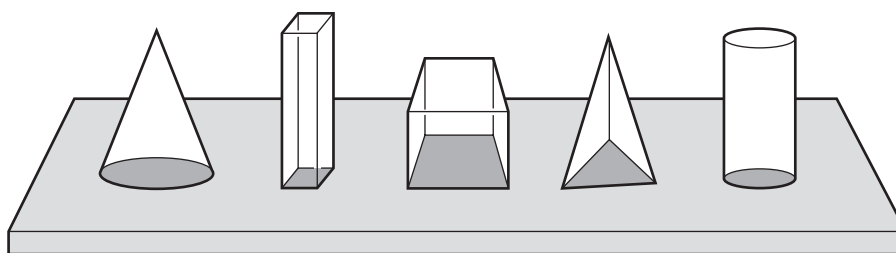


Part of the work done increases the gravitational potential energy E of the crate and the rest is work done W against friction.

What are the values of E and W ?

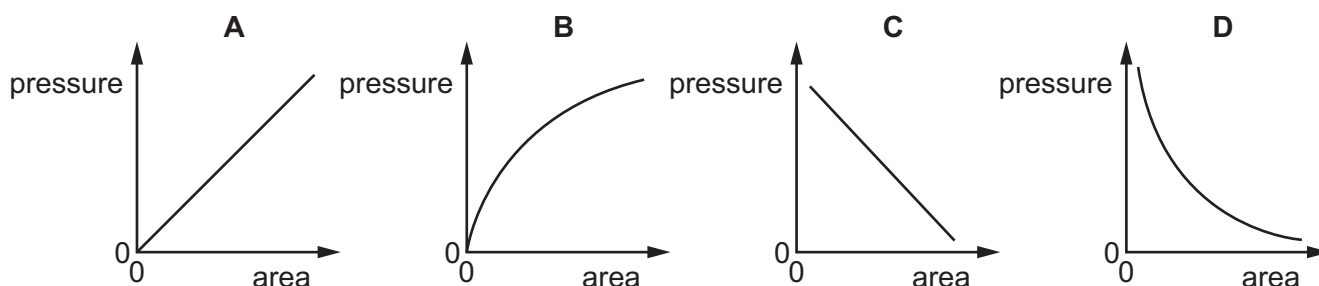
	E/J	W/J
A	1500	500
B	1500	2000
C	2000	2500
D	3500	500

- 16** Five blocks have the same mass but different base areas. They all rest on a horizontal table.



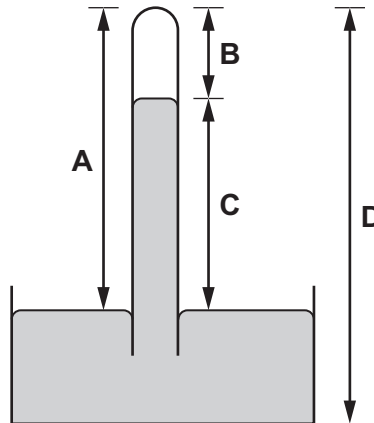
A graph is plotted to show the relationship between the pressure exerted on the table and the base area of the block.

Which graph shows this relationship?



- 17 The diagram shows a simple mercury barometer.

Which height is a measure of the atmospheric pressure?



- 18 A sealed packet containing air and a snack is purchased at an airport. The sealed packet is taken on board an aircraft. During the flight the packet becomes larger.

What causes the packet to become larger?

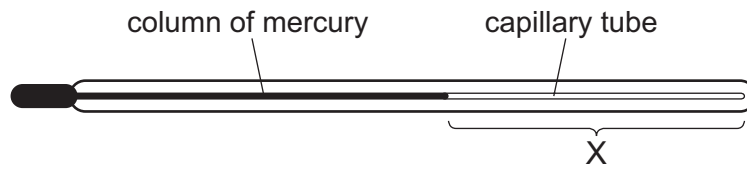
- A The density of the air inside the packet increases.
 - B The mass of the packet increases.
 - C The pressure of the air outside the packet decreases.
 - D The volume of the air inside the packet decreases.
- 19 A sealed container of gas is heated and the pressure inside increases.
- What happens to the molecules of the gas to cause this increase in pressure?
- A Their kinetic energy decreases.
 - B They become heavier.
 - C They expand.
 - D They hit the container more frequently.
- 20 Equal masses of copper and water are heated to the same temperature. As they cool down, the copper and the water lose thermal energy at the same rate.

The temperature of the copper falls faster.

Why is this?

- A Copper has a larger specific heat capacity.
- B Copper has a larger specific latent heat.
- C Copper has a smaller specific heat capacity.
- D Copper has a smaller specific latent heat.

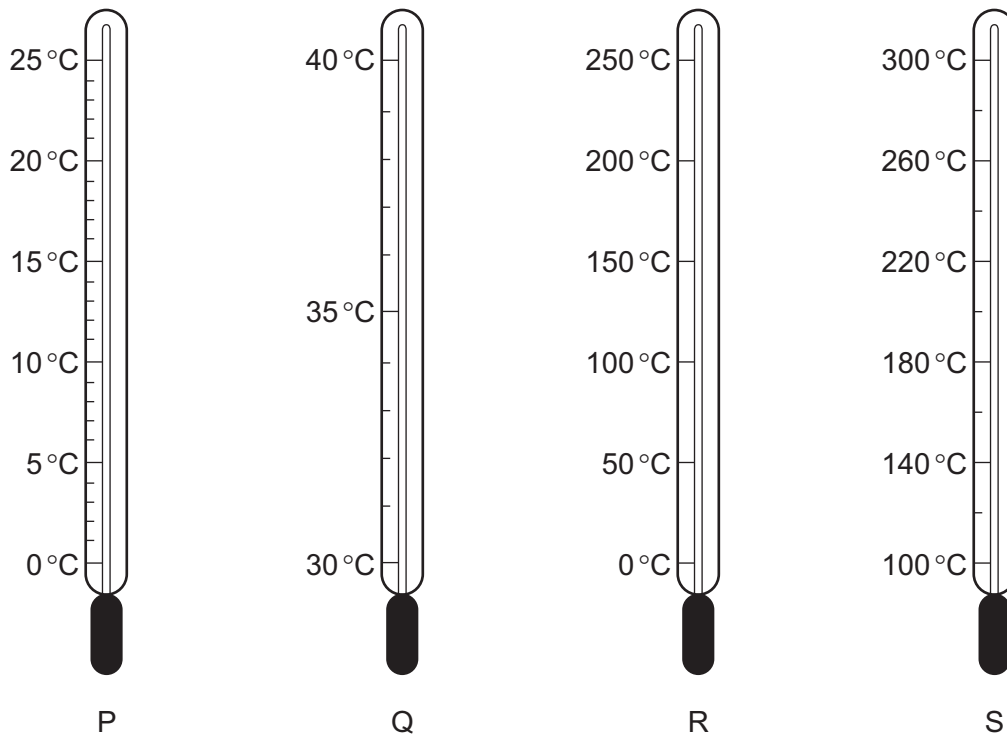
21 The diagram shows a thermometer.



What is in region X of the capillary tube?

- A** a vacuum
- B** alcohol
- C** steam
- D** water

22 The diagrams represent four thermometers.



Which thermometer has the greatest sensitivity and which thermometer has the greatest range?

	greatest sensitivity	greatest range
A	P	R
B	P	S
C	Q	R
D	Q	S

- 23 The resistance R of a wire increases uniformly with temperature. The values of R at the fixed points are shown in the table.

	0 °C	100 °C
R/Ω	100	250

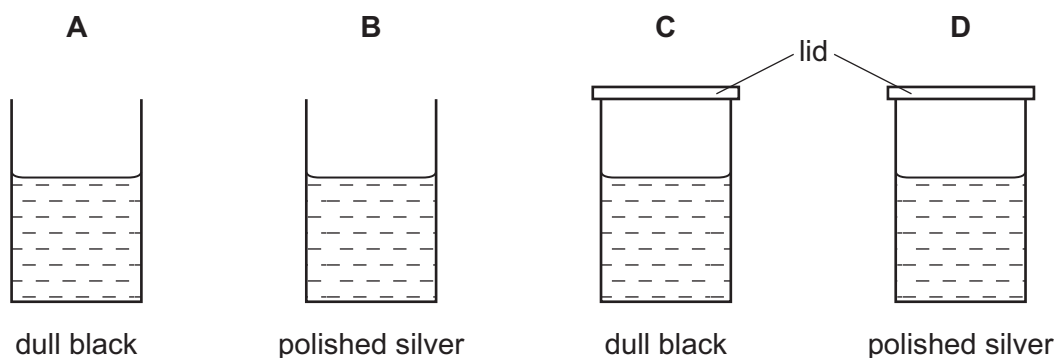
What is the temperature when $R = 160\Omega$?

- A 24 °C B 40 °C C 60 °C D 64 °C
- 24 Samples of four different materials at room temperature are heated from below and heat is transferred upwards.

In which material is all of the heat transferred by the vibration of molecules?

- A air
B mercury
C rubber
D water
- 25 The diagrams show four identical cans with their outside surfaces either polished silver or painted dull black. Each can contains the same volume of water, initially at 80 °C.

After five minutes in a cool room, which can contains the coolest water?



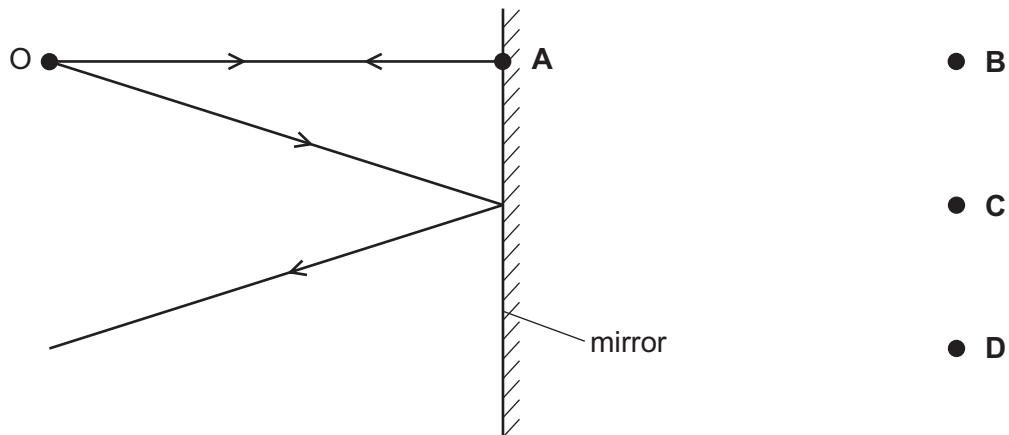
- 26 A communications system uses microwaves with a wavelength of 15 cm. The speed of light is 3.0×10^8 m/s.

What is the frequency of the microwaves?

- A 2.0×10^7 Hz B 4.5×10^7 Hz C 2.0×10^9 Hz D 4.5×10^9 Hz

- 27 The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

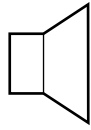
At which position is the image formed?



- 28 Which statement about blue light is correct?
- A Blue light has a smaller frequency than red light.
 - B Blue light has a longer wavelength than red light.
 - C Blue light has the same speed in glass as red light.
 - D Blue light is refracted more by a glass prism than red light.
- 29 Which device uses total internal reflection?
- A magnifying glass
 - B optical fibre
 - C photographic enlarger
 - D projector
- 30 A television controller emits an infra-red beam.
- Which statement about infra-red radiation is correct?
- A It causes ionisation.
 - B It consists of longitudinal waves.
 - C It has a higher frequency than ultra-violet light.
 - D It travels at the speed of light.

- 31 The diagram shows a loudspeaker that is producing a continuous sound wave of frequency 200 Hz in air.

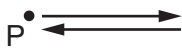
loudspeaker



P

Which diagram best shows how the sound wave causes a molecule at P to move during $\frac{1}{200}$ s ?

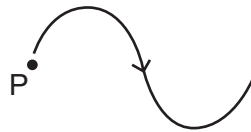
A



B



C



D



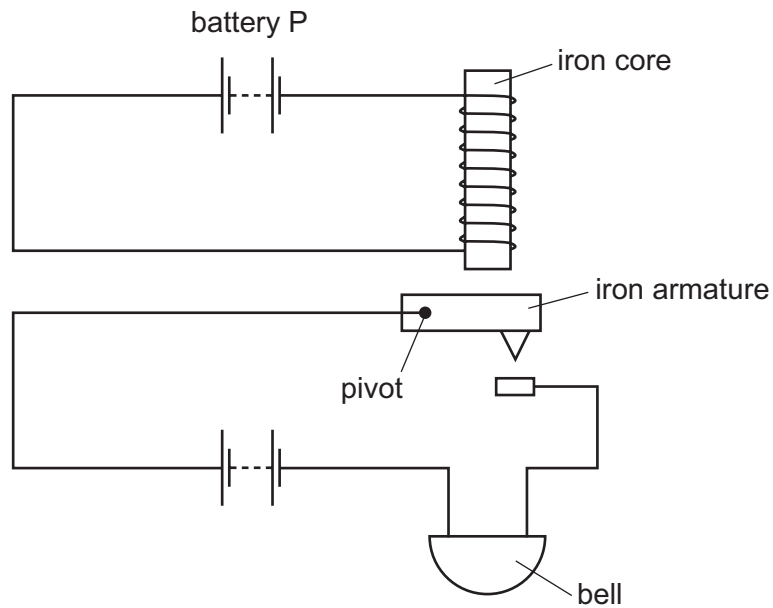
- 32 Which frequency is in the range heard by a healthy human ear?

A 12 mHz B 12 Hz C 12 kHz D 12 MHz

- 33 What is one of the uses of ultrasound?

A cleaning jewellery
 B satellite communication
 C fluorescent tubes
 D optical fibres

34 The diagram shows an alarm system.



What happens when battery P is disconnected?

	iron armature	bell
A	falls	rings
B	falls	stops ringing
C	moves up	rings
D	moves up	stops ringing

35 V is a potential difference, I is a current, R is a resistance, and t is a time.

Which expression has units of energy?

- A** It **B** I^2R **C** VIt **D** $\frac{V^2}{R}$

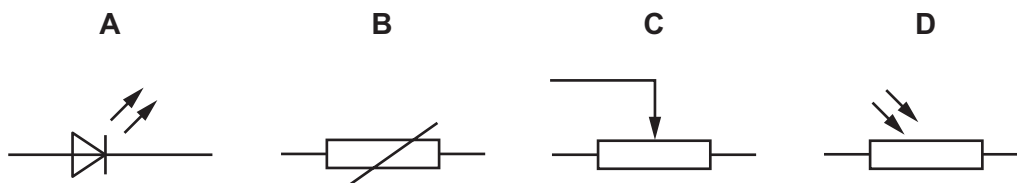
36 A step-down transformer has a primary coil and a secondary coil wound on a soft-iron core.

The primary coil is connected to a 6.0 V direct current (d.c.) supply.

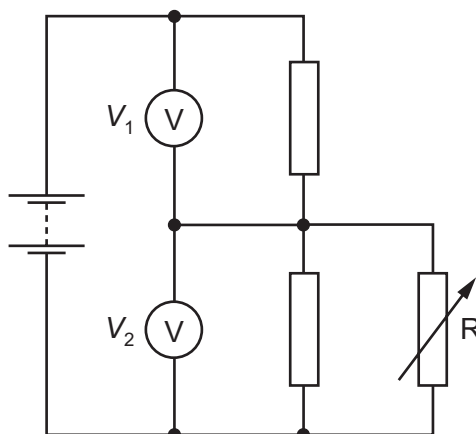
Which statement about the transformer is correct?

- A** The output voltage is equal to 6.0 V.
B The output voltage is greater than 6.0 V.
C The output voltage is less than 6.0 V but more than zero.
D There is no output voltage.

37 Which symbol is used for a light-dependent resistor?



38 The circuit diagram shows a variable resistor R connected in parallel to the lower half of a potential divider.

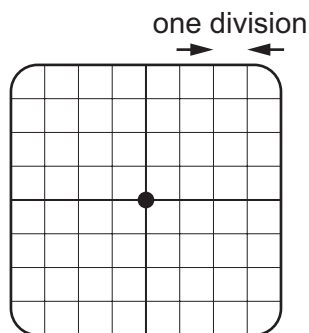


The resistance of R increases.

What happens to the two voltmeter readings?

	V_1	V_2
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

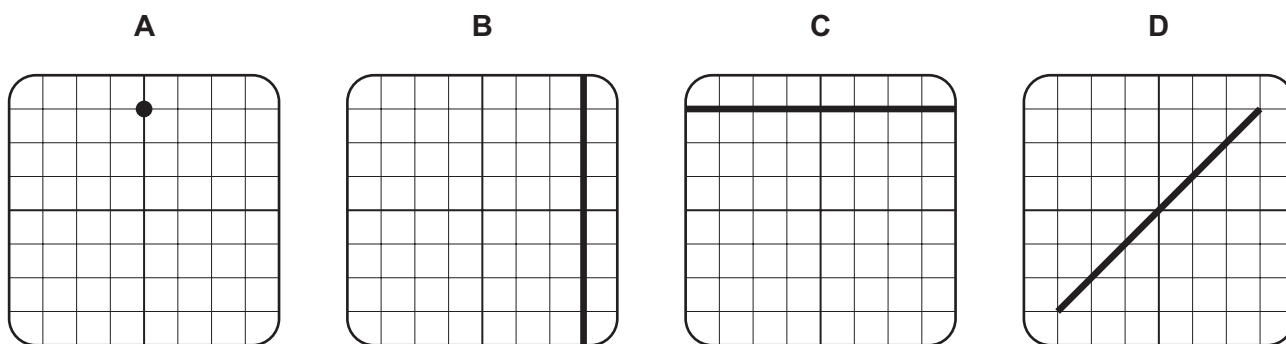
- 39 An oscilloscope is used to measure potential difference (p.d.). The trace with no input connected is shown.



A 1.5 V d.c. supply is connected to the oscilloscope.

The Y-gain is set at 0.5 V/div. The time-base is set at 0.5 ms/div.

Which trace shows a supply of 1.5 V d.c.?



- 40 How many nucleons are in one neutral atom of the krypton isotope $^{84}_{36}\text{Kr}$?

A 36 B 48 C 84 D 120

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/11

May/June 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

This document consists of **19** printed pages and **1** blank page.

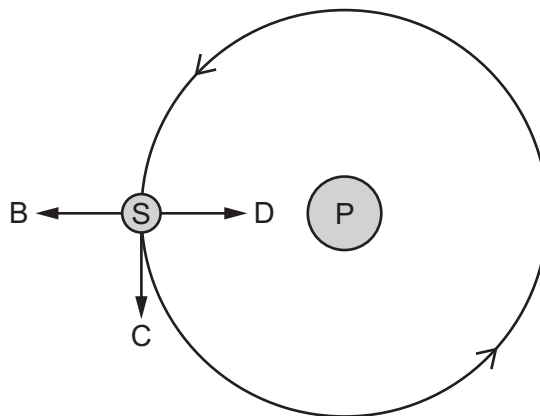


- 1 Which statement about electromagnetic waves is correct?
- A All electromagnetic waves have speeds in air of approximately 3×10^8 m/s.
 - B In air, some electromagnetic waves travel faster than light.
 - C The electromagnetic waves with the largest wavelength are in the infra-red region.
 - D The electromagnetic waves with the smallest wavelength are in the X-ray region.

- 2 A student wishes to measure directly the circumference of a football.

Which is the most suitable instrument to use?

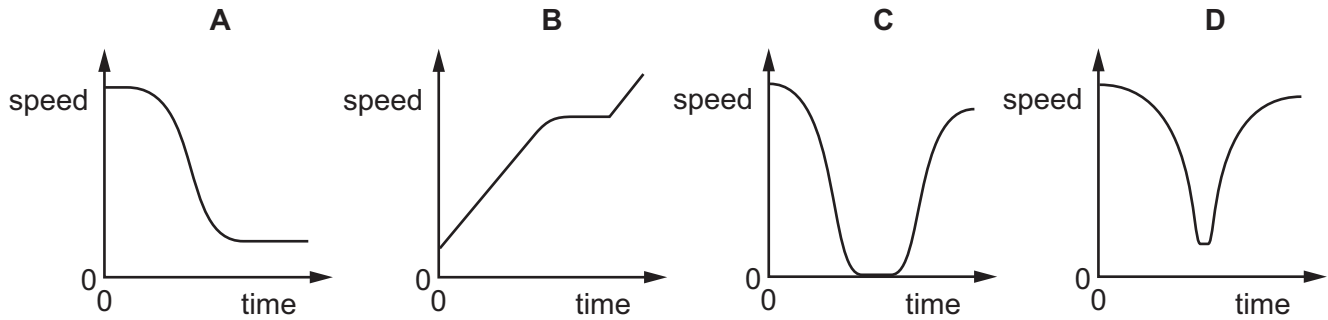
- A calipers
 - B a measuring tape
 - C a micrometer
 - D a ruler
- 3 The diagram shows a satellite S travelling at a constant speed in a circular orbit around a planet P.



Which statement is correct?

- A The resultant force on the satellite is zero.
- B The resultant force on the satellite is in direction B.
- C The resultant force on the satellite is in direction C.
- D The resultant force on the satellite is in direction D.

- 4 Which speed-time graph represents the motion of a railway train making a short stop at a station?



- 5 The minimum braking distance for a car is tested on a dry road.

The test is then repeated on a wet road.

What happens to the braking distance and to the frictional force between the tyres and the road?

	braking distance	frictional force
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 6 Which piece of equipment is used to measure mass?

- A** balance
- B** manometer
- C** measuring cylinder
- D** newton meter

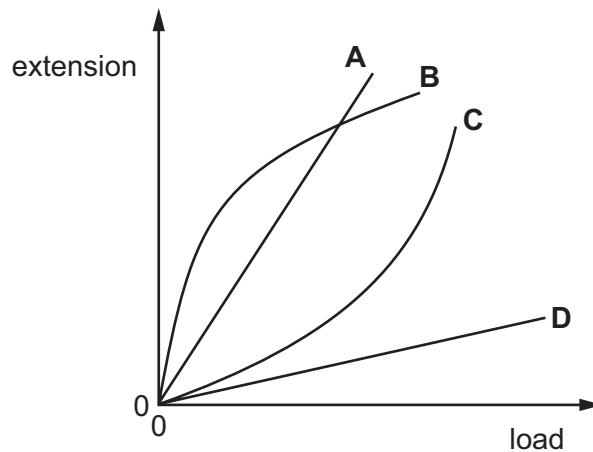
- 7 A body of mass 10 kg falling freely in the gravitational field close to the Moon's surface has an acceleration of 1.6 m/s^2 .

What is the gravitational field strength on the Moon?

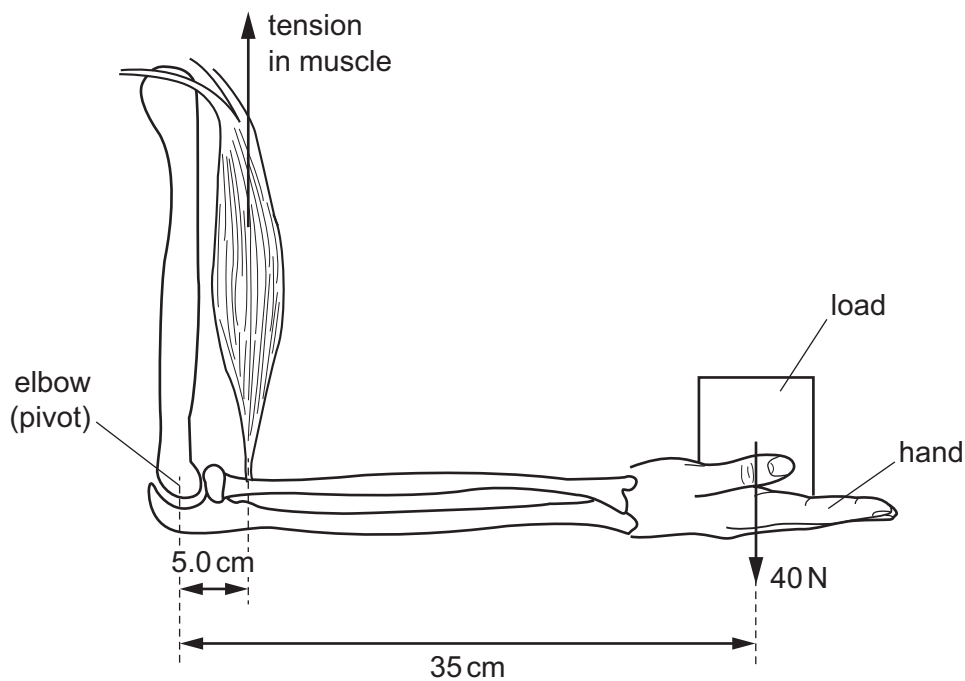
- A** 0 N/kg
- B** 1.6 N/kg
- C** 10 N/kg
- D** 16 N/kg

- 8 The graph shows extension-load curves for four fibres.

Which fibre is the most difficult to stretch over the range of loads shown?



- 9 The diagram shows a muscle and bones in a person's arm. The hand holds a load of weight 40 N. The elbow acts as a pivot and the tension in the muscle keeps the lower part of the arm horizontal.

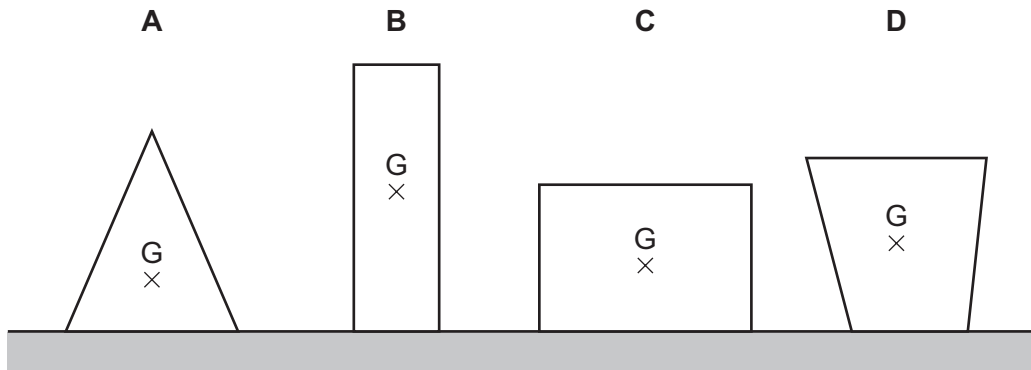


What is the tension in the muscle due to the load?

- A** 200 N **B** 240 N **C** 280 N **D** 1400 N

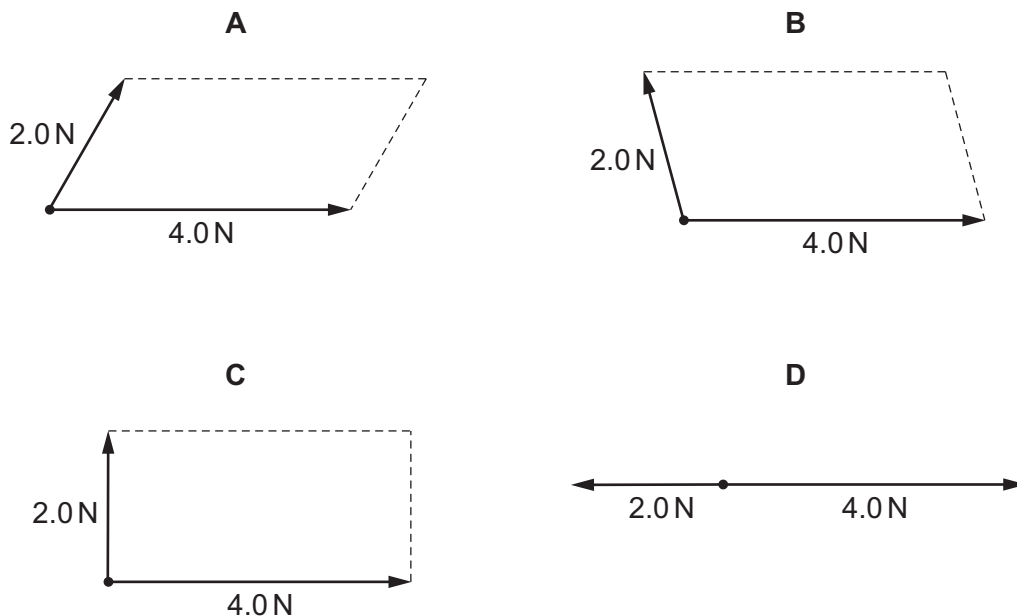
- 10 Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

Which object is the least stable?



- 11 Forces of 4.0 N and 2.0 N act at a point.

Which scale diagram shows the forces that have a resultant of 4.0 N?



- 12 A student calculates his power in running up a flight of stairs. He measures the vertical height of the stairs, the time taken to run up the stairs and his weight.

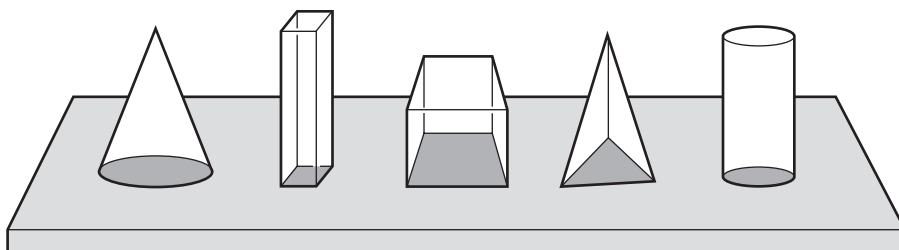
How does he calculate his power?

- A height $\times$ time $\times$ weight
 B $\frac{\text{height} \times \text{weight}}{\text{time}}$
 C $\frac{\text{time} \times \text{weight}}{\text{height}}$
 D $\frac{\text{weight}}{\text{height} \times \text{time}}$

13 How does an oil-fired power station differ from a nuclear power station?

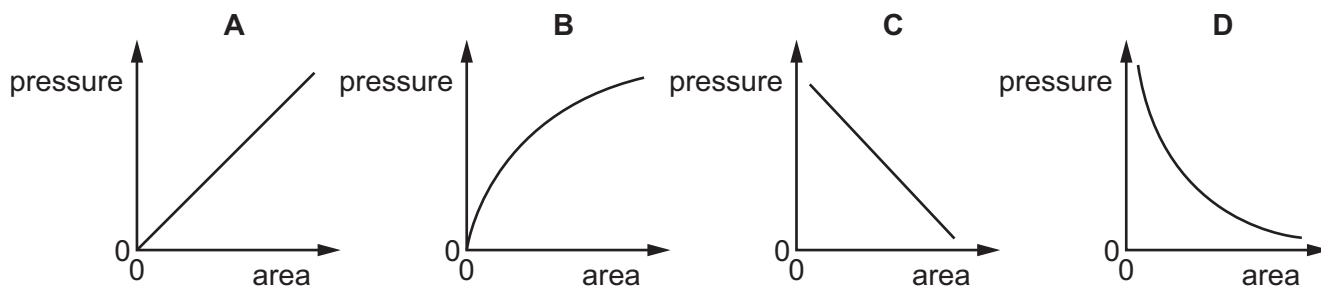
- A** Gases emitted by hot fuel are emitted into the atmosphere.
- B** Steam is produced in a boiler using hot fuel.
- C** The hot steam is used to turn a turbine.
- D** Turbines are used to drive an electric generator.

14 Five blocks have the same mass but different base areas. They all rest on a horizontal table.



A graph is plotted to show the relationship between the pressure exerted on the table and the base area of the block.

Which graph shows this relationship?



15 Each tyre of a car has an area of 100 cm^2 in contact with the ground.

The car has a mass of 1600 kg . The weight of the car is equally distributed amongst the four tyres.

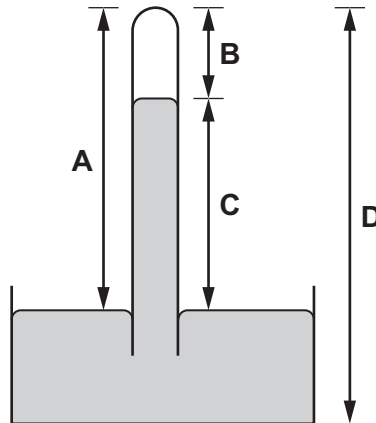
The gravitational field strength g is 10 N/kg .

What is the pressure exerted on the ground?

- A** 4.0 N/cm^2
- B** 16 N/cm^2
- C** 40 N/cm^2
- D** 160 N/cm^2

16 The diagram shows a simple mercury barometer.

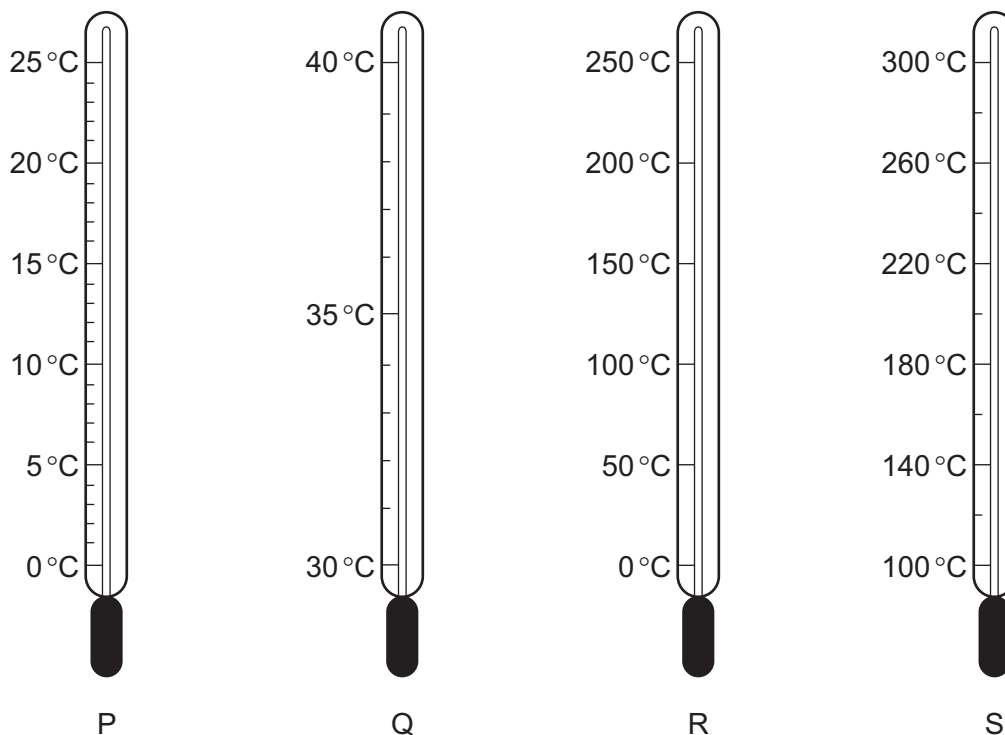
Which height is a measure of the atmospheric pressure?



17 What is described as ‘the escape of more energetic molecules from the surface of a liquid’?

- A** condensation
- B** evaporation
- C** melting
- D** radiation

18 The diagrams represent four thermometers.



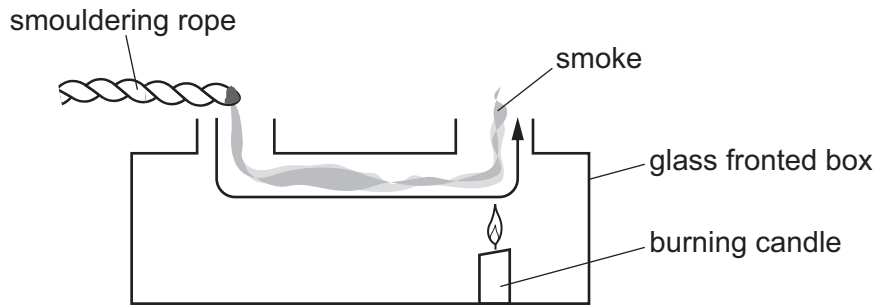
Which thermometer has the greatest sensitivity and which thermometer has the greatest range?

	greatest sensitivity	greatest range
A	P	R
B	P	S
C	Q	R
D	Q	S

19 What happens when a solid is heated and expands?

- A** The molecules do not change size and the spaces between the molecules become larger.
- B** The molecules expand and the spaces between the molecules become larger.
- C** The molecules expand and the spaces between the molecules become smaller.
- D** The molecules expand and the spaces between the molecules do not change size.

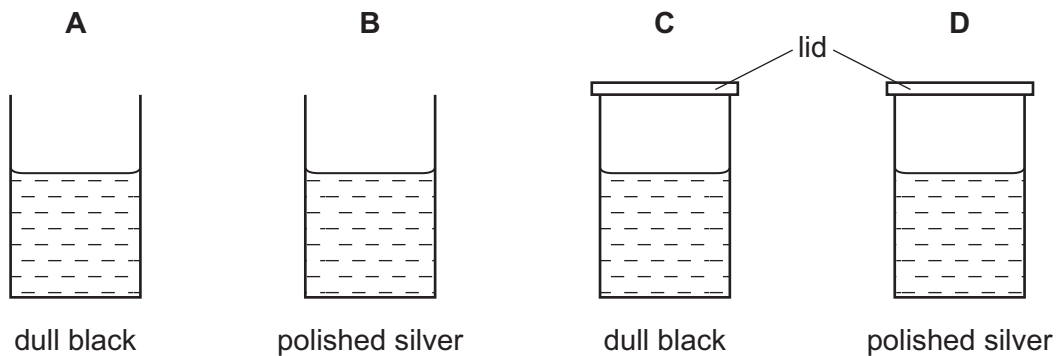
- 20** When a piece of smouldering rope is held at the opening of the box in the diagram, smoke moves in the direction indicated.



What is responsible for the movement of the smoke?

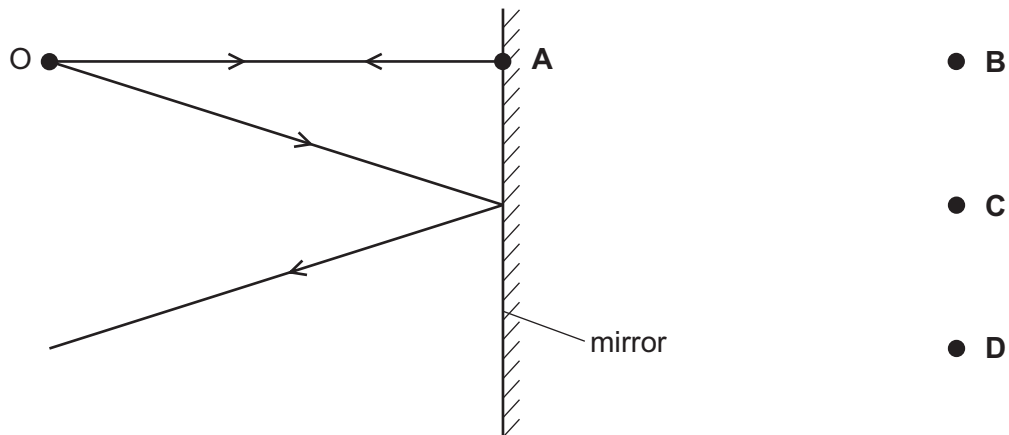
- A** convection
 - B** movement of free electrons
 - C** radiation
 - D** vibration of molecules
- 21** The diagrams show four identical cans with their outside surfaces either polished silver or painted dull black. Each can contains the same volume of water, initially at 80 °C.

After five minutes in a cool room, which can contains the coolest water?



- 22 The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

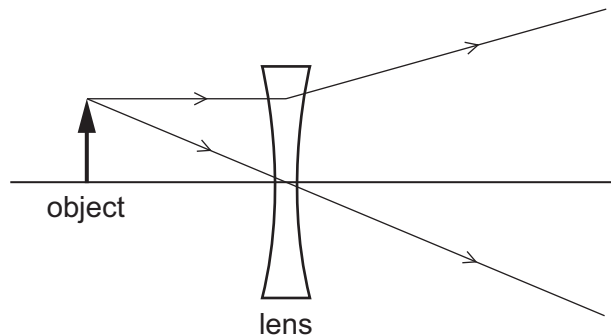
At which position is the image formed?



- 23 Which statement is correct?

- A Total internal reflection only occurs when light travels from air into glass.
- B The larger the refractive index of glass, the larger is the critical angle.
- C When total internal reflection occurs, the angle of incidence is equal to the angle of reflection.
- D When total internal reflection occurs, the angle of incidence is less than the critical angle.

- 24 The diagram shows the paths of two rays from the top of an object through a lens.



The object is viewed from the opposite side of the lens to the object.

How does the image compare with the object?

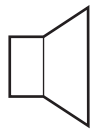
- A larger and inverted
- B larger and the same way up
- C smaller and inverted
- D smaller and the same way up

- 25** A young person with healthy ears can hear a range of frequencies.

What is the approximate range of frequencies?

- A** 2 Hz to 2000 Hz only
 - B** 2 Hz to 20 000 Hz only
 - C** 20 Hz to 2000 Hz only
 - D** 20 Hz to 20 000 Hz only
- 26** The diagram shows a loudspeaker that is producing a continuous sound wave of frequency 200 Hz in air.

loudspeaker



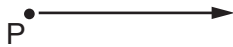
P

Which diagram best shows how the sound wave causes a molecule at P to move during $\frac{1}{200}$ s?

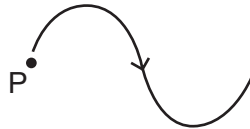
A



B



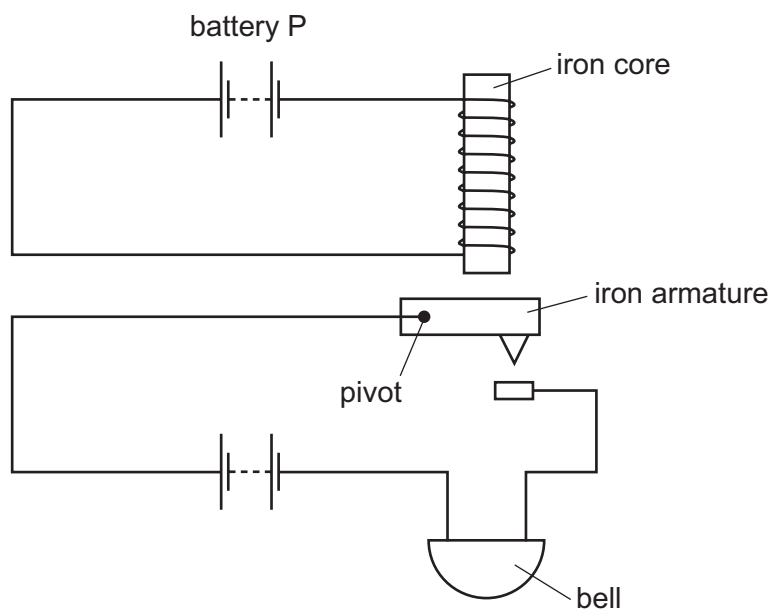
C



D



27 The diagram shows an alarm system.

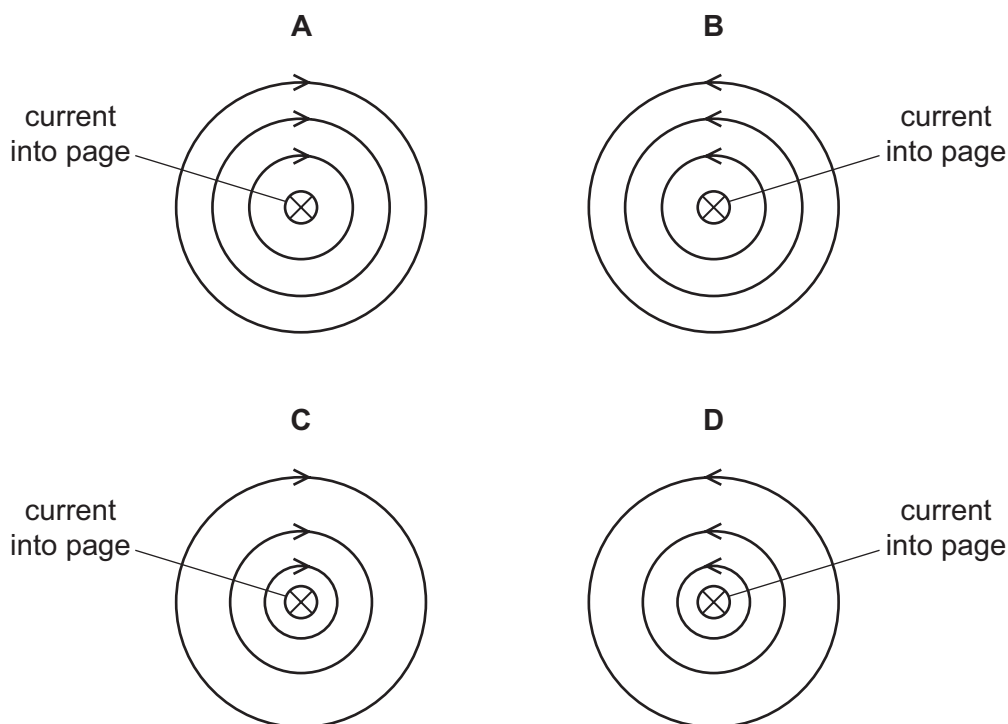


What happens when battery P is disconnected?

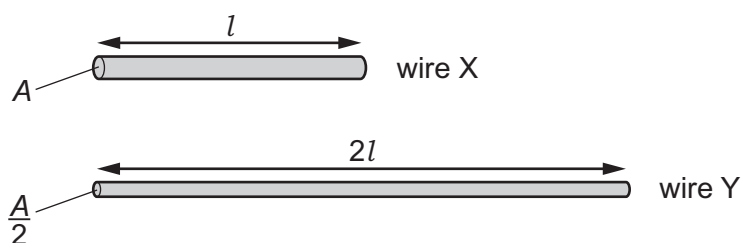
	iron armature	bell
A	falls	rings
B	falls	stops ringing
C	moves up	rings
D	moves up	stops ringing

- 28 An electric current in a wire is into the page.

Which diagram shows the shape and direction of the magnetic field around the wire?



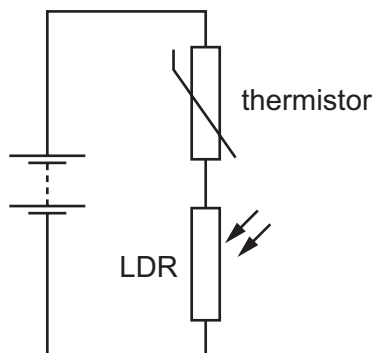
- 29 A copper wire X has resistance R . Another copper wire Y has twice the length and half the cross-sectional area of X.



What is the resistance of Y?

- A $\frac{R}{2}$ B R C $2R$ D $4R$

- 30** In the circuit shown, the temperature of the room and the amount of light affect the current.



Under which conditions is the current in the circuit the largest?

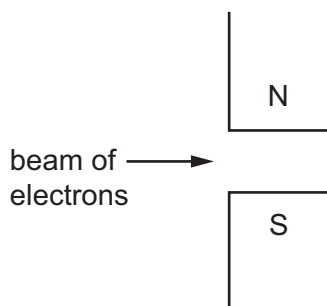
	temperature	amount of light
A	high	in bright light
B	high	in the dark
C	low	in bright light
D	low	in the dark

- 31** An electrical appliance is plugged into a socket in the wall. The plug contains a fuse.

What is the main purpose of the fuse?

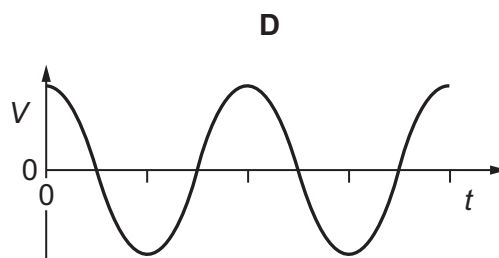
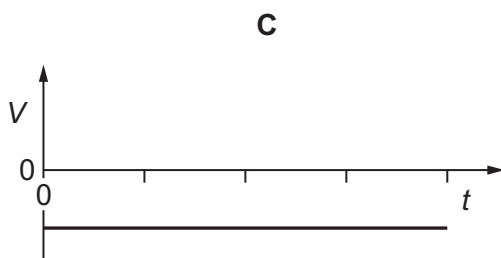
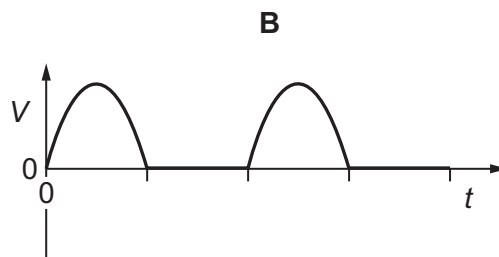
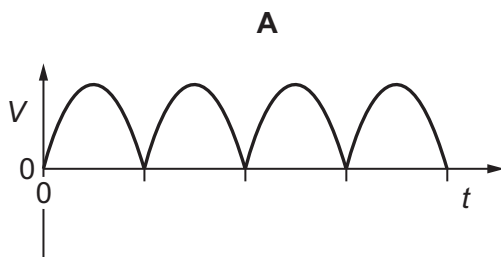
- A** to earth the appliance
- B** to earth the plug
- C** to protect the user from electric shock
- D** to protect the wiring from overheating

- 32 A horizontal beam of electrons passes between the two poles of a magnet.



In which direction is the beam deflected?

- A into the page
 - B out of the page
 - C towards the north pole
 - D towards the south pole
- 33 Which graph shows the voltage output V against time t for an a.c. generator?

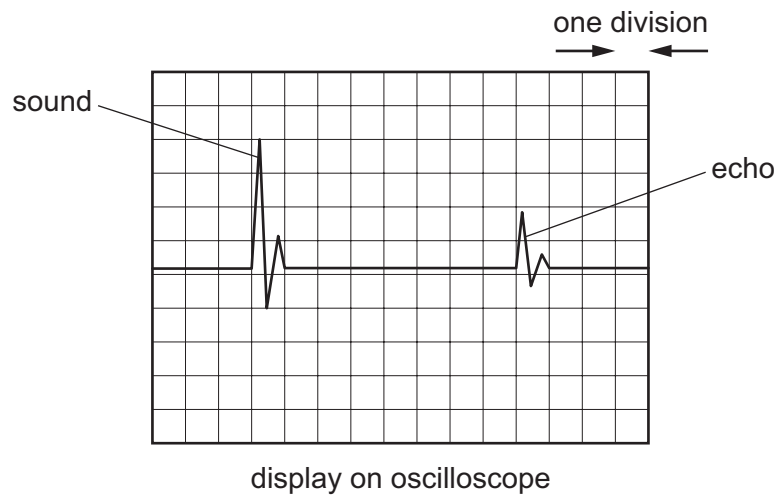


- 34** A loudspeaker and a microphone are placed in front of a wall.



The loudspeaker makes a sound which is detected by the microphone.

The microphone is connected to an oscilloscope which is set so that each division on the screen represents 0.01 s. The microphone detects the original sound and the echo.

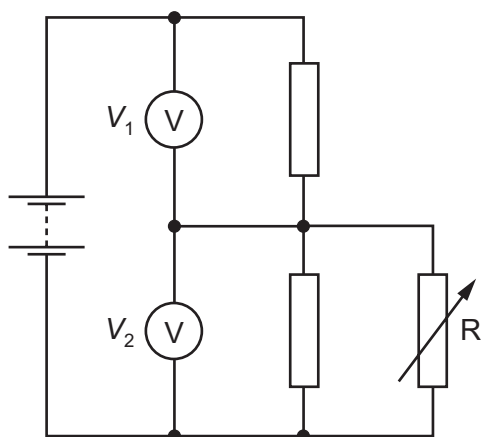


The speed of sound in air is 300 m/s.

What is the distance between the loudspeaker and the wall?

- A** 6.0 m **B** 12 m **C** 24 m **D** 48 m

- 35 The circuit diagram shows a variable resistor R connected in parallel to the lower half of a potential divider.



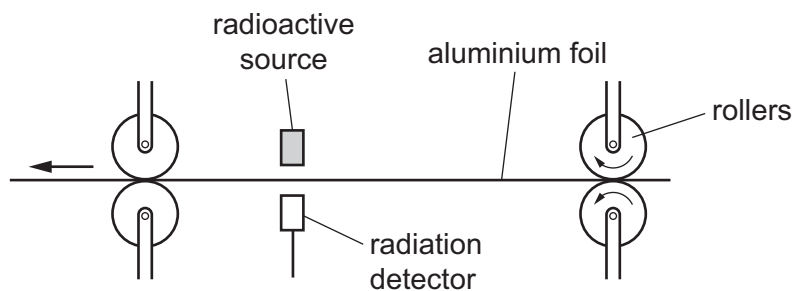
The resistance of R increases.

What happens to the two voltmeter readings?

	V_1	V_2
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 36** Thin aluminium foil is used in cooking.

The diagram shows a system used to control the thickness of aluminium foil being made in a factory.



What is the most suitable radioactive source for the system?

	source	half-life of source
A	beta	1 hour
B	beta	6 years
C	gamma	1 hour
D	gamma	6 years

- 37** Which of alpha, beta and gamma radiations are waves?

- A** alpha only
- B** beta only
- C** gamma only
- D** alpha and beta

- 38** A radioactive isotope of hydrogen has a half-life of 12 years.

A sample contains 40 million atoms of this isotope.

How many atoms of this isotope are left after 24 years?

- A** 20 million
- B** 10 million
- C** 5 million
- D** zero

39 How many nucleons are in one neutral atom of the krypton isotope $^{84}_{36}\text{Kr}$?

- A 36 B 48 C 84 D 120

40 A radioactive nucleus is represented as $^{230}_{92}\text{U}$.

Which row represents another isotope of the same element?

	number of protons in nucleus	number of neutrons in nucleus
A	90	140
B	91	138
C	92	139
D	93	136

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Cambridge Assessment International Education
Cambridge Ordinary Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2018

MARK SCHEME

Maximum Mark: 40

Published

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This document consists of **3** printed pages.



Cambridge Assessment
International Education

Question	Answer	Marks
1	C	1
2	B	1
3	C	1
4	A	1
5	D	1
6	A	1
7	D	1
8	B	1
9	B	1
10	B	1
11	D	1
12	D	1
13	B	1
14	A	1
15	A	1
16	D	1
17	C	1
18	C	1
19	D	1
20	C	1
21	A	1
22	C	1
23	B	1
24	C	1
25	A	1
26	C	1
27	B	1
28	D	1

Question	Answer	Marks
29	B	1
30	D	1
31	A	1
32	C	1
33	A	1
34	A	1
35	C	1
36	D	1
37	D	1
38	B	1
39	C	1
40	C	1



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Question	Answer	Marks
1	A	1
2	B	1
3	D	1
4	C	1
5	C	1
6	A	1
7	B	1
8	D	1
9	C	1
10	B	1
11	B	1
12	B	1
13	A	1
14	D	1
15	C	1
16	C	1
17	B	1
18	C	1
19	A	1
20	A	1
21	A	1
22	B	1
23	C	1
24	D	1
25	D	1
26	A	1
27	A	1
28	C	1

Question	Answer	Marks
29	D	1
30	A	1
31	D	1
32	B	1
33	D	1
34	B	1
35	B	1
36	B	1
37	C	1
38	B	1
39	C	1
40	C	1



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/12

May/June 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

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Electronic calculators may be used.

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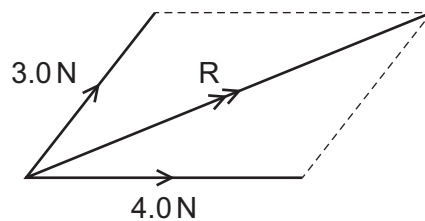


- 1 A car accelerates at 5.0 m/s^2 along a straight, horizontal road and reaches a velocity of 20 m/s in a time of 4.0 s .

During this time, its total displacement is 40 m .

Which quantity is a scalar?

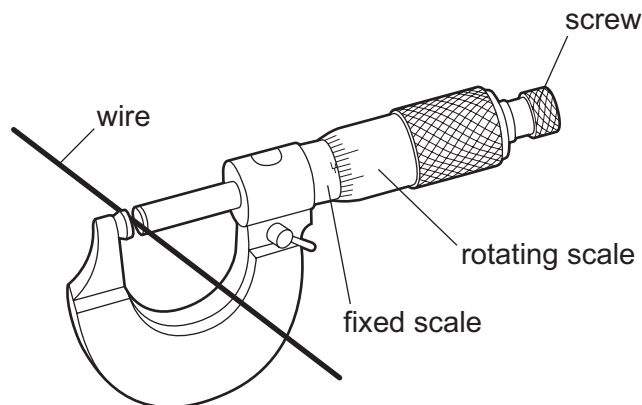
- A a displacement of 40 m
 B a time of 4.0 s
 C a velocity of 20 m/s
 D an acceleration of 5.0 m/s^2
- 2 The diagram shows the resultant R of a 3.0 N force and a 4.0 N force acting at a point.



The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90° .

Which value of R is **not** possible?

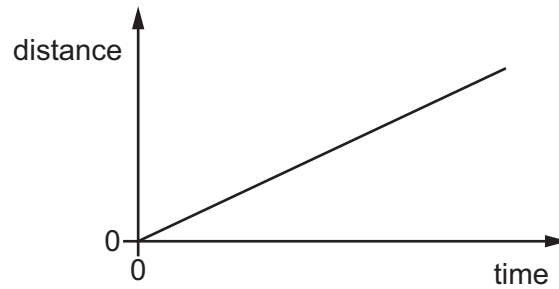
- A 4.0 N B 5.0 N C 6.0 N D 7.0 N
- 3 A micrometer is used to measure the diameter of a uniform wire.



What is done to obtain an accurate answer?

- A Make the micrometer horizontal and then use the scales to find the reading.
 B Subtract the fixed-scale reading from the rotating-scale reading.
 C Subtract the rotating-scale reading from the fixed-scale reading.
 D Use the scales to find the reading and add or subtract any zero error.

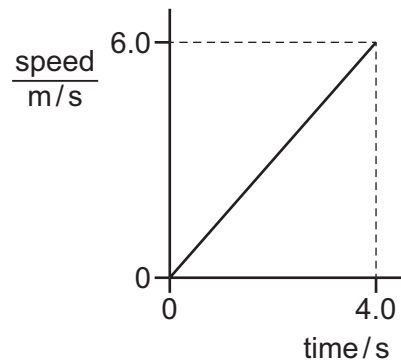
- 4 The diagram shows the distance-time graph of a car.



The car is travelling along a straight road up a hill.

Which quantity for the car is constant and greater than zero?

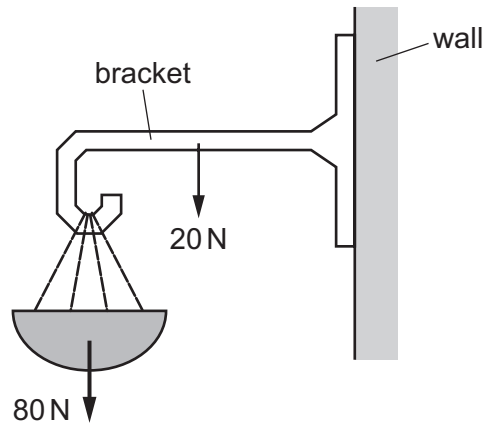
- A acceleration
 - B gravitational potential energy
 - C kinetic energy
 - D resultant force
- 5 The diagram shows a speed-time graph for an object moving with uniform acceleration.



What is the distance travelled in the first 4.0 s?

- A 0.67 m B 1.5 m C 12 m D 24 m

- 6 A hanging basket is fixed to a wall by a bracket.



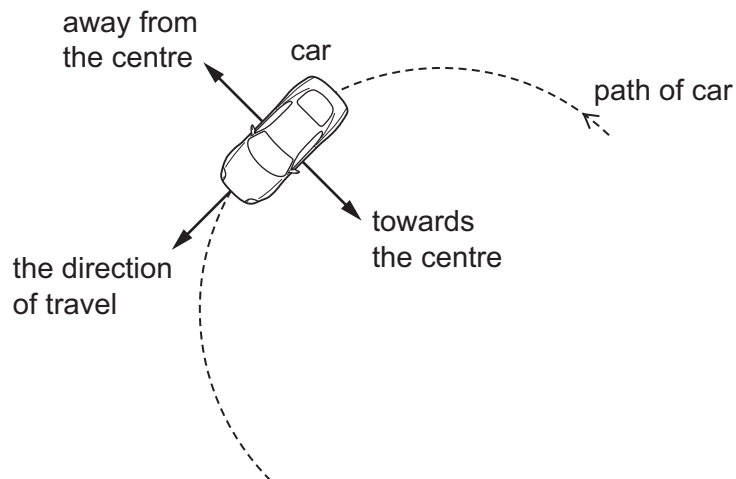
The weight of the basket is 80 N. The weight of the bracket is 20 N.

What is the size of the upwards force exerted on the bracket by the wall?

- A** 60 N **B** 70 N **C** 90 N **D** 100 N
- 7 A train of mass 240 000 kg is travelling at a speed of 60 m/s. The brakes are applied and it decelerates for 10 minutes until it comes to rest.

What is the average resultant force?

- A** 24 000 N **B** 40 000 N **C** 480 000 N **D** 720 000 N
- 8 A car travels at a constant speed along a circular, horizontal path.

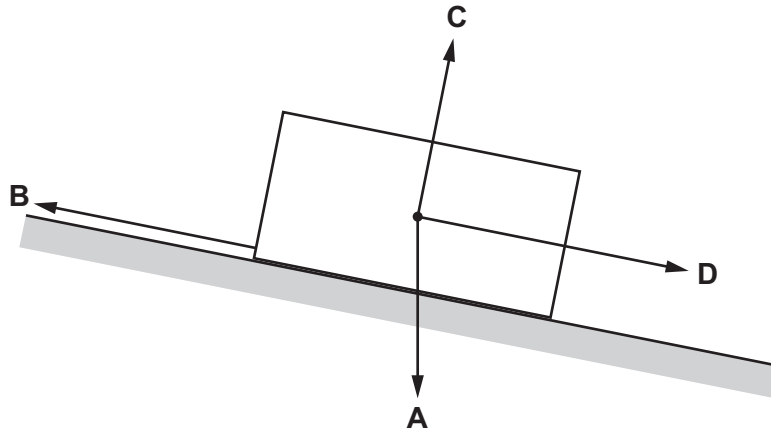


Which statement describes the forces acting on the car?

- A** They are balanced as the car is moving at constant speed.
B They are unbalanced with a resultant in the direction away from the centre.
C They are unbalanced with a resultant in the direction of travel of the car.
D They are unbalanced with a resultant in the direction towards the centre.

- 9 The diagram shows a block of wood resting on a sloping board.

Which arrow shows the direction of the gravitational force acting on the block?



- 10 On the Earth's surface, the gravitational field strength is 10 N/kg .

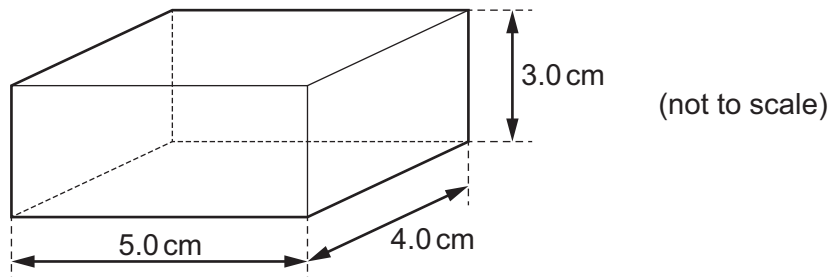
On the surface of Mars, the gravitational field strength is 3.8 N/kg .

A robot vehicle has a weight of 2000 N on Earth.

What is the weight of the robot vehicle on Mars?

- A** 200 N **B** 760 N **C** 7600 N **D** 76000 N

- 11 The diagram shows a rectangular block.

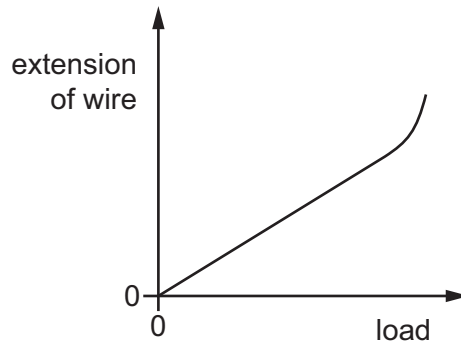


The density of the block is 2.5 g/cm^3 .

What is the mass of the block?

- A** 18 g **B** 24 g **C** 50 g **D** 150 g

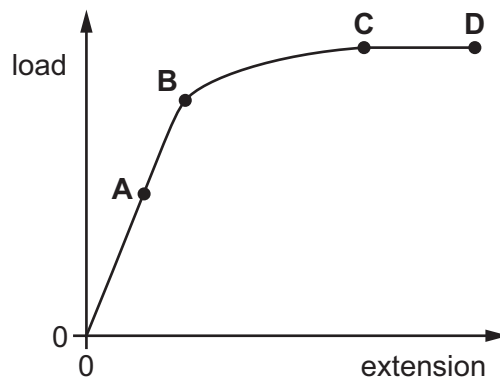
- 12 The graph shows the extension of a piece of copper wire as the load on it is increased.



What does the graph show?

- A At a certain load, the wire becomes easier to extend.
 - B At a certain load, the wire becomes harder to extend.
 - C The load and the extension are directly proportional for all loads.
 - D The load and the extension are inversely proportional for all loads.
- 13 The diagram shows a load-extension graph for a wire.

Which point is the limit of proportionality?

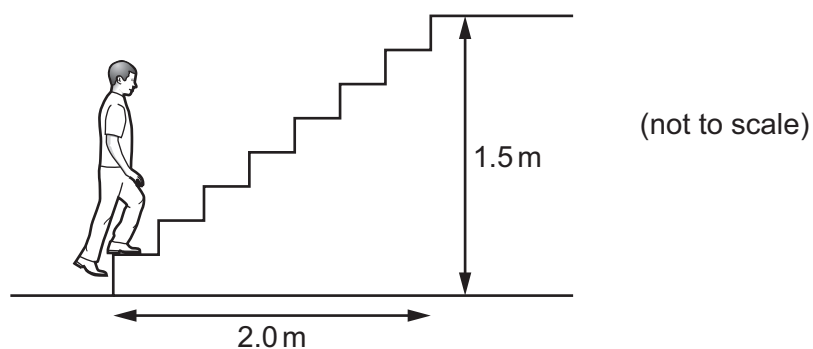


- 14 A small motor has an input power rating of 10 W and is run for 5.0 minutes.

What is the electrical energy input to the motor in this time?

- A 2.0 J
- B 50 J
- C 300 J
- D 3000 J

- 15** A student of mass 60 kg climbs some steps. He travels a horizontal distance of 2.0 m and a vertical distance of 1.5 m. The gravitational field strength g is 10 N/kg.



What is the work done against the force of gravity?

- A** 90 J **B** 120 J **C** 900 J **D** 1200 J
- 16** Which energy source is non-renewable?
- A** nuclear energy
B solar energy
C tidal energy
D wind energy
- 17** In a coal-fired power station, the coal is burnt and thermal energy (heat) is produced. The thermal energy is used to produce electrical energy.

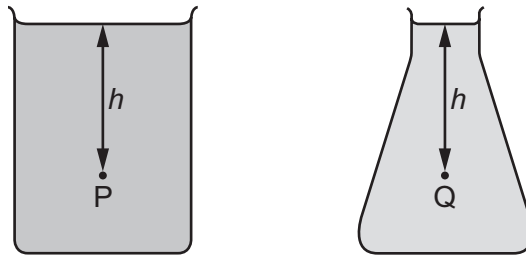
In which order does the energy pass through parts of the power station?

- A** boiler → generator → turbine
B boiler → turbine → generator
C turbine → boiler → generator
D turbine → generator → boiler

- 18 Two glass containers filled with different liquids are placed next to each other.

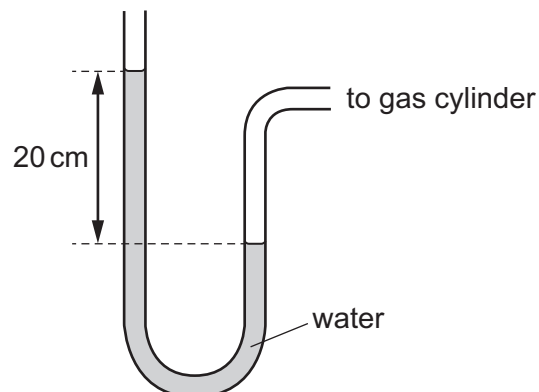
Point P is a distance h below the surface of the liquid in one container.

Point Q is a distance h below the surface of the liquid in the other container.



Why is the pressure at P different from the pressure at Q?

- A The atmospheric pressure is different at P.
 - B The densities of the liquids are different.
 - C The gravitational field strength is different at P.
 - D The shapes of the containers are different.
- 19 The pressure of a gas in a cylinder is measured using a water manometer.

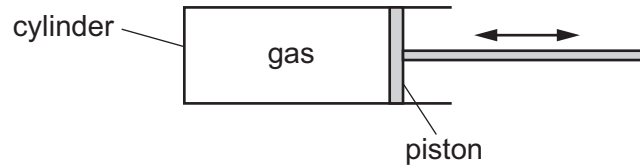


The density of water is 1000 kg/m^3 and the gravitational field strength g is 10 N/kg .

What is the pressure, above atmospheric pressure, of the gas in the cylinder?

- A 200 Pa
- B 2000 Pa
- C 20 000 Pa
- D 200 000 Pa

- 20 A fixed mass of gas is enclosed in a cylinder with a movable piston.



The gas is initially at pressure p_1 and has a volume V_1 .

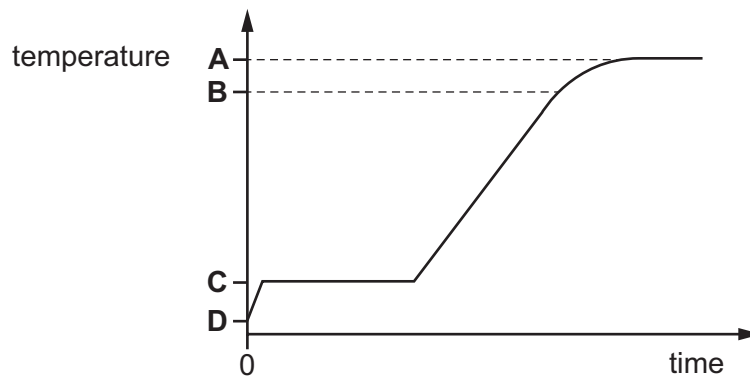
The temperature is kept constant. The piston is moved so that the pressure becomes p_2 and the volume becomes V_2 .

Which equation is correct?

- A** $\frac{p_1}{V_1} = \frac{p_2}{V_2}$
- B** $\frac{p_1}{p_2} = \frac{V_1}{V_2}$
- C** $p_1V_1 = p_2V_2$
- D** $p_1V_2 = p_2V_1$
- 21 A substance consists of particles that are close together and moving past each other at random. The average speed of the particles is gradually increasing.
- What best describes the substance?
- A** a gas being heated
- B** a liquid being heated
- C** a liquid undergoing solidification
- D** a solid being heated

- 22** Some ice cubes are taken from a freezer and heated in a container. The readings of temperature and time are recorded on the graph.

Which temperature is 0°C ?



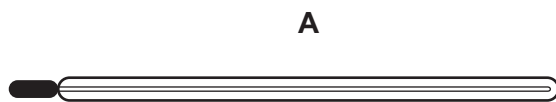
- 23** A tray for making ice cubes holds 30 g of water. The specific latent heat of fusion of ice is $3.3 \times 10^5 \text{ J/kg}$.

How much thermal energy is removed from the water at 0°C to change it into ice at 0°C ?

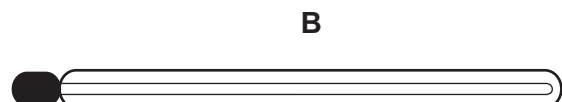
- A** $9.9 \times 10^3 \text{ J}$ **B** $1.1 \times 10^4 \text{ J}$ **C** $9.9 \times 10^6 \text{ J}$ **D** $1.1 \times 10^7 \text{ J}$

- 24** The bulbs of the four thermometers shown all contain mercury.

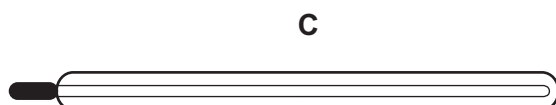
Which thermometer is the most sensitive?



small bulb, narrow tube



large bulb, wide tube



small bulb, wide tube

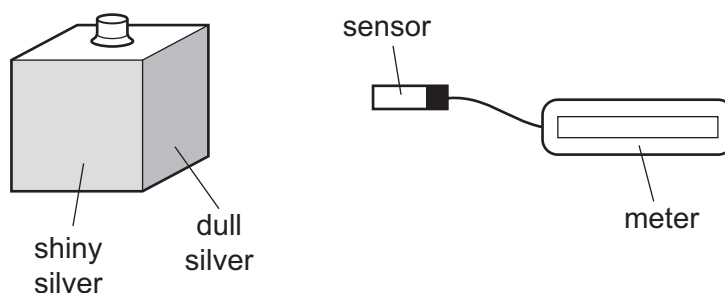


large bulb, narrow tube

25 A metal box has four different surfaces; dull black, shiny black, dull silver and shiny silver.

The box is filled with boiling water so that each surface is at the same temperature.

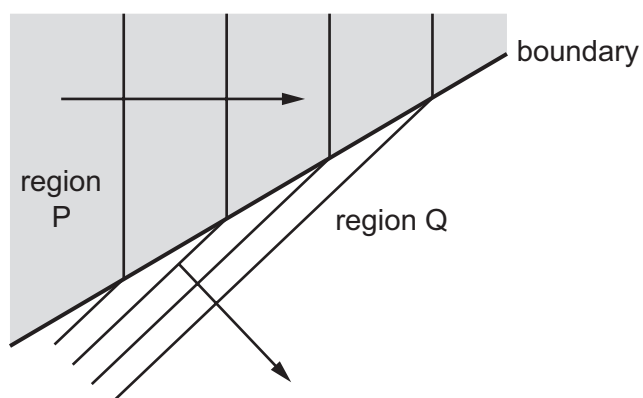
A sensor measures the amount of radiation from each surface.



Which surface emits the least radiation and which surface emits the most radiation?

	least	most
A	dull black	shiny silver
B	dull silver	shiny black
C	shiny black	dull silver
D	shiny silver	dull black

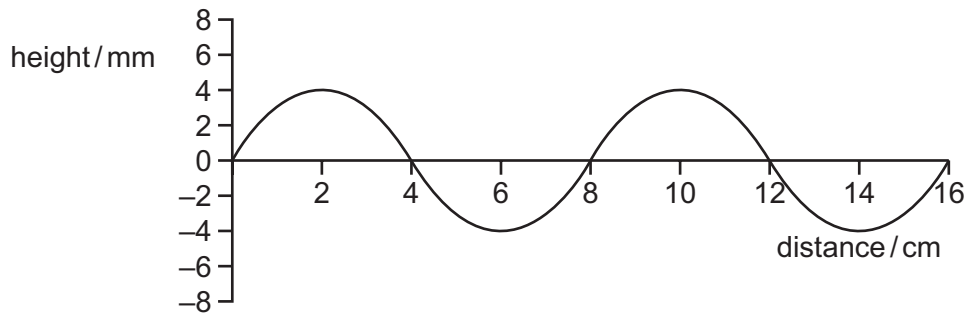
- 26 The diagram shows a water wave in a ripple tank being refracted at the boundary between region P and region Q.



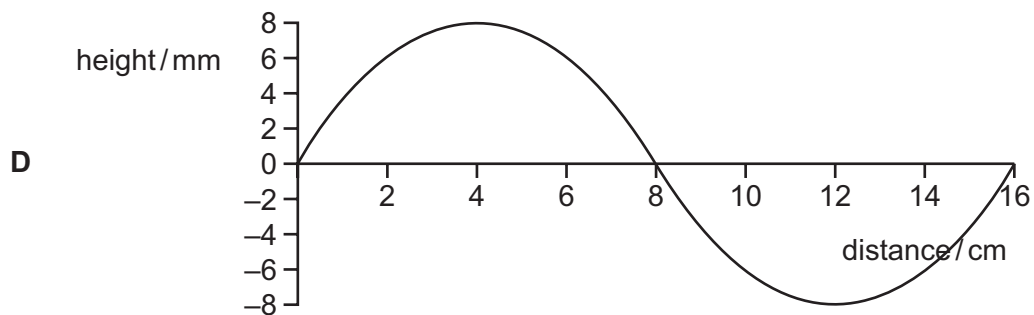
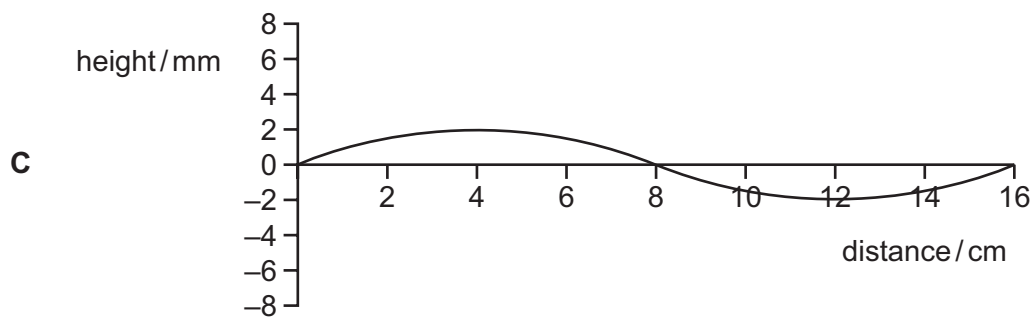
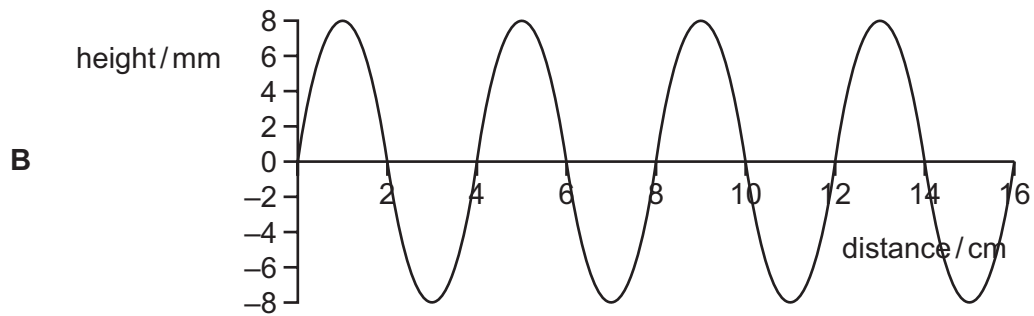
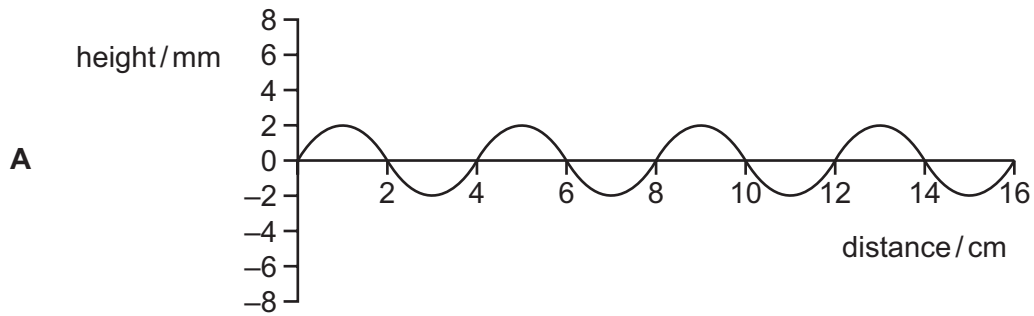
How does the wave in region P compare with the wave in region Q?

	wavelength of wave in region P	speed of wave in region P
A	larger	faster
B	larger	slower
C	smaller	faster
D	smaller	slower

27 The graph shows how the height of a water wave varies with distance along the wave.

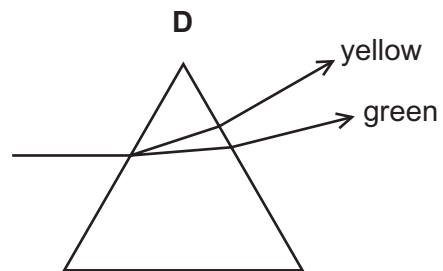
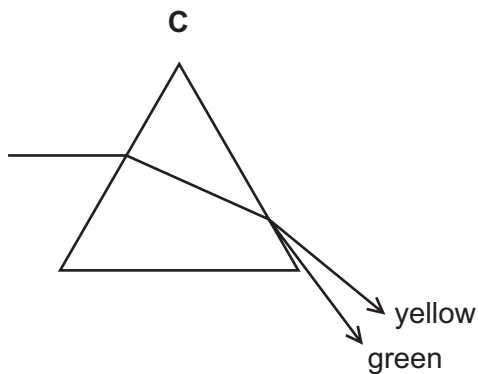
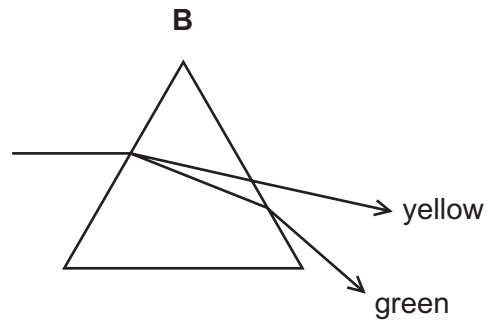
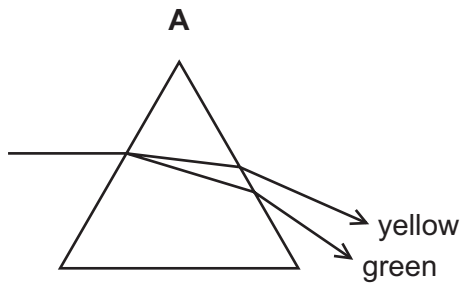


Which graph shows a wave with twice the amplitude, half the frequency, and the same speed?



28 A narrow beam of yellow and green light is separated as it passes through a prism.

Which ray diagram is correct?



29 Which type of wave is used to send television signals to a satellite?

- A** infra-red waves
- B** light waves
- C** microwaves
- D** sound waves

30 What proves that a metal bar is a permanent magnet?

- A** It attracts both ends of a compass needle.
- B** It attracts one end of another magnet.
- C** It conducts electricity.
- D** It repels one end of another magnet.

31 Which action demagnetises a bar magnet?

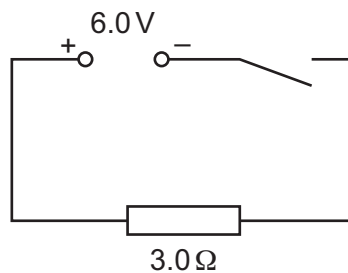
- A placing the magnet inside a coil connected to a d.c. supply
- B placing the magnet inside a coil connected to an a.c. supply
- C withdrawing the magnet from inside a coil connected to a d.c. supply
- D withdrawing the magnet from inside a coil connected to an a.c. supply

32 Electrical apparatus is protected from magnetic fields by placing the apparatus in a box.

What is the box made from?

- A aluminium
- B iron
- C rubber
- D steel

33 The circuit shown is switched on for 1.0 minute.

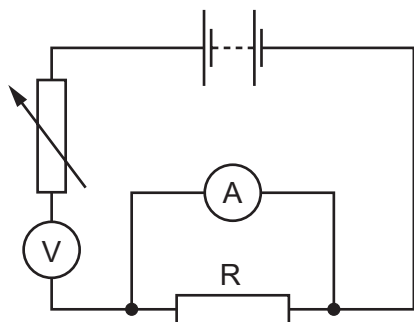
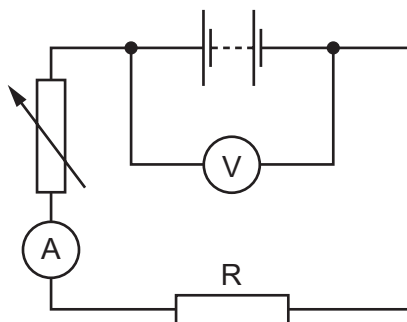
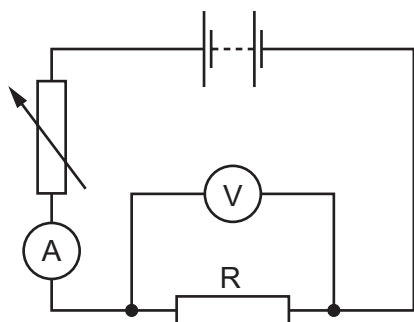
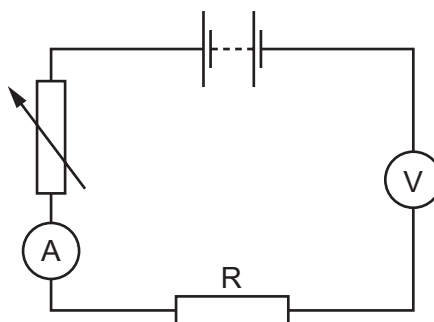


How much charge passes through the $3.0\,\Omega$ resistor?

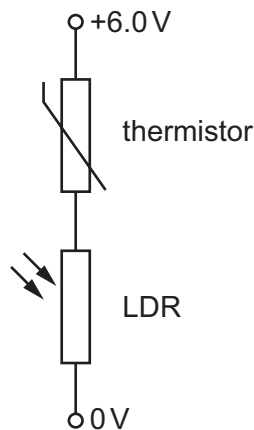
- A 2.0 C
- B 12 C
- C 120 C
- D 720 C

34 The readings of an ammeter and a voltmeter are used to calculate the resistance of resistor R.

Which circuit shows the ammeter and voltmeter in the correct positions?

A**B****C****D**

- 35 A thermistor and a light-dependent resistor (LDR) are connected in series. A potential difference (p.d.) of 6.0 V is applied across them as shown.

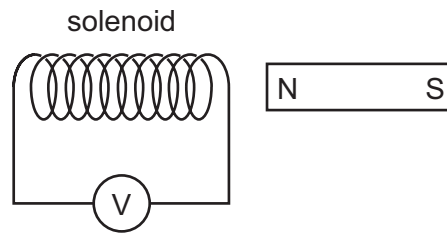


The thermistor has a resistance of $6000\ \Omega$ in a cold room and $1000\ \Omega$ in a warm room. The LDR has a resistance of $2000\ \Omega$ in dim light and $500\ \Omega$ in bright light.

When is the p.d. across the LDR equal to 2.0 V?

- A in a cold room with bright light
 - B in a cold room with dim light
 - C in a warm room with bright light
 - D in a warm room with dim light
- 36 What is the effect of using a split-ring commutator?
- A it ensures that the current is the same in all parts of a series circuit
 - B it generates an alternating electric current
 - C it produces a force on a current-carrying coil
 - D it reverses the direction of the current in the coil of a motor

- 37 The N-pole of a magnet is moved into a solenoid and an e.m.f. is induced.



What causes an increase in the induced e.m.f.?

- A moving the magnet more quickly
 - B moving the magnet more slowly
 - C pulling the magnet out instead of pushing it in
 - D using the S-pole of the magnet instead of the N-pole
- 38 Which material is used for the core of a transformer?
- A copper
 - B iron
 - C plastic
 - D steel
- 39 Which row is correct for nuclear fission and for nuclear fusion?

	fission	fusion
A	produces larger nuclei	is the energy source of a star
B	produces larger nuclei	releases energy in a power station
C	produces smaller nuclei	is the energy source of a star
D	produces smaller nuclei	releases energy in a power station

- 40 What happens to the proton number (atomic number) of a nucleus as it emits an alpha-particle?
- A It decreases by one.
 - B It decreases by two.
 - C It decreases by four.
 - D It does not change.

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/11

May/June 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet
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Soft pencil (type B or HB is recommended)

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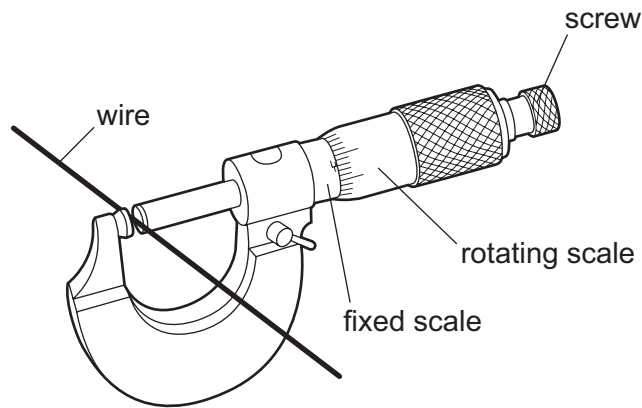
Any rough working should be done in this booklet.

Electronic calculators may be used.

This document consists of **15** printed pages and **1** blank page.



- 1 A micrometer is used to measure the diameter of a uniform wire.

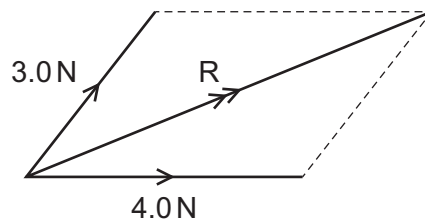


What is done to obtain an accurate answer?

- A** Make the micrometer horizontal and then use the scales to find the reading.
B Subtract the fixed-scale reading from the rotating-scale reading.
C Subtract the rotating-scale reading from the fixed-scale reading.
D Use the scales to find the reading and add or subtract any zero error.
- 2 A car's acceleration and maximum speed are improved by using an engine of smaller mass and greater driving force.

How many of the underlined quantities are vectors?

- A** 1 **B** 2 **C** 3 **D** 4
- 3 The diagram shows the resultant R of a 3.0 N force and a 4.0 N force acting at a point.

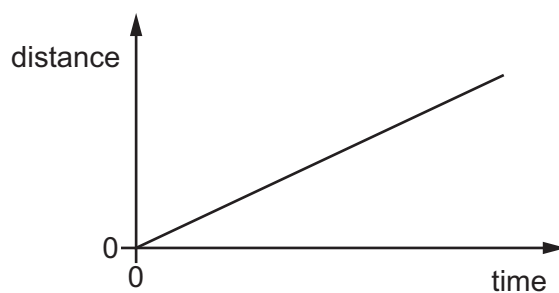


The angle between the 3.0 N force and the 4.0 N force can be any value from 0° to 90° .

Which value of R is **not** possible?

- A** 4.0 N **B** 5.0 N **C** 6.0 N **D** 7.0 N

- 4 The diagram shows the distance-time graph of a car.



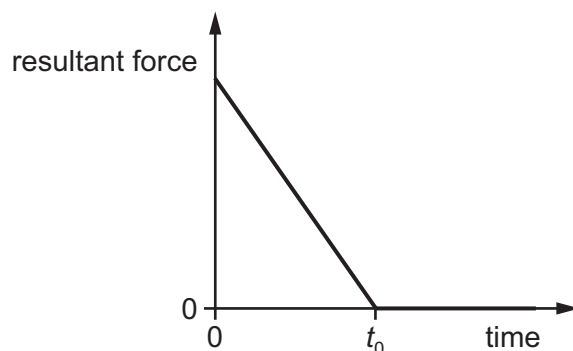
The car is travelling along a straight road up a hill.

Which quantity for the car is constant and greater than zero?

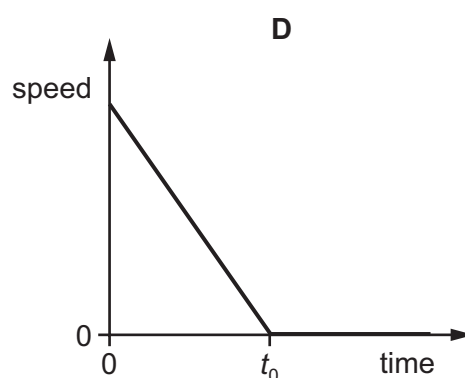
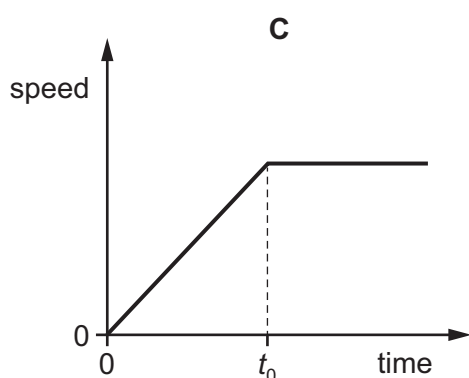
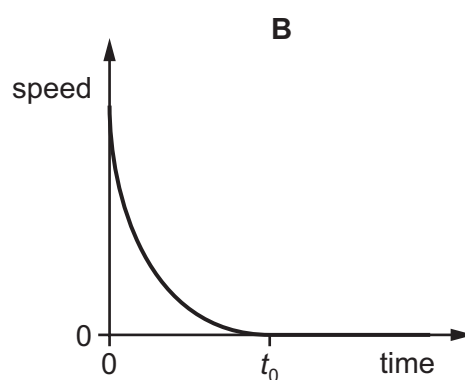
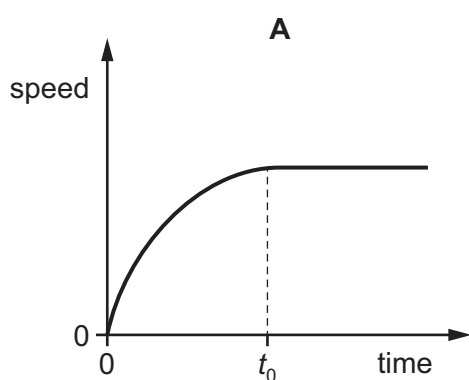
- A** acceleration
- B** gravitational potential energy
- C** kinetic energy
- D** resultant force

- 5 A resultant force acts on an object and causes it to move in a straight line.

The graph shows how the resultant force varies with time.



Which graph is the speed-time graph for the object?



- 6 A skydiver jumps from an aeroplane. After a few seconds, he reaches a terminal velocity without opening his parachute.

Why does he reach terminal velocity?

- A** Air resistance becomes greater than his weight and slows him down.
- B** Air resistance decreases and he speeds up.
- C** Air resistance increases and balances his weight so that his acceleration is zero.
- D** His weight decreases and balances the air resistance.

- 7 A satellite is orbiting the Earth in a circular orbit.

Which two quantities are always in the same direction as each other?

- A the acceleration of the satellite and the displacement of the satellite
- B the acceleration of the satellite and the velocity of the satellite
- C the resultant force on the satellite and the acceleration of the satellite
- D the resultant force on the satellite and the velocity of the satellite

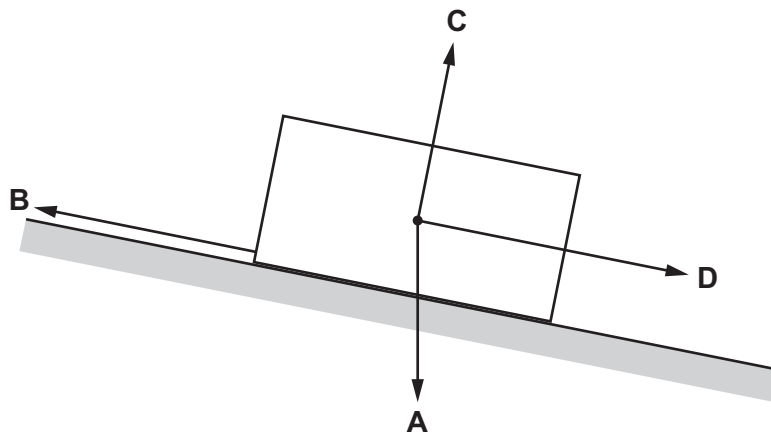
- 8 A satellite is in a circular orbit around a planet.

Which statement is correct?

- A Its acceleration is constant in direction but not in size.
- B Its acceleration is constant in size but not in direction.
- C Its gravitational potential energy varies.
- D Its velocity is constant.

- 9 The diagram shows a block of wood resting on a sloping board.

Which arrow shows the direction of the gravitational force acting on the block?

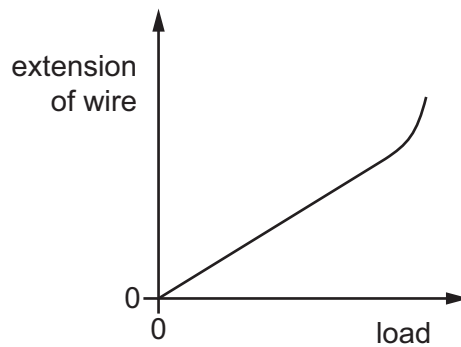


- 10 An astronaut travels to the International Space Station.

Which row describes how his mass and his weight compare with their sizes on Earth?

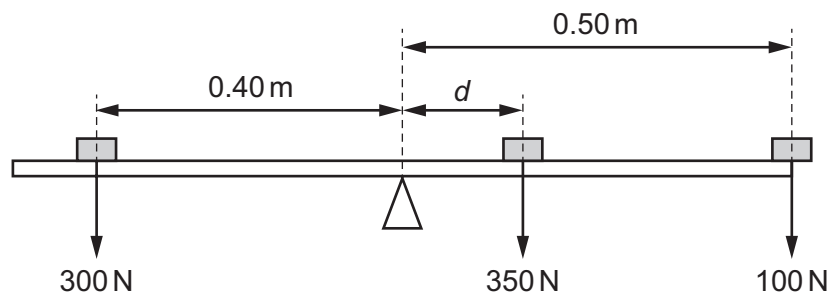
	mass	weight
A	different	different
B	different	the same
C	the same	different
D	the same	the same

- 11 The graph shows the extension of a piece of copper wire as the load on it is increased.



What does the graph show?

- A** At a certain load, the wire becomes easier to extend.
B At a certain load, the wire becomes harder to extend.
C The load and the extension are directly proportional for all loads.
D The load and the extension are inversely proportional for all loads.
- 12 A uniform beam is pivoted at its centre. The beam is balanced by three weights in the positions shown.



What is the length d ?

- A** 0.020 m **B** 0.050 m **C** 0.20 m **D** 0.48 m

- 13** The work done by a force on a body is calculated by multiplying the force by a quantity.

Which quantity?

- A** the distance travelled in the direction of the force
- B** the distance travelled perpendicular to the direction of the force
- C** the speed in the direction perpendicular to the force
- D** the velocity in the direction of the force

- 14** A small motor has an input power rating of 10 W and is run for 5.0 minutes.

What is the electrical energy input to the motor in this time?

- A** 2.0 J
- B** 50 J
- C** 300 J
- D** 3000 J

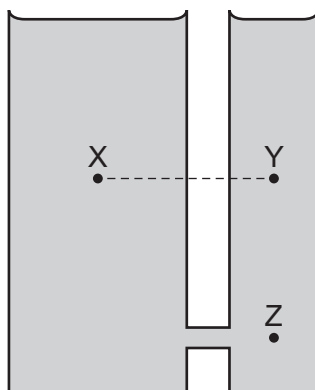
- 15** A small hydroelectric power station diverts water from a river. Every second, 20 kg of water flows through a pipe and falls through a vertical drop of 15 m. The efficiency of the power station is 0.60 (60%).

The gravitational field strength g is 10 N/kg.

What is the power output?

- A** 0.18 kW
- B** 1.8 kW
- C** 3.0 kW
- D** 180 kW

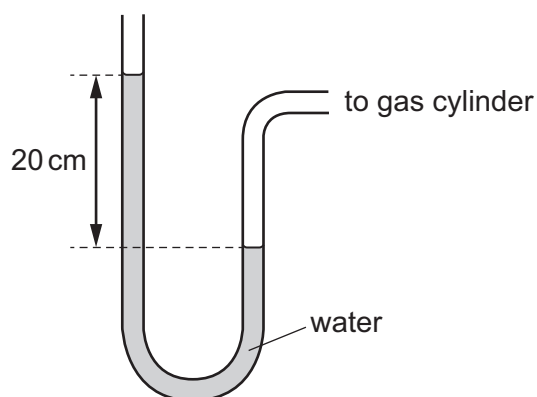
- 16** Two cylindrical vessels are joined together and filled with water as shown.



How does the pressure at point X compare to the pressure at points Y and Z?

	compared to Y	compared to Z
A	pressure at X is higher	pressure at X is lower
B	pressure at X is higher	pressure at X is the same
C	pressure at X is the same	pressure at X is lower
D	pressure at X is the same	pressure at X is the same

- 17 The pressure of a gas in a cylinder is measured using a water manometer.



The density of water is 1000 kg/m^3 and the gravitational field strength g is 10 N/kg .

What is the pressure, above atmospheric pressure, of the gas in the cylinder?

- A** 200 Pa **B** 2000 Pa **C** 20 000 Pa **D** 200 000 Pa

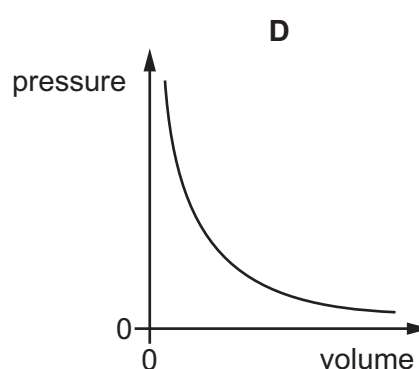
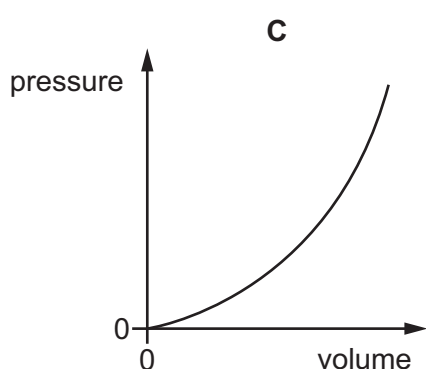
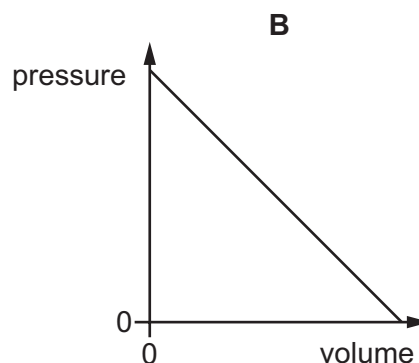
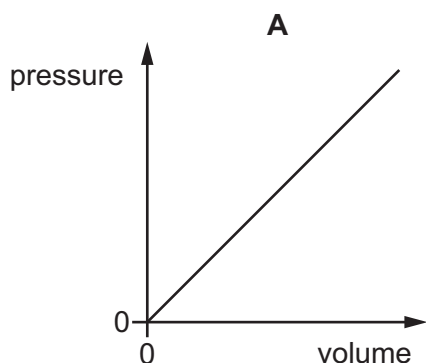
- 18 Four beakers contain equal volumes of water at two different temperatures. The beakers are placed in the open air.

From which beaker does water evaporate the fastest?

	surface area of beaker/ cm^2	temperature of water/ $^{\circ}\text{C}$
A	20	20
B	20	40
C	40	20
D	40	40

- 19** A fixed mass of gas undergoes a change of volume at constant temperature.

Which diagram shows the relationship between the volume and the pressure of the gas?



- 20** A substance consists of particles that are close together and moving past each other at random. The average speed of the particles is gradually increasing.

What best describes the substance?

- A** a gas being heated
- B** a liquid being heated
- C** a liquid undergoing solidification
- D** a solid being heated

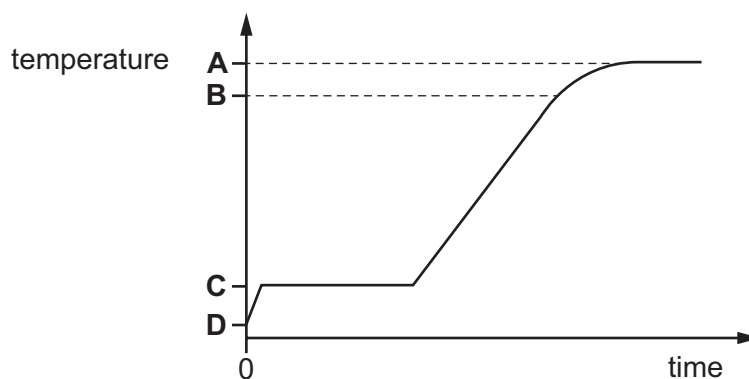
- 21 The table shows the melting points and the boiling points of some elements.

Which element is a liquid over the largest range of temperature?

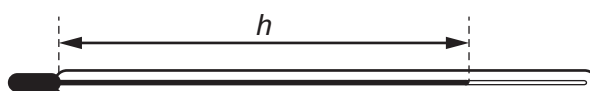
	element	melting point / $^{\circ}\text{C}$	boiling point / $^{\circ}\text{C}$
A	bromine	-7	59
B	chlorine	-102	-34
C	fluorine	-220	-188
D	iodine	114	184

- 22 Some ice cubes are taken from a freezer and heated in a container. The readings of temperature and time are recorded on the graph.

Which temperature is 0°C ?



- 23 The mercury-in-glass thermometer shown has a linear scale.



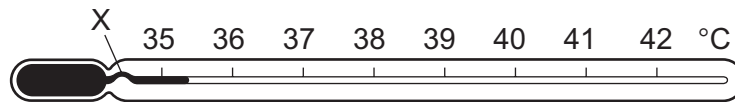
At a temperature of 100°C , h has a value of 28 cm.

At 80°C , h has a value of 24 cm.

What is the value of h when the temperature is 0°C ?

- A** 0.0 cm **B** 2.8 cm **C** 4.0 cm **D** 8.0 cm

- 24 The diagram shows a clinical thermometer.



How does the diagram show that this is a **sensitive** thermometer?

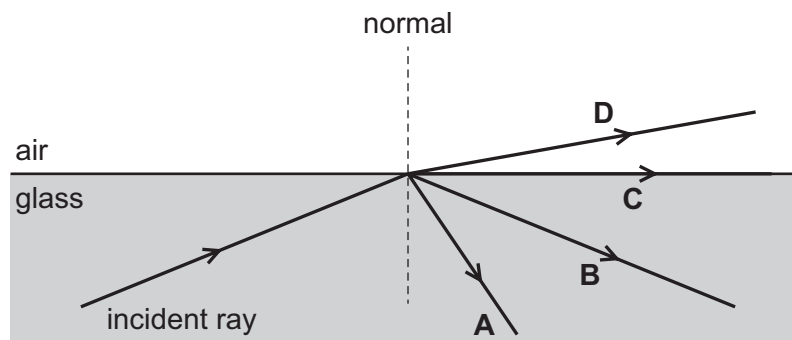
- A It has a range of 7 °C.
 - B There are equal distances between each °C mark on the scale.
 - C There is a constriction in the tube at X.
 - D There is a large distance between each °C mark on the scale.
- 25 There is a tight metal lid on a glass jar. The lid is loosened by pouring hot water over it.

Which physics property explains this?

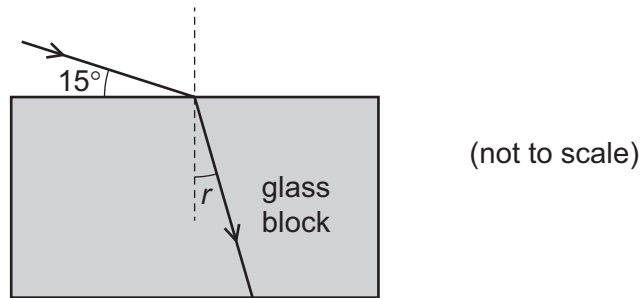
- A convection
 - B evaporation
 - C expansion
 - D radiation
- 26 Which waves are longitudinal?
- A sound waves in water
 - B ultra-violet waves in air
 - C waves on the surface of water
 - D X-rays in a vacuum

- 27 Light travelling in glass is incident on a glass-air boundary. The angle of incidence of the light is greater than the critical angle.

Which arrow shows the direction of the light after it is incident on the boundary?



- 28 Light strikes the top surface of a glass block at an angle of 15° as shown.



The refractive index of glass is 1.5.

What is the angle of refraction r ?

- A** 10° **B** 23° **C** 40° **D** 50°
- 29 A wall poster showing the electromagnetic spectrum is displayed in a laboratory.



A section of the electromagnetic spectrum has been accidentally ripped from this wall poster.

Which piece is missing?

- A** infra-red visible light microwave
- B** microwave infra-red visible light
- C** visible light infra-red microwave
- D** visible light microwave infra-red
- 30 Which type of wave is used to send television signals to a satellite?
- A** infra-red waves
B light waves
C microwaves
D sound waves

- 31** A ship that is stationary on the surface of the sea sends pulses of sound vertically downwards towards the sea bed.

Each pulse that reflects from the sea bed is received 1.0 s after it is sent out.

A whale swims under the boat and a pulse is received 0.60 s after it is sent out.

The speed of sound in sea water is 1500 m/s.

What is the distance of the whale above the sea bed?

- A** 300 m **B** 450 m **C** 600 m **D** 750 m

- 32** A charged plastic ball is at rest.

Which fields are found in the region surrounding the ball?
(Ignore the magnetic field of the Earth.)

- A** electric, gravitational and magnetic
B electric and gravitational only
C electric and magnetic only
D gravitational and magnetic only

- 33** What proves that a metal bar is a permanent magnet?

- A** It attracts both ends of a compass needle.
B It attracts one end of another magnet.
C It conducts electricity.
D It repels one end of another magnet.

- 34** Electrical apparatus is protected from magnetic fields by placing the apparatus in a box.

What is the box made from?

- A** aluminium
B iron
C rubber
D steel

- 35** A shoe becomes positively charged by friction when it rubs against a carpet.

What happens as the shoe becomes charged?

- A** Negative electrons are transferred to the carpet.
- B** Negative electrons are transferred to the shoe.
- C** Positive electrons are transferred to the carpet.
- D** Positive electrons are transferred to the shoe.

- 36** During a thunderstorm, a lightning flash occurs when charge flows between a cloud and the Earth. On average, 20 C of charge flows during each flash and there are three flashes every minute.

What is the average current between the cloud and the Earth during the thunderstorm?

- A** 1.0 A **B** 36 A **C** 60 A **D** 1800 A
- 37** Why are batteries sometimes made from several identical cells arranged in parallel?
- A** The battery has a greater e.m.f. than a single cell on its own.
 - B** The battery has a lower e.m.f. than a single cell on its own.
 - C** The battery lasts longer than a single cell on its own.
 - D** The battery supplies a smaller current than a single cell on its own.

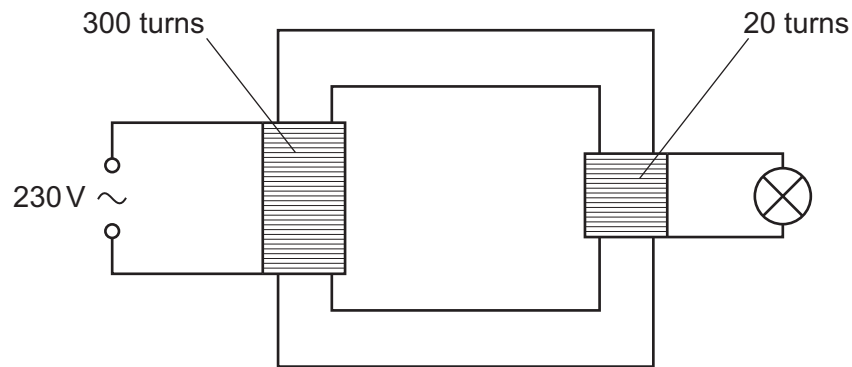
- 38** A cylinder of conducting material has resistance R .

A second cylinder of the same material is twice as long but has half the cross-sectional area.

What is the resistance of the second cylinder?

- A** $\frac{R}{2}$ **B** R **C** $2R$ **D** $4R$

- 39** A student uses a transformer to light a filament lamp using a 230 V a.c. supply. The lamp has a maximum voltage rating of 6.0 V.



What happens when the circuit is switched on?

- A** The lamp does not light at all.
 - B** The lamp lights dimly.
 - C** The lamp lights at normal brightness.
 - D** The lamp lights up brightly and then goes out.
- 40** Which row is correct for nuclear fission and for nuclear fusion?

	fission	fusion
A	produces larger nuclei	is the energy source of a star
B	produces larger nuclei	releases energy in a power station
C	produces smaller nuclei	is the energy source of a star
D	produces smaller nuclei	releases energy in a power station

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2017

MARK SCHEME

Maximum Mark: 40

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Question	Answer	Marks
1	B	1
2	A	1
3	D	1
4	C	1
5	C	1
6	D	1
7	A	1
8	D	1
9	A	1
10	B	1
11	D	1
12	A	1
13	B	1
14	D	1
15	C	1
16	A	1
17	B	1
18	B	1
19	B	1
20	C	1
21	B	1
22	C	1
23	A	1
24	D	1
25	D	1
26	A	1
27	D	1
28	A	1

Question	Answer	Marks
29	C	1
30	D	1
31	D	1
32	B	1
33	C	1
34	C	1
35	C	1
36	D	1
37	A	1
38	B	1
39	C	1
40	B	1



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

5054/11

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1	D	1
2	B	1
3	A	1
4	C	1
5	A	1
6	C	1
7	C	1
8	B	1
9	A	1
10	C	1
11	A	1
12	C	1
13	A	1
14	D	1
15	B	1
16	C	1
17	B	1
18	D	1
19	D	1
20	B	1
21	D	1
22	C	1
23	D	1
24	D	1
25	C	1
26	A	1
27	B	1
28	C	1

Question	Answer	Marks
29	C	1
30	C	1
31	A	1
32	B	1
33	D	1
34	B	1
35	A	1
36	A	1
37	C	1
38	D	1
39	D	1
40	C	1



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/12

May/June 2016

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

This document consists of **16** printed pages.



- 1 A student determines the circumference of a golf ball.

Which instrument gives a reading that is the circumference of the golf ball?

- A calipers
- B micrometer
- C rule
- D tape

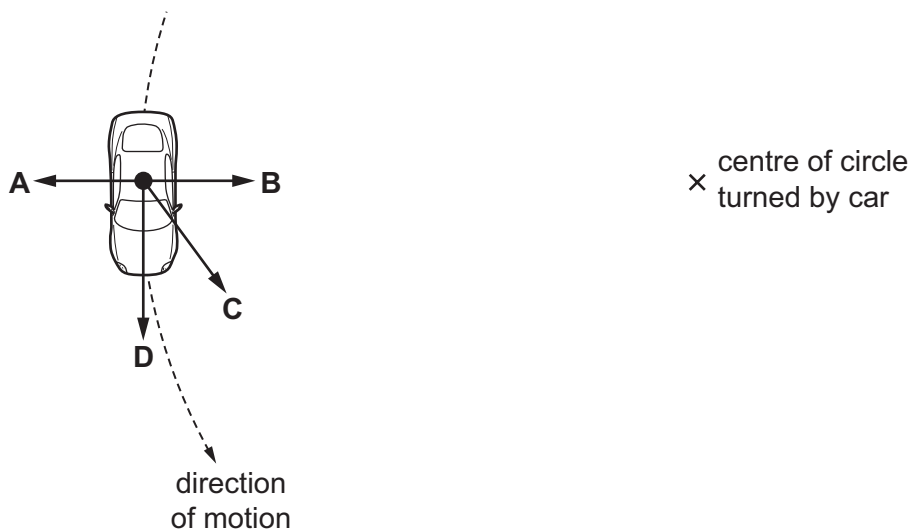
- 2 A man pulls a sledge of mass 25 kg across level ground with a horizontal force of 60 N. A constant force of friction of 20 N acts on the sledge.

What is the acceleration of the sledge?

- A 0.63 m/s^2 B 1.6 m/s^2 C 2.4 m/s^2 D 3.2 m/s^2

- 3 A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?

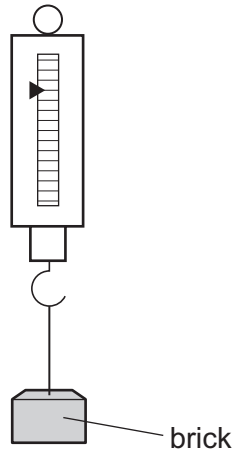


- 4 Which property of a body resists change from a state of rest or of motion?

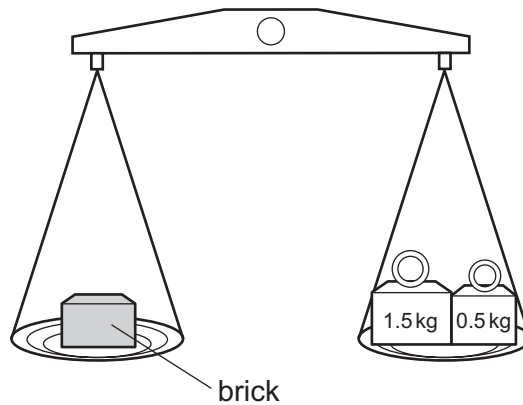
- A density
- B mass
- C volume
- D weight

- 5 A brick is placed on a newton meter and then on a beam balance.

newton meter



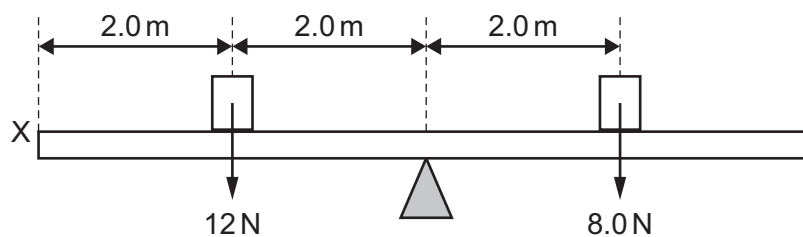
beam balance



What is measured by each instrument?

	newton meter	beam balance
A	mass	mass
B	mass	weight
C	weight	mass
D	weight	weight

- 6 A uniform plank is pivoted at its mid-point. Two weights are added to the plank, one weight on each side of the pivot in the positions shown.



A vertical force is applied at point X to balance the plank.

What is the size and direction of this force?

	size / N	direction
A	2.0	downwards
B	2.0	upwards
C	4.0	downwards
D	4.0	upwards

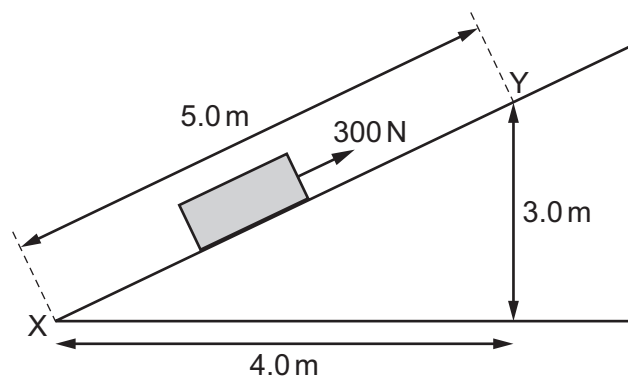
7 Which quantity is a vector?

- A energy
- B force
- C speed
- D time

8 Which process in the Sun produces energy?

- A burning
- B nuclear fission
- C nuclear fusion
- D radiation

9 A 300 N force is applied to a box to move it up a ramp, as shown.

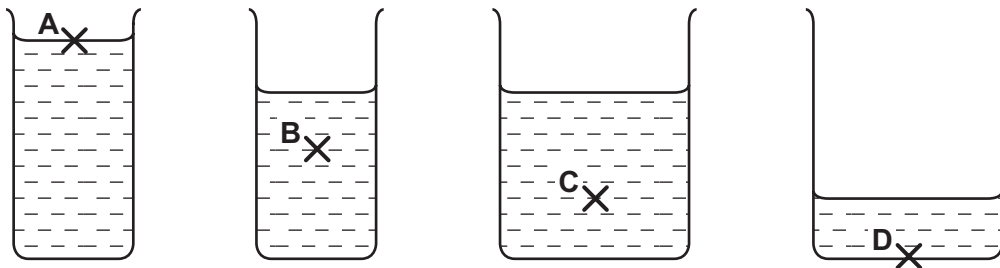


How much work is done by the force when moving the box from X to Y?

- A 900 J
- B 1200 J
- C 1500 J
- D 3000 J

10 Four beakers contain the same liquid.

At which point is the pressure the greatest?



- 11 A block of weight W rests on a side of area A . The gravitational field strength is g .

What is the pressure exerted on the ground due to the block?

- A WA B $\frac{W}{A}$ C $\frac{WA}{g}$ D $\frac{W}{g}$

- 12 Water of depth 10 m exerts a pressure equal to atmospheric pressure.

An air bubble rises to the surface of a lake which is 20 m deep. When the bubble reaches the surface, its volume is 6.0 cm^3 .

What is the volume of the air bubble at the bottom of the lake?

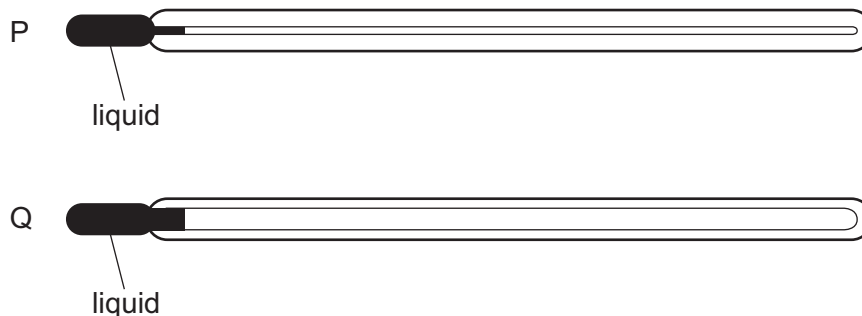
- A 2.0 cm^3 B 3.0 cm^3 C 12 cm^3 D 18 cm^3

- 13 A gas in a container of fixed volume is heated.

What happens to the molecules of the gas?

- A They collide less frequently.
B They expand.
C They move faster.
D They move further apart.

- 14 The diagram shows two liquid-in-glass thermometers P and Q.

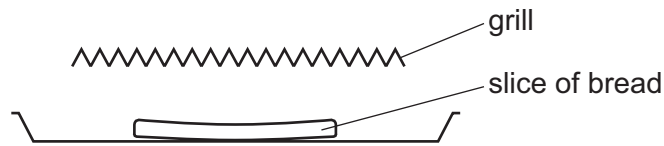


The thermometers are identical except that Q has a capillary tube with a larger internal diameter than P.

Which thermometer has the larger range and which thermometer has the greater sensitivity?

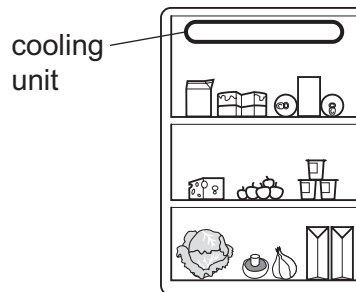
	larger range	greater sensitivity
A	P	P
B	P	Q
C	Q	P
D	Q	Q

- 15 A slice of bread is placed under a red-hot electric grill to make toast.



How does heat energy reach the bread?

- A conduction and convection
 - B conduction only
 - C convection and radiation
 - D radiation only
- 16 The diagram shows the inside of a refrigerator.



When the refrigerator is first switched on, what happens to the air near the cooling unit?

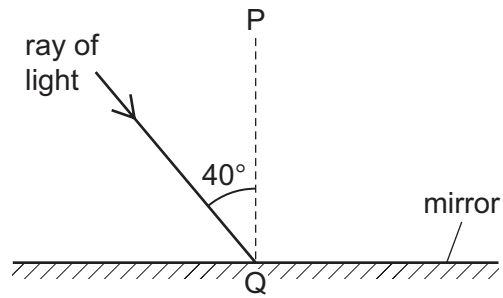
	the particles of this air	the density of this air
A	become smaller	decreases
B	become smaller	increases
C	move closer together	decreases
D	move closer together	increases

- 17 A water wave in a ripple tank refracts as it passes from deep water to shallow water.

Which properties change as the wave refracts?

- A frequency and amplitude
- B frequency and wavelength
- C speed and frequency
- D speed and wavelength

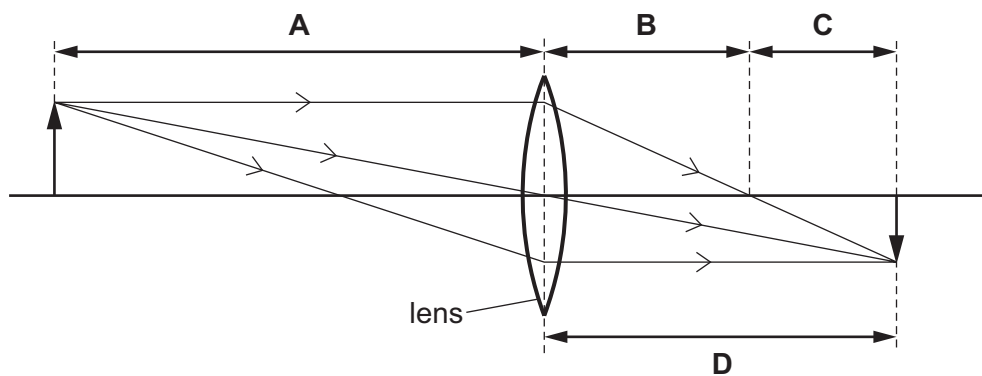
18 The diagram shows light incident on a plane mirror.



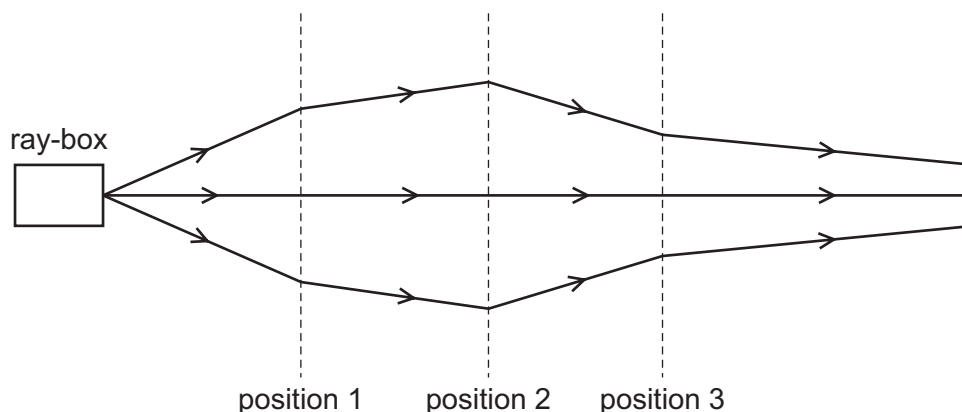
Which row gives the angle of reflection and the name of line PQ?

	angle of reflection	the line PQ is called the
A	40°	normal
B	40°	reflected ray
C	50°	normal
D	50°	reflected ray

19 Which length is the focal length of the lens shown in the diagram?



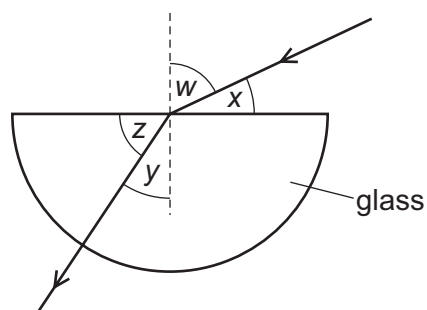
- 20 The rays of light from a ray-box pass through three lenses placed at positions 1, 2 and 3.



What type of lens is used at each position?

	position 1	position 2	position 3
A	converging	converging	converging
B	converging	converging	diverging
C	diverging	converging	diverging
D	diverging	diverging	converging

- 21 Light passes from air into a block of glass, as shown.

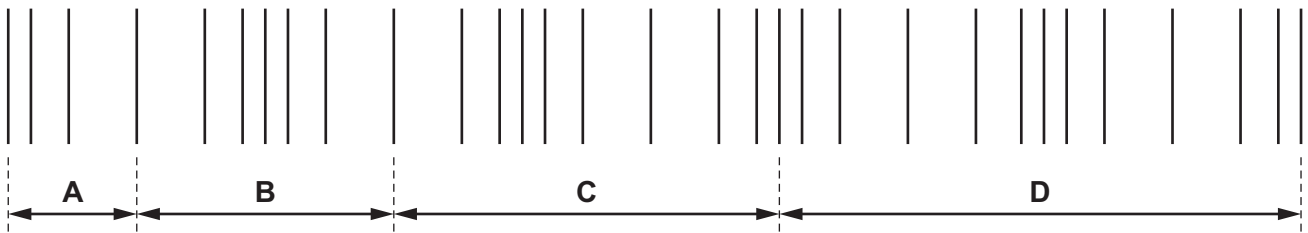


Which expression is equal to the refractive index of glass?

- A** $\frac{\sin w}{\sin y}$ **B** $\frac{\sin w}{\sin z}$ **C** $\frac{\sin y}{\sin w}$ **D** $\frac{\sin z}{\sin x}$

- 22** A sound wave travels through air. The lines in the diagram show the positions of layers of air at one particular time.

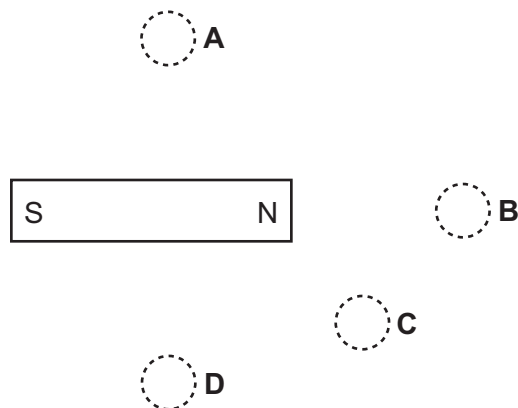
Which distance shows the wavelength of the wave?



- 23** Four plotting compasses are placed near a bar magnet. You may ignore any effects of the Earth's magnetic field.

One compass appears like this  .

What is a possible position for this compass?



- 24** A polythene rod is rubbed with a cloth and the cloth becomes positively charged.

Which statement describes the transfer of charge?

- A** Negative charges are transferred from the cloth to the polythene.
- B** Negative charges are transferred from the polythene to the cloth.
- C** Positive charges are transferred from the cloth to the polythene.
- D** Positive charges are transferred from the polythene to the cloth.

- 25** A charge of 45 C flows through an electric appliance in 3.0 minutes.

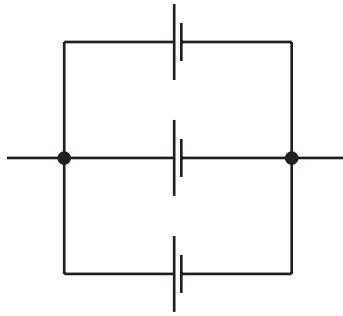
What is the average current in the appliance?

- A** 0.25 A
- B** 4.0 A
- C** 15 A
- D** 135 A

26 How can *one volt* also be expressed?

- A one coulomb per ampere
- B one coulomb per joule
- C one joule per ampere
- D one joule per coulomb

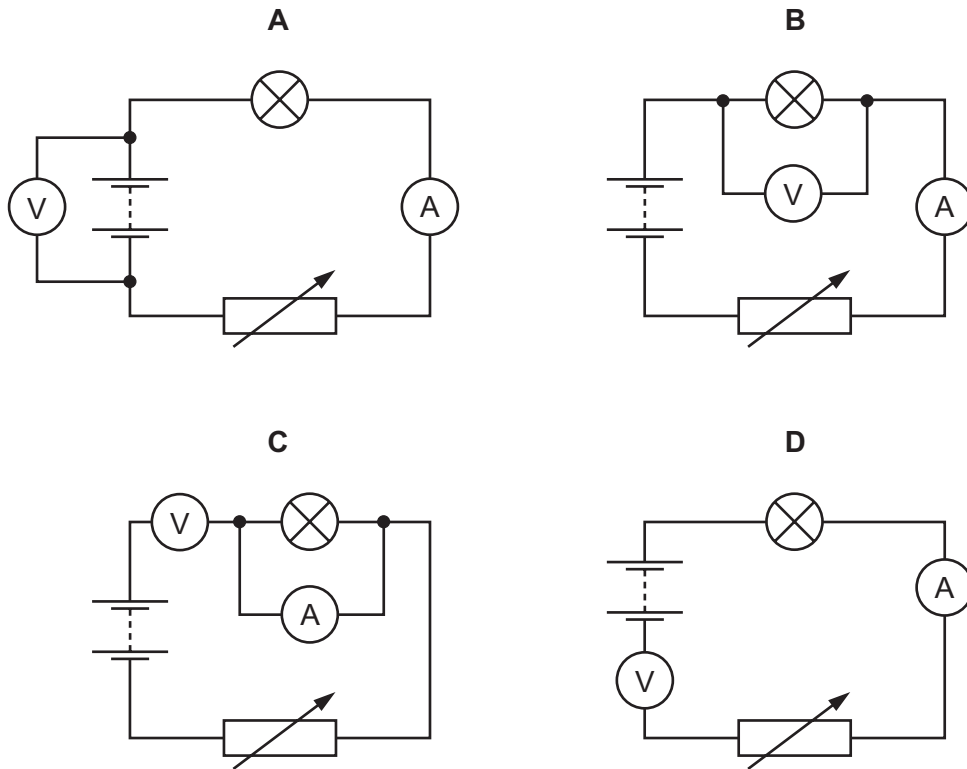
27 The diagram shows a battery of three 1.5 V cells.



What is an advantage of this arrangement of cells?

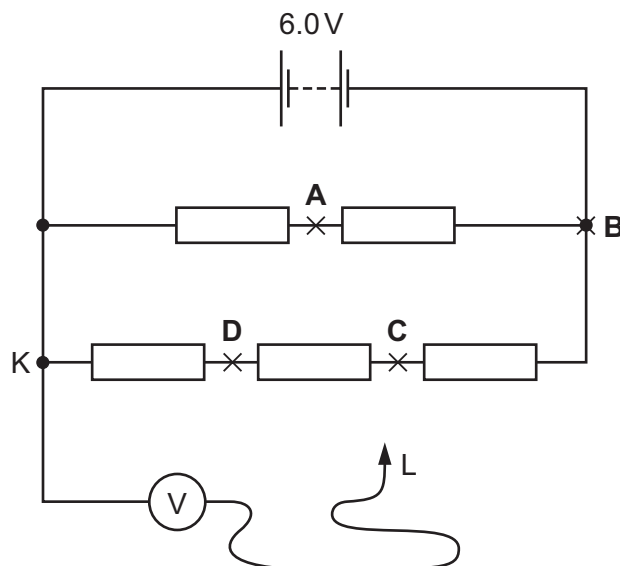
- A The battery can supply a current for a longer time than a single 1.5 V cell.
- B The battery can supply any e.m.f. between 0 V and 4.5 V.
- C The battery supplies more energy to each coulomb of charge than a single 1.5 V cell.
- D The e.m.f. of the battery is 4.5 V.

- 28 Which diagram shows the arrangement of the ammeter and voltmeter to obtain readings to find the power of a lamp?

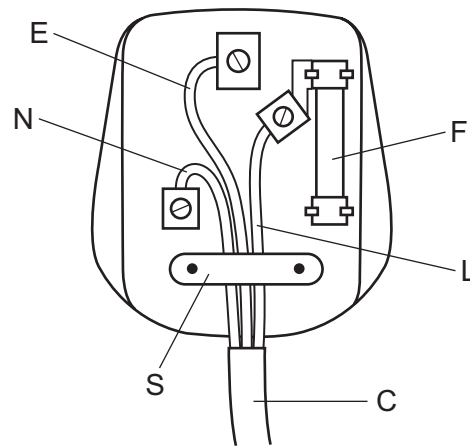


- 29 A 6.0V battery is connected to a network containing five identical resistors. A voltmeter has one lead connected to point K as shown.

At which point should lead L be connected so that the voltmeter reads 3.0V?



- 30 The diagram shows the wiring of a three-pin mains plug. There is an error in the diagram.



What is the error?

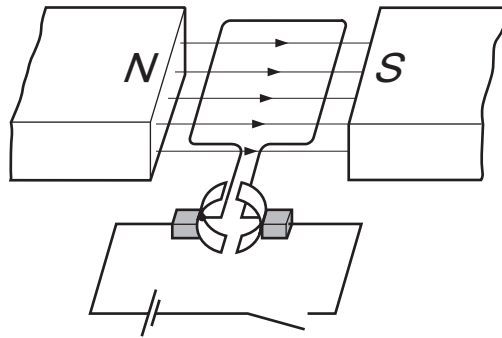
- A The cable cover C is not under the clip S.
 - B The earth wire E is connected to the wrong terminal.
 - C The fuse F is connected to the live wire L.
 - D The live wire L is connected to the wrong end of the fuse F.
- 31 A 100 W lamp is switched on for five hours each day for three weeks.

The cost of one unit of electricity is \$0.24.

How much does it cost to run the lamp for this time?

- A \$0.36
- B \$0.84
- C \$2.52
- D \$25.20

32 The diagram shows a simple d.c. motor.

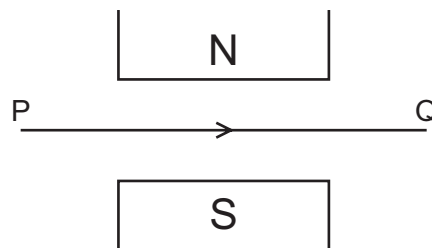


The switch is closed and the coil rotates.

Which change makes the coil rotate in the opposite direction **and** at a faster rate?

- A increase the current in the coil and increase the number of turns in the coil
- B reverse both the magnetic field and the current in the coil
- C reverse the magnetic field and decrease the current in the coil
- D reverse the magnetic field and increase the current in the coil

33 The diagram shows a wire PQ between the N-pole and the S-pole of a magnet. There is a current in the wire in the direction of the arrow.



What is the direction of the force on the wire PQ?

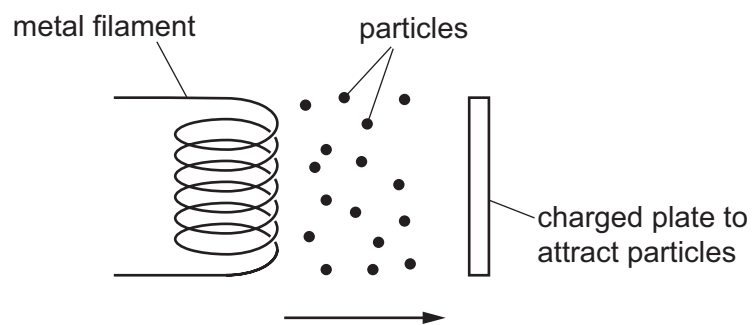
- A into the page
- B out of the page
- C towards the N-pole
- D towards the S-pole

34 Electrical power is transmitted by cables over long distances at very high voltages.

What are the effects of using a high voltage transmission system?

	power loss in the cables	current in the cables
A	high	high
B	high	low
C	low	high
D	low	low

35 The diagram represents thermionic emission from a metal filament.

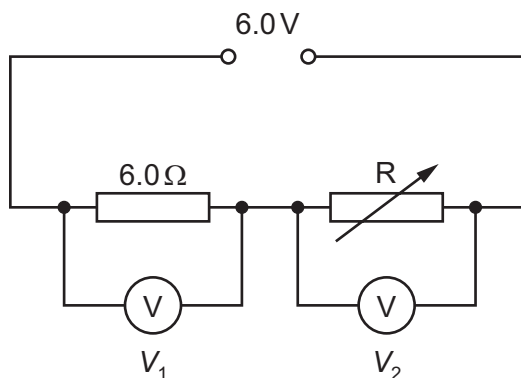


The particles are attracted by a charged plate.

Which row shows the charge on the plate, and the temperature of the metal filament?

	charge on plate	temperature of metal filament
A	negative	high
B	negative	low
C	positive	high
D	positive	low

- 36 A potential divider is connected across the terminals of a 6.0 V supply.



When R is adjusted to $6.0\ \Omega$, the voltmeter readings V_1 and V_2 are equal.

What happens to the readings when the resistance of R is then increased?

	V_1	V_2
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 37 Which nucleus is produced when americium-241 ($^{241}_{95}\text{Am}$) emits an alpha-particle?

A $^{237}_{93}\text{Np}$

B $^{237}_{97}\text{Bk}$

C $^{245}_{93}\text{Np}$

D $^{245}_{97}\text{Bk}$

- 38 Which states the three types of radiation emitted by radioactive isotopes in order of their ionising effect from highest to lowest?

A alpha-particles, beta-particles, gamma-rays

B alpha-particles, gamma-rays, beta-particles

C beta-particles, gamma-rays, alpha-particles

D gamma-rays, beta-particles, alpha-particles

- 39 Which statement about the half-life of a radioactive isotope is correct?

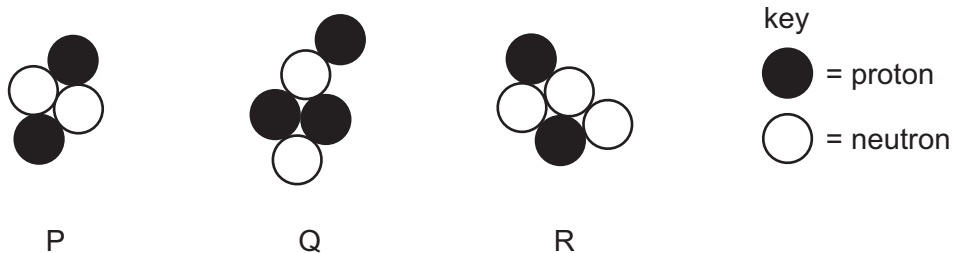
A Half-life changes as the isotope decays.

B Half-life is the time it takes for the nucleon number of the isotope to halve.

C Half-life is half the time it takes for the number of nuclei of the isotope to decrease to zero.

D Half-life is the time it takes for the number of nuclei of the isotope to decrease by half.

40 The diagrams represent the nuclei of three atoms.



Which are isotopes of the same element?

- A P and Q only
- B P and R only
- C Q and R only
- D P, Q and R

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

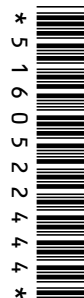
Paper 1 Multiple Choice

5054/11

May/June 2016

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

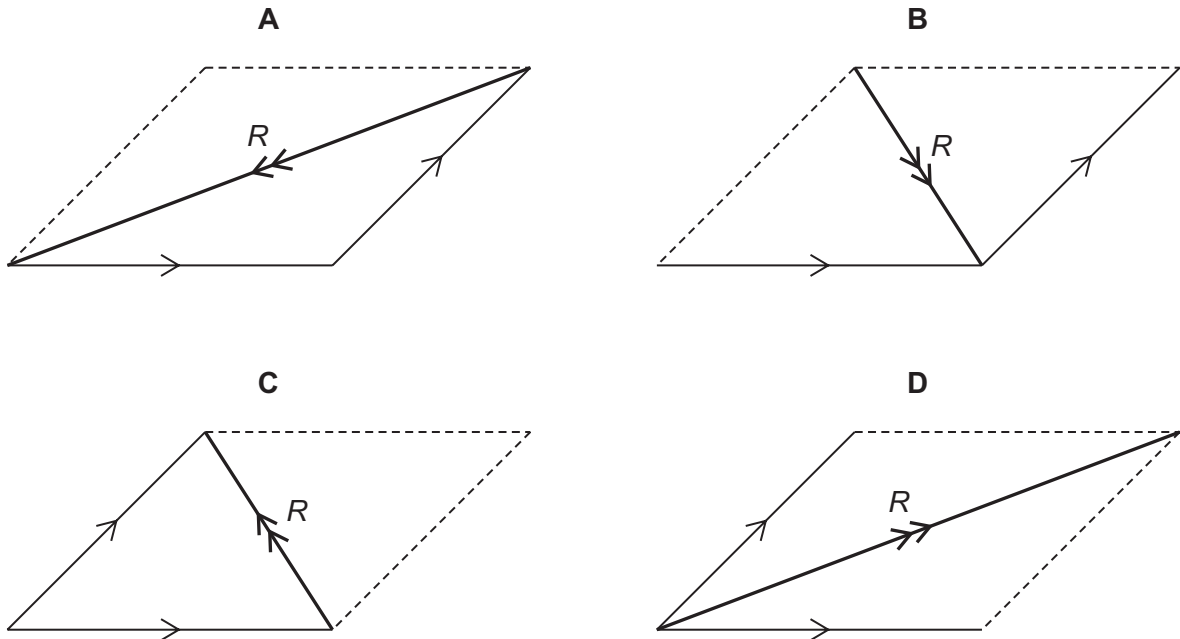
This document consists of **16** printed pages.



- 1 The diagram shows arrows representing two vector quantities.



Which diagram shows the resultant R of these two vectors?

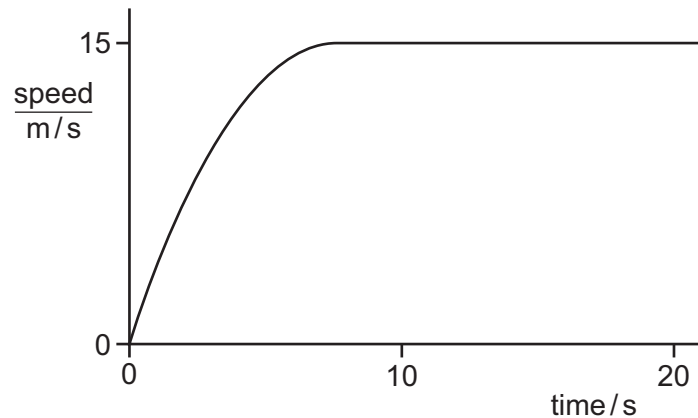


- 2 Which set of quantities are all vectors?
- A acceleration, displacement, velocity
 - B chemical energy, mass, power
 - C extension, force, gravitational potential energy
 - D weight, kinetic energy, work
- 3 A student determines the circumference of a golf ball.

Which instrument gives a reading that is the circumference of the golf ball?

- A calipers
- B micrometer
- C rule
- D tape

- 4 The graph shows how the speed of a car varies with time.



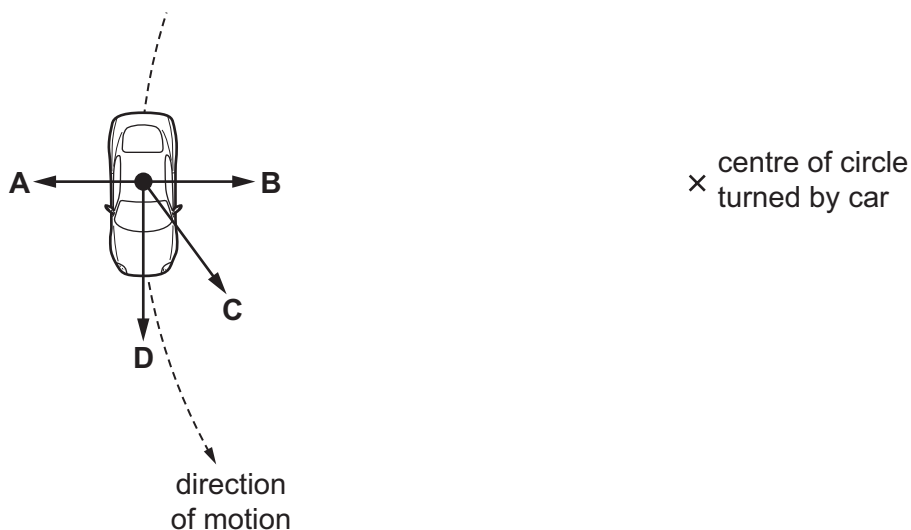
Which statement about the acceleration of the car between 10 s and 20 s is correct?

- A The acceleration decreases.
 - B The acceleration increases.
 - C The acceleration is constant, but not zero.
 - D The acceleration is zero.
- 5 A man pulls a sledge of mass 25 kg across level ground with a horizontal force of 60 N. A constant force of friction of 20 N acts on the sledge.

What is the acceleration of the sledge?

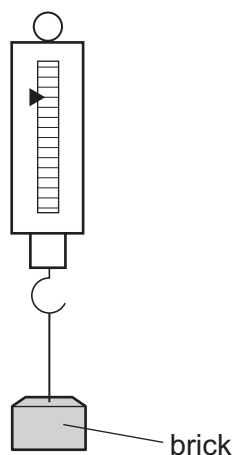
- A 0.63 m/s^2
 - B 1.6 m/s^2
 - C 2.4 m/s^2
 - D 3.2 m/s^2
- 6 A car moves in a circle at constant speed.

What is the direction of the resultant force acting on the car?

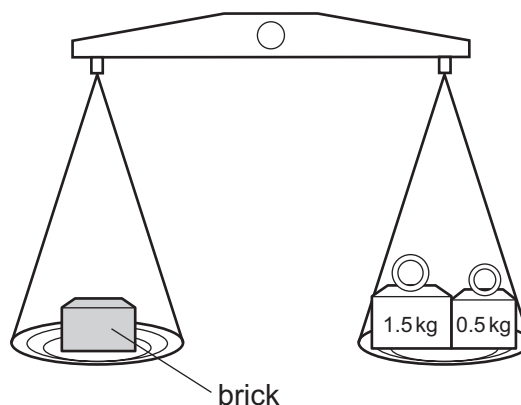


- 7 A brick is placed on a newton meter and then on a beam balance.

newton meter



beam balance



What is measured by each instrument?

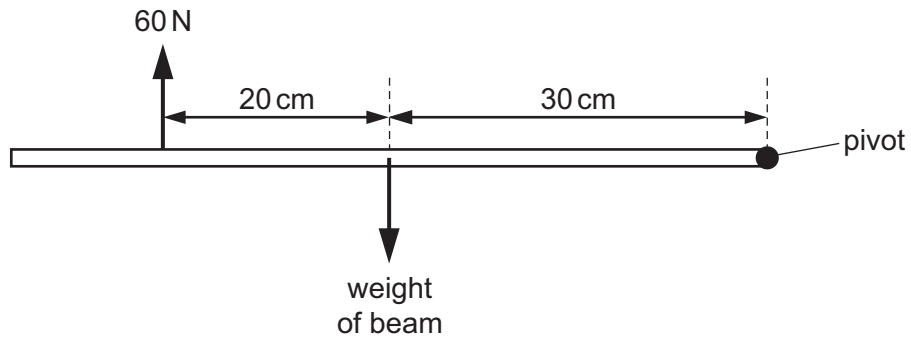
	newton meter	beam balance
A	mass	mass
B	mass	weight
C	weight	mass
D	weight	weight

- 8 A force acts on a body.

Which list contains only quantities that can be changed by the force?

- A** mass, shape, velocity
- B** mass, shape, volume
- C** mass, velocity, volume
- D** shape, velocity, volume

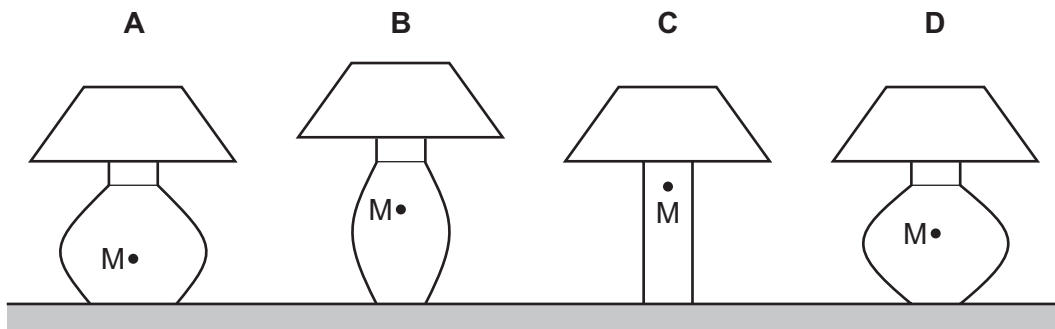
- 9 A uniform horizontal beam, pivoted at its right-hand end, is in equilibrium. A force of 60 N acts vertically upwards on the beam as shown.



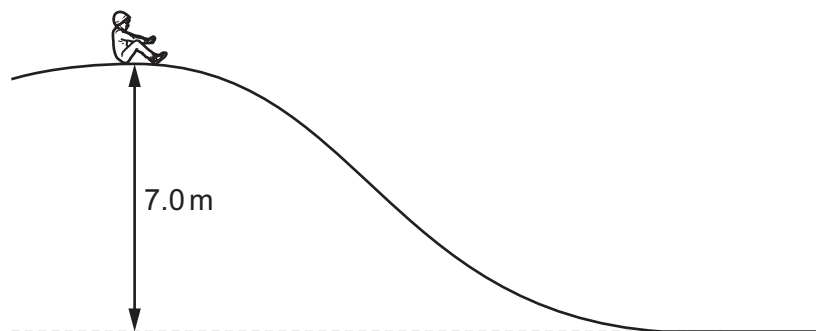
What is the weight of the beam?

- A 36 N B 40 N C 90 N D 100 N
- 10 Four table lamps are shown along with the position M of the centre of mass in each case.

Which lamp is the most stable?



- 11 A child slides down a slide.



The weight of the child is 250 N. The height of the slide is 7.0 m. The work done against friction as the child travels down the slide is 1300 J.

What is the change in gravitational potential energy and what is the final kinetic energy of the child?

	change in gravitational potential energy	final kinetic energy
A	1750	450
B	1750	1750
C	17 500	16 200
D	17 500	17 500

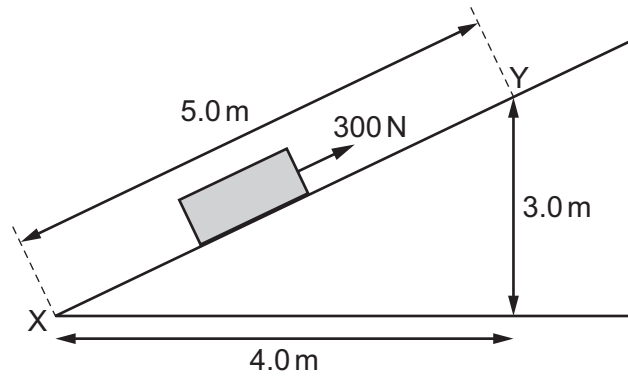
- 12 What uses non-renewable energy?

- A** a geothermal heating system
- B** a nuclear power station
- C** a solar panel
- D** a wind turbine

- 13 Which process in the Sun produces energy?

- A** burning
- B** nuclear fission
- C** nuclear fusion
- D** radiation

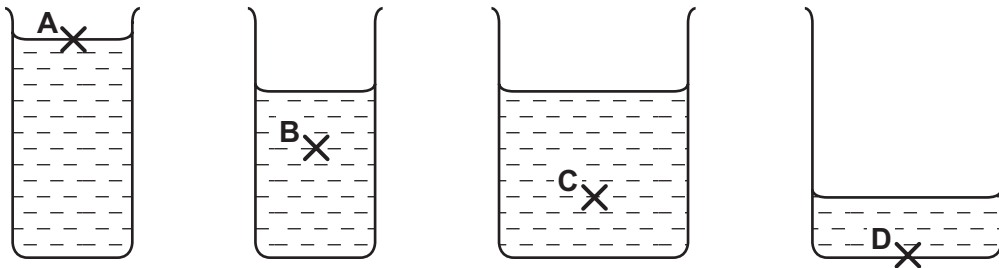
- 14 A 300 N force is applied to a box to move it up a ramp, as shown.



How much work is done by the force when moving the box from X to Y?

- A 900 J B 1200 J C 1500 J D 3000 J
- 15 Four beakers contain the same liquid.

At which point is the pressure the greatest?



- 16 Water of depth 10 m exerts a pressure equal to atmospheric pressure.

An air bubble rises to the surface of a lake which is 20 m deep. When the bubble reaches the surface, its volume is 6.0 cm^3 .

What is the volume of the air bubble at the bottom of the lake?

- A 2.0 cm^3 B 3.0 cm^3 C 12 cm^3 D 18 cm^3
- 17 A gas in a container of fixed volume is heated.
- What happens to the molecules of the gas?
- A They collide less frequently.
 B They expand.
 C They move faster.
 D They move further apart.

18 Which statement about thermal radiation is correct?

- A In a vacuum, thermal radiation travels at the speed of light.
- B Thermal radiation is a longitudinal wave.
- C Thermal radiation travels as an ultra-violet wave.
- D White surfaces are better emitters of thermal radiation than black surfaces.

19 For a transverse wave, what is a *wavefront*?

- A a line joining all points on the same crest of a wave
- B a line showing the displacement of a wave
- C the energy content of a wave
- D the first part of a wave to reach a point

20 A longitudinal wave passes along a spring. The coils of the spring vibrate from side to side.

The diagram shows the positions of the coils at one particular time.



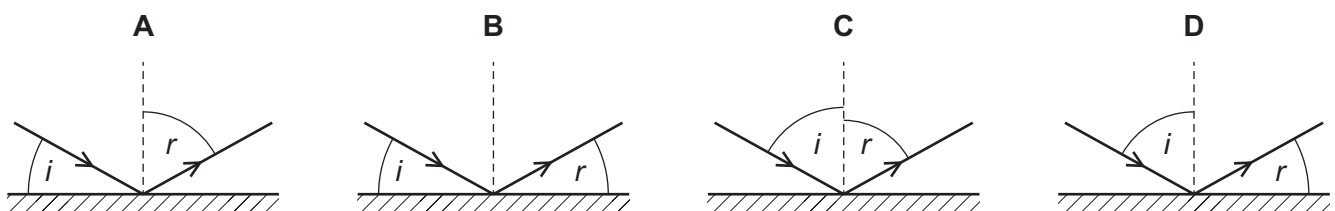
Which positions are one wavelength apart?

- A W and X
- B W and Z
- C X and Z
- D Y and Z

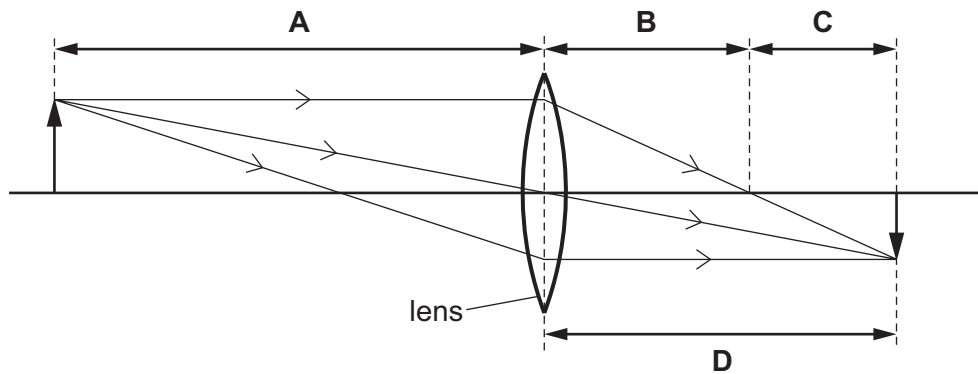
21 Light is incident on a mirror. The light is reflected from the mirror.

The angle of incidence is i and the angle of reflection is r .

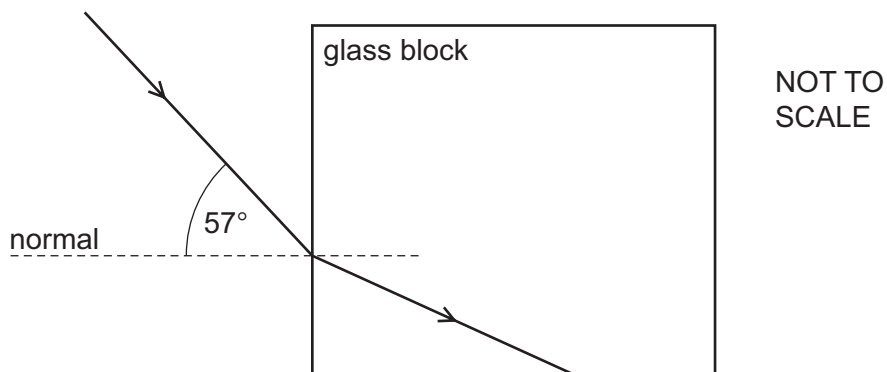
Which diagram correctly shows i and r ?



22 Which length is the focal length of the lens shown in the diagram?



23 Light passes from air into a glass block of refractive index 1.5, as shown.



What is the angle of refraction in the glass and what is the critical angle?

	angle of refraction	critical angle
A	34°	42°
B	34°	60°
C	38°	42°
D	38°	60°

24 Microwaves are used to transmit television signals to and from a satellite.

Which statement about microwaves is correct?

- A** They have a longer wavelength than radio waves.
- B** They penetrate the atmosphere without significant loss of energy.
- C** They travel much faster than radio waves in a vacuum.
- D** They warm the satellite and stop it freezing.

25 Where are gamma-rays used?

- A in fluorescent tubes
- B in killing cancerous cells
- C in pre-natal scanning
- D in sunbeds

26 An intruder alarm is adjusted to give a quieter sound without affecting the pitch of the note.

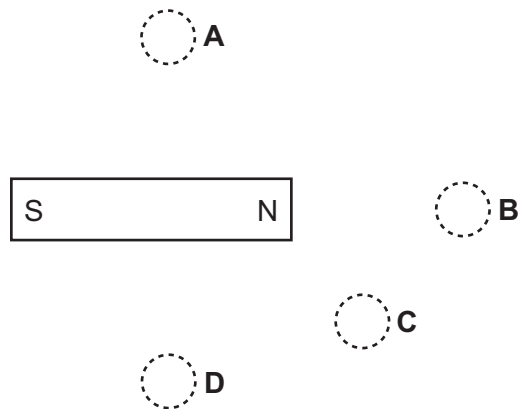
How are the amplitude and the frequency of the sound affected?

	amplitude	frequency
A	lower	lower
B	lower	same
C	same	lower
D	same	same

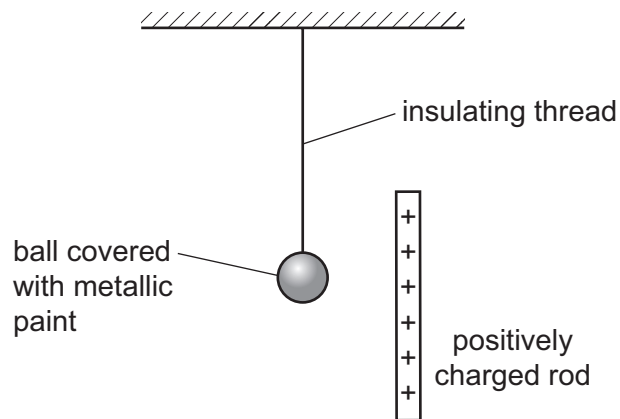
27 Four plotting compasses are placed near a bar magnet. You may ignore any effects of the Earth's magnetic field.

One compass appears like this .

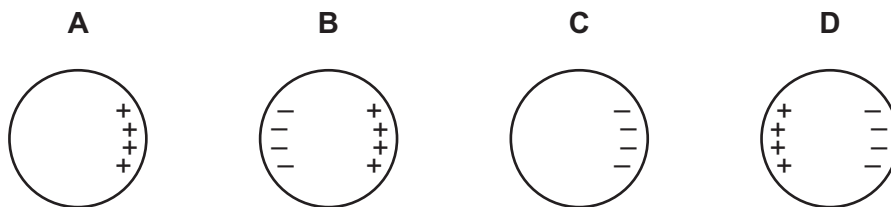
What is a possible position for this compass?



- 28 The diagram shows an uncharged ball coated with metallic paint. The ball is suspended from an insulating thread. It is placed near a positively charged rod.



Which diagram shows the charge distribution on the ball?



- 29 The current in a car headlamp is 2.0 A. The headlamp is switched on for 4.0 minutes.

How much charge passes through the headlamp?

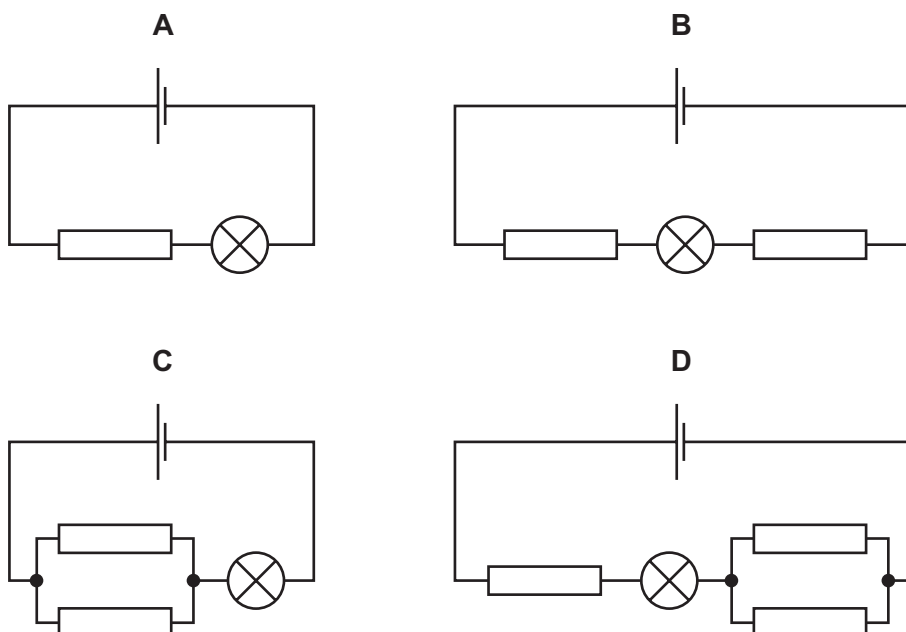
- A 8.0 C B 30 C C 120 C D 480 C

- 30 How can *one volt* also be expressed?

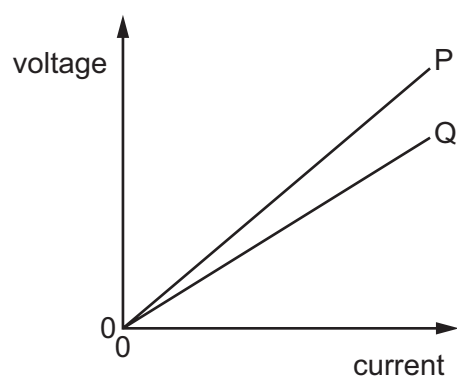
- A one coulomb per ampere
 B one coulomb per joule
 C one joule per ampere
 D one joule per coulomb

31 The cells, lamps and resistors in the circuits are identical.

In which circuit is the lamp the brightest?



32 The graph is the voltage-current graph for two resistance wires P and Q.



The wires are made from the same material and have equal lengths.

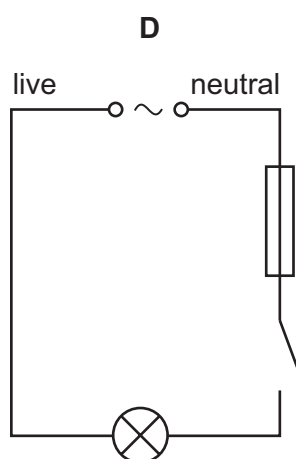
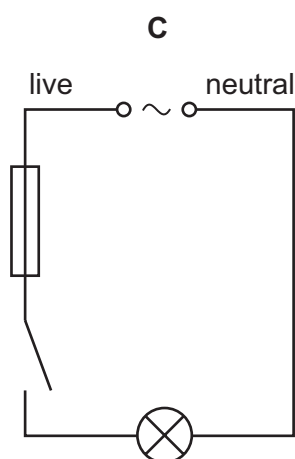
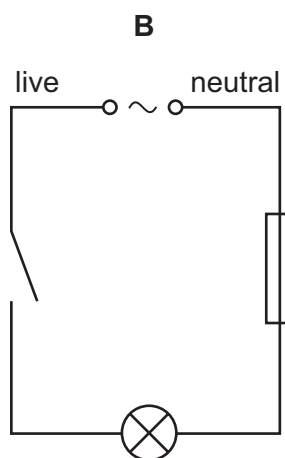
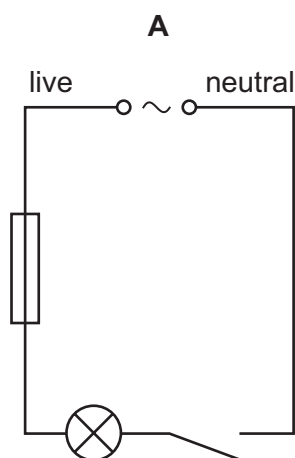
The resistances of the wires and their cross-sectional areas are different.

Which wire has the greater resistance and which wire has the larger cross-sectional area?

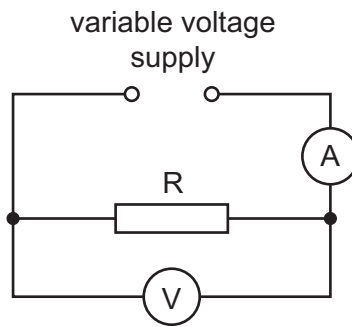
	greater resistance	larger cross-sectional area
A	P	P
B	P	Q
C	Q	P
D	Q	Q

33 A lamp is connected to the a.c. mains supply in series with a switch and a fuse.

Which circuit shows these components wired correctly?



- 34 A student is investigating resistance using the circuit shown.



The resistance of R is approximately 5Ω .

What are the most suitable ranges for the voltmeter and for the ammeter?

	voltmeter range	ammeter range
A	0–2 V	0–0.5 A
B	0–2 V	0–2 A
C	0–5 V	0–5 A
D	0–10 V	0–5 A

- 35 A split-ring commutator is used in a simple d.c. motor. It reverses the current in the coil.

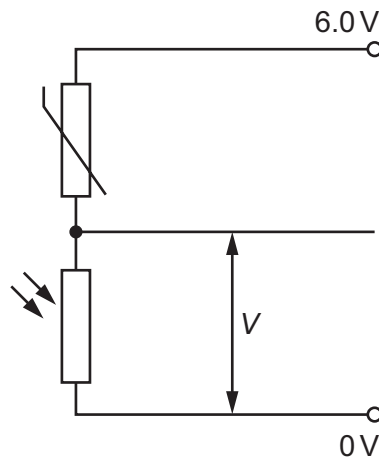
How often does it reverse the current?

- A** every quarter turn
- B** every half turn
- C** every full turn
- D** every two turns

- 36 Which component, when used in a circuit, allows current to pass in only one direction?



- 37 A potential divider consists of a thermistor and a light-dependent resistor (LDR).



Which conditions give the smallest voltage V across the LDR?

- A cold and dark
 - B cold and light
 - C hot and dark
 - D hot and light
- 38 Which type of radiation consists of electrons?
- A alpha-particles
 - B beta-particles
 - C gamma-rays
 - D X-rays
- 39 A sample of a radioactive isotope produces a count rate of 10 000 counts per minute. The half-life of the isotope is one day.
- What was the count rate of the sample two days ago?
- A 2500 counts per minute
 - B 5000 counts per minute
 - C 20 000 counts per minute
 - D 40 000 counts per minute

- 40** Which statement about a nucleus of $^{15}_7\text{N}$ is correct?
- A** The nucleus contains 7 neutrons and 8 protons.
 - B** The nucleus contains 7 neutrons and 15 protons.
 - C** The nucleus contains 7 protons and 8 neutrons.
 - D** The nucleus contains 7 protons and 15 neutrons.

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

5054/12

Paper 1 Multiple Choice

May/June 2016

MARK SCHEME

Maximum Mark: 40

Published

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Cambridge is publishing the mark schemes for the May/June 2016 series for most Cambridge IGCSE[®], Cambridge International A and AS Level components and some Cambridge O Level components.

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2016	5054	12

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	D	21	A
2	B	22	B
3	B	23	C
4	B	24	A
5	C	25	A
6	B	26	D
7	B	27	A
8	C	28	B
9	C	29	A
10	C	30	A
11	B	31	C
12	A	32	D
13	C	33	A
14	C	34	D
15	D	35	C
16	D	36	B
17	D	37	A
18	A	38	A
19	B	39	D
20	B	40	B



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

5054/11

Paper 1 Multiple Choice

May/June 2016

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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2	A	22	B
3	D	23	A
4	D	24	B
5	B	25	B
6	B	26	B
7	C	27	C
8	D	28	D
9	D	29	D
10	A	30	D
11	A	31	C
12	B	32	B
13	C	33	C
14	C	34	A
15	C	35	B
16	A	36	A
17	C	37	B
18	A	38	B
19	A	39	D
20	D	40	C



Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/12

May/June 2015

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

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Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

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This document consists of **15** printed pages and **1** blank page.



1 Which quantity is a scalar?

- A acceleration
- B force
- C temperature
- D velocity

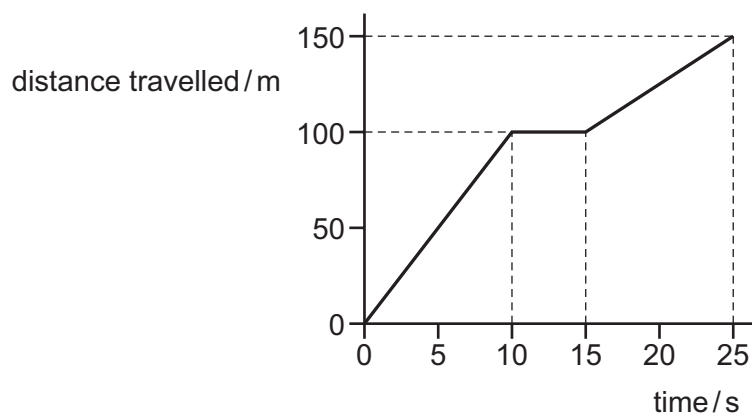
2 The diameter and the length of a thin wire, approximately 50 cm in length, are measured as precisely as possible.

What are the best instruments to use?

	diameter	length
A	micrometer	rule
B	micrometer	vernier calipers
C	rule	tape
D	vernier calipers	rule

3 A cyclist takes a ride lasting 25 s.

The diagram shows how her distance travelled from the starting position varies with time.



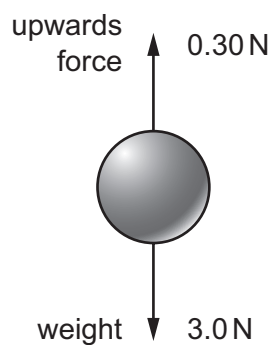
What is her average speed for the whole ride?

- A 6.0 m/s
- B 7.5 m/s
- C 10.0 m/s
- D 11.0 m/s

- 4 A car begins to move. It speeds up until it reaches a constant speed. It continues to travel at this constant speed for the rest of the journey.

What happens to the acceleration and what happens to the velocity of the car during the journey?

- A Both the acceleration and the velocity change.
 B Only the acceleration changes.
 C Only the velocity changes.
 D Neither the acceleration nor the velocity changes.
- 5 A metal ball of mass 0.30 kg and weight 3.0 N is held so that it is below the surface of oil. It experiences an upwards force of 0.30 N.

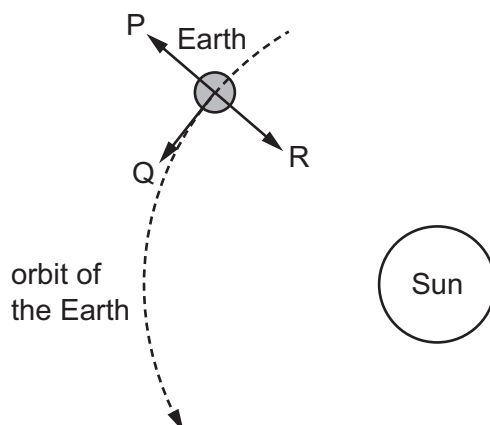


When the ball is released, what is its initial acceleration?

- A 1.0 m/s^2 B 9.0 m/s^2 C 10 m/s^2 D 11 m/s^2
- 6 A student drops, from rest, a table-tennis ball in air.
- What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

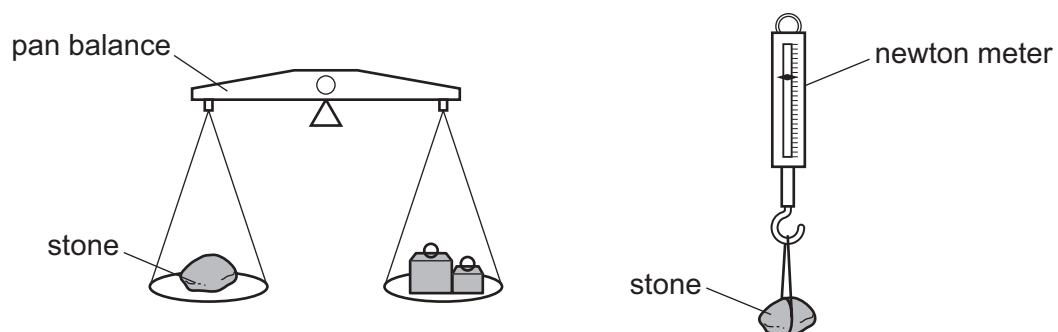
- 7 The Earth travels in a circular orbit around the Sun at constant speed.



Which arrows show the direction of the acceleration of the Earth and the direction of the velocity of the Earth?

	direction of acceleration	direction of velocity
A	P	Q
B	Q	P
C	Q	R
D	R	Q

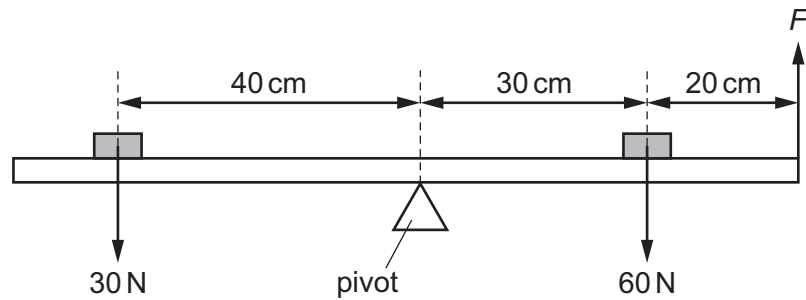
- 8 The mass of a stone is found on Earth using a pan balance. The weight of the stone is found using a newton meter.



Are the readings the same or different on the Moon?

	reading on pan balance	reading on newton meter
A	different	different
B	same	different
C	different	same
D	same	same

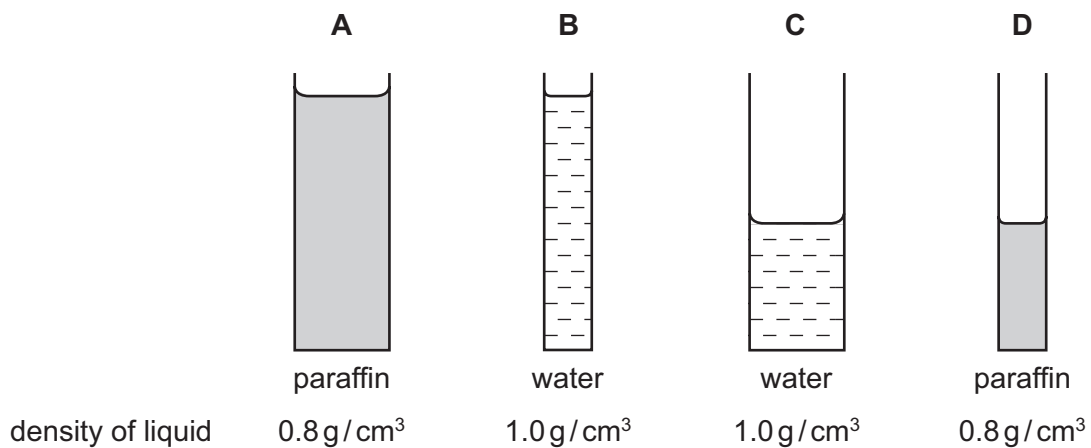
- 9 A uniform beam is pivoted at its centre. Two weights are placed on the beam in the positions shown and the beam is balanced by an upward force F .



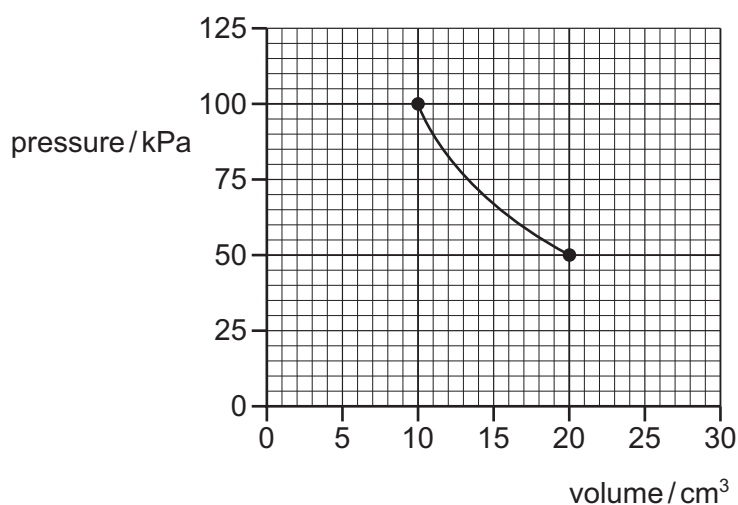
What is the size of F ?

- A 6 N B 12 N C 30 N D 60 N
- 10 The diagrams show liquids in containers.

Which column of liquid exerts the greatest pressure on the base of its container?



- 11 The graph shows how the pressure of a fixed mass of gas varies with volume at constant temperature.



What is the volume of the gas when the pressure is 25 kPa?

- A** 2.5 cm³ **B** 10 cm³ **C** 30 cm³ **D** 40 cm³
- 12 A rocket of total mass M is travelling at a speed v . The engine of the rocket is fired and fuel is used up. The mass of the rocket decreases to $M/2$ and its speed increases to $2v$.

What happens to the kinetic energy of the rocket?

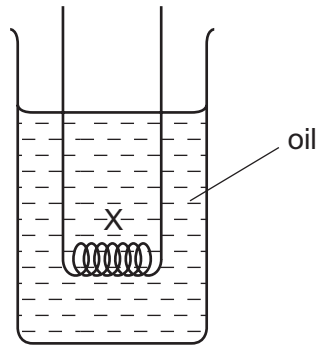
- A** it doubles
B it halves
C it increases by a factor of four
D it stays the same
- 13 A builder lifts eight slabs from the ground on to the back of a lorry 1.5 m high.

The total time taken is 48 s and each slab weighs 200 N.

How much useful power does the builder produce?

- A** 50 W **B** 400 W **C** 2400 W **D** 3200 W
- 14 A solid bar is heated at one end.
- How is thermal energy transferred to the other end of the bar?
- A** Heated molecules move along the bar, carrying energy to the other end.
B Heated molecules move along the bar, giving energy to others along the bar.
C Heated molecules stay completely still, but give energy to other molecules.
D Heated molecules vibrate more rapidly and pass energy to other molecules.

- 15 An electrical heater is placed in a beaker of cold oil, as shown.



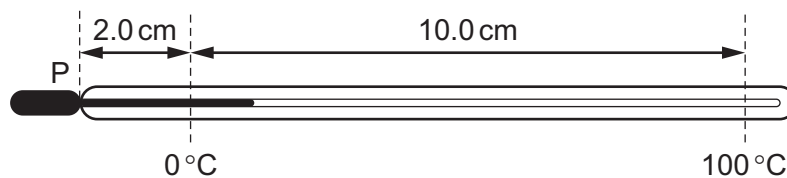
The heater is switched on.

What happens to the liquid at X?

- A It becomes less dense and falls.
 - B It becomes less dense and rises.
 - C It becomes more dense and falls.
 - D It becomes more dense and rises.
- 16 The tubes inside solar heating panels use the Sun's radiation to warm water.

Why are the tubes painted black?

- A Black surfaces absorb radiation well.
 - B Black surfaces conduct heat well.
 - C Black surfaces emit radiation well.
 - D Black surfaces reflect radiation well.
- 17 In a liquid-in-glass thermometer, the liquid column is 2.0 cm long at 0 °C and it expands 10.0 cm when heated to 100 °C.



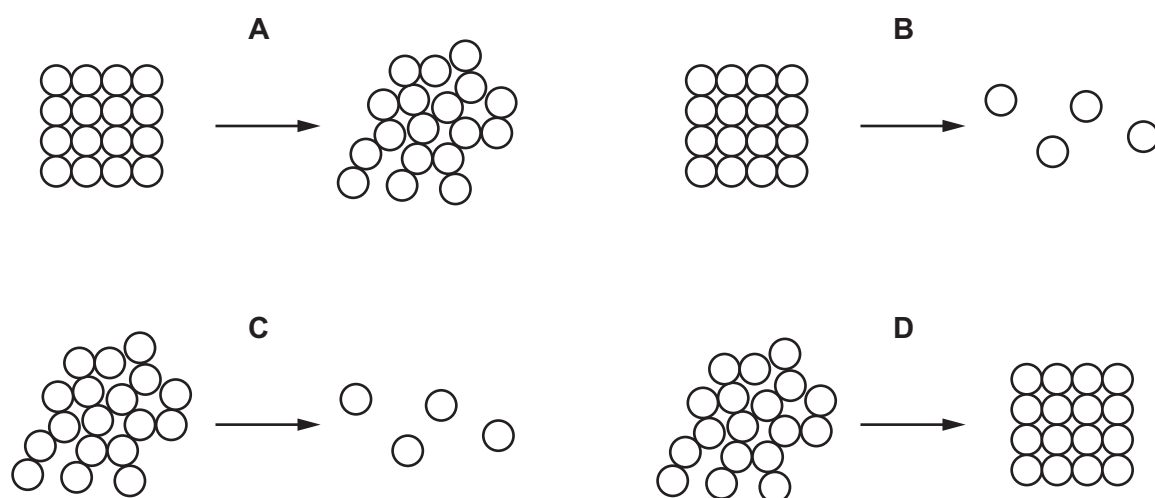
Measuring from P, how long is the liquid column at 30 °C?

- A 2.3 cm
- B 3.0 cm
- C 5.0 cm
- D 7.0 cm

18 Which substance in the table is liquid at 20 °C?

	melting point / °C	boiling point / °C
A	–218	–183
B	–39	357
C	44	280
D	119	444

19 Which diagram represents the change in the arrangement of the molecules in a solid as the substance melts?



20 Some gas is trapped in a closed container. The gas is cooled and the volume of the container is kept constant.

What happens to the gas molecules?

- A** They collide with the walls more often.
- B** They contract.
- C** They get closer together.
- D** They move more slowly.

- 21 In a liquid, some energetic molecules break free from the surface even when the liquid is too cold for bubbles to form.

What is the name of this process?

- A boiling
- B condensation
- C convection
- D evaporation

- 22 A pupil adds 37 g of ice at 0°C to 100 g of water at 30°C . The final temperature of the water and melted ice is 0°C . No heat is lost to, or gained from, the surroundings.

The specific heat capacity of water is $4.2\text{ J}/(\text{g }^{\circ}\text{C})$.

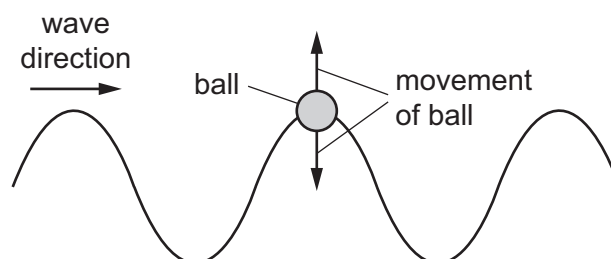
What is the specific latent heat of ice?

- A 47 J/g
- B 341 J/g
- C 4700 J/g
- D 12600 J/g

- 23 The heat capacity of an object, of mass 2.0 kg , is C . The energy needed to

- A increase the temperature of the whole object by Δt is $C\Delta t$.
- B increase the temperature of unit mass of the object by Δt is $C\Delta t$.
- C melt the whole object is C .
- D melt unit mass of the object is C .

- 24 A ball floating in a ripple tank begins to move vertically up and down as a wave passes beneath it. The ball does not move horizontally.

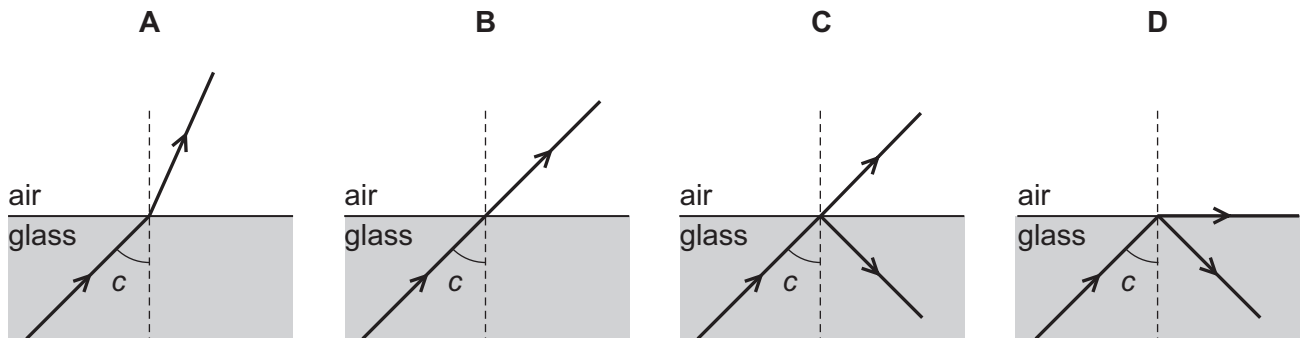


Which statement is correct?

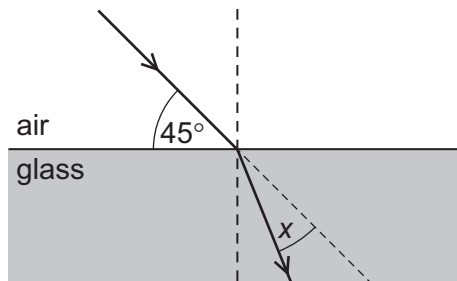
- A Both energy and water are transferred in the wave direction.
- B Energy is not transferred in the wave direction but water is.
- C Energy is transferred in the wave direction but water is not.
- D Neither energy nor water is transferred in the wave direction.

- 25** A ray of light in glass is incident on the surface at an angle c . The angle c is the critical angle.

Which diagram shows what happens to the light?



- 26** A ray of light is incident on the surface of a glass block, as shown in the diagram below.



The refractive index of the glass is 1.5.

The light ray changes direction when entering the glass.

What is the angle x through which the ray moves?

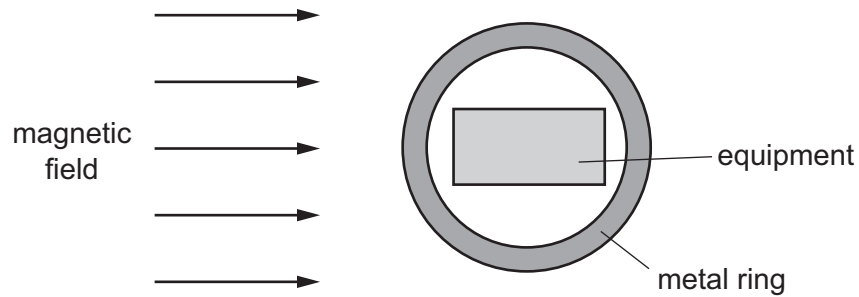
- A** 30° **B** 28° **C** 17° **D** 15°
- 27** An image is formed by a thin converging lens when it is used as a magnifying glass.

What is the correct description of the image?

- A** real and erect
B real and inverted
C virtual and erect
D virtual and inverted

- 28 Which component of the electromagnetic spectrum is used for television transmission from satellites?
- A microwaves
 - B radio waves
 - C ultra-violet
 - D X-rays
- 29 In which situation do sound waves **not** travel?
- A from a satellite in space to Earth
 - B from a ship to a submarine
 - C from an explosion underground to the surface
 - D through a balloon filled with helium gas
- 30 Two sound waves X and Y are compared.
- X has the greater frequency.
- Y has the greater amplitude.
- How do the loudness and pitch of sound wave Y compare to those of X?
- A Y is louder and higher pitch.
 - B Y is louder and lower pitch.
 - C Y is quieter and higher pitch.
 - D Y is quieter and lower pitch.

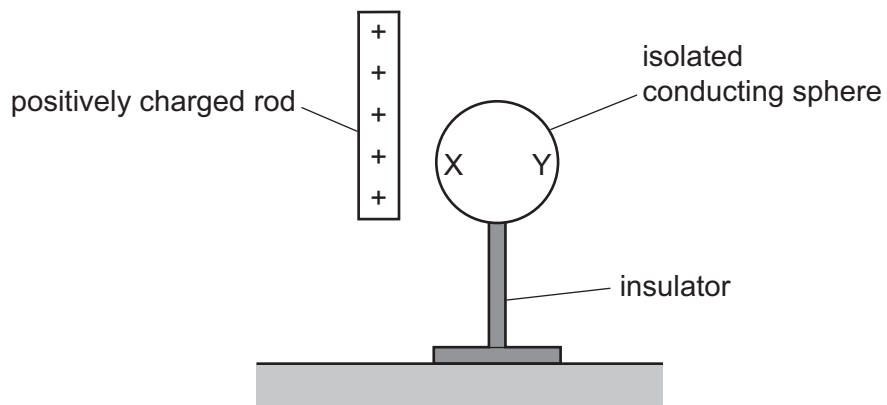
- 31 A metal ring screens a piece of equipment from a magnetic field.



Which metal should be used for the ring, and why?

	metal	reason
A	copper	the metal carries the field lines around the equipment
B	copper	the metal is non-magnetic
C	iron	the metal carries the field lines around the equipment
D	iron	the metal is non-magnetic

- 32 A positively charged rod is brought near to an isolated uncharged conducting sphere.

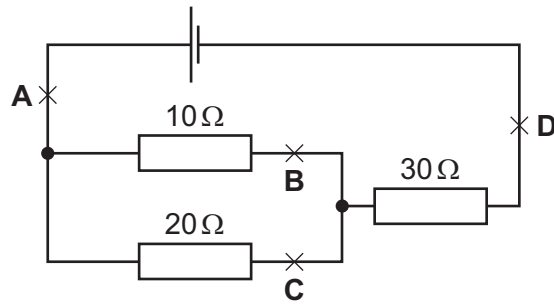


What are the charges on sides X and Y of the sphere?

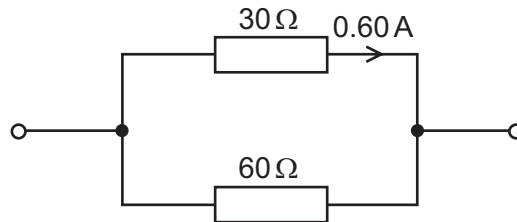
- A** Both X and Y are positively charged.
- B** Both X and Y are negatively charged.
- C** X is positively charged and Y is negatively charged.
- D** X is negatively charged and Y is positively charged.

- 33 The diagram shows a circuit.

Where must an ammeter be connected to measure the smallest current?



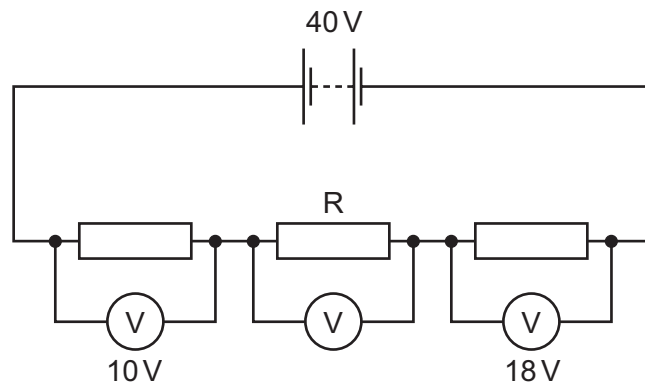
- 34 Two resistors of resistances $30\ \Omega$ and $60\ \Omega$ are arranged in parallel. The current in the $30\ \Omega$ resistor is $0.60\ \text{A}$.



What is the potential difference across the $60\ \Omega$ resistor?

- A** $9.0\ \text{V}$ **B** $18\ \text{V}$ **C** $36\ \text{V}$ **D** $54\ \text{V}$

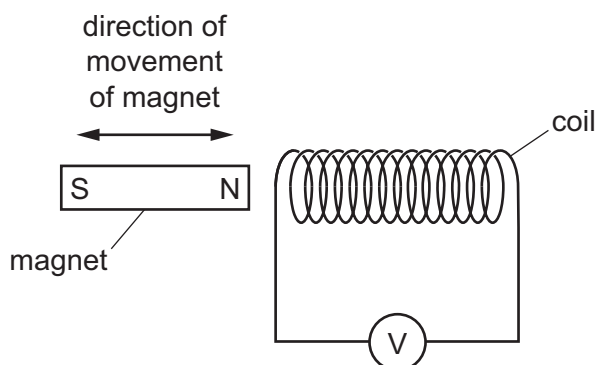
- 35 The circuit shows three resistors in series connected to a battery. Each resistor has a voltmeter across it and two of the voltages are shown.



What is the potential difference (p.d.) across the resistor R?

- A** $12\ \text{V}$ **B** $22\ \text{V}$ **C** $30\ \text{V}$ **D** $68\ \text{V}$

- 36 A teacher moves a magnet into and out of a coil of wire, as shown, in order to demonstrate electromagnetic induction.



Which statement is correct?

- A** As the magnet is moved into the coil the left-hand end of the coil becomes a S-pole.
- B** As the magnet is taken out of the coil the left-hand end of the coil becomes a N-pole.
- C** Increasing the speed at which the magnet enters the coil, increases the induced voltage.
- D** Increasing the speed at which the magnet leaves the coil decreases the induced voltage.

- 37 A transformer consists of two coils which are wound on to a metallic core.

Which type of voltage is supplied to the transformer and which metal is used to make the core?

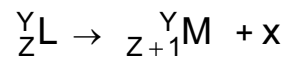
	supply voltage	metal
A	alternating	iron
B	alternating	steel
C	direct	iron
D	direct	steel

- 38 A nucleus of $^{215}_{84}\text{Po}$ decays by emitting an alpha-particle and the resulting nucleus then decays by emitting a beta-particle.

What is the nucleon number and proton number of the final nucleus?

	nucleon number	proton number
A	211	81
B	211	83
C	212	81
D	212	83

- 39 A radioactive material decays by this process:



What is particle x?

- A an electron
 - B a helium nucleus
 - C a neutron
 - D a proton
- 40 The half-life of a radioactive isotope is 24 hours. A sample of this isotope produces an initial count rate of 720 counts per second.

How long does it take for the count rate to fall to 90 counts per second?

- A 3 hours
- B 72 hours
- C 96 hours
- D 192 hours

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Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS

Paper 1 Multiple Choice

5054/11

May/June 2015

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



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Electronic calculators may be used.

This document consists of **15** printed pages and **1** blank page.



- 1 Is mass a scalar or a vector, and is acceleration a scalar or a vector?

	mass	acceleration
A	scalar	scalar
B	scalar	vector
C	vector	scalar
D	vector	vector

- 2 The diameter and the length of a thin wire, approximately 50 cm in length, are measured as precisely as possible.

What are the best instruments to use?

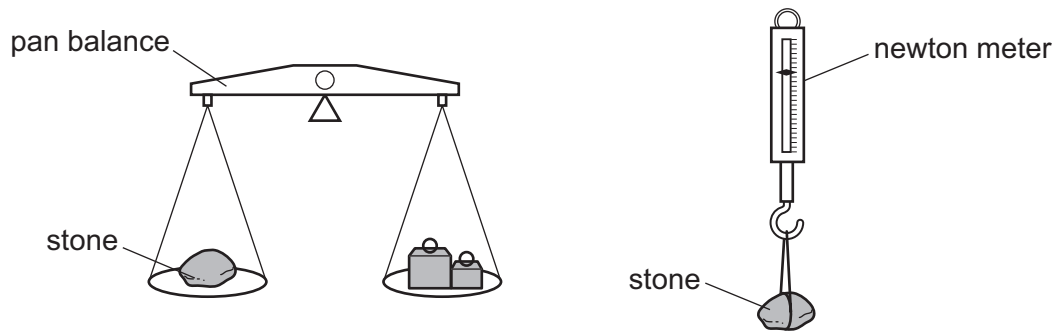
	diameter	length
A	micrometer	rule
B	micrometer	vernier calipers
C	rule	tape
D	vernier calipers	rule

- 3 A student walks at a constant speed. He takes 100 s to walk 160 paces. The length of each pace is 0.80 m.

How far does the student walk in 50 s?

- A** 64 m **B** 80 m **C** 128 m **D** 256 m

- 4 The mass of a stone is found on Earth using a pan balance. The weight of the stone is found using a newton meter.



Are the readings the same or different on the Moon?

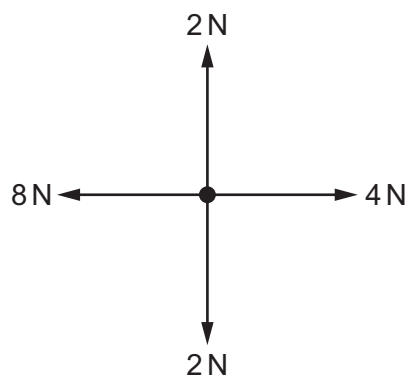
	reading on pan balance	reading on newton meter
A	different	different
B	same	different
C	different	same
D	same	same

- 5 The table shows the weights and masses of four objects on different planets.

On which planet is the gravitational field strength the largest?

	weight / N	mass / kg
A	2.0	20
B	4.0	30
C	6.0	40
D	8.0	50

- 6 Four forces act at a point as shown.



What is the size of the resultant force?

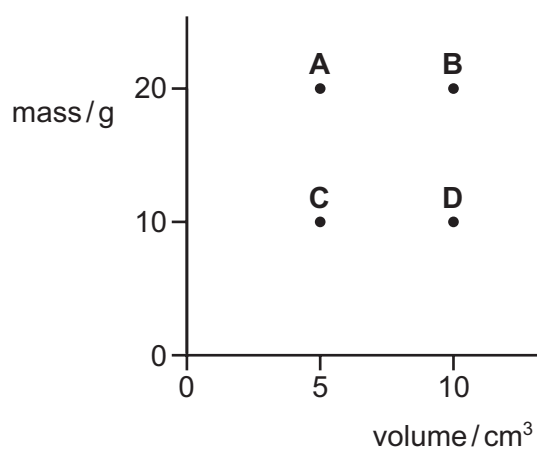
- A** 0 N **B** 4 N **C** 6 N **D** 8 N
- 7 A student drops, from rest, a table-tennis ball in air.

What happens to the velocity and to the acceleration of the ball during the first few seconds after release?

	velocity	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 8 The mass and volume of four different objects are plotted as shown.

Which object has the smallest density?



- 9 Newton's third law involves two quantities which are equal in size and opposite in direction.

What is the unit for these two quantities?

A J

B m/s^2

C N

D W

- 10 When a car turns a corner at speed, it risks toppling over. Two factors affecting the stability of a car are the height of its centre of mass and the distance between its front wheels.

Which factors make the car most stable?

	centre of mass	distance between front wheels
A	high	small
B	high	large
C	low	small
D	low	large

- 11 A cyclist travels along a horizontal track at constant speed.

The work done by the cyclist is equal to

A the change in kinetic energy.

B the force of air resistance.

C the force of friction in the bicycle.

D the thermal energy (heat) produced.

- 12 A 2.0 kg mass has 300 J of kinetic energy.

What is the speed of the mass?

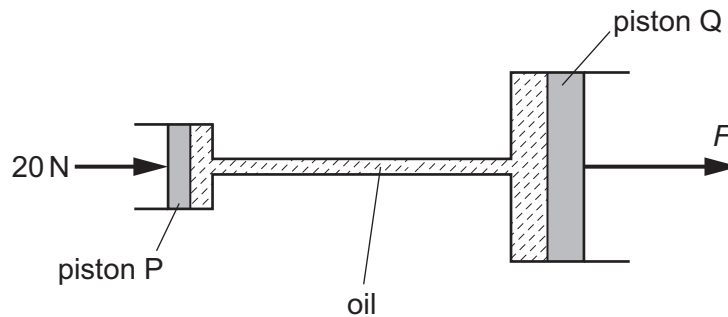
A 8.7 m/s

B 12 m/s

C 17 m/s

D 24 m/s

- 13 The diagram shows a simple model of the braking system of a car. A force of 20 N is applied to piston P. As a result, there is a force F on piston Q.



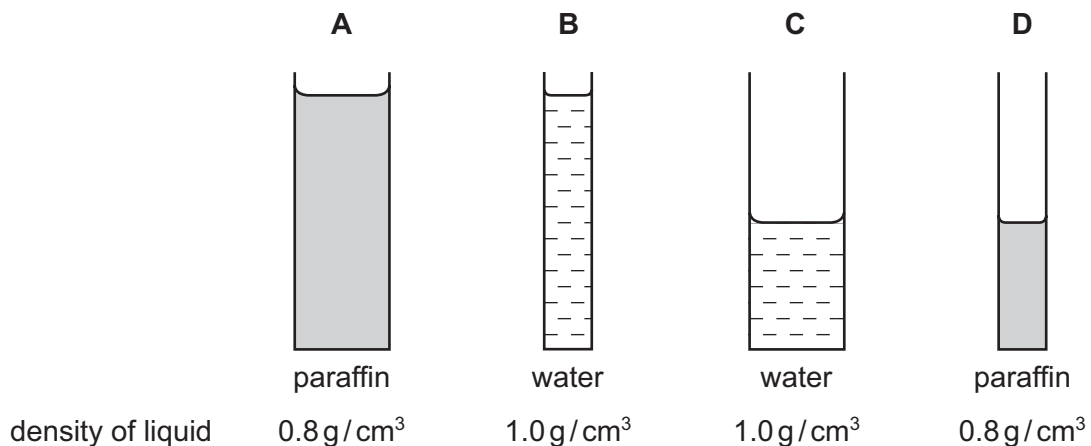
Piston P has an area of 5.0 cm^2 and piston Q has an area of 25 cm^2 .

What is the force F ?

- A** 4.0 N **B** 20 N **C** 100 N **D** 500 N

- 14 The diagrams show liquids in containers.

Which column of liquid exerts the greatest pressure on the base of its container?



- 15 A solid bar is heated at one end.

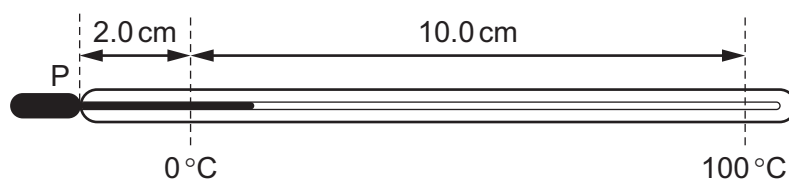
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- A** Heated molecules move along the bar, carrying energy to the other end.
B Heated molecules move along the bar, giving energy to others along the bar.
C Heated molecules stay completely still, but give energy to other molecules.
D Heated molecules vibrate more rapidly and pass energy to other molecules.

- 16 The tubes inside solar heating panels use the Sun's radiation to warm water.

Why are the tubes painted black?

- A Black surfaces absorb radiation well.
 - B Black surfaces conduct heat well.
 - C Black surfaces emit radiation well.
 - D Black surfaces reflect radiation well.
- 17 Why is there a constriction in a clinical thermometer?
- A to give the thermometer a smaller temperature range
 - B to make the thermometer more sensitive
 - C to prevent the mercury breaking the bulb when it expands
 - D to stop the mercury from going back to the bulb
- 18 In a liquid-in-glass thermometer, the liquid column is 2.0 cm long at 0°C and it expands 10.0 cm when heated to 100°C.



Measuring from P, how long is the liquid column at 30°C?

- A 2.3 cm
 - B 3.0 cm
 - C 5.0 cm
 - D 7.0 cm
- 19 Which substance in the table is liquid at 20°C?

	melting point /°C	boiling point /°C
A	-218	-183
B	-39	357
C	44	280
D	119	444

20 In which situation do sound waves **not** travel?

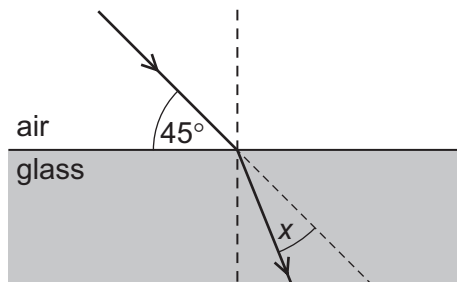
- A from a satellite in space to Earth
- B from a ship to a submarine
- C from an explosion underground to the surface
- D through a balloon filled with helium gas

21 A wave in a ripple tank passes from a deeper to a shallower region and refracts.

Which wave properties decrease as the wave enters the shallow region?

- A frequency only
- B speed only
- C frequency and wavelength
- D speed and wavelength

22 A ray of light is incident on the surface of a glass block, as shown in the diagram below.



The refractive index of the glass is 1.5.

The light ray changes direction when entering the glass.

What is the angle x through which the ray moves?

- A 30°
- B 28°
- C 17°
- D 15°

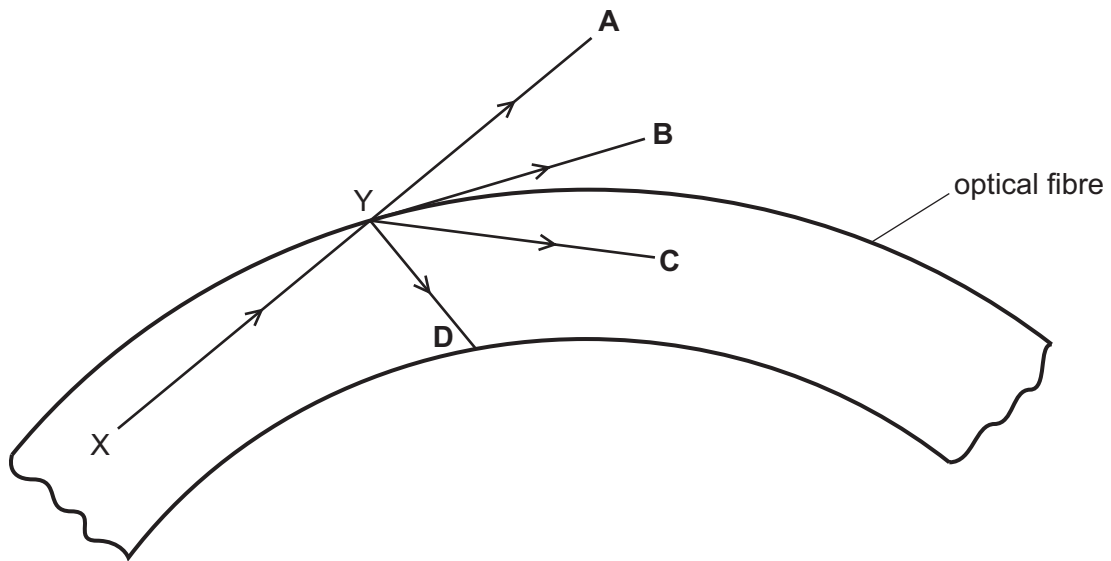
23 An image is formed by a thin converging lens when it is used as a magnifying glass.

What is the correct description of the image?

- A real and erect
- B real and inverted
- C virtual and erect
- D virtual and inverted

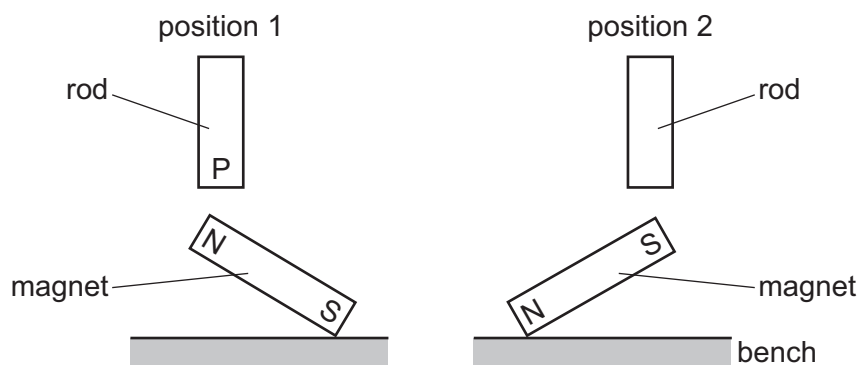
- 24** A ray of light travels from X to Y along an optical fibre. The angle of incidence at Y is greater than the critical angle.

In which direction does the ray of light travel after reaching point Y?



- 25** One end of a rod picks up the N-pole of a bar magnet when in position 1.

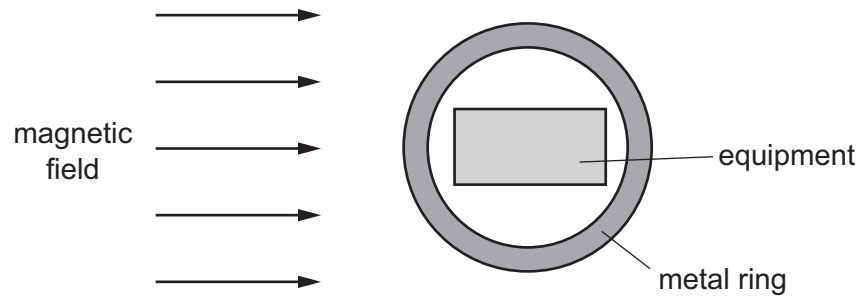
The same end of the rod picks up the S-pole of the bar magnet when in position 2.



Which material is the rod made from and what is the pole at end P of the rod when in position 1?

	material	pole at P
A	iron	N-pole
B	iron	S-pole
C	steel	N-pole
D	steel	S-pole

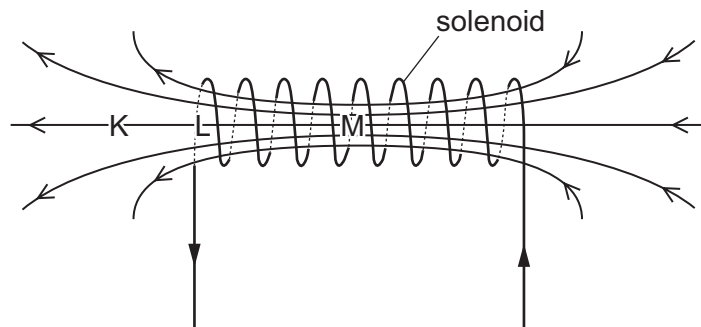
26 A metal ring screens a piece of equipment from a magnetic field.



Which metal should be used for the ring, and why?

	metal	reason
A	copper	the metal carries the field lines around the equipment
B	copper	the metal is non-magnetic
C	iron	the metal carries the field lines around the equipment
D	iron	the metal is non-magnetic

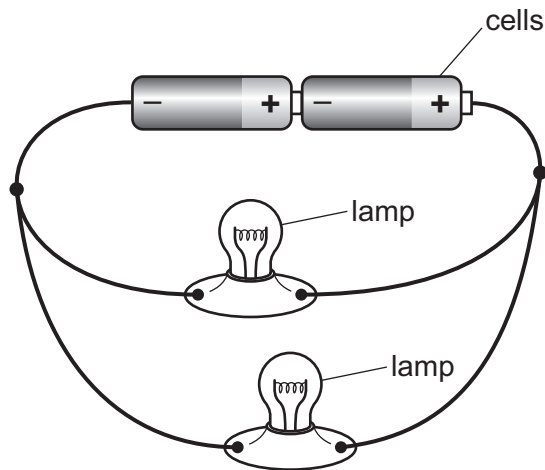
27 The diagram shows the magnetic field pattern of a current in a solenoid.



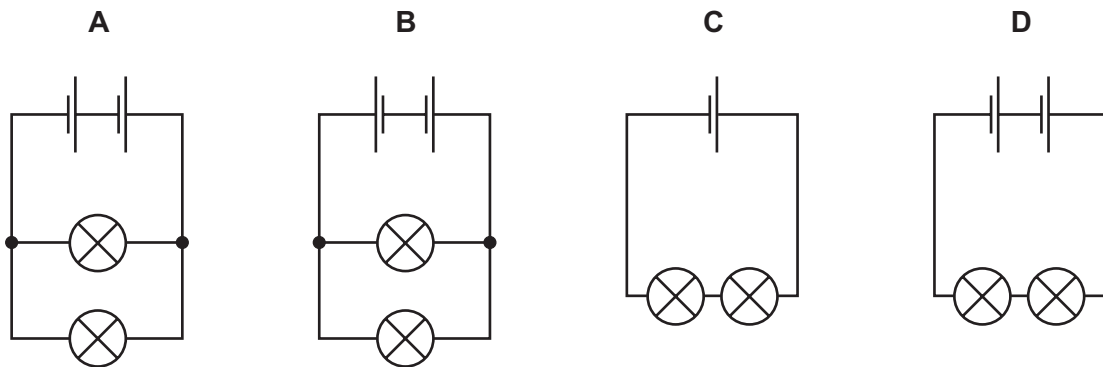
When the current in the solenoid is increased, where is there an increase in the magnetic field strength?

- A** K, L and M
- B** K and L only
- C** M and L only
- D** M only

28 The diagram shows a circuit.



Which circuit diagram shows this circuit?



29 A charge of 7.5 C flows through a resistor in 5.0 s . A student has ammeters with different ranges that he can use to measure the current in the resistor.

Which ammeter range is the most appropriate?

- A** 0-1 A **B** 0-2 A **C** 0-5 A **D** 0-40 A

30 Three identical cells are connected in parallel to a resistor.

What is the advantage of using three cells in parallel, rather than using a single cell?

- A** Each cell produces more energy.
B Each cell supplies more charge.
C Each cell takes longer to run down.
D The total electromotive force (e.m.f.) is larger.

- 31** When the flash on a camera is used, a charge of 1.5 C flows through the flash-tube in 0.0030 s .
The average voltage across the flash-tube is 3600 V .

What is the electrical energy supplied to the flash-tube and what is the average power supplied?

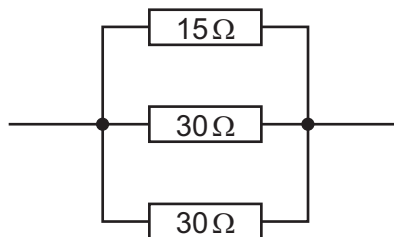
	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

- 32** A 1.0 m length of wire of cross-sectional area 0.40 mm^2 has a resistance of $2.0\ \Omega$.

What is the resistance of a 0.50 m length of wire, of the same material, with a cross-sectional area of 0.80 mm^2 ?

- A** $0.5\ \Omega$ **B** $1.0\ \Omega$ **C** $4.0\ \Omega$ **D** $8.0\ \Omega$

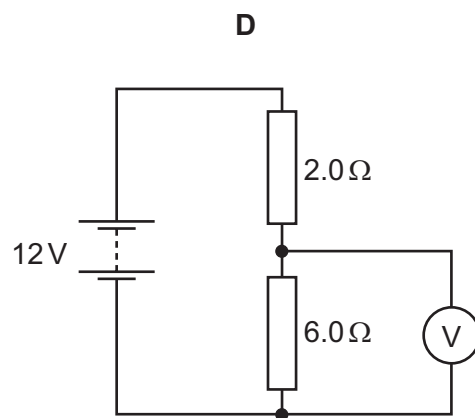
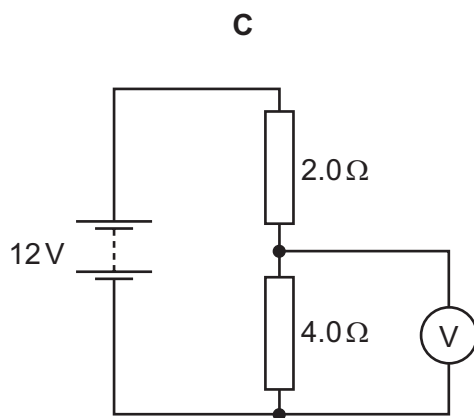
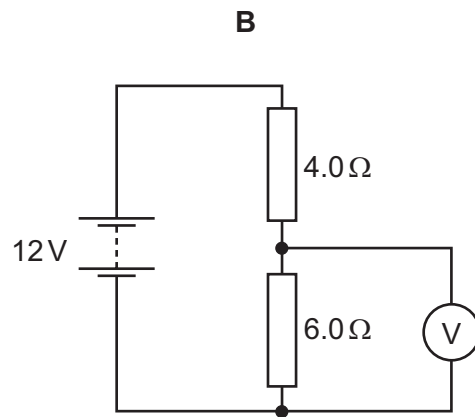
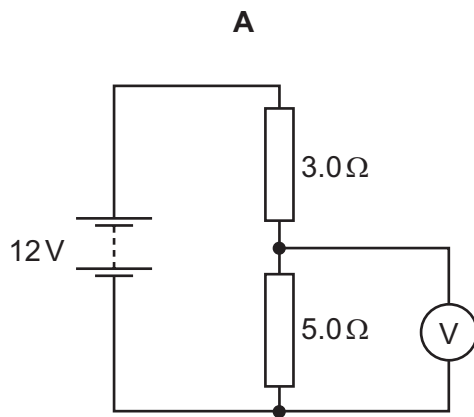
- 33** The diagram shows three resistors in parallel.



What is the combined resistance?

- A** $7.5\ \Omega$ **B** $15\ \Omega$ **C** $30\ \Omega$ **D** $75\ \Omega$

34 In which circuit is the voltmeter reading 7.2 V?



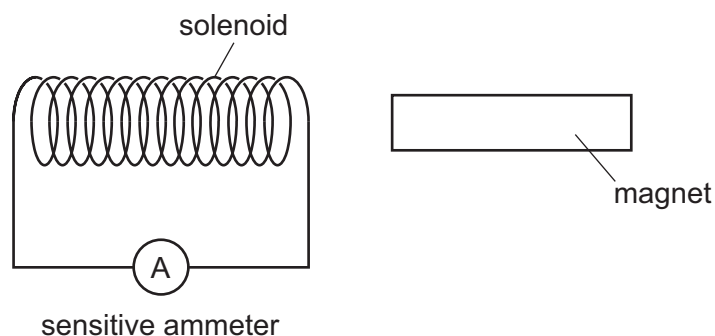
35 The diagram shows the information found on an electric kettle.

240 V	50 Hz
600 W	700 cm ³

What is the frequency of the electrical supply used to power the kettle?

- A** 50 Hz **B** 240 V **C** 600 W **D** 700 cm³

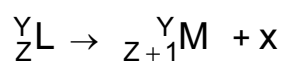
- 36 A magnet is placed near to a solenoid that is connected to a sensitive centre-zero ammeter.



The magnet is pushed towards the solenoid. It accelerates, then moves at constant speed, then decelerates and stops inside the solenoid.

When is the reading on the ammeter zero?

- A when the magnet is accelerating
 - B when the magnet is moving at constant speed
 - C when the magnet is decelerating
 - D when the magnet is stationary
- 37 A radioactive material decays by this process:



What is particle x?

- A an electron
 - B a helium nucleus
 - C a neutron
 - D a proton
- 38 A factory continuously produces plastic sheets. A radioactive isotope and a detector are used to check the thickness of the sheets.

What is the most suitable source to use?

- A an alpha source with a half-life of a few minutes
- B an alpha source with a half-life of several years
- C a beta source with a half-life of a few minutes
- D a beta source with a half-life of several years

- 39** A source contains a radioactive material.

Without the radioactive source present, a detector records a background count rate of 20 counts per minute.

This source is placed in a fixed position near the detector. Initially a count rate of 520 per minute is recorded.

What count rate is recorded after a time of **two** half-lives of the radioactive source?

- A** 125 counts per minute
- B** 130 counts per minute
- C** 135 counts per minute
- D** 145 counts per minute

- 40** A nucleus contains 94 protons and 240 nucleons. It emits an alpha-particle.

How many protons and how many neutrons are in the nucleus produced?

	number of protons	number of neutrons
A	90	144
B	90	236
C	92	144
D	92	236

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1	C	21	D
2	A	22	B
3	A	23	A
4	A	24	C
5	B	25	D
6	C	26	C
7	D	27	C
8	B	28	A
9	B	29	A
10	B	30	B
11	D	31	C
12	A	32	D
13	A	33	C
14	D	34	B
15	B	35	A
16	A	36	C
17	C	37	A
18	B	38	B
19	A	39	A
20	D	40	B

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6	B	26	C
7	C	27	A
8	D	28	A
9	C	29	B
10	D	30	C
11	D	31	D
12	C	32	A
13	C	33	A
14	B	34	B
15	D	35	A
16	A	36	D
17	D	37	A
18	C	38	D
19	B	39	D
20	A	40	C